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Edited by **Kenji Suzuki**

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**Artificial Neural Networks
- Methodological Advances and Biomedical Applications**

Edited by Kenji Suzuki

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Preface

Artificial neural networks may probably be the single most successful technology in the last two decades which has been widely used in a large variety of applications in various areas. An artificial neural network, often just called a neural network, is a mathematical (or computational) model that is inspired by the structure and function of biological neural networks in the brain. An artificial neural network consists of a number of artificial neurons (i.e., nonlinear processing units) which are connected each other via synaptic weights (or simply just weights). An artificial neural network can “learn” a task by adjusting weights. There are supervised and unsupervised models. A supervised model requires a “teacher” or desired (ideal) output to learn a task. An unsupervised model does not require a “teacher,” but it learns a task based on a cost function associated with the task. An artificial neural network is a powerful, versatile tool. Artificial neural networks have been successfully used in various applications such as biological, medical, industrial, control engineering, software engineering, environmental, economical, and social applications. The high versatility of artificial neural networks comes from its high capability and learning function. It has been theoretically proved that an artificial neural network can approximate any continuous mapping by arbitrary precision. Desired continuous mapping or a desired task is acquired in an artificial neural network by learning.

The purpose of this book series is to provide recent advances of artificial neural network applications in a wide range of areas. The series consists of two volumes: the first volume contains methodological advances and biomedical applications of artificial neural networks; the second volume contains artificial neural network applications in industrial and control engineering.

This first volume begins with a section of fundamentals of artificial neural networks which covers an introduction, design, and optimization of artificial neural networks. The fundamental concept, principles, and theory in the section help understand and use an artificial neural network in a specific application properly as well as effectively. A section of advanced architectures for biomedical applications follows. Researchers have developed advanced architectures for artificial neural networks specifically for biomedical applications. Such advanced architectures offer improved performance and desirable properties. Sections continue with biological applications such as gene, plant biology, and stem cell, medical applications such as skin diseases, sclerosis, anesthesia, and physiotherapy, and clinical and other applications such as clinical outcome, telecare, and pre-med student failure prediction.

Thus, this book will be a fundamental source of recent advances and applications of artificial neural networks in biomedical areas. The target audience of this book includes professors, college students, and graduate students in engineering and medical schools, engineers in biomedical companies, researchers in biomedical and health sciences, medical doctors such as radiologists, cardiologists, pathologists, and surgeons, healthcare professionals such as radiology technologists and medical physicists. I hope this book will be a useful source for readers and inspire them.

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Part 1

Fundamentals

Introduction to the Artificial Neural Networks

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1. Introduction

An Artificial Neural Network (ANN) is a mathematical model that tries to simulate the structure and functionalities of biological neural networks. Basic building block of every artificial neural network is artificial neuron, that is, a simple mathematical model (function). Such a model has three simple sets of rules: multiplication, summation and activation. At the entrance of artificial neuron the inputs are weighted what means that every input value is multiplied with individual weight. In the middle section of artificial neuron is sum function that sums all weighted inputs and bias. At the exit of artificial neuron the sum of previously weighted inputs and bias is passing through activation function that is also called transfer function (Fig. 1.).

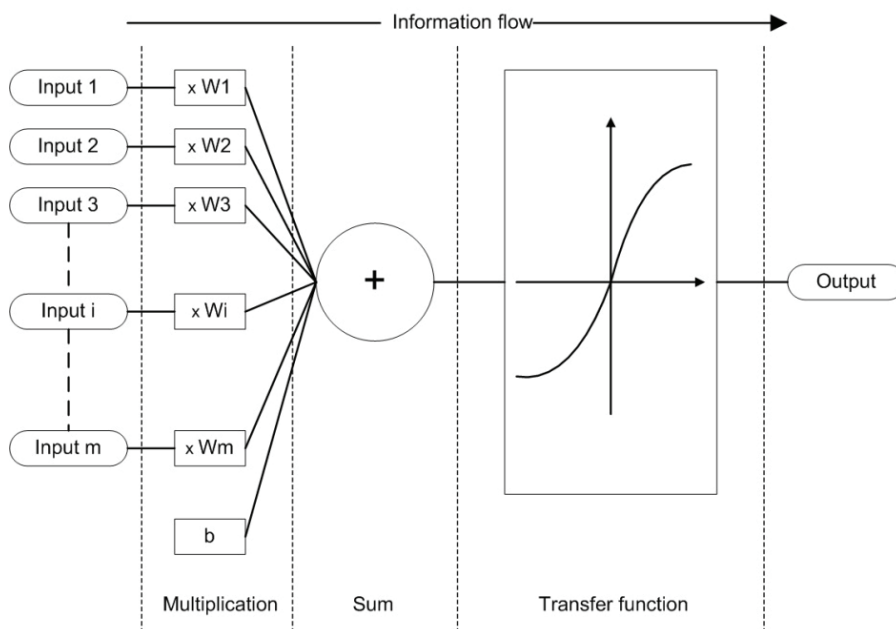


Fig. 1. Working principle of an artificial neuron.

Although the working principles and simple set of rules of artificial neuron looks like nothing special the full potential and calculation power of these models come to life when we start to interconnect them into artificial neural networks (Fig. 2.). These artificial neural networks use simple fact that complexity can grown out of merely few basic and simple rules.

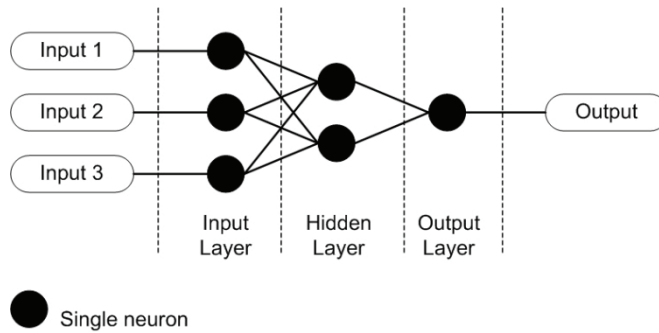


Fig. 2. Example of simple artificial neural network.

In order to fully harvest the benefits of mathematical complexity that can be achieved through interconnection of individual artificial neurons and not just making system complex and unmanageable we usually do not interconnect these artificial neurons randomly. In the past, researchers have come up with several “standardised” topographies of artificial neural networks. These predefined topographies can help us with easier, faster and more efficient problem solving. Different types of artificial neural network topographies are suited for solving different types of problems. After determining the type of given problem we need to decide for topology of artificial neural network we are going to use and then fine-tune it. We need to fine-tune the topology itself and its parameters.

Fine tuned topology of artificial neural network does not mean that we can start using our artificial neural network, it is only a precondition. Before we can use our artificial neural network we need to teach it solving the type of given problem. Just as biological neural networks can learn their behaviour/responses on the basis of inputs that they get from their environment the artificial neural networks can do the same. There are three major learning paradigms: supervised learning, unsupervised learning and reinforcement learning. We choose learning paradigm similar as we chose artificial neuron network topography - based on the problem we are trying to solve. Although learning paradigms are different in their principles they all have one thing in common; on the basis of “learning data” and “learning rules” (chosen cost function) artificial neural network is trying to achieve proper output response in accordance to input signals.

After choosing topology of an artificial neural network, fine-tuning of the topology and when artificial neural network has learn a proper behaviour we can start using it for solving given problem. Artificial neural networks have been in use for some time now and we can find them working in areas such as process control, chemistry, gaming, radar systems, automotive industry, space industry, astronomy, genetics, banking, fraud detection, etc. and solving of problems like function approximation, regression analysis, time series prediction, classification, pattern recognition, decision making, data processing, filtering, clustering, etc., naming a few.

As topic of artificial neural networks is complex and this chapter is only informative nature we encourage novice reader to find detail information on artificial neural networks in (Gurney, 1997; Kröse & Smagt 1996; Pavešić, 2000; Rojas 1996).

2. Artificial neuron

Artificial neuron is a basic building block of every artificial neural network. Its design and functionalities are derived from observation of a biological neuron that is basic building block of biological neural networks (systems) which includes the brain, spinal cord and peripheral ganglia. Similarities in design and functionalities can be seen in Fig.3. where the left side of a figure represents a biological neuron with its soma, dendrites and axon and where the right side of a figure represents an artificial neuron with its inputs, weights, transfer function, bias and outputs.

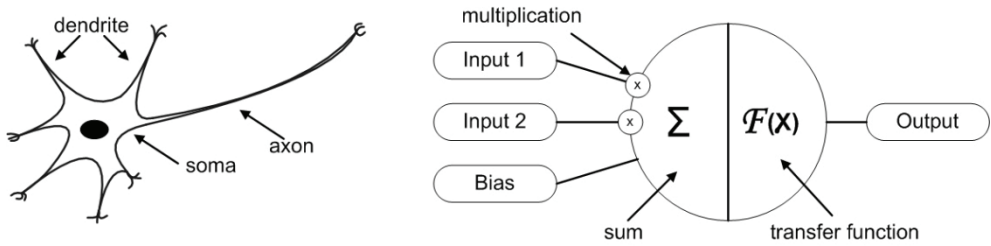


Fig. 3. Biological and artificial neuron design.

In case of biological neuron information comes into the neuron via dendrite, soma processes the information and passes it on via axon. In case of artificial neuron the information comes into the body of an artificial neuron via inputs that are weighted (each input can be individually multiplied with a weight). The body of an artificial neuron then sums the weighted inputs, bias and “processes” the sum with a transfer function. At the end an artificial neuron passes the processed information via output(s). Benefit of artificial neuron model simplicity can be seen in its mathematical description below:

$$y(k) = F \left(\sum_{i=0}^m w_i(k) \cdot x_i(k) + b \right) \quad (1)$$

Where:

- $x_i(k)$ is input value in discrete time k where i goes from 0 to m ,
- $w_i(k)$ is weight value in discrete time k where i goes from 0 to m ,
- b is bias,
- F is a transfer function,
- $y_i(k)$ is output value in discrete time k .

As seen from a model of an artificial neuron and its equation (1) the major unknown variable of our model is its transfer function. Transfer function defines the properties of artificial neuron and can be any mathematical function. We choose it on the basis of problem that artificial neuron (artificial neural network) needs to solve and in most cases we choose it from the following set of functions: *Step function*, *Linear function* and *Non-linear (Sigmoid) function*.

Step function is binary function that has only two possible output values (e.g. zero and one). That means if input value meets specific threshold the output value results in one value and if specific threshold is not met that results in different output value. Situation can be described with equation (2).

$$y = \begin{cases} 1 & \text{if } w_i x_i \geq \text{threshold} \\ 0 & \text{if } w_i x_i < \text{threshold} \end{cases} \quad (2)$$

When this type of transfer function is used in artificial neuron we call this artificial neuron *perceptron*. Perceptron is used for solving classification problems and as such it can be most commonly found in the last layer of artificial neural networks. In case of linear transfer function artificial neuron is doing simple linear transformation over the sum of weighted inputs and bias. Such an artificial neuron is in contrast to perceptron most commonly used in the input layer of artificial neural networks. When we use non-linear function the sigmoid function is the most commonly used. Sigmoid function has easily calculated derivate, which can be important when calculating weight updates in the artificial neural network.

3. Artificial Neural Networks

When combining two or more artificial neurons we are getting an artificial neural network. If single artificial neuron has almost no usefulness in solving real-life problems the artificial neural networks have it. In fact artificial neural networks are capable of solving complex real-life problems by processing information in their basic building blocks (artificial neurons) in a non-linear, distributed, parallel and local way.

The way that individual artificial neurons are interconnected is called topology, architecture or graph of an artificial neural network. The fact that interconnection can be done in numerous ways results in numerous possible topologies that are divided into two basic classes. Fig. 4. shows these two topologies; the left side of the figure represent simple feed-forward topology (acyclic graph) where information flows from inputs to outputs in only one direction and the right side of the figure represent simple recurrent topology (semi-cyclic graph) where some of the information flows not only in one direction from input to output but also in opposite direction. While observing Fig. 4. we need to mention that for easier handling and mathematical describing of an artificial neural network we group individual neurons in layers. On Fig. 4. we can see input, hidden and output layer.

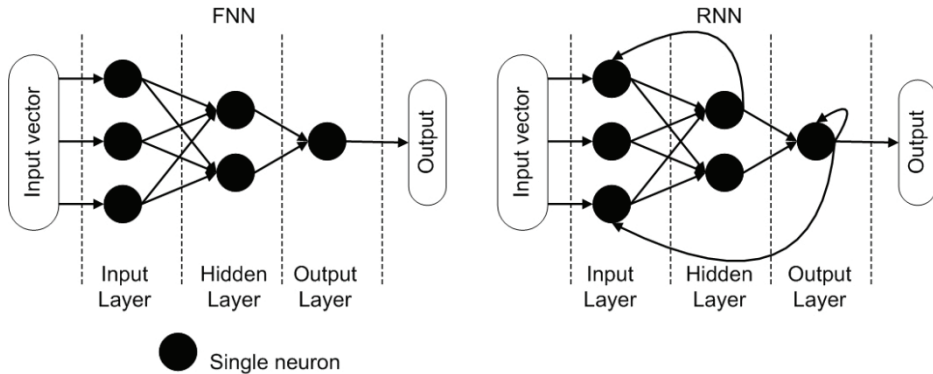


Fig. 4. Feed-forward (FNN) and recurrent (RNN) topology of an artificial neural network.

When we choose and build topology of our artificial neural network we only finished half of the task before we can use this artificial neural network for solving given problem. Just as biological neural networks need to learn their proper responses to the given inputs from the environment the artificial neural networks need to do the same. So the next step is to learn proper response of an artificial neural network and this can be achieved through learning (supervised, un-supervised or reinforcement learning). No matter which method we use, the task of learning is to set the values of weight and biases on basis of learning data to minimize the chosen cost function.

3.1 Feed-forward Artificial Neural Networks

Artificial neural network with feed-forward topology is called *Feed-Forward* artificial neural network and as such has only one condition: information must flow from input to output in only one direction with no back-loops. There are no limitations on number of layers, type of transfer function used in individual artificial neuron or number of connections between individual artificial neurons. The simplest feed-forward artificial neural network is a single perceptron that is only capable of learning linear separable problems. Simple multi-layer feed-forward artificial neural network for purpose of analytical description (sets of equations (3), (4) and (5)) is shown on Fig. 5.

$$\begin{aligned} n_1 &= F_1(w_1x_1 + b_1) \\ n_2 &= F_2(w_2x_2 + b_2) \\ n_3 &= F_2(w_2x_2 + b_2) \\ n_4 &= F_3(w_3x_3 + b_3) \end{aligned} \quad (3)$$

$$\begin{aligned} m_1 &= F_4(q_1n_1 + q_2n_2 + b_4) \\ m_2 &= F_5(q_3n_3 + q_4n_4 + b_5) \end{aligned} \quad (4)$$

$$y = F_6(r_1m_1 + r_2m_2 + b_6)$$

$$y = F_6 \left[r_1(F_4[q_1F_1[w_1x_1 + b_1] + q_2F_2[w_2x_2 + b_2]] + b_4) + \dots \right. \\ \left. \dots + r_2(F_5[q_3F_2[w_2x_2 + b_2] + q_4F_3[w_3x_3 + b_3] + b_5]) + b_6 \right] \quad (5)$$

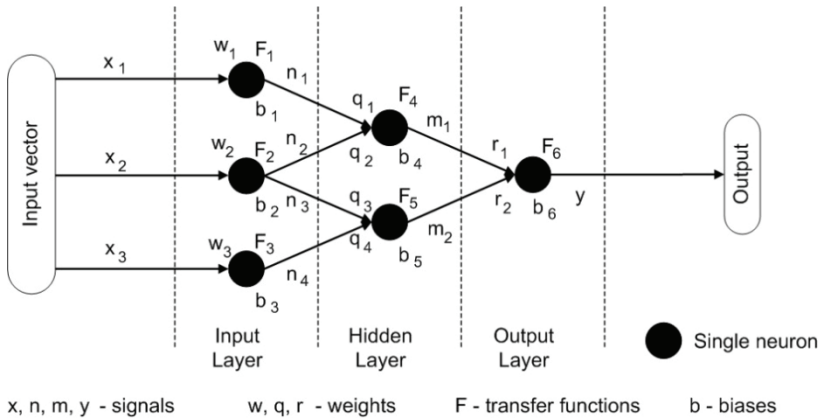


Fig. 5. Feed-forward artificial neural network.

As seen on Fig. 5 and corresponding analytical description with sets of equations (3), (4) and (5) the simple feed-forward artificial neural network can lead to relatively long mathematical descriptions where artificial neural networks' parameters optimization problem solving by hand is impractical. Although analytical description can be used on any complex artificial neural network in practise we use computers and specialised software that can help us build, mathematically describe and optimise any type of artificial neural network.

3.2 Recurrent Artificial Neural Networks

Artificial neural network with the recurrent topology is called *Recurrent* artificial neural network. It is similar to feed-forward neural network with no limitations regarding back-loops. In these cases information is no longer transmitted only in one direction but it is also transmitted backwards. This creates an internal state of the network which allows it to exhibit dynamic temporal behaviour. Recurrent artificial neural networks can use their internal memory to process any sequence of inputs. Fig. 6. shows small *Fully Recurrent* artificial neural network and complexity of its artificial neuron interconnections.

The most basic topology of recurrent artificial neural network is fully recurrent artificial network where every basic building block (artificial neuron) is directly connected to every other basic building block in all direction. Other recurrent artificial neural networks such as Hopfield, Elman, Jordan, bi-directional and other networks are just special cases of recurrent artificial neural networks.

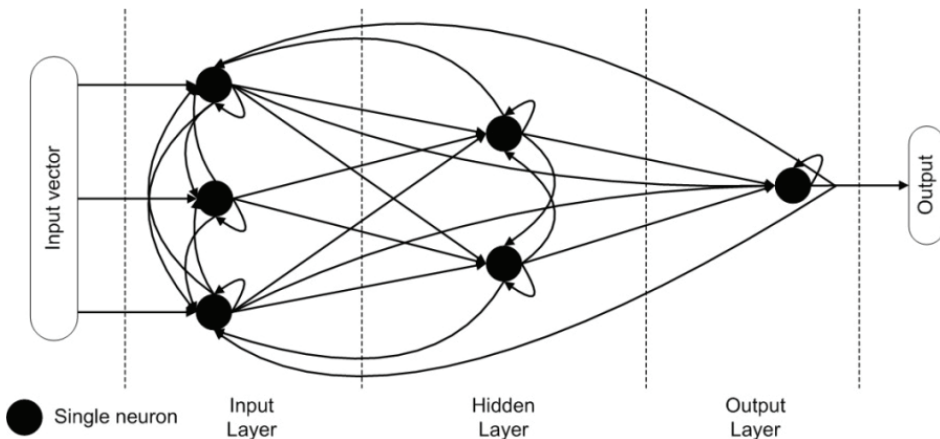


Fig. 6. Fully recurrent artificial neural network.

3.3 Hopfield Artificial Neural Network

A *Hopfield* artificial neural network is a type of recurrent artificial neural network that is used to store one or more stable target vectors. These stable vectors can be viewed as memories that the network recalls when provided with similar vectors that act as a cue to the network memory. These binary units only take two different values for their states that are determined by whether or not the units' input exceeds their threshold. Binary units can take either values of 1 or -1, or values of 1 or 0. Consequently there are two possible definitions for binary unit activation a_i (equation (6) and (7)):

$$a_i = \begin{cases} -1 & \text{if } \sum_j w_{ij}s_j > \theta_i, \\ 1 & \text{otherwise.} \end{cases} \quad (6)$$

$$a_i = \begin{cases} 0 & \text{if } \sum_j w_{ij}s_j > \theta_i, \\ 1 & \text{otherwise.} \end{cases} \quad (7)$$

Where:

- w_{ij} is the strength of the connection weight from unit j to unit i ,
- s_j is the state of unit j ,
- θ_i is the threshold of unit i .

While talking about connections w_{ij} we need to mention that there are typical two restrictions: no unit has a connection with itself (w_{ii}) and that connections are symmetric $w_{ij} = w_{ji}$.

The requirement that weights must be symmetric is typically used, as it will guarantee that the energy function decreases monotonically while following the activation rules. If non-symmetric weights are used the network may exhibit some periodic or chaotic behaviour. Training a Hopfield artificial neural network (Fig. 7.) involves lowering the energy of states that the artificial neural network should remember.

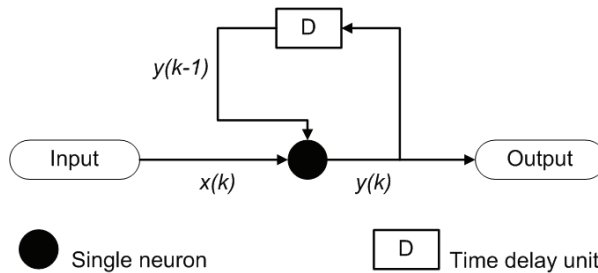


Fig. 7. Simple “one neuron” Hopfield artificial neural network.

3.4 Elman and Jordan Artificial Neural Networks

Elman network also referred as *Simple Recurrent Network* is special case of recurrent artificial neural networks. It differs from conventional two-layer networks in that the first layer has a recurrent connection. It is a simple three-layer artificial neural network that has back-loop from hidden layer to input layer through so called context unit (Fig. 8.). This type of artificial neural network has memory that allowing it to both detect and generate time-varying patterns.

The Elman artificial neural network has typically sigmoid artificial neurons in its hidden layer, and linear artificial neurons in its output layer. This combination of artificial neurons transfer functions can approximate any function with arbitrary accuracy if only there is enough artificial neurons in hidden layer. Being able to store information Elman artificial neural network is capable of generating temporal patterns as well as spatial patterns and

responding on them. Jordan network (Fig. 9.) is similar to Elman network. The only difference is that context units are fed from the output layer instead of the hidden layer.

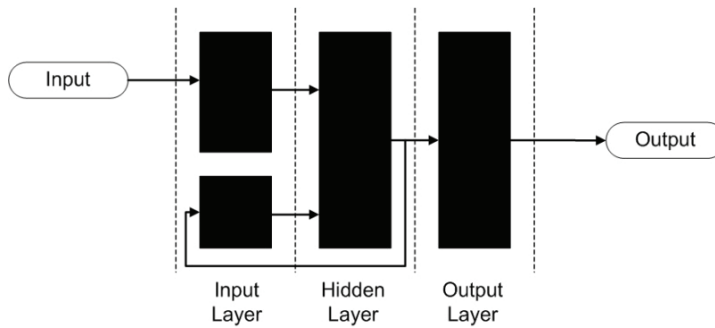


Fig. 8. Elman artificial neural network.

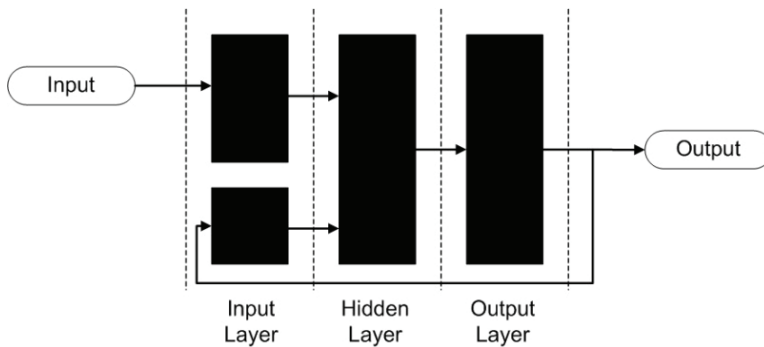


Fig. 9. Jordan artificial neural network.

3.5 Long Short Term Memory

Long Short Term Memory is one of the recurrent artificial neural networks topologies. In contrast with basic recurrent artificial neural networks it can learn from its experience to process, classify and predict time series with very long time lags of unknown size between important events. This makes Long Short Term Memory to outperform other recurrent artificial neural networks, Hidden Markov Models and other sequence learning methods.

Long Short Term Memory artificial neural network is build from Long Short Term Memory blocks that are capable of remembering value for any length of time. This is achieved with gates that determine when the input is significant enough remembering it, when continue to remembering or forgetting it, and when to output the value.

Architecture of Long Short Term Memory block is shown in Fig. 10 where input layer consists of sigmoid units. Top neuron in the input layer process input value that might be

sent to a memory unit depends on computed value of second neuron from the top in the input layer. The third neuron from the top in the input layer decide how long will memory unit hold (remember) its value and the bottom most neuron determines when value from memory should be released to the output. Neurons in first hidden layer and in output layer are doing simple multiplication of their inputs and a neuron in the second hidden layer computes simple linear function of its inputs. Output of the second hidden layer is fed back into input and first hidden layer in order to help making decisions.

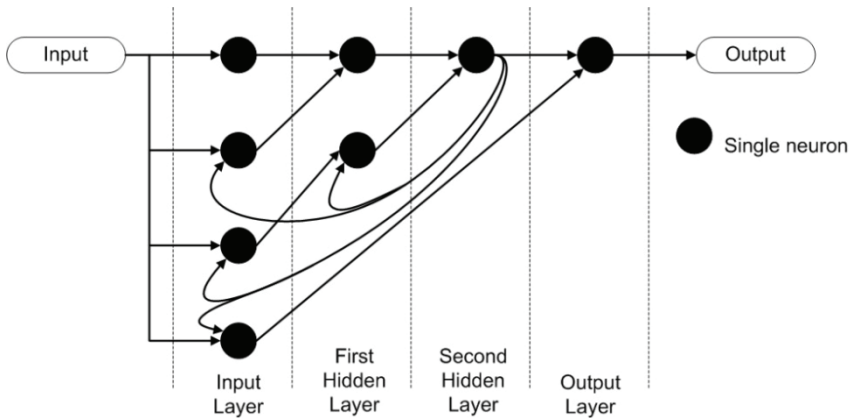


Fig. 10. Simple Long Short Term Memory artificial neural network (block).

3.6 Bi-directional Artificial Neural Networks (Bi-ANN)

Bi-directional artificial neural networks (Fig. 11.) are designed to predict complex time series. They consist of two individual interconnected artificial neural (sub) networks that performs direct and inverse (bidirectional) transformation. Interconnection of artificial neural sub networks is done through two dynamic artificial neurons that are capable of remembering their internal states. This type of interconnection between future and past values of the processed signals increase time series prediction capabilities. As such these artificial neural networks not only predict future values of input data but also past values. That brings need for two phase learning; in first phase we teach one artificial neural sub network for predicting future and in the second phase we teach a second artificial neural sub network for predicting past.

3.7 Self-Organizing Map (SOM)

Self-organizing map is an artificial neural network that is related to feed-forward networks but it needs to be told that this type of architecture is fundamentally different in arrangement of neurons and motivation. Common arrangement of neurons is in a hexagonal or rectangular grid (Fig. 12.). Self-organizing map is different in comparison to other artificial neural networks in the sense that they use a neighbourhood function to preserve the topological properties of the input space. They uses unsupervised learning paradigm to

produce a low-dimensional, discrete representation of the input space of the training samples, called a map what makes them especially useful for visualizing low-dimensional views of high-dimensional data. Such networks can learn to detect regularities and correlations in their input and adapt their future responses to that input accordingly.

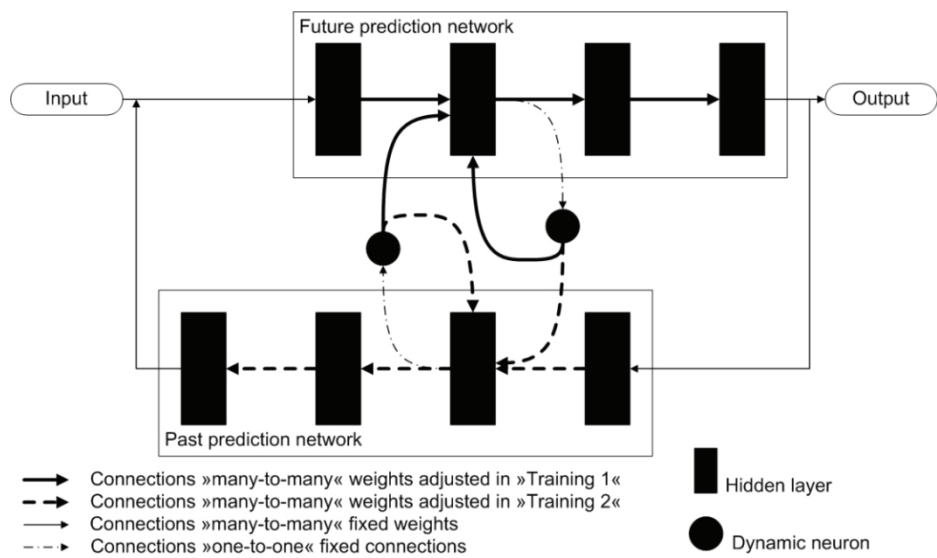


Fig. 11. Bi-directional artificial neural network.

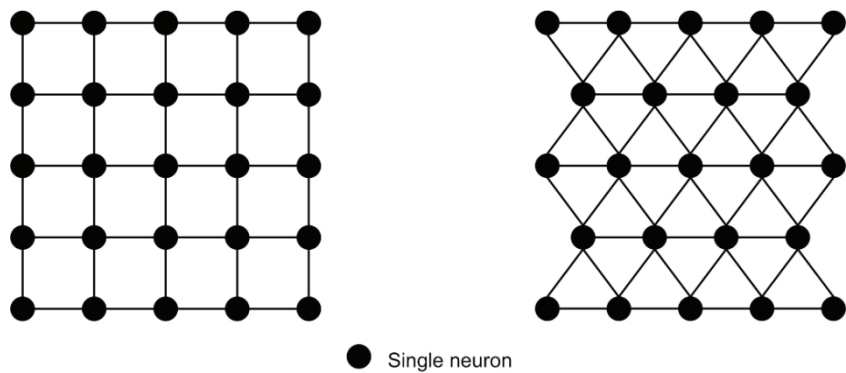


Fig. 12. Self-organizing Map in rectangular (left) and hexagonal (right) grid.

Just as others artificial neural networks need learning before they can be used the same goes for self-organizing map; where the goal of learning is to cause different parts of the artificial neural network to respond similarly to certain input patterns. While adjusting the weights of the neurons in the process of learning they are initialized either to small random values or sampled evenly from the subspace spanned by the two largest principal component eigenvectors. After initialization artificial neural network needs to be fed with large number of example vectors. At that time Euclidean distance to all weight vectors is computed and the neuron with weight vector most similar to the input is called the best matching unit. The weights of the best matching unit and neurons close to it are adjusted towards the input vector. This process is repeated for each input vector for a number of cycles. After learning phase we do so-called mapping (usage of artificial neural network) and during this phase the only one neuron whose weight vector lies closest to the input vector will be winning neuron. Distance between input and weight vector is again determined by calculating the Euclidean distance between them.

3.8 Stochastic Artificial Neural Network

Stochastic artificial neural networks are a type of an artificial intelligence tool. They are built by introducing random variations into the network, either by giving the network's neurons stochastic transfer functions, or by giving them stochastic weights. This makes them useful tools for optimization problems, since the random fluctuations help it escape from local minima. Stochastic neural networks that are built by using stochastic transfer functions are often called Boltzmann machine.

3.9 Physical Artificial Neural Network

Most of the artificial neural networks today are software-based but that does not exclude the possibility to create them with physical elements which base on adjustable electrical current resistance materials. History of physical artificial neural networks goes back in 1960's when first physical artificial neural networks were created with memory transistors called memistors. Memistors emulate synapses of artificial neurons. Although these artificial neural networks were commercialized they did not last for long due to their incapability for scalability. After this attempt several others followed such as attempt to create physical artificial neural network based on nanotechnology or phase change material.

4. Learning

There are three major learning paradigms; supervised learning, unsupervised learning and reinforcement learning. Usually they can be employed by any given type of artificial neural network architecture. Each learning paradigm has many training algorithms.

4.1 Supervised learning

Supervised learning is a machine learning technique that sets parameters of an artificial neural network from training data. The task of the learning artificial neural network is to set the value of its parameters for any valid input value after having seen output value. The training data consist of pairs of input and desired output values that are traditionally represented in data vectors. Supervised learning can also be referred as classification, where we have a wide range of classifiers, each with its strengths and weaknesses. Choosing a

suitable classifier (Multilayer perceptron, Support Vector Machines, k-nearest neighbour algorithm, Gaussian mixture model, Gaussian, naive Bayes, decision tree, radial basis function classifiers,...) for a given problem is however still more an art than a science.

In order to solve a given problem of supervised learning various steps has to be considered. In the first step we have to determine the type of training examples. In the second step we need to gather a training data set that satisfactory describe a given problem. In the third step we need to describe gathered training data set in form understandable to a chosen artificial neural network. In the fourth step we do the learning and after the learning we can test the performance of learned artificial neural network with the test (validation) data set. Test data set consist of data that has not been introduced to artificial neural network while learning.

4.2 Unsupervised learning

Unsupervised learning is a machine learning technique that sets parameters of an artificial neural network based on given data and a cost function which is to be minimized. Cost function can be any function and it is determined by the task formulation. Unsupervised learning is mostly used in applications that fall within the domain of estimation problems such as statistical modelling, compression, filtering, blind source separation and clustering.

In unsupervised learning we seek to determine how the data is organized. It differs from supervised learning and reinforcement learning in that the artificial neural network is given only unlabeled examples. One common form of unsupervised learning is clustering where we try to categorize data in different clusters by their similarity. Among above described artificial neural network models, the Self-organizing maps are the ones that the most commonly use unsupervised learning algorithms.

4.3 Reinforcement learning

Reinforcement learning is a machine learning technique that sets parameters of an artificial neural network, where data is usually not given, but generated by interactions with the environment. Reinforcement learning is concerned with how an artificial neural network ought to take actions in an environment so as to maximize some notion of long-term reward. Reinforcement learning is frequently used as a part of artificial neural network's overall learning algorithm.

After return function that needs to be maximized is defined, reinforcement learning uses several algorithms to find the policy which produces the maximum return. Naive brute force algorithm in first step calculates return function for each possible policy and chooses the policy with the largest return. Obvious weakness of this algorithm is in case of extremely large or even infinite number of possible policies. This weakness can be overcome by *value function approaches* or *direct policy estimation*. Value function approaches attempt to find a policy that maximizes the return by maintaining a set of estimates of expected returns for one policy; usually either the current or the optimal estimates. These methods converge to the correct estimates for a fixed policy and can also be used to find the optimal policy. Similar as value function approaches the direct policy estimation can also find the optimal policy. It can find it by searching it directly in policy space what greatly increases the computational cost.

Reinforcement learning is particularly suited to problems which include a long-term versus short-term reward trade-off. It has been applied successfully to various problems, including

robot control, telecommunications, and games such as chess and other sequential decision making tasks.

5. Usage of Artificial Neural Networks

One of the greatest advantages of artificial neural networks is their capability to learn from their environment. Learning from the environment comes useful in applications where complexity of the environment (data or task) make implementations of other type of solutions impractical. As such artificial neural networks can be used for variety of tasks like classification, function approximation, data processing, filtering, clustering, compression, robotics, regulations, decision making, etc. Choosing the right artificial neural network topology depends on the type of the application and data representation of a given problem. When choosing and using artificial neural networks we need to be familiar with theory of artificial neural network models and learning algorithms. Complexity of the chosen model is crucial; using to simple model for specific task usually results in poor or wrong results and over complex model for a specific task can lead to problems in the process of learning. Complex model and simple task results in memorizing and not learning. There are many learning algorithms with numerous tradeoffs between them and almost all are suitable for any type of artificial neural network model. Choosing the right learning algorithm for a given task takes a lot of experiences and experimentation on given problem and data set. When artificial neural network model and learning algorithm is properly selected we get robust tool for solving given problem.

5.1 Example: Using bi-directional artificial neural network for ICT fraud detection

Spread of Information and Communication Technologies results in not only benefits for individuals and society but also in threats and increase of Information and Communication Technology frauds. One of the main tasks for Information and Communication Technology developers is to prevent potential fraudulent misuse of new products and services. If protection against fraud fails there is a vital need to detect frauds as soon as possible. Information and Communication Technology frauds detection is based on numerous principles. One of such principle is use of artificial neural networks in the detection algorithms. Below is an example of how to use bi-directional artificial neural network for detecting mobile-phone fraud.

First task is to represent problem of detecting our fraud in the way that can be easily understand by humans and machines (computers). Each individual user or group of users behave in specific way while using mobile phone. By learning their behaviour we can teach our system to recognize and predict users' future behaviour to a certain degree of accuracy. Later comparison between predicted and real-life behaviour and potential discrepancy between them can indicate a potential fraudulent behaviour. It was shown that mobile-phone usage behaviour can be represented in the form of time series suitable for further analysis with artificial neural networks (Krenker et al., 2009). With this representation we transform the behaviour prediction task in time series prediction task. Time series prediction task can be realized with several different types of artificial neural networks but as mentioned in earlier chapters some are more suitable then others. Because we expect long and short time periods between important events in our data representation of users' behaviour the most obvious artificial neural networks to use are Long Short Term Memory and bi-directional

artificial neural networks. On the basis of others researchers' favourable results in time series prediction with bi-directional artificial neural network (Wakuya & Shida, 2001) we decided to use this artificial neural network topology for predicting our time series.

After we choose artificial neural network architecture we choose the type of learning paradigm; we choose supervised learning where we gather real life data form telecommunication system. Gathered data was divided into two sub-sets; training sub-set and validation subset. With training data sub-set artificial neural network learn to predict future and past time series and with validation data sub-set we simulate and validate the prediction capabilities of designed and fine-tuned bi-directional artificial neural networks. Validation was done with calculation of the *Average Relative Variance* that represents a measure of similarity between predicted and expected time series.

Only after we gathered information about mobile-phone fraud and after choosing representation of our problem and basic approaches for solving it we could start building the overall model for detecting mobile-phone fraud (Fig. 13.).

On Fig. 13. we can see that mobile-phone fraud detection model is build out of three modules; input module, artificial neural network module and comparison module. *Input Module* gathers users' information about usage of mobile-phone from telecommunication system in three parts. In first part it is used for gathering learning data from which *Artificial Neural Network Module* learn it-self. In second part *Input Module* gathers users' data for purpose of validating the *Artificial Neural Network Module* and in the third part it collects users' data in real time for purpose of using deployed mobile-phone fraud system. *Artificial Neural Network Module* is bi-directional artificial neural network that is learning from gathered data and later when the mobile-phone fraud detection system is deployed continuously predicts time series that represents users' behaviour. *Comparison module* is used for validation of *Artificial Neural Network Module* in the process of learning and later when the mobile-phone fraud detection system is deployed it is used for triggering alarms in case of discrepancies between predicted and real-life gathered information about users' behaviour.

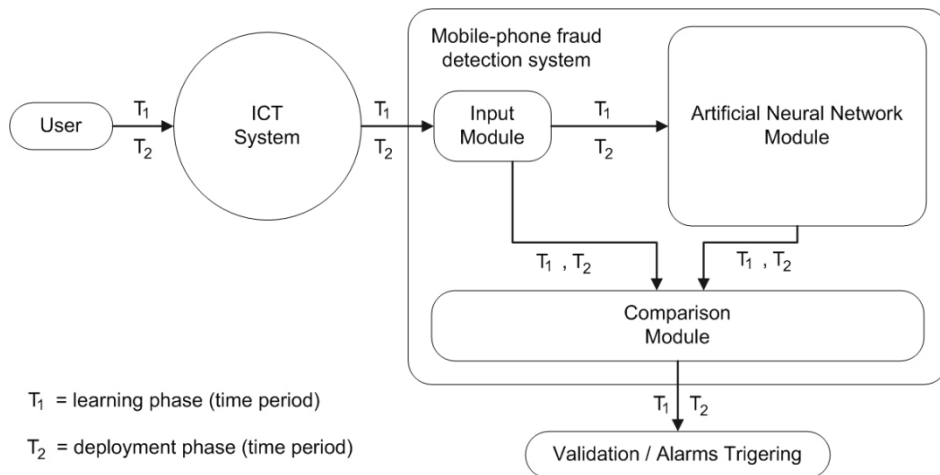


Fig. 13. Mobile-phone fraud detection model.

Although mobile-phone fraud detection system described above is simple and straight forward reader needs to realize that majority of work is not in creating and later implementing desired systems but in fine-tuning of data representation and artificial neural network architecture and its parameters that is strongly dependant on type of input data.

6. Conclusions

Artificial neural networks are widely spread and used in everyday services, products and applications. Although modern software products enable relatively easy handling with artificial neural networks, their creation, optimisation and usage in real-life situations it is necessary to understand theory that stands behind them. This chapter of the book introduces artificial neural networks to novice reader and serves as a stepping stone for all of those who would like to get more involved in the area of artificial neural networks.

In the *Introduction* in order to lighten the area of artificial neural networks we briefly described basic building blocks (artificial neuron) of artificial neural networks and their "transformation" from single artificial neuron to complete artificial neural network. In the chapter *Artificial Neuron* we present basic and important information about artificial neuron and where researchers borrowed the idea to create one. We show the similarities between biological and artificial neuron their composition and inner workings. In the chapter *Artificial Neural Networks* we describe basic information about different, most commonly used artificial neural networks topologies. We described *Feed-forward*, *Recurrent*, *Hopfield*, *Elman*, *Jordan*, *Long Short Term Memory*, *Bi-directional*, *Self Organizing Maps*, *Stochastic* and *Physical* artificial neural networks. After describing various types of artificial neural networks architectures we describe how to make them useful by learning. We describe different learning paradigms (supervised, unsupervised and reinforcement learning) in chapter *Learning*. In the last chapter *Usage of Artificial Neural Networks* we describe how to handle artificial neural networks in order to make them capable of solving certain problems. In order to show what artificial neural networks are capable of, we gave a short example how to use bi-directional artificial neural network in mobile-phone fraud detection system.

7. References

- Gurney, K. (1997). *An Introduction to Neural Networks*, Routledge, ISBN 1-85728-673-1 London
- Krenker A.; Volk M.; Sedlar U.; Bešter J.; Kos A. (2009). Bidirectional artificial neural networks for mobile-phone fraud detection. *ETRI Jurnal.*, vol. 31, no. 1, Feb. 2009, pp. 92-94, COBISS.SI-ID 6951764
- Kröse B.; Smagt P. (1996). *An Introduction to Neural Networks*, The University of Amsterdam, Amsterdam.
- Pavešić N. (2000). *Razpoznavanje vzorcev: uvod v analizo in razumevanje vidnih in slušnih signalov*, Fakulteta za elektrotehniko, ISBN 961-6210-81-5, Ljubljana
- Rojas R. (1996). *Neural Networks: A Systematic Introduction*, Springer, ISBN 3-540-60505-3, Germany.

Wakuya H.; Shida K.. (2001). Bi-directionalization of neural computing architecture for time series prediction. III. Application to laser intensity time record "Data Set A". *Proceedings of International Joint Conference on Neural Networks*, pp. 2098 – 2103, ISBN 0-7803-7044-9, Washington DC, 2001, Washington DC.

Review of Input Variable Selection Methods for Artificial Neural Networks

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1. Introduction

The choice of input variables is a fundamental, and yet crucial consideration in identifying the optimal functional form of statistical models. The task of selecting input variables is common to the development of all statistical models, and is largely dependent on the discovery of relationships within the available data to identify suitable predictors of the model output. In the case of parametric, or semi-parametric empirical models, the difficulty of the input variable selection task is somewhat alleviated by the *a priori* assumption of the functional form of the model, which is based on some physical interpretation of the underlying system or process being modelled. However, in the case of artificial neural networks (ANNs), and other similarly data-driven statistical modelling approaches, there is no such assumption made regarding the structure of the model. Instead, the input variables are selected from the available data, and the model is developed subsequently. The difficulty of selecting input variables arises due to (i) the number of available variables, which may be very large; (ii) correlations between potential input variables, which creates redundancy; and (iii) variables that have little or no predictive power.

Variable subset selection has been a longstanding issue in fields of applied statistics dealing with inference and linear regression (Miller, 1984), and the advent of ANN models has only served to create new challenges in this field. The non-linearity, inherent complexity and non-parametric nature of ANN regression make it difficult to apply many existing analytical variable selection methods. The difficulty of selecting input variables is further exacerbated during ANN development, since the task of selecting inputs is often delegated to the ANN during the learning phase of development. A popular notion is that an ANN is adequately capable of identifying redundant and noise variables during training, and that the trained network will use only the salient input variables. ANN architectures can be built with arbitrary flexibility and can be successfully trained using any combination of input variables (assuming they are good predictors). Consequently, allowances are often made for a large number of input variables, with the belief that the ability to incorporate such flexibility and redundancy creates a more robust model. Such pragmatism is perhaps symptomatic of the popularisation of ANN models through machine learning, rather than statistical learning theory. ANN models are too often developed without due consideration given to the effect that the choice of input variables has on model complexity, learning difficulty, and performance of the subsequently trained ANN.

Recently, ANN modellers have become increasingly aware of the need to undertake input variable selection (IVS), and a myriad of methods employed to undertake the IVS task are described within reported ANN applications—some more suited to ANN development than others. This review therefore serves to provide some guidance to ANN modellers, by highlighting some of the key issues surrounding variable selection within the context of ANN development, and survey some the alternative strategies that can be adopted within a general framework, and provide some examples with discussion on the benefits and disadvantages in each case.

2. The input variable selection problem

Recall that for an unknown, steady-state input-output process, the development of an ANN provides the non-linear transfer function

$$Y = F(X) + \varepsilon, \quad (1)$$

where the model output Y is some variable of interest, X is a k -dimensional input vector, whose component variables are denoted by $X_i (i = 1, \dots, k)$, and ε is some small random noise. Let C denote the set of d variables that are available to construct the ANN model. The I_{d-k} problem of input variable selection (IVS) is to choose a set of k variables from C to form X (Battiti, 1994; Kwak & Choi, 2002) that leads to the optimal form of the model, F .

Dynamic processes will require the development of an ANN to provide a time-series model of the general form

$$Y(t+k) = F(Y(t), \dots, Y(t-p), X(t), \dots, X(t-p)) + \varepsilon(t). \quad (2)$$

Here, the output variable is predicted at some future time $t+k$, as a function of past values of both input X and output Y . Past observations of each variable are referred to as *lags*, and the model order p defines the maximum lag of the model. The model order reflects the persistence of dynamics within the system. In comparison to the steady-state model formulation, the number of variables in the candidate set C is now multiplied by the model order. Consequently, for systems with strong persistence, the number of candidate variables is often quite large.

ANN models may be specified with insufficient, or uninformative input variables (under-specified); or more inputs than is strictly necessary (over-specified), due to the inclusion of superfluous variables that are uninformative, weakly informative, or redundant. Defining what constitutes an optimal set of ANN input variables first requires some consideration of the impact that the choice of input variables has on model performance. The following arguments summarise the key considerations:

- *Relevance.* Arguably the most obvious concern is that too few variables are selected, or that the selected set of input variables is not sufficiently informative. In this case, the outcome is a poorly performing model, since some of the behaviour of the output remains unexplained by the selected input variables. In most cases, it is reasonable to assume that a modeller will have some expert knowledge of the system under consideration; will have surveyed the available data, and will have arrived at a reasonable set of candidate input variables. The *a priori* assumption of model development is that at least one or more of the available candidate variables is capable of describing some, if not all, of the output behaviour, and that it is the nature and relative strength of these relationships that is unknown (which is, of course, the motivation behind the development of non-parametric

models). Should it happen that none of the available candidates are good predictors, then the problem of model development is intractable, and it may be necessary to reconsider the available data and the choice of model output, and to undertake further measurements or observations before revisiting the task of model development.

- *Computational Effort.* The immediately obvious effect of including a greater number of input variables is that the size of an ANN increases, which increases the computational burden associated with querying the network—a significant influence in determining the speed of training. In the case of the multilayer perceptron (MLP), the input layer will have an increased number of incoming connection weights. In the case of kernel-based generalised regression neural network (GRNN) and radial basis function (RBF) networks, the computation of distance to prototype vectors is more expensive due to higher dimensionality. Furthermore, additional variables place an increased burden on any data pre-processing steps that may be undertaken during ANN development.
- *Training difficulty.* The task of training an ANN becomes more difficult due to the inclusion of redundant and irrelevant input variables. The effect of redundant variables is to increase the number of local optima in the error function that is projected over the parameter space of the model, since there are more combinations of parameters that can yield locally optimal error values. Algorithms such as the back-propagation algorithm, which are based on gradient descent, are therefore more likely to converge to a local optimum resulting in poor generalisation performance. Training of the network is also slower because the relationship between redundant parameters and the error is more difficult to map. Irrelevant variables add noise into the model, which also hinders the learning process. The training algorithm may expend resources adjusting weights that have no bearing on the output variable, or the noise may mask the important input-output relationships. Consequently, many more iterations of the training algorithm may be required to determine a near-global optimum error, which adds to the computational burden of model development.
- *Dimensionality.* The so-called *curse of dimensionality* (Bellman, 1961) is that, as the dimensionality of a model increases linearly, the total volume of the modelling problem domain increases exponentially. Hence, in order to map a given function over the model parameter space with sufficient confidence, an exponentially increasing number of samples is required (Scott, 1992). Alternatively, where a finite number of data are available (as is generally the case in real-world applications), it can be said that the confidence or certainty that the true mapping has been found will diminish. ANN architectures like the MLP are particularly susceptible to the curse due to the rapid growth in the number of connection weights as input variables are added. Table 1 illustrates the growth in the sample size required to maintain a constant error associated with estimates of the input probability, as determined by the pattern layer of a GRNN. Some ANN architectures can also circumvent the curse of dimensionality through their handling of redundancy and their ability to simply ignore irrelevant variables (Sarle, 1997). Others, such as RBF networks and GRNN architectures, are unable to achieve this without significant modifications to the behaviour of their kernel functions, and are particularly sensitive to increasing dimensionality (Specht, 1991).
- *Comprehensibility.* In many applications, such as in the case of ANN transfer functions for process modelling, it will often suffice to regard an ANN as a “black-box” model. However, ANN modellers are increasingly concerned with the development of ANN

Dimension, d	Sample size, N
1	4
2	19
3	67
4	223
5	768
6	2790
7	10 700
8	43 700
9	180 700
10	842 000

Table 1. Growth of sample size with increasing dimensionality required to maintain a constant standard error of the probability of an input estimated in the GRNN pattern layer (Silverman, 1986).

models for knowledge discovery from data (KDD) and data mining (Craven & Shavlik, 1998). The goal of KDD is to train an ANN based on observations of a process, and then interrogate the ANN to gain further understanding of the process behaviour it has learned. Rule-extraction from ANN models can be useful for a number of purposes, including: (i) defining input domains that produce certain ANN outputs, which can be useful knowledge in itself; (ii) validation of the ANN behaviour (e.g. verifying that input-output response trends make sense), which increases confidence in the ANN predictions; and (iii) the discovery of new relationships, which reveals previously unknown insights into the underlying physical process (Craven & Shavlik, 1998; Darbari, 2000). Reducing the complexity of the ANN architecture, by minimising redundancy and the size of the network, can significantly improve the performance of data mining and rule extraction algorithms.

Based on the arguments presented, a desirable input variable is a highly informative explanatory variable (i.e a good predictor) that is dissimilar to other input variables (i.e. independent). Consequently, the optimal input variable set will contain the fewest input variables required to describe the behaviour of the output variable, with a minimum degree of redundancy and with no uninformative (noise) variables. Identification of an optimal set of input variables will lead to a more accurate, efficient, cost-effective and more easily interpretable ANN model.

The fundamental importance of the IVS issue is evident from the depth of literature surrounding the development and discussion of IVS algorithms in fields such as classification, machine learning, statistical learning theory, and many other fields where ANN models are applied. In a broad context, reviews of IVS approaches have been presented by Kohavi & John (1997), Blum & Langley (1997) and more recently, by Guyon & Elisseeff (2003). However, in many examples of the application of ANNs to modelling and data analysis applications, the importance of IVS is often understated. In other cases, the task is given only marginal consideration and this often results in the application of *ad hoc* or inappropriate methods. Reviews by Maier & Dandy (2000) and Bowden (2003) examined the IVS methods that have been applied to ANN applications in engineering and concluded that there was a need for a more considered approach to the IVS task. Certainly, no consensus has been reached regarding

suitable methods for undertaking the IVS task in the development of ANN regression or time-series forecasting models (Bowden, 2003).

3. Taxonomy of algorithms

Figure 1 presents a taxonomy, which provides some examples of the various approaches that have been proposed within ANN literature. IVS algorithms can be broadly classified into three main classes: *wrapper*, *embedded* or *filter* algorithms (Blum & Langley, 1997; Guyon & Elisseeff, 2003; Kohavi & John, 1997), as shown in Figure 1. These different conceptual approaches to IVS algorithm design are illustrated in Figure 2. Wrapper algorithms, as shown in Figure 2(a), approach the IVS task as part of the optimisation of model architecture. The optimisation searches through the set, or a subset, of all possible combinations of input variables, and selects the set that yields the optimal generalisation performance of the trained ANN. As the name indicates, embedded algorithms (Figure 2(b)) for IVS are directly incorporated into the ANN training algorithm, such that the adjustment of input weights considers the impact of each input on the performance of the model, with irrelevant and/or redundant weights progressively removed as training proceeds. In contrast, IVS filters (Figure 2(c)) distinctly separate the IVS task from ANN training and instead adopt an auxiliary statistical analysis technique to measure the relevance of individual, or combinations of, input variables.

Given the general basis for the formulation of both IVS wrapper and filter designs, the diversity of implementations that can possibly be conceived is immediately apparent. However, designs for wrappers and filters share the same overall components, in that, in addition to a measure of the informativeness of input variables, each class of selection algorithms requires:

1. a criterion or test to determine the influence of the selected input variable(s), and
2. a strategy for searching among the combinations of candidate input variables.

3.1 Optimality Criteria

The optimality criterion defines the interpretation of the arguments presented in Section 2 into an expression for the optimal size k and composition of the input vector, X . Optimality criteria for wrapper selection algorithms are derived from, or are exactly the same as, criteria that are ultimately used to assess the predictive performance of the trained ANN. Essentially, the wrapper approach treats the IVS task as a model selection exercise, where each model corresponds to a unique combination of input variables. Recall that the most commonly adopted measure of predictive performance for ANNs is the mean squared error (MSE), which is given by

$$\text{MSE} = \frac{1}{n} \sum_{j=1}^n (y_j - \hat{y}_j)^2 \quad (3)$$

where y_j and $\hat{y}_j$ are the actual and predicted outputs, which correspond to a set of test data. Following the development of m models, a simple strategy is to select the model that corresponds to the minimum MSE. However, the drawback of this criterion is that the “best” performing model, in terms of the MSE, is not necessarily the “optimal” model, since models with a large number of input variables tend to be biased as a result of over-fitting. Consequently, it is more common to adopt an optimality criterion such as Mallows’ C_p (Mallows, 1973), or the Akaike information criterion (AIC) (Akaike, 1974), which penalise overfitting. Both Mallows’ C_p and the AIC determine the optimal number of input variables

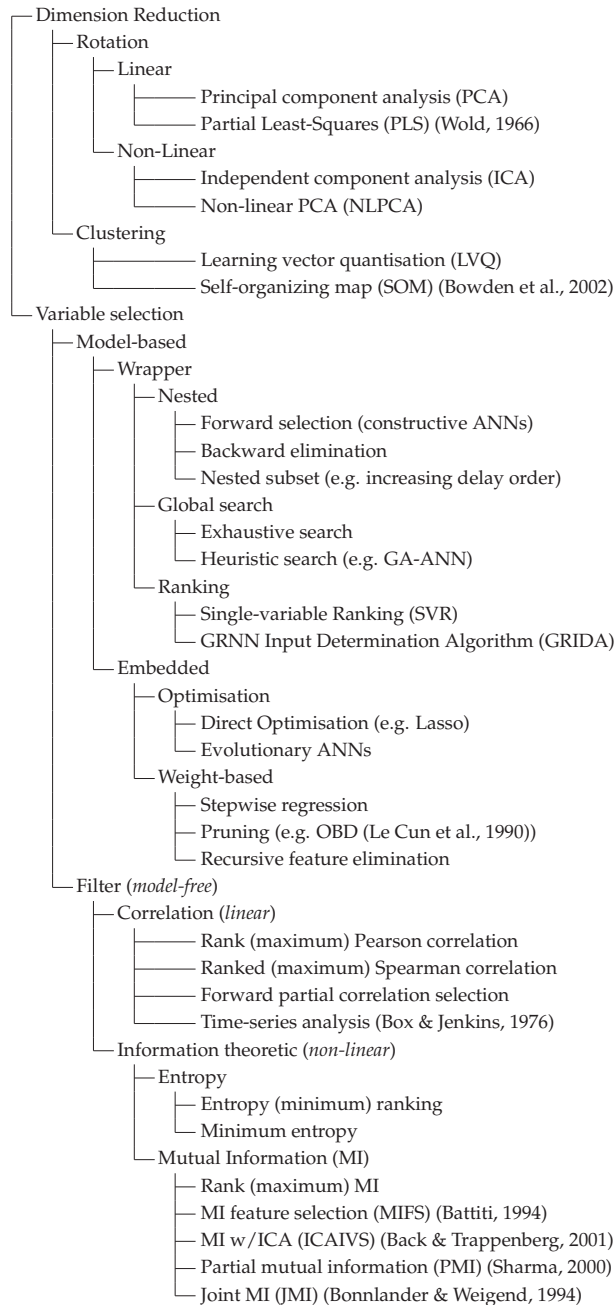
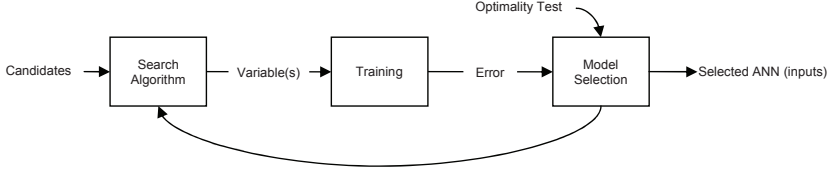
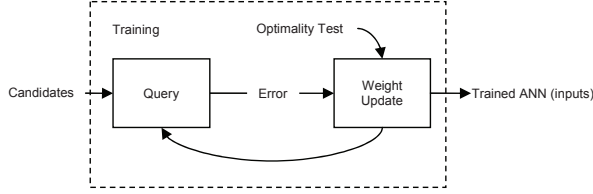


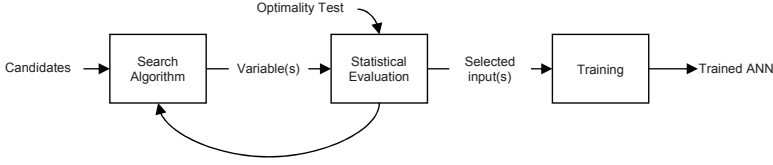
Fig. 1. Taxonomy of IVS Strategies and Algorithms



(a) Wrapper



(b) Embedded



(c) Filter

Fig. 2. Conceptual IVS approach using a (a) wrapper, (c) embedded, or (b) filter algorithm.

by defining the optimal trade-off between model size and accuracy by penalising models with an increasing number of parameters. In fact, the C_p criterion is considered to be a special case of the AIC.

Mallows' C_p is defined as

$$C_p = \frac{\sum_{j=1}^n (y_j - \hat{y}_j(k))^2}{\sigma_d^2} - n + 2p, \quad (4)$$

where $y_j(k)$ are the outputs generated by a model using p parameters, and σ_d^2 are residuals for a full model trained using all d possible input variables. C_p measures the relative bias and variance of a model with p variables. The theoretical value of C_p for an unbiased (optimal) model will be p , and in model selection, the model with the C_p value that is closest to p is selected.

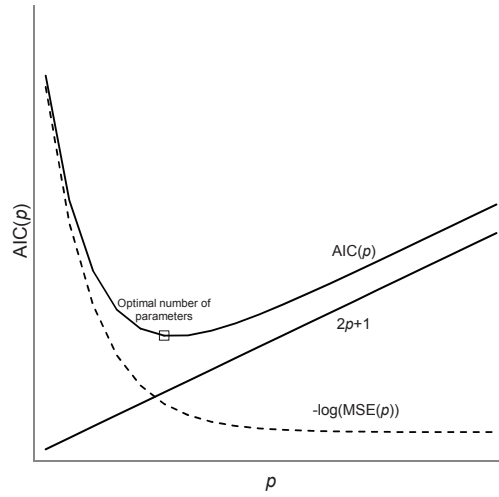


Fig. 3. The Akaike Information Criterion determines the optimum trade-off between model error and size

The AIC is defined as

$$\text{AIC} = -n \log \frac{\sum_{j=1}^n (y_j - \hat{y}_j(k))^2}{n} + 2(p+1). \quad (5)$$

Here, the accuracy is determined by the log-likelihood, which is a function of the MSE. The complexity of the model is determined by the term $p+1$, where p is the number of model parameters. Typically, the regression error decreases with increasing p , but since the model is more likely to be over-fit for a fixed sample size, the increasing complexity is penalised. At some point an optimal AIC is determined, which represents the optimal trade-off between model accuracy and model complexity. The optimum model is determined by minimising the AIC with respect to the number of model parameters, p . This is illustrated in Figure 3.

Other model selection criteria have also been similarly derived, such as the Bayesian information criterion (BIC) (Schwarz, 1978), which is similar to the AIC, although it applies a more severe penalty of $(k \ln n)$ to the number of model parameters. The expression for the AIC in (5) assumes a linear regression model, but can be extended to non-linear regression. However, it should be noted that in this case, $p+1$ no longer sufficiently describes the complexity of the model and other measures are required. Such measures include the *effective number of parameters*, or Vapnik-Chervonovskis dimension. The values of these measures are a function of the class of regression model that is estimated and the training data. The effective number of parameters, d can be determined by $\text{trace}(S)$, where S is a matrix defined by the expression

$$\hat{y} = Sy. \quad (6)$$

For kernel regression, the hat matrix, S , is equal to $K^T K$, where the elements of K correspond to each $K_j(x, h)$, and the complexity is therefore given by $\text{trace}(K^T K)$. Factors affecting

complexity include the number of data, the dimension of the data, and the number of basis functions. The VC-dimension is similarly defined as the number of data points that can be *shattered* by the model (i.e. how many points in space can be uniquely separated by the regression function). However, calculating the VC-dimension of complex regression functions can be difficult (Hastie et al., 2001). For MLP architectures, the VC-dimension is related to the number of connection weights, and for RBF networks the VC-dimension depends on the number of basis functions and their respective bandwidths, if different value are used for each basis function. Both the effective number of parameters and the VC-dimension revert to the value of $p + 1$ for linear models.

In filter algorithm designs, the optimality criterion is embedded in the statistical analysis of candidate variables, which defines the interpretation of “good” input variables. In general, selection filters search amongst the candidate variables and identify suitable input variables according to the following criteria:

1. maximum relevance (MR),
2. minimum redundancy (mR), and
3. minimum redundancy–maximum Relevance (mRMR).

The criterion of maximum relevance ensures that the selected input variables are highly informative by searching for variables that have a high degree of correlation with the output variable. Input ranking schemes are a prime example of MR techniques, in which the relevance is determined for each input variable with the output variable. Greedy selection can be applied to select the k most relevant variables, or a threshold value can be applied to select inputs that are relevant, and reject those which are not.

The issue with MR criteria is that the selection of the k most relevant candidate variables does not strictly yield an optimal ANN. Here, Kohavi & John (1997) make the distinction between relevance and usefulness by observing that redundancy between variables can render highly relevant variables useless as predictors. Consequently, a criterion of minimum redundancy aims to find inputs that are maximally dissimilar from one another, in order to select the most useful set of relevant variables. The application of an additional mR criterion with the existing MR criterion leads to mRMR selection criteria, where input variables are evaluated with the dual consideration of relevance, with respect to the output variable; and independence (dissimilarity), with respect to the other candidate variables (Ding & Peng, 2005).

Embedded IVS considers regularisation (reducing the size or number) of the weights of a regression to minimise the complexity, while maintaining predictive performance. This involves the formulation of a training algorithm that simultaneously finds the minimum model error and model complexity, somewhat analogous to finding the optimum the AIC. If the model architecture is linear-in-the-parameters, the resulting expression can be solved directly. Depending on the model complexity term, this approach gives rise to various embedded selection algorithms, such as the Lasso (Tibshirani, 1996). However, the non-linear and non-parametric nature of ANN regression does not lend itself to this approach (Guyon & Elisseeff, 2003; Tikka, 2008). Instead, embedded selection is typically applied in ANN model development in the form of a *pruning* strategy, where the connection weights of a network are assessed, and insignificant weights are removed from the network. Pruning algorithms were originally developed to address the computational burden associated with training fully connected networks, given that many of the weights may be only marginally important due to redundancy within the ANN architecture. However, the strategy also offers the means of selectively removing inputs, since an input variable is eliminated by eliminating all connection

weights between an input and the first hidden layer (Tikka, 2008). A criterion is required to identify which connection weights should be pruned, and several different approaches can be used to determine how weights are removed (Guyon & Elisseeff, 2003):

1. Analysis of sensitivity of training error to elimination of weights, or
2. Elimination of variables based on weight magnitude.

Where the first approach has been used, different expressions for the sensitivity of the error to the weights have led to various different algorithms. The use of derivatives of error functions or transfer functions at hidden nodes with respect to the weights are common strategies, and lead to examples of pruning algorithms such as the optimal brain damage (OBD) algorithm (Le Cun et al., 1990).

3.2 Search strategies

Search strategies applied to IVS algorithms seek to provide an efficient method for searching through the many possible combinations of input variables and determining an optimal, or near optimal set, while working within computational constraints. Searches may be global, and consider many combinations; or local methods, which begin at a start location and move through the search space incrementally. The latter are also commonly referred to as *nested subset* techniques, since the region they explore comprises overlapping (i.e. nested) sets by incrementally adding variables.

3.2.1 Exhaustive search

Exhaustive search simply evaluates all of the possible combinations of input variables and selects the best set according to a predetermined optimality criteria. The method is the only selection technique that is guaranteed to determine the optimal set of input variables for a given ANN model (Bonnlander & Weigend, 1994). Given the combinatorial nature of the IVS problem, the number of possible subsets that form the search space is equal to 2^d , with subsets ranging in size from single input variables, to the set of all available input variables. Exhaustive evaluation of all of these possible combinations may be feasible when the dimensionality of the candidate set is low, but quickly becomes infeasible as dimensionality increases.

3.2.2 Forward selection

Forward selection is a linear incremental search strategy that selects individual candidate variables one at a time. In the case of wrappers, the method starts by training d single-variable ANN models and selecting the input variable that maximises the model performance-based optimality criterion. Selection then continues by iteratively training $d - 1$ bivariate ANN models, in each case adding a remaining candidate to the previously selected input variable. Selection is terminated when the addition of another input variable fails to improve the performance of the ANN model. In filter designs, the single most relevant candidate variable is selected first, and then forward selection proceeds by iteratively identifying the next most relevant candidate and evaluating whether the variable should be selected, until the optimality criterion is satisfied.

The approach is computationally efficient overall, and tends to result in the selection of relatively small input variable sets, since it considers the smallest possible models, and trials increasingly larger input variable sets until the optimal set is reached. However, because forward selection does not consider all of the possible combinations, and only searches a

small subset, it is possible that the algorithm may encounter a locally optimum set of input variables and terminate prematurely. Also, due to the incremental nature of the forward search, the algorithm may ignore highly informative combinations of input variables that are only marginally relevant individually (Guyon & Elisseeff, 2003).

3.2.3 Step-wise selection

Forward selection is said to have *fidelity*, in that once an input variable is selected, the selection can not be undone. Step-wise selection is an extension of the forward selection approach, where input variables may also be removed at any subsequent iteration. The formulation of the step-wise approach is aimed at handling redundancy between candidate variables. For example, a variable X_a may be selected initially due to high relevance, but is later found to be inferior to the combination of two other variables, X_b and X_c , which only arises at a subsequent iteration. The initially selected input variable X_a is now redundant, and can be removed in favour of the pair X_b and X_c .

A common example of this approach is step-wise regression, which is widely used for the development of linear regression models. In this wrapper approach, linear models are iteratively constructed by adding an input variable to the model, and re-estimating the model coefficients. Input variables are retained based on analysis of the coefficients of the newly developed model. The selection process continues until the model satisfies some optimality criterion, such as the AIC (see Section 3.1), that is, when $k + 1$ input variables are no better than the preceding k variables.

3.2.4 Backward elimination

Backward elimination is essentially the reverse of the forward selection approach. In this case, all d input variables are initially selected, and then the most unimportant variables are eliminated one-by-one. In wrapper selection strategies, the relative importance of an input variable may be determined by removing an input variable X_i and evaluating the effect on the model that is retrained without it; or, by examining the influence of each of the input variables on the output y through some sensitivity analysis. In filter strategies, the least relevant candidates are iteratively removed until the optimality criterion is satisfied.

In general, backward elimination is inefficient in comparison with forward selection, as it can require the development and evaluation of many large ANN models before reaching the optimal model. Since all input variables are initially included, it may be more difficult to determine the relative importance of an individual input variable than in forward selection, which starts with a single input variable. Also, wrapper algorithms based on backward elimination may potentially be biased by overfitting of large models.

3.2.5 Heuristic search

Heuristic search techniques are widely used in optimisation problems where the search space is large. Heuristic search algorithms are particularly adept at efficiently finding global, or near-global optimum solutions within large search spaces by exploiting the common attributes of good solutions. In general, the various algorithms each implement a search that combines random evaluation of solutions throughout the entire search space, with a mechanism to increase the focus of the search in regions that lead to good solutions. Examples of heuristic search algorithms applied to IVS include evolutionary algorithms (EAs), such as genetic algorithms (GAs) (Bowden, 2003) and ant colony optimization (ACO) (Izrailev & Agrafiotis, 2002; Marcoulides & Drezner, 2003; Shen et al., 2005).

The application of heuristic optimisation techniques to IVS overcomes the significant computational requirement of exhaustive search, while maintaining the desirable characteristic of providing a global (or, near-global) optimum. Moreover, EA-based IVS wrappers are an attractive option because they can also be included as part of evolutionary ANN training algorithms, which also seek to determine optimal ANN parameter values by minimising the ANN cross-validation error. However, the application of heuristic search techniques requires calibration of search algorithm parameters, which is itself not a trivial task. In general, setting the search parameters involves a trade-off between the amount the search space that is explored, and the rate at which the algorithm converges to a final solution. Finally, heuristic algorithms retain a certain degree of randomness, and although they search more solutions in comparison to sequential selection algorithms, there is still no guarantee that the sub-space explored will include the globally optimal solution.

4. Dimensionality reduction

The taxonomical classification in Figure 1 includes dimensionality reduction algorithms as a class of algorithms reducing the number of variables within a dataset. Dimensionality reduction is often performed in order to reduce the computational effort associated with data processing, or to identify a suitable subset of variables to include in the analysis. Although the goal of dimension reduction differs from that of variable selection, it is a closely related area and is a regularly employed data pre-processing step in many multivariate data analysis applications, and it is worth including some discussion since many dimensionality reduction techniques are employed for IVS. The following considers some common examples of dimension reduction algorithms. Comprehensive surveys of dimensionality reduction techniques can be found in Carreira-Perpinan (1997) and Fodor (2002).

4.1 Principal component analysis

Principal component analysis (PCA) is a commonly adopted technique for reducing the dimensionality of a dataset X . PCA achieves dimensionality reduction by expressing the p variables $(x_1, \dots, x_p)$ as d feature vectors (or, *principal components* (PCs)), where $d < p$. The PCs are a set of orthogonal, linear combinations of the original variables within the dataset. Essentially, PCA can be considered a data pre-processing algorithm that determines an optimal rotational transformation of the dataset, X , that maximises the amount of variance of the output Y that is explained by the PCs (Fodor, 2002).

Considering a given dataset X , PCA is performed as follows:

- i. Subtract the mean value of each variable, to ensure that $\bar{x}_i = 0$ for each $x_i \in X$.
- ii. Find the covariance matrix $\Sigma = \text{Cov}(X) = X^T X$.
- iii. Determine the unit eigenvectors $e_1, \dots, e_p$ of Σ .
- iv. Determine the corresponding eigenvalues $\lambda_1, \dots, \lambda_p$.
- v. Rank the eigenvectors according to their eigenvalues.
- vi. Select the d PCs according to their eigenvalues.

Selection of PCs is based on examining the eigenvalues of each PC, which correspond to the amount of variance explained by each PC, and thereby including only the significant PCs as input features. A common selection method is to rank the PCs and select all PCs whose eigenvalues exceed some threshold λ_0 , or generate a plot of the cumulative eigenvalue as a function of the number of PCs, k , to ensure the selected components explain the desired

amount of variance of Y . Another technique is to use and generate a *scree* plot of the percentage contribution of each k^{th} PC and to visually identify an optimal value of k (Fodor, 2002).

PCA has been used as the basis for IVS for the development of ANN models (see, for example, Olsson et al. (2004), Gibbs et al. (2006), and Bowden (2003)). However, the mixing of input variables is assumed to be linear, as is the relationship between principal components and the output. Consequently, the application of PCA in this case is flawed, since it will fail to identify any non-linear relationships within the data. Although non-linear versions of the PCA algorithm exist, the transformations of the data can be highly complex, and interpretation of the PCs is much more difficult. An additional disadvantage of PCA is that the algorithm identifies important component vectors, rather than variables. Consequently, although PCA may be useful in removing noise from the data, it is not possible to distinguish the unique contributions of individual variables to the variance in the output.

4.2 Independent component analysis

Independent component analysis (ICA) seeks to determine a set of d independent component vectors within a dataset X . The approach is conceptually similar to PCA, although it relaxes the orthogonality constraint on component vectors. Furthermore, where PCA determines the optimal transformation of the data by considering covariance and identifying uncorrelated PCs based on covariance, ICA considers statistically independent combinations of variables where the order of the statistic that is used can be arbitrary (Fodor, 2002). ICA is therefore not restricted to linear correlations, and is more widely applicable to non-linear datasets (Back & Trappenberg, 2001). However, like PCA, ICA cannot discriminate unique variables as predictors, and is restricted to determining independent feature vectors.

4.3 Vector quantization

Vector quantization (VQ) refers to techniques that describe a larger set of n vectors by c codebook, or prototype vectors. VQ is closely associated with data clustering and is more commonly associated with algorithms for data compression, in terms of length n . Bowden (2003) demonstrates the potential for the self-organising map (SOM) to perform vector quantisation as an alternative to PCA for data dimensionality reduction. In this case, the d vectors of the candidate set are represented by the prototype vectors of the SOM. Similar candidate variables will be identified by the formation of groups, which have the closest proximity (defined by some distance measure) to the same prototype vector. However, the distance metric needs to be carefully selected, since the Euclidean distance used in SOM applications is not strictly a measure of correlation. Using linear correlation or covariance will cluster based on the strength of linear dependence between variables. Other measures, such as entropy or mutual information (MI), would be more suitable for clustering ANN variables, since they will measure non-linear dependence.

5. Wrappers

Wrapper algorithms are the first of the three main classes of variable selection algorithm shown according to Figure 1. Wrapper algorithms are the simplest IVS algorithm to formulate. Essentially, the algorithm that results is defined by the choice of the induction algorithm (i.e. model architecture). The efficiency of a wrapper algorithm will depend on the ability of the model to represent relationships within the data; and how efficiently trial models can be constructed and evaluated.

5.1 Single variable regression (SVR)

The notion of ranking individual candidate variables according to correlation can be extended by implementing a wrapper approach in order to relax the assumption of linearity in correlation analysis (Guyon & Elisseeff, 2003). In this approach, a single variable regression¹ (SVR) is constructed using each candidate variable, which is then ranked according to the model performance-based optimality criterion, such as the cross-validation error. In comparison to ranking filters, SVR can potentially suffer from overfitting due to the additional flexibility in the regression model.

The GRNN input determination algorithm (GRIDA) (Bowden et al., 2006) is a recent example of an SVR wrapper for input variable ranking, which proceeds as follows:

- i. Let $X \rightarrow C$. (Initialisation)
- ii. For each $x \in X$,
- iii. Train a GRNN and determine MSE_x .
- iv. For $b = 1$ to 100, (Bootstrap)
- v. Randomly shuffle $x \rightarrow \varepsilon$.
- vi. Estimate $MSE_{\varepsilon,b}$.
- vii. Estimate $MSE_{\varepsilon}^{(95)}$.
- viii. If $MSE_x > MSE_{\varepsilon}^{(95)}$ or $MSE_x > \Theta$ (Selection),
- ix. Remove x from X .
- x. Return X .

where $MSE_{\varepsilon}^{(95)}$ is the 95th percentile, and Θ is some threshold value.

Considering each variable in turn, a GRNN is trained, and then the MSE of the model is determined for a set of test data. However, rather than greedy selection of the k best variables, each variable is compared to a bootstrap estimate of a confidence bound for the randomised model error, $MSE_{\varepsilon}^{(95)}$. A variable is rejected immediately if the model error exceeds the randomised error, since it is no better predictor than a random noise variable. Further strictness on selections is imposed through the heuristic error threshold, Θ . However, a suitable value for Θ needs to be determined first. The number of variables selected for a given value of Θ will be dependent on several factors, including the degree of noise in the data, the error function used, and the distribution of the error over the candidate variables. Consequently, optimal values for Θ can only be determined for each dataset by trial and error. The estimation of the confidence bound on the error for each SVR is a significant computational requirement. Given the assumed constraint $0 < \Theta < MSE_{\varepsilon}^{(95)}$, the estimation of the bootstrap may not even be necessary to perform IVS. However, the method does provide useful information in discriminating noise variables from weakly informative ones. SVR does not consider interactions between variables, and may select redundant variables. In order to overcome this, dimensionality reduction is required as a pre-processing step, in order to obtain an independent set of candidate variables, prior to variable selection. Such an approach was used in the example of the GRNN-based SVR IVS algorithm (Bowden et al., 2006), where a SOM was used to achieve dimension reduction.

¹ The term has been adapted from the term single variable classifier (SVC), which is more often referred to within literature due to its application in classification

5.2 GA-ANN

Heuristic search techniques are ideally suited to IVS wrapper designs, since they provide an efficient search of the combinations of candidate input variables, with the simplicity of the black-box wrapper approach. Bowden et al. (2005) utilised an evolutionary wrapper strategy for IVS that combined a genetic algorithm (GA) optimisation with a generalised regression neural network (GRNN). The method exploits the fast GRNN training times, and the fixed architecture of the GRNN, which avoids the need to optimise the internal architecture and training algorithm parameters. These are required for the development of other architectures, such as the MLP. A simple binary GA (a GA with decisions encoded as 1 or 0 within a binary string) was utilised, with the objective of minimising the MSE obtained by hold-out validation on a set of test data. In order to overcome the inability of the wrapper methodology to detect interactions between candidate variables, as with GRIDA, Bowden et al. (2005) adopted SOM-based dimensionality reduction as a pre-processing stage to reduce the candidate variables to a subset of independent variables.

6. Embedded algorithms

Embedded algorithms are a distinct class of IVS algorithm where the selection of variables is *embedded* within the ANN training algorithm. The distinction between wrapper and embedded algorithms is not always obvious, since embedded selection also involves iterative update and evaluation of the model parameters based on model performance. The key difference is that the evaluation of input variables occurs within the training algorithm, and only a single model is trained. Embedded algorithms also consider the impact of each individual variable on the performance of the model, while wrapper algorithms simply consider model performance for a given set of input variables as a whole.

6.1 Recursive feature elimination

Recursive feature elimination (RFE) is an implementation of backward-elimination as an embedded IVS algorithm (Guyon & Elisseeff, 2003). The RFE approach involves an iterative process of training an ANN model, initially using all candidate input variables, and then removing one or more input variables at each iteration based on the rank magnitude of the weights corresponding to each input variable. The technique has been developed and successfully applied using support vector machines (SVMs), but is extensible to other ANN architectures by providing a suitable expression for the overall connection weight for a given input variable.

6.2 Evolutionary ANNs

Evolutionary ANNs (EANNs) utilise an evolutionary algorithm to determine the optimal set of weights, where the optimisation is formulated with ANN weights as the decision variables, and an objective function incorporating model error, with penalties for model complexity. Due to the complexity of the ANN error surface that is projected over the weight-space, evolutionary algorithms have been found to be a good alternative to gradient descent algorithms. EAs are robust and able to more reliably find a near-globally optimum solution, even for highly non-linear functions, with many multiple local optima; whereas gradient descent algorithms have greater potential to converge prematurely at a local minimum. By applying penalty terms to avoid large weights, or by using a model sparsity term within the objective function, the optimisation can effectively determine the optimum trade-off between model error and model complexity during training (Tikka, 2008).

IVS is embedded within the EANN approach, since the optimisation will implicitly exclude input variables by setting the input connection weight values close to, or equal to zero. Alternatively, the approach may also be made more explicit by extending the optimisation formulation to include the inclusion/exclusion of input variables within the ANN architecture as a separate binary decision variable. This approach is therefore similar to the GA-ANN wrapper approach, but extends the optimisation to the weight parameters.

7. Filters

In the taxonomy shown in Figure 1, filter algorithms represent the second sub-class of variable selection algorithms and represent an alternative to the wrapper approach. The design of filter algorithms is typically defined by the measure of relevance that is used to distinguish the important input variables, as well as the optimality criteria, as they have been previously defined for filters in Section 3.1. Incremental search strategies tend to dominate filter designs, since the relevance measure is usually a bivariate statistic, which necessitates evaluating each candidate-output relationship. Currently, two broad classes of filters have been considered: those based on linear correlation; and those based on information theoretic measures, such as mutual information.

7.1 Rank correlation

Arguably the most commonly used relevance measure in multivariate statistics is the Pearson correlation. The Pearson correlation (also called *linear correlation*, or *cross-correlation*), R , is defined by

$$R_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (7)$$

where R_{XY} is the short-hand notation for $R(X, Y)$. In (7), the numerator is simply the sample covariance, $Var(X, Y)$; and the two terms in the denominator are the square-root of the sample variances, $Var(X)$ and $Var(Y)$. The application of correlation analysis to variable selection originates from linear regression analysis. The squared correlation, R_{XY}^2 , is the coefficient of determination, and if X and Y have been standardised to have a zero mean, R^2 is the equivalent to the coefficient of a linear fit between X and Y .

Input variable ranking based on the Pearson correlation is one of the most widely used IVS methods. The selection of candidate variables that are sorted by order of decreasing correlation is based either on greedy selection of the first k variables, or upon all variables for which the correlation is significantly different from zero. The significance of the Pearson correlation can be determined directly, since the error associated with estimation of correlation from a sample is defined by the t -distribution. A rule of thumb (for large n) is that variables with an absolute correlation greater than $2/\sqrt{n}$ are significant.

Identification of significant correlations is a common technique in data mining applications, such as gene expression analysis, where the goal is simply to mark potentially important genes for further investigation. However, in terms of IVS algorithms, the method is classed as an MR filter, and does not consider interactions between variables. Redundancy is particularly problematic for multivariate time-series forecasting, which considers lagged values that are often highly correlated (auto-correlated).

7.2 Partial correlation

In the case where candidate variables are themselves correlated, redundancy becomes an important issue. In such cases, a correlation ranking approach is likely to select too many variables, since many candidates will each provide the same information regarding the output variable. Given three variables X , Y and Z , the partial correlation, $R'(X, Y|Z)$ measures the correlation between X and Y after the relationship between Y and Z has been discounted. The partial correlation can be determined from the Pearson correlation using the equation:

$$R_{XY \cdot Z} = \frac{R_{XY} - R_{XZ}R_{YZ}}{\sqrt{(1 - R_{XZ}^2)(1 - R_{YZ}^2)}} \quad (8)$$

where $R_{XY \cdot Z}$ and R_{XY} etc. are the short-hand notation for $R'(X, Y|Z)$ and R_{XY} etc. Partial correlation is similar to stepwise multiple linear regression. The subtle difference is that in stepwise MLR, successive models are fitted with additional input variables, and variables are selected (or later rejected) based on the estimated model coefficients. However, in partial correlation analysis, the magnitude of R' for each variable is not necessarily equal to the regression coefficients for a fitted MLR model, since redundancy between variables means that the solution to the MLR parameter estimation is a line (two redundant coefficients) or a surface, that is, there will be infinite combinations of equivalent model coefficients. The partial correlations obtained are in fact one specific solution to the MLR parameter estimation. Another difference is that forward selection is used in partial correlation analysis, because once the most salient variable has been selected, it will not be rejected later, and the partial correlations of subsequent variables will be dependent on those already selected.

7.3 Box-Jenkins

Box-Jenkins time-series analysis (Box & Jenkins, 1976), which considers the development of linear auto-regressive, moving-average (ARMA) models to represent dynamic processes, is the most common approach to the development of time-series and process transfer functions. ARMA models are described by the general form

$$y(t+1) = \sum_{k=0}^p \alpha_k y(t-k) + \sum_{k=0}^q \beta_k u(t-k) \quad (9)$$

where α_k and β_k are coefficients and p and q denote the order of the autoregressive (AR) and moving-average (MA) components of the model, respectively. Identification of the optimal model parameters p and q forms the goal of Box-Jenkins model identification, and hence variable selection. The autocorrelation function (ACF), $R(Y(t-k), Y(t))$, determines q and the partial autocorrelation function (PACF) determines p . The ACF is determined for a given time-series sample by

$$R_k = \frac{\sum_{i=1}^{n-k} (x_i - \bar{x})(x_{i-k} - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (10)$$

where R_k is the short-hand notation for the auto-correlation of a time-series with a delay of k . The PACF at a delay of k is denoted by ϕ_{kk} , and is estimated from the ACF based on the following series of equations

$$\phi_{11} = R_1 \quad (11)$$

$$\phi_{22} = \frac{R_2 - R_1^2}{1 - R_1^2} \quad (12)$$

$$\phi_{kj} = \phi_{k-1,j} - \phi_{kk}\phi_{k-1,k-j}, \text{ for } k \geq 2 \text{ and } j \geq 1, \quad (13)$$

$$\phi_{kk} = \frac{R_k - \sum_{j=1}^{k-1} \phi_{k-1,j}R_{k-j}}{1 - \sum_{j=1}^{k-1} \phi_{k-1,j}R_j}, \text{ for } k \geq 3. \quad (14)$$

The Box-Jenkins methodology can be used to similarly identify optimal linear autoregressive with exogenous inputs (ARX) models. In this case, the partial cross-correlation is used to identify the relevant lags of the exogenous variables.

Box-Jenkins and partial autocorrelation analysis have been used as the basis for IVS in the development of ANN models. In some examples, ANNs have been developed based on an optimal set determined for an ARX model. The ANNs were found to produce better predictions than the ARX model, and this has often provided the justification for ANN modelling in favour of conventional time-series techniques (Rodriguez et al., 1997). However, although this demonstrated the additional flexibility of ANN architectures to describe more complex behaviour, the ANN developed may not have been optimal, since the selection of inputs was based on the identification of a linear model. It may be the case that variables that are highly informative, but non-linearly correlated with the output variable, will be overlooked and excluded from the ANN model.

7.4 Mutual information

The limitations of linear correlation analysis have created interest in alternative statistical measures of dependence, which are more adept at identifying and quantifying dependence that may be chaotic or non-linear; and which may therefore be more suitable for the development of ANN models. Mutual information (MI) is a measure of dependence that is based on information theory and the notion of entropy Shannon (1948), and is determined by the equation

$$I(X;Y) = \iint p(x,y) \log \frac{p(x,y)}{p(x)p(y)} dx dy, \quad (15)$$

where I denotes the MI between X and Y . MI measures the quantity of information about a variable Y that is provided by a second variable X . However, it is often convenient to simply regard MI as a more general measure of correlation, since despite originating from information theory, rather than statistics, MI is not entirely unrelated to Pearson correlation. In fact, it can be shown that in the case of noise-free, Gaussian data, MI will be related to linear correlation according the relationship:

$$I(X;Y) = \frac{1}{2} \log \left(1 - R_{XY}^2 \right). \quad (16)$$

The advantage of MI over linear correlation is that MI is based solely on probability distributions within the data and is therefore an arbitrary measure, which makes no assumption regarding the structure of the dependence between variables. It has also been found to be robust due to its insensitivity to noise and data transformations (Battiti, 1994;

Darbellay, 1999; Soofi & Retzer, 2003). Consequently, MI has recently been found to be a more suitable measure of dependence for IVS during ANN development. Torkkola (2003) also discusses the merit of analysing MI, given that it provides an approximation to the Bayes error rate. Bayes' theorem, which gives the most general form of statistical inference, is given by

$$p(y|x \in X) = \frac{p(y \in Y)p(x \in X|Y)}{p(x \in X)} \quad (17)$$

Bayes' theorem can be used to determine the expectation $E(y|x \in X)$, assuming the probability distributions are known. MI provides an approximation to the error associated with the Bayes estimate of $E(y|X \in X)$, since it can be shown that the minimum error will be achieved for a maximal value of $I(X;Y)$. In effect, MI provides a generic estimation of the *modellability* of an output variable Y , which therefore makes MI an attractive measure of relevance in determining an optimal set of input variables, since we would seek the set of input variables that maximises the JMI, that is, the MI between the output and the input variable set.

The MIFS algorithm is a forward selection filter proposed by Battiti (1994) to address shortcomings with algorithms based on linear correlation. Considering the candidate set C and output variable Y , the MIFS algorithm proceeds as follows:

- i. Let $X \rightarrow \phi$.
- ii. While $|X| < k$,
- iii. For each $c \in C$,
- iv. Estimate $I(c, Y|X) = I(c, Y) - \beta \sum_{x \in X} I(c; x)$.
- v. Find c_s that maximises $I(c, Y|X)$.
- vi. Move c_s to X .
- vii. Return X .

MIFS defines a MR filter and identifies suitable candidates according to the estimated bivariate MI between candidate variables and the output variable. The MI between the most salient candidate c_s and the already selected variables in X is estimated and subtracted from the relevance in order to achieve minimum redundancy. The heuristic weighting β determines the degree of redundancy checking within MIFS. If $\beta = 0$, then MIFS will neglect relationships between candidates and MIFS is reduced to a MI ranking filter. Increasing β increases the influence of candidate redundancy on selections, however if β is too large, then the redundancy is overstated and candidate interactions dominate the selection of variables, rather than the input-output relationships (Kwak & Choi, 2002). Battiti (1994) recommends that a weighting of 0.5–1.0 is appropriate. A criticism of the forward selection approach is that the JMI of the input variable set must be considered in order to correctly determine the optimality of the input variables (Bonnlander & Weigend, 1994). However, in MIFS, the forward selection procedure considers variables individually, and optimality of the JMI is inferred by the mRMR selection. The heuristic redundancy parameter β provides only an approximation to the conditional dependence and does not necessarily relate to the JMI.

7.5 Partial mutual information

Sharma (2000) proposed an IVS filter that is structured similarly to MIFS, but is based instead upon direct estimation of partial mutual information (PMI). The algorithm has been successfully applied to select predictors for hydrological models (Sharma, 2000) and ANN water quality forecasting models (Bowden et al., 2002; Kingston, 2006). The PMI-based filter also incorporates a mechanism for testing the significance of candidate variables, so that the termination point of the algorithm is optimally determined, which is an improvement over the greedy selection of k variables in MIFS. In this case, the termination criterion is based upon the distribution of the error in PMI estimation, which is numerically approximated by a bootstrap approach (Sharma, 2000). The significance of the most relevant candidate is determined by direct comparison to the upper confidence bound on the estimation error.

The details of the algorithm, are as follows (May et al., 2009a):

- 1: Let $\mathbf{X} \rightarrow \phi$ (Initialisation)
- 2: While $C \neq \phi$ (Forward selection)
- 3: Construct kernel regression estimator $\hat{m}_Y(\mathbf{X})$
- 4: Calculate residual output $u = Y - \hat{m}_Y(\mathbf{X})$
- 5: For each $c \in C$
- 6: Construct kernel regression estimator $\hat{m}_c(\mathbf{X})$
- 7: Calculate residual candidate $v = c - \hat{m}_c(\mathbf{X})$
- 8: Estimate $I(v; u)$
- 9: Find candidate c_s (and v_s) that maximises $I(v; u)$
- 10: For $b = 1$ to B (Bootstrap)
- 11: Randomly shuffle v_s to obtain v_s^*
- 12: Estimate $I_b = I(v_s^*; u)$
- 13: Find confidence bound $I_b^{(95)}$
- 14: If $I(v_s, u) > I_b^{(95)}$ (Selection/termination)
- 15: Move c_s to $\mathbf{X}$
- 16: Else
- 17: Break
- 18: Return selected input set $\mathbf{X}$.

Here, B is the bootstrap size; and $I_b^{(95)}$ denotes the 95th percentile bootstrap estimate of the randomised PMI, I_b . The algorithm is structured in a similar fashion to MIFS (Battiti, 1994), but has two advantages. First, PMIS inherently handles redundancy within the candidate set through the direct estimation of PMI, whereas MIFS approximates the effect of selected inputs by means of a heuristic weighting factor. Second, while MIFS uses greedy selection of a pre-specified number of input variables, PMIS includes a criterion that automatically determines the optimum point at which to terminate the selection procedure. The optimality of the input variable set is ensured because PMI is directly estimated, and the JMI can be

determined as a result of the MI chain-rule decomposition, which is given as (Cover & Thomas, 1991)

$$I(x_1, \dots, x_p; y) = I(x_1; y) + I(x_2; y|x_1) + \dots + I(x_p; y|x_1, \dots, x_{p-1}). \quad (18)$$

Recall that in MIFS, the JMI cannot be directly approximated because redundancy is only approximated by a heuristic weighting factor. The termination criterion in PMIS automatically determines the optimal number of input variables, since the increase in JMI is additive, and once the contribution of an additional input variable is insignificant, the selection process terminates and the JMI will be maximised.

An additional benefit of the PMI-based approach is that the information yield during IVS provides a useful indication of the contribution of each input variable to the prediction of the output variable. Several methods for determining the usefulness of input variables based on analysis of the trained model have been described and range from sensitivity analysis, to aggregation of the weights associated with each input variable. However, the relative importance of an input variable can be determined statistically from the MI between each input and the output variable (Soofi & Retzer, 2003). The PMI estimated for a given variable can potentially also be used to classify input variables as informative, or weakly informative, as defined by Kohavi & John (1997), by considering the conditional relevance. Kingston (2006) considered several techniques for determining the relative importance (RI) of input variables and found that the method based on PMI yielded similar estimates of RI as methods based on analysis of the connection weights for a trained MLP. The method for estimating RI was based on the formula

$$RI(i) = \frac{I'(x_i; y)}{\sum_{x \in X} I'(x; y)}, \quad (19)$$

where I' denotes the PMI estimated for candidate variable x during PMIS.

The usefulness of RI is that it provides an indication of the way in which the ANN generates predictions. Although it is assumed that the ANN is using all of the input variables, it may in fact only require some small subset of the available input variables to generate predictions. In this case, further refinements to the ANN can be made based on this interpretation. Such considerations might be important when considering the cost of data collection that is associated with ongoing deployment of ANN models. One might consider sacrificing model accuracy in favour of cost reductions in ANN maintenance by reducing the number of input variables even further, which would reduce data requirements. The RI of variables may also encourage increased efforts toward the development of measurement techniques to ensure data quality for important variables.

The main limitation of the PMI filter is the computational effort associated with the bootstrap estimation of MI. Even obtaining a single estimate of MI is an expensive computation due to the $O\{n^2\}$ density estimation, and computational efficiency is therefore influenced by the sample size, n . The use of a bootstrap-based test for significance further adds to the overall computational burden by increasing the number of estimations of MI required to implement IVS. Sharma (2000) restricts the size of the bootstrap to 100 in order to maintain reasonable analysis times. However, a small bootstrap of this size might compromise the accuracy of the termination criterion, since potentially the confidence bounds may be poorly estimated. May et al. (2009b) introduce several alternative termination criteria that eliminate the bootstrap significance-test from the algorithm. These were found to provide a significant improvement in both the accuracy and computational effort associated with the PMI-based algorithm.

Fernando et al. (2009) also present a fast implementation of the PMI-based filter design using an average-shifted histogram (ASH) (Silverman, 1986), rather than kernel density estimation, for the computation of PMI during selection.

7.6 ICAIVS

A hybrid ICA and IVS filter algorithm (ICAIVS) was proposed by Back & Trappenberg (2001), which consists of two main steps: (Trappenberg et al., 2006)

- i. Produce a set of candidates which are as independent as possible (ICA).
- ii. Measure relevance between the independent candidate variables and the desired output variables (IVS).

Here, the statistical analysis of input relevance is based on estimation of the joint dependence of combinations of input variables and considers all combinations from $c(x_1^p, y)$ through to $c(x_1^p, \dots, x_n^p, y)$, where p denotes the order of the dependence that is measured. The IVS procedure then compares the relevance for each subset of variables, with respect to the average dependence for all subsets, and a subset is selected if the dependence exceeds some threshold value K .

A drawback of ICAIVS is that the algorithm does not scale well, given the large number $(3^n - 1)$ of statistical tests that must be performed, considering only second-order statistics. Recently, an improved version of ICAIVS was described that utilised MI as the statistical measure of dependence (Trappenberg et al., 2006). This reduced the number of statistical tests by considering only first order MI. However, the problem of specifying a suitable threshold value still remains. Both Back & Trappenberg (2001) and Trappenberg et al. (2006) used a value of 0.2 for this threshold. However, in this case the threshold value is heuristically determined, and a suitable value may vary depending on the dataset.

8. Summary

Input variable selection remains an important part of ANN model development, due to the negative impact that poor selection can have on the performance of ANNs during training and deployment post-development. The brief taxonomy of variable selections indicates the different classes of algorithms that have been employed to perform the IVS task, of which many more variations within each class are described within ANN literature. A summary of the attributes of different IVS algorithms is given in Table 2.

Several key considerations should guide a modeller in determining the most appropriate approach to take for IVS in any given circumstance. The first consideration is whether the algorithm is suitable for identifying non-linear relationships, which is a fundamental requirement for the development of ANN models. Many methods applied successfully within the context of linear regression are not suited to ANN development, where there is a presumption of non-linearity, and important relationships within the data may not be identified by a linear selection algorithm.

The choice between model-free and model-based IVS algorithms is also a key consideration. The restriction imposed by wrapper designs on the choice of model architecture, and the computational requirements associated with training multiple model instances, will dictate which approach to use. Where portability between ANN architectures is not required, the use of wrapper or embedded algorithms is likely to be straightforward to implement, and are a pragmatic approach to take. However, in all cases a wrapper that includes a regularisation

Algorithm	Criterion	Type	Non-linear Redundancy		Search
Rank-R	Correlation	Filter	No	No	Greedy
Box-Jenkins	ACF/PACF	Filter	No	Yes	Greedy
R'	Correlation	Filter	No	Yes	Stepwise
SVR	Rank error	Wrapper	Yes	No	Greedy
Forward selection	Error	Wrapper	Yes	Yes	Nested
Backward selection	Error	Wrapper	Yes	Yes	Nested
GA-ANN	Error	Wrapper	Yes	Yes	Heuristic
MIFS	MI	Filter	Yes	Yes	Nested
PMIS	PMI	Filter	Yes	Yes	Nested
RFE	Weight size	Embedded	Yes	Yes	Nested
EANN	Error	Embedded	Yes	Yes	Heuristic

Table 2. Characteristics of input variable selection algorithms

term, or considers the impact of model size, should be used to achieve an optimal trade-off between model performance and size.

High computational requirements are the main drawback of model-based techniques. The training time for a single instance of an ANN will be a strong indication of the feasibility of a model-based approach. Popular methods for wrappers and embedded algorithms are typically applied where the number of training examples is relatively small, and the number of candidate input variables is large. Alternatively, fixed ANN architectures like the GRNN provide fast-training, which increases the suitability of model-based IVS. The search strategy employed in a wrapper approach also represents a balance between the number of unique solutions considered, and the computational effort expended. Backward elimination algorithms, such as RFE, train relatively few ANN models and can be highly efficient, but will restrict the search due to nesting behaviour. Brute force search is likely to be infeasible in most cases, but evolutionary search approaches can provide a suitable compromise that allows for greater coverage of combinations of input variables. If evolutionary optimisation is to be used for ANN training, then the embedded EANN approach may be appropriate, as an extension of weight optimisation.

In contrast to model-based IVS, filter algorithms offer a fast, model-free approach to variable selection, and are particularly suited to applications where independence from a specific ANN architecture is required; or, where computational efficiency is sought. A generic estimation of input variable importance and redundancy avoids the traps of model over-fitting, or idiosyncrasies of a specific ANN architecture. The ability to identify an optimal set of input variables prior to training an ANN eliminates the computational burden associated with training and model selection, which can reduce the overall effort of ANN development.

Most filter designs are based on bivariate statistics, since estimation of statistics for multivariate data is often too inaccurate to facilitate IVS due to finite-sample error, and typically use a ranking scheme or sequential forward selection scheme. Simple ranking schemes are greedy and select more input variables than necessary, by ignoring redundancy of candidates, and are not ideally suited to multivariate ANN regression. Forward selection provides an efficient search, and greediness can be overcome provided that adequate redundancy checking is incorporated into the statistical analysis of variables. This ensures that the approach selects the most informative input set with the smallest number of variables.

Mutual information provides a generic measure of the relevance of a candidate input variable and is highly suitable for ANN development because it measures the strength of both linear and non-linear relationships. MI is also less sensitive to data transformations than correlation, which makes it more reliable, even for linear data analysis applications. The PMI-based forward selection approach provides a forward selection strategy that both minimises redundancy and maximises relevance. The approach determines the optimal set of input variables by estimating the maximum joint mutual information. Estimation of MI is more computationally intensive than estimation of correlation, due to the need to estimate density functions of the data. However, several refinements related to estimation of critical values, and the use of faster methods for density estimation can significantly reduce the computational effort of this approach.

9. References

- Akaike, H. (1974). A new look at the statistical model identification, *IEEE Transactions of Automatic Control* 19: 716–723.
- Back, A. D. & Trappenberg, T. P. (2001). Selecting inputs for modeling using normalized higher order statistics and independent component analysis, *IEEE Transactions on Neural Networks* 12(3): 612–617.
- Battiti, R. (1994). Using mutual information for selecting features in supervised neural net learning, *IEEE Transactions on Neural Networks* 5(4): 537–550.
- Bellman, R. (1961). *Adaptive control processes: a guided tour*, Princeton University Press, New Jersey.
- Blum, A. & Langley, P. (1997). Selection of relevant features and examples in machine learning, *Artificial Intelligence* 97(1–2): 245–271.
- Bonnlander, B. V. & Weigend, A. S. (1994). Selecting input variables using mutual information and nonparametric density estimation, *International Symposium on Artificial Neural Networks*, Taiwan, pp. 42–50.
- Bowden, G. J. (2003). *Forecasting water resources variables using artificial neural techniques*, Ph.d, University of Adelaide.
- Bowden, G. J., Dandy, G. C. & Maier, H. R. (2005). Input determination for neural network models in water resources applications. part 1 - background and methodology, *Journal of Hydrology* 301(1–4): 75–92.
- Bowden, G. J., Maier, H. R. & Dandy, G. C. (2002). Optimal division of data for neural network models in water resources applications, *Water Resources Research* 38(2): 1–11.
- Bowden, G. J., Nixon, J. B., Dandy, G. C., Maier, H. R. & Holmes, M. (2006). Forecasting chlorine residuals in a water distribution system using a general regression neural network, *Mathematical and Computer Modelling* 44(5–6): 469–484.
- Box, G. E. P. & Jenkins, G. M. (1976). *Time Series Analysis, Forecasting and Control*, Holden-Day Inc., San Francisco.
- Carreira-Perpinan, M. A. (1997). A review of dimension reduction techniques, *Technical report*, Dept. of Computer Science, University of Sheffield.
- Cover, T. M. & Thomas, J. A. (1991). *Elements of information theory*, Wiley series in telecommunications, John Wiley & Sons, Inc., New York.
- Craven, M. W. & Shavlik, J. W. (1998). Using neural networks for data mining, *Future Generation Computer Systems* 13(2–3): 211–229.
- Darbari, A. (2000). Rule extraction from trained ANN: A survey, *Technical report*, Institute of Artificial Intelligence, Dept. of Computer Science, TU Dresden.

- Darbellay, G. A. (1999). An estimator of the mutual information based on a criterion for independence, *Computational Statistics & Data Analysis* 32: 1–17.
- Ding, C. & Peng, H. (2005). Minimum redundancy feature selection from microarray gene expression data, *Journal of Bioinformatics and Computational Biology* 3(2): 185–205.
- Fernando, T. M. K. G., Maier, H. R. & Dandy, G. C. (2009). Selection of input variables for data driven models: An average shifted histogram partial mutual information estimator approach., *Journal of Hydrology* 367(3–4): 165–176.
- Fodor, I. K. (2002). A survey of dimension reduction techniques, *Technical report*, Center for Applied Scientific Computing, Lawrence Livermore National Laboratory.
- Gibbs, M. S., Morgan, N., Maier, H. R., Dandy, G. C., Nixon, J. B. & Holmes, M. (2006). Investigation into the relationship between chlorine decay and water distribution parameters using data driven methods, *Mathematical and Computer Modelling* 44(5-6): 485–498.
- Guyon, I. & Elisseeff, A. (2003). An introduction to variable and feature selection, *The Journal of Machine Learning Research* 3: 1157–1182.
- Hastie, T., Tibshirani, R. & Friedman, J. (2001). *The Elements of Statistical Learning. Data Mining, Inference and Prediction*, Springer Series in Statistics, Springer, New York.
- Izrailev, S. & Agrafiotis, D. K. (2002). Variable selection for QSAR by artificial ant colony systems, *SAR and QSAR in Environmental Research* 13(3–4): 417–423.
- Kingston, G. B. (2006). *Bayesian Artificial Neural Networks in Water Resources Engineering*, Ph.d., The University of Adelaide.
- Kohavi, R. & John, G. (1997). Wrappers for feature selection, *Artificial Intelligence* 97(1–2): 273–324.
- Kwak, N. & Choi, C.-H. (2002). Input feature selection for classification problems, *IEEE Transactions on Neural Networks* 13(1): 143–159.
- Le Cun, Y., Denker, J. S. & Solla, S. A. (1990). Optimal brain damage, *Advances in Neural Information Processing Systems*, Morgan Kaufmann, pp. 598–605.
- Maier, H. R. & Dandy, G. C. (2000). Application of neural networks to forecasting of surface water quality variables: Issues, applications and challenges, *Environmental Modelling & Software* 15(1): 101–124.
- Mallows, C. L. (1973). Some comments on Cp, *Technometrics* 15: 661–675.
- Marcoulides, G. A. & Drezner, Z. (2003). Model specification searches using ant colony optimization algorithms, *Structural Equation Modeling: A Multidisciplinary Journal* 10(1): 154–164.
- May, R. J., Maier, H. R. & Dandy, G. C. (2009a). Data splitting for artificial neural networks using SOM-based stratified sampling, *Neural Networks* 23: 283–294.
- May, R. J., Maier, H. R. & Dandy, G. C. (2009b). Development of artificial neural networks for water quality modelling and analysis, in G. Hanrahan (ed.), *Modelling of Pollutants in Complex Environmental Systems*, Vol. 1, ILM Publications, London, UK, pp. 27–62.
- Miller, A. J. (1984). Selection of subsets of regression variables, *Journal of The Royal Statistical Society. Series A*. 147(3): 389–425.
- Olsson, J., Uvo, C. B., Jinno, K., Kawamura, A., Nishiyama, K., Koreeda, N., Nakashima, T. & Morita, O. (2004). Neural networks for forecasting rainfall by atmospheric downscaling, *Journal of Hydraulic Engineering, ASCE* 9(1): 1–12.
- Rodriguez, M. J., Serodes, J.-B. & Cote, P. (1997). Advanced chlorination control in drinking water systems using artificial neural networks, *Water Supply* 15(2): 159–168.

- Sarle, W. (1997). Neural network FAQ. Periodic posting to the Usenet newsgroup comp.ai.neural-nets.
- Schwarz, G. (1978). Estimating the dimension of a model, *Annals of Statistics* 6(2): 461–464.
- Scott, D. W. (1992). *Multivariate density estimation: theory, practice and visualisation*, John Wiley and Sons, New York.
- Shannon, C. E. (1948). A mathematical theory of communication, *Bell System Technical Journal* 27: 379–423.
- Sharma, A. (2000). Seasonal to interannual rainfall probabilistic forecasts for improved water supply management: Part 1 - a strategy for system predictor identification, *Journal of Hydrology* 239: 232–239.
- Shen, Q., Jiang, J.-H., Tao, J.-C., Shen, G.-L. & Yu, R.-Q. (2005). Modified ant colony optimization algorithm for variable selection in QSAR modeling : QSAR studies of cyclooxygenase inhibitors, *Journal of Chemical Information and Modeling* 45(4): 1024–1029.
- Silverman, B. W. (1986). *Density estimation for statistics and data analysis*, Chapman and Hall, London.
- Soofi, E. S. & Retzer, J. J. (2003). Information importance of explanatory variables, *IEE Conference in Honor of Arnold Zellner: Recent Developments in the Theory, Method and Application of Entropy Econometrics.*, Washington.
- Specht, D. F. (1991). A general regression neural network, *IEEE Transactions on Neural Networks* 2(6): 568–576.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso, *Journal of the Royal Statistical Society. Series B (Methodological)* 58(1): 267–288.
- Tikka, J. (2008). *Input variable selection methods for construction of interpretable regression models*, Ph.d., Helsinki University of Technology.
- Torkkola, K. (2003). Feature extraction by non-parametric mutual information maximization, *Journal of Machine Learning Research* 3: 1415–1438.
- Trappenberg, T. P., Ouyang, J. & Back, A. D. (2006). Input variable selection: mutual information and linear mixing measures, *IEEE Transactions on Knowledge and Data Engineering* 18(1): 37–46.
- Wold, H. (1966). Estimation of principal components and related models by iterative least squares, in P. Krishnaiah (ed.), *Multivariate Analysis*, Academic Press, New York.

Artificial Neural Networks and Efficient Optimization Techniques for Applications in Engineering

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1. Introduction

This chapter proposal describes some artificial neural network (ANN) neuromodeling techniques used in association with powerful optimization tools, such as natural optimization algorithms and wavelet transforms, which can be used in a variety of applications in Engineering, for example, Electromagnetism (Cruz, 2009), Signal Processing (Peixoto et al., 2009b) and Pattern Recognition and Classification (Magalhães et al., 2008).

The application of ANN models associated with RF/microwave devices (Cruz et al., 2009a, 2009b; Silva et al., 2010a) and/or pattern recognition (Lopes et al., 2009) becomes usual. In this chapter, we present neuromodeling techniques based on one or two hidden layer feedforward neural network configurations and modular neural networks – trained with efficient algorithms, such as Resilient Backpropagation (RPROP) (Riedmiller & Braun, 1993), Levenberg-Marquardt (Hagan & Menhaj, 1999) and other hybrid learning algorithms (Magalhães et al., 2008), in order to find the best training algorithm for such investigation, in terms of convergence and computational cost. The mathematical formulation and implementation details of neural network models, wavelet transforms and natural optimization algorithms are also presented.

Natural optimization algorithms, which are stochastic population-based global search methods inspired in nature, such as genetic algorithm (GA) and particle swarm optimization (PSO) are effective for optimization problems with a large number of design variables and inexpensive cost function evaluation (Kennedy & Eberhart, 1995; R. Haupt & S. Haupt, 2004). However, the main computational drawback for optimization of nonlinear devices relies on the repetitive evaluation of numerically expensive cost functions (Haupt & Werner, 2007; Rahmat-Samii, 2003). Finding a way to shorten the optimization cycle is highly desirable. In case of GA, for example, several schemes are available in order to improve its performance, such as: the use of fast full-wave methods, micro-genetic algorithm, which aims to reduce the population size, and parallel GA using parallel computation (R. Haupt & S. Haupt, 2004; Haupt & Werner, 2007). Therefore, this chapter

also describes some hybrid EM-optimization methods, using continuous-GA and PSO algorithms, blended with multilayer perceptrons (MLP) artificial neural network models. These methods are applied to design spatial mesh filters, such as frequency selective surfaces (FSSs). Moreover, the MLP model is used for fast and accurate evaluation of cost function into continuous GA and PSO simulations, in order to overcome the computational requirements associated with full wave numerical simulations.

Wavelets and artificial neural networks (ANN) have generated enormous interest in recent years, both in science and practical applications (Peixoto et al., 2009c). The big advantage of using wavelets is the fact of these functions make a local behavior, not only in the frequency domain but also in the field space and time. ANNs are capable of learning from a training set, which makes it useful in many applications, especially in pattern recognition. The purpose of using wavelet transforms is to find an easier way to compress and extract the most important features present in images, thereby creating a vector of descriptors that should be used to optimize the pattern recognition by a neural network. The array of descriptors contains elements whose values accurately describe the image content, and should take up less space than a simple pixel by pixel representation. The greatest difficulty found in this process is the generation of this vector, where the image content of interest should be very well described, in order to show really relevant features. Thus, this chapter proposal also shows a way to use a multiresolution technique, such as the wavelet transforms, to perform filtering, compression and extraction of image descriptors for a later classification by an ANN.

This chapter is organized in six sections. Section 2 presents the most important fundamentals of artificial neural networks and the methodology used for the investigated applications. In section 3, artificial neural networks are optimized using wavelet transforms for applications in image processing (extraction and compression). Section 4 presents an EM-optimization using artificial neural networks and natural optimization algorithms for the optimal synthesis of stop-band filters, such as frequency selective surfaces. Section 5 shows a modular artificial neural network implementation used for pattern recognition and classification. Finally, section 6 presents important considerations about the artificial neural network models used in association with efficient optimization tools for applications in Engineering.

2. Artificial Neural Networks

2.1 Fundamentals

Rosenblatt (1958) perceptron is the most used artificial neuron in neural network configurations and is based on the nonlinear model proposed by McCulloch and Pitts (1943). In this model, neurons are signal processing units composed by a set of input connections (weights), an adder (for summing the input signals, weighted by the respective synapses of a neuron, constituting a linear combiner) and an activation function, that can be linear or nonlinear, as shown in Fig. 1(a). The input signals are defined as x_i , $i = 0, 1, \dots, N_i$, whose result corresponds to the level of internal activity of a neuron net_j , as defined in (1), where $x_0 = +1$ is the polarization potential (or bias) of the neurons. The output signal y_j is the activation function response $\phi(\cdot)$ to the activation potential net_j , as shown in (2) (Silva et al., 2010b).

$$net_j = \sum_{i=0}^{N_i} w_{ji} \cdot x_i \quad (1)$$

$$y_j = \phi(net_j) \quad (2)$$

For a feedforward neural network (FNN), the artificial neurons are set into layers. Each neuron of a layer is connected to those of the previous layer, as illustrated in Fig. 1(b). Signal propagation occurs from input to output layers, passing through the hidden layers of the FNN. Hidden neurons represent the input characteristics, while output neurons generate the neural network responses (Haykin, 1999).

Modular artificial neural network is based on a principle commonly used: divided to conquer. This concept aims to divide a large and complex task in a set of sub-tasks that are easier to be solved. The modular artificial neural network could be defined, in summary, as a set of learning machines, also called experts, whose decisions are combined to achieve a better answer than the answers achieved individually, that is, a machine with a better performance.

In the past few years, one of the main areas of learning machine is the characterization of methods capable to design this kind of machines. There are two types of these machines: static and dynamic structures. Modular neural networks, as seen in Fig. 1(c), is a dynamic type. The input signal is used by the gating network to design the global response. An advantage of modular artificial neural networks, when compared with other neural networks, is the learning speed. The machine learning process is accelerated in case of problems where it is observed a natural decomposition of data at simple functions. To develop the modular machine architecture and to implement the experts, it is usual to apply multilayer perceptrons (MLP) neural networks.

2.2 Methodology

Generally, the design of a neural network is composed by three main steps: configuration – how layers are organized and connected; learning – how information is stored; generalization – how neural network produces reasonable outputs for inputs not found in the training (Haykin, 1999). In this work, we use feedforward and modular neural networks associated with supervised learning to develop neural network models.

In the computational simulation of supervised learning process, a training algorithm is used for the adaptation of neural network synaptic weights. The instantaneous error $e(n)$, as defined in (3), represents the difference between the desired response, $d(n)$, and the neural network output, $z(n)$, at the iteration n , corresponding to the presentation of the n th training pattern $[x(n);(d(n))]$ (Silva et al., 2010b).

$$e(n) = z(n) - d(n) \quad (3)$$

Supervised learning can be illustrated through the block diagram of Fig. 2(a) and has as objective the minimization of the mean square error $E(t)$, given in (4), where the index t represents the number of training epochs (one complete presentation of all training examples, $n = 1, 2, \dots, N$, where N is the total number of examples, called an epoch) (Silva et al., 2010b).

$$E(t) = \frac{1}{N} \sum_{n=1}^N \frac{1}{2} e(n)^2 \quad (4)$$

Currently, there are several algorithms for the training of neural networks that use different optimization techniques (Peixoto et al., 2009a). The most popular training algorithms are those derived from backpropagation algorithm (Rumelhart et al., 1986). Among the family of backpropagation algorithms, the RPROP shows to be very efficient in the solution of complex electromagnetic problems. In this work, the stop criteria are defined in terms of the maximum number of epochs and/or the minimum error and the activation function used was the sigmoid tangent (Haykin, 1999).

After training, the neural network is submitted to a test, in order to verify its capability of generalizing to new values that do not belong to the training dataset, for example, parts of the region of interest where there is not enough knowledge about the modeled device/circuit. Therefore, the neural network operates like a “black box” model that is illustrated in Fig. 2(b) (Silva et al., 2010b).

Resilient backpropagation algorithm is a first-order local adaptive learning scheme. The basic principle of RPROP is to eliminate the harmful influence of the partial derivative size in the weight update. Only the sign of the derivative is considered to indicate the direction of the weight update, Δw_{hp} , as given in (5).

$$\Delta w_{hp}(t) = \begin{cases} -\Delta_{hp}(t), & \text{if } \frac{\partial E}{\partial w_{hp}}(t) > 0 \\ +\Delta_{hp}(t), & \text{if } \frac{\partial E}{\partial w_{hp}}(t) < 0 \\ 0, & \text{elsewhere} \end{cases} \quad (5)$$

The second step of RPROP algorithm is to determine the new update-values $\Delta_{hp}(t)$. This is based on a sign-dependent adaptation process, similar to the learning-rate adaptation shown by Jacobs (1988). The changes in the weight size are exclusively determined by a weight ‘update-value’, Δ_{hp} , as given in (6).

$$\Delta_{hp}(t) = \begin{cases} \eta^+ \cdot \Delta_{hp}(t-1), & \frac{\partial E}{\partial w_{hp}}(t-1) \cdot \frac{\partial E}{\partial w_{hp}}(t) > 0 \\ \eta^- \cdot \Delta_{hp}(t-1), & \frac{\partial E}{\partial w_{hp}}(t-1) \cdot \frac{\partial E}{\partial w_{hp}}(t) < 0 \\ \Delta_{hp}(t-1), & \text{elsewhere} \end{cases} \quad (6)$$

Here, the following RPROP parameters were employed: $\eta^+ = 1.2$ and $\eta^- = 0.5$. The update-values were restricted to the range $10^{-6} \leq \Delta_{hp} \leq 50$.

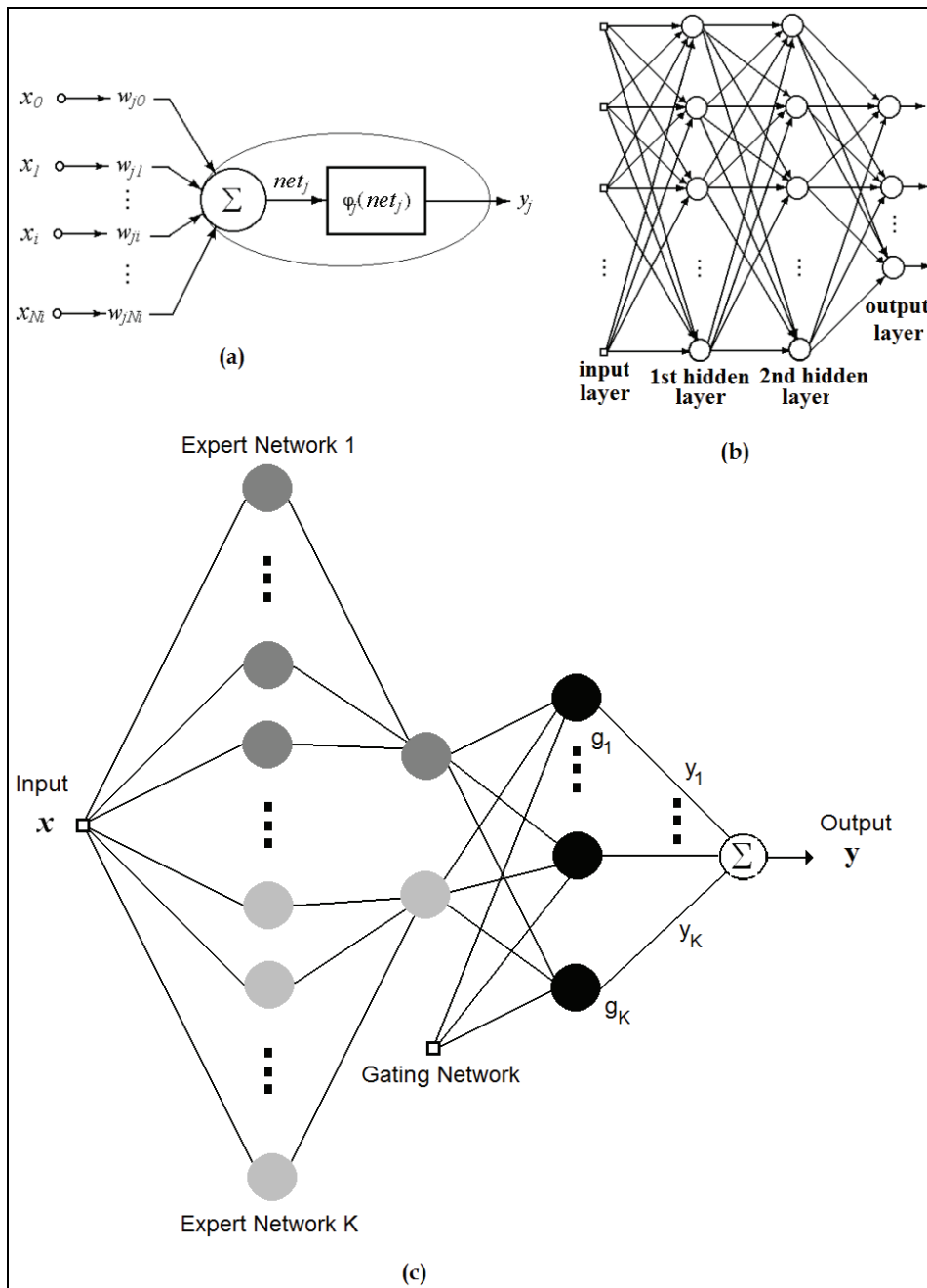


Fig. 1. (a) Nonlinear model of an artificial neuron; (b) FNN configuration with two hidden layers; (c) Extended modular neural network configuration with K experts

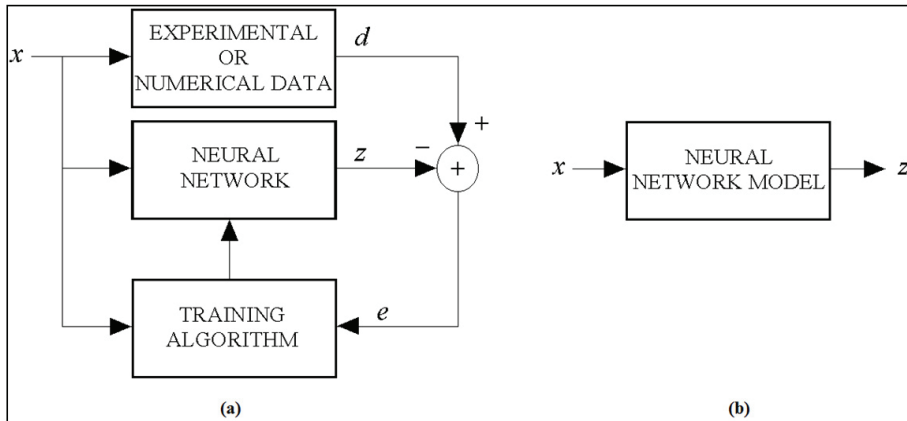


Fig. 2. (a) Block diagram of supervised learning; (b) neural network “black box” model

3. Artificial Neural Networks optimization using wavelet transforms

This section introduces the main concepts about wavelet transforms and some of their most important features used to optimize artificial neural networks: the extraction of characteristic information descriptors in order to improve training and pattern classification mechanisms. Moreover, a model that can be used to solve various problems related to pattern recognition is presented and a neural classifier is implemented to validate the importance of such optimization. Transforms are widely used mathematical tools to understand and analyze different signal behaviors. The objective of this analysis is to extract important information (or features) that can essentially represent the signal from some decomposition or transformation performed on it.

3.1 Wavelet transforms

Wavelet transforms have generated enormous interest from scientists, resulting in the development of applications in various areas, such as computer vision (Wang et al., 2009), seismology (Parolai et al., 2008), radar (Masnadi-Shirazi et al., 2009), astronomy (Ottensamer et al., 2008), image compression (Bhatia et al., 2009), signal filtering (Vimal et al., 2009), system optimization (Pinto, 2009) and many others. In general, the major advantage of using wavelet transforms is the possibility of applying it to non-stationary signals, which allows the study of function local behaviors, in both frequency and time-scale domains.

3.1.1 Advantages of use

Traditional methods of signal analysis based on Fourier transform can determine all the frequencies present in the signal, however its relationship to the time domain does not exist. To overcome this problem, it was created the Gabor transform (or STFT - Short Time Fourier Transform); the main idea of this transform is to introduce a new measure of local frequency as the local transformation observed the signal through a narrow window within which the signal remains nearly stationary (Oliveira, 2007). The problems in time and frequency domain resolution are result of a physical phenomenon known as Heisenberg's uncertainty principle (it is impossible to know the exact frequency and time that a signal occurs). This

phenomenon is independent of the transformation used (Oliveira, 2007). Therefore, the wavelet transform was developed as an alternative to the Gabor transform to solve the resolution problem.

Wavelets are mathematical transformations that separate signals in different components and extract each one of them with a resolution appropriated to its corresponding scale. According to the transformation characteristics, there is the Continuous Fourier Transform (CFT), that can be expressed as:

$$F(w) = \int_{-\infty}^{+\infty} f(t) e^{(-j2\pi ft)} dt \quad (7)$$

Knowing the spectrum $F(w)$ of a signal, it is possible to obtain it in time domain, using the inverse transform concept, according to (8):

$$f(t) = \int_{-\infty}^{+\infty} \frac{1}{2\pi} F(w) e^{(j2\pi ft)} dw \quad (8)$$

On the other hand, the Continuous Wavelet Transform (CWT) is given by:

$$CWT(\tau, a) = \int_{-\infty}^{+\infty} f(t) \frac{1}{\sqrt{a}} \psi^* \left(\frac{t - \tau}{a} \right) dt \quad (9)$$

and its corresponding inverse can be expressed according to (10):

$$f(t) = \frac{1}{C_\psi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} CWT(a, b) \left\{ \frac{1}{\sqrt{a}} \psi \left(\frac{t - \tau}{a} \right) \right\} \frac{dad b}{a^2} \quad (10)$$

where $\Psi(t)$ is the mother wavelet, τ and a are the translation and scale parameters, respectively.

3.1.2 Discrete wavelets

The continuous wavelet transform is calculated by performing continuous translations and scalings of a function over a signal. In practice, this transformation is not feasible because it is necessary to make endless translations and scalings, requiring much time, effort and computational redundancy. Discrete wavelets were introduced to overcome this problem and therefore they are used in this work. They are not translated or scaled continuously but in discrete steps, which is achieved from a modification of the continuous wavelet:

$$\psi_{s,\tau}(t) = \frac{1}{\sqrt{|s|}} \psi \left(\frac{t - \tau}{s} \right) \quad (11)$$

$$\psi_{j,k}(t) = \frac{1}{\sqrt{|s_0^j|}} \psi \left(\frac{t - k\tau_0 s_0^j}{s_0^j} \right) \quad (12)$$

where j and k are integers; $s_0 > 1$ is a fixed dilation parameter; τ_0 is the translation factor, which depends on the dilation factor.

Generally, it is chosen $s_0 = 2$ in order to have a sampling frequency called dyadic sampling $\tau_0 = 1$ is chosen for the temporal sampling, also dyadic This can be shown in (13) (Oliveira, 2007):

$$\psi_{j,k}(t) = \sqrt{2^j} \psi(2^j t - k) \quad (13)$$

When discrete wavelets are used to analyze a signal, the result is a series of wavelet coefficients, also called the series of wavelet decomposition (Oliveira, 2007). As a wavelet can be viewed as a bandpass filter, the scaled wavelet series can be seen as a set of bandpass filters, with a Q factor (set of filters fidelity factor). In practice, there is a discretized wavelet, with upper and lower limits for translations and scales. The wavelet discretization, associated with the idea of passing the signal through a filter set, results in the well known subband coding (Oliveira, 2007).

3.1.3 Multiresolution analysis

Mathematical transformations are used in a dataset to obtain additional information not available in the primitive data model. For example, it may be necessary to use a transformation that detects changes in color tones of a pixel neighborhood and its corresponding spatial location, and even that efficiently transposes these changes in a multiresolution space (Castleman, 1996). The multiresolution analysis using wavelet transforms has become increasingly popular with the release of JPEG-2000 standard (Weeks, 2007) and consists of a signal processing strategy where it is used a set of specialized filters to extract signal information, such as the range of frequencies present in it and their location as a function of the signal duration at different resolutions (Castleman, 1996). A brief description of the multiresolution analysis enables to display two functions responsible for generating the entire wavelet system: the scale function and the primary wavelet (or mother wavelet). The term mother comes from the fact that functions with different sizes are used in the process of transformation and all of them are originated from a specific or mother wavelet.

Scale function $\phi_{j,k}$ and the primary wavelets $\psi_{j,k}$ are considered orthogonal to follow the condition shown in (14):

$$\int_{-\infty}^{+\infty} \phi_{j,k}(x) \psi_{j,k}(x) dx = 0 \quad (14)$$

where $j \in \mathbb{Z}$ corresponds to the scale function parameter and $k \in \mathbb{Z}$ corresponds to the translation of $\frac{k}{2^j}$ in relation to the scale function and the primary wavelet, given by $j = 0$

and $k = 0$, respectively. Both scale and wavelet functions are defined in the set of real ($\mathbb{R}$) numbers, by scalings and translations of the mentioned functions. The translation parameter corresponds to time information in the transform domain and the scaling parameter is the process of compression and expansion of the signal (Mallat, 2009). In Fig. 3 is shown an example of a scale function and a primary Haar wavelet.

Therefore, one can say that multiresolution analysis using discrete time wavelets corresponds to successive band-pass filtering, through which signals are decomposed at

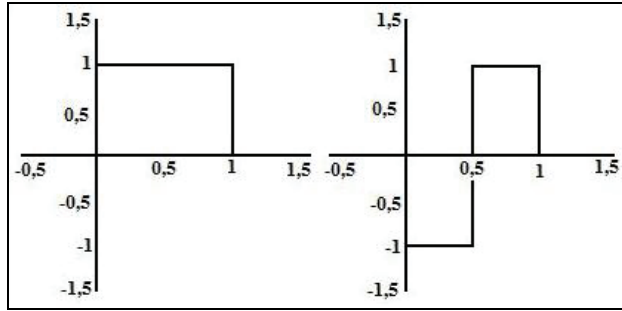


Fig. 3. Scale function and the primary Haar wavelet

each step, in terms of approximation and details. Fig. 4 illustrates this procedure being applied to an input image, where filters are used in successive rows and columns, creating the new scales. The reverse process, which performs the sum of subspaces, can reconstruct the original image. The dilation equations expressed by $h(k)$ represent the low-pass filters that generate the approximations of the original image. However, the translation equations $g(k)$ represent the high-pass filters and are responsible for obtaining the details of the original image. The decomposition of detail functions consists of details on vertical (high-pass filter at the rows and low-pass filters at the columns), details on horizontal (low-pass filter at the rows and high-pass filters at the columns), details on diagonal (high-pass filter at the rows and columns).

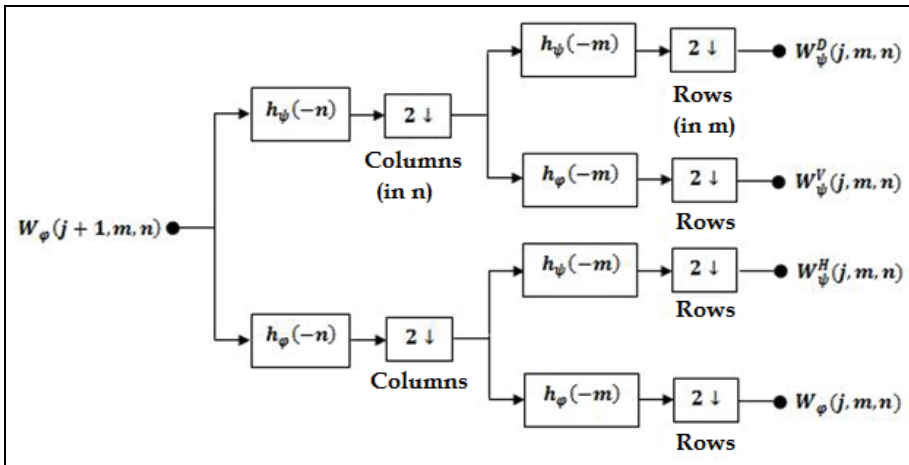


Fig. 4. Wavelet decomposition

Then, using an input image with scale of $(j+1)$, with m rows and n columns, it is shown in Fig. 5 a two level image decomposition example.

The original image and its approximations are the lighter areas of the picture. The other three remaining subpictures correspond to the three detail functions of the original image. In the second step, the lightest part of the picture is decomposed again, generating a new image approximation and three new detail subpictures. Thus, a twice smaller scale image

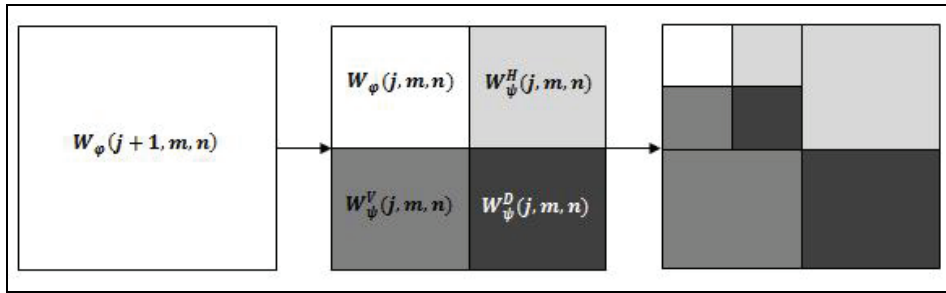


Fig. 5. Multiresolution representation of an image

was generated. Fig. 6 shows an initial image and the degree of refinement obtained with the wavelet transform. This form of decompose and reconstruct images can be implemented quickly and effectively, because of the use of wavelet transforms.

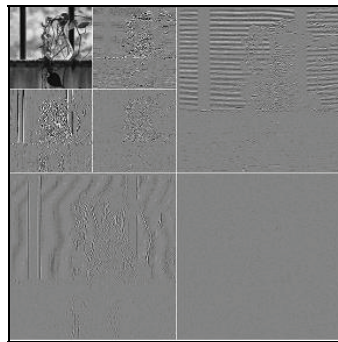


Fig. 6. Multiresolution analysis of an image

3.2 Wavelet pre-processing

Pre-processing step aims to improve the performance of the next steps. Then, the refinement of these data that will be used for training and classification are of fundamental importance. Designing a neural classifier consists of choosing architecture, training algorithm, quality of training dataset, among other aspects that must be optimized, in order to reduce the time used in network training at the same time that the accuracy of the results is increased. This optimization can be performed using wavelet transforms. The block diagram described in Fig. 7 shows that the data input used by the neural network may be submitted to a wavelet pre-processing or not. Here, both the situations are analyzed in parallel.

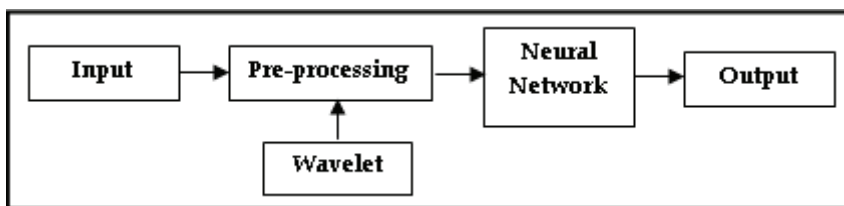


Fig. 7. Block diagram of a neural network system

3.3 Neural network implementation

A simple but effective neural classifier has been implemented at this step, using the classic aspects of this type of project. It is important to know that even robust classifiers as SVM (Support Vector Machine) can not function properly without a pre-processing. Fig. 8 shows a sample training set used in the experiment. This consists of 50 images containing the vowels A, E, I, O, U, with slight 2D dislocation and no noise.



Fig. 8. Training dataset model

Fig. 9 shows a sample of the test dataset (composed by 25 images), where there is the inclusion of uncorrelated noise to images up to four levels, aiming to difficult the neural network generalization.



Fig. 9. Test dataset model

Here, the neural network configuration used the following parameters: a multilayer perceptron neural network; the Levenberg-Marquardt training algorithm; number of epochs (150) or Error ($10e-5$) as stop criteria; two hidden layers and the sigmoid tangent as activation function.

The results are shown in Table 01, according to two categories of pre-processing: without wavelet - larger training set (designed pixel by pixel) and higher time to neural network learning; with wavelet (Daubechies, level 2) - smaller and more efficient training dataset due to the choice of better wavelet descriptors (higher intensity), which provides a faster network learning. It is important to observe that the stop criterion was the number of epochs while the error was kept around $10e-4$ and the generalization result was about 92%, which enabled to recognize images of the vowels a, e, i, o, u with small 2D dislocation levels ($<5\%$) and noise uncorrelated ($<50\%$).

PP	CT	TT (min)	Error (e-04)	GE
Without wavelet	4096x50	4.27	1.8120	94%
With wavelet	1000x50	1.91	7.4728	96%

Table 1. Results: PP - Pre-processing; TD - Training Dataset; TT -Time of Training; GE - Generalization

The MLP Neural Network was used efficiently in the presented pattern classification, generalizing in both cases above 94%. The extraction process of features was simplified and improved by the use of wavelet transform at the pre-processing step in order to optimize all

the classification step, since the time of training until the generalization of results. It is worth to say that not only the method, but also its ability of reducing the matrix of data, preserving the meaning of the information, was of extreme importance to reduce the computational cost more than twice.

4. Hybrid EM-optimization method for optimal design of FSS

4.1 Frequency selective surfaces

Frequency selective surface (FSS) (Wu, 1995; Munk, 2000; Campos, 2009; Cruz, 2009) is a two-dimensional periodic array of elements (patches or apertures) in a conducting screen, which could be either freestanding or supported by dielectric substrates, as shown in Fig. 10(a). FSSs are used for a variety of applications, such as: hybrid radomes (Cruz et al., 2010), polarizers and dichroic reflectors (Munk, 2000), waveguide filters (Monorchio, 2005), artificial magnetic conductors (Genovesi et al., 2006) and microwave absorbers (Liu et al., 2004). To provide high-performance filtering properties at microwave bands, electromagnetic engineers have investigated various types of FSS devices: reconfigurable FSS screens, fractal self-similar geometries (Cruz et al., 2009a), pixelized FSS screens, multilayered FSS structures, as well as FSSs on anisotropic substrates (Silva et al., 2007; Campos et al., 2001).

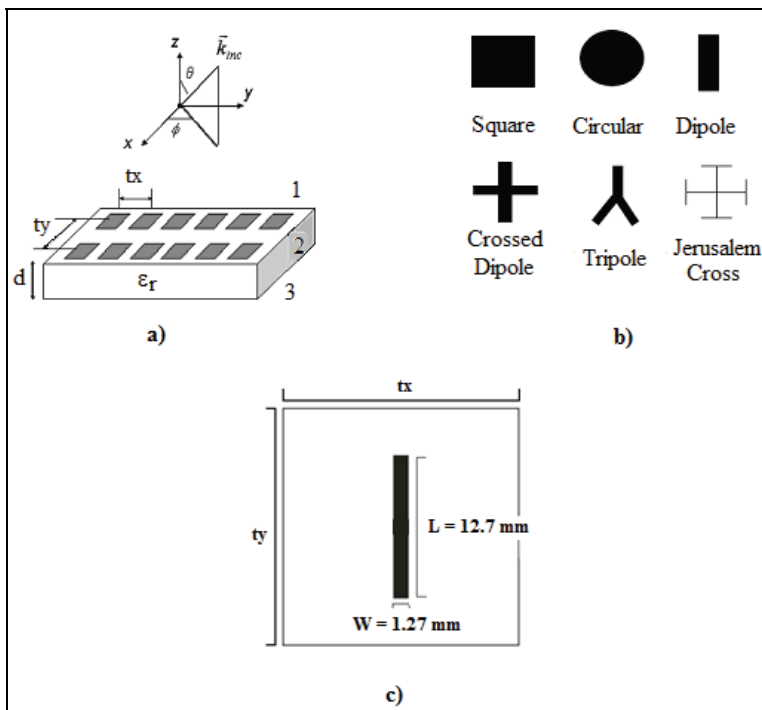


Fig. 10. (a) The conventional geometry of a FSS; (b) some patch element shapes; (c) thin dipole unit cell configuration

There is no closed form solution directly from a given desired frequency response to the corresponding FSS (Hwang et al, 1990). The analysis of scattering characteristics from FSS devices requires the application of rigorous full-wave techniques. Besides that, due to the computational complexity of using a full-wave simulator to evaluate the FSS scattering variables, many electromagnetic engineers still use a trial-and-error process until to achieve a given design criterion. Obviously, this procedure is very laborious and human dependent. Therefore, optimization techniques are required to design practical FSSs with desired filter specifications. Some authors have employed neural networks, PSO and GA for FSS design and optimization (Cruz et al., 2009b; Silva et al., 2010a; 2010b).

4.2 Natural optimization algorithms

Natural optimization algorithms are stochastic population-based global search methods that start with an initial population of candidate individuals for an optimal solution. Assuming an optimization problem with N_{var} input variables and N_{pop} individuals, the population at the k th iteration is a matrix $P(k)_{N_{pop} \times N_{var}}$ of floating-point elements, denoted by $p_{m,n}^k$, with each row corresponding to an individual (Cruz, 2009; Silva et al., 2010b). Under GA and PSO jargons, for example, individuals are named *chromosomes* and *particles* (or *agents*), respectively. The implementation details of these algorithms are described.

4.2.1 Continuous Genetic Algorithm

Continuous genetic algorithm is very similar to the binary-GA but works with floating-point variables. Continuous-GA chromosomes are defined by (15), where each one corresponds to a vector with N_{var} floating-point optimization variables. Each chromosome is evaluated by means of its associated cost, which is computed through the cost function E given in (4) (Cruz, 2009; Silva et al., 2010b).

$$chromosome(k,m) = [p_{m,1}^k, p_{m,2}^k, \dots, p_{m,N_{var}}^k], \quad m = 1, 2, \dots, N_{pop} \quad (15)$$

$$cost(k,m) = E(chromosome(k,m)) \quad (16)$$

Based on the cost associated to each chromosome, the population evolves through generations with the application of genetic operators, such as: selection, crossover and mutation. Fig. 11(a) gives a “big picture” overview of continuous-GA. The mating step includes roulette wheel selection presented in (Haupt & Werner, 2007; R. Haupt & S. Haupt, 2004). Population selection is performed after the N_{pop} chromosomes are ranked from lowest to highest costs. Then, the N_{keep} most-fit chromosomes are selected to form the mating pool and the rest are discarded to make room for the new offspring. Mothers and fathers pair in a random fashion through the blending crossover method (R. Haupt & S. Haupt, 2004). Each pair produces two offspring that contain traits from each parent. In addition, the parents survive to be part of the next generation. After mating, a fraction of chromosomes in the population will suffer mutation. Then, the chromosome variable selected for real-value mutation is added to a normally distributed random number (Cruz, 2009; Silva et al., 2010b).

4.2.2 Particle Swarm Optimization

Particle swarm optimization algorithm was first formulated by Kennedy and Eberhart (1995). The thought process behind the algorithm was inspired by the social behavior of animals, such as bird flocking or fish schooling. PSO algorithm is similar to continuous-GA since it begins

with a random population matrix. Unlike GA, PSO has no evolution operators such as crossover and mutation. Each particle moves along the cost surface with an individual velocity. A flow chart of the PSO algorithm is shown in Fig. 11(b). The implemented PSO algorithm updates the velocities and positions of the particles based on the best local and global solutions according to (17) and (18), respectively (Cruz, 2009; Silva et al., 2010b).

$$v_{m,n}^{k+1} = C \left[r_0 v_{m,n}^k + \Gamma_1 \cdot r_1 \cdot (p_{m,n}^{local\ best(k)} - p_{m,n}^k) + \Gamma_2 \cdot r_2 \cdot (p_{m,n}^{global\ best(k)} - p_{m,n}^k) \right] \quad (17)$$

$$p_{m,n}^{k+1} = p_{m,n}^k + v_{m,n}^{k+1} \quad (18)$$

Here, $v_{m,n}$ is the particle velocity, $p_{m,n}$ is the particle variables, r_0 , r_1 and r_2 are independent uniform random numbers; Γ_1 and Γ_2 are the cognitive and social parameters, respectively, $p_{m,n}^{local\ best(k)}$ and $p_{m,n}^{global\ best(k)}$ are the best local and global solutions, respectively; C is the constriction parameter (Kennedy & Eberhart, 1995). If the best local solution has a cost less than the best cost of the current global solution, then the best local solution replaces the best global solution. PSO is a very simple natural optimization algorithm, easy to implement and with few parameters to adjust.

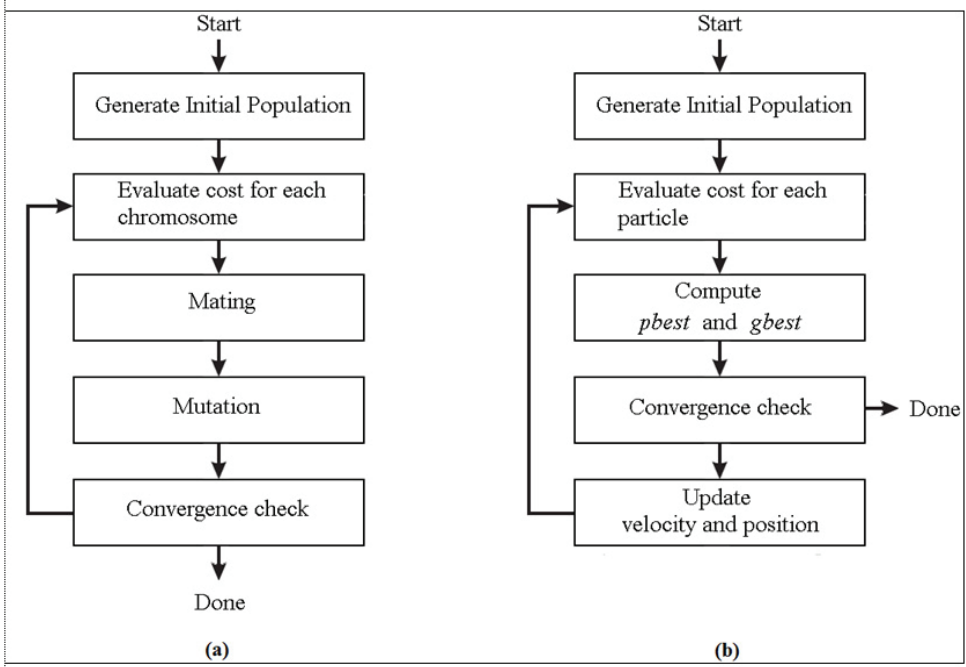


Fig. 11. Flow charts of the natural optimization algorithms: (a) GA; (b) PSO

4.3 EM-optimization method

The FSS design methodology, which blends MLP models, GA and PSO algorithms, is applied for a thin dipole FSS synthesis, considering electromagnetic parameters such as

resonant frequency (f_r) and -3dB bandwidth (BW) as a function of the substrate thickness (d) and the periodicity of elements ($t = t_x = t_y$) corresponding to a square unit cell. The dimensions of the thin dipole elements used in this work are shown in Fig. 10(c). The width (W) and length (L) of the patch remain the same. The region of interest (or search space) defined by desired input variables is a rectangle: $15.24 \leq t \leq 19.05$ mm and $0.1 \leq d \leq 2.0$ mm.

The FSS optimization problem consists of obtaining an optimal solution (d^* ; t^*) in the search space that results in a minimal thickness FSS with the desired specifications for resonant frequency and bandwidth. After the definition of the FSS filter-type, design input variables and search space, a parametric analysis is performed in order to observe the FSS EM-behavior. At this step, the FSS transmission coefficients at 6.0–14.0 GHz band were obtained by means of Method of Moments (MoM) simulations considering substrate anisotropy (Campos et al., 2001). Representative EM-datasets must be obtained from the parametric analysis for supervised learning of FSS synthesis MLP models.

The second step consists of modeling f_r and BW responses by the synthesis MLP network using the conventional neuromodeling technique (Zhang & Gupta, 2000). The third step just consists of implementing the natural optimization algorithms for FSS optimal synthesis. In particular, the continuous-GA and PSO algorithms were used.

Through the FSS neuromodeling, we avoid the very intensive application of the CPU for MoM analysis in GA and PSO simulations. The ANN model was developed for FSS synthesis using one hidden layer MLP configuration. The MLP learning processes were carried out using the RPROP training algorithm, with standard parameters (Riedmiller & Braun, 1993), in order to minimize the mean square error as defined in (4). In this case, the RPROP training is performed until the mean square error reaches a minimum pre-established value. The synthesis MLP network configuration (composed by three inputs: -1 , d , t ; twenty hidden neurons and two outputs: f_r , BW) simultaneously approaches both the mappings $f_r(d, t)$ and $BW(d, t)$. The MLP configuration is similar to that presented in Fig. 1(b). To generate the synthesis training dataset were assumed the vectors given in (19) for the design input variables d and t , adding up 72 training examples.

$$\begin{cases} d = [0.1 & 0.3 & 0.5 & 0.7 & 0.9 & 1.0 & 1.2 & 1.4 & 1.5 & 1.6 & 1.8 & 2.0] \text{ mm} \\ tx = ty = t = [15.24 & 15.87 & 16.51 & 17.14 & 18.41 & 19.05] \text{ mm} \end{cases} \quad (19)$$

GA and PSO algorithms were implemented in Matlab® for optimal FSS synthesis. The assumed design input variables (substrate thickness and periodicity) were symbolized by d_m^k and t_m^k to take into account the m -th individual at the k -th iteration. Given a desired FSS filter specification, ($f_r^{desired}$, $BW^{desired}$), the aim is the minimization of the quadratic cost function as defined in (20) in terms of absolute percent errors.

$$\text{cost}(k, m) = \left(\frac{\left| f_r^{desired} - f_r(d_m^k, t_m^k) \right|}{f_r^{desired}} + \frac{\left| BW^{desired} - BW(d_m^k, t_m^k) \right|}{BW^{desired}} \right)^2, \quad (20)$$

$$m=1, 2, \dots, Npop$$

To evaluate the cost function shown in (20), the synthesis MLP approach the EM relations $f_r(d_m^k, t_m^k)$ and $BW(d_m^k, t_m^k)$. When the population evolves, each individual is constrained to the region of interest using (21), where the dummy variable ξ can be replaced by d or t .

$$\xi_m^k = \min\left(\max\left(\xi_m^k, \xi_{\min}\right), \xi_{\max}\right) \quad (21)$$

The parameters used for continuous-GA and PSO simulations are shown in Table 2.

Continuous-GA	PSO
initial population = 50	initial population = 25
maximum iteration number (generations) = 500	maximum iteration number = 200
crossover probability = 0.9	constriction parameter C = 0.8
mutation rate = 0.01	cognitive parameter, $\Gamma_1 = 2$
	cognitive parameter, $\Gamma_2 = 2$

Table 2. GA and PSO parameters used in the simulations

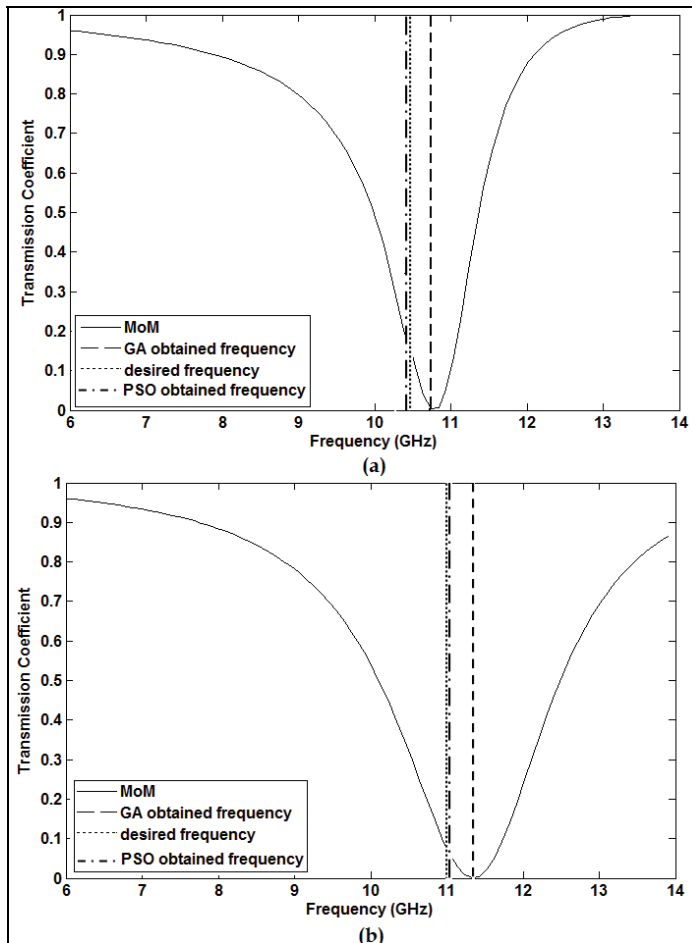


Fig. 12. Transmission coefficients of the thin dipole FSS: (a) first and (b) second examples

In this section, examples that use the hybrid EM-optimization technique for the optimal synthesis of FSSs are presented. Random values for the resonant frequency and bandwidth were chosen within the search space used for the MoM simulations of the synthesis MLP network. The first example corresponds to output data $(f_r; BW) = (10.5; 1.5)$ GHz and the second example corresponds to output data $(f_r; BW) = (11.0; 2.5)$ GHz.

Figs. 12(a) and 12(b) show the transmission coefficients obtained from MoM, GA and PSO simulations for the first and second examples, respectively.

It is observed that although GA algorithm response is closer to MoM simulations, PSO algorithm presents the best solutions of resonant frequency and bandwidth, compared to the desired values. Table 3 shows the input/output values obtained from both algorithms.

The cost surface contours, the initial, intermediate, and final populations, as well as the best path for both algorithms can be found in (Silva et al., 2010a; 2010b). One can also find a new version of GA algorithm called improved GA in (Cruz, 2009).

First example investigated - $(f_r; BW) = (10.5; 1.5)$ GHz		
Parameters	Results	
	GA	PSO
Thickness of the substrate d (mm)	0.7685	0.5401
Periodicity of elements t (mm)	18.193	15.283
obtained f_r (GHz)	10.7407	10.4917
obtained BW (GHz)	1.3827	1.8632
Second example investigated - $(f_r; BW) = (11.0; 2.5)$ GHz		
Parameters	Results	
	GA	PSO
Thickness of the substrate d (mm)	0.195	1.9457
Periodicity of elements t (mm)	15.826	19.006
obtained f_r (GHz)	11.4444	11.1121
obtained BW (GHz)	2.2716	2.2924

Table 3. Input/output values obtained from GA and PSO algorithms

5. Hybrid and high processing pattern classification

In recent years, literature (Dieterich, 1998) indicated some development directions in machine learning which have a highlight in this research scene, including the committee machines and also methods for scaling up supervised learning algorithms. Ensembles studies, which deal with aggregation among many learning machines on the resolution of others complex problems, had an increased effort with empiric data, formalization and new methods (Valentini & Masulli, 2002; Kuncheva, 2004; Dimitrakakis & Bengio, 2005). New training methods, such as hierarchy and parallel processing, have been developed and evaluated in several applications.

This section describes a pattern classification problem that deal with electric power line systems. The quality of the energy provided by an electric system is one of the greatest points of interest for concessionaires and electric energy consumers. Literature presents distinct approaches in the acquisition, characterization and classification of disturbs that may occur in high potential transmission nets. Some pre-processing methods, like wavelet transform, are used for characterization of voltage or current signals.

Considering high performance and hybrid architectures neural networks as a development perspective, this work treats a learning algorithm for extended modular neural networks. The motivation is the unsatisfactory results in pattern classifications and function approximations commonly used (Jacobs et al, 1991). In such cases, modular neural experts are composed by a single linear neuron layer and training process is performed through associative gaussian mixture models.

The architecture proposed for solution of this problem uses multilayer perceptrons experts, as shown in Fig. 1(c). The modular neural network has K experts, L layers of q neurons with linear or nonlinear activation function and bias, as well as a gating network with similar architecture.

5.1 The hybrid algorithm

The modified modular artificial neural network is trained with a developed algorithm adapted according to (Jacobs et al., 1991), for the model of Gaussian mixing associative and also the error backpropagation algorithm, by including the calculation of the descend gradient. The main steps of this algorithm is in this section; more details can be found in (Magalhães et al., 2008).

In this description, some conventions are utilized: the structure has MLP experts (indexed by $i = 1, \dots, k$) with L^{Esp_i} layers (indexed by $l = 1, \dots, L^{Esp_i}$), with $q^{(l)Esp_i}$ neurons in each layer (indexed by $j = 1, \dots, q^{(l)Esp_i}$). Also, there is a MLP gating network with L^{Pas} layers (indexed by $l = 1, \dots, L^{Pas}$), with $q^{(l)Pas}$ neurons in each layer (indexed by $j = 1, \dots, q^{(l)Pas}$). The neurons activation functions in all networks can be linear or non-linear. The processing algorithm can be divided in three steps.

5.1.1 First step

The first step of the algorithm is the calculation of *a priori* probability associated with the i -th output layer of the gating network, as expressed by (22):

$$g_i(n) = \frac{\exp(u_i^{(l)}(n))}{\sum_{j=1}^K \exp(u_j^{(l)}(n))} \quad (22)$$

where $u_i^{(l)}(n)$ is the i -th output neuron of the l -th layer of the gating network and is giving by:

$$u_i^l(n) = \phi_{pi}^{(l)}(v_{pi}^{(l)}(n)) \quad (23)$$

where $\phi_{pi}^{(i)}$ is the derivative function from the activation potential $v_{pi}^{(i)}$, from the i -th neuron of the l -th layer of the gating network, as shown in (24):

$$v_{pi}^{(l)}(n) = \sum_{j=0}^{q^{(l)Pas}} a_{ij}^{(l)} u_i^{l-1}(n) \quad (24)$$

where a_{ij} is the weight associated to the j -th input neuron i of the l -th layer of the gating network.

5.1.2 Second step

The second step of the algorithm is responsible for evaluating the values of *posteriori* probabilities $h_i(n)$, associated to the i -th output neuron of the output gating network. The *posteriori* probabilities can be expressed as:

$$h_i(n) = \frac{g_i(n) \exp\left(\frac{-1}{2} |d(n) - y_{e_i^{(k)}}(n)|^2\right)}{\sum_{j=1}^K g_j(n) \exp\left(\frac{-1}{2} |d(n) - y_{e_j^{(k)}}(n)|^2\right)} \quad (25)$$

where $d(n)$ is the desired response and $y_{e_i^{(k)}}(n)$ is the actual response of the i -th neuron of the l -th layer of the K -th expert, as shown in (26):

$$y_{e_i^{(k)}}^{(l)}(n) = \varphi_{e_i^{(k)}}^{(l)}(v_{e_i^{(k)}}^{(l)}(n)) \quad (26)$$

where $\varphi(l)$ is the activation potential derivative $v(l)$ from the i -th neuron, l -th layer and k -th expert.

5.1.3 Third step

The principal point in this algorithm is the incrementation in synaptic weights of the modular network, that is done with the multiple layers. The synaptic weights from the networks experts are updated according to:

$$w_{e_i^{(k)}j}^{(l)}(n+1) = w_{e_i^{(k)}j}^{(l)}(n) + \eta \delta_{e_i^{(k)}}^{(l)}(n) y_j^{(l-1)}(n) \quad (27)$$

where η is the learning rate, and the gradient $\delta_{e_i^{(k)}}^{(l)}$ for the output layer neurons is obtained by:

$$\delta_{e_i^{(k)}}^{(l)}(n) = h_i(n) e_i(n) \varphi_{e_i^{(k)}}^{(l)}(v_{e_i^{(k)}}^{(l)}(n)) \quad (28)$$

where e_i is the difference between d_i and $y_e(K)$. The gradient for the neurons of hidden layers is computed as:

$$\delta_{e_i^{(k)}}^{(l)}(n) = \varphi_{e_i^{(k)}}^{(l)}(v_{e_i^{(k)}}^{(l)}(n)) \sum_{m=1}^{q^{(l+1)Esp_i}} \delta_m^{(l+1)}(n) w_{me_i^{(k)}}^{(l+1)}(n) \quad (29)$$

The synaptic weight update of the gating network is accomplished according to (30):

$$a_{p,j}^{(l)}(n+1) = a_{p,j}^{(l)}(n) + \eta \delta_{p_i}^{(l)}(n) y_j^{(l-1)}(n) \quad (30)$$

where the output layer gradient is obtained by:

$$\delta_{p_i}^{(l)}(n) = [h_i(n) - g_i(n)] \varphi_{p_i}^{(l)}(v_{p_i}^{(l)}(n)) \quad (31)$$

The error is the difference between h_i and g_i . The gradient of hidden layers is calculated by:

$$\delta_{pi}^{(l)}(n) = \varphi_{pi}^{(l)}(v^{(l)}) \sum_{m=1}^q \delta_m^{(l+1)}(n) a_{mpi}^{(l+1)}(n) \quad (32)$$

Thus, the error backpropagates from the gating network to its hidden layers.

5.2 The application and results

As mentioned before, the application implemented was a pattern classification for power line disturbances. The implementation methodology, data files and information were the same presented in (Medeiros et al., 2007). The application evaluated the performance of an intelligent system classifier, in this case, a modular artificial neural network, in electric disturbances classification. The approach is performed into four main steps: getting the signal, pre-processing, definition and classification of descriptors.

First, the electrical signals were obtained using the oscillograph network of São Francisco Hydro Electric Company (CHESF) and also from the simulation via Transient Alternative Program (ATP). The electric network consists of 370 oscillographs operation with a sampling rate ranging between 20 and 256 samples/cycle. Here, the signals were collected in voltage line levels of 69, 230 and 500 kV, with a rate of 128 samples/cycle during 14 cycles. The pre-processing stage aimed to suggest descriptors that characterize the signal variations when diverted from a certain standard. The third step, which deals with the descriptor definitions, is performed by the decomposition of signals from the previous step. Once obtained the descriptors, four disturbances classes were defined as: Voltage Sag, Voltage Swell, Harmonics and Transitories. The last step (classification) is performed by the application of classifiers based on modular artificial neural networks. Several different architectures were tested, as shown in Table 4.

Net	MOD-0	MOD-1	MOD-2	MOD-3
Number of Experts	3	3	3	3
Expert Architecture	10:3:4	10:5:4	10:10:4	10:15:4
Gating Architecture	10:5:4	10:5:4	10:10:4	10:15:4
Classification (%)	98,46	99,48	100	100

Table 4. Modular Neural Network Architecture

The classification step used two datasets, one for the training at computer, and the other for validation at multiprocessor on chip, utilizing a FPGA system architecture. The training set consisted of 800 patterns formed by 4 output classes. To validate the modular neural network, 344 input patterns were used with their respective expected responses, consisting only of data obtained from the oscillographs.

Table 4 shows that the modular neural network with the proposed algorithm reaches a high amount of accuracy, approximately 100%. Besides, the utilization of two processors on a FPGA system increased the optimization speed-up by 47%, while using four processors, the speed-up was increased by 87%.

6. Conclusion

This chapter described some of the neuromodeling methodology used for applications in various areas of Engineering, in particular, EM-optimization, Signal Processing and Pattern Classification and Recognition. Some original contributions were shown, such as the hybrid EM-optimization for optimal design of FSS, the artificial neural network optimization using wavelet transforms and the modular neural network used to pattern classification and recognition.

The choice of the activation function of a FNN or a modular neural network strongly influences the neural model performance. However, there is not an universal activation function that can be used to solve any kind of nonlinear modeling problem. Using additional information from the hybrid neural models, as well as sharing the training process with modular neural networks, increased the efficiency and therefore the generalization ability and model reliability of the resulting neural models.

The fast and accurate obtained results for all the applications demonstrated the improvements with the utilization of these models, and proved that the MLP network global approximations are able to generalize. In addition, the idea of blending artificial neural networks and natural optimization algorithms, as well as mathematical transforms (for EM-optimization and pattern recognition, respectively) shows to be very interesting. PSO algorithm needs less individuals and reaches the best solution at a few iterations, compared to GA algorithm. The PSO implementation is simpler than GA, since it is not required the presence of genetic operations such as crossover and mutation. Therefore, PSO algorithm has proved to be an interesting EM-optimization tool, with few parameters to adjust and low computational cost.

The characteristics of ANN models (precision, CPU efficiency and flexibility) can be perfectly used in association with these optimization techniques in order to develop powerful soft computing tools. A good point in combining these complementary tools is the possibility of multiprocessing application, using parallel processors and computers, not only to increase performance and execution speed, but also to enable an improved type of competition and collaboration. Each individual tool allows solving the problem or part of the problem and, at the same time, they can collaborate one each other to improve the solution in a higher level. This is the intelligent computing that this chapter wants to apply in efficient engineering.

7. References

- Beale, E. M. L. (1972). A derivation of conjugate gradients, In: *Numerical Methods for Nonlinear Optimization*, F. A. Lootsma, (Ed.), Academic Press, London.
- Bhatia, P.; Bharti, S. & Gupta, R. (2009). Comparative analysis of image compression techniques: a case study on medical images, In: *International Conference on Advances in Recent Technologies in Communication and Computing*, pp. 820-822.
- Campos, A. L. P. de S. (2009). *Superfícies seletivas em frequência: análise e projeto*, IFRN, 198 p., Natal, RN, Brazil.
- Campos, A. L. P. S.; D'Assunção, A. G. & Melo, M. A. B. (2001). Investigation of scattering parameters for frequency selective surfaces using thin dipoles on anisotropic layers. *Proceedings of the SBMO/IEEE MTT-S International Microwave and Optoelectronics Conference*, Vol. 1, pp. 413-416.

- Castleman, K. R. (1996). *Digital Imaging Processing*. Prentice Hall, 1st Ed, Englewood Cliffs, New Jersey.
- Cruz, R. M. S. (2009). Análise e otimização de superfícies seletivas de frequência utilizando redes neurais artificiais e algoritmos de otimização natural, Ph.D. Dissertation, Federal University of Rio Grande do Norte, Natal, RN, Brazil.
- Cruz, R. M. S.; Santos, A. F. & D'Assunção, A. G. (2010). Spectral analysis of electromagnetic waves in uniaxial anisotropic dielectric materials, *Proceedings of 14^o SBMO – Simpósio Brasileiro de Microondas e Optoeletrônica and 9^o CBMag – Congresso Brasileiro de Eletromagnetismo, MOMAG2010*, pp. 440-444, Vila Velha, ES, Brazil.
- Cruz, R. M. S.; Silva, P. H. da F. & D'Assunção, A. G. (2009a). Neuromodeling stop-band properties of Koch island patch elements for FSS filter design. In: *Microwave and Optical Technology Letters*, Vol. 51, No. 12, pp. 3014-3019.
- . (2009b). Synthesis of crossed dipole frequency selective surfaces using genetic algorithms and artificial neural networks, *Proceedings of the 2009 International Joint Conference on Neural Networks*, pp. 627-633, ISBN 978-1-4244-3553-1, Atlanta, GA, USA.
- Dietterich, T. G. (1998). Machine-learning research: four current directions, *The AI Magazine*, Vol. 4, pp. 97-136.
- Dimitrakakis, C. & Bengio, S. (2005). Online adaptive policies for ensemble classifiers. *Neurocomputing*, Vol. 64, pp. 211-221.
- Genovesi, S. et al. (2006). Particle swarm optimization of frequency selective surfaces for the design of artificial magnetic conductors, In: *IEEE Antennas and Propagation Society International Symposium*, Vol. 9, No. 14, pp. 3519-3522.
- Hagan, M. T. & Menhaj, M. (1999). Training feed-forward networks with the Marquardt algorithm. *IEEE Transactions on Neural Networks*. Vol. 5, No. 6, pp. 989-993.
- Haupt, R. L. & Haupt, S. (2004). *Practical genetic algorithms*, Wiley & Sons, Inc., 2nd Ed, New Jersey.
- Haupt, R. L. & Werner, D. H. (2007). *Genetic algorithms in electromagnetic*, John Wiley & Sons, Inc., 1st Ed, New York.
- Hwang, J. N.; Chan, C. H. & Marks, R. J. (1990). Frequency selective surface design based on iterative inversion of neural networks. *Proceedings of the International Joint Conference on Neural Networks*, Vol. 1, pp. 39-44.
- Haykin, S. (1999). *Neural networks: a comprehensive foundation*, Prentice-Hall, 2nd Ed, Upper Saddle River, New Jersey.
- Jacobs, R. A.; Jordan, M. I.; Nowlan, S. J. & Hinton, G. E. (1991). Adaptive mixture of local experts. *Neural Computation*, MIT Press, Vol. 3, pp. 79-87.
- Kennedy, L. & Eberhart, R. C. (1995). Particle swarm optimization. *Proceedings of IV IEEE Conference On Neural Networks*, Piscataway, New Jersey.
- Kuncheva, L. I. (2004). *Combining pattern classifiers: methods and algorithms*, John Wiley & Sons, Inc., New Jersey.
- Liu, J. C. et al. (2004). Implementation of broadband microwave absorber using FSS screens coated with Ba(MnTi)Fe₁₀O₁₉ ferrite, In: *Microwave and Optical Technology Letters*, Vol. 41, No. 4, pp. 323-326.
- Lopes, D. C. et al. (2009). Implementation of a Modular Neural Network in a Multiple Processor System on FPGA to Classify Electric Disturbance. In: *Industrial Electronics, 2009. IECON '09. 35th Annual Conference of IEEE*, Porto, Portugal.

- Magalhães, R. M. et al. (2008). Application of a Hybrid Algorithm in the Modular Neural Nets Training with Multilayers Specialists in Electric Disturbance Classification. In: *INES 2008 12th International Conference on Intelligent Engineering Systems*, Miami, Florida.
- Mallat, S. (2009). *A Wavelet Tour of Signal Processing*, Elsevier Inc, 3rd ed.
- Masnadi-Shirazi, R.; Mohseni, A. & Sheikhi, M. A. (2009). Efficient compression of wavelet packet OFDM radar signals.
- McCulloch, W. S. & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*, Vol. 5, pp. 115-133.
- Medeiros, M. F. et al. (2007). Influence of signal processing in the Efficiency of algorithms Based on Neural Networks for Disturbance Classification, *Computacional Intelligence in Image and Signal Processing*, pp. 95 -100.
- Monorchio, A. et al. (2005). Design of waveguide filters by using genetically optimized frequency selective surfaces, In: *IEEE Microwave Wireless Component Letters*, Vol. 15, No. 6, pp. 407-409.
- Munk, B. A. (2000). *Frequency selective surfaces - theory and design*, John Wiley & Sons, Inc., ISBN 0-471-37047-9, New York, USA.
- Oliveira, H. M. (2007). *Análise de Fourier e Wavelets: sinais estacionários e não estacionários*. Ed. Universitária UFPE, 1^a Ed, Recife.
- Ottensamer, J.; Bobin, J. & Starck, R. (2008). Compressed sensing in astronomy, In: *IEEE Journal of Selected Topics in Signal Processing*, Vol. 2, No. 5, pp. 718-726.
- Parolai, J. J.; Galiana-Merino, J. L. & Rosa-Herranz, S. (2008). Seismic P phase picking using a kurtosis-based criterion in the stationary wavelet domain, In: *IEEE Transactions on Geoscience and Remote Sensing*, Vol. 46, No. 11, pp. 3815-3826.
- Peixoto, H. M.; Cruz, R. M. S.; Guerreiro, A. M. G.; Dória Neto, A. D. & D'Assunção, A. G. (2009a). A comparison of multilayer perceptrons training algorithms for the optimization of frequency selective surfaces, *Proceedings of 8th International Information and Telecommunication Technologies Symposium*, Florianópolis, SC, Brazil.
- Peixoto, H. M.; Guerreiro, A. M. G. & Dória Neto, A. D. (2009b). Image processing for eye detection and classification of the gaze direction, *Proceedings of the 2009 International Joint Conference on Neural Networks*, pp. 2475-2480, ISBN 978-1-4244-3553-1, Atlanta, GA, USA.
- Peixoto, H. M.; Rego, J. B. A.; Guerreiro, A. M. G. & Dória Neto, A. D. (2009c). Otimizando redes neurais artificiais com transformadas *wavelets*, *Anais do IX Congresso Brasileiro de Redes Neurais – Inteligência Computacional*, Ouro Preto, MG, Brazil.
- Pinto, L. F. Q.; Silveira, L. G. Q. S.; Junior, F. M. & Assis, E. L. (2009). Analysis and optimization of wavelet-coded communication systems, Vol. 8, No. 2, pp. 563-657.
- Rahmat-Samii, Y. (2003). Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) in Engineering Electromagnetics, *Proceedings of 17th International Conference on Applied Electromagnetics and Communications*, Dubrovnik, Croatia.
- Riedmiller, M. & Braun, H. (1993). A direct adaptive method for faster backpropagation learning: the RPROP algorithm. *Proceedings of IEEE International Conference on Neural Networks*, pp. 586-591, San Francisco, USA.
- Rosenblatt, F. (1958). The Perceptron: A probabilistic model for information storage and organization in the brain. *Psychological Review*, Vol. 65, No. 6, pp. 386-408. Retrieved September 19, 2010, from

- http://www.ling.upenn.edu/courses/Fall_2007/cogs501/Rosenblatt1958.pdf.
- Rumelhart, D. E.; Hinton, G. E. & Williams, R. J. (1986). Learning internal representations by error propagation. In: *Parallel Distributed Processing: Explorations in the microstructure of cognition*, D. E. Rumelhart & J. L. McClelland (Ed.), MIT Press, Vol. 1, pp. 318-362, Cambridge, MA.
- Silva, P. H. da F. & Campos, A. L. P. S. (2008). Fast and accurate modeling of frequency selective surfaces using a new modular neural network configuration of multilayer perceptrons, *Microwaves, Antennas e Propagation*, IET, Vol. 2, No. 5, pp. 503-511.
- Silva, P. H. da F.; Cruz, R. M. S & D'Assunção, A. G. (2010a). Blending PSO and ANN for optimal design of FSS filters with Koch island patch elements, *IEEE Transactions on Magnetics*, Vol. 46, No. 8.
- _____. (2010b). Neuromodeling and natural optimization of nonlinear devices and circuits, In: *System and Circuit Design for Biologically-Inspired Intelligent Learning*, Turgay Temel (Ed.), IGI Global, ISBN 9781609600181, Hershey PA. No prelo.
- Silva, P. H. da F.; Lacouth, P.; Fontgalland, G.; Campos, A. L. P. S. & D'Assunção, A. G. (2007). Design of frequency selective surfaces using a novel MoM-ANN-GA technique. *Proceedings of the SBMO/IEEE MTT-S International Microwave and Optoelectronics Conference*, Vol. 1, pp. 275-279.
- Valentini, G. & Masulli, F. (2002). Ensemble of learning machines, in Neural Nets WIRN Vietri-02, *Series Lectures Notes in Computer Sciences*, Springer-Verlag.
- Vimal, P. S.; Kumar, R. & Arumugabathan, C. (2009). Wavelet based ocular artifact removal from EEG signals using arma method and adaptive filtering, pp. 667-671.
- Wang, X. Q. G.; Liu, D. G. Z.; Li, Z. & Wang, H. (2009). Object categorization using hierarchical wavelet packet texture descriptors, In: *11th IEEE International Symposium on Multimedia*, pp. 44-51.
- Weeks, M. (2007). *Digital Signal Processing Using MATLAB and Wavelets*, Infinity Science Press LLC Hingham, Georgia State University, Massachusetts.
- Wu, T. K. (1995). *Frequency selective surface and grid array*, John Wiley & Sons, Inc., ISBN 0-471-31189-8, New York, USA.
- Zhang, Q. J. & Gupta, C. (2000). *Neural networks for RF and microwaves design*, Artech House, Norwood, MA.

Part 2

Advanced Architectures for Biomedical Applications

Pixel-Based Artificial Neural Networks in Computer-Aided Diagnosis

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1. Introduction

Artificial neural networks (ANNs) play an essential role in the medical imaging field, including medical image analysis and computer-aided diagnosis (CAD) (Doi 2005; Giger and Suzuki 2007), because objects such as lesions and organs in medical images may not be represented accurately by a simple equation. For example, a lung nodule is generally modeled as a solid sphere, but there are nodules of various shapes and nodules with internal inhomogeneities, such as spiculated nodules and ground-glass nodules. A polyp in the colon is modeled as a bulbous object, but there are also polyps which exhibit a flat shape (Lostumbo, Wanamaker et al. 2010). Thus, diagnostic tasks in medical images essentially require “learning from examples (or data).”

One of the most popular uses of ANNs in medical image analysis is the classification of objects such as lesions into certain classes (e.g., abnormal or normal, lesions or non-lesions, and malignant or benign). The task of ANNs here is to determine “optimal” boundaries for separating classes in the multi-dimensional feature space which is formed by input features (e.g., contrast, area, and circularity) obtained from object candidates. Machine-learning algorithms for classification include linear discriminant analysis (Fukunaga 1990), quadratic discriminant analysis (Fukunaga 1990), multilayer perceptron (Rumelhart, Hinton et al. 1986), and support vector machines (Vapnik 1995). Such machine-learning algorithms were applied to lung nodule detection in chest radiography (Shiraishi, Li et al. 2006) and thoracic CT (Armato, Giger et al. 2001; Arimura, Katsuragawa et al. 2004), classification of lung nodules into benign or malignant in chest radiography (Aoyama, Li et al. 2002) and thoracic CT (Aoyama, Li et al. 2003), detection of microcalcifications in mammography (Wu, Doi et al. 1992), classification of masses into benign or malignant in mammography (Huo, Giger et al. 1998), polyp detection in CT colonography (Yoshida and Nappi 2001; Jerebko, Summers et al. 2003), determining subjective similarity measure of mammographic images (Muramatsu, Li et al. 2005; Muramatsu, Li et al. 2006; Muramatsu, Li et al. 2007), and detection of aneurysms in brain MRI (Arimura, Li et al. 2006).

Recently, as available computational power increased dramatically, pixel/voxel-based ANNs (PANNs) emerged in medical image processing/analysis which use pixel/voxel values in images directly instead of features calculated from segmented regions as input

information; thus, feature calculation or segmentation is not required. Because the PANN can avoid errors caused by inaccurate feature calculation and segmentation, the performance of the PANN can potentially be higher than that of common classifiers. In this chapter, PANNs are surveyed and reviewed to make clear a) classes of PANNs, b) the similarities and differences within different PANNs and those between PANNs and ordinary classifiers, c) the advantages and limitations of PANNs, and d) their applications in medical imaging.

2. Pixel/voxel-based Artificial Neural Network (PANN)

2.1 Overview

PANNs have been developed for tasks in medical image processing/analysis and computer vision. Table 1 summarizes classes of PANNs, their functions, and their applications. There are three classes of PANNs: neural filters (Suzuki, Horiba et al. 2002; Suzuki, Horiba et al. 2002) (including neural edge enhancers (Suzuki, Horiba et al. 2003; Suzuki, Horiba et al. 2004)), convolution neural networks (NNs) (Lo, Lou et al. 1995; Lo, Chan et al. 1995; Lin, Lo et al. 1996; Sahiner, Chan et al. 1996; Lawrence, Giles et al. 1997; Neubauer 1998; Lo, Li et al. 2002) (including shift-invariant NNs (Zhang, Doi et al. 1994; Wei, Nishikawa et al. 1996; Zhang, Doi et al. 1996)), and massive-training ANNs (MTANNs) (Suzuki, Armato et al. 2003; Suzuki, Abe et al. 2006; Suzuki, Yoshida et al. 2006; Oda, Awai et al. 2009; Suzuki 2009) (including multiple MTANNs (Suzuki, Horiba et al. 2002; Suzuki, Horiba et al. 2002; Suzuki, Armato et al. 2003; Arimura, Katsuragawa et al. 2004; Suzuki, Li et al. 2005; Suzuki, Shiraishi et al. 2005), a mixture of expert MTANNs (Suzuki, Yoshida et al. 2008; Suzuki, Rockey et al. 2010), a multi-resolution MTANN (Suzuki, Abe et al. 2006), a Laplacian eigenfunction MTANN (LAP-MTANN) (Suzuki, Zhang et al. in press), and a massive-training support vector regression (MTSVR) (Xu and Suzuki 2010 (in press))). The class of neural filters has been used for image-processing tasks such as edge-preserving noise reduction in radiographs and other digital pictures (Suzuki, Horiba et al. 2002; Suzuki, Horiba et al. 2002), edge enhancement from noisy images (Suzuki, Horiba et al. 2003), and enhancement of subjective edges traced by a physician in left ventriculograms (Suzuki, Horiba et al. 2004). The class of convolution NNs has been applied to classification tasks such as false-positive (FP) reduction in CAD schemes for detection of lung nodules in chest radiographs (CXRs) (Lo, Lou et al. 1995; Lo, Chan et al. 1995; Lin, Lo et al. 1996), FP reduction in CAD schemes for detection of microcalcifications (Lo, Li et al. 2002) and masses (Sahiner, Chan et al. 1996) in mammography, face recognition (Lawrence, Giles et al. 1997), and character recognition (Neubauer 1998). The class of MTANNs has been used for classification, such as FP reduction in CAD schemes for detection of lung nodules in CXR (Suzuki, Shiraishi et al. 2005) and CT (Suzuki, Armato et al. 2003; Arimura, Katsuragawa et al. 2004; Li, Arimura et al. 2005), distinction between benign and malignant lung nodules in CT (Suzuki, Li et al. 2005), and FP reduction in a CAD scheme for polyp detection in CT colonography (Suzuki, Yoshida et al. 2006; Suzuki, Yoshida et al. 2008; Suzuki, Rockey et al. 2010; Xu and Suzuki 2010 (in press); Suzuki, Zhang et al. in press). The MTANNs have also been applied to pattern enhancement and suppression such as separation of bone from soft tissue in CXR (Suzuki, Abe et al. 2006; Oda, Awai et al. 2009), and enhancement of lung nodules in CT (Suzuki 2009).

PANNs	Functions	Applications
Neural filters (including neural edge enhancers)	Image processing	Edge-preserving noise reduction [20, 21]. Edge enhancement from noisy images [22]. Enhancement of subjective edges traced by a physician [23].
Convolution neural networks (including shift-invariant neural networks)	Classification	FP reduction in CAD for lung nodule detection in CXR [24-26]. FP reduction in CAD for detection of microcalcifications [27] and masses [28] in mammography. Face recognition [29]. Character recognition [30].
Massive-training artificial neural networks (MTANNs, including a mixture of expert MTANNs, and a LAP-MTANN)	Classification (image processing + scoring), pattern enhancement and suppression, object detection (pattern enhancement followed by thresholding or segmentation),	FP reduction in CAD for detection of lung nodules in CXR [39] and CT [9, 34, 45]. Distinction between benign and malignant lung nodules in CT [40]. FP reduction in CAD for polyp detection in CT colonography [35, 41-44]. Bone separation from soft tissue in CXR [36, 37]. Enhancement of lung nodules in CT [38].

Table 1. Classes of PANNs, their functions, and their applications

2.2 Neural filters

In the field of signal/image processing, supervised nonlinear filters based on a multilayer ANN, called neural filters, have been studied (Suzuki, Horiba et al. 2002; Suzuki, Horiba et al. 2002). The neural filter employs a linear-output ANN model as a convolution kernel of a filter. The inputs to the neural filter are an object pixel value and spatially/spatiotemporally adjacent pixel values in a subregion (or local window). The output of the neural filter is a single pixel value. The neural filter is trained with input images and corresponding "teaching" (desired or ideal) images. The training is performed by a linear-output back-propagation algorithm which is a back-propagation algorithm modified for the linear-output ANN architecture. Neural filters can acquire the functions of various linear and nonlinear filtering through training. Neural filters have been applied to reduction of the

quantum noise in x-ray fluoroscopic and radiographic images (Suzuki, Horiba et al. 2002; Suzuki, Horiba et al. 2002). It was reported that the performance of the neural filter was superior to that of well-known nonlinear filters such as an adaptive weighted averaging filter (Ozkan, Sezan et al. 1993). A study (Suzuki, Horiba et al. 2002) showed that adding features from the subregion to the input information improved the performance of the neural filter. Neural filters have been extended to accommodate the task of enhancement of edges, and a supervised edge enhancer (detector), called a neural edge enhancer, was developed (Suzuki, Horiba et al. 2003). The neural edge enhancer can acquire the function of a desired edge enhancer through training. It was reported that the performance of the neural edge enhancer in the detection of edges from noisy images was far superior to that of well-known edge detectors such as the Canny edge detector (Canny 1986), the Marr-Hildreth edge detector (Marr and Hildreth 1980), and the Huckel edge detector (Hueckel 1971). In its application to the contour extraction of the left ventricular cavity in digital angiography, it has been reported that the neural edge enhancer can accurately replicate the subjective edges traced by a cardiologist (Suzuki, Horiba et al. 2004).

2.3 Massive-Training Artificial Neural Network (MTANN)

An MTANN was developed by extension of neural filters to accommodate various pattern-recognition tasks (Suzuki, Armato et al. 2003). A two-dimensional (2D) MTANN was first developed for distinguishing a specific opacity (pattern) from other opacities (patterns) in 2D images (Suzuki, Armato et al. 2003). The 2D MTANN was applied to reduction of FPs in computerized detection of lung nodules on 2D CT slices in a slice-by-slice way (Suzuki, Armato et al. 2003; Arimura, Katsuragawa et al. 2004; Li, Arimura et al. 2005) and in CXR (Suzuki, Shiraishi et al. 2005), the separation of ribs from soft tissue in CXR (Suzuki, Abe et al. 2006; Oda, Awai et al. 2009), and the distinction between benign and malignant lung nodules on 2D CT slices (Suzuki, Li et al. 2005). For processing of three-dimensional (3D) volume data, a 3D MTANN was developed by extending the structure of the 2D MTANN, and it was applied to 3D CT colonography data (Suzuki, Yoshida et al. 2006; Suzuki, Yoshida et al. 2008; Suzuki, Rockey et al. 2010; Xu and Suzuki 2010 (in press); Suzuki, Zhang et al. in press).

The generalized architecture of an MTANN which unifies 2D and 3D MTANNs is shown in Fig. 1. An MTANN consists of an ANN model such as a linear-output ANN regression model and a support vector regression model, which is capable of operating on pixel/voxel data directly (Suzuki, Horiba et al. 2003). The linear-output ANN regression model employs a linear function instead of a sigmoid function as the activation function of the unit in the output layer because the characteristics of an ANN were improved significantly with a linear function when applied to the continuous mapping of values in image processing (Suzuki, Horiba et al. 2003). Note that the activation functions of the units in the hidden layer are a sigmoid function for nonlinear processing, and those of the unit in the input layer are an identity function, as usual. The pixel/voxel values of the input images/volumes may be normalized from 0 to 1. The input to the MTANN consists of pixel/voxel values in a subregion/sub-volume, R , extracted from an input image/volume. The output of the MTANN is a continuous scalar value, which is associated with the center voxel in the subregion, and is represented by

$$O(x, y, z \text{ or } t) = NN \{ I(x - i, y - j, z - k \text{ or } t - k) | (i, j, k) \in R \}, \quad (1)$$

where x, y , and z or t are the coordinate indices, $NN(\cdot)$ is the output of the ANN model, and $l(x, y, z \text{ or } t)$ is a pixel/voxel value of the input image/volume. The entire output image/volume is obtained by scanning with the input subvolume of the MTANN on the entire input image/volume. The input subregion/subvolume and the scanning with the MTANN can be analogous to the kernel of a convolution filter and the convolutional operation of the filter, respectively.

The MTANN is trained with input images/volumes and the corresponding “teaching” images/volumes for enhancement of a specific pattern and suppression of other patterns in images/volumes. The “teaching” images/volumes are ideal or desired images for the corresponding input images/volumes. For enhancement of lesions and suppression of non-lesions, the teaching volume contains a map for the “likelihood of being lesions.”

To enrich the training samples, a training region, R_T , extracted from the input images is divided pixel by pixel into a large number of overlapping subregions. Single pixels are extracted from the corresponding teaching images as teaching values. The MTANN is massively trained by use of each of a large number of input subregions together with each of the corresponding teaching single pixels; hence the term “massive-training ANN.” The error to be minimized by training of the MTANN is represented by

$$E = \frac{1}{P} \sum_c \sum_{(x, y, z \text{ or } t) \in R_T} \{T_c(x, y, z \text{ or } t) - O_c(x, y, z \text{ or } t)\}^2, \quad (2)$$

where c is a training case number, O_c is the output of the MTANN for the c^{th} case, T_c is the teaching value for the MTANN for the c^{th} case, and P is the number of total training voxels in the training region for the MTANN, R_T . The expert 3D MTANN is trained by a linear-output back-propagation (BP) algorithm (Suzuki, Horiba et al. 2003) which was derived for the linear-output ANN model by use of the generalized delta rule (Rumelhart, Hinton et al. 1986). After training, the MTANN is expected to output the highest value when a lesion is located at the center of the subregion of the MTANN, a lower value as the distance from the subregion center increases, and zero when the input subregion contains a non-lesion.

A scoring method is used for combining output pixels from the trained MTANNs. A score for a given region of interest (ROI) from the MTANN is defined as

$$S = \sum_{(x, y, z \text{ or } t) \in R_E} f_W(x, y, z \text{ or } t) \times O(x, y, z \text{ or } t), \quad (3)$$

where f_W is a weighting function for combining pixel-based output responses from the trained MTANN into a single score, which may often be the same distribution function used in the teaching images, and with its center corresponding to the center of the region for evaluation, R_E ; and O is the output image of the trained MTANN, where its center corresponds to the center of R_E . This score represents the weighted sum of the estimates for the likelihood that the ROI (e.g., a lesion candidate) contains a lesion near the center, i.e., a higher score would indicate a lesion, and a lower score would indicate a non-lesion. Thresholding is then performed on the scores for distinction between lesions and non-lesions.

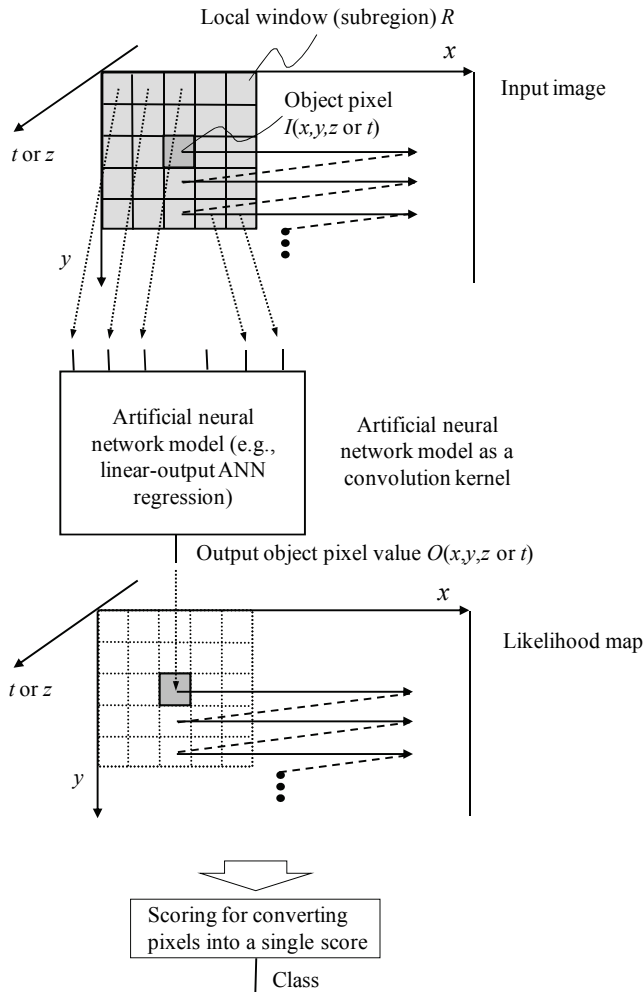


Fig. 1. Generalized architecture of an MTANN consisting of an ANN model (e.g., linear-output ANN regression and support vector regression) with sub-region input and single-pixel output. All pixel values in a sub-region extracted from an input image are entered as input to the ANN model. The ANN model outputs a single pixel value for each sub-region, the location of which corresponds to the center pixel in the sub-region. Output pixel value is mapped back to the corresponding pixel in the output image.

2.4 Convolution Neural Network (NN)

A convolution NN has first been proposed for handwritten ZIP-code recognition (LeCun, Boser et al. 1989). The architecture of a convolution NN is illustrated in Fig. 2. The convolution NN can be considered as a simplified version of the Neocognitron model which was proposed to simulate the human visual system in 1980 (Fukushima 1980). The input and output of the convolution NN are images and nominal class labels, respectively. The

convolution NN consists of one input layer, several hidden layers, and one output layer. The layers are connected with local shift-invariant inter-connections (or convolution with a local kernel). Unlike the Neocognitron, the convolution NN has no lateral interconnections or feedback loops, and the error BP algorithm (Rumelhart, Hinton et al. 1986) is used for training of the convolution NN. Units (neurons) in any hidden layer are organized in groups. Each unit in a subsequent layer is connected with the units of a small region in each group in the preceding layer. The groups between adjacent layers are interconnected by weights that are organized in kernels. For obtaining the shift-invariant responses, connection weights between any two groups in two layers are constrained to be shift-invariant; in other words, forward signal propagation is similar to a shift-invariant convolution operation. The signals from the units in a certain layer are convolved with the weight kernel, and the resulting value of the convolution is collected into the corresponding unit in the subsequent layer. This value is further processed by the unit through an activation function and produces an output signal. The activation function between two layers is a sigmoid function. For deriving the training algorithm for the convolution NN, the generalized delta rule (Rumelhart, Hinton et al. 1986) is applied to the architecture of the convolution NN. For distinguishing an ROI containing a lesion from an ROI containing a non-lesion, a class label (e.g., 1 for a lesion, 0 for a non-lesion) is assigned to an output unit. Variants of the convolution NN have been proposed. The dual-kernel approach, which employs central kernels and peripheral kernels in each layer (Lo, Chan et al. 1995), was proposed for distinction between lung nodules and non-nodules in chest radiographs (Lo, Lou et al. 1995; Lo, Chan et al. 1995) and distinction between microcalcifications and other anatomic structures in mammograms (Lo, Chan et al. 1995). This dual-kernel-based convolution NN has several output units (instead of one or two output units in the standard convolution NN) for two-class classification. The fuzzy association was employed for transformation of output values from the output units to two classes (i.e., nodules or non-nodules; microcalcifications or other anatomic structures). A convolution NN which has subsampling layers has been developed for face recognition (Lawrence, Giles et al. 1997).

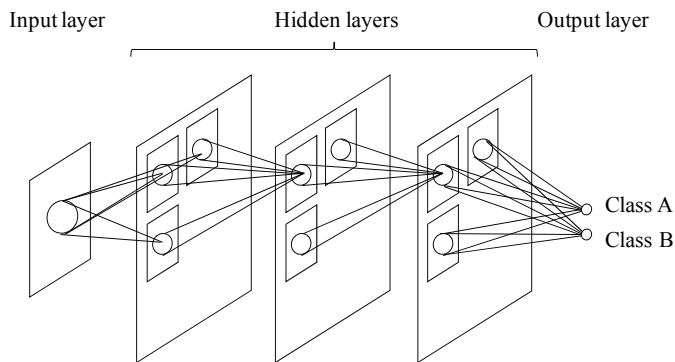


Fig. 2. Architecture of a convolution NN. The convolution NN can be considered as a simplified version of the Neocognitron model, which was proposed to simulate the human visual system. The layers in the convolution NN are connected with local shift-invariant inter-connections (or convolution with a local kernel). The input and output of the convolution NN are images and nominal class labels (e.g., Class A and Class B), respectively.

Some convolution NNs have one output unit (Neubauer 1998; Gurcan, Sahiner et al. 2001), some have two output units (Chan, Lo et al. 1995), and some have more than two output units (Lo, Lou et al. 1995; Lo, Chan et al. 1995; Lawrence, Giles et al. 1997; Lo, Li et al. 2002) for two-class classification.

Shift-invariant NNs (Zhang, Doi et al. 1994; Zhang, Doi et al. 1996) are mostly the same as convolution NNs except for the output layer, which outputs images instead of classes. The shift-invariant NNs were used for localization (detection) of lesions in images, for example, detection of microcalcifications in mammograms (Zhang, Doi et al. 1994; Zhang, Doi et al. 1996), and detection of the boundaries of the human corneal endothelium in photomicrographs (Hasegawa, Itoh et al. 1996).

2.5 Non-PANN – Feature-based classifiers

One of most popular uses of ANNs would probably be classification. In this use, an ANN is called a classifier. A standard classification approach based on a multilayer perceptron is illustrated in Fig. 3. First, target objects are segmented by use of a segmentation method. Next, features are extracted from the segmented objects. Then, extracted features are entered as input to an ANN model such as linear discriminant analysis (Fukunaga 1990), quadratic

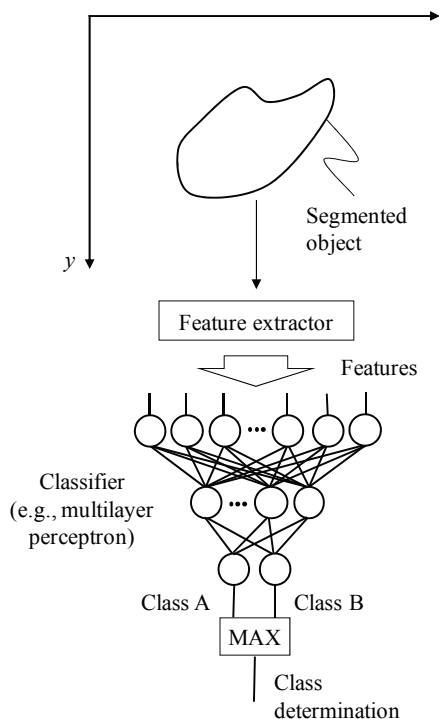


Fig. 3. Standard classifier approach to classification of an object. Features (e.g., contrast, effective diameter, and circularity) are extracted from a segmented object in an image. Those features are entered as input to a classifier such as a multilayer perceptron. Class determination is made by taking the class of the output unit with the maximum value.

discriminant analysis (Fukunaga 1990), a multilayer perceptron (Rumelhart, Hinton et al. 1986), and a support-vector machine (Vapnik 1995). The ANN model is trained with sets of input features and correct class labels. A class label of 1 is assigned to the corresponding output unit when a training sample belongs to that class, and 0 is assigned to the other output units. After training, the class of the unit with the maximum value is determined to be the corresponding class to which an unknown sample belongs. For details of feature-based classifiers, refer to one of many textbooks in pattern recognition such as (Rumelhart, Hinton et al. 1986; Fukunaga 1990; Bishop 1995; Vapnik 1995)

3. Similarities and differences

3.1 Within different PANN algorithms

MTANNs (Suzuki, Armato et al. 2003) were developed by extension of neural filters to accommodate various pattern-recognition tasks. In other words, neural filters are a subclass or a special case of MTANNs. The applications and functions of neural filters are limited to noise reduction (Suzuki, Horiba et al. 2002; Suzuki, Horiba et al. 2002) and edge enhancement (Suzuki, Horiba et al. 2003; Suzuki, Horiba et al. 2004), whereas those of MTANNs were extended to include classification (Suzuki, Armato et al. 2003; Suzuki, Li et al. 2005; Suzuki, Shiraishi et al. 2005; Suzuki, Abe et al. 2006; Suzuki, Yoshida et al. 2006; Suzuki, Yoshida et al. 2008; Suzuki, Rockey et al. 2010; Xu and Suzuki 2010 (in press); Suzuki, Zhang et al. in press), pattern enhancement and suppression (Suzuki, Abe et al. 2006), and object detection (Suzuki 2009). The input information to MTANNs, which is the pixel values in a subregion, is the same as that to neural filters. However, the output of (thus, teacher for) neural filters is the desired pixel values in a given image, whereas that of MTANNs is a map for the likelihood of being a specific pattern in a given image.

Both convolution NNs and the perceptron used for character recognition are in the class of PANN. Input information to the convolution NNs and the perceptron is the pixel values in a given image, whereas the output of (thus, teacher for) both algorithms is a nominal class label for the given image. Thus, the input and output information are the same for both algorithms. However, the input images for the perceptron for character recognition are limited to be binary, although the perceptron itself is capable of processing gray-scale images. The major difference between convolution NNs and the perceptron used for character recognition is their internal architectures. Units in layers of the perceptron are fully connected, whereas the connections in the convolution NN are spatially (locally) limited. Because of this architecture, forward signal propagation in the convolution NN is realized by a convolution operation. This convolution operation offers a shift-invariant property which is desirable for image classification. The applications and functions of the perceptron are limited to character recognition such as zip code recognition and optical character recognition, whereas those of convolution NNs are general classification of images into known classes such as classification of lesion candidates into lesions or non-lesions (Lo, Lou et al. 1995; Lo, Chan et al. 1995; Lin, Lo et al. 1996; Sahiner, Chan et al. 1996; Lo, Li et al. 2002), classification of faces (Lawrence, Giles et al. 1997), and classification of characters (Neubauer 1998).

Shift-invariant NNs are mostly the same as convolution NNs except for the output layer, which outputs images instead of classes. The shift-invariant NNs can be used for localization (detection) of objects in images in addition to classification (Zhang, Doi et al. 1994; Zhang, Doi et al. 1996).

Convolution NNs, shift-invariant NNs, and MTANNs perform convolution operations. In convolution NNs and shift-invariant NNs, convolution operations are performed within the network, as shown in Fig. 2, whereas the convolutional operation is performed outside the network in the MTANN, as shown in Fig. 1.

3.2 Between PANN algorithms and ordinary classifiers

The major difference between PANNs and ordinary classifiers (i.e., feature-based classifiers) is the input information. Ordinary classifiers use features extracted from a segmented object in a given image, whereas PANNs use pixel values in a given image as the input information. Although the input information to PANNs can be features (see addition of features to the input information to neural filters in (Suzuki, Horiba et al. 2002), for example), these features are obtained pixel by pixel (rather than by object). In other words, features for PANNs are features at each pixel in a given image, whereas features for ordinary classifiers are features from a segmented object. In that sense, feature-based classifiers may be referred to as object-based classifiers. Because PANNs use pixel/voxel values in images directly instead of features calculated from segmented objects as the input information, feature calculation or segmentation is not required. Although the development of segmentation techniques has been studied for a long time, segmentation of objects is still challenging, especially for complicated objects, subtle objects, and objects in a complex background. Thus, segmentation errors may occur for complicated objects. Because with PANNs, errors caused by inaccurate feature calculation and segmentation can be avoided, the performance of PANNs can be higher than that of ordinary classifiers for some cases, such as complicated objects.

The output information from ordinary classifiers, convolution NNs, and the perceptron used for character recognition is nominal class labels, whereas that from neural filters, MTANNs, and shift-invariant NNs is images. With the scoring method in MTANNs, output images of the MTANNs are converted to likelihood scores for distinguishing among classes, which allow MTANNs to do classification. In addition to classification, MTANNs can perform pattern enhancement and suppression as well as object detection, whereas the other PANNs cannot.

4. Applications of PANN algorithms in medical images

4.1 Bone separation from soft tissue in chest radiographs (CXRs) by use of MTANNs

CXR is the most frequently used diagnostic imaging examination for chest diseases such as lung cancer, tuberculosis, and pneumonia. More than 9 million people worldwide die annually from chest diseases (Murray and Lopez 1997). Lung cancer causes 945,000 deaths, and is the leading cause of cancer deaths in the world (Murray and Lopez 1997) and in countries such as the United States, the United Kingdom, and Japan (Goodman 2002). Lung nodules (i.e., potential lung cancers) in CXR, however, can be overlooked by radiologists in from 12 to 90% of cases that have nodules visible in retrospect (Austin, Romney et al. 1992; Shah, Austin et al. 2003). Studies showed that 82 to 95% of the missed lung cancers were partly obscured by overlying bones such as ribs and/or a clavicle (Austin, Romney et al. 1992; Shah, Austin et al. 2003). To address this issue, dual-energy imaging has been investigated (Glocker and Frohnmayer 1925; Jacobson and Mackay 1958). Dual-energy imaging uses the energy dependence of the x-ray attenuation by different materials; it can

produce two tissue-selective images, i.e., a “bone” image and a “soft-tissue” image (Ishigaki, Sakuma et al. 1986; Ishigaki, Sakuma et al. 1988; Stewart and Huang 1990). Major drawbacks of dual-energy imaging, however, are that (a) the radiation dose can be double, (b) specialized equipment for obtaining dual-energy x-ray exposures is required, and (c) the subtraction of two-energy images causes an increased noise level in the images.

For resolving the above drawbacks with dual-energy images, MTANNs have been developed as an image-processing technique for separation of ribs from soft tissue (Suzuki, Abe et al. 2006). The basic idea is to train the MTANN with soft-tissue and bone images acquired with a dual-energy radiography system. For separation of ribs from soft tissue, the MTANN was trained with input CXRs and the corresponding “teaching” dual-energy bone images, as illustrated in Fig. 4 (a). Figure 4 (b) shows a non-training original CXR and a soft-tissue image obtained by use of the trained MTANN. The contrast of ribs is suppressed substantially in the MTANN soft-tissue image, whereas the contrast of soft tissue such as lung vessels is maintained.

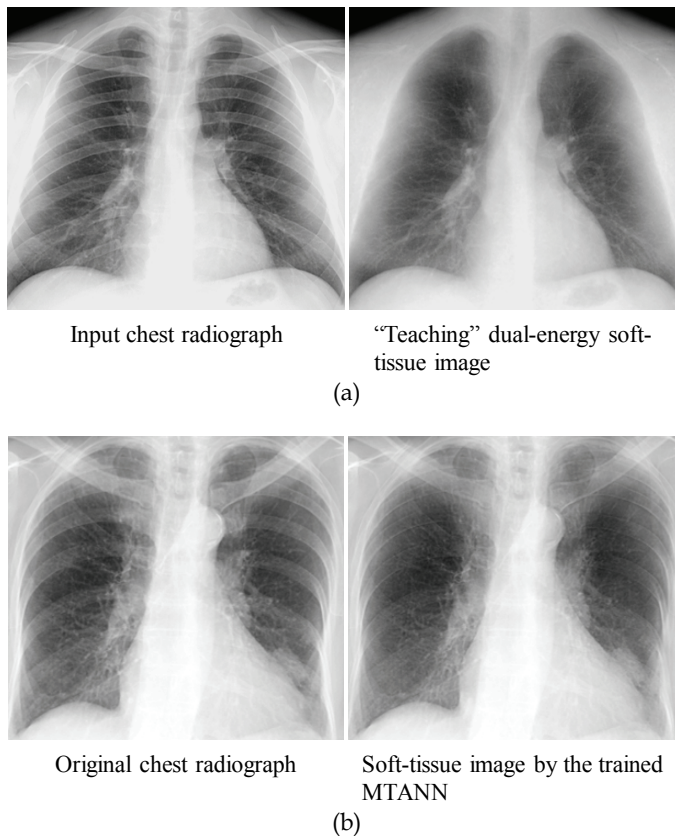


Fig. 4. Separation of bones from soft tissue in CXRs by use of an MTANN. (a) Images used for training the MTANN. (b) Result of an application of the trained MTANN to a non-training CXR.

4.2 Enhancement and detection of lesions by use of MTANNs

Computer-aided diagnosis (CAD) has been an active area of study in medical image analysis (Doi 2005; Giger 2005; Doi 2007; Giger and Suzuki 2007). Some CAD schemes employ a filter for enhancement of lesions as a preprocessing step for improving sensitivity and specificity; but some do not employ such a filter. The filter enhances objects similar to a model employed in the filter; e.g., a blob-enhancement filter based on the Hessian matrix enhances sphere-like objects (Frangi, Niessen et al. 1999). Actual lesions, however, often differ from a simple model, e.g., a lung nodule is generally modeled as a solid sphere, but there are nodules of various shapes and inhomogeneous nodules such as nodules with spiculation and ground-glass nodules. Thus, conventional filters often fail to enhance such actual lesions.

To address this issue, a “lesion-enhancement” filter based on MTANNs has been developed for enhancement of actual lesions in a CAD scheme for detection of lung nodules in CT (Suzuki 2009). For enhancement of lesions and suppression of non-lesions in CT images, the teaching image contains a map for the “likelihood of being lesions.” For enhancement of a nodule in an input CT image, a 2D Gaussian distribution was placed at the location of the nodule in the teaching image, as a model of the likelihood of being a lesion. For testing of the performance, the trained MTANN was applied to non-training lung CT images. As shown in Fig. 5, the nodule is enhanced in the output image of the trained MTANN filter, while normal structures such as lung vessels are suppressed. Note that small remaining regions due to vessels can easily be separated from nodules by use of their area information which can be obtained by use of connected-component labeling (Suzuki, Horiba et al. 2003; He, Chao et al. 2008; He, Chao et al. 2009).

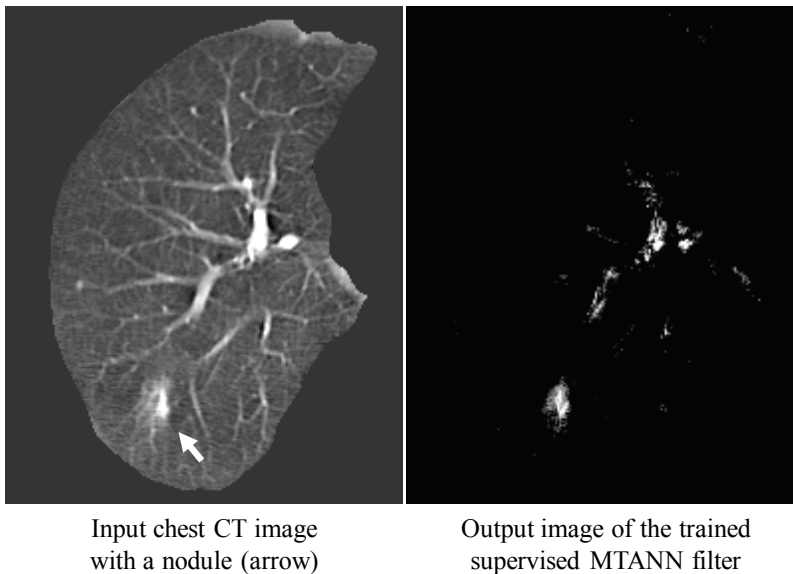


Fig. 5. Enhancement of a lesion by use of the trained lesion-enhancement MTANN filter for a non-training case. (a) Original chest CT image of the segmented lung with a nodule (indicated by an arrow). (b) Output image of the trained lesion-enhancement MTANN filter.

4.3 Classification between lesions and non-lesions by use of different PANN Algorithms

1. MTANNs

A major challenge in CAD development is to reduce the number of FPs, because there are various normal structures similar to lesions in medical images. To address this issue, an FP-reduction technique based on an MTANN has been developed for a CAD scheme for lung nodule detection in CT (Suzuki, Armato et al. 2003). For enhancement of nodules (i.e., true positives) and suppression of non-nodules (i.e., FPs) on CT images, the teaching image contains a distribution of values that represent the "likelihood of being a nodule." For example, the teaching volume contains a 3D Gaussian distribution with standard deviation σ_T for a lesion and zero (i.e., completely dark) for non-lesions, as illustrated in Fig. 6. This distribution represents the "likelihood of being a lesion":

$$T(x, y, z \text{ or } t) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma_T} \exp\left\{-\frac{(x^2 + y^2 + z^2 \text{ or } t^2)}{2\sigma_T^2}\right\} & \text{for a lesion} \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

A 3D Gaussian distribution is used to approximate an average shape of lesions. The MTANN involves training with a large number of subvolume-voxel pairs, which is called a massive-subvolumes training scheme.

A scoring method is used for combining of output voxels from the trained MTANNs. A score for a given ROI from the MTANN is defined as

$$S = \sum_{(x, y, z \text{ or } t) \in R_E} f_W(x, y, z \text{ or } t) \times O(x, y, z \text{ or } t), \quad (5)$$

where

$$f_W(x, y, z \text{ or } t) = f_G(x, y, z \text{ or } t; \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2 + y^2 + z^2 \text{ or } t^2}{2\sigma^2}} \quad (6)$$

is a 3D Gaussian weighting function with standard deviation σ , and with its center corresponding to the center of the volume for evaluation, R_E ; and O is the output image of the trained MTANN, where its center corresponds to the center of R_E . The use of the 3D Gaussian weighting function allows us to combine the responses (outputs) of a trained MTANN as a 3D distribution. A 3D Gaussian function is used for scoring, because the output of a trained MTANN is expected to be similar to the 3D Gaussian distribution used in the teaching images. This score represents the weighted sum of the estimates for the likelihood that the ROI (lesion candidate) contains a lesion near the center, i.e., a higher score would indicate a lesion, and a lower score would indicate a non-lesion. Thresholding is then performed on the scores for distinction between lesions and non-lesions.

An MTANN was trained with typical nodules and typical types of FPs (non-nodules) and corresponding teaching images. The trained MTANN was applied to 57 true positives (nodules) and 1,726 FPs (non-nodules) produced by a CAD scheme (Suzuki, Armato et al.

2003). Figure 7 shows various types of nodules and non-nodules and the corresponding output images of the trained MTANN. Nodules such as a solid nodule, a part-solid (mixed-ground-glass) nodule, and a non-solid (ground-glass) nodule are enhanced, whereas non-nodules such as different-sized lung vessels and soft-tissue opacity are suppressed around the centers of ROIs. For combining output pixels into a single score for each nodule candidate, a scoring method was applied to the output images for distinction between a nodules and a non-nodule. Thresholding of scores was done for classification of nodule candidates into nodules or non-nodules. Free-response receiver operating characteristic (FROC) analysis (Bunch, Hamilton et al. 1978) was carried out for evaluation of the performance of the trained MTANN. The FROC curve for the MTANN indicates 80.3% overall sensitivity (100% classification performance) and a reduction in the FP rate from 0.98 to 0.18 per section, as shown in Fig. 8.

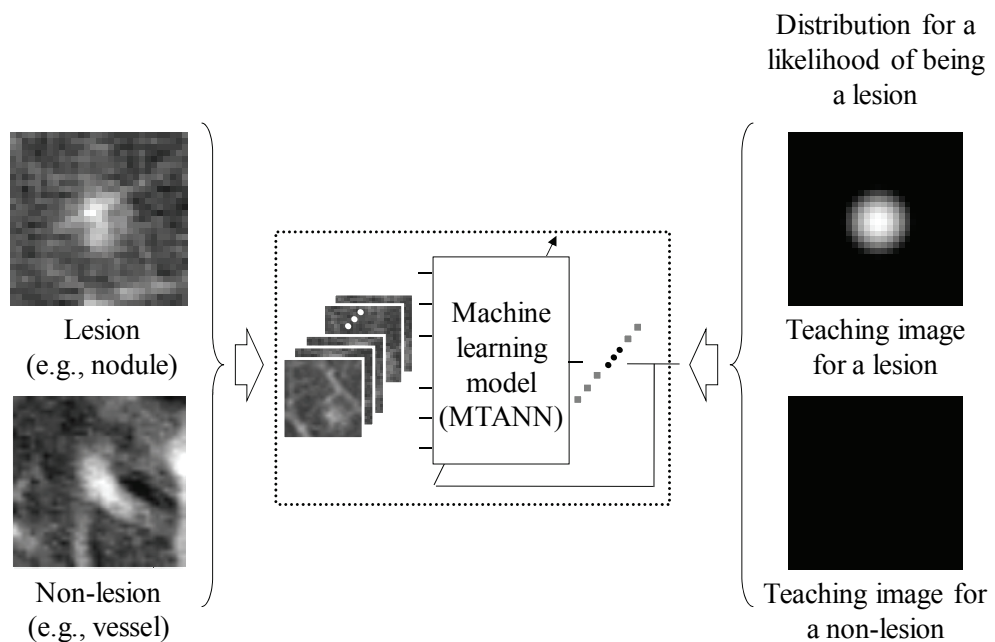


Fig. 6. Training of an MTANN for distinction between lesions and non-lesions in a CAD scheme for detection of lesions in medical images. The teaching image for a lesion contains a Gaussian distribution; that for a non-lesion contains zero (completely dark). After the training, the MTANN expects to enhance lesions and suppress non-lesions.

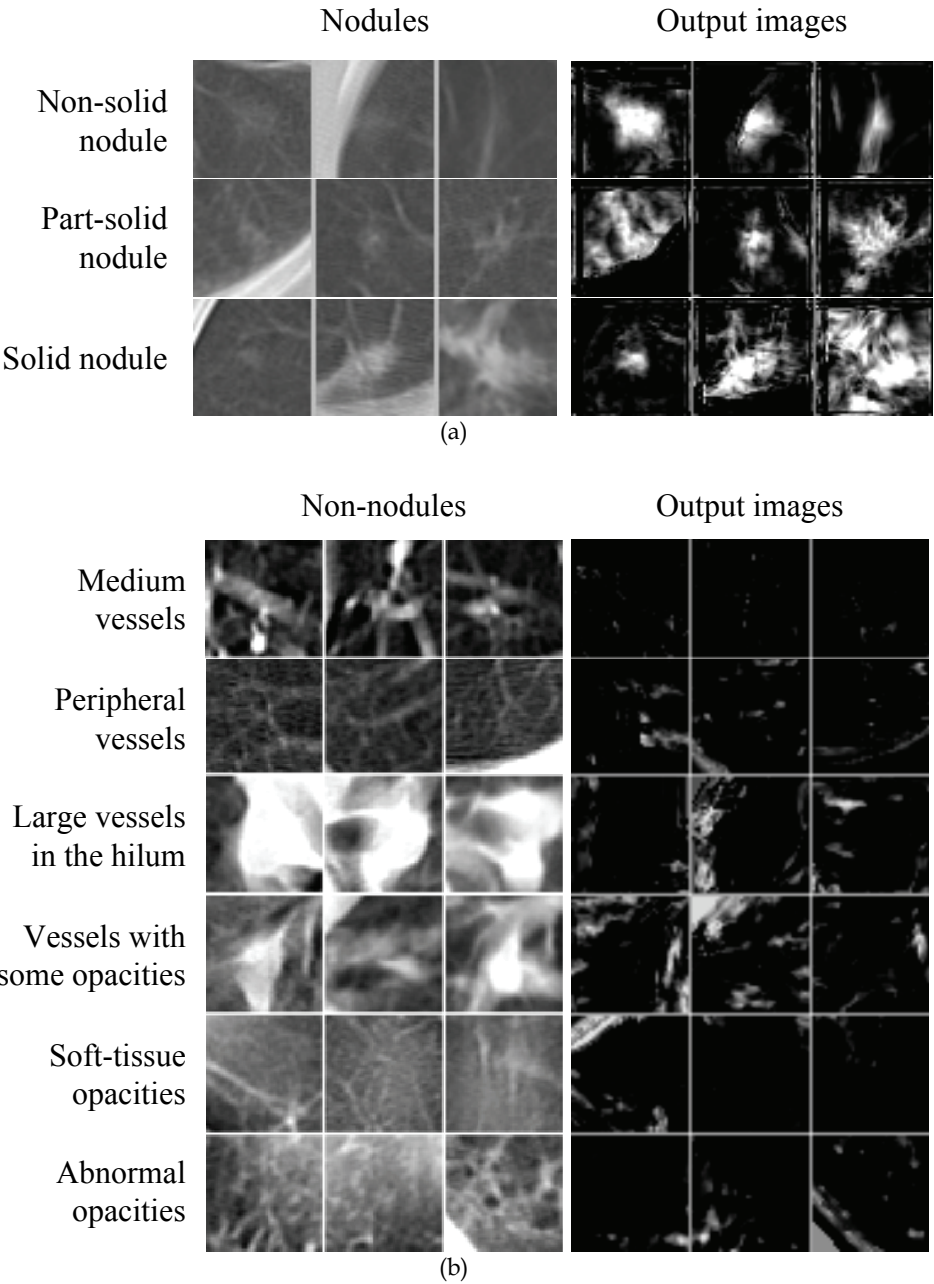


Fig. 7. Illustrations of various types of non-training nodules and non-nodules and corresponding output images of the trained MTANN. Nodules are represented by bright pixels, whereas non-nodules are almost dark around the centers of ROIs.

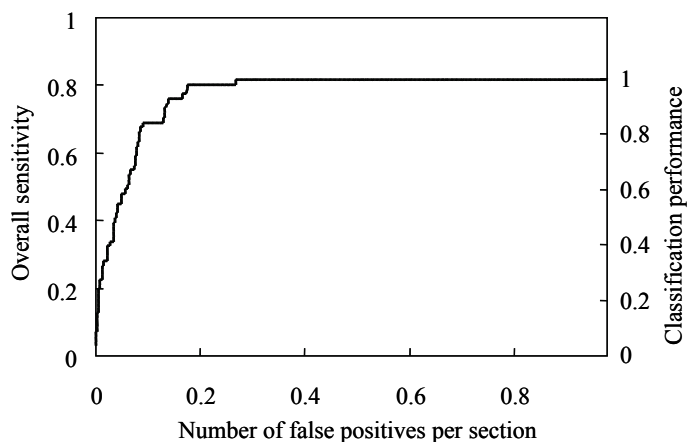


Fig. 8. FROC curve indicating the performance of the MTANN in distinction between 57 true positives (nodules) and 1.726 FPs (non-nodules).

2. Convolution NNs and shift-invariant NNs

Convolution NNs have been used for FP reduction in CAD schemes for lung nodule detection in CXRs (Lo, Lou et al. 1995; Lo, Chan et al. 1995; Lin, Lo et al. 1996). A convolution NN was trained with 28 chest radiographs for distinguishing lung nodules from non-nodules (i.e., FPs produced by an initial CAD scheme). The trained convolution NN reduced 79% of FP detections (which is equivalent to 2-3 FPs per patient), while 80% of true-positive detections were preserved. Convolution NNs have been applied to FP reduction in CAD schemes for detection of microcalcifications (Lo, Li et al. 2002) and masses (Sahiner, Chan et al. 1996) in mammography. A convolution NN was trained with 34 mammograms for distinguishing microcalcifications from FPs. The trained convolution NN reduced 90% of FP detections, which resulted in 0.5 FP detections per image, while a true-positive detection rate of 87% was preserved (Lo, Li et al. 2002).

Shift-invariant NNs have been used for FP reduction in CAD for detection of microcalcifications (Zhang, Doi et al. 1994; Zhang, Doi et al. 1996). A shift-invariant NN was trained to detect microcalcifications in ROIs. Microcalcifications were detected by thresholding of the output images of the trained shift-invariant NN. When the number of detected microcalcifications was greater than a predetermined number, the ROI was considered as a microcalcification ROI. With the trained shift-invariant NN, 55% of FPs was removed without any loss of true positives.

5. Advantages and limitations of PANN algorithms

As described earlier, the major difference between PANNs and ordinary classifiers is the direct use of pixel values with PANN. In other words, unlike ordinary classifiers, feature calculation from segmented objects is not necessary. Because the PANN can avoid errors caused by inaccurate feature calculation and segmentation, the performance of the PANN can potentially be higher than that of ordinary feature-based classifiers for some cases. PANNs learn pixel data directly, and thus all information on pixels should not be lost before the pixel data are entered into the PANN, whereas ordinary feature-based classifiers learn the features extracted from segmented lesions and thus important information can be lost

with this indirect extraction; also, inaccurate segmentation often occurs for complicated patterns. In addition, because feature calculation is not required for PANN, development and implementation of segmentation and feature calculation, and selection of features are unnecessary.

Ordinary classifiers such as linear discriminant analysis, ANNs, and support vector machines cannot be used for image processing, detection (localization) of objects, or enhancement of objects or patterns, whereas MTANNs can do those tasks. For example, MTANNs can separate bones from soft tissue in CXRs (Suzuki, Abe et al. 2006), and MTANN can enhance and detect lung nodules on CT images (Suzuki 2009).

The characteristics of PANNs which use pixel data directly should differ from those of ordinary feature-based classifiers. Therefore, combining an ordinary feature-based classifier with a PANN would yield a higher performance than that of a classifier alone or a PANN alone. Indeed, in previous studies, both classifier and PANN were used successfully for classification of lesion candidates into lesions and non-lesions (Zhang, Doi et al. 1994; Sahiner, Chan et al. 1996; Wei, Nishikawa et al. 1996; Zhang, Doi et al. 1996; Lo, Li et al. 2002; Suzuki, Armato et al. 2003; Arimura, Katsuragawa et al. 2004; Li, Arimura et al. 2005; Suzuki, Li et al. 2005; Suzuki, Yoshida et al. 2006; Suzuki, Yoshida et al. 2008; Suzuki, Rockey et al. 2010; Xu and Suzuki 2010 (in press); Suzuki, Zhang et al. in press).

A limitation of PANNs is the relatively long time for training because of the high dimensionality of input data. Because PANNs use pixel data in images directly, the number of input dimensions is generally large. For example, a 3D MTANN for 3D CT data requires 171 dimensions for its input (Suzuki, Yoshida et al. 2006; Suzuki, Yoshida et al. 2008). The ordinary feature-based classifiers are more efficient than PANNs. In an application of PANNs and feature-based classifiers to CAD schemes, a feature-based classifier should be applied first, because the number of lesion candidates that need to be classified is larger at an earlier stage. After the number of lesion candidates is reduced by use of the feature-based classifier, a PANN should be applied for further reduction of FPs. Indeed, previous studies employed this strategy (Suzuki, Armato et al. 2003; Arimura, Katsuragawa et al. 2004; Suzuki, Li et al. 2005; Suzuki, Yoshida et al. 2006; Suzuki, Yoshida et al. 2008; Suzuki, Rockey et al. 2010; Suzuki, Zhang et al. in press).

To address the issue of training time for PANN, dimensionality reduction methods for PANN have been proposed (Suzuki, Zhang et al. in press). With the use of the Laplacian-eigenfunction-based dimensionality reduction of the input vectors to a 3D MTANN, the training time was reduced by a factor of 8.5.

6. Conclusion

In this paper, PANNs were surveyed and compared with each other as well as with other non-PANN algorithms (i.e., ordinary feature-based classifiers) to make the similarities, differences, advantages, and limitations clear. The major difference between PANNs and non-PANN algorithms (e.g., classifiers) is a need for segmentation and feature calculation with non-PANN algorithms. The major advantage of PANNs over non-PANN algorithms is that no information is lost due to inaccurate segmentation and feature calculation, which would result in a higher performance for some cases such as complicated patterns. With the combination of PANNs with non-PANN algorithms, the performance of a system can be improved substantially. In addition to a classification task, MTANNs can be used for enhancement (and suppression) and detection (i.e., localization) of objects (or patterns) in images.

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8. References

- [1] M. L. Giger and K. Suzuki, "Computer-Aided Diagnosis (CAD)," in *Biomedical Information Technology*, D. D. Feng, Ed.: Academic Press, 2007, pp. 359-374.
- [2] K. Doi, "Current status and future potential of computer-aided diagnosis in medical imaging," *Br J Radiol*, vol. 78 Spec No 1, pp. S3-S19, 2005.
- [3] A. Lostumbo, C. Wanamaker, J. Tsai, K. Suzuki, and A. H. Dachman, "Comparison of 2D and 3D views for evaluation of flat lesions in CT colonography," *Acad. Radiol.*, vol. 17, no. 1, pp. 39-47, 2010.
- [4] K. Fukunaga, *Introduction to Statistical Pattern Recognition*, 2nd edition ed. San Diego: Academic Press, 1990.
- [5] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors," *Nature*, vol. 323, pp. 533-536, 1986.
- [6] V. N. Vapnik, *The Nature of Statistical Learning Theory*. Berlin: Springer-Verlag, 1995.
- [7] J. Shiraishi, Q. Li, K. Suzuki, R. Engelmann, and K. Doi, "Computer-aided diagnostic scheme for the detection of lung nodules on chest radiographs: localized search method based on anatomical classification," *Med. Phys.*, vol. 33, no. 7, pp. 2642-53, 2006.
- [8] S. G. Armato, 3rd, M. L. Giger, and H. MacMahon, "Automated detection of lung nodules in CT scans: preliminary results," *Med. Phys.*, vol. 28, no. 8, pp. 1552-1561, 2001.
- [9] H. Arimura, S. Katsuragawa, K. Suzuki, F. Li, J. Shiraishi, et al., "Computerized scheme for automated detection of lung nodules in low-dose computed tomography images for lung cancer screening," *Acad. Radiol.*, vol. 11, no. 6, pp. 617-629, 2004.
- [10] M. Aoyama, Q. Li, S. Katsuragawa, H. MacMahon, and K. Doi, "Automated computerized scheme for distinction between benign and malignant solitary pulmonary nodules on chest images," *Med. Phys.*, vol. 29, no. 5, pp. 701-8, 2002.
- [11] M. Aoyama, Q. Li, S. Katsuragawa, F. Li, S. Sone, et al., "Computerized scheme for determination of the likelihood measure of malignancy for pulmonary nodules on low-dose CT images," *Med. Phys.*, vol. 30, no. 3, pp. 387-94, 2003.
- [12] Y. Wu, K. Doi, M. L. Giger, and R. M. Nishikawa, "Computerized detection of clustered microcalcifications in digital mammograms: applications of artificial neural networks," *Med Phys*, vol. 19, no. 3, pp. 555-60, 1992.
- [13] Z. Huo, M. L. Giger, C. J. Vyborny, D. E. Wolverton, R. A. Schmidt, et al., "Automated computerized classification of malignant and benign masses on digitized mammograms," *Acad Radiol*, vol. 5, no. 3, pp. 155-68, 1998.
- [14] H. Yoshida and J. Nappi, "Three-dimensional computer-aided diagnosis scheme for detection of colonic polyps," *IEEE Trans. Med. Imaging*, vol. 20, no. 12, pp. 1261-74, 2001.

- [15] A. K. Jerebko, R. M. Summers, J. D. Malley, M. Franaszek, and C. D. Johnson, "Computer-assisted detection of colonic polyps with CT colonography using neural networks and binary classification trees," *Medical Physics*, vol. 30, no. 1, pp. 52-60, 2003.
- [16] C. Muramatsu, Q. Li, R. A. Schmidt, J. Shiraishi, K. Suzuki, et al., "Determination of subjective similarity for pairs of masses and pairs of clustered microcalcifications on mammograms: comparison of similarity ranking scores and absolute similarity ratings," *Med. Phys.*, vol. 34, no. 7, pp. 2890-5, 2007.
- [17] C. Muramatsu, Q. Li, R. Schmidt, K. Suzuki, J. Shiraishi, et al., "Experimental determination of subjective similarity for pairs of clustered microcalcifications on mammograms: observer study results," *Med. Phys.*, vol. 33, no. 9, pp. 3460-8, 2006.
- [18] C. Muramatsu, Q. Li, K. Suzuki, R. A. Schmidt, J. Shiraishi, et al., "Investigation of psychophysical measure for evaluation of similar images for mammographic masses: preliminary results," *Med. Phys.*, vol. 32, no. 7, pp. 2295-304, 2005.
- [19] H. Arimura, Q. Li, Y. Korogi, T. Hirai, S. Katsuragawa, et al., "Computerized detection of intracranial aneurysms for three-dimensional MR angiography: feature extraction of small protrusions based on a shape-based difference image technique," *Med Phys*, vol. 33, no. 2, pp. 394-401, 2006.
- [20] K. Suzuki, I. Horiba, N. Sugie, and M. Nanki, "Neural filter with selection of input features and its application to image quality improvement of medical image sequences," *IEICE Trans. Inf. Syst.*, vol. E85-D, no. 10, pp. 1710-1718, 2002.
- [21] K. Suzuki, I. Horiba, and N. Sugie, "Efficient approximation of neural filters for removing quantum noise from images," *IEEE Trans. Signal Process.*, vol. 50, no. 7, pp. 1787-1799, 2002.
- [22] K. Suzuki, I. Horiba, and N. Sugie, "Neural edge enhancer for supervised edge enhancement from noisy images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 12, pp. 1582-1596, 2003.
- [23] K. Suzuki, I. Horiba, N. Sugie, and M. Nanki, "Extraction of left ventricular contours from left ventriculograms by means of a neural edge detector," *IEEE Trans. Med. Imaging*, vol. 23, no. 3, pp. 330-339, 2004.
- [24] S. B. Lo, S. A. Lou, J. S. Lin, M. T. Freedman, M. V. Chien, et al., "Artificial convolution neural network techniques and applications for lung nodule detection," *IEEE Trans Med Imaging*, vol. 14, no. 4, pp. 711-8, 1995.
- [25] S. C. B. Lo, H. P. Chan, J. S. Lin, H. Li, M. T. Freedman, et al., "Artificial convolution neural network for medical image pattern recognition," *Neural Networks*, vol. 8, no. 7-8, pp. 1201-1214, 1995.
- [26] J. S. Lin, S. B. Lo, A. Hasegawa, M. T. Freedman, and S. K. Mun, "Reduction of false positives in lung nodule detection using a two-level neural classification," *IEEE Trans Med Imaging*, vol. 15, no. 2, pp. 206-17, 1996.
- [27] S. C. Lo, H. Li, Y. Wang, L. Kinnard, and M. T. Freedman, "A multiple circular path convolution neural network system for detection of mammographic masses," *IEEE Trans Med Imaging*, vol. 21, no. 2, pp. 150-8, 2002.
- [28] B. Sahiner, H. P. Chan, N. Petrick, D. Wei, M. A. Helvie, et al., "Classification of mass and normal breast tissue: a convolution neural network classifier with spatial domain and texture images," *IEEE Trans Med Imaging*, vol. 15, no. 5, pp. 598-610, 1996.

- [29] S. Lawrence, C. L. Giles, A. C. Tsoi, and A. D. Back, "Face recognition: A convolutional neural-network approach," *IEEE Transactions on Neural Networks*, vol. 8, no. 1, pp. 98-113, 1997.
- [30] C. Neubauer, "Evaluation of convolutional neural networks for visual recognition," *IEEE Transactions on Neural Networks*, vol. 9, no. 4, pp. 685-696, 1998.
- [31] D. Wei, R. M. Nishikawa, and K. Doi, "Application of texture analysis and shift-invariant artificial neural network to microcalcification cluster detection," *Radiology*, vol. 201, pp. 696-696, 1996.
- [32] W. Zhang, K. Doi, M. L. Giger, R. M. Nishikawa, and R. A. Schmidt, "An improved shift-invariant artificial neural network for computerized detection of clustered microcalcifications in digital mammograms," *Med Phys*, vol. 23, no. 4, pp. 595-601, 1996.
- [33] W. Zhang, K. Doi, M. L. Giger, Y. Wu, R. M. Nishikawa, et al., "Computerized detection of clustered microcalcifications in digital mammograms using a shift-invariant artificial neural network," *Med Phys*, vol. 21, no. 4, pp. 517-24, 1994.
- [34] K. Suzuki, S. G. Armato, 3rd, F. Li, S. Sone, and K. Doi, "Massive training artificial neural network (MTANN) for reduction of false positives in computerized detection of lung nodules in low-dose computed tomography," *Med. Phys.*, vol. 30, no. 7, pp. 1602-17, 2003.
- [35] K. Suzuki, H. Yoshida, J. Nappi, and A. H. Dachman, "Massive-training artificial neural network (MTANN) for reduction of false positives in computer-aided detection of polyps: Suppression of rectal tubes," *Med. Phys.*, vol. 33, no. 10, pp. 3814-24, 2006.
- [36] K. Suzuki, H. Abe, H. MacMahon, and K. Doi, "Image-processing technique for suppressing ribs in chest radiographs by means of massive training artificial neural network (MTANN)," *IEEE Trans. Med. Imaging*, vol. 25, no. 4, pp. 406-16, 2006.
- [37] S. Oda, K. Awai, K. Suzuki, Y. Yanaga, Y. Funama, et al., "Performance of radiologists in detection of small pulmonary nodules on chest radiographs: effect of rib suppression with a massive-training artificial neural network," *AJR. Am. J. Roentgenol.*, vol. 193, no. 5, pp. W397-402, 2009.
- [38] K. Suzuki, "A supervised 'lesion-enhancement' filter by use of a massive-training artificial neural network (MTANN) in computer-aided diagnosis (CAD)," *Phys. Med. Biol.*, vol. 54, no. 18, pp. S31-45, 2009.
- [39] K. Suzuki, J. Shiraishi, H. Abe, H. MacMahon, and K. Doi, "False-positive reduction in computer-aided diagnostic scheme for detecting nodules in chest radiographs by means of massive training artificial neural network," *Acad. Radiol.*, vol. 12, no. 2, pp. 191-201, 2005.
- [40] K. Suzuki, F. Li, S. Sone, and K. Doi, "Computer-aided diagnostic scheme for distinction between benign and malignant nodules in thoracic low-dose CT by use of massive training artificial neural network," *IEEE Trans. Med. Imaging*, vol. 24, no. 9, pp. 1138-1150, 2005.
- [41] K. Suzuki, D. C. Rockey, and A. H. Dachman, "CT colonography: Advanced computer-aided detection scheme utilizing MTANNs for detection of "missed" polyps in a multicenter clinical trial," *Med. Phys.*, vol. 30, pp. 2-21, 2010.
- [42] K. Suzuki, H. Yoshida, J. Nappi, S. G. Armato, 3rd, and A. H. Dachman, "Mixture of expert 3D massive-training ANNs for reduction of multiple types of false positives

- in CAD for detection of polyps in CT colonography," *Med. Phys.*, vol. 35, no. 2, pp. 694-703, 2008.
- [43] K. Suzuki, J. Zhang, and J. Xu, "Massive-training artificial neural network coupled with Laplacian-eigenfunction-based dimensionality reduction for computer-aided detection of polyps in CT colonography," *IEEE Trans. Med. Imaging*, in press.
- [44] J.-W. Xu and K. Suzuki, "False-positive reduction in computer-aided detection of polyps in CT colonography: a massive-training support vector regression approach," in MICCAI Workshop on Computational Challenges and Clinical Opportunities in Virtual Colonoscopy and Abdominal Imaging, Beijing, China, 2010 (in press).
- [45] F. Li, H. Arimura, K. Suzuki, J. Shiraishi, Q. Li, et al., "Computer-aided detection of peripheral lung cancers missed at CT: ROC analyses without and with localization," *Radiology*, vol. 237, no. 2, pp. 684-90, 2005.
- [46] M. K. Ozkan, M. I. Sezan, and A. M. Tekalp, "Adaptive motion-compensated filtering of noisy image sequences," *IEEE transactions on circuits and systems for video technology*, vol. 3, no. 4, pp. 277-290, 1993.
- [47] J. Canny, "A computational approach to edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 8, no. 6, pp. 679-698, 1986.
- [48] D. Marr and E. Hildreth, "Theory of edge detection," *Proceedings of the Royal Society of London. Series B, Biological Sciences*, vol. 207, no. 1167, pp. 187-217, 1980.
- [49] M. H. Hueckel, "An operator which locates edges in digitized pictures," *Journal of the ACM (JACM)*, vol. 18, no. 1, pp. 113-125, 1971.
- [50] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, et al., "Backpropagation applied to handwritten zip code recognition," *Neural Comput.*, vol. 1, no. 4, pp. 541-551, 1989.
- [51] K. Fukushima, "Neocognitron: a self organizing neural network model for a mechanism of pattern recognition unaffected by shift in position," *Biol Cybern*, vol. 36, no. 4, pp. 193-202, 1980.
- [52] M. N. Gurcan, B. Sahiner, H. P. Chan, L. Hadjiiski, and N. Petrick, "Selection of an optimal neural network architecture for computer-aided detection of microcalcifications--comparison of automated optimization techniques," *Med Phys*, vol. 28, no. 9, pp. 1937-48, 2001.
- [53] H. P. Chan, S. C. Lo, B. Sahiner, K. L. Lam, and M. A. Helvie, "Computer-aided detection of mammographic microcalcifications: pattern recognition with an artificial neural network," *Med Phys*, vol. 22, no. 10, pp. 1555-67, 1995.
- [54] A. Hasegawa, K. Itoh, and Y. Ichioka, "Generalization of shift invariant neural networks: image processing of corneal endothelium," *Neural Networks*, vol. 9, no. 2, pp. 345-356, 1996.
- [55] C. M. Bishop, "An example - character recognition," in *Neural Networks for Pattern Recognition*, C. M. Bishop, Ed. New York: Oxford University Press, 1995, pp. 1-4.
- [56] D. F. Michaels, "Internal Organization of Classifier Networks Trained by Backpropagation," in *Neural Networks in Vision and Pattern Recognition*, J. Skrzypek and W. Karplus, Eds. Singapore: World Scientific, 1992.
- [57] C. M. Bishop, "Neural Networks for Pattern Recognition." New York: Oxford University Press, 1995.
- [58] R. O. Duda, P. E. Hart, and D. G. Stork, in *Pattern Recognition*, 2nd ed. Hoboken, NJ: Wiley Interscience, 2001, pp. 267-272.

- [59] C. J. Murray and A. D. Lopez, "Mortality by cause for eight regions of the world: Global Burden of Disease Study," *Lancet*, vol. 349, no. 9061, pp. 1269-1276, 1997.
- [60] G. E. Goodman, "Lung cancer. 1: prevention of lung cancer," *Thorax*, vol. 57, no. 11, pp. 994-999, 2002.
- [61] J. H. Austin, B. M. Romney, and L. S. Goldsmith, "Missed bronchogenic carcinoma: radiographic findings in 27 patients with a potentially resectable lesion evident in retrospect," *Radiology*, vol. 182, no. 1, pp. 115-122, 1992.
- [62] P. K. Shah, J. H. Austin, C. S. White, P. Patel, L. B. Haramati, et al., "Missed non-small cell lung cancer: radiographic findings of potentially resectable lesions evident only in retrospect," *Radiology*, vol. 226, no. 1, pp. 235-241, 2003.
- [63] R. Glocker and W. Frohnmayer, "Über die röntgenspektroskopische Bestimmung des Gewichtsanteiles eines Elementes in Gemengen und Verbindungen," *Ann. Physik.*, vol. 76, pp. 369-395, 1925.
- [64] B. Jacobson and R. S. Mackay, "Radiological contrast enhancing methods," *Adv. Biol. Med. Phys.*, vol. 6, pp. 201-261, 1958.
- [65] T. Ishigaki, S. Sakuma, Y. Horikawa, M. Ikeda, and H. Yamaguchi, "One-shot dual-energy subtraction imaging," *Radiology*, vol. 161, no. 1, pp. 271-273, 1986.
- [66] T. Ishigaki, S. Sakuma, and M. Ikeda, "One-shot dual-energy subtraction chest imaging with computed radiography: clinical evaluation of film images," *Radiology*, vol. 168, no. 1, pp. 67-72, 1988.
- [67] B. K. Stewart and H. K. Huang, "Single-exposure dual-energy computed radiography," *Med. Phys.*, vol. 17, no. 5, pp. 866-875, 1990.
- [68] K. Doi, "Computer-aided diagnosis in medical imaging: Historical review, current status and future potential," *Comput. Med. Imaging Graph.*, vol. 31, no. 4-5, pp. 198-211, 2007.
- [69] M. L. Giger, "Update on the potential role of CAD in radiologic interpretations: are we making progress?" *Acad. Radiol.*, vol. 12, no. 6, pp. 669-70, 2005.
- [70] A. F. Frangi, W. J. Niessen, R. M. Hoogeveen, T. van Walsum, and M. A. Viergever, "Model-based quantitation of 3-D magnetic resonance angiographic images," *IEEE Trans. Med. Imaging*, vol. 18, no. 10, pp. 946-56, 1999.
- [71] L. He, Y. Chao, K. Suzuki, and K. Wu, "Fast connected-component labeling," *Pattern Recognit.*, vol. 42, pp. 1977-1987, 2009.
- [72] L. He, Y. Chao, and K. Suzuki, "A run-based two-scan labeling algorithm," *IEEE Trans Image Process*, vol. 17, no. 5, pp. 749-56, 2008.
- [73] K. Suzuki, I. Horiba, and N. Sugie, "Linear-time connected-component labeling based on sequential local operations," *Comput. Vis. Image Underst.*, vol. 89, no. 1, pp. 1-23, 2003.
- [74] P. C. Bunch, J. F. Hamilton, G. K. Sanderson, and A. H. Simmons, "A free-response approach to the measurement and characterization of radiographic-observer performance," *Journal of Applied Photographic Engineering*, vol. 4, pp. 166-171, 1978.

Applied Artificial Neural Networks: from Associative Memories to Biomedical Applications

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1. Introduction

Due to remarkable capabilities of artificial neural networks (ANNs) such as generalization and nonlinear system modeling, ANNs have been extensively studied and applied in a wide variety of applications (Amiri et al., 2007; Davande et al., 2008). The rapid development of ANN technology in recent years has led to an entirely new approach for the solution of many data processing-based problems, usually encountered in real applications (Hosseini et al., 2007).

ANNs are characterized in principle by a network topology, a connection pattern, neural activation properties and a training strategy to process data. In this section, a brief explanation for four types of ANNs including Multi-Layer Perceptron (MLP), Radial Basis Function Network (RBFN), Generalized Regression Neural Network (GRNN) and Self-Feedback Neural Network (SFNN) is provided. The MLP, RBFN and GRNN belong to a feed-forward class of neural networks (FFNN), while the SFNN belongs to the other important class of neural networks that is recurrent neural networks (RNN). Next, we investigate the diverse and innovative applications of these neural networks such as associative neural networks for recognition of analog and digital patterns, estimating the release profile of betamethasone (BTM) and betamethasone acetate (BTMA) and optimization of drug delivery system formulation. Regarding the first application, we propose a hybrid model consists of the SFNN in parallel with the GRNN. In the proposed hybrid model, storing of desired patterns is performed by employing a new one-shot learning algorithm put forward in the chapter. It will be shown that this new hybrid model is able to perform essential properties found in associative memories such as generalization, completion and recognition of corrupted patterns. Moreover, a number of case studies are performed for the purpose of performance comparison between the hybrid model and others from different classes. For the recurrent associative memory class, the comparison is

made by NDRAM. For comparison with the feed-forward class, the MLP and for the competitive class, the ART2 are used.

Regarding the second application, ANNs are used in pharmaceutical and pharmacokinetic areas to model complex interactions and predict the nonlinear relationship between causal factors and response variables. Specifically, several experiments are performed and an implant controlled-release system for corticosteroid drug delivery based on biodegradable polymer is designed. Next, the MLP, GRNN and RBFN are employed to model the release data and to predict the release profile of BTM and BTMA where in situ forming systems consist of poly (Lactide-co-glycolide), N-methyl-1-2-pyrrolidon and ethyl heptanoat as a polymer, solvent and additive, respectively. Several simulations are presented to compare the potential of each neural network. It is demonstrated that the MLP, as a data modeling tool, is more reliable and efficient than RBFN and GRNN for estimating the release profile of BTM and BTMA. At the end of this chapter, we investigate the application of the GRNN and MLP for optimization of drug delivery system formulation. It would appear that GRNN is promising in providing better solutions for determining drug formulation. Therefore, the application of the ANNs in biomedical research will definitely increase in the near future. However, the point which is noteworthy is the fact that there is no single modeling approach to address all requirements.

1.1 Multi-Layer Perceptron (MLP)

The schematic diagram of an MLP illustrated in Fig.1. In the conventional structure of an MLP, a neuron receives its input either from other neurons or from external inputs (input vector). A weighted sum of these inputs constitutes the argument of a nonlinear *activation function*. The resulting value of the activation function is the neural output. In this structure, the weights correspond to the synapses in a biological neuron, while the activation function is associated with the intracellular current conduction mechanism in the soma. An artificial neuron is an oversimplified but useful approximation of the biological neuron. This simple model ignores many of the characteristics of its biological counterpart, e.g. it does not take into account the time delays that affect the dynamics of the system (Amiri et al., 2009b).

In Fig.1, the output Y of the MLP is a vector with n components determined in the terms of m components of an input vector X and l components of the hidden layer. The mathematical representation may be expressed as:

$$y_i = \sum_{j=1}^l \left[v_{ij} g \left(\sum_{k=1}^m w_{jk} x_k + b_{wj} \right) + b_{vi} \right] \quad i = 1, \dots, n \quad (1)$$

where v_{ij} and w_{jk} are synaptic weights, x_k is k^{th} element of the input vector, $g(\cdot)$ is an activation function and b is the bias which has the effect of increasing or decreasing the net input of the activation function depending on whether it is positive or negative, respectively. It has been shown that the MLP with a *tanh* nonlinearity or other monotonic nonlinearities is a universal approximator to any arbitrary input-output mappings provided that some reasonable conditions on the nonlinear mapping are satisfied (Chen and Chen 1995).

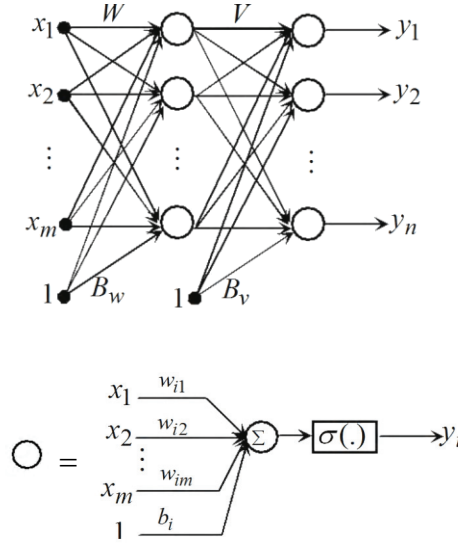


Fig. 1. The MLP structure.

1.2 Radial Basis Function Network (RBFN)

Radial basis function neural networks are special classes of the feed-forward neural network models. RBF network is a three-layer network, where each hidden unit implements a radial activation function (a nonlinear transfer function) and each output unit implements a weighted sum of hidden units' outputs. The structure of the RBF network is shown in Fig. 2. The output of i^{th} neuron in the output layer of the RBF network is determined as follows:

$$y_i(x) = \sum_{j=1}^M w_{ij} \varphi(\|x - c_j\|) \quad ; \quad i = 1, \dots, m \quad (2)$$

where $\varphi(\cdot)$ is the basis function which is described using $\|x - c_j\|$, c_j is the center vector for hidden neuron j and w_{ij} is the weight between the node j of the hidden layer and the node i of the output layer, m is the number of nodes in the output layer. The norm is typically taken to be the Euclidean distance and the basis function is taken to be Gaussian:

$$\varphi(\|x - c_j\|) = \exp\left\{-\frac{\|x - c_j\|^2}{2\sigma_j^2}\right\} \quad (3)$$

where σ_j is the width parameter of the j^{th} hidden unit in the hidden layer (Amiri et al., 2009).

In an RBF network there are three types of parameters that need to be chosen to adapt the network for a particular task: the center vectors c_j , the output weights w_{ij} , and the RBF width parameters σ_j . In this way, the training process is usually divided into two steps:

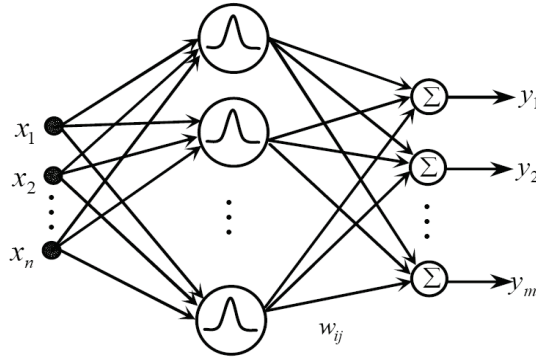


Fig. 2. The RBFN architecture.

First, the center and width parameters of the hidden layer are determined using only the input data set and by utilizing unsupervised training algorithm such as K-means (Moody and Darken, 1991), decision trees (Kubat, 1998) and self-organizing feature maps (Robert and Hewlett, 2001). Second, the output weights (connecting the hidden layer with the output layer) are determined using both input and output data and by Singular Value Decomposition (SVD) or Least Mean Squared (LMS) algorithms (Bing and Xingshi, 2006). Both steps are relatively fast when compared to back-propagation training algorithm. The number of basis functions controls the complexity and the generalization ability of the RBF network. RBF networks with too few basis functions cannot fit the training data adequately due to limited flexibility.

1.3 Generalize Regression Neural Network (GRNN)

GRNNs belong to the class of neural networks widely used for the continuous function mapping. The main function of a GRNN is to estimate a linear or nonlinear regression surface on independent variables (input vectors) U , given the dependent variables (desired output vectors) X . That is, the network computes the most probable value of an output, $\hat{x}$, given only training vectors U . Specifically, the network computes the joint probability density function of U and X . Then the expected value of X given U is expressed as (Wachowiak et al., 2001):

$$E[X|U] = \frac{\int_{-\infty}^{\infty} X f(U, X) dX}{\int_{-\infty}^{\infty} f(U, X) dX} \quad (4)$$

An important advantage of the GRNN is its simplicity and fast approximation procedure. Another attractive feature is that, unlike back-propagation based neural networks (BP-NN), GRNN does not converge to local minima (Specht, 1991). In addition, the training process with a GRNN-type algorithm is much more efficient than with the BP-NN algorithm (Huang and Williamson, 1994).

The topology of a GRNN is described in Fig. 2, and it consists of the following four parts. First, there is an input layer that is fully connected to the pattern layer. Second, there is a pattern layer that has one unit for each pattern. It computes the pattern Gaussian function expressed by

$$h_i = \exp[-D_i^2 / 2\sigma^2] ; \quad D_i^2 = (u - U_i)^T (u - U_i) \quad (5)$$

where σ denotes the smoothing parameter, u is the input presented to the network and U_i is each of the training vector. Third, there is a summation layer that has two units N and P . The first unit computes the weighted sum of the hidden layer outputs. The second unit has weights equal to "1", and therefore sums exponential terms (h_i) alone. Fourth, there is an output unit that divides N by P to provide the prediction result ($\hat{x}$).

$$\hat{x} = \sum_{i=1}^n h_i X_i / \sum_{i=1}^n h_i \quad (6)$$

In fact, this is the neural implementation of (4) (Amrouche and Rouvaen, 2006). Overall, the GRNN has only a single parameter (σ) that needs to be determined.

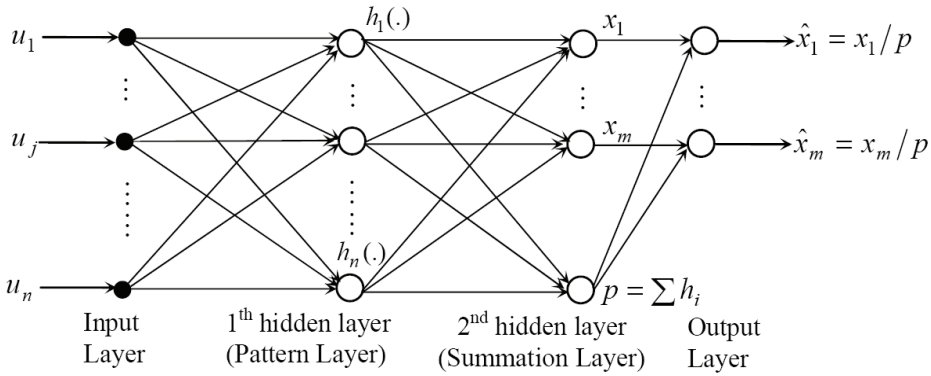


Fig. 3. The GRNN architecture.

1.4 Self-Feedback Neural Network (SFNN)

A self-feedback neural network (SFNN) is a simple recurrent, two-layer network, where the output layer contains self-feedback units. In this model, there are no interlinks among units in the feedback layer. The self-feedback connection of units ensures that the output of the SFNN contains the complete past information of the system. Since there are no interlinks among units in the feedback layer, the SFNN has considerably fewer weights than the fully recurrent neural network and the network is noticeably simplified (Ku & Lee, 1995).

The architecture of the SFNN model is depicted in Fig. 4. The mathematical description is as follows (Ku and Lee, 1995; Amiri et al., 2007):

$$S_j(k) = \sum_{i=1}^n W_{ij}^I u_i(k) + W_j^D X_j(k-1) \quad (7)$$

$$X_j(k) = f(S_j(k)) \quad (8)$$

where $u_i(k)$ ($i=1, \dots, n$) denotes the external input, and $S_j(k)$, $X_j(k)$ ($j=1, \dots, m$) are the state variable and output of the j^{th} unit of the output layer, respectively. $f(\lambda)$ is the sigmoid activation function defined as $f(\lambda) = 1 / (1 + e^{-\lambda})$. W_{ij}^I , W_j^D are connection weights from input to output layer and within the output layer, respectively.

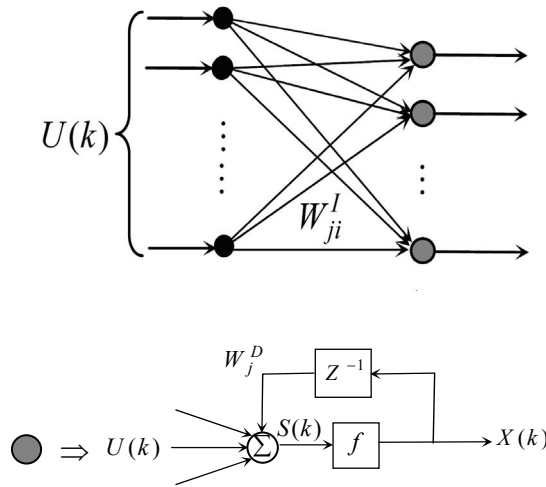


Fig. 4. The Structure of the SFNN model.

2. Associative memory

Associative neural networks (AsNN) are dynamical nonlinear systems capable of processing information through the evolution of its state in a high-dimensional state space (Amiri et al., 2008a). The aim of such networks is to retrieve a previously learned pattern from an example which is similar to, or a noisy version of, one of the previously presented patterns. Therefore, these networks have the property of being robust to noisy patterns or partial information. The main requirement associated with AsNNs is that every given memory should be an asymptotically stable equilibrium (attractor) of the system (Amiri, Menhaj, & Yazdanpanah, 2008b; Atiya & Abu-Mostafa, 1993). These memory patterns are generally represented by binary (digital) or real-valued (analog) vectors. If the learning is performed adequately, such networks are able to generalize to new stimuli. In this way, they can retrieve a previously learned pattern from an example that is similar to one of the previously presented patterns (Chartier et al., 2009). In other words, AsNNs provide distributed storage of information, within which every neuron stores fragments of information needed to retrieve any stored data record. This property of associative neural networks makes them suitable for a variety of applications such as image segmentation (Cheng, et al., 1996) and recognition of chemical substances (Reznik, et al., 2005).

This subject has received most research attention after the study of Hopfield (1982), so that many networks have been proposed to store and properly recall patterns and images. However, few networks can store both analog and digital patterns simultaneously. An example of a model with analog pattern storage capabilities is the nonlinear dynamic recurrent associative memory (NDRAM) (Chartier & Proulx, 2005), which is based on an unsupervised time-difference covariance matrix. This model is able to develop both analog and bipolar attractors. Moreover, the model is able to develop less spurious attractors and has a better recall performance under random noise than many Hopfield-type neural networks (Chartier & Proulx, 2005).

SFNNs are simple recurrent neural networks that have difficulties learning and storing analog and digital patterns as associative memories (Amiri, et al., 2008b). On the other hand, GRNNs can find a solution for any given problem, but lack a recurrent structure to filter noise. Therefore, in this chapter a hybrid model of SFNN and GRNN for associative recall of analog and digital patterns is proposed.

2.1 The hybrid model

Fig. 5 illustrates the hybrid model of the SFNN and GRNN. This model is used as an associative neural network to store the desired pattern representations as asymptotically stable equilibrium points of the SFNN. The new hybrid model can then perform classification over noisy versions of the original patterns. To achieve this, first the number of output units is determined as a function of the number of inputs. Based on the mathematical analysis presented by Amiri and colleagues (2007, 2008, 2010), each self-feedback unit can store a maximum of two asymptotically stable fixed points, a network with m self-feedback unit can store 2^m stable fixed points. Therefore, the number of stable fixed points (2^m) should be greater or equal to the number of desired patterns. In other words, m should be set to the smallest integer value such that $m \geq \log_2$ (number of patterns). The selection of network parameter values is performed based on the training algorithm which will be presented in the next section. Next, we proceed with storing pattern representations in the SFNN and selecting the initial conditions of its dynamical equations. In our proposed approach, first, lower dimension representations of the patterns are stored as the asymptotically stable fixed points of the SFNN. Then, we utilize the input patterns and corresponding desired initial conditions, i.e., lower dimension representations, of the SFNN as the input and desired output vectors of the GRNN, respectively. These desired initial conditions are obtained by selecting an arbitrary point in the attraction domain of each asymptotically stable equilibrium point. In the recognition stage, each new pattern is applied first to the GRNN in order to extract the corresponding approximate initial condition ($\hat{x}_j$) that will be used for the SFNN ($\hat{x}_j(0) = \hat{x}_j$). Next, the new input pattern is applied to the SFNN in conjunction with appropriate initial condition (given by GRNN). Then, the output is fed back to the input; (7) and (8) are computed recursively until a predefined threshold is reached. The initial states of the system are set equal to output given by the GRNN approximation.

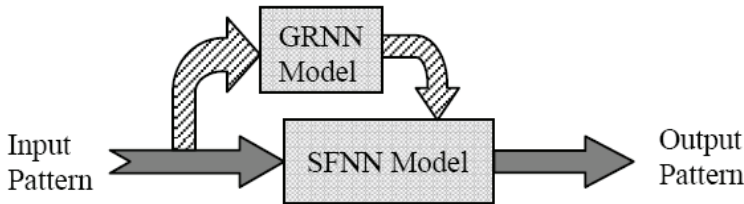


Fig. 5. The structure of the proposed hybrid model. It is used as an associative neural network.

2.2 Learning and retrieval procedures

Based on the stability analysis performed by Amiri and collaborators (2007, 2008, 2010), a simple and efficient procedure for storing and recovering the desired patterns for the

proposed hybrid model is presented. This algorithm is written for the general case where there are n input nodes and m self-feedback units in the output layer. It should be mentioned that m should be set to the smallest integer value such that $2^m \geq$ number of patterns. The one-shot learning algorithm is as follows:

1. Choose an arbitrary value greater than 4 for the self-feedback coefficient of each self-feedback unit ($w_j^D > 4$).
2. Regarding the selected value for w_j^D calculate following terms for each self-feedback unit:

$$b_{j1} = \frac{2}{w_j^D} \ln \left(\frac{\sqrt{w_j^D} + \sqrt{w_j^D - 4}}{2} \right) - \frac{\sqrt{w_j^D} + \sqrt{w_j^D - 4}}{2\sqrt{w_j^D}}$$

$$b_{j2} = \frac{2}{w_j^D} \ln \left(\frac{\sqrt{w_j^D} - \sqrt{w_j^D - 4}}{2} \right) - \frac{\sqrt{w_j^D} - \sqrt{w_j^D - 4}}{2\sqrt{w_j^D}} \quad (9)$$

($j = 1, \dots, m$)

3. To use the maximum capacity of the SFNN, adjust the value of parameter b_j as $b_{j1} < b_j < b_{j2}$. In this way, each self-feedback unit will code two patterns, and resulting in a small network even for a large number of patterns. To do this, compute $b_j = \alpha b_{j1} + (1 - \alpha) b_{j2}$ ($0 < \alpha < 1$) to ensure that $b_{j1} < b_j < b_{j2}$. Furthermore, w_j^D and α can be set to any arbitrary value satisfying those conditions mentioned previously (i.e., $w_j^D > 4$ and $0 < \alpha < 1$). Even, these values can be same for all self-feedback units.
4. To estimate the boundaries of attraction domains for each attractor in each dimension use the following equations:

$$\bar{s}_{j1} > w_j^D \left(\frac{\sqrt{w_j^D} + \sqrt{w_j^D - 4}}{2\sqrt{w_j^D}} + b_j \right) \quad (j = 1, \dots, m)$$

(10)

$$\bar{s}_{j2} < w_j^D \left(\frac{\sqrt{w_j^D} - \sqrt{w_j^D - 4}}{2\sqrt{w_j^D}} + b_j \right)$$

5. Based on these selected values and for every j , set

$w_{ij}^l = (b_j w_j^D) / \sum_{l=1}^n u_l ; (w_{(i+1)j}^l = w_{ij}^l ; i = 1, \dots, n ; j = 1, \dots, m)$ to update the input weight matrix (W) for each input vector. This will end the process of SFNN model training.

6. In the recovering stage, (2) and (3) are used to calculate the output of the SFNN model. It is noted that for each input vector, these recursive equations should be computed until the SFNN converges to one of its stable states, i.e., a predefined threshold is reached.

3. Simulation results

In this section, simulations on auto-associative tasks are presented in order to assess the efficiency of the hybrid model. We have studied two tasks. The first task consists of learning and retrieving a numeral data set (bipolar representation) while the second task consists of learning a pictorial data set (analog representation). Gaussian basis functions with constant smoothing parameters ($\sigma=0.85$) were used for the GRNN throughout this study. All experiments were implemented in MATLAB (ver.7) on a personal computer with AMD Athlon 64 bit Processor 3500+.

3.1 Digital numbers

As shown in Fig. 6, the training set used here consists of ten noise-free images. The images represent 8x8 digital numbers (0 to 9). Each pattern has the value of $u(0)$ initially. Using the training algorithm described previously, these patterns are stored in the SFNN. Since there are ten patterns, the smallest integer value that satisfies $2^m \geq 10$ is $m=4$, therefore, four self-feedback units are needed. Parameters $w_1^D \dots w_4^D$ can be determined based on step 1 of the training algorithm. In the examples, shown in Fig.7, the parameter values were set to $w_1^D=10$, $w_2^D=20$, $w_3^D=30$, $w_4^D=40$. The connections of the input weight matrix will then be selected using these values and the ones from parameters b_j . Based on steps 2 and step 3 of the training algorithm, the values of the parameters b_j can be calculated and are given in Table 1. It is noted that the fifth and sixth rows of this table are not applicable to this simulation, because only four self-feedback units are needed. Finally, as described in step 5, the values of the weight matrix can be computed using the input patterns and the selected network parameters.

Storage of the desired patterns in the SFNN is continued by selecting the initial conditions of its dynamical equations. In this way, we select an arbitrary value from the basin of attraction of the attractor. The boundaries of each attraction domain are estimated using (10) and are given in Table 1. In order to select an arbitrary initial condition, it is sufficient to add a small arbitrary value, such as 1.5, to each of these calculated borders. It should be mentioned that sixteen 4-component initial conditions can be generated based on different combinations of s_{jk} , but since there are only ten patterns to be stored, only ten initial conditions are needed. These selected initial conditions play the role of desired outputs for the GRNN. Considering the desired patterns (8x8 digital numbers) as the input vectors and these ten 4-component initial conditions as the desired output vectors, GRNN is trained to approximate the mapping function between input and desired output.

In the retrieval stage (step 6), each new pattern is applied first to the GRNN. The output will then be used as the approximate initial conditions of that pattern. These initial conditions are used to initiate the dynamical equations of the SFNN. Next, this new input pattern is applied to the SFNN which is driven by appropriate initial conditions. Equations (7) and (8) are computed recursively until the difference between each iteration is less than 10^{-5} .

The numeral data set was tested for several cases of noisy patterns. The noise in each case was measured by the percentage of pixels that were altered from 0 to 1 and vice versa. Thus, if the noise was 25%, 16 out of the 64 randomly chosen pixel values were given opposite values. Several noise levels were tested, but in the interest of simplicity, we merely mention the case of 25% of noise, as presented in Fig.7. In this situation, the percent of recognition (PR) is 100%. As can be seen from this simulation, the results are remarkable, especially when one observe the extremely poor quality of the testing samples.



Fig. 6. Ten stored digital patterns. Each of them is an 8x8 image.

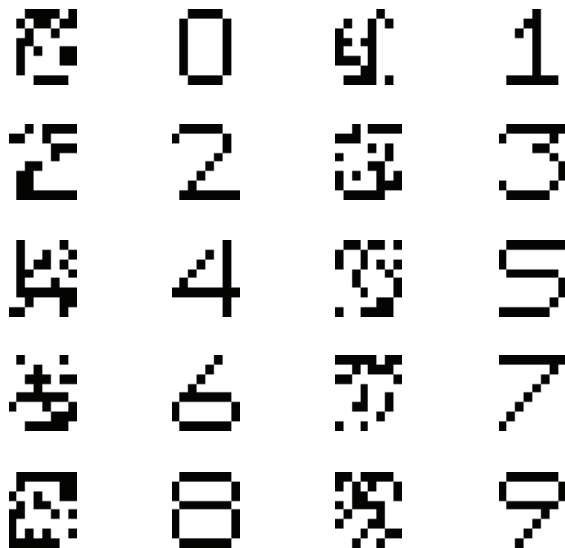


Fig. 7. Some noisy versions of each digital number are provided for the hybrid model. The percent of noise for this example is 25%.

b_{ij}			s_{ij}		
$b_{11}=-0.681$	$b_{12}=-0.319$	$b_1=-0.464$	$s_{11}>4.23$; $s_{12}<-3.51$	$s_{11}=5.73$	$s_{12}= -5.01$
$b_{21}=-0.803$	$b_{22}=-0.197$	$b_2=-0.439$	$s_{21}>10.15$; $s_{22}<-7.73$	$s_{21}=11.65$	$s_{22}=-9.23$
$b_{31}=-0.854$	$b_{32}=-0.146$	$b_3=-0.429$	$s_{31}>16.09$; $s_{32}<-11.84$	$s_{31}=17.59$	$s_{32}=-13.34$
$b_{41}=-0.883$	$b_{42}=-0.117$	$b_4=-0.423$	$s_{41}>22.04$; $s_{42}<-15.9$	$s_{41}=23.54$	$s_{42}=-17.41$
$b_{51}=-0.902$	$b_{52}=-0.098$	$b_5=-0.420$	$s_{51}>28.00$; $s_{52}<-19.96$	$s_{51}=29.50$	$s_{52}=-21.46$
$b_{61}=-0.915$	$b_{62}=-0.085$	$b_6=-0.417$	$s_{61}>33.97$; $s_{62}<-24.00$	$s_{61}=35.47$	$s_{62}=-25.50$

Table 1. Parameter values for three sets of simulations.

For each percent of noise (PN), 100 noisy patterns are randomly generated and the PR for each noisy pattern is calculated. Performance was then averaged. Fig. 11 illustrates the results for digit “1” and “8”. As can be observed, when the PN is small, the model can recognize all patterns (PR=100%). As PN increases, the PR decreases. Even when the PN of each digit is about 35%, the PR is still greater than 80%.

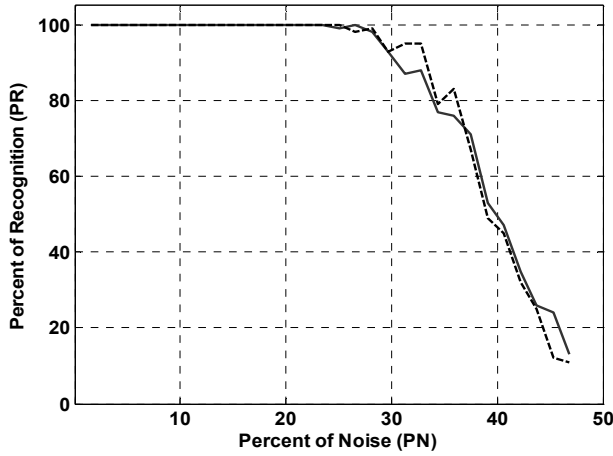


Fig. 8. Correct retrieval percent versus percent of noise for stored patterns by using the proposed hybrid model. For each PN value, simulations on 100 randomly generated patterns were performed. Solid line is for digit "1" and the dashed line is for digit "8".

3.2 Grayscale pictorials

The last set of simulations for evaluating the abilities of the proposed hybrid model in recognition of corrupted patterns are carried out in this section. Similar to the previous example, the six patterns shown in Fig. 9 are stored as stable memories in the SFNN by the learning algorithm developed previously. Since there are six patterns, we need 3 self-feedback neurons for the SFNN. The parameter values of the SFNN are the same as preceding examples and are given in Table 1 (the first three rows of the Table 1). Several simulations for recovering various corrupted patterns with different Euclidean Distance

(ED) were performed. ED is defined as $ED = \sqrt{\frac{1}{N} \left(\sum_{i=1}^N (op_i - cp_i)^2 \right)}$ where op_i and cp_i are

values of the i^{th} element of an original pattern (noise-free pattern) and a corrupted pattern, respectively. Several simulations for retrieving various corrupted images with different EDs were performed. The results of one of these simulations are depicted in Fig.10. As can be deduced from this experiment, the results are significant, since the ED value of noisy patterns is 0.34 which completely destroy the original images and produce extremely poor quality patterns. Fig. 11 shows some corrupted versions of the "Lena" and "Man with Camera" patterns and their corresponding ED values. Similar to previous examples, for each selected ED, we randomly generate 100 patterns. Each time, the model runs from a generated pattern and we check whether it is correctly recognized (we compute PRs). Next, the average value of these 100 PRs is calculated for "Lena" image. The result is illustrated in Fig.12. As can be realized from this figure, increasing the ED value leads to reducing the PR value. When the ED value is about 0.46, the PR value is still greater than 80% and reveals the advantages of the proposed model.

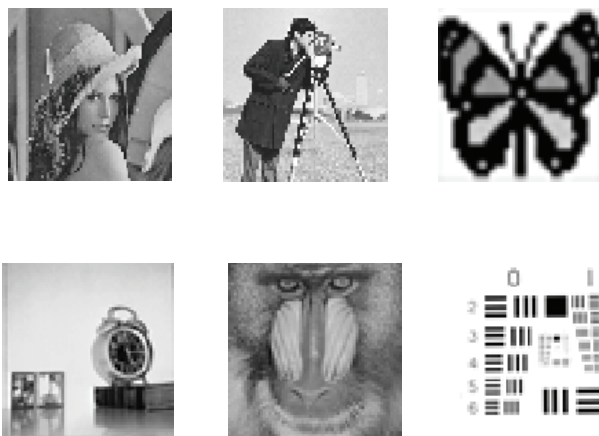


Fig. 9. Six stored patterns. Each of them is a 256 gray-level 32x32 image.

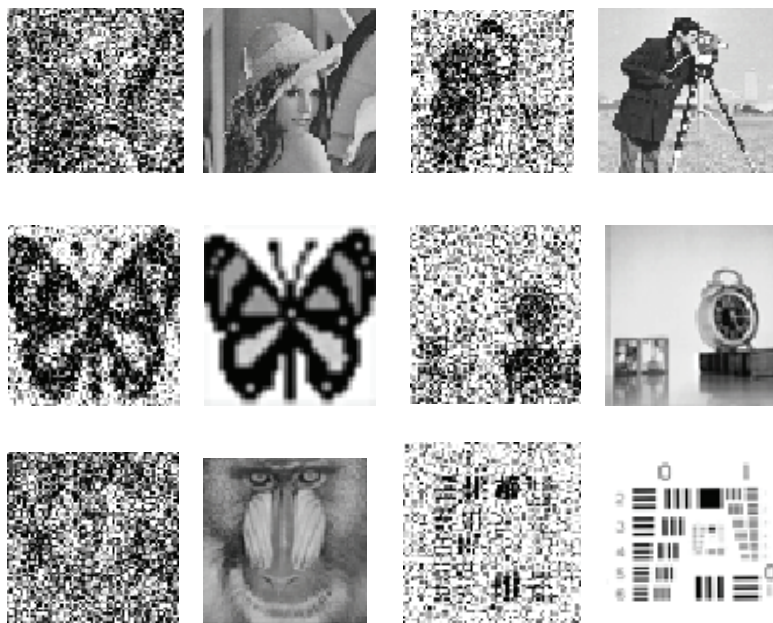


Fig. 10. Some noisy versions of each grayscale pictorial are provided for the hybrid model. (ED=0.34)

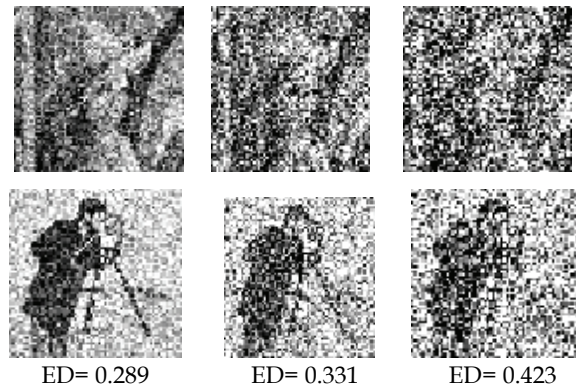


Fig. 11. Examples of corrupted patterns "Lena" and "Man with Camera" and their ED values.

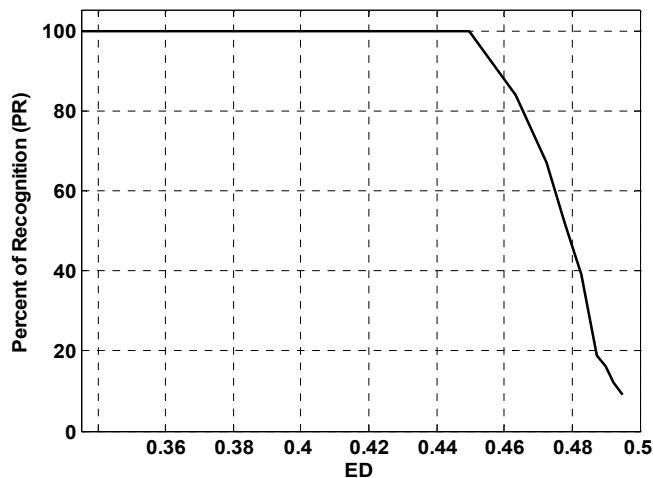


Fig. 12. Correct retrieval percent versus the ED for "Lena" image by using the proposed model. For each ED value, simulations on 100 randomly generated patterns are performed.

3.3 Comparison with other models

In this section, a number of case studies are performed for the purpose of performance comparison between the proposed hybrid model and others from different classes including Hopfield-type network (NDRAM (Chartier and Proulx, 2005), Storkey et. al, (Storkey and Valabregue, 1999)) and competitive model (ART2 (Carpenter and Grossberg, 1987, 2003)). In addition, the model was also compared with MLP, Bégin and Proulx (Bégin and Proulx, 1996) and Hopfield (Hopfield, 1982). Since the results of these networks were poorer than the previous three models, and in some cases they could only be used for one type of simulation (for instance, binary patterns), for the sake of simplicity we did not include their results in the chapter. Interested readers can refer to Amiri and colleagues (2007, 2008, 2009, 2010).

In these simulations, all the neural network models learn forty correlated binary patterns placed in 64x64 grids. Those patterns present a good variety of correlation with each other.

Fig. 13(a) illustrates the patterns. Each pattern is converted into a vector of 4096 elements. For NDRAM and Storkey et al. models, a white pixel is given a value of -1 and a black pixel is given a value of +1, while, for the remaining models a white pixel is given a value of 0 and a black pixel is given a value of +1. For NDRAM model, all the parameters are set according to (Chartier and Proulx, 2005). Based on the preceding examples, six self-feedback units are required for the hybrid model to store these forty patterns. The parameter values of the SFNN are given in Table 1. For ART2 and for the Storkey and colleagues, the parameters are set according to (Freeman, 1993) and (Storkey and Valabregue, 1999), respectively. Both models have to find the weights in a 4096×4096 matrix.

After the learning, the network's performance is evaluated using two different recall tasks. The first task consisted of recalling noisy inputs. A noisy input is obtained by generating a random vector normally distributed with a mean of zero and a variance of P added to a given learned prototype. For the simulation, the proportion of noise varied from 0.5 to 19. The second recall task consists of testing the network with random pixel flip noise. For this trial, we flip a fixed number of pixels randomly and let the network self-stabilize. Fig. 13(b) depicts a noisy version of the picture "man" versus the variance of the applied noise. Fig. 13(c) illustrates noisy versions of the picture of an "octopus" as a function of the number of pixel flips. It is obvious that occluded or partially altered patterns are similar to pixel flips. In addition, the flipped pixels situation is more close to reality; since flipped pixels could be interpreted as misclassified features of an attribute vector.

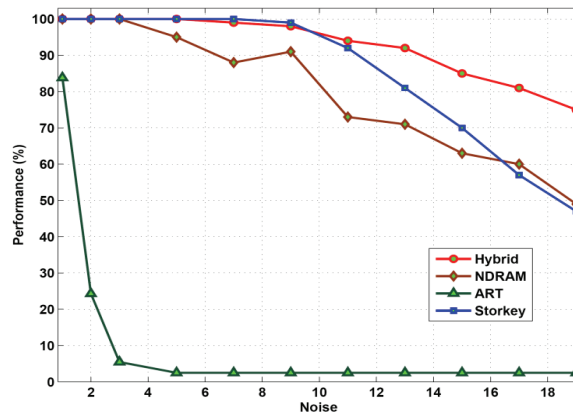
Each recall trial is accomplished according to the following procedure:

1. A distorted patterns bank is constructed by an addition of normally distributed noise or a random pixel flips to each pattern.
2. A pattern is randomly selected from the distorted patterns bank.
3. Each model is run to check whether the corrupted pattern is properly recognized (calculation of PR).
4. Steps (1) to (3) are repeated for 100 times and the PR for each trial is computed.
5. The average value of these 100 PRs is considered as the factual value of recognition for that percent of noise.
6. Steps (1) to (5) are repeated for the next value of noise.

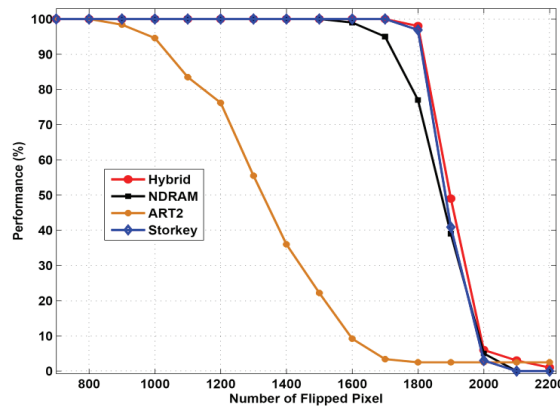
Results and discussion: Fig. 14(a) shows that the performance of the hybrid model is better than those of NDRAM, Storkey and ART models when the recall is accomplished from the normally distributed random noise. For example, even under a noise proportion of 15.0, the hybrid model still has a performance of about 85%, rather than only 63% for the NDRAM model 2.5% for the ART and 70% for Storkey. If we look at the performance for the random flip noise task (Fig. 14(b)), one can easily notice that again the hybrid model performs better than NDRAM, Storkey and ART2 models. To illustrate, at a noise proportion of 44% (1800 pixel flips), the hybrid model still reaches 98% of good recall, rather than 77% for the NDRAM model, 2.5% for the ART and 96.9% for Storkey. In this random flip noise task, the hybrid model and Storkey model closely matched each other. It should be mentioned that according to (Chartier and Proulx, 2005) the performance of the NDRAM is much better than other models such as Kanter and Sompolsky (Kanter and Sompolsky, 1987), Diederich and Oppel (Diederich and Oppel, 1987).

In comparison to the above-mentioned models, the hybrid model has at least four significant advantages: First, just like any one-shot learning procedure, the training process of the hybrid model is much faster than NDRAM, and Storkey. This leads to the reduction of computational time and cost. Second and more important is that, contrarily to Hopfield-type

associative memory, the hybrid model does not have any spurious attractor (like ART2) since the initial conditions which are used to initiate the dynamical equations of the SFNN are determined through equations (10). In other words, using a mapping from the original input into lower mutually exclusive attraction domain enables the hybrid model to overcome the problem of spurious attractors. The third advantage of the hybrid model is that, unlike in Hopfield-type networks (e.g. NDRAM and Storkey), the performance does not decrease as the memory load increases. In the hybrid model (as well as the other models) the memory load will affect the performances indirectly by the degree of correlation between the stimuli. The fourth distinct characteristic is the lack of any additional adjustment when the hybrid model encounters a digital or real-valued vector as the input pattern, unlike the ART2 model that needs a 5-layered preprocessing step to deal with real value vectors.



(a)



(b)

Fig. 14. (a) Performance percentage in function of random noise proportion (b) Performance of correct categorization in function of the number of pixels flips. As these simulations shows, the hybrid model performs much better than NDRAM, ART2 and Storkey in recognition of corrupted patterns.

4. Biomedical applications

Recently there has been increased interest in applications of artificial neural networks (ANNs) in biomedical researches (Hosseini et al., 2007, Amiri et al., 2009, Rafienia et al., 2010). ANNs are used in pharmaceutical and pharmacokinetic areas to model complex relationships and to predict the nonlinear relationship between causal factors and response variables. The distinct features of the ANN make this approach very useful in situations where the functional dependence between the inputs and outputs is not clear. The basic concepts of the multiobjective simultaneous optimization technique of drug formulations, by utilizing ANN, were reviewed by Takayama and colleagues (2003). The applicability of the ANN in modeling and predicting drug release profiles was investigated to evaluate an experimental study in transdermal iontophoresis (Lim et al. 2003). An ANN-based system was reported to predict peaks and troughs of gentamicin serum concentrations based on a set of empirical data, and the results were comparable with those using nonlinear mixed effect modeling (Brier et al. 1995). Furthermore, some researchers focused on developing pharmacokinetic models to predict plasma drug concentration based on ANNs and calculate the estimated concentrations of heparin for patients undergoing hemodialysis treatment (Valafar & Valafar, 1999). In the following, we review and discuss two recent applications of ANNs in biomedical applications including estimation of Betamethasone release profiles from an in situ forming system based on the biodegradable polymer (PLGA 75/25) and optimization of a new drug delivery formulation.

4.1 Estimation of the release profile of drugs

Considering the recent researches published by Amiri and colleagues (2009) and Rafienia and collaborators (2010), here, we presented an ANN based approach to estimate the nonlinear correlation between the drug loaded formulations and the release profiles. Specifically, the potential of three FFNNs including MLP, RBFN and GRNN to estimate the release profiles of two kinds of drugs, i.e. betamethasone and betamethasone acetate is compared and discussed. The 22 data samples used to train each network were collected from the in vitro experiments of drug release evaluations where in situ forming systems consist of poly (lactide-co-glycolide), N -methyl-1-2-pyrrolidon, and ethyl heptanoat as a polymer, solvent, and additive, respectively. The nonlinear principal component analysis (NLPCA) feature extraction technique was utilized to extract three features from each release graphs. In this way, as proposed by Kramer, a 3-hidden layer auto-associative neural network is used as the NLPCA structure which is shown in Fig. 15.

In a 3-hidden layer NLPCA (Fig.15), the input vector is transferred to the *encoding* neurons in the first hidden layer. The hyperbolic tangent function ($\tanh$) is used as the transfer function in all of the neurons in hidden layers and output layer. Basically, a nonlinear function maps from the higher dimension input space to the low dimensional *bottleneck* space, followed by an inverse transform mapping from the *bottleneck* space back to the original space represented by the *outputs*. This is an auto-associative network, where the target and input data sets are the same. The cost function, which is the mean square error (MSE) between the outputs and the inputs, is minimized by adjusting the weight matrices of the neural network. Data compression is achieved by the *bottleneck*. The *bottleneck* neurons in Fig.15 give the nonlinear principal components of the input vectors. The numbers of encoding and decoding neurons are adjustable for the optimal fit, but are set the same for simplicity. The NLPCA in Fig.15 with 15, 7, 3, 7, 15 neurons in its 5 layers will be referred to a 15-7-3-7-15 model.

In this research, using NLPCA, three features are extracted from each release graphs which are shown in Fig 16. To do this, each release is fed into the NLPCA with structure of 15-7-3-7-15. Since each neuron in the output layer has a *tanh* transfer function, the output of the network is at the range of (-1, 1). To overcome this limitation and also maintaining the symmetry of the network, a possible technique is to normalize the input data set to this range. Another possibility is to add a layer which contains linear neurons. In this case, which is used in this research, the output of the network can be extended to an arbitrary range depending on the input data sets. The dashed lines in the Fig. 15 illustrate the augmented layer to the standard NLPCA structure. By utilizing these three features, we have converted and reduced the released graphs into three more effective values which are used in the simulations later.

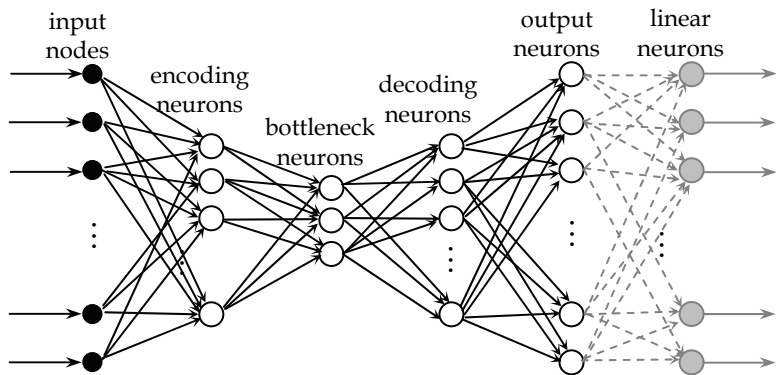


Fig. 15. NLPCA architecture.

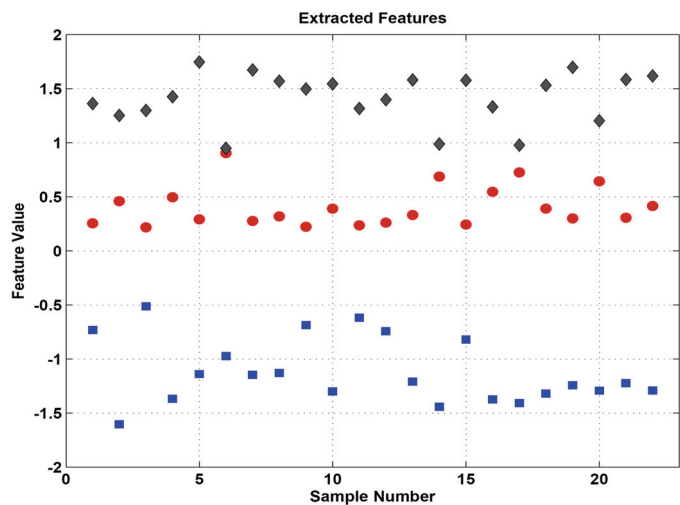


Fig. 16. Three extracted features of each release graph. Red circles, blue squares and black diamonds are first, second and third features, respectively.

In order to train an ANN model, the most common approach is to divide the data samples collected from experiments into two groups, the training and validation data sets. The training group is used to train the ANN model by adjusting the weight matrices of the network model. The validation group is used to ensure that the ANN has properly learned the relationship between inputs and outputs and has been able to generalize the results. This data set should include samples which are not included in the training data set. This method is suitable when there are enough data samples to train the neural network. Therefore due to lack of enough data samples, another training approach, i.e., a *cross validation algorithm* was used. In this method, the data are divided into k subsets. The ANN is trained where each time one of the k subsets is used as the validation set and the other $(k-1)$ subsets are put together to form a training set. The average error across all k trials is computed. The advantage of this method is that every subset appears once in a validation set and $(k-1)$ times in a training set. This is known as *k-fold cross-validation*. The variance of the estimated results decreases as k increases. *Leave-one-out* (L.O.O.) cross-validation is a k -fold cross validation where k is equal to the total number (n) of the data samples. This means that one data sample is used for validation and the remaining samples are used for training, and the process is repeated n separate times. As before, the average error is computed and used to evaluate the model. In this case, each of the neural networks takes four input variables consisting of drug concentration, gamma irradiation, additive substance, type of drug (BTMA and BTM), and provides three output variables, which are the extracted features by NLPKA. In order to train the MLP network, resilient back-propagation (RP) was utilized as a learning algorithm. The initial weight matrices are randomly selected and the learning process continues until a sufficiently low MSE (10^{-3}) on validation data is achieved. A *tanh* function is used as the activation function in all of the neurons in hidden and output layers. Gaussian basis functions with constant smoothing parameters were used for the RBFN and GRNN. After several trial-and-error simulations and to improve generalization, we selected $\sigma=0.2$ for RBF network and $\sigma=1.05$ for GRNN. It is noted that training processes in RBFN and GRNN use optimized number of hidden neurons that in turn allow for efficient approximation of the mapping function between the input and output spaces. In this technique, neurons are added to the network until the sum-squared error falls beneath an error goal or a maximum number of neurons have been reached. RBFN and GRNN require more neurons than MLP network, but they can be designed in a fraction of the time that it takes to train MLP network. Next, we used a L.O.O. cross-validation training algorithm with 22 data samples. To evaluate the precision of estimations for each data set, according to Rafienia and colleagues (2010) , we calculated the mean prediction error (MPE) as defined by:

$$MPE = \frac{\sum (x_i - \hat{x}_i)}{n} \quad (11)$$

where x and $\hat{x}$ are the target value and the estimated value of the variable, respectively and n is the number of data set. This quantity was computed for each of the testing data set. In this case, we had 22 MPEs, in which their mean value was used as an index to evaluate the trained neural network performance. For this criterion, the mean value error of the each extracted feature for each network is shown in Table 2. The average mean prediction error (AMPE) for each network is also shown in Table 1. Since, different executions of the MLP network leads to the different MPE, this process should be repeated several times to

guarantee the accuracy of the neural network response. Fig. 17 shows these 10 mean value errors computed for each of the 10 trials and for each feature. The variance of the aforementioned process for the first, second and the third feature respectively are about 0.1%, 0.28%, and 0.12%. This confirms the accuracy of the estimations for the MLP network. Noteworthy that there is no need to carry out this process for RBFN and GRNN since the performance of these two networks will not changed during different trails. In fact, this is one the main advantages of these networks. They don't depend on different executions.

	MPE of the first feature	MPE of the second feature	MPE of the third feature	Average MPE
MLP	0.1109	0.2389	0.1334	0.1611
RBFN	0.1417	0.2045	0.2001	0.1821
GRNN	0.1438	0.2109	0.1792	0.1780

Table 2. Mean prediction errors of the each feature for each neural network

As can be seen from Table 2, the performance of the MLP network in estimation of release profile is better than the RBFN and GRNN. To verify this, the estimated features are used as the input vectors for the next part of the NLPCA network (from bottleneck neurons toward output neurons). Fig. 18 shows the performance of each neural network and compares the release profiles calculated by the ANN with the release profiles measured by HPLC. It is apparent that MLP is more reliable and has better performance in estimation of BTM and BTMA release profiles than GRNN and RBF networks.

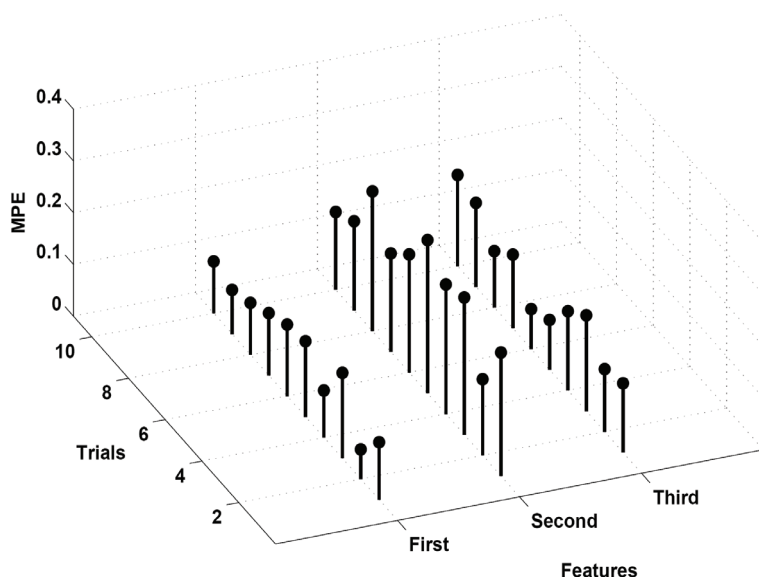


Fig. 17. Ten mean value errors computed for each of the ten trials for each extracted features.

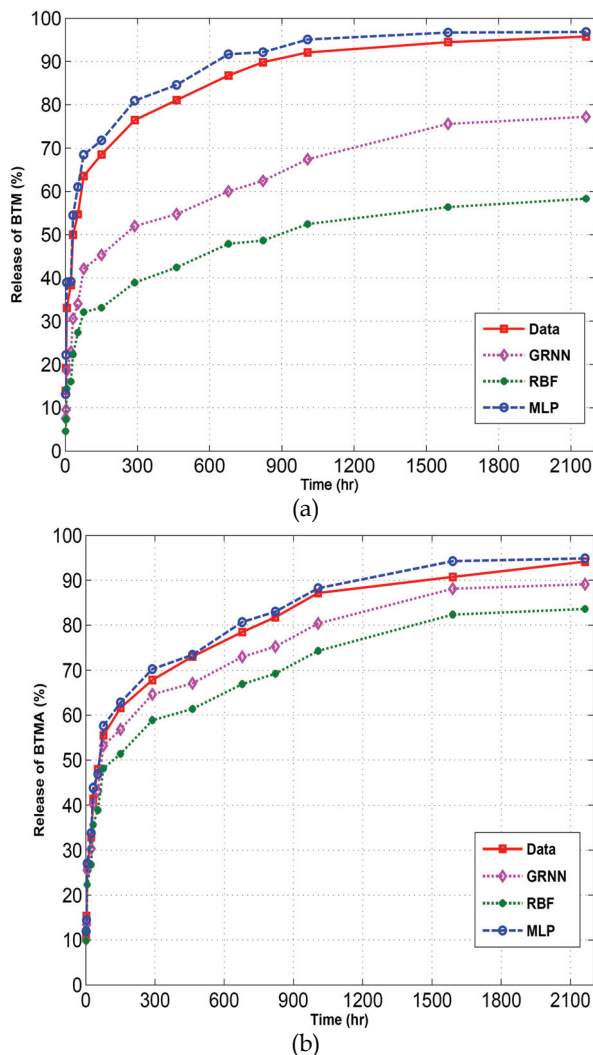


Fig. 18. Estimated release profiles and experimental release profiles of (a) BTM (b) BTMA for one sample.

4.2 Optimization of drug delivery system formulation

Cardiovascular disease (CVD) has recently become the leading cause of death in developing countries (Mackay et al., 2004). Several drugs have been used to control these diseases. Furosemide 5-(aminosulfonyl)-4-chloro-2-[(furanylmethyl) amino] benzoic acid is a potent diuretic and antihypertensive drug which belongs to class an IV of Biopharmaceutical Classification System (BCS) (Lindenberg et al., 2004). This drug has poor and erratic absorption after orally administration, and inter- subject variation in pharmacokinetic parameters (Hua et al., 2003; Derakhshandeh et al., 2007). In the last few years, new drug

delivery systems such as microparticles, liposomes, nanocapsules, micellar systems and conjugates have become increasingly important, because these systems can overcome these pharmacokinetic problems, and maintain the pharmacological effect for an appropriate extended time. Controlled release therapeutic systems present some advantages over traditional pharmaceutical preparations due to the fact, less active drug is necessary for similar results and consequently less secondary effects are present (Dadashzadeh et al., 2008; Derakhshandeh et al., 2010a, 2010b, 2010c, 2010d). Since these systems are often polymeric and submicron in size, they can in general be used to provide targeted delivery (cellular/tissue) of drugs, to improve bioavailability, to sustain drug effect in target tissue, to solubilize drug for intravascular delivery and to improve the stability of therapeutic agents from enzymatic degradation (Derakhshandeh et al., 2010b).

Sodium alginate as a biocompatible and biodegradable compound have lower toxicity compared to other polymers when taken orally and as a carrier could be suitable for encapsulation of drug. In this section, sodium alginate microparticle for oral delivery of furosemide was designed whether the encapsulation into microparticles might improve the oral absorption of this potent loop diuretic. To prepare an optimum formulation, MLP and GRNN are employed. The drug loaded formulation parameters are the input vectors of each network and are listed in Table 3.

The microparticles drug loading (Y1), size of microspheres (Y2) and the amount of drug release in 2 h (Y3) constitute the output vector of GRNN and MLP. In this way, ANN was trained to investigate the functional dependence of input variables on the output response.

Factors	Low level	High level
X ₁ concentration of sodium alginate (%)	0.8	4
X ₂ concentration of CaCl ₂ (%)	1	6
X ₃ volume t of internal phase (ml)	15	30
X ₄ volume of external phase (ml)	20	75

Table 3. Factorial design parameters and experimental conditions.

Microspheres were prepared by ionotropic gelation technique (Chan et al., 2002). Sodium alginate was dissolved in distilled water with agitation to have different concentrations of 1.5 to 4% (w/v). The drug was added to aqueous solutions of sodium alginate and the solution was dropped using a hypodermic syringe into a second solution, containing CaCl₂ with different concentration to cure for 15 min. After the microspheres formed, were separated, washed with distilled water, and dried in oven for 48 h.

Based on preliminary study of the effect of parameters on the drug loading, size and preparation method yield of microparticle, 20 formulations were prepared.

In vitro drug release study was carried out in USP XXII basket type dissolution test apparatus using Phosphate buffer solution (pH 7.4), simulated gastric fluid (SGF, pH 1.2) and simulated intestinal fluid (SIF, pH 7.4) as dissolution medium. Volume of dissolution medium was 500 ml and bath temperature was maintained at (37±10°C) throughout study. Basket speed was adjusted to 50 rpm. An interval of 1 hr, 5 ml of sample was withdrawn with replacement of fresh medium and analyzed for furosemide content by UV-Visible spectrophotometer at 273 nm. All the experimental units were analyzed in triplicate (n=3). At the end of this procedure, 20 formulations were prepared and are listed in Table 4. Indeed, the data of this table constitute the training and validation data sets which are used

to obtain the optimum formulation. The variables X_1 - X_4 are the four input variables of the ANN while the variables Y_1 - Y_3 represent the output variables of the network. In other words, the network has four inputs and three outputs. In this research, MLP and GRNN are employed to approximate the function between input and output spaces and consequently to obtain the optimum values of the input variables.

Run	X_1 (%)	X_2 (%)	X_3 (ml)	X_4 (ml)	Y_1 (%)	Y_2 (μ m)	Y_3 (%)
F1	1.20	1.00	15.00	20.00	10.00 $\pm$ 0.5	500.15 $\pm$ 51.20	41.26 $\pm$ 4.66
F2	1.50	2.00	25.00	50.00	52.00 $\pm$ 0.64	550.11 $\pm$ 50.10	53.24 $\pm$ 2.56
F3	2.00	1.50	25.00	50.00	61.22 $\pm$ 0.75	600.14 $\pm$ 49.25	68.29 $\pm$ 1.25
F4	2.00	2.00	25.00	50.00	63.10 $\pm$ 0.91	621.14 $\pm$ 50.05	68.15 $\pm$ 3.85
F5	2.00	2.50	25.00	50.00	64.12 $\pm$ 0.95	650.45 $\pm$ 50.17	69.23 $\pm$ 6.59
F6	2.50	1.00	25.00	50.00	56.12 $\pm$ 0.85	670.27 $\pm$ 52.41	60.48 $\pm$ 7.65
F7	2.50	1.50	25.00	50.00	60.81 $\pm$ 0.94	700.09 $\pm$ 33.15	42.14 $\pm$ 3.34
F8	2.50	2.00	25.00	50.00	65.45 $\pm$ 0.87	705.34 $\pm$ 28.46	63.26 $\pm$ 3.78
F9	2.50	2.50	25.00	50.00	74.25 $\pm$ 0.92	750.24 $\pm$ 42.15	62.49 $\pm$ 5.61
F10	2.50	3.00	25.00	50.00	79.58 $\pm$ 0.75	800.07 $\pm$ 35.67	55.58 $\pm$ 5.59
F11	2.50	2.50	15.00	50.00	60.23 $\pm$ 0.91	850.27 $\pm$ 34.25	59.45 $\pm$ 4.25
F12	2.50	2.50	25.00	75.00	62.21 $\pm$ 0.94	854.11 $\pm$ 43.65	49.26 $\pm$ 7.21
F13	2.50	6.00	25.00	25.00	58.10 $\pm$ 0.89	873.27 $\pm$ 41.27	54.33 $\pm$ 6.98
F14	3.00	1.50	25.00	50.00	49.21 $\pm$ 0.92	850.78 $\pm$ 40.25	46.85 $\pm$ 6.31
F15	3.00	2.00	25.00	50.00	54.21 $\pm$ 0.94	907.58 $\pm$ 39.57	52.18 $\pm$ 5.36
F16	3.00	2.50	25.00	50.00	60.52 $\pm$ 0.84	935.67 $\pm$ 43.57	45.54 $\pm$ 2.52
F17	3.00	3.00	25.00	50.00	64.61 $\pm$ 0.95	1010.25 $\pm$ 74.21	40.27 $\pm$ 3.56
F18	3.50	2.00	25.00	50.00	50.09 $\pm$ 0.87	1205.08 $\pm$ 69.74	35.64 $\pm$ 2.25
F19	4.00	3.00	30.00	50.00	44.9 $\pm$ 0.94	1350.08 $\pm$ 97.25	28.67 $\pm$ 4.50
F20	4.00	3.00	20.00	50.00	70.27 $\pm$ 0.90	1200.89 $\pm$ 49.67	36.33 $\pm$ 7.21

Table 4. Experimental design and percentage of drug loading, size and microparticle yield responses ($n = 3$)

To find the optimum formulation, at first, GRNN and MLP are trained using the data sample listed in Table 4. In so doing, we divide the data samples into two groups, the training and validation data sets. The training group contains 17 randomly selected samples from Table 4 and the validation group includes the other three remaining samples. There are significant variations in the scales of the values of the input variables. These different scales of the inputs led to ill-conditioning of the problem and hence the ANN could not be trained efficiently. To avoid this problem, all the data listed in Table 4 are normalized to the range of $[-1, 1]$ before training of the networks. The selection of suitable network architecture is another important factor, since it affects the network convergence as well as the accuracy of estimations (Simon and Frenandes, 2004). There exist no analytical methods to determine the optimum number of neurons required for a specific problem (Hosseini et al., 2007). Several rules of thumb to select the number of hidden neurons in an ANN have been proposed by various researchers (Amiri et al., 2009; Rafeinia et al., 2010). It should be mentioned that the number of hidden neurons generally depends on many factors, especially the distribution of training data and the number of data samples. However, in this work, MLP network was trained with a different number (from three to seven) of hidden neurons. The error value is high when the number of neurons

is low; in addition, increasing the number of hidden neurons decreases the final error value and makes the network move towards the global minimum. However, when we have more than four neurons in the hidden layer, over-fitting occurs. In this case the number of network parameters that should be adjusted through the learning algorithm is more than the required ones. Therefore, four neurons are required in the hidden layer which leads to the fast convergence and stable minimum error. Resilient back-propagation (RP) was utilized as a learning algorithm. In this optimization technique, the sign of the derivative is only used to determine the direction of the weight updating and the magnitude of the derivative has no effect on the weight updating (Riedmiller and Braun, 1993). Furthermore, like the other gradient based approaches, this learning algorithm will finally settle on the local minimum. This drawback certainly introduces inaccuracies into the results.

To train GRNN, after several simulations and based on trial and error method, $\sigma=0.5$ was selected for GRNN. This results in better generalization for new input vectors occurring between input vectors used in the design. When the MLP and GRNN are trained, we used the lower and upper limits of the input variables given in Table 3 to partition the input space into smaller regions. In this way, the interval between upper and lower limits of each input variable is divided into several segments. In this research, for the first input variable X_1 , the interval $[0.8, 4]$ is divided into 32 segments with each step length equal to 0.1. For the second, third and fourth input variables the step lengths are 0.1, 2.5 and 5 respectively. Next, the values of these segments constitute the input vector of trained MLP or trained GRNN to produce the corresponding outputs. Since we want to optimize the output variables such that the drug loading (Y_1) to be maximized and at the same time the size of microspheres (Y_2) and the amount drug release in 2 h (Y_3) to be minimized, we define the following cost function:

$$J = Y_2 + Y_3 - Y_1 \quad (12)$$

Based on this definition, minimizing the cost function J corresponds to the minimization of Y_2 and Y_3 and maximization of Y_1 . Next, we use an exhaustive search method to search in the partitioned input space. In this way, we will find the optimum value of the input variables (X_1 - X_4) such that the output variables (Y_1 - Y_3) which produced by the trained MLP or GRNN, minimize the cost function (12). Since the performance of the trained GRNN to minimize (12) is superior to the performance of the trained MLP, the results of this network are only mentioned in Table 5. Therefore, in this application, GRNN can effectively approximate the function between input and output vectors to find the optimum formulation and is more reliable than MLP.

X4: Ext. Phase (ml)	X3: Int. Phase (ml)	X2: CaCl ₂ (%)	X1: Alg. (%)	RUN
50	20	2.5	3.1	F*

Table 5. The optimum formulation of alginate microsphere obtained by the GRNN.

The optimum formulation showed a narrow size distribution with an average diameter of $700 \pm 50 \mu\text{m}$, and drug loading of more than 75%. The drug release profile showed a sustained released pattern about 36% in 2 hour. In vitro drug release rate for microspheres was found to be sustained over 24 hours, obeying Higushi order kinetic with good entrapment efficiency. The results of this section showed that utilizing ANN to obtain optimum formulation needs fewer experiments which may present new opportunities for the development of easy, reproducible and cost effective method in drug delivery applications.

5. Conclusion

In this chapter, we provided a brief description of four types of ANNs including SFNN, GRNN, RBFN and MLP. Next, we investigated the diverse and innovative applications of these neural networks such as associative neural networks for recognition of analog and digital patterns, estimating the release profile of the betamethasone (BTM) and betamethasone acetate (BTMA) and optimizing the forusemide microcarrier formulation. Considering the first application, a hybrid model consists of SFNN in parallel with GRNN was proposed. SFNN is a simple recurrent neural network which has difficulty learning and storing analog and digital patterns as associative memories (Amiri et al., 2008). GRNN can find solutions for any given problem, but lack a recurrent structure to filter noise. Therefore, the hybrid model of SFNN and GRNN was proposed and a new one-shot learning algorithm for training the hybrid model put forwarded in the chapter. It was shown that this new hybrid model is able to perform essential properties found in associative memories such as generalization, completion and recognition of corrupted patterns. Moreover, a number of case studies were performed for the purpose of performance comparison between the hybrid model and others from different classes such as NDRAM, ART2 and Storkey. It was discussed that in comparison to classic associative memory models, the hybrid model has at least three significant advantages. First, the learning and recalling processes in the hybrid model are very short and efficient which make the hybrid model act much faster compared to the other networks. This is very helpful when either the dimensionality or the number of patterns to be stored is large, which results in significant reduction of computational time and cost. Second, more importantly the hybrid model does not have any spurious attractor. The third distinct feature is that the hybrid model not only realizes association of binary patterns but can also realize association of analog patterns without any preprocessing (Amiri et al., 2007, 2008, 2010). Consequently, we believe that this hybrid model constitutes a serious candidate for associative recall of analog and digital patterns which should be explored further in future studies.

Regarding the biomedical application, the MLP, GRNN and RBFN are employed to model the release data and to predict the release profile of the BTM and the BTMA where in situ forming systems consist of poly (Lactide-co-glycolide), N-methyl-1-2-pyrrolidone and ethyl heptanoate as a polymer, solvent and additive, respectively. Several simulations were presented to compare the potential of each neural network. NLPCA feature extraction technique was utilized to extract three features from each release graph, constituting the outputs of the neural network. By utilizing these three features, we converted and reduced the released graphs into three more effective values. Training the networks was carried out using L.O.O. cross-validation approach. This approach allows the training algorithm to use the entire data set for training and at the same time to test the performance of the trained network on new data which has not already seen by the network. It was demonstrated that the MLP as a data modeling tool, is more reliable and efficient tool than RBFN and GRNN, in order to estimate the release profile of BTM and BTMA drugs. Furthermore, we investigated the application of the GRNN and MLP for optimization of drug delivery system formulation. It would appear that performance of the trained GRNN to minimize the cost function is superior to the performance of the trained MLP. In this way, GRNN is promising to determine the optimum drug formulation. In sum, the application of the ANNs in biomedical research will definitely increase in the near future. However, the point which is noteworthy is the fact that there is no single modeling approach to address all requirements.

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7. References

- Amiri, M., Menhaj, M. B., & Fallah, A., (2007). Stability Analysis of self-feedback neural network structures, *Amirkabir Journal of Science and Technology*, 14(66A), 103-109.
- Amiri, M., Davandeh, H., Sadeghian, A., & Seyyedsalehi, S.A., (2007). Auto-associative neural network based on new hybrid model of SFNN and GRNN. *In Proceedings of the International Joint Conference on Neural Networks (IJCNN)*, 2664-2670, USA.
- Amiri, M., Davandeh, H., Sadeghian, A., & Chartier, S., (2010) Feedback associative memory based on a new hybrid model of generalized regression and self-feedback neural networks, *Neural Network*, 23(7), 892-904.
- Amiri, M., Menhaj, M. B., & Yazdanpanah, M.J. (2008a). A neural-network-based controller for a single-link flexible manipulator: comparison of FFNN and DRNN controllers. *IEEE World Congress on Computational Intelligence (WCCI)*, 1687-92, Hong Kong.
- Amiri M., Rafienia, M., Sadeghian A., (2009) Estimation of betamethasone release profiles from an in situ forming system based on the biodegradable polymer using artificial neural networks, *IFMBE Proceedings*, (2016-2019), Germany.
- Amiri, M., Sadeghian, A., & Chartier, S., (2010) One-shot training algorithm for self feedback neural networks, *Annual Conference of the North American Fuzzy Information Processing Society - NAFIPS*, no. 5548272, Toronto, Canada.
- Amiri, M., Saeb, S., Yazdanpanah, M.J., & Seyyedsalehi, S.A., (2008b). Analysis of the dynamical behavior of a feedback auto-associative memory. *Neurocomputing*, 71, 486-494.
- Amiri, M., Bahrami, F. & Janahmadi, M. Functional modeling of astrocytes in epilepsy: a feedback system perspective, *Neural Computing & Applications*, DOI 10.1007/s00521-010-0479-0
- Amrouche, A., & Rouvaen, J.M., (2006). Efficient system for speech recognition using general regression neural network. *International Journal of Computer Systems Science and Engineering*, 1(2), 183-189.
- Atiya, A., & Abu-Mostafa, Y.S., (1993). An analog feedback associative memory. *IEEE Transactions on Neural Networks*, 2, 117-126.
- Bégin, J., & Proulx, R., (1996). Categorization in unsupervised neural networks: The Eidos model. *IEEE Transactions on Neural Networks*, 7, 147-154.
- Bing Yu , & Xingshi He (2006) Training Radial Basis Function Networks with Differential Evolution, *Proceedings of World Academi of Science, Engineering and Technology*, 11 , ISSN 1307-6884.
- Brier, E., Zurada, J.M., & Aronoff G.R., (1995) Neural network predicted peak and trough of gentamicin concentrations. *Pharm. Res.* 12, 406-412.
- Carpenter, G.A., & Grossberg, S., (2003) Adaptive resonance theory, In M.A. Arbib (Ed.), *The Handbook of Brain Theory and Neural Networks*, Second Edition, Cambridge, MA: MIT Press, 87-90.
- Carpenter, G.A., & Grossberg, S., (1987) ART 2: Stable self-organization of pattern recognition codes for analog input patterns. *Applied Optics*, 26, 4919-4930.

- Chan L.W., Lee H.Y., Heng P.W.S. (2002). Production of alginate microspheres by internal gelation using an emulsification method. *International Journal of Pharmaceutics*, 242 , 259-260.
- Chartier, S., & Proulx, R., (2005) NDRAM: Nonlinear dynamic recurrent associative memory for learning bipolar and nonbipolar correlated patterns, *IEEE Transactions on Neural Networks*, 16(6), 1393-1400.
- Chartier, S., Boukadoum, M., & Amiri, M., (2009) BAM learning of nonlinearly separable tasks by using an asymmetrical output function and reinforcement learning, *IEEE Transactions on Neural Networks*, 20(8), 1281-1292.
- Chen, T., & Chen. H., (1995) Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical system. *IEEE Trans. Neural Networks* 6, 911-917.
- Cheng, K.S., Lin, J.S., & Mao, C.W., (1996) The application of competitive Hopfield neural network to medical image segmentation. *IEEE Transactions on Medical Imaging*, 15(4), 560-567.
- Dadashzadeh S., Derakhshandeh K., Shirazi F.H., (2008) 9-Nitrocamptothecin polymeric nanoparticles: Cytotoxicity and pharmacokinetic studies of lactone and total forms of drug in rats. *Anticancer Drugs*, 19(8), 805-811.
- Davande, H., Amiri, M., Sadeghian, A., & Chartier, S., (2008). Associative memory based on a new hybrid model of SFNN and GRNN: performance comparison with NDRAM, ART2 and MLP," *IEEE World Congress on Computational Intelligence (WCCI)*, (1699-1704), Hong Kong.
- Derakhshandeh K., Erfan M., Dadashzadeh S., (2007) Encapsulation of 9-nitrocamptothecin, a novel anticancer drug, in biodegradable nanoparticles: factorial design, characterization and release kinetics. *Eur J Pharm biopharm* 66, 34-41.
- Derakhshandeh K., Dadashzadeh S., Soheili M., (2010a) Preparation and in vitro characterization of 9-nitrocamptothecin-loaded long circulating nanoparticles for delivery in cancer patients. *International Journal of Nanomedicine*. 463-471.
- Derakhshandeh K., Fashi M., Seifoleslami S., (2010b) Thermosensitive Pluronic® hydrogel: prolonged injectable formulation for drug abuse. *Drug Design, Development and Therapy*, 4, 255-262.
- Derakhshandeh K, Soleimani M., (2010c) Formulation and in vitro evaluation of nifedipine controlled release tablet: influence of combination of hydrophilic and hydrophobic matrix forms. *Asian Journal of Pharmaceutics*. *in press*.
- Derakhshandeh K, Nikmohammadi M, Hosseinalizadeh A. (2010d) Factorial effect of process parameters on pharmaceutical characteristics of biodegradable PLGA microparticles. *International Journal of Drug delivery*. *in press*
- Diederich, S., & Oppen, M., (1987). Learning of correlated pattern in spin-glass networks by local learning rules. *Physical Review Letter*, 58, 949-952.
- Freeman, J.A., (1993). *Simulating neural networks with mathematica*, 1st Ed. Addison-Wesley Longman Publishing Co., Inc.
- Hopfield, J.J., (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceeding of Natural Academic Science*, 79, 2554-2558.
- Hosseini, S.M., Amiri, M., Najarian, S., & Dargahi, J. (2007) Application of artificial neural networks for the estimation of tumour characteristics in biological tissues, *International Journal of Medical Robotics and Computer Assisted Surgery* 3 (3), 235-244.
- Hua A., Steven A. J., Melgardt M. D., Yuri M. L. (2003) Nano-encapsulation of furosemide microcrystals for controlled drug release. *Journal of Controlled Release*, 86, 59-62.

- Huang, Z., & Williamson, M.A., (1994) Geological pattern recognition and modeling with a general regression neural network. *Canadian Journal of Exploration Geophysics*, 30(1), 60-68.
- Kamat HV, & Rao DH. (1995) Direct adaptive control of non-linear systems using a dynamic neural network. *IEEE Transactions on Automatic Control*. 39, 987-991.
- Kanter, I., & Sompolinsky, H., (1987) Associative recall of memory without errors. *Physical Review A*, 35(1), 380-392.
- Kramer, M.A. (1991) Nonlinear principal component analysis using auto-associative neural networks. *AIChE J.* 37,233-243.
- Kubat, M. (1998) Decision Trees Can Initialize Radial-Basis Function Networks. *IEEE Transactions Neural Networks*, 813-821.
- Ku, C.C., & Lee, K.Y., (1995) Diagonal recurrent neural network for dynamic system control. *IEEE Transactions on Neural Networks*, 6, 144-155.
- Lim, C. P., Quek, S.S., & Peh, K.K. (2003) Prediction of drug release profiles using an intelligent learning system: an experimental study in transdermal iontophoresis. *J. Pharma. Biomed. Anal.* 31, 159-168.
- Lindenberg, M., Kopp, S., Dressman B. (2004) Classification of orally administered drugs on the World Health Organization Model list of Essential Medicines according to the biopharmaceutics classification system. *Eur J Pharm Biopharm*, 58, 265-2782.
- Mackay, J., Mensah GA, Mendis S, Greenlund K. The atlas of heart disease and stroke. *Brighton: Myriad Edition Limited*; 2004.
- Moody, J., & Darken, C. (1991) Fast Learning Networks of Locally-Tuned Processing Units. *Neural Computation*. 579-588.
- Rafienia, M., Amiri M., Janmaleki M., & Sadeghian A., (2010) Application of Artificial Neural Networks in controlled drug delivery Systems, *Applied Artificial Intelligence*. 24, 807-820.
- Reznik, A., Galinskaya, A., Dekhtyarenko, O., & Nowicki, D. (2005). Preprocessing of matrix qcm sensors data for the classification by means of neural network. *Sensors and Actuators B.*, 158-163.
- Riedmiller, M., & Braun H., (1993) A direct adaptive method for faster backpropagation learning: the RPROP algorithm. *IEEE Int. Conf. on Neural Networks* 1, 586-591.
- Robert, J., & Hewlett L.C.J. (2001) Radial Basis Function Networks 2: *New Advances in Design*.
- Simon, L., & Frenandes, M. (2004) Neural network-based prediction and optimization of estradiol release from ethylene-vinyl acetate membranes. *Comp. Chem. Eng.* 28, 2407-2419.
- Specht, D.E. (1991) A general regression neural network. *IEEE Transactions on Neural Networks*, 2(6), 568-576.
- Storkey, A.J. & Valabregue, R., (1999) The basins of attraction of a new Hopfield learning rule. *Neural Networks*, 12, 869-876.
- Takayama, K., Fujikawa, M., Obata, Y., & Morishita, M., (2003) Neural network based optimization of drug formulations. *Adv. Drug Deliv. Rev.* 55, 1217-1231.
- Valafar, H., & Valafar, F., (1999) Prediction of a patient's response to a specific drug treatment using artificial neural networks. *IEEE* 3694-3697.
- Wachowiak, M.P., Elmaghraby, A.S., Smolikova, R., & Zurada, J.M., (2001). Generalized regression neural networks for biomedical image interpolation. *Int. Joint Conf. on Neural Networks (IJCNN)*, 2133-2138, USA.

Medical Image Segmentation Using Artificial Neural Networks

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1. Introduction

Segmentation of tissues and structures from medical images is the first step in many image analysis applications developed for medical diagnosis. Development of treatment plans and evaluation of disease progression are other applications. These applications stem from the fact that diseases affect specific tissues or structures, lead to loss, atrophy (volume loss), and abnormalities. Consequently, an accurate, reliable, and automatic segmentation of these tissues and structures can improve diagnosis and treatment of diseases. Manual segmentation, although prone to rater drift and bias, is usually accurate but is impractical for large datasets because it is tedious and time consuming. Automatic segmentation methods can be useful for clinical applications if they have: 1) ability to segment like an expert; 2) excellent performance for diverse datasets; and 3) reasonable processing speed.

Artificial Neural Networks (ANNs) have been developed for a wide range of applications such as function approximation, feature extraction, optimization, and classification. In particular, they have been developed for image enhancement, segmentation, registration, feature extraction, and object recognition. Among these, image segmentation is more important as it is a critical step for high-level processing such as object recognition. Multi-Layer Perceptron (MLP), Radial Basis Function (RBF), Hopfield, Cellular, and Pulse-Coupled neural networks have been used for image segmentation. These networks can be categorized into feed-forward (associative) and feedback (auto-associative) networks. MLP, Self-Organized Map (SOM), and RBF neural networks belong to the feed-forward networks while Hopfield, Cellular, and Pulse-Coupled neural networks belong to the feedback networks.

This chapter is organized as follows. Section 2 explains methods that benefit from feedback networks such as Hopfield, Cellular, and Pulse-Coupled neural networks for image segmentation. In Section 3, we review the methods that use feedforward networks such as MLP and RBF neural networks. Then, we present our recent method. In this method, deep brain structures are segmented using Geometric Moment Invariants (GMIs) and MLP neural networks.

2. Medical image segmentation using feedback Neural Networks

Feedback or recurrent networks include feedback loops. These networks are very powerful and can get extremely complicated. Hopfield, Cellular, and Pulse-Coupled neural networks described in this section belong to this category of networks.

2.1 Hopfield Neural Network

Hopfield Neural Network (HNN), introduced by Hopfield, (1982), consists of a pool of neurons with connections between each unit. Every neuron connects to other neurons with a weight in the network. All neurons are both input and output neurons. The output is a binary value (0,1 or -1,1). In the original form of HNN, the state of each neuron is determined by (Hopfield, 1982),

$$V_i \rightarrow 1 \quad \text{if} \quad \sum_{i \neq j} T_{ij} V_j > U_i \quad (1)$$

$$V_i \rightarrow 0 \quad \text{if} \quad \sum_{i \neq j} T_{ij} V_j < U_i \quad (2)$$

where V_i is the output of neuron i and U_i is a threshold value. T_{ij} is the strength of the connection between neurons i and j . HNN has a scalar value associated with the given state of the network, called energy, E , of the network. It is defined as

$$E = -\frac{1}{2} \sum \sum_{i \neq j} T_{ij} V_i V_j \quad (3)$$

The constraint $T_{ij} = T_{ji}$ (i.e., symmetric weights) guarantees that the energy function decreases monotonically (Hopfield, 1982; Amartur et al., 1992). In this condition, initial values of neurons lead the network to converge at a local minimum. In some version of HNN, a bias for each neuron is considered and the network energy is determined by

$$E = -\frac{1}{2} \sum \sum_{i \neq j} T_{ij} V_i V_j - \sum_k I_k V_k \quad (4)$$

where I_k is a bias term.

The energy function minimization is obtained by solving a set of motion equations (Amartur et al., 1992)

$$\frac{\partial U_i}{\partial t} = -\frac{\partial E}{\partial V_i} \quad (5)$$

where U_i is the input of the i -th neuron. If the input-output function decreases monotonically and the motion equations are satisfied, the energy function decreases as time passes and converges at a local minimum (Amartur et al., 1992). The energy function is non-convex and has more than one local minimum.

A primary application of HNN is an associative memory (Wu and Pados, 2000; Giménez-Martínez, 2000). The network is able to save a given pattern by choosing a proper set of

weights. From image segmentation point of view, HNN consists of $N \times M$ neurons with the pixels as the rows and the classes as the columns. HNN is used as a map between the image pixels and their labels (Amartur et al., 1992) (i.e., assigning N pixels to M classes). The assignment of the pixels minimizes the energy function. The criterion function or weights can be based on a metric measure between a pixel and a class. In (Amartur et al., 1992), generalized distance measure between the k -th pixel and the centroid of class l is determined by

$$R_{kl} = \|X_k - \bar{X}_l\|_{A_l^{-1}} \quad (6)$$

where X_k is the P -dimensional feature vector of the k -th pixel, A_l is a positive definite weighting matrix, and $\bar{X}_l$ is the P -dimensional centroid for class l . The objective function to minimize is given by

$$E = -\frac{1}{2} \sum_{k=1}^N \sum_{l=1}^M R_{kl} V_{lk}^2 \quad (7)$$

where R_{kl} is a symmetric distance measure. By substitution (7) into (5), the neuron dynamics can be described by

$$\frac{dU_{kl}}{dt} = -R_{kl} V_{kl} \quad (8)$$

In (Amartur et al., 1992), the winner-takes-all neuron is used to construct the input-output function for the k -th pixel.

$$V_{kn}(t+1) = \begin{cases} 1 & \text{if } U_{kn} = \text{MAX}(U_{kl}; \forall l) \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

The algorithm for clustering an image is as follows:

1. Initialize the inputs of the neurons randomly.
2. Using (9), calculate the output of the neurons. Assign pixels to classes.
3. Compute the centroid and the covariance matrix for class l as described in (Amartur et al., 1992).
4. Solve equation (8) using Euler's approximation.
5. If there is a significant change in the input of each neuron, repeat from step 2), else, terminate.

If the number of clusters is large, the network may over-classify the image into many disjoint regions. In this condition, a similarity or dissimilarity measure should be used to merge them. Fig. 1 shows an example of clustering a magnetic resonance image (MRI) of the head (Amartur et al., 1992). This network is easily implemented in hardware and its high speed makes it appropriate for real-time applications.

The maximum operator generates a crisp classification. To generate a fuzzy classification, (Lin et al., 1996a; Lin et al. 1996b; Lin et al. 1996c) integrated a fuzzy c-means strategy with the HNN and introduced a Fuzzy Hopfield Neural Network (FHNN). They modified the energy function to include the fuzzy parameters. In this network, all of the neurons on the same row compete and eventually the neuron with the maximum membership to a given

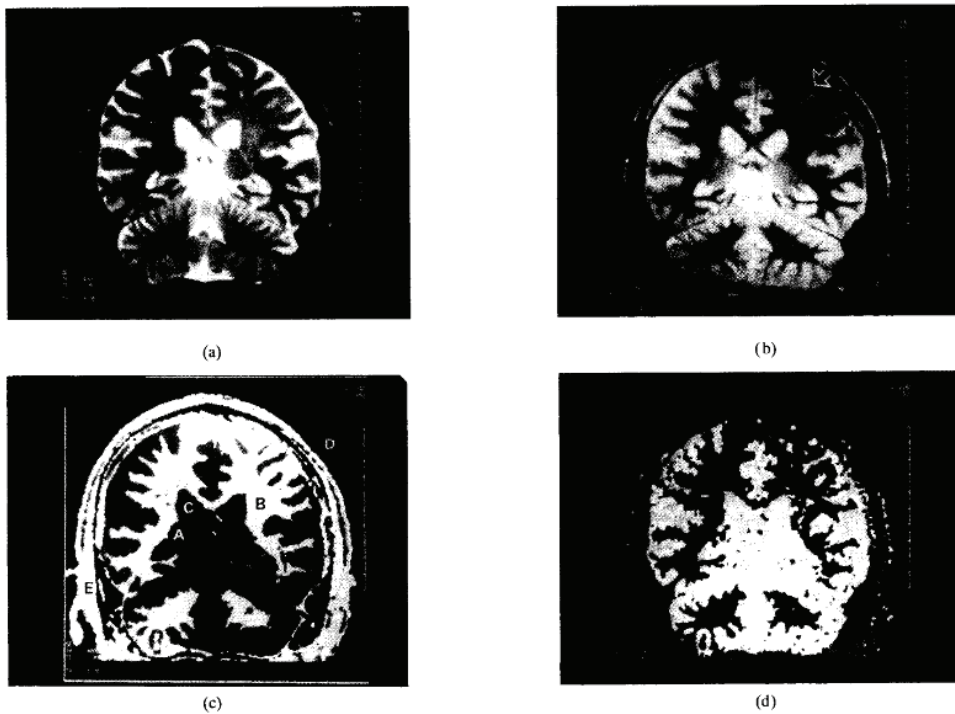


Fig. 1. Magnetic resonance images of the head of a normal subject: (a) T2-weighted at level 1; (b) proton density-weighted at level 1; (c) classified into 5 clusters; and (d) classified into 7 clusters (From Amartur et al., 1992 with permission).

class wins. In this approach, there is no need to determine sensitivities associated with the weighting factors (Lin et al., 1996a; Lin et al. 1996b) that are difficult to determine. Another technique was proposed by Cheng et al., (1996) as Competitive Hopfield Neural Network (CHNN) where the competitive learning rule, Winner-Takes-All (WTA), was utilized to update the weights.

(Chang and Chung, 2001) describe another difficulty of the HNN: the original HNN cannot incorporate contextual information into the segmentation process. They also mention that the robustness of the HNN to noise is low. To overcome this limitation, they introduce Contextual-Constraint-based Hopfield Neural Cube (CCBHNC). The network benefits from a three-dimensional architecture. They incorporated the pixel's feature and its surrounding contextual information into the third dimension.

2.2 Cellular Neural Network

Cellular Neural Network (CNN) was proposed by Chua et al. (1988). It is made of a pool of cells that communicate with each other through their neighbours. Consider an $M \times N$ cellular neural network, arranged in M rows and N columns. Each cell connects to the nearest $r \times r$ neighbourhood and the output of each cell is connected with the inputs of the cells in the same neighbourhood. A typical example is presented in Fig. 2. Each cell is

modelled as an electric circuit. Complete details about the model are found in (Chua et al., 1988a; Chua et al., 1988b; Chua et al., 1988c). The cell equations are as follows (Chua, 1988c). State equation:

$$C \frac{dv_{xij}(t)}{dt} = -\frac{1}{R_x} v_{xij}(t) + \sum_{C(k,l) \in N(i,j)} A(i,j;k,l) v_{ykl}(t) + \sum_{C(k,l) \in N(i,j)} B(i,j;k,l) v_{ukl}(t) + I \quad (10)$$

Output equation:

$$v_{yij}(t) = \frac{1}{2} \left(|v_{xij}(t) + 1| - |v_{xij}(t) - 1| \right) \quad (11)$$

Input equation:

$$v_{uij}(t) = E_{ij} \quad (12)$$

Constraint conditions:

$$|v_{xij}(0)| \leq 1, \quad |v_{uij}| \leq 1 \quad (13)$$

where the subscripts u , x , and y denote input, state, and output, respectively. For example, v_{xij} denotes the state voltage of the cell (i,j) , R_x is a linear resistor, I is an independent current source and E_{ij} is a time-invariant independent voltage source. Matrixes A and B are considered as feedback template and control template, respectively.

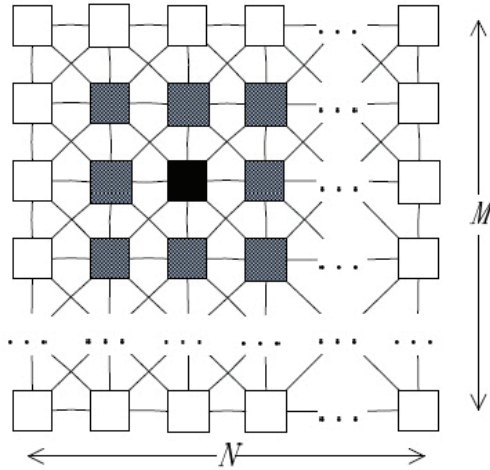


Fig. 2. Cellular Neural Network with a 3×3 neighbourhood (from Zhang et al., 2007 with permission).

Satisfying the constraints (13) guarantees that the solution always converges to a constant steady state. This means that any transient driven by a given input image vanishes and the network generates a constant dc output. In (Chua et al., 1988b), two medical applications of

CNN are described: noise removal and feature extraction (edge detection). CNN has advantages like: 1) parallel calculation on hardware; 2) high calculation speed (almost real-time); and 3) being independent of the image size (Chua et al., 1988b).

CNN is used to develop image segmentation methods (Zhang et al., 2007; Luan et al., 2007; Grassi and Vecchio, 2006). Zhang et al. (2007; 2008) segment urinary images using CNN where the image is segmented after it is enhanced. Selvathi et al. (2010) segment brain tumors in magnetic resonance images using CNN and gradient information.

Yang et al. (1996) and Colodro and Torralba (1996) proposed a fuzzy version of Cellular Neural Networks (FCNN) which is a generalization of CNN to incorporate the uncertainties in human cognitive processes. Chua et al. (1988b) presented applications of FCNN in mathematical morphology and fuzzy inference rule by else action (FIRE) operators like FIRE edge detecting operator. In medical imaging, FCNN was utilized for white blood cell segmentation (Shitong and Min, 2006) and CT liver image segmentation (Wang et al., 2007). In (Shitong and Min, 2006), FCNN was combined with Threshold Segmentation followed by Mathematical Morphology (TSM) to detect white blood cells. Moreover, Wang et al. (2007) proposed an advanced version of FCNN to segment the CT liver images.

Finally, Gacsádi and Szolgay (2010) proposed a new variational computing based medical segmentation method using CNN. Over the last decade, variational method and partial equations have been widely used for medical image segmentation in algorithms such as level-set (Angelini et al., 2005). This method needs a multi-layered CNN to solve an optimization problem to produce segmentation results.

2.3 Pulse-coupled Neural Network

Pulse-Coupled Neural Network (PCNN) is inspired from a cat's visual cortex. A model of the cat's visual cortex was proposed by Eckhorn et al. (1989). Over the past decade, PCNNs have been used for a variety of image processing applications such as segmentation, feature and face extraction, motion detection, and noise reduction (Lindblad and Kinser, 2005). PCNN consists of a pool of neurons where each neuron corresponds to a pixel in the image receiving its local information and the stimuli from its neighbouring neurons. These stimuli are accumulated in the neuron. The neuron fires when the average of stimuli exceeds a dynamic threshold. The dynamic threshold leads to the pulsing nature of PCNN. This network differs from other neural networks in its structure and operation. In particular, the original PCNN does not contain a training phase.

A PCNN neuron shown in Fig. 3 consists of three compartments (Lindblad and Kinser, 2005): the dendritic tree; the modulation; and the pulse generator. There are two types of inputs: Feeding and Linking. Feeding receives an input stimulus from an external source and other neuron outputs while linking only receives them from other neuron outputs. The related formulas are:

$$F_{ij}[n] = e^{-\alpha_F \delta_n} F_{ij}[n-1] + S_{ij} + V_F \sum_{kl} M_{ijkl} Y_{kl}[n-1] \quad (14)$$

$$L_{ij}[n] = e^{-\alpha_L \delta_n} L_{ij}[n-1] + V_L \sum_{kl} W_{ijkl} Y_{kl}[n-1] \quad (15)$$

where L_{ij} and F_{ij} are the linking and feeding of the (i, j) neuron corresponding to the (i, j) -th pixel in a 2D image. Both compartments contain a memory of the previous state with

decays, $e^{-\alpha_L \delta_n}$ and $e^{-\alpha_F \delta_n}$. V_L and V_F are normalizing constants. W and M matrices include the synaptic weights where each neuron communicates with its neighbouring neurons.

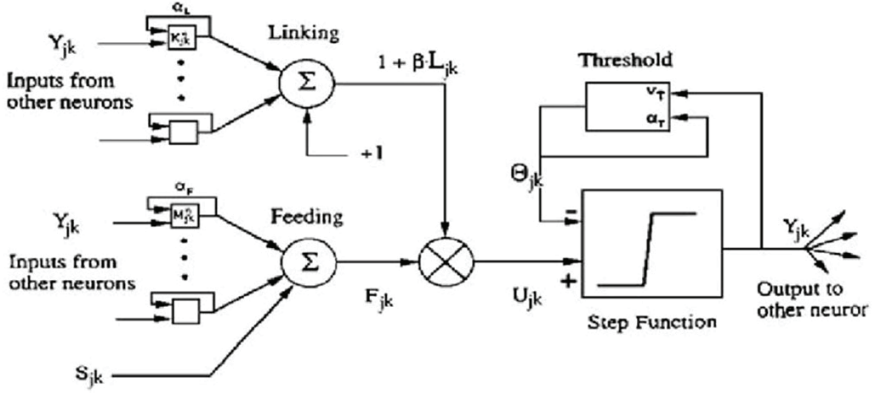


Fig. 3. Schematic of a neuron in PCNN (from Lindblad and Kinser, 2005 with permission).

In the modulation section, the outputs of feeding and linking combine to form a second order equation. The output of modulation section is determined by

$$U_{ij}[n] = F_{ij}[n] \{1 + \beta L_{ij}[n]\} \quad (16)$$

where β is the linking strength.

In the pulse generator, the output of the previous section is compared to a dynamic threshold, θ , to produce the output of the neuron, Y . The dynamic threshold of each iteration is determined by

$$\Theta_{ij}[n] = e^{-\alpha_\Theta \delta_n} \Theta_{ij}[n-1] + V_\Theta Y_{ij}[n] \quad (17)$$

where V_Θ is a large constant.

At the beginning, a pixel or a set of neighbouring pixels in the image is considered as a neuron in the PCNN. The values of outputs are all set to zero. A neuron that has an external stimulus will fire in the first iteration. In this situation, the threshold values are high and it will take several iterations in which neurons are allowed to fire again. This iterative process will continue until the user stops it. There is no automatic procedure to stop the PCNN; this is a disadvantage of PCNN. Another disadvantage is a large number of parameters that should be determined appropriately.

The goal of PCNNs for a segmentation process is managing the network such that when a neuron fires, it shows a specific feature like an edge. In addition, similarities between the input pixels may lead to fire the corresponding neurons simultaneously thus showing texture or similar objects.

Keller and McKinnon's (1999) showed that PCNN works well when the segments are almost uniform in intensity. Two kinds of images, magnetic resonance images of the brain and the abdomen and a set of nuclear scintigraphic images of lung, are used for testing the effectiveness of PCNN. They also showed that PCNN did a good job for contrast enhancement. In addition, they found that PCNN was unable to perform a segmentation

task when the intensity in each segment was not distributed uniformly. They mentioned selection of optimal parameter values as another difficulty of PCNN. For example, it is possible that a set of parameters segment objects properly in a particular image but they do not perform well on a similar image. To this end, Hassanien and Ali (2004) used a Fuzzy Histogram Hyperbolization (FHH) algorithm as a contrast improvement filter for the digital mammogram image segmentation. Then a PCNN performed the segmentation process.

In (Ma et al., 2004), Shannon entropy was used as a criterion to stop the network automatically. In their work, the binary image was used and the information of the original image was not utilized. To overcome to this limitation, Xiao et al. (2009) integrated PCNN with fuzzy mutual information to stop the segmentation process with optimal results. They segmented the CT and ultrasound images with strong robustness against noise. Their method consisted of the following steps:

1. Set initial parameters and iteration number to 1.
2. Use normalized gray image as an external input to PCNN.
3. Iterate, $n=n+1$.
4. Segment the image by PCNN.
5. Compute the value of Fuzzy Mutual Information (FMI). If $FMI < FMI_{max}$, go to Step 3. Otherwise, stop the process and report the final result.

In (Shi et al., 2009), a new method was proposed based on improved PCNN and Tsallis entropy. Their method can automatically segment medical images without choosing the PCNN parameters. In the improved PCNN, each neuron is activated only once.

Finally, Fu et al. (2010) proposed an automatic method that integrated the statistical exception maximization (EM) model and PCNN for MRI segmentation. In their method, the EM model does two major works: evaluation of the PCNN image segmentation; and adaptive adjustment of the PCNN parameters for optimal segmentation. Their method has two stages: pre-processing and segmentation. In the former, they retrieve intracranial tissues by a watershed algorithm and in the latter, PCNN segments the image and its parameters are adjusted by the EM model.

3. Medical image segmentation using feed-forward Neural Networks

3.1 Self organized map Neural Network

Self Organized Map (SOM), introduced by Kohonen (2001), is a feed-forward neural network that has been utilized for medical image segmentation. Sometimes SOM is called Kohonen map. It projects a high dimensional space into a low dimensional space. SOM consists of a 2D array of nodes. Each node is associated with a weight vector, w_i , of the same dimension and a position in the map space. An unsupervised, iterative learning process finds the weight vector for each node. In each iteration, the most similar node, called the best matching unit (BMU), is found by a similarity measure. A training process considers the neighboring nodes of the BMU and updates their corresponding weight vector as follows (Kohonen, 2001).

$$w_i(t+1) = w_i(t) + \alpha(t)h_{ci}(t)(x(t) - w_i(t)) \quad (18)$$

where $\alpha(t)$ is a monotonically decreasing learning coefficient, $h_{ci}(t)$ is the neighborhood function, typically considered as a Gaussian function. The process repeats for a large number of iterations. SOM preserves the most important topological and metric relationships of the primary data items.

Although a primary application of SOM is dimension reduction, it has been utilized for medical image segmentation. Rickard et al. (2004) segmented the mammogram images using multi-scale analysis and SOM. Chang and Teng (2007) proposed a two-stage SOM to identify dominant color components and segment a medical image. SOM has been utilized for color reduction of color medical images, although due to the noisy nature of medical images, this method may not be effective. To overcome this problem, Chang and Teng (2007) proposed a two-stage SOM shown in Fig. 4. The first SOM identifies a number of prehensive colors while the second SOM determines the final dominant colors. A sample result of this method is shown in Fig. 5.

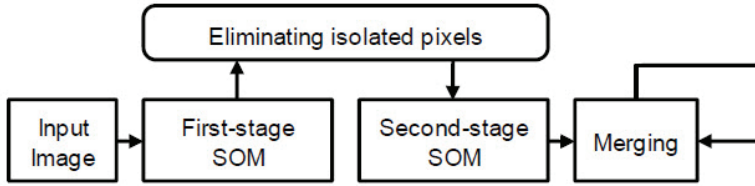


Fig. 4. A flowchart of the two-stage SOM method (from Chang and Teng, 2007 with permission).

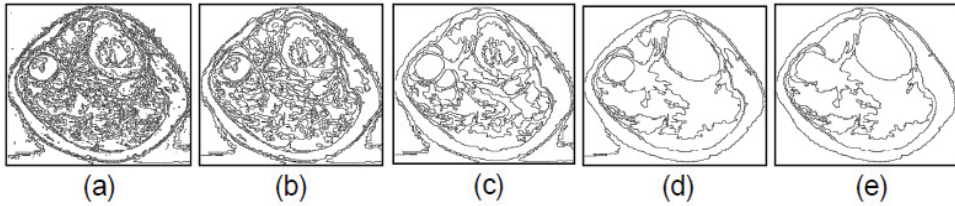


Fig. 5. Identified segments using merging steps: (a) before merging (b) first merging (c) second merging (d) third merging (e) fourth merging (from Chang and Teng, 2007 with permission).

3.2 Radial basis function network

Radial Basis Function (RBF) neural network is another type of feed-forward neural network that uses radial basis functions as activation functions. RBF network consists of three layers: input layer; hidden layer with radial basis function as activation function; and output layer. The output of an RBF network is a linear combination of weighted radial basis functions.

$$\varphi(\mathbf{x}) = \sum_{i=1}^N a_i \exp(-\beta_i \|\mathbf{x} - \mathbf{c}_i\|) \quad (19)$$

where N is the number of neurons in the hidden layer, $\mathbf{c}_i$ is the center of neuron i , and a_i is weight of the linear output neuron. In the training phase, a_i and β_i of each neuron are estimated by a least squares objective function or a regularized objective function. The number of the neuron is set manually.

RBF has been utilized for function approximation and classification (Bishop, 2006). For medical image segmentation, RBF extracts features and classifies image pixels (Sing et al.,

2005). In (Kovacevic, 1997), an RBF network was utilized to segment CT images of the head. In (Halkiotis, 2007), an RBF network was used to detect clustered microclasifications in digital mammograms automatically.

3.3 Multi-layer perceptron Neural Network

Multi-layer Perceptron Artificial Neural Network (MLP-ANN) is used in a wide range of applications including function approximation, feature extraction, optimization, compression, and classification (Bishop, 2006). It is appropriate for medical imaging applications because of the following reasons: 1) For an MLP-ANN with a hidden layer, the output is a nonlinear function of a linear combination of the neuron outputs in the hidden layer which themselves are nonlinear functions of linear combinations of the input variables. This allows the network to construct parametric nonlinear functions of the input features. The parameters of the network are estimated by optimizing an objective function (usually the mean square error between the desired and actual outputs) by a back-propagation algorithm. 2) Due to the correlation between the input features, the intrinsic dimensionality of the input feature space is usually smaller than the number of input features. When the number of neurons in the hidden layer is smaller than that of the input layer, it means that the hidden layer represents a lower dimensional feature space, which can have the same number of dimensions as that of the intrinsic feature space. 3) Depending on the desired output, some of the features may be irrelevant. An MLP-ANN automatically ignores them by assigning zero (or very small weights) to them. In a sense, it does the job of feature selection.

Requirements and limitations of MLP-ANN are the following. First, a proper configuration of the network with best generalization should be established by choosing proper numbers of hidden layers and neurons. Secondly, the objective function of the network is not always convex. Therefore, the gradient descend algorithm may find a local minimum instead of the global minimum. To overcome the first problem and achieve best generalization, available data is divided into training, testing, and validation datasets, where the error is controlled for the validation data.

Some image segmentation algorithms have benefited from MLP-ANN (Jabarouti Moghaddam et al., 2009; Jabarouti Moghaddam and Soltanian-Zadeh, 2009; Magnotta et al., 1999; Powell et al., 2008). The method developed by Magnotta et al. (1999) uses Image Intensity Values (IIVs) of the neighboring voxels as the image features. They classify the voxels into two classes – being inside and outside of a desired structure. They design an MLP-ANN for each structure. Since their image features do not contain a shape representation, they need to use a large training set. The IIVs solve the segmentation problem for the high-resolution MRI they utilized. Recently, Powell et al. (2008) have further developed their previous algorithm (Magnotta et al., 1999) using 9 IIVs along with the largest gradient, a probabilistic map, and the IIVs along each of the three orthogonal axes. They use high-resolution images the same as (Magnotta et al., 1999) for the segmentation of the brain structures (putamen, caudate, thalamus, and cerebellar regions of interest). They use 15 image volumes, collected using two protocols, T1-weighted and T2-weighted, in the training set. Their approach does not include any shape representations. For dimension reduction of the IIV features, the IIVs along the largest gradient are utilized. In low resolution MRI with intensity inhomogeneity inside the structures, these features may not work well.

Jabarouti Moghaddam et al. (2009) used Geometric Moment Invariants (GMIs) (Lo and Don, 1989) to improve the differentiation between the brain regions compared to the IIVs. In (Mangin et al., 2004), shape descriptors were defined based on the moments of the voxel coordinates for morphometry of the cortical sulci. GMIs characterize the underlying anatomical structures calculated for each voxel. They are eleven moment invariants calculated from the first-order, second-order, and third-order 3D-regular moments. They are invariant to translation and rotation. They designed an MLP-ANN to segment putamen. The image features were the GMIs for white matter (WM) and gray matter (GM) tissues along with the neighboring IIVs. The output of the network was a scalar that took two values (-1, 1) for indicating inside of the structure and its outside. The GMIs were calculated only for one scale. However, considering only one scale may not be sufficient to distinguish all parts of the putamen. In addition, the method depends on the tissue segmentation results as it calculates the GMIs of the WM and the GM maps.

Jabarouti Moghaddam and Soltanian-Zadeh (2009) developed a two-stage method for the segmentation of the brain structures as shown in Fig. 6. In the first stage, GMIs were used along voxel intensity values as an input feature and signed distance function of a desired structure as an output of the network. In this stage, there is a network for each scale of GMIs. Fig. 7 shows the MLP-ANN used for the first stage.

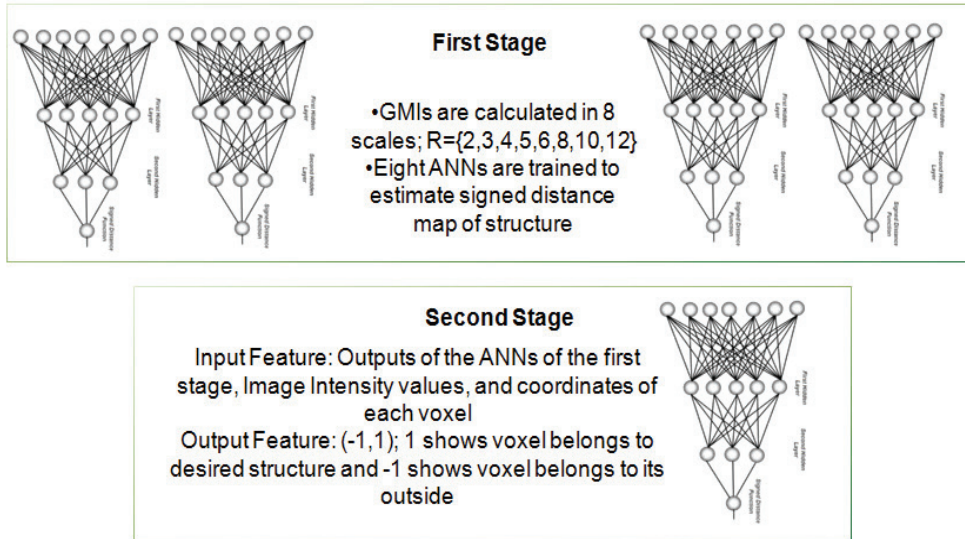


Fig. 6. A flowchart of the two-stage method.

In the first stage, each input feature vector includes the neighboring voxel intensity values, GMIs, and coordinates of voxels. More formally, at voxel x , the input feature vector $a(x)$ is composed of three vectors $a(x)=[a_1(x) a_2(x) a_3(x)]$ where $a_1(x)$ is for voxel intensity values, $a_2(x)$ is for coordinate of voxel x , and $a_3(x)$ is for GMIs.

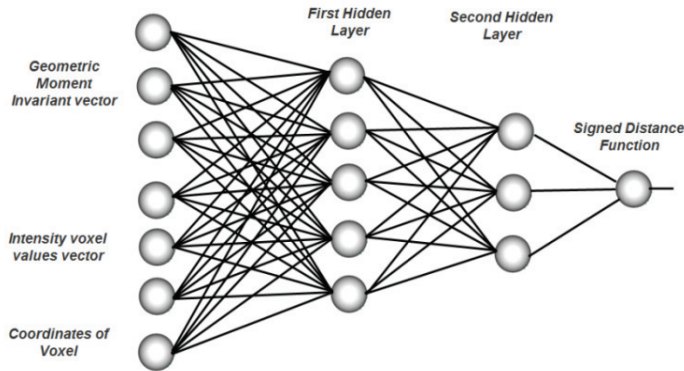


Fig. 7. MLP-ANN architecture of the first stage. Input feature contains three components – GMIs, voxel intensity values, and coordinates of voxels. There are two hidden layers. There are 33 neurons in the input layer, 17 neurons in the first hidden layer, 7 neurons in the second hidden layer, and a single neuron in the output. The output is designed to approximate the signed distance function of the structure.

The vector $a_1(x)$ contains the neighboring voxel intensity values of voxel x . We define $a_1(x)$ by considering the neighbouring voxels along the orthogonal axes (± 3 voxels in each direction) of the 3D-space. Therefore, the dimension of $a_1(x)$ is 13 (Fig. 8).

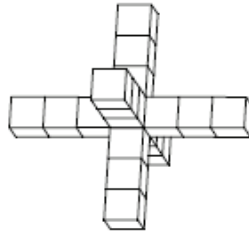


Fig. 8. Neighboring voxel intensity values that are considered for each voxel of the training data sets as a part of the input feature vector.

The vector $a_2(x)$ consists of the coordinates of voxels in the Cartesian coordinate system. This can help the network to find relations between the voxels of a specific structure. The brain structures may have discontinuous image intensities inside the structure. Therefore, considering coordinates of the voxels in the input feature vector can help with the training of the network.

The vector $a_3(x)$ contains the GMIs at voxel x . These features include rich geometric properties, represent the underlying anatomical structures, and can distinguish voxels that are inside the structure from those that are outside of the structure. Lo and Don (1989) explicitly define 11 moment invariants consisting of the first, second, and third order 3D-regular moments. The input feature vector is made of 33 features: 19 features for voxel intensity values; 11 features for GMIs; and 3 features for voxel coordinates in the training data sets.

For shape representation, a signed distance function of the structure of interest is used as the output. The signed distance function is a subset of implicit functions in shape representation. It is defined to be positive in the exterior, negative in the interior, and zero on the boundary. Near the boundaries, the value of this function decreases and far from the boundaries, the value of this function increases.

The network is trained using the backpropagation algorithm and voxels inside the structure of interest. Mean Square Error (MSE) is calculated and compared until it reaches an asymptotic value. The range of MSE in the first stage is from 0.02 to 0.03. Each training network takes 1 day so we need almost 9 days for training the networks used in both stages. Although this is a relatively long time but it is only performed once.

Outputs of the first stage can distinguish different parts of the structure. The 8 outputs of the first stage must be combined to generate the final result. As such, another MLP-ANN was used to combine the 8 outputs of the networks in the first stage. In the second stage, we use the network as a classifier, not as a function approximator. Our task in this stage is to classify voxels into two classes – being inside or outside of the structure. The results of the networks in the first stage are smoothed using a mean filter with a kernel size of 3. This eliminates the outlier voxels. The input features consist of the 8 smoothed versions of the outputs of the first stage, 19 image intensity values of the neighboring voxels, and 3 voxel coordinates in the Cartesian coordinate system. They generate a 30-dimensional feature vector. The network setting used in this stage is the same as that used in the first stage. The lower the error in this stage, the better the results. The resulting network activation function is thresholded at 0 to define the structure of interest. The voxels with outputs larger than 0 define the structure of interest and the voxels with outputs smaller than 0 define the outside region.

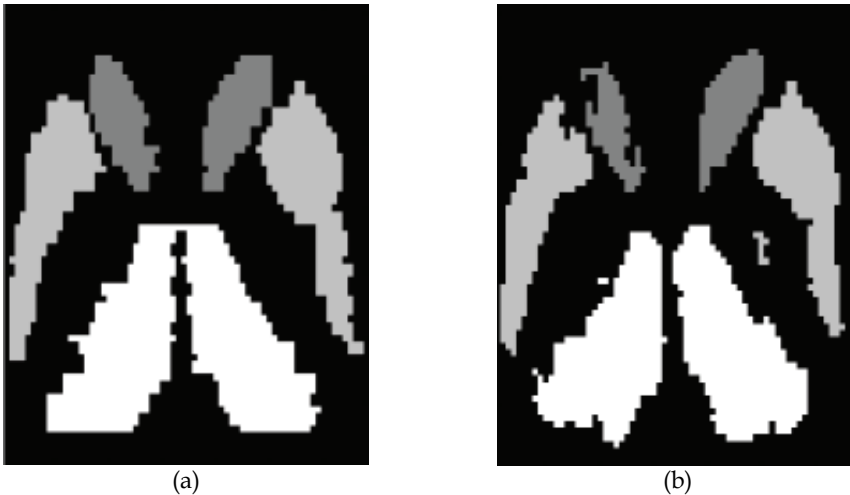


Fig. 9. (a) Manual segmentation by experts. (b) Outputs of the second stage showing the final segmentation result for a sample slice.

The results of the second stage and a 3D visualization of the putamen, caudate, and thalamus are shown in Figures 9 and 10, respectively.

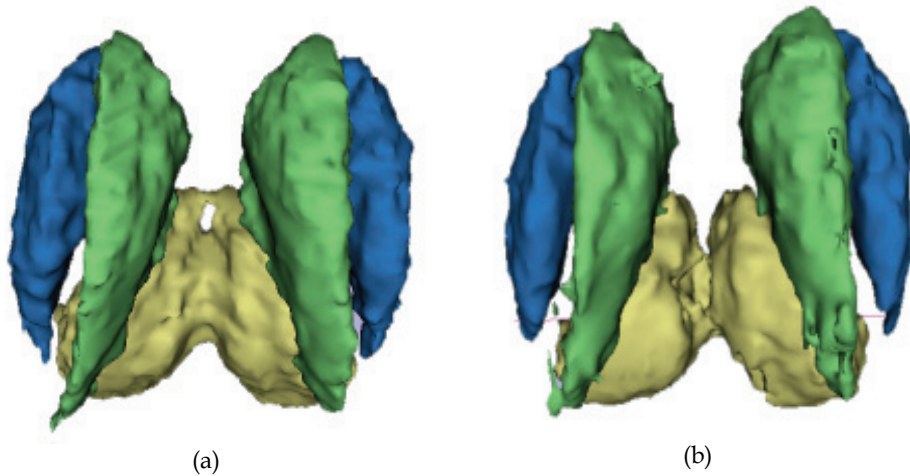


Fig. 10. 3D visualization of the outputs of three networks used for three structures. Caudate is shown in green, putamen is shown in blue, and thalamus is shown in yellow.

4. Conclusion

In this chapter, we presented several methods for medical image segmentation based on artificial neural networks. The networks were categorized into feedback and feed-forward networks. Among the feedback networks, Hopfield, Cellular, and Pulse-Coupled networks have been used. Among the feed-forward neural networks, Self Organizing Map (Kohonen), Multi-Layer Perceptron, and Radial Basis Function neural networks have been utilized.

In Section 2, the Hopfield network was described as a map between the image pixels and their labels. The network is easily implemented in hardware and used in real-time applications. On the other hand, the network is fully connected and it is not applicable for 3D or huge 2D medical image segmentation due to its large number of parameters. In addition, for object recognition, it is not essential that each pixel connects to all other pixels. Indeed, it may be ineffective to incorporate information of all other pixels to label a pixel. Next, Cellular Neural Network was discussed. In this network, the information of the neighboring pixels is included in pixel segmentation process which is an advantage relative to the Hopfield network. Due to this advantage, it is applicable for 3D or huge 2D medical image segmentation. At the end of the section, the Pulse Coupled Neural Network was explained. This network differs from other neural networks in its structure and operation. In particular, the original PCNN does not contain a training phase. The lack of a stopping method is a challenge for PCNN.

In Section 3, the SOM neural network was explained as a feed-forward network. It is usually utilized for dimension reduction. As discussed, it has been used for color reduction of

medical images. The next network discussed was the RBF neural network. This type of neural network is usually useful for function approximation and classification. Finally, applications of the Multi-Layer Perceptron neural network were discussed. This type of neural network is effective for segmenting deep brain structures in 3D. Its high speed makes it appropriate for real-time applications. Due to its speed and accuracy, it has a bright future for medical image segmentation applications.

5. References

- Amartur, S. C.; Piraino, D. & Takefuji, Y. (1992). Optimization neural networks for the segmentation of magnetic resonance images. *IEEE Transactions on Medical Imaging*, 11(2): 215-220.
- Angelini, E. D.; Jin, Y. & Laine, A. (2005). State-of-the-art of levelset methods in segmentation and registration of medical imaging modalities. *The Handbook of Medical Image Analysis* 3.
- Bishop, C. M. (2006). Pattern Recognition and Machine Learning, *Springer*.
- Chang, C. Y. & Chung, P. C. (2001). Medical image segmentation using a contextual-constraint-based Hopfield neural cube. *Image and Vision Computing*, 19(9-10): 669-678.
- Chang, P. L. and W. G. Teng (2007). Exploiting the self-organizing map for medical image segmentation, *20th IEEE international Symposium on Computer-Based Medical Systems (CBMS'07)*, Maribor, 281:288.
- Cheng, K.-S.; Lin, J.-S. & Mao, C.-W. (1996). Application of competitive Hopfield neural network to medical image segmentation. *IEEE Transactions on Medical Imaging* 15(4): 560-561.
- Chua, L. O. & Yang, L. (1988a). Cellular neural networks, *IEEE international Symposium, Circuits and Systems*.
- Chua, L. O. & Yang, L. (1988b). Cellular neural networks: Applications. *IEEE transactions on circuits and systems*, 35(10): 1273-1290.
- Chua, L. O. & Yang, L. (1988c). Cellular neural networks: Theory. *IEEE transactions on circuits and systems*, 35(10): 1257-1272.
- Colodro, F. & Torralba, A. (1996). Cellular neuro-fuzzy networks (CNFNs), a new class of cellular networks, *Proceeding of 5th IEEE on Fuzzy Systems*, New Orleans, LA, USA.
- Eckhorn, R.; Reitboeck, H. J., Arndt, M. & Dicke, P. (1989). Feature linking via stimulus-evoked oscillations: Experimental results from cat visual cortex and functional implications from a network model, *International Joint Conference on Neural Network*, Washington, DC, USA.
- Fu, J. C.; Chen, C. C., Wong S.T.C. & Li, I.C. (2010). Image segmentation by EM-based adaptive pulse coupled neural networks in brain magnetic resonance imaging. *Computerized Medical Imaging and Graphics* 34(4): 308-320.
- Gacsádi, A. & Szolgay, P. (2010). Variational computing based segmentation methods for medical imaging by using CNN, *12th international Workshop on Cellular Neural Networks and Their Application (CNNA)*, 1:6.

- Giménez-Martínez, V. (2000). A modified Hopfield auto-associative memory with improved capacity. *IEEE Transactions on Neural Networks*, 11(4): 867-878.
- Grassi, G. and P. Vecchio (2006). Cellular neural networks for object-oriented segmentation. *Research in Microelectronics and Electronics 2006*, Otranto.
- Halkiotis, S.; Botsis, T. & Rangoussi, M. (2007). "Automatic detection of clustered microcalcifications in digital mammograms using mathematical morphology and neural networks." *Signal Processing*, 87(7): 1559-1568.
- Hassanien, A. E. & Ali J.M. (2004). Digital mammogram segmentation algorithm using pulse coupled neural networks, *Third International Conference on Image and Graphics (ICIG'04)*, Hong Kong.
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences of the United States of America*, 79(8): 2554-2558.
- Jabarouti Moghaddam, M.; Rahmani, R. & Soltanian-Zadeh, H. (2009). Automatic segmentation of putamen using geometric moment invariants. *The 15th Iranian Conference on Biomedical Engineering, Mashad, Iran*.
- Jabarouti Moghaddam, M. & Soltanian-Zadeh, H. (2009). Automatic segmentation of brain structures using geometric moment invariants and artificial neural networks. *Information processing in medical imaging : 21st International Conference on Information Processing in Medical Imaging (IPMI'09)*, Williamsburg, VA, 21: 326-337.
- Keller, P. E. & McKinnon, A.D. (1999). Segmentation of medical imagery with pulse-coupled neural networks, *International Joint Conference on Neural Network*, Washington, DC, USA, IEEE.
- Kohonen, T. (2001). *Self-Organizing Maps*, Springer.
- Kovacevic, D. & Loncaric, S. (1997). Radial basis function-based image segmentation using a receptive field, *Processing of 10th IEEE Symposium on Computer-Based Medical Systems*, Maribor, Slovenia.
- Lin, J.-S.; Cheng, K.-S. & Mao, C.-W. (1996a). Modified Hopfield neural network with fuzzy c-means technique for multispectral MR image segmentation, *Processings of International IEEE Conf., Lausanne, Switz*, 1,327:330.
- Lin, J.-S.; Cheng, K.-S. & Mao, C.-W. (1996b). a fuzzy hopfield neural network for medical image segmentation, *IEEE Transactions on Nuclear Science* 43(4 PART 2): 2389-2398.
- Lin, J.-S.; Cheng, K.-S. & Mao, C.-W. (1996c). "Multispectral magnetic resonance images segmentation using fuzzy Hopfield neural network." *International Journal of Bio-Medical Computing* 42(3): 205-214.
- Lindblad, T. & Kinser, J.M. (2005). *Image Processing Using Pulse-Coupled Neural Networks*, Springer.
- Lo, C.-H. & Don, H.-S. (1989). 3-D moment forms: Their construction and application to object identification and positioning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(10): 1053-1064.

- Luan, Q.X.; Li, X., Kang, J.Y. & Zhi, Z. (2007). Analysis of dental plaque by using cellular neural network-based image segmentation. *Chinese journal of stomatology* 42(12): 720-722.
- Ma, Y. D.; Liu, Q. & Zhi-Bai, Q. (2004). Automated image segmentation using improved PCNN model based on cross-entropy, *Proceeding of 2004 International Symposium on Intelligent Multimedia, Video and Speech Processing*, Hong Kong, China.
- Magnotta, V. A.; D. Heckel, Andreasen, N.C., Corson, P.W., Ehrhardt, J.C. & Yuh, W.T.C. (1999). Measurement of brain structures with artificial neural networks: Two- and three-dimensional applications. *Radiology* 211(3): 781-790.
- Mangin, J. F.; F. Poupon, Duchesnay, E., Riviere, D., Cachia, A., Collins, D.L., Evans, A.C., Regis, J. (2004). Brain morphometry using 3D moment invariants. *Medical Image Analysis*, 8(3): 187-196.
- Powell, S.; V. A. Magnotta, Johnson, H., Jammalamadaka, V.K., Pierson, R. & Andreasen, N.C. (2008). Registration and machine learning-based automated segmentation of subcortical and cerebellar brain structures. *NeuroImage* 39(1): 238-247.
- Rickard, H. E.; Tourassi, G. D. & Elmaghraby, A.S. (2004). Breast segmentation in screening mammograms using multiscale analysis and self-organizing maps, *Proceedings of the 26th Annual International Conference of the IEEE EMBS*, San Francisco, CA.
- Selvathi, D.; Selvaraj, H. & Selvi, S.T. (2010). Hybrid approach for brain tumor segmentation in magnetic resonance images using cellular neural networks and optimization techniques. *International Journal of Computational Intelligence and Applications* 9(1): 17-31.
- Shi, W.; Miao, Y., Zhanfang, C. & Hongbiao, Z. (2009). Research of automatic medical image segmentation algorithm based on Tsallis entropy and improved PCNN, *International Conference on Mechatronics and Automation*, Changchun.
- Shitong, W. & Min, W. (2006). A new detection algorithm (NDA) based on fuzzy cellular neural networks for white blood cell detection. *IEEE Transactions on Information Technology in Biomedicine*, 10(1): 5-10.
- Sing, J. K., D. K. Basu, Nasipuri, M. & Kundu, M. (2005). Self-adaptive RBF neural network-based segmentation of medical images of the brain, *Proceedings of International Conference on Intelligent Sensing and Information Processing*.
- Wang, S.; Fu, D., Xu, M. & Dwen, H.U. (2007). Advanced fuzzy cellular neural network: Application to CT liver images." *Artificial Intelligence in Medicine* 39(1): 65-77.
- Wu, Y. & Pados, D.A. (2000). A feedforward bidirectional associative memory. *IEEE Transactions on Neural Networks*, 11(4): 859-866.
- Xiao, Z.; Shi, J. & Chang, Q. (2009). Automatic image segmentation algorithm based on PCNN and fuzzy mutual information, *IEEE international Conference on Computer and Information Technology*, 241:245.
- Yang, T.; Yang, L.-B., Chai, W.W. & Chua, L.O. (1996). Fuzzy cellular neural networks: theory, 4th *IEEE International Workshop on Cellular Neural Network*, Seville, Spain.

- Zhang, Z.; Xia, S.R. & Duan, H.L. (2007). Cellular neural network based urinary image segmentation. *3th International Conference on Natural Computation (ICNC)*, Haikou, Hainan.
- Zhang, Z. C.; Xia, S.R. & Duan, H.L. (2008). Cellular neural network based urinary sediment image segmentation. *Zhejiang Daxue Xuebao (Gongxue Ban)/Journal of Zhejiang University (Engineering Science)* 42(12): 2139-2144.

Artificial Neural Networks and Predictive Medicine: a Revolutionary Paradigm Shift

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1. Introduction

Physicians every day try to accomplish their mission: to give the better care and assistance to their individual patients. They know perfectly that an individual subject is typically "unique". He or she has specific characteristics, qualities and features, different perceptions of risk value, different values scale, different familiar environments and social roles, all together interacting in a complex way. In other words no subject is equivalent to another, nor even mono-ovular twins.

But physician has to take decisions continuously, in front to every single individual, facing always with uncertainty. Will my diagnosis be confirmed later on? Will my treatment work in this patient? Will this patient follow my indications? These are questions that most physicians ask themselves every day.

Physicians know that an individual patient is not the average representative of the population. Rather he or she is a person with unique characteristics. What they not always know is that an intervention may be effective for a population but not necessarily for the individual patient. The recommendation of a guideline may not be right for a particular patient because it is not what he or she wants, and implementing the recommendation will not necessarily mean a favourable outcome.

The fundamental resource for them still remain their experience and their intelligence in using experience as a guide. In addition, increasing expectations of the highest quality of health care and the rapid growth of ever more detailed medical knowledge leave the physician without adequate time to devote to each patient – who all-too-often becomes a case or a file rather than being a person. So, for lack of time, most medical decisions must be based on rapid judgments of the "case" relying on the physician's unaided memory. Only in rare situations can a literature search or other extended investigations be undertaken to assure the doctor (and the patient) that the latest knowledge is being brought to bear on any particular malady. Continued training and recertification procedures encourage the physician to keep more of the relevant information constantly in mind, but fundamental limitations of human memory and recall coupled with the growth of available and accessible knowledge assures that most of what is known cannot be known by most individuals.

The spectacular development of information technology and molecular biology has increased exponentially the quantity of the information available for the busy physician. It is not unusual to have at hand, especially when faced with treatment planning for a chronic degenerative disorder, hundreds of different variables, consisting of clinical history data, objective findings, symptoms, multi-item scales of different meanings, laboratory examinations, etc. With the increased availability and use of functional genomics and digital imaging we now tentatively have at our disposition thousands of data per subject. This creates a paradox: in comparison with 10-20 years ago we are now able to collect more data per subject than subjects per study.

2.The emergence of complexity in medicine

Non linearity, complexity, fuzzy interaction are emerging features of chronic degenerative diseases which account for most morbidity and mortality in western world. Unfortunately even the most powerful and well established statistical methods were developed in the first half of the past century when the scenario was dominated by acute infective diseases and the available information was much more simple, or at maximum “complicated” rather than “complex”. comparison with today.

More features imply more information and potentially higher accuracy. Unfortunately more features we have, the more difficult information extraction is. In this high dimensional space, the hyper points corresponding to single individuals are sparse and the notion of proximity fails to retain its meaningfulness. For this reason clustering become extremely hard to be performed. In this situation we are dealing with flat, rectangular data set, a sort of telescope data set. This kind of data set are intractable from a traditional statistics point of view due to the fact that the excessive amount of degrees of freedom allows any kind of data interpolation most of the time meaningless.

A part from quantitative features, non linearity, complexity, fuzzy interaction are new emerging qualitative features of chronic degenerative diseases which account for most morbidity and mortality in western world. Unfortunately even the most powerful and well established statistical methods were developed in the first half of the past century when the scenario was dominated by acute infective diseases and the available information was much more simple, or at maximum “complicated” rather than “complex”. comparison with today.

There is now the reason to ask a fundamental question: is the mathematics used in medicine what it should be given the complexity of the chronic degenerative diseases?

There are a number of different reasons to apply complex systems mathematics on predictive medicine and some of them are listed in the table 1.

Processes are based on complex networks of interacting genes and proteins.
Health status is the consequence of dynamic processes that regulate these networks
Non linear critical thresholds link to pathology
The predictions have to be applied at individual patient level.
Huge amount of data per subject hamper statistical tests

Table 1. Motivations to apply complex systems mathematics on predictive medicine

Complexity is based on small elementary units working together in small populations of synchronous processes. In a complex system each component changes, over time, losing its identity outside of the system. Complexity needs a different kind of mathematics, able to handle chaotic behaviour, non linear dynamics, and fractal geometry (Kaplan & Glass, 1995; Goldberger, 1996).

Computational and mathematical medicine needs different statistical approaches, based on new mathematical and logic assumptions broadly belonging to complex theory setting allow to tame these intractable data sets. Seen in this perspective computer science is now playing the role which mathematics did from the seventeenth through the twentieth centuries: providing an orderly, formal framework and exploratory apparatus for knowledge progress.

Actually the coupling of computer science and these new theoretical bases coming from complex systems mathematics allows the creation of “intelligent” agents able to adapt themselves dynamically to problem of high complexity: the Artificial Neural Networks (ANNs).

3. Artificial Neural Networks

Artificial Neural Networks are adaptive models for the analysis of data which are inspired by the functioning processes of the human brain. They are systems which are able to modify their internal structure in relation to a function objective and are particularly suited for solving problems of the non linear type, being able to reconstruct the approximate rules that put a certain set of data - which describing the underlying problem with a particular target (specific diagnosis, outcome etc.).

They are particularly suited for solving problems of the non linear type, being able to reconstruct the approximate rules that put a certain set of data - which describes the problem being considered - with a set of data which provides the solution. For a detailed description of these models and tools we refer to recent reviews (Grossi & Buscema, 2007; Grossi & Buscema, 2006)

The base elements of the ANNs are the nodes, also called processing elements, and the connections. Each node has its own input, from which it receives communications from other nodes and/or from the environment and its own output, from which it communicates with other nodes or with the environment. Finally, each node has a function f through which it transforms its own global input into output (see figure 1).

Each connection is characterized by the strength with which pairs of nodes are excited or inhibited. Positive values indicate excitatory connections, the negative ones inhibitory connections.

The connections between the nodes can modify themselves over time. This dynamic starts a learning process in the entire ANNs. The way through which the nodes modify themselves is called “Law of Learning”. The total dynamic of an ANNs is tied to time. In fact, for the ANNs to modify its own connections, the environment has to necessarily act on the ANNs more times. Data are the environment which acts on the ANNs.

The learning process is, therefore, one of the key mechanisms that characterize the ANNs, which are considered adaptive processing systems. The learning process is one way to adapt the connections of an ANNs to the data structure that make up the environment and, therefore, a way to “understand” the environment and the relations that characterize it.

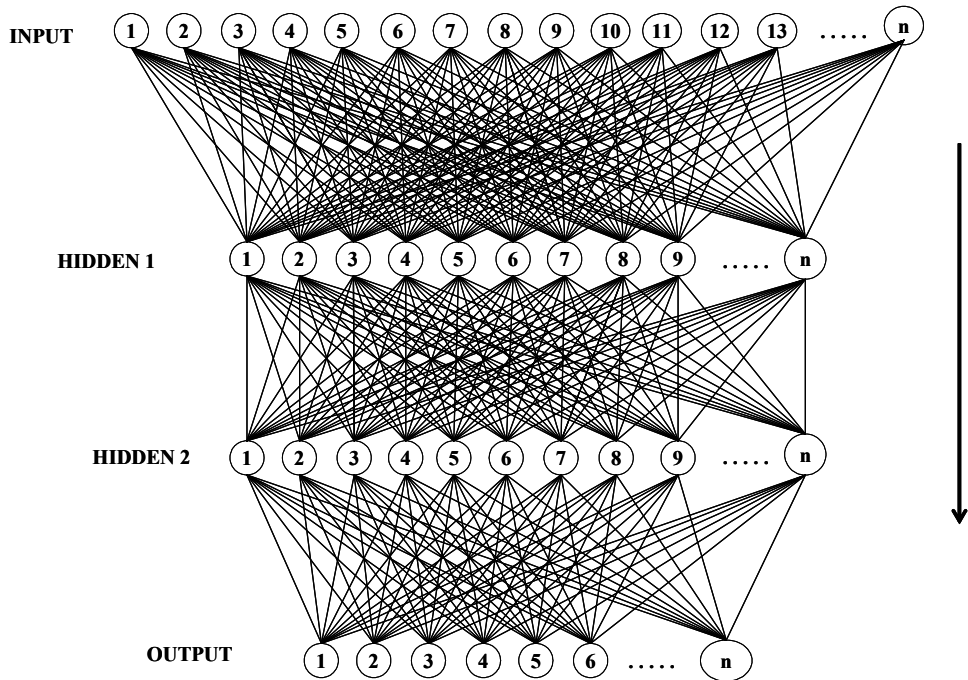


Fig. 1. Example of Supervised artificial neural network. The architecture is that of back propagation neural network with one layer of input nodes, corresponding to independent variables, two layers of hidden nodes allowing the integration of non linear functions, and one layer of output nodes corresponding to the target variables.

The raw indicate the direction of flow of data along the network.

4. Evolutionary algorithms

At variance with neural networks which are adaptive systems able to discover the optimal hidden rules explaining a certain data set, Evolutionary Algorithms (EA) are Artificial Adaptive Systems able to find optimal data when fixed rules or constraints have to be respected. They are in other words optimisation tools which become fundamental when the space of possible states in a dynamic system tends toward infinitum.

An EA for example can help to distribute the original sample in two or more sub-samples with the aim of obtaining the maximum performance possible from an ANN that is trained on the first sample and tested on the second. It is possible, in order to limit eventual optimistic polarizations in the evaluation of the performance, to invert the two samples and to consider the mean between the two approximations obtained as fitness of the algorithm and as an estimate of the model's quality.

Also the problem to select among the different variables available those most related to a particular outcome without the recur to linear correlation parameters can be approached with EA. When linear systems are used, an instrument exists, correlation index, which indicates the degree of relationship existing between the input and output variables of the

system, in this way suggesting which of the variables available to use to build a model of the problem. The problem of selecting a subset of variables on which to build the model for the process under examination stems from the fact that, when data are gathered to build the Data Base, the relationship between the collected variables and the function of the process being examined is not known. In this case the natural approach is to include all of the variables that may have a connection with the event being studied. The result of this approach is that often a series of variables which do not contain any information regarding the process being examined are present. These variables, inserted in the model, cause an increase of the noise and therefore a greater difficulty for the ANN to learn the data correctly.

The coupling of ANN and EA and brings to the concept of artificial organisms, able to optimize the classification performance and prediction ability.

Training & Testing and Input Selection which have been combine in a single system called TWIST (Buscema, 2007-2008) are example of such artificial organisms which are devoted to the problems above discussed respectively.

5. Basic philosophy of Artificial Neural Networks

The basic principle which is proposed in ANNs is very simple: all the biological signals from all the sources available are analyzed together -and not individually- both in time and space. The reason for such an approach is quite simple and self-explaining: the instant value of the system in any recording source depends, in fact, upon its previous and following values (how many, and in which amount for each previous state?), upon the previous and following values of all the other recording sources (how many, and in which amount for each previous state?).

In summary, the aim of the “analyzer” is not to analyze the language of each individual variable, but to evaluate the meta-language which considers the holistic contribution of all the recorded variables. We, in fact, believe that the equilibrium of each individual subject is defined by a specific background signal model, distributed in time and in the space. Such a model is a set of background invariant features able to specify the quality of the immune activity for example. The system that we propose to apply in this research context completely ignores the subject’s contingent characteristics. It utilizes a recurrent procedure which squeezes at progressive steps the significant signal and progressively eliminates the non-significant noise.

6. The paradigm shifts of Artificial Neural Networks in predictive medicine

The use of ANNs and in particular Neural Networks are already emerging as new trends in medical statistics. Although these methods are not yet in widespread use, they have already had a clinical impact in specific areas, notably cervical cytology, x-Ray mammography and early detection of acute myocardial infarction where large-scale prospective multicenter studies have been carried out. Extensive reviews on this subject has been published (Lisboa, 2002).

There are in the literature many examples of successful application of ANN in outcome research.

Our group has proved the usefulness of the added predictive value gained with the use of advanced artificial neural networks coupled in a number of medical fields, ranging from

heart diseases, gastroenterology and neurology with special regard to Alzheimer disease, stroke and Amyotrophic Lateral Sclerosis.

Our group has proved the usefulness of the added predictive value gained with the use of advanced artificial neural networks coupled with evolutionary algorithms in a number of medical fields, ranging from heart diseases, gastroenterology and neurology with special regard to Alzheimer disease, stroke and Amyotrophic Lateral Sclerosis (Street *et al.*, 2008; Grossi, 2006).

ANNs bring a number of revolutionary paradigm shifts which will have a strong impact in predictive medicine. They are listed in the table 2.

No limitation in the amount of data processed
No limitation in the different nature of data processed
No limitation in the degree of complexity of data processed
Bottom - up computation: models are data driven
Interactions among different factors are easily picked-up
Inference takes place at individual level
Internal validity of modelling ensured with validation protocols
Fuzzy logic allows to escape from the probability theory trap

Table 2. Paradigm shift introduced by ANNs in medicine

ANNs are able to reproduce the dynamical interaction of multiple factors simultaneously, allowing the study of complexity; this is very important for the researcher interested to deep the knowledge of a specific disease or to better understand the possible implications relative to strange associations among variables. This has to do to what is called "intelligent data mining". But one the other hand ANNs can also help medical doctors in making decisions under extreme uncertainty and to draw conclusions on individual basis and not as average trends. The modern patient wants to be treated as an individual person and not just as a statistics. Patients want to know their own risk, not just a parameter regarding a class of people similar to them just for some aspects. ANNs are very powerful in modelling at single individual level, and by combining several parallel ANNs trained on the same data set is possible to make multiple statistics on a single subject, allowing in this way the calculation of the confidence interval of the prediction estimate. Finally ANNs make possible to treat huge amount of information without squeezing arbitrarily the data and without losing complexity. This contributes to a new holistic vision of the human subject contrasting the statistical reductionism, which tends to squeeze or even delete the single subject sacrificing him to his group of belongingness. A remarkable contribution to this individual approach comes from Fuzzy Logic, according to which there are no sharp limits between opposite things, like health and disease. This approach allows to partially escape from probability theory trap in situations where is fundamental to express a judgment based on a single case and favours a novel humanism directed to the management of the patient as individual subject.

7. The unfulfilled dream: prediction of individual response to treatments

Making predictions for specific outcomes (diagnosis, risk assessment, prognosis) represents a fascinating aspect of medical science. Different statistical approaches have been proposed to define models to identify factors that are predictive for the outcome of interest. Studies

have been performed to define the clinical and biological characteristics that could be helpful in predicting who will benefit from an anti-obesity drug for example, but results have been limited (Padwal *et al.*, 2003).

Traditional statistical approaches encounter problems when the data show big variability and not easily normalized for inherent nonlinearity. More-advanced analysis techniques, such as dynamic mathematical models, can be useful because they are particularly suitable for solving nonlinear problems frequently associated with complex biological systems.

Use of ANNs in biological systems has been proposed for different purposes, including studies on deoxyribonucleic acid sequencing (Parbhane *et al.*, 2000) and protein structure (Jagla & Schuchhardt, 2000).

ANNs have been used in different clinical settings to predict the effectiveness of instrumental evaluation (echocardiography, brain single photon emission computed tomography, lung scans, prostate biopsy) in increasing diagnostic sensitivity and specificity and in laboratory medicine in general (Tafeit & Reibnegger, 1999). Also, they have proven effective in identifying gastro-oesophageal reflux patients on the sole basis of clinical data (Pace *et al.*, 2005). But the most promising application of ANNs relates to prediction of possible clinical outcomes with specific therapy. ANNs have proven effective in detecting responsiveness to methadone treatments of drug addicts (Massini & Shabtay, 1998), to pharmacological treatment in Alzheimer disease (Mecocci *et al.*, 2002), to clozapine in schizophrenic patients (Lin *et al.*, 2008) and in various fields of psychiatric research (Politi *et al.*, 1999).

The use of ANNs for predictive modelling in obesity dates back to a decade ago, where it was proposed to model the waist-hip ratio from 13 other health parameters (Abdel-Aal & Mangoud, 1997). Later, it has been proposed as a tool for body composition research (Linder *et al.*, 2003).

One of the main factors preventing a more efficient use of new pharmacological treatments for chronic diseases like for example hypertension, cancer, Alzheimer disease or obesity is represented by the difficulty of predicting “a priori” the chance of response of the single patient to a specific drug. A major methodological setback in drawing inferences and making predictions from data collected in the real world setting, such as observational studies, is that variability in the underlying biological substrates of the studied population and the quality and content of medical intervention influence outcomes. Because there is no reason to believe that these, like other health factors, work together in a linear manner, the traditional statistical methods, based on the generalized linear model, have limited value in predicting outcomes such as responsiveness to a particular drug.

Most studies have shown that up to 50% of patients treated with new molecules given in monotherapy or as an adjunct to standard treatments may show an unsatisfactory response. As a matter of fact, when time comes for the physician to decide about type of treatment, there is very little evidence that can help her/him in drug treatment choice. Take for example obesity. Here only scanty data are available on predictive factors to the specific treatment, and attempts at developing models for predicting response to the drug by using traditional techniques of multiple regression have showed an unsatisfactory predictive capacity (i.e. inferior to 80% of total variance). (Hansen *et al.*, 2001; Hainer *et al.*, 2005). A possible explanation could be that obesity is a so-called complex disease, where different factors interact with multiple interactions among variables, positive and negative feedback loops, and non-linear system dynamics. Another good example is Alzheimer Disease.

Clinical trials have established the efficacy of cholinesterase inhibitor drugs (ChEI), such as tacrine, (Knopman *et al.*, 1996) donepezil, (Rogers *et al.*, 1998) and rivastigmine (Rösler *et al.*, 1999) based on improvement in cognitive aspects and in overall functioning using the Alzheimer's Disease Scale—Cognitive subscale (ADAS-Cog) and the Clinician's Interview Based Impression of Change (CIBIC), respectively. Although the mean score of treated patients in both scales was significantly higher than the placebo group, many subjects under active treatment showed little or no improvement (non-responders).

However it is not possible to estimate which patients are likely to respond to pharmacological therapy with ChEI. This prediction would be an important decision-making factor in improving the use of healthcare resources.

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Because there is no reason, *a priori*, to believe that these, like other health factors, work together in a linear manner, the traditional statistical methods, based on the generalized linear model, have limited value in predicting outcomes such as responsiveness to a particular drug.

A possible alternative approach to the solution of the problem is represented by the use of Neural Networks.

Although ANNs have been applied to various areas of medical research, they have not been employed intensively clinical pharmacology as they might deserve.

In our personal experience ANNs proved to be a useful method to discriminate between responders and non-responders, better than traditional statistical methods in our three experimental studies carried out with donepezil in Alzheimer disease and with sibutramine in obesity and with infliximab in Crohn disease.

In a paper published in 2002 (Mecocci, 2002) we have evaluated the accuracy of artificial neural networks compared with discriminant analysis in classifying positive and negative response to the cholinesterase inhibitor donepezil in a opportunistic group of 61 old patients of both genders affected by Alzheimer's disease (AD) patients in real world setting along three months follow-up.

Accuracy in detecting subjects sensitive (responders) or not (non responders) to therapy was based on the standard FDA criterion standard for evaluation of efficacy: the scores of Alzheimer's Disease Assessment Scale—Cognitive portion and Clinician's Interview Based Impression of Change—plus scales. In this study ANNs were more effective in discriminating between responders and nonresponders than other advanced statistical methods, particularly linear discriminant analysis. The total accuracy in predicting the outcome was 92.59%.

In a second study we evaluated the performance of ANN in predicting response to Sibutramine treatment in obese patients (Petroni *et al.*, in press). Out of 162 patients treated in an open-label multicentre study with 6-month Sibutramine 10 mg/day, increased to 15 mg/day if weight loss $\leq 5\%$ after 3 months, we evaluated 92 obese subjects (10 M, 82 F, mean age 44.2 ± 12.4) with full data set available. Patients were considered responders if body weight loss was $\geq 5\%$ at the end of the study. At descriptive analysis, non-responders ($n=26$) only differed from responders ($n=66$) in having significantly lower values of TSH and in having a more frequent history of previous dietary or pharmacological treatment for weight loss. Out of 84 baseline variables, after a preliminary analysis, 33 variables which

best discriminated between responders and non-responders were selected; a logistic regression (LR) was also performed on same variables. The overall mean accuracy of ANN detection of responders was 79.1%, being correctly recognised in 90.5%; predictive model could not be developed by LR. Among baseline predictive variables, those with the highest "input relevance" were: previous drug treatment, android fat distribution, alcohol consumption, HDL cholesterol, hypertension; variables that also significantly contributed to the model were: previous dietary treatment, pregnancies, abnormal lipid profile, physical activity, FT4, heart rate, diastolic blood pressure, obesity in siblings, previous smoking, triglycerides, age, basal energy expenditure, weight at 20 years, current smoking, TSH, creatinine. In Conclusion ANN proved to be a useful method to discriminate between responders and non-responders, better than traditional statistical methods.

Finally in a third study we evaluated the use of artificial neural networks in predicting response to infliximab treatment in patients with Crohn's disease (Kohn *et al.*, 2005).

In this pilot study, different ANN models were applied to a data sheet with demographic and clinical data from 76 patients with steroid resistant/dependant or fistulizing CD treated with Infliximab to compare accuracy in classifying responder and non responder subjects with that of linear discriminant analysis.

Eighty one outpatients with CD (31 men, 50 women; mean age $\pm$ standard deviation 39.9 ± 15 range: 12-81) participating to an Italian Multicentric Study (Parbhane *et al.*, 2000), were enrolled in the study. All patients were treated, between April 1999 and December 2003, with a dose of Infliximab 5 mg/kg of body weight for luminal refractory (CDAI > 220-400) (43 patients), fistulizing CD (19 patients) or both of them (14 patients).

The final data sheet consisted of 45 independent variables related to the anagraphic and anamnestic data (sex, age at diagnosis, age at infusion, smoking habit, previous Crohn's related abdominal surgery [ileal or ileo-cecal resections] and concomitant treatments including immunomodulators and corticosteroids) and to clinical aspects (location of disease, perianal disease, type of fistulas, extraintestinal manifestations, clinical activity at the first infusion [CDAI], indication for treatment). Smokers were defined as those smoking a minimum of 5 cigarettes per day for at least 6 months before their first dose of Infliximab. Non smokers were defined as those who had never smoked before, those who had quit smoking at least 6 months before their first dose of Infliximab, or those who smoked fewer than 5 cigarettes per day. Concomitant immunosuppressive use was defined as initiation of methotrexate before their first Infliximab infusion or initiation of 6-mercaptopurine (6-MP) or azathioprine more than 3 months before their first Infliximab infusion.

Assessment of response was determined by clinical evaluation 12 weeks after the first infusion for all patients. Determination of response in patients with inflammatory CD was based on the Crohn's Disease Activity Index (CDAI). For clear-cut estimate clinical response was evaluated as complete response or partial / no response.

Complete response was defined as (a) clinical remission (CDAI < 150) in luminal refractory disease and (b) temporary closure of all draining fistulas at consecutive visits in the case of enterocutaneous and perianal fistulas; entero-enteric fistulas were evaluated by small bowel barium enema and vaginal-bladder fistula by lack of drainage at consecutive visits. For patients with both indications the outcome was evaluated independently for each indication.

Two different experiments were planned following an identical research protocol. The first one included all 45 independent variables including frequency and intensity Crohn disease symptoms, plus numerous other social and demographic characteristics, clinical features

and history. In the second experiment the IS system coupled to the T&T system automatically selected the most relevant variables and therefore 22 variables were included in the model.

Discriminant analysis was also performed on the same data sets to evaluate the predictive performance of this advanced statistical method by a statistician blinded to ANN results. Different models were assessed to optimise the predictive ability. In each experiment the sample was randomly divided into two sub-samples, one for the training phase and the other for the testing phase, with the same record distributions used for ANN validation. ANNs reached an overall accuracy rate of 88% while LDA performance was only of 72%.

8. Conclusions

New mathematical laws coupled by computer programming allow today to perform predictions which till few years ago were considered impossible. The prediction of response to a specific treatment in individual patients with supervised artificial neural networks has been documented and validated extensively in the literature. Less experience exists in predicting major unwanted effects of a new drugs after the commercialisation. In doing this task a close cooperation between regulatory agency scientific team and bio mathematicians experts in the use of new generations artificial adaptive system has to be strongly recommended.

9. References

- Abdel-Aal, R.E., Mangoud, A.M. (1997). Modeling obesity using abductive networks, *Comput Biomed Res.* Vol.(30): 451-71.
- Buscema, M. (2007). Twist: Input Search and T&T reverse, version 2.0, *Semeion Software #39*, Rome, Italy.
- Buscema, M. (2008). Meta Auto Associative ANNs, version 1.0, *Semeion Software #50*, Rome, Italy.
- Goldberger, A.L. (1996). Nonlinear dynamics for clinicians: chaos theory, fractals, and complexity at the bedside, *Lancet* Vol. (347): 1312-1314.
- Grossi, E. (2006). How artificial intelligence tools can be used to assess individual patient risk in cardiovascular disease: problems with the current methods. *BMC Cardiovasc Disord.* Vol. (6): 20.
- Grossi, E., Buscema, M. (2006). Artificial Intelligence and Outcome Research, *Drug Development Research* Vol.(67): 227-244
- Grossi, E., Buscema, M. (2007). Introduction to artificial neural networks, *Eur J Gastroenterol Hepatol.*, Vol (19): 1046-1054.
- Grossi, E., Mancini, A., Buscema, M. (2007) International experience on the use of artificial neural networks in gastroenterology, *Dig Liver Dis.* Vol. (39): 278-85.
- Hainer, V., Kunesova, M., Bellisle, F., Hill, M., Braunerova, R., Wagenknecht, M. (2005) Psychobehavioral and nutritional predictors of weight loss in obese women treated with sibutramine. *Int J Obes*, Vol (29): 208-16.
- Hansen, D., Astrup, A., Toubro, S., Finer, N., Kopelman, P., Hilsted, J., Rössner, S., Saris, W., Van Gaal, L., James, W., Goulder, M. For The STORM Study Group. (2001). Predictors of weight loss and maintenance during 2 years of treatment by sibutramine in obesity. Results from the European multi-centre STORM trial.

- Sibutramine Trial of Obesity Reduction and Maintenance, *Int J Obes Relat Metab Disord.* Vol. (25): 496-501.
- Helgason, C.M., Grossi, E., Pandey, D. *et al.* (2008). Platelet aggregation and recruitment with aspirin-clopidogrel therapy, *Cerebrovasc Dis.* Vol. (25):392-400.
- Jagla, B., Schuchhardt, J. (2000). Adaptive encoding neural networks for the recognition of human signal peptide cleavage site, *Bioinformatics* Vol. (16): 245-250.
- Kaplan, D., Glass, L. Understanding nonlinear dynamics.(1995) *Springer-Verlag*, New York
- Knopman, D., Schneider, L., Davis, K. *et al.* (1996). Long-term tacrine (Cognex) treatment. Effects on nursing home placement and mortality. Tacrine Study Group, *Neurology* Vol. (47): 166-177.
- Kohn, A., Grossi, E., Mangiarotti, R., Prantera, C. (2005). Use of Artificial Neural Networks (ANN) in predictor response to infliximab treatment in patients with Crohn's disease, *Dig Liver Dis.* Vol. (37) (suppl.1): S51.
- Lahner, E., Intraligi, M., Buscema, M. *et al.* (2008). Artificial neural networks in the recognition of the presence of thyroid disease in patients with atrophic body gastritis, *World J. Gastroenterol.* Vol. (14): 563-8.
- Licastro, F., Porcellini, E., Chiappelli, M. *et al.* (2010). Multivariable network associated with cognitive decline and dementia. *Neurobiol. Aging.* Vol. (31): 257-269
- Lin, C.C., Wang, Y.C., Chen, J.Y., Liou, Y.J., Bai, Y.M., Lai, I.C., Chen, T.T., Chiu, H.W., Li, Y.C. (2008). Artificial neural network prediction of clozapine response with combined pharmacogenetic and clinical data. *Comput. Methods Programs Biomed.* Vol. (91): 91-9.
- Linder, R., Mohamed, EI., De Lorenzo, A., Pöpl, S.J. (2003). The capabilities of artificial neural networks in body composition research, *Acta Diabetol.* Vol. (40), Suppl 1: S9-14
- Lisboa, P.J.G. (2002), A review of evidence of health benefit from artificial neural networks in medical intervention. *Neural Netw.* Vol. (15): 11-39
- Massini, G., Shabtay, L. (1998). Use of a constraint satisfaction network model for the evaluation of the methadone treatments of drug addicts, *Subst. Use Misuse* Vol. (33): 625-656.
- Mecocci, P., Grossi, E., Buscema, M., Intraligi, M., Savarè, R., Rinaldi, P., Cherubini, A., Senin, U. (2002). Use of Artificial Networks in Clinical Trials: A Pilot Study to predict Responsiveness to Donepezil in Alzheimer's Disease, *J Am Geriatr. Soc.* Vol. (50): 1857-1860.
- Pace, F., Buscema, M., Dominici, P., Intraligi, M., Baldi, F., Cestari, R., Passaretti, S., Bianchi-Porro, G., Grossi, E. (2005). Artificial neural networks are able to recognize gastro-oesophageal reflux disease patients solely on the basis of clinical data. *Eur J Gastroenterol. Hepatol.* Vol. (6): 605-10
- Padwal, R.S., Rucker, D., Li, S.K., Curioni, C., Lau, D.C.W. (2003). Long-term pharmacotherapy for obesity and overweight. *Cochrane database of Systematic Reviews*, Issue 4, Art. No.: CD004094. DOI:10.1002/14651858.CD004094.pub2
- Parbhane, R.V., Tambe, S.S., Kulkarni, B.D. (2000). ANN modeling of DNA sequences: New strategies using DNA shape code, *Comput. Chem.* Vol. (24): 699-711.
- Penco, S., Buscema, M., Patrosso, M.C. *et al.* (2008). New application of intelligent agents in sporadic amyotrophic lateral sclerosis identifies unexpected specific genetic background. *BMC Bioinformatics.* Vol. (30): 9-254.

- Petroni, M.L., Rivolta, G., Grossi, E., Cerutti, R., Fatati, G. on behalf of ADI Neural Networks Study Group. Factors predicting response to sibutramine treatment in obesity: a neural network analysis. *Obesity*, in press
- Politi, E., Balduzzi, C., Bussi, R. *et al.* (1999). Artificial neural network: A study in clinical psychopharmacology. *Psychiatry Res.* Vol. (87): 203–215.
- Rogers, S.L., Farlow, M.R., Doody, R.S., *et al.* (1998). A 24-week, double-blind, placebocontrolled trial of donepezil in patients with Alzheimer's disease. *Neurology* Vol. (50): 136–145.
- Rösler, M., Anand, R., Cicin-Sain, A. *et al.* (1999). Efficacy and safety of rivastigmine in patients with Alzheimer's disease: International randomised controlled trial, *BMJ* Vol. (318): 633–638.
- Rossini, P.M., Buscema, M., Capriotti, M. *et al.* (2008). Is it possible to automatically distinguish resting EEG data of normal elderly vs. mild cognitive impairment subjects with high degree of accuracy? *Clin. Neurophysiol.* Vol. (119): 1534-1545
- Street, M.E., Grossi, E., Volta, C. *et al.* (2008). Placental determinants of fetal growth: identification of key factors in the insulin-like growth factor and cytokine systems using artificial neural networks. *BMC Pediatr.* Vol. (7): 8-24.
- Tafeit, E., Reibnegger, G. (1999). Artificial neural networks in laboratory medicine and medical outcome prediction. *Clin. Chem. Lab. Med.* Vol. (37): 845–853.

Reputation-Based Neural Network Combinations

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1. Introduction

The exercise of combining classifiers is primarily driven by the desire to enhance the performance of a classification system. There may also be problem-specific rationale for integrating several individual classifiers. For example, a designer may have access to different types of features from the same study participant. For instance, in the human identification problem, the participant's voice, face, and handwriting provide different types of features. In such instances, it may be sensible to train one classifier on each type of feature (Jain et al., 2000). In other situations, there may be multiple training sets, collected at different times or under slightly different circumstances. Individual classifiers can be trained on each available data set (Jain et al., 2000; Xu et al., 1992). Lastly, the demand for classification speed in online applications may necessitate the use of multiple classifiers (Jain et al., 2000).

Optimal combination of multiple classifiers is a well-studied area. Traditionally, the goal of these methods is to improve classification accuracy by employing multiple classifiers to address the complexity and non-uniformity of class boundaries in the feature space. For example, classifiers with different parameter choices and architectures may be combined so that each classifier focuses on the subset of the feature space where it performs best. Well-known examples of these methods include bagging (Breiman, 1996a) and boosting (Bauer & Kohavi, 1999).

Given the universal approximation ability of neural networks such as multilayer perceptrons and radial basis functions (Haykin, 1994), there is theoretical appeal to combine several neural network classifiers to enhance classification. Indeed, several methods have been developed for this purpose, including, for example, optimal linear combinations (Ueda, 2002) and mixture of experts (Jacobs et al., 1991), and negative correlation (Chen & Yao, 2009) and evolving neural network ensembles (Yao & Islam, 2008). In these methods, all base classifiers are generally trained on the same feature space (either using the entire training set or subsets of the training set). While these methods have proven effective in many applications, they are associated with numerical instabilities and high computational complexity in some cases (Bauer & Kohavi, 1999).

Another approach to classifier combination is to train the base classifiers on different feature spaces. This approach is advantageous in combating the undesirable effects of associated with high-dimensional feature spaces (curse of dimensionality). Moreover, the feature sets can be chosen to minimize the correlation between the individual base classifiers to further improve the overall accuracy and generalization power of classification (Ueda, 2002). These methods are also highly desirable in situations where heterogeneous feature combinations are used.

Combination of classifiers based on different feature has been generally accomplished through fixed classification rules. These rules may select one classifier output among all available outputs (for example, using the minimum or maximum operator), or they may provide a classification decision based on the collective outputs of all classifiers (for example, using the mean, median, or voting operators) (Bloch, 2002; Kuncheva, 2002). Among the latter, the simplest and most widely applied rule is the majority vote (Hull et al., 1988; Suen et al., 1990). Many authors have demonstrated that classification performance improves beyond that of the single classifier scenario when multiple classifier decisions are combined via a simple majority vote (Nadal et al., 1990; Xu et al., 1992). Xu et al. (1992) further introduced the notion of weighted majority voting by incorporating classifier-specific beliefs which reflect each classifier's uncertainty about a given test case. Unfortunately, this method does not deal with the degenerate case when one or more beliefs are zero - a situation likely to occur in multi-class classification problems. Moreover, this classifier relies on the training data set to derive beliefs values for each classifier. This approach, therefore, risks overfitting the classifier to the training set and a consequent degradation in generalization power.

In this chapter, we describe a method for combining several neural network classifiers in a manner which is computationally inexpensive and does not demand additional training data beyond that needed to train individual classifiers. Our reputation method will build on the ideas of (Xu et al., 1992). In the following, we present notation that is used throughout the manuscript and detail the majority voting algorithm using this notation. The following presentation is adapted from (Nikjoo et al., 2011).

1.1 Notation

Assume the time series, S , is the pre-processed version of an acquired signal. Also let $\Theta = \{\theta_1, \theta_2, \dots, \theta_L\}$ be a set of $L \geq 2$ classifiers and $\Omega = \{\omega_1, \omega_2, \dots, \omega_c\}$ be a set of $c \geq 2$ class labels, where $\omega_j \neq \omega_k, \forall j \neq k$. Without loss of generality, $\Omega \subset \mathbb{N}$. The input of each classifier is the feature vector $x \in \mathcal{R}^{n_i}$, where n_i is the dimension of the feature space for the i^{th} classifier θ_i , whose output is a class label $\omega_j, j = 1, \dots, c$. In other words, the i^{th} classifier, $i = 1, \dots, L$, is a functional mapping, $\mathcal{R}^{n_i} \rightarrow \Omega$, which for each input x gives an output $\theta_i(x) \in \Omega$. Generally, the classifier function could be linear or non-linear. It is assumed that for the i^{th} classifier, a total number of d_i subjects are assigned for training. The main goal of combining the decisions of different classifiers is to increase the accuracy of class selection.

1.2 Majority voting algorithm

In a multi-classifier system, the problem is to arrive at a global decision $\Theta^*(x) = \omega_j$ given a number of local decisions $\theta_i(x) \in \Omega$, where generally (Xu et al., 1992):

$$\theta_1(x) = \theta_2(x) = \dots = \theta_L(x). \quad (1)$$

In the literature, a classical approach for solving this problem is *majority voting* (Hull et al., 1988) (Suen et al., 1990). To express this idea mathematically, we define an indicator function

$$I_i(x, \omega_j) = I(\theta_i(x) - \omega_j) = \begin{cases} 1, & \text{when } \theta_i(x) = \omega_j, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Now, using (2), the majority voting rule can be expressed as follows:

$$\Theta^*(x) = \begin{cases} \omega_{\max}, & \text{if } \max_{\omega_j} I_{\Omega}(x, \omega_j) > L/2, \\ Q, & \text{otherwise,} \end{cases} \quad (3)$$

where $\omega_{\max} = \arg \max_{\omega_j} I_{\Omega}(x, \omega_j)$, $I_{\Omega}(x, \omega_j) = \sum_{i=1}^L I_i(x, \omega_j)$, $j = 1, \dots, c$, and $Q \notin \Omega$ is the rejection state. In other words, given a feature vector, each classifier votes for a specific class. The class with the majority of votes is selected as the *candidate class*. If the candidate class earns more than half of the total votes, it is selected as the final output of the system. Otherwise, the feature vector is rejected by the system.

The majority voting algorithm is computationally inexpensive, simple to implement and applicable to a wide array of classification problems (Lam & Suen, 1997) (Jain et al., 2000). Despite its simplicity, majority voting can significantly improve upon the classification accuracies of individual classifiers (Lam & Suen, 1997). However, this method suffers from a major drawback; the decision heuristic is strictly *democratic*, meaning that the votes from different classifiers are always equally weighted, regardless of the past performance of individual classifiers. Therefore, votes of *weak classifiers*, i.e., classifiers whose performance only slightly exceeds that of the random classifier, can diminish the overall performance of the system when they have the majority. To exemplify this issue, consider a classification system with $c = 2$ classes, $\Omega = \{\omega_1, \omega_2\}$, and $L = 3$ classifiers, $\Theta = \{\theta_1, \theta_2, \theta_3\}$, where two are weak classifiers with 51% average accuracy while the remaining one is a strong classifier with 99% average accuracy. Now assume that for a specific feature vector both the weak classifiers vote for ω_1 but the strong classifier votes for ω_2 . Based on the majority voting rule, ω_1 is preferred over ω_2 , which is mostly likely an incorrect classification.

As it is discussed in (Tresp & Taniguchi, 1995) (Jacob et al., 1991) (Jain et al., 2000), in practice, there are various situations in which the majority vote may be suboptimal. Motivated by the above, we propose a novel algorithm for combining several classifiers based on their individual *reputations*, numerical weights that reflect each classifier's past performance. The algorithm is detailed in the next section and again is adapted from Nikjoo et al. (2011).

2. Reputation-based voting algorithm

To mitigate the risk of the overall decision being unduly influenced by poorly performing classifiers, we propose a novel fusion algorithm which extends the majority voting concept to acknowledge the past performance of classifiers. To measure the past performance of the i^{th} classifier, we define a measure called *reputation*, $r_i \in \mathbb{R}$, $0 \leq r_i \leq 1$. For a classifier with high performance, the reputation is close to 1 while a weak classifier would have a reputation value close to 0. For each feature vector, both the majority vote and the reputation of each classifier contribute to the final global decision. The collection of reputation values for L classifiers constitutes the reputation set, $R = \{r_1, r_2, \dots, r_L\}$. Each classifier is mapped to a real-valued reputation, r_i , namely,

$$r_i = r(\theta_i) = \alpha, i = 1, \dots, L, \quad (4)$$

where $r : \Theta \rightarrow [0, 1]$, $\alpha \in \mathbb{R}$ and $0 \leq \alpha \leq 1$. To determine the reputation of each classifier, we utilize a *validation set* in addition to the classical training and test sets. Specifically, the performance of the trained classifiers on the validation data determines their reputation values. Now, we have all the necessary tools to explain the proposed algorithm.

1. For a classification problem with $c \geq 2$ classes, we design and develop $L \geq 2$ individual classifiers. The proposed algorithm is especially useful if the individual classifiers are independent. This condition can be guaranteed by using different training sets or using various resampling techniques such as bagging (Breiman, 1996b) and boosting (Schapire, 1990). Unlike some of the previous work (Lee & Srihari, 1993) (Hull et al., 1988), there is no restriction on the number of classifiers L and this value can be either an odd or an even number. Also, it should be noted here that, in general, the feature space dimension, n_i , of each classifier could be different and the number of training exemplars, d_i , for each classifier could be unique.
2. After training the L classifiers individually, the respective performance of each is evaluated using the validation set and a reputation value is assigned to each classifier. The validation sets are disjoint from the training sets. It is important to note that here we use two different validation sets. The first one is the traditional validation set which is used repeatedly until the classifier is satisfactorily trained (Duda et al., 2000). In contrast, the second validation set is used to calculate the reputation values of individual classifiers. The accuracy of each classifier is estimated with the corresponding validation set and normalized to $[0, 1]$ to generate a reputation value. For instance, a classifier, θ_i , with 90% accuracy (on the latter validation set mentioned above) has a reputation $r_i = 0.9$.
3. Now, for each feature vector, x , in the test set, L decisions are made using L distinctive classifiers:

$$\Omega(x) = \{\theta_1(x), \theta_2(x), \dots, \theta_L(x)\}. \quad (5)$$

4. To arrive at a final decision, we consider the votes of the classifiers with high reputations rather than simply selecting the majority class. First, we sort the reputation values of the classifiers in descending order,

$$R^* = \{r_{1^*}, r_{2^*}, \dots, r_{L^*}\}, \quad (6)$$

such that $r_{1^*} \geq r_{2^*} \geq \dots \geq r_{L^*}$. Then, using this set, we rank the classifiers to obtain a reputation-ordered set of classifiers, Θ^* .

$$\Theta^* = \begin{pmatrix} \theta_{1^*} \\ \theta_{2^*} \\ \vdots \\ \theta_{L^*} \end{pmatrix}. \quad (7)$$

The first element of this set corresponds to the classifier with the highest reputation.

5. Next, we examine the votes of the first m elements of the reputation-ordered set of classifiers, with

$$m = \begin{cases} \frac{L}{2}, & \text{if } L \text{ is even,} \\ \frac{L+1}{2}, & \text{if } L \text{ is odd.} \end{cases} \quad (8)$$

If the top m classifiers vote for the same class, ω_j , we accept the majority vote and take ω_j as the final decision of the system. However, if the votes of the first m classifiers are not equal, we consider the classifier's individual reputations (Step 2).

6. Let $p(\omega_j)$ be the *prior probability* of class ω_j . As before, $\Theta(x) = \{\theta_1(x), \theta_2(x), \dots, \theta_L(x)\}$ represents the local decisions made by different classifiers about the input vector x . The probability that the combined classifier decision is ω_j given the input vector x and the individual local classifier decisions is denoted as the *posterior probability*,

$$p(\omega_j | \theta_1(x), \theta_2(x), \dots, \theta_L(x)) \quad (9)$$

Clearly, we should choose the class which maximizes this probability.

$$\arg \max_{\omega_j} p(\omega_j | \theta_1(x), \theta_2(x), \dots, \theta_L(x)), \quad j = 1, \dots, c. \quad (10)$$

To estimate the posterior probability we use *Bayes formula*. For notational simplicity we drop the argument x from the local decisions.

$$p(\omega_j | \theta_1, \dots, \theta_L) = \frac{p(\theta_1, \dots, \theta_L | \omega_j) p(\omega_j)}{p(\theta_1, \dots, \theta_L)}, \quad (11)$$

where $p(\theta_1, \dots, \theta_L | \omega_j)$ is the *likelihood* of ω_j and $p(\theta_1, \dots, \theta_L)$ is the *evidence* factor, which is estimated using the *law of total probability*

$$p(\theta_1, \dots, \theta_L) = \sum_{j=1}^c p(x, \theta_1, \dots, \theta_L | \omega_j) p(\omega_j). \quad (12)$$

By assuming that the classifiers are independent of each other, the likelihood can be written as

$$p(\theta_1, \dots, \theta_L | \omega_j) = p(\theta_1 | \omega_j) \dots p(\theta_L | \omega_j). \quad (13)$$

Substituting (12) into the Bayes rule (11) and then replacing the likelihood term with (13), we obtain,

$$p(\omega_j | \theta_1, \dots, \theta_L) = \frac{\prod_{i=1}^L p(\theta_i | \omega_j) p(\omega_j)}{\sum_{t=1}^c \prod_{i=1}^L p(\theta_i | \omega_t) p(\omega_t)}. \quad (14)$$

To calculate the local likelihood functions, $p(\theta_i | \omega_j)$, we use the reputation values calculated in Step 6. When the correct class is ω_j and classifier θ_i classifies x into the class ω_j , i.e., $\theta_i(x) = \omega_j$, we can write

$$p(\theta_i = \omega_j | \omega_j) = r_i. \quad (15)$$

In other words, $p(\theta_i = \omega_j | \omega_j)$ is the probability that the classifier θ_i correctly classifies x into class ω_j when x actually belongs to this class. This probability is exactly equal to the reputation of the classifier. On the other hand, when the classifier categorizes x incorrectly, i.e., $\theta_i(x) \neq \omega_j$ given that the correct class is ω_j , then

$$p(\theta_i \neq \omega_j | \omega_j) = 1 - r_i. \quad (16)$$

When there is no known priority among classes, we can assume equal prior probabilities. Hence,

$$p(\omega_1) = p(\omega_2) = \dots = p(\omega_c) = \frac{1}{c}. \quad (17)$$

Finally, for each class, ω_j , we compute the a posteriori probabilities as given by (14) using (15), (16), and (17). The class with the highest posterior probability is selected as the final decision of the system and the input subject x is categorized as belonging to this class.

The advantage of the reputation-based algorithm over the majority voting algorithm lies in the fact that the former has a higher probability of correct consensus and a faster rate of convergence to the peak probability of correct classification (Nikjoo et al., 2011).

3. Discriminating between healthy and abnormal swallows

We apply the proposed algorithm to the problem of swallow classification. Specifically, the problem is to differentiate between safe and unsafe swallowing on the basis of dual-axis accelerometry (Damouras et al., 2010; Sejdic, Komisar, Steele & Chau, 2010). The basic idea is to decompose a high dimensional classification problem into 3 lower dimensional problems, each with a unique subset of features and a dedicated classifier. The individual classifier decisions are then melded according to the proposed reputation algorithm.

3.1 Signal acquisition and pre-processing

In this chapter, we consider a randomly selected subset of 100 healthy swallows and 100 dysphagic swallows from the larger database originally introduced and analyzed in (Lee et al., 2009). Briefly, dual-axis swallowing accelerometry data were collected at 10 kHz from 24 adult patients (mean age 64.8 ± 18.6 years, 2 males) with dysphagia and 17 non-dysphagic persons (mean age 46.9 ± 23.8 years, 8 males). Patients provided an average number of 17.8 ± 8.8 swallows while non-dysphagic participants completed 19 swallow sequences each. Both groups swallowed boluses of different consistencies. For more details of the instrumentation and swallowing protocol, please see (Lee et al., 2009). It has been shown in (Lee et al., 2008) that the majority of power in a swallowing vibration lies below 100Hz. Therefore, we downsampled all signals to 1KHz. Then, individual swallows were segmented according to previously identified swallow onsets and offsets (Lee et al., 2009). Each segmented swallow was denoised using a 5! -level discrete Daubechies-5 wavelet transform. To remove low-frequency motion artifacts due to bolus intake and participant motion, each signal was subjected to a 4th-order highpass Butterworth filter with a cutoff frequency of 1Hz.

3.2 Feature extraction

Let S be a pre-processed acceleration time series, $S = \{s_2, s_2, \dots, s_n\}$. As in previous accelerometry studies, signal features from three different domains were considered (Lee et al., 2010; 2009). The different genres of features are summarized below.

1. Time-Domain Features

- Mean: The sample mean of a distribution is an unbiased estimation of the location of that distribution. The sample mean of the time series S can be calculated as

$$\mu_s = \frac{1}{n} \sum_{i=1}^n s_i. \quad (18)$$

- Variance: The variance of a distribution measures its spread around the mean and reflects the signal's power. The unbiased estimation of variance can be obtained as

$$\sigma_s^2 = \frac{1}{n-1} \sum_{i=1}^n (s_i - \mu_s)^2. \quad (19)$$

- Skewness: The skewness of a distribution is a measure of the symmetry of a distribution. This factor can be computed as follows

$$\gamma_{1,s} = \frac{\frac{1}{n} \sum_{i=1}^n (s_i - \mu_s)^3}{\left(\frac{1}{n} \sum_{i=1}^n (s_i - \mu_s)^2\right)^{1.5}}. \quad (20)$$

- Kurtosis: This feature reflects the peakedness of a distribution and can be found as

$$\gamma_{2,s} = \frac{\frac{1}{n} \sum_{i=1}^n (s_i - \mu_s)^4}{\left(\frac{1}{n} \sum_{i=1}^n (s_i - \mu_s)^2\right)^2}. \quad (21)$$

2. Frequency-Domain Features

- The peak magnitude value of the Fast Fourier Transform (FFT) of the signal S is used as a feature. All the FFT coefficients are normalized by the length of the signal, n .
- The centroid frequency of the signal S (Sejdic, Komisar, Steele & Chau, 2010) can be estimated as

$$\hat{f} = \frac{\int_0^{f_{max}} f |F_s(f)|^2 df}{\int_0^{f_{max}} |F_s(f)|^2 df}, \quad (22)$$

where $F_s(f)$ is the Fourier transform of the signal S and f_{max} is the Nyquist frequency (5KHz in this study).

- The bandwidth of the spectrum can be computed using the following formula

$$BW = \sqrt{\frac{\int_0^{f_{max}} (f - \hat{f})^2 |F_s(f)|^2 df}{\int_0^{f_{max}} |F_s(f)|^2 df}}. \quad (23)$$

3. Information-Theory-Based Features

- Entropy Rate (Porta et al., 2001): Porta et al. (2001) introduced a new method for measuring the entropy rate in a signal which quantifies the extent of regularity in that signal. They showed that this rate is useful for signals with some relationship among consecutive signal points. Lee et al. (Under review) used the entropy rate for the classification of healthy and abnormal swallowing. Following their approach, we first normalized the signal S to zero-mean and unit variance. Then, we quantized the normalized signal into 10 equally spaced levels, represented by the integers 0 to 9, ranging from the minimum to maximum value. Now, the sequence of U consecutive points in the quantized signal, $\hat{S} = \{\hat{s}_1, \hat{s}_2, \dots, \hat{s}_3\}$, was coded using the following equation

$$a_i = \hat{s}_{i+U-1} \cdot 10^{U-1} + \dots + \hat{s}_i \cdot 10^0, \quad (24)$$

with $i = 1, 2, \dots, n - U + 1$. The coded integers comprised the coding set $A_U = \{a_1, \dots, a_{n-U+1}\}$. Using the Shannon entropy formula, we estimated entropy

$$E(U) = - \sum_{t=0}^{10^U-1} P_{A_U}(t) \ln P_{A_U}(t), \quad (25)$$

where $p_{A_U}(t)$ represents the probability of observing the value t in A_U , approximated by the corresponding sample frequency. Then, the entropy rate was normalized using the following equation

$$NE(U) = \frac{E(U) - E(U-1) + E(1) \cdot \beta}{E(1)}, \quad (26)$$

where β was the percentage of the coded integers in A_L that occurred only once. Finally, the regularity index $\rho \in [0, 1]$ was obtained as

$$\rho = 1 - \min NE(U), \quad (27)$$

where a value of ρ close to 0 signifies maximum randomness while ρ close to 1 indicates maximum regularity.

- Memory (Lee et al., 2010): To calculate the memory of the signal, its autocorrelation function was computed from zero to the maximum time lag. Then, it was normalized such that the autocorrelation at zero lag was unity. The memory was estimated as the time duration from zero to the point where the autocorrelation decays to $1/e$ of its zero lag value.
- Lemple-Ziv (L-Z) complexity (Lempel & Ziv, 1976): The L-Z complexity measures the predictability of a signal. To compute the L-Z complexity for signal S , first, the minimum and the maximum values of signal points were calculated and then, the signal was quantized into 100 equally spaced levels between its minimum and maximum values. Then, the quantized signal, $B_1^n = \{b_1, b_2, \dots, b_n\}$, was decomposed into T different blocks, $B_1^n = \{\psi_1, \psi_2, \dots, \psi_T\}$. A block ψ was defined as

$$\Psi = B_j^\ell = \{b_j, b_{j+1}, \dots, b_\ell\}, 1 \leq j \leq \ell \leq n. \quad (28)$$

The values of the blocks can be calculated as follows:

$$\Psi = \begin{cases} \psi_m = b_1, & \text{if } m=1, \\ \psi_{m+1} = B_{h_m+1}^{h_{m+1}}, & m \geq 1, \end{cases} \quad (29)$$

where h_m is the ending index for ψ_m , such that ψ_{m+1} is a unique sequence of minimal length within the sequence $B_1^{h_{m+1}-1}$. Finally, the normalized L-Z complexity was calculated as

$$LZ = \frac{T \log_{100} n}{n}. \quad (30)$$

3.3 Classification

We trained 3 separate back-propagation neural network (NN) classifiers (Duda et al., 2000), one for each genre of signal feature outlined above. Hence, the feature space dimensionalities

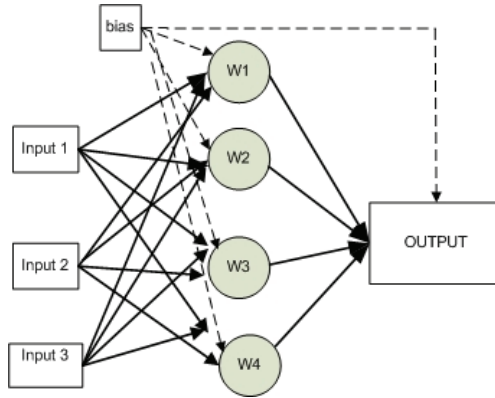


Fig. 1. The schematic of back-propagation neural network with 3 inputs and 4 hidden layers

for the classifiers were 4 (NN with time features), 3 (NN with frequency features) and 3 (NN with information-theoretic features). Each neural net classifier had 2 inputs, 4 hidden units and 1 output. Figure 1 shows the schematic of one NN classifier used in our work. Although it is possible to invoke different classifiers for each genre of signal feature, we utilized the same classifiers here to facilitate the evaluation of local decisions. The use of different feature sets for each classifier ensures that the classifiers will perform independently (Xu et al., 1992).

Figure 2 is a block diagram of our proposed algorithm. First, the three *small* neural networks, classify their inputs independently. Then, using the outputs of these classifiers and their respective reputation values, the reputation-based algorithm determines the correct label of the input.

Classifier accuracy was estimated via a 10-fold cross validation with a 90-10 split. However, unlike classical cross-validation, we further segmented the 'training' set into an actual training set and a validation set. In other words, in each fold, 160 (80%) swallows were used for training, 20 (10%) for validation and 20 (10%) reserved for testing. Among the 20 swallows of the validation set, 10 were used as a traditional validation set and 10 were used for computation of the reputation values. After training, classifier reputations were estimated using this second validation set. Classifiers were then ranked according to their reputation values. Without loss of generality, assume $r_1 \geq r_2 \geq r_3$. If θ_1 and θ_2 cast the same vote about a test swallow, their common decision was accepted as the final classification. However, if they voted differently, the a posteriori probability of each class was computed using (14) and the maximum a posteriori probability rule was applied to select the final classification.

To better understand the difference between the multiple classifier system and a single, all encompassing classifier, we also trained a multilayer neural network via back-propagation with all 10 features, i.e., using the collective inputs of all three smaller classifiers. This all encompassing classifier, from hereon referred to as the *grand* classifier, also had 4 hidden units. We also statistically compared the accuracies of the individual classifiers against

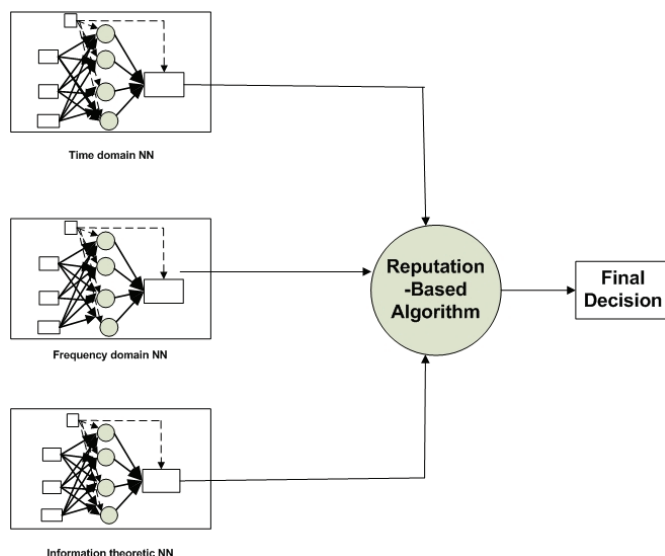


Fig. 2. The block diagram of the proposed algorithm

those of a majority vote classifier combination and a reputation-based classifier combination (Section 4.1).

To understand the knowledge representation of the individual small classifiers, we plotted Hinton diagrams for the input-to-hidden unit weight matrices (Section 4.2). Subsequently, we qualitatively compared the training performance of the small and grand classifiers to ascertain potential benefits of training a collection of small classifiers (Section 4.3). Through a systematic perturbation analysis, we quantified the local robustness of the reputation-based neural network combination (Section 4.4). In particular, we qualitatively examined the change in the classifier output and accuracy as the magnitude of the input perturbation increased. Finally, we estimated the breakdown point of the reputation-based neural network combination using increasing proportion of contaminants in the overall data set (Section 4.5).

4. Results and discussion

4.1 Classification accuracy

Table 1 tabulates the local and global classification results. On average, the frequency domain classifier appears best among the individual NNs while the information-theoretic NN fares worst. Also, it is clear from this table that by combining the local decisions of the classifiers, using reputation based algorithm, the overall performance of the system increases dramatically. The result of the grand classifier is statistically the same as the small classifiers. However, training this classifier is more difficult and requires more time. Hence, there appears to be no justification of considering an all encompassing classifier in this application. Collectively, these results indicate that there is merit in combining neural network classifiers

Classifier	Average Performance (%)
Time domain NN	67.5 ± 12.5
Frequency domain NN	69.5 ± 10.3
Information theoretic NN	65.5 ± 11.2
Grand classifier	70.0 ± 8.5
Majority vote	74.5 ± 8.9
Combined classifier decision	78.0 ± 8.2

Table 1. The average performance of the individual classifiers and their reputation-based combination.

in this problem domain. The accuracy of the majority vote neural network combination did not significantly differ from that of the individual ($p > 0.11$) and grand classifiers ($p = 0.16$). On the other hand, the reputation-based combination led to further improvement in accuracy over the time domain ($p = 0.04$) and information-theoretic ($p = 0.05$) classifiers, but did not significantly surpass the grand ($p = 0.09$) and frequency domain networks ($p = 0.09$).

The reputation-based scheme yields accuracies better than those reported in (Lee et al., Under review) (74.7%). Moreover, in (Lee et al., Under review), the entire database was required and the maximum feature space dimension was 12. Here, we only considered a fraction of the database and no classifier had a feature space dimensionality greater than 4. Therefore, our system offers the advantages of computational efficiency and less stringent demands on training data.

4.2 Internal neural network representations

Figures 3, 4 and 5 are the Hinton graphs for the input to hidden layer weight matrices for the time, frequency, and information theoretic domain neural networks, respectively. In these figures, the weight matrix of each classifier is represented using a grid of squares. The area of each square represents the weight magnitude, while the shading reveals the sign. Shaded squares signify negative weights. The first column denotes the hidden unit biases while the subsequent columns are the weights on the connections projecting from each input unit. For instance, the frequency domain neural network uses 3 input features and 1 bias, resulting in 4 columns. Given that there are 4 hidden units, the weight matrix is represented as a 4×4 grid. In Figure 3, we see that the first neuron has a very large negative weight for kurtosis and a sizable one for variance. This suggests that this neuron represents swallows with low variance and platykurtic distributions. The second neuron seems to primarily represent swallows with leptokurtic distributions given its positive weight on the kurtosis input. By the same token, then third neuron appears to internally denote swallows with large positive means and leptokurtic distributions. Finally, the last neuron captures swallows primarily with high variance. Overall, the strongest weights are found on variance and kurtosis features, suggesting that they play the most important role in distinguishing between healthy and unhealthy swallows in our sample. Resonating with our findings here, Lee et al.

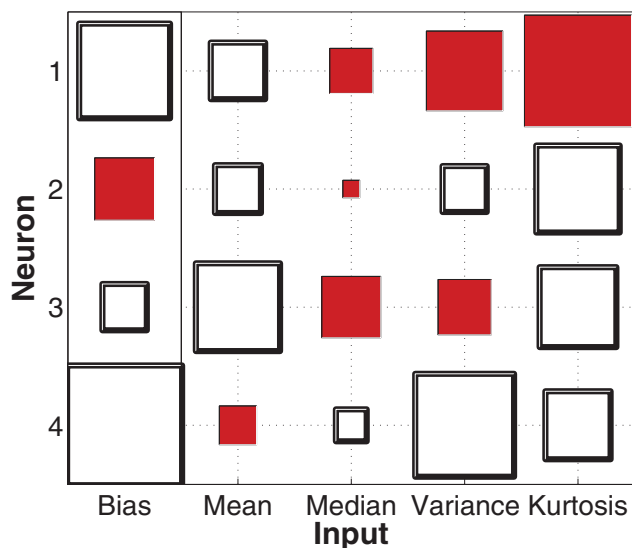


Fig. 3. The Hinton graph of the weight matrix for the time domain classifier

(2006) identified a dispersion-type measure as discriminatory between healthy swallows and swallows. Similarly, Lee et al. (2009) determined that the kurtosis of swallow accelerometry signals tended to be axial-specific and thus potentially discriminatory between different types of swallows.

Moving on to Figure 4, we notice that neurons one and two seem to have captured inverse dependencies of spectral centroid and bandwidth features. While neuron one embodies swallows with lower spectral centroid but broad bandwidth, neuron two captures swallows high frequency narrow band swallows. The peak FFT feature seems to be the least important spectral information, which is consoling in some senses as this suggests that decisions are not based upon signal strength, which may vary greatly across swallows regardless of swallowing health.

In the information theoretic neural network (Figure 5), we find that the memory feature seems to have a distributed representation across the 4 neurons, with three favoring weak memory or rapidly decaying autocorrelations. Neuron one almost uniformly considers all three information theoretic features, specifically epitomizing swallows with low complexity, low entropy rate and minimal memory. This characterization might reflect 'noisy' swallows. Interestingly, neuron three focuses on positive complexity and memory. We can interpret this neuron as representing swallows which have strong predictability and hence longer memory. In short, it appears that each individual neural network has internally represented some unique flavors of swallows. This apportioned representation across neural networks suggests that there is indeed sound rationale to combine classifiers, in order to comprehensively characterize the diversity of swallows.

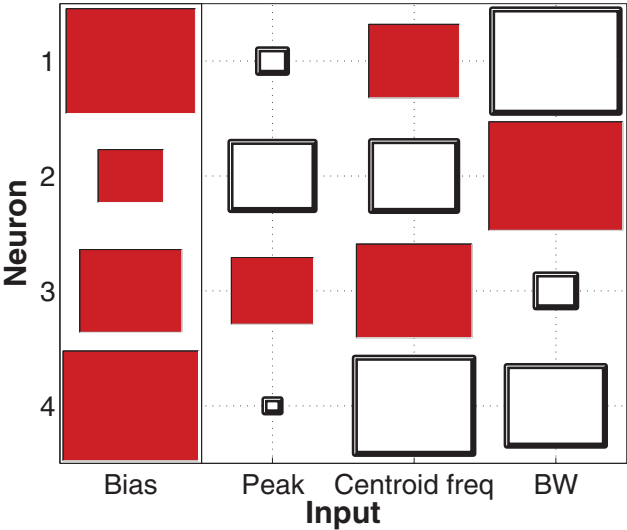


Fig. 4. The Hinton graph of the weight matrix for the frequency domain classifier (Peak - peak value of FFT; BW - bandwidth)

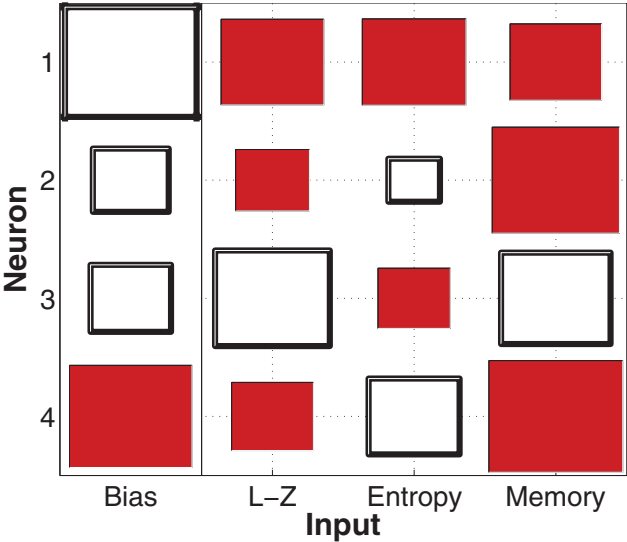


Fig. 5. The Hinton graph of the weight matrix for the information theoretic classifier.

4.3 Training error and convergence

Figure 6 pits the training performance of the small classifiers against the grand classifier as the number of training epochs increase. After 12 epochs, the small classifiers have lower

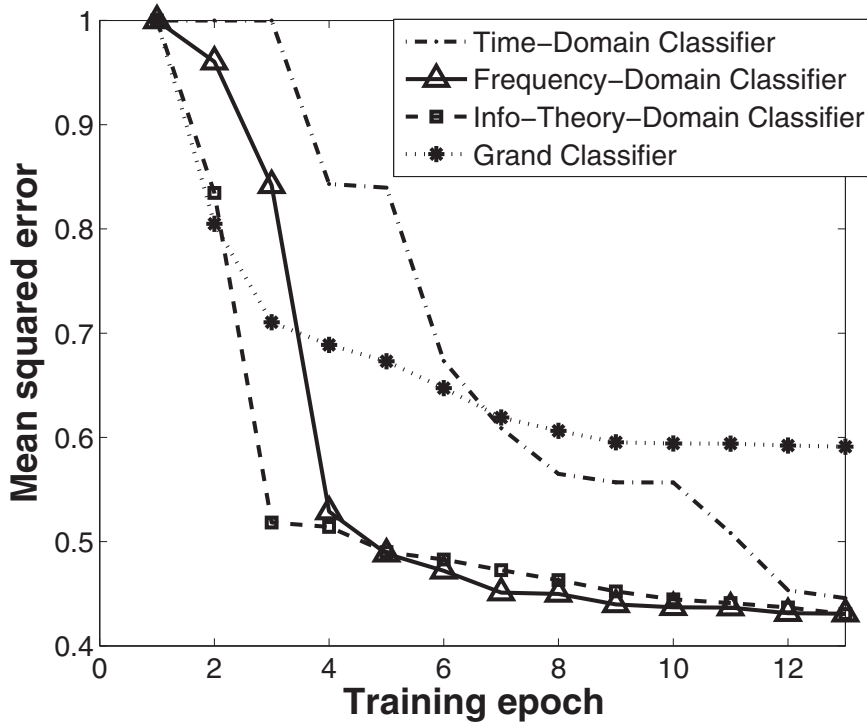


Fig. 6. The training error of the different neural network classifiers versus the number of training epochs

normalized mean squared errors than the grand classifier. This is one of the main advantages of using a multiple classifier system over a single all encompassing classifier; the rate of convergence during training is often faster with smaller classifiers, i.e., those with fewer input features, and in many cases lower training error can be achieved.

4.4 Local robustness

To gauge one aspect of the local robustness of the proposed neural network combination, we measured the sensitivity of the system to a local perturbation of the input. Recall that the reputation algorithm yields a class label rather than a continuous number. Thus, to facilitate sensitivity analysis, we calculated the reputation-weighted average of the outputs of the small classifiers for a specific input. For semantic convenience, we will just call this the reputation-weighted output. The unperturbed input sample was the mean vector of all the features in the test set. Perturbed inputs were created by adding varying degrees of positive and negative offsets to every feature of the mean vector. The sensitivity of the system to a given perturbation was defined as the difference between the reputation-weighted output for the unperturbed input and that for the perturbed input. At each iteration, the amount of perturbation was proportional to the range of the features in the test set. For instance, in the first iteration, 5% of the range of each feature in the test set was added to the respective

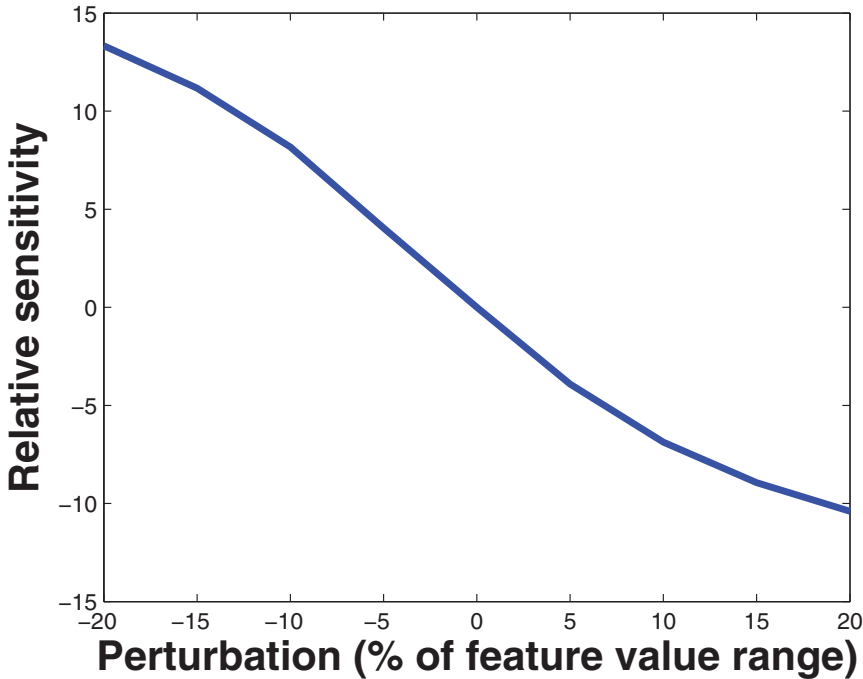


Fig. 7. Sensitivity to varying magnitudes of perturbation of the input vector

feature. Figure 7 shows the relative sensitivity of the system versus the magnitude of input perturbation. Between $\pm 10\%$, the relative sensitivity is less than 10% of the output value, suggesting that the reputation-based classifier while not robust in the strict statistical sense, can tolerate a modest level perturbation at the inputs.

To examine the effect of a local perturbation on the final decision of our algorithm, we again added/subtracted noise to the mean input vector and computed the output label using the reputation-based algorithm. For the present problem, the output was binary and without loss of generality, denoted arbitrarily as '1' or '2'. The unperturbed input belonged to class 1. As portrayed in Figure 8, the decision of the proposed algorithm is robust to negative perturbations up to 20% of the range of the features and positive perturbations up to 10% of the range of the features. However, for a positive perturbation higher than 10% of the range of the features, the reputation algorithm misclassifies the input. For practical purposes, this means that the reputation-based neural network combination can tolerate a simultaneous 10% perturbation in all its input features before making a wrong decision in the binary classification case. In the specific domain of dual-axis accelerometry, head movement induces high magnitude perturbations Sejdic, Steele & Chau (2010) which, according to our analysis here, will likely cause classification errors.

We also investigated the effect of local perturbations on the accuracy of the proposed algorithm. We perturbed all 20 samples in the test set. The amount of perturbation ranged from 0 to 100% of the range of the features in the test set. For each contamination value,

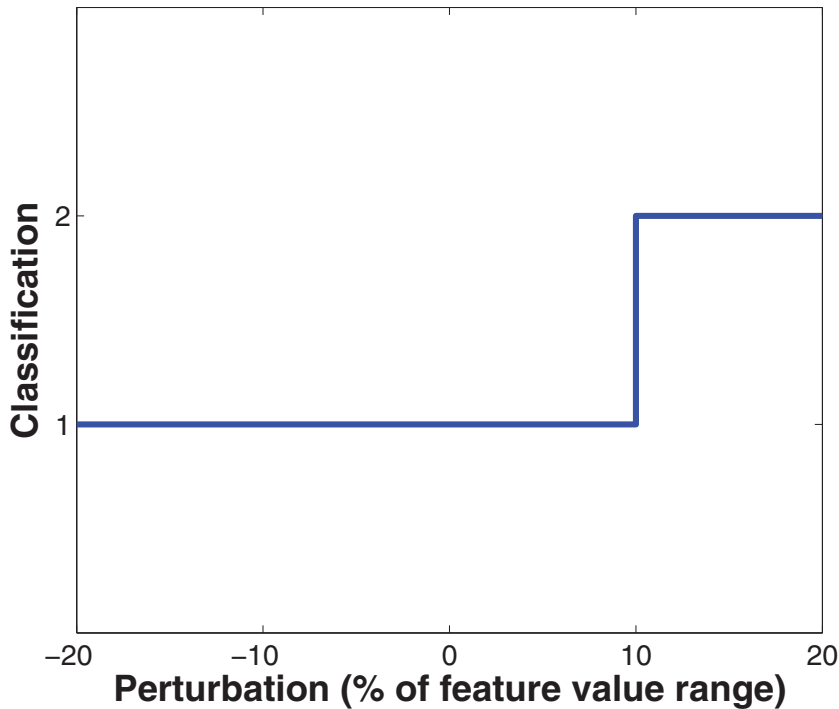


Fig. 8. The output decision versus magnitude of input perturbation

we calculated the accuracy of the proposed algorithm using the perturbed test set. Figure 9 illustrates the effect of varying levels of perturbation on the accuracy of the proposed algorithm. Based on this figure, the accuracy of the proposed algorithm decreased with increasing magnitude of perturbation in the test set. The initial accuracy of the proposed algorithm, for the unperturbed test set, was 78% and decreased to 50% for full-range (100% of the range of the features) perturbations. It is interesting to note that the decay in accuracy is quite steep for the first 20%, indicating that accuracy will take a hit with any non-zero amount of perturbation. Intuitively, this finding makes sense as the resemblance between test and training data diminishes as the magnitude of perturbation increases.

4.5 Global robustness

The sensitivity curve only offers local information about the robustness of the classifier. To measure the robustness of the system globally, we estimated the *breakdown point* for the proposed algorithm. For this matter, we substituted some feature vectors from among the 200 initial samples with contaminated versions. Contaminated feature vectors were created by sampling from a normal distribution with mean equal to that of the feature vector but with 3 times the standard deviation. The number of contaminants ranged from 20 to 100, i.e., 10 to 50% of the original data set. Using 10-fold cross validation, we divided the samples into 3 sets: training, testing, and validation. Therefore, it was possible that contaminations

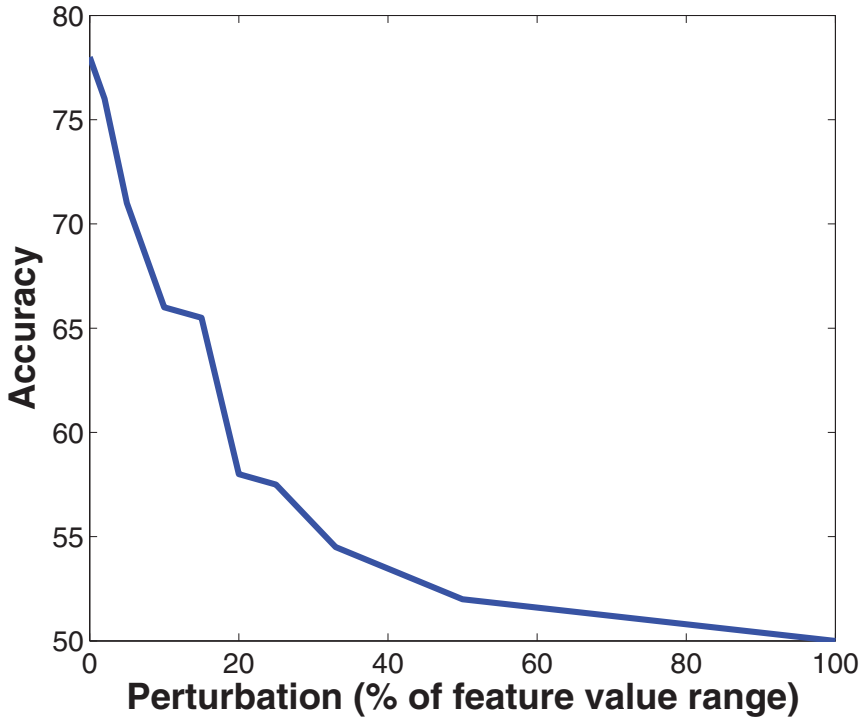


Fig. 9. The accuracy of the proposed classifier with increasing magnitude of perturbation in the test set

appeared in any or all of the training, testing, and validation sets. Figure 10 plots the accuracy of the proposed algorithm for different numbers of contaminated samples. Error-bars in this figure depict the standard deviation of each accuracy obtained from the cross-validation. To estimate the breakdown point for this system, we used the *rank sum test* to test for a significant difference between accuracies with and without varying levels of contamination. At a 5% significance level, we identified the first significant departure from the uncontaminated distribution of accuracy at 80 contaminated samples ($p = 0.043$). Given that there were 200 samples, the breakdown point was thus identified as $\frac{80}{200} = 0.4$.

5. Conclusion

We have presented the formulation of a reputation-based neural network combination. The method was demonstrated using a dysphagia dataset. We noted that generally the reputation-based classifier either achieved higher accuracies than single classifiers or exhibited comparable accuracies to the best single classifiers. Interpreting the weight matrices of the neural networks, we observed that many different aspects of time, frequency and information-theoretic characteristics of swallows were encoded. Finally, we empirically characterized the local and global robustness of the reputation-based classifier, showing that

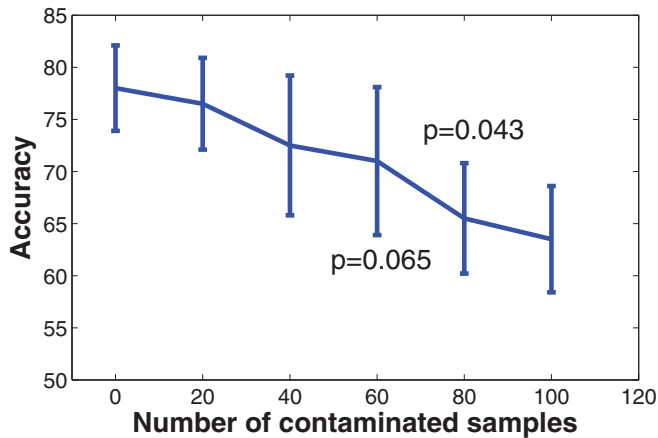


Fig. 10. The accuracy of the proposed classifier as the number of contaminated samples increase. The p-values arise from a comparison of the accuracy between the uncontaminated sample and samples with varying levels of contamination.

there exists a certain tolerance (approximately 10% of the range of a feature value) to input perturbations. However, large magnitude perturbations, such as those observed in head movement, would likely lead to erroneous classification of the swallowing accelerometry input feature vector.

6. References

- Bauer, E. & Kohavi, R. (1999). An empirical comparison of voting classification algorithms: Bagging, boosting, and variants, *Machine learning* 36(1): 105–139.
- Bloch, I. (2002). Information combination operators for data fusion: A comparative review with classification, *Systems, Man and Cybernetics, Part A: Systems and Humans, IEEE Transactions on* 26(1): 52–67.
- Breiman, L. (1996a). Bagging predictors, *Machine learning* 24(2): 123–140.
- Breiman, L. (1996b). Bagging predictors, *Machine Learning* 24(2): 123–140.
- Chen, H. & Yao, X. (2009). Regularized negative correlation learning for neural network ensembles, *Neural Networks, IEEE Transactions on* 20(12): 1962–1979.
- Damouras, S., Sejdic, E., Steele, C. & Chau, T. (2010). An on-line swallow detection algorithm based on the quadratic variation of dual-axis accelerometry, *IEEE Transactions on Signal Processing* 58(6): 3352–3359.
- Duda, R., Hart, P. & Stork, D. (2000). *Pattern Classification*, 2nd edn, Wiley-Interscience.
- Haykin, S. (1994). *Neural Networks*, Macmillan College Publishing Company, New York.
- Hull, J., Srihari, N., Cohen, E., Kuan, C., Cullen, P. & Palumbo, P. (1988). A blackboard-based approach to handwritten zip code recognition, *Proceedings of the US Postal Service Advanced Technology Conference*, Washington, DC, pp. 1018–1032.
- Jacob, R., Jordan, M., Nowlan, S. & Hinton, G. (1991). Adaptive mixtures of local experts, *Neural Computation* 3(5): 79–87.

- Jacobs, R., Jordan, M., Nowlan, S. & Hinton, G. (1991). Adaptive mixtures of local experts, *Neural computation* 3(1): 79–87.
- Jain, A., Duin, R. & Mao, J. (2000). Statistical pattern recognition: A review, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 22(1): 4–37.
- Kuncheva, L. (2002). A theoretical study on six classifier fusion strategies, *Pattern Analysis and Machine Intelligence, IEEE Transactions on* 24(2): 281–286.
- Lam, L. & Suen, C. (1997). Application of majority voting to pattern recognition: An analysis of its behavior and performance, *IEEE Transactions on Systems, Man, and Cybernetics* 27(5): 553–568.
- Lee, D. & Srihari, S. (1993). Handwritten digit recognition: A comparison of algorithms, *3rd Intl. Workshop Frontiers Handwriting Recognition*, pp. 153–162.
- Lee, J., Blain, S., Casas, M., Berall, G., Kenny, D. & Chau, T. (2006). A radial basis classifier for the automatic detection of aspiration in children with dysphagia, *Journal of Neuroengineering and Rehabilitation* 3(14): 1–17.
- Lee, J., Sejdic, E., Steele, C. & Chau, T. (2010). Effects of liquid stimuli on dual-axis swallowing accelerometry signals in a healthy population, *Biomedical Engineering OnLine* 9(7): 10 pp.
- Lee, J., Steele, C. & Chau, T. (2008). Time and time-frequency characterization of dual-axis swallowing accelerometry signals, *Physiological Measurement* 29(9): 1105–1120.
- Lee, J., Steele, C. & Chau, T. (2009). Swallow segmentation with artificial neural networks and multi-sensor fusion, *Medical Engineering & Physics* 31(9): 1049–1055.
- Lee, J., Steele, C. & Chau, T. (Under review). Classification of healthy and abnormal swallows based on accelerometry and nasal airflow signals, *Artificial Intelligence in Medicine* xx(yy): zz.
- Lempel, A. & Ziv, J. (1976). On the complexity of finite sequences, *IEEE Transactions on Information Theory* 22: 75–81.
- Nadal, C., Legault, R. & Suen, C. (1990). Complementary algorithms for the recognition of totally unconstrained handwritten numerals, *10th Int. Conf. Pattern Recognition* A: 434–449.
- Nikjoo, M., Steele, C. & Chau, T. (2011). Automatic discrimination between safe and unsafe swallowing using a reputation-based classifier, *IEEE Transactions on Biomedical Engineering* submitted(?): ?
- Porta, A., Guzzetti, S., Montano, N., Furlan, R., Pagani, M., Malliani, A. & Cerutti, S. (2001). Entropy, entropy rate, and pattern classification as tools to typify complexity in short heart period variability series, *IEEE Transactions on Biomedical Engineering* 48(11): 1282–1291.
- Schapire, R. (1990). The strength of weak learnability, *Machine Learning* 5(2): 197–227.
- Sejdic, E., Komisar, V., Steele, C. & Chau, T. (2010). Baseline characteristics of dual-axis cervical accelerometry signals, *Annals of Biomedical Engineering* 38(3): 1048–1059.
- Sejdic, E., Steele, C. & Chau, T. (2010). The effects of head movement on dual-axis cervical accelerometry signals, *BMC Research Notes* ?(?): in press.
- Suen, C., Nadal, C., Mai, T., Legault, R. & Lam, L. (1990). Recognition of totally unconstrained handwritten numerals based on the concept of multiple experts, in C. Suen (ed.), *Proceedings of the International Workshop on Frontiers in Handwriting Recognition*, pp. 131–143.

- Tresp, V. & Taniguchi, M. (1995). Combining estimators using non-constant weighting functions, in G. Tesauro, D. Touretzky & T. Leen (eds), *Advances in Neural Information Processing Systems*, Vol. 7, MIT Press, Cambridge, MA, pp. 418–435.
- Ueda, N. (2002). Optimal linear combination of neural networks for improving classification performance, *Pattern Analysis and Machine Intelligence, IEEE Transactions on* 22(2): 207–215.
- Xu, L., Kryzak, A. & Suen, C. (1992). Methods of combining multiple classifiers and their applications to handwriting recognition, *IEEE Transactions on Systems, Man and Cybernetics* 22(3): 418–435.
- Yao, X. & Islam, M. (2008). Evolving artificial neural network ensembles, *IEEE Computational Intelligence Magazine* 3(1): 31–42.

Part 3

Biological Applications

Prioritising Genes with an Artificial Neural Network Comprising Medical Documents to Accelerate Positional Cloning in Biological Research

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1. Introduction

Linkage analysis is used to identify genes with a certain phenotype or genetic defect and determine chromosomal intervals containing several tens to hundreds of candidate genes for positional cloning. Before further experiments are performed, the candidate genes must be prioritised using as much biological knowledge as possible. For this purpose, it is an ambitious challenge to create an artificial superbrain that has learned a vast amount of knowledge stored as various omics data.

As conventionally such omics data have been published in different data structures and semantics, a data platform that integrates heterogeneous data into a single machine-readable data set is desired. The Semantic Web is a framework for knowledge description and discovery by inferences; it uses relationships given as semantic links between two entities denoted by Uniform Resource Identifiers (URIs) (Berners-Lee et al., 2001). A goal of the Semantic Web is the realisation of human-machine communication by adding metadata describing the semantic links of entities based on Resource Description Framework (RDF) (Manola & Miller, 2004). In biomedical fields, some datasets using common ontologies shared by people on the Semantic Web, such as the Gene Ontology (Ashburner et al., 2000, The Gene Ontology Consortium, 2006) and the uniprot RDF (<http://dev.isb-sib.ch/projects/uniprot-rdf/>) have been published in RDF. However, because the task of generating consistent RDF triples (subject, predicate and object) against a vast amount of biomedical content is too expensive, an information space that covers our entire exhaustive biomedical knowledge on the Semantic Web has not been realised.

For the practical use of published biomedical data in the Semantic Web, especially of the data that is difficult to utilise because of the lack of semantic links, it is beneficial to reinforce the acquisition of such data by supplying a hybrid methodology combining not only inferences over the knowledge described with RDF but also those supported by statistical significance over multiple raw documents. For instance, MEDLINE, a biomedical document repository, includes more than 18 million reports; the entire knowledge amount of represented by MEDLINE cannot be consistently reconstructed as well-formed ontology-based knowledge.

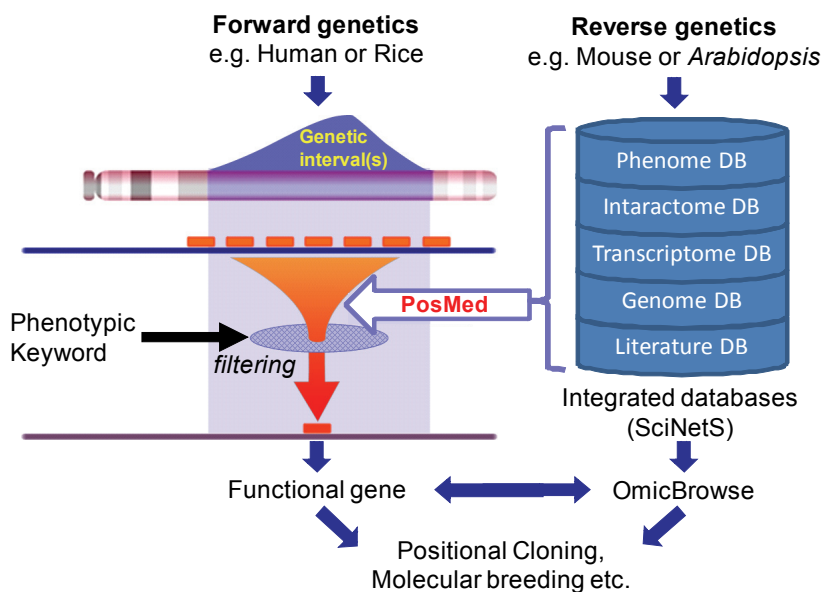


Fig. 1. PosMed accelerates forward genetics gene discoveries (left) by integrating the omics knowledge collected from reverse genetics as an integrated databases named Scientists' Network System (SciNetS; Masuya et al., 2010) (right). PosMed helps users narrow down the candidate responsible genes from those existing within chromosomal intervals. OmicBrowse (Matsushima et al., 2009; Toyoda et al., 2007) helps users look into every piece of detailed information for each candidate gene. The entire system is designed to coherently support positional cloning studies, plant molecular breeding research and plant-upgrading science.

However, the current framework of the Semantic Web cannot handle numerical criteria such as relationship strength or the result of statistical tests of relationships. To effectively use both well-formed RDF datasets and a vast number of biomedical documents, the extension of current query languages should support not only Boolean relationships but also statistically evaluated relationships.

In this chapter, we will discuss the development of a web-based tool named Positional Medline (PosMed) that can immediately suggest genes related to a certain phenotype by accessing a Semantic Web based databases over omics entities named the Scientists' Networking System (SciNetS; Masuya et al., 2010) and document databases (Fig. 1). We initially developed a semantic link database for each entity, which holds relationships between the entity and other entities, including documents such as orthologue relationships and document co-citation relationships. We then developed a search engine named General and Rapid Association Study Engine (GRASE) and an associated query language named General and Rapid Association Study Query Language (GRASQL) (Kobayashi & Toyoda, 2008). GRASQL is a powerful language for expressing the statistical analysis of data retrievable by the RDF query language SPARQL (Prud'Hommeaux & Seaborne, 2008) in a Semantic Web manner. The current implementation of GRASE is optimised to efficiently calculate the statistical prioritisation of candidate genes on the bases of more than 18 million

medical and biological documents and to facilitate quick return of the results within a few seconds of computational time.

Several software tools that have been developed for prioritising positional candidate genes are based on functional annotation, gene expression patterns, protein-protein interaction (PPI) and/or sequence-based features (Adie et al., 2006; Aerts et al., 2006; GeneSniffer; Köhler et al., 2008; Seelow et al., 2008; Van Driel et al., 2005). The evaluation of two of these tools and PosMed using their data sets has demonstrated the effectiveness of PosMed, which showed an accuracy of 88.7%, the highest among the three tools (Thornblad et al., 2007).

Currently, PosMed supports prioritisation of candidate genes for positional cloning in the human, mouse and rat, and prioritisation of other entities not having genomic positions such as metabolites, drugs, diseases and researchers (Yoshida et al., 2009). Further, a plant service version of PosMed named Positional MEDLINE for plant-upgrading science (PosMed-plus)¹ was implemented as the first cross-species integrated database that inferentially prioritises candidate genes for forward genetics approaches in plant science supporting Arabidopsis and rice (Makita et al., 2009).

PosMed and PosMed-plus are available at <http://omicspace.riken.jp/>.

2. Data model for gene prioritisation in PosMed

2.1 Neural Network representation of statistical algorithm for searching complex semantic web data

PosMed prioritises candidate genes for positional cloning by employing our original database search engine GRASE. As an example of this prioritisation against mouse genes, GRASE is used to execute an inferential process similar to that of an artificial neural network comprising documental neurons (or 'documentrons') that represent each document contained in databases such as MEDLINE (Fig. 2). Given a user-specified query, PosMed initially performs a full-text search of each documentron in the first-layer artificial neurons and then calculates the statistical significance of the connections between the hit documents and the second-layer artificial neurons representing each mouse gene. When a chromosomal interval(s) in mice is specified, PosMed explores the second- and third-layer artificial neurons representing genes within the chromosomal interval by evaluating the combined significance of the connections from the hit documentrons to the genes.

When a chromosomal interval(s) in human is specified, PosMed further explores the fourth-layer artificial neurons representing human genes within the chromosomal interval by using orthologous correspondences between mouse genes and human genes. For the output, PosMed displays the ranked genes with evidence documents in which the user's keyword is highlighted.

PosMed is, therefore, a powerful tool that immediately ranks the candidate genes by connecting them to user's keywords, with connections representing both gene-gene interactions and other biological interactions such as metabolite-gene, drug-gene, disease-gene, phenotype-gene, subcellular localisation-gene, co-expression, PPI, and orthologue and paralogue data. By using orthologous and paralogous connections, PosMed facilitates the ranking of genes based on evidence found in other species.

¹ In this article, PosMed-plus is included in PosMed unless otherwise stated.

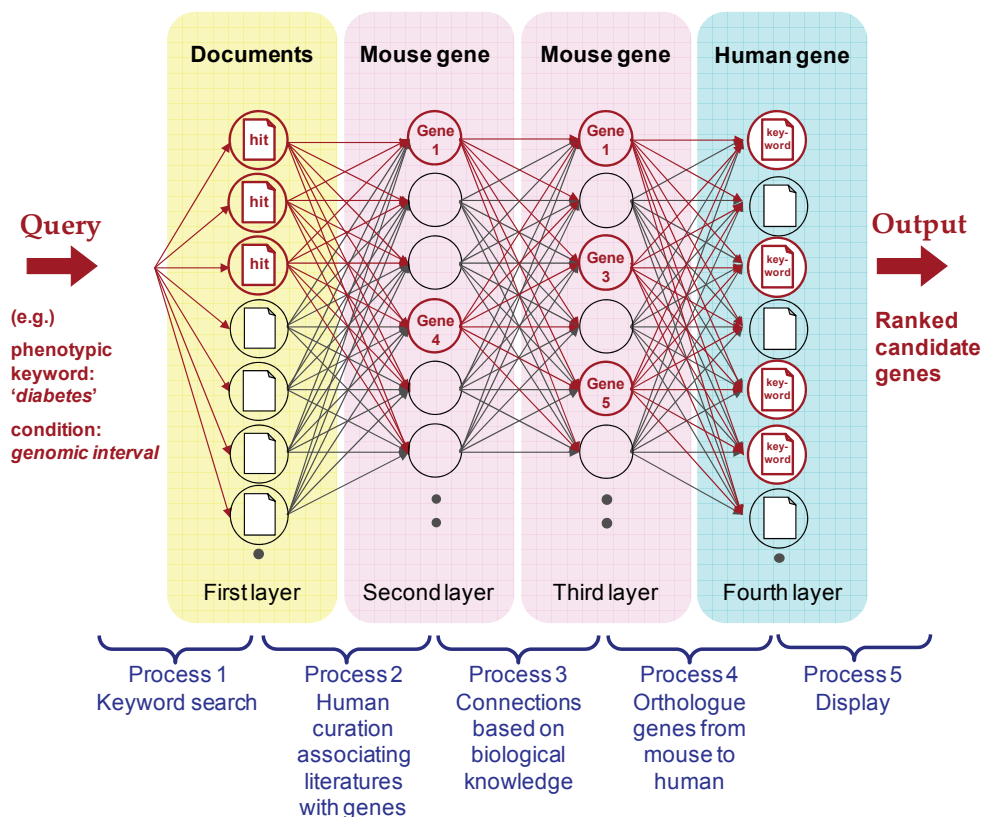
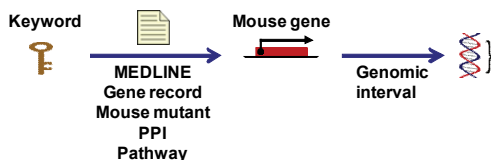


Fig. 2. Neural network model for the PosMed gene search algorithm. As an example, the user's keyword 'diabetes' can be found in several documents, including some in MEDLINE (Process 1). These documents are mapped to genes that are supported by manual curation (Process 2). Using biological knowledge (e.g. protein-protein interaction and co-citation of document sets), PosMed can also suggest genes that do not have the user's keyword 'diabetes' in their associated documents (Process 3). PosMed then returns the candidate genes that are located within the user's specified genomic interval using orthologous relationships (Process 4). Thereafter, the resultant genes are displayed with documents in which the user's keyword is highlighted (Process 5).

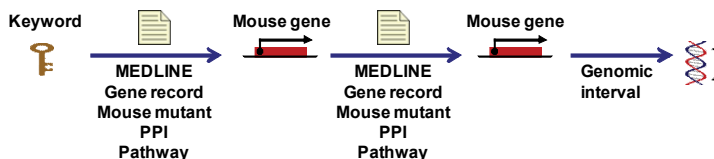
2.2 Manual curation work connecting genes to the literature

The accuracy of PosMed is strongly correlated with its ability to make correct associations between genes and documents. This is because GRASE uses these associations to execute direct searches and inference searches that are supported by co-citations. To increase the accuracy of PosMed, we employed manual curation to connect genes and papers by semantic links. Our original curation method is based on named entity recognition (NER; see Section 4.2. for details). Rather than connecting every literature reference to genes, specialised curators create search rules to retrieve all the correct references from document titles, abstracts and MeSH terms.

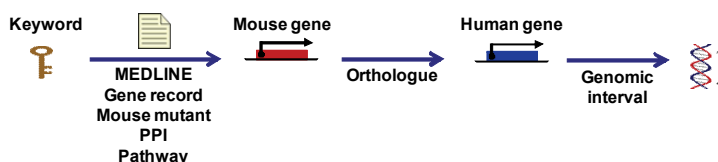
(A) Direct search



(B) Inference search



(C) Cross-species search



(D) Cross-species inference search

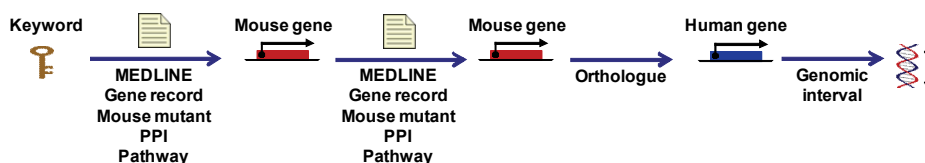


Fig. 3. Sequential data flow representation of PosMed search paths. (A and B) Data flow of PosMed search and comparison of direct search and inference search, respectively. (C and D) Data flow of PosMed cross-species searches.

2.3 Search paths of the PosMed Neural Network

Using the search functionalities of GRASE, PosMed supports the following four types of search:

- i. Direct search: GRASE searches genes located in the user's chromosomal interval by performing a full-text search against the set of databases with the user's keyword; i.e. the following search path is realised: *keyword* → *document* → *gene* → *chromosomal interval* (Fig. 3(A)).
- ii. Inference search: By applying gene-gene relationships over the genes extracted by a direct search outside the user's chromosomal interval, GRASE discovers further genes that are indirectly related to the keyword via gene-gene relationships; i.e. the following

search path is realised: $keyword \rightarrow document \rightarrow gene_1 \rightarrow gene_2 \rightarrow chromosomal \ interval$.

The link between $gene_1$ and $gene_2$ is supported by omics data (Fig. 3(B)).

- iii. Cross-species search: This is an extension of the direct search (i) to the human genome. The connections from mouse genes to human genes are supported by orthologue data (Fig. 3(C)).
- iv. Cross-species inference search: This is an extension of the inference search (ii) to the human genome. As for (iii) above, orthologue data connect mouse genes to human genes (Fig. 3(D)).

In the final stage, these types of search result are integrated into a ranked gene list by species.

3. Statistical query language and its processor

As mentioned above, the core data processing software component of PosMed, which performs statistical inference searching, is GRASE. GRASE is implemented as a prototype of a language processor that interprets and executes a program written in a query language named GRASQL. GRASQL is our extension of SPARQL, which is highly rated because it seems intuitively understandable for typical biologists who are not familiar with programming languages, but does not adequately support statistical evaluation of semantic links.

GRASQL is designed as a language for ranking resultant entities such as genes to discover entities statistically associated with a user's keyword. It does this by considering statistical values computed for each entity on the basis of sets of RDF triples hit by RDF graph pattern matching. In the rest of this section, we present an overview of GRASQL and discuss the programs in PosMed that use it.

3.1 Overview of GRASQL

We start with researcher ranking problems as introductory programs in GRASQL to show how a statistical evaluation is integrated with the existing RDF search.

The first example is researcher ranking using an index called the h index to characterise a researcher's scientific output (Hirsch, 2005). The h index is introduced as follows: 'A scientist has index h if h of his or her N_p papers have at least h citations each, and the other $(N_p - h)$ papers have at most h citations each', where N_p is the number of papers published over n years. Figure 4 shows a GRASQL query of this problem, which uses MEDLINE abstracts and citation relationships.

First, we obtain a set $\overline{d_r}$ of documents published by each researcher r in the MEDLINE abstracts published in 2005 or later, and then documents $\overline{c_d}$ that each cite document $d \in \overline{d_r}$. This search implements the RDF graph pattern matching specified in the WHERE clause in Fig. 4, as shown in Fig. 5. Then, we compute the h index for each researcher. This statistical step cannot be realised with RDF graph pattern matching unless an external procedure against the sequences of solutions obtained in the first step is used. In Fig. 4, the EVALUATE clause, which is newly introduced in GRASQL, specifies a method of computing the h index $?h$ for each researcher $?researcher$ using the external statistical function `ris:hIndex` given in Fig. 6. Since MEDLINE does not contain citation information, the `rip:hasCitation` links should be generated on the bases of other resources, such as Google Scholar (Noruzi, 2005).

```

@prefix rio: <http://omicspace.riken.jp/GRASQL/>
@prefix rip: <http://omicspace.riken.jp/GRASQL/predicate>
@prefix ris: <http://omicspace.riken.jp/GRASQL/statistics>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@let %documentSet rio:MEDLINE
@let %researcher rio:Researcher
SELECT ?researcher ?h
WHERE {
    ?researcher    rdf:type          rio:Researcher ;
                  rip:hasDocument   ?doc .
    ?docCite       rip:hasCitation   ?doc ;
                  rdf:type          %documentSet .
    ?doc           rip:publishYear   ?year ;
                  rdf:type          %documentSet .
    FILTER (?year >= 2005)
}
EVALUATE ?h FOR ?researcher {
    ?h = ris:hIndex([?doc,?docCite]);
}
ORDER BY DESC(?h)

```

Fig. 4. GRASQL query that ranks researchers by the h index using MEDLINE abstracts and citation relationships. The LET statement is written as @let %constantName value, where % constantName is the name of a constant that starts with %, and value is its constant value. The statistical function ris:hIndex in the EVALUATE clause is called for each ?researcher value containing the sequences of solutions obtained by RDF graph pattern matching specified in the WHERE clause. [?doc, ?docCite] is a sequence of pairs of ?doc and ?docCite included in the sub-sequences of solutions concerning the value of ?researcher when ris:hIndex is called.

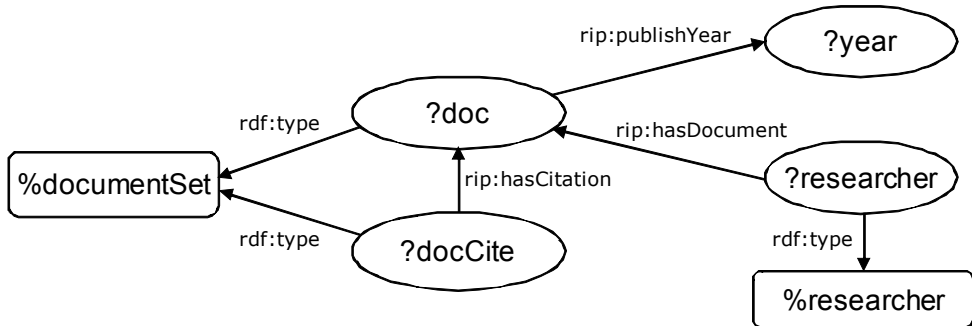


Fig. 5. RDF graph pattern specified in the WHERE clause in Fig. 4.

The second example is for ranking researchers in a topic specified with a keyword. That is, we would like to rank researchers by considering $N_{r,k}$ number of documents written by a researcher r including a keyword k . We call the number $N_{r,k}$ the k index of researcher r . Figure 7 shows a query for this example in GRASQL. The documents ?doc written by the researcher ?r including the keyword %keyword are obtained not only by RDF graph pattern matching but also by calling an external program specified in the WHERE clause

shown in Fig. 8. The predicate `rix:hasWord` of this example is used to call a full-text search engine for finding MEDLINE abstracts including the keyword, and the results are cached in

```
Statistic Function rix:hIndex(D): integer
{ Input D is a sequence of pair of document doc and document docCite
  which sites document doc.}
begin
  Count number Np of documents including the first element doc of D without overlap ;
  val h:=0 ;
  repeat
    begin
      h:=h+1;
      Count number L of documents cited at least h documents utilising D ;
      Count number M of documents cited at most h documents utilising D ;
      if (L ≥ h and (Np − h) ≤ M ) then break ;
    end ;
  return h ;
end ;
```

Fig. 6. Algorithm that implements the statistical function `rix:hIndex`, which computes the *h* index with a sequence of pairs of document *doc* and document *docCite* that cites document *doc*.

```
@prefix rio: <http://omicspace.riken.jp/GRASQL/>
@prefix rip: <http://omicspace.riken.jp/GRASQL/predicate>
@prefix rix: <http://omicspace.riken.jp/GRASQL/procedure>
@prefix ris: <http://omicspace.riken.jp/GRASQL/statistics>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@let %documentSet rio:MEDLINE
@let %researcher rio:Researcher
@let %keyword "arabidopsis"
SELECT ?researcher ?k
WHERE {
  ?researcher rdf:type %researcher ;
  rip:hasDocument ?doc .
  ?doc EXT:rix:hasWord %keyword ;
  rip:publishYear ?year ;
  rdf:type %documentSet .
  FILTER (?year >= 2005)
}
EVALUATE ?k FOR ?researcher {
  ?h = ris:hIndex([?doc,?docCite]);
}
ORDER BY DESC(?k)
```

Fig. 7. GRASQL query that ranks researchers by *k* index using MEDLINE abstracts.

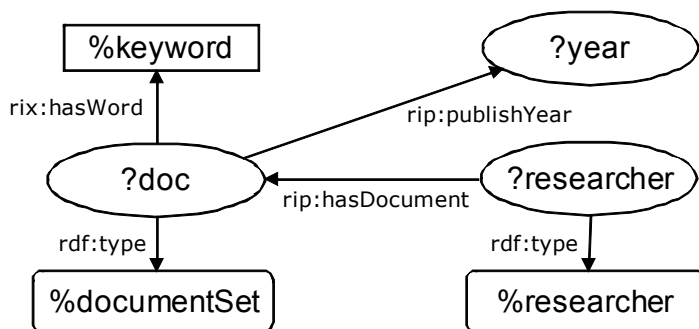


Fig. 8. RDF graph pattern specified in the WHERE clause in Fig. 7.

RDF graphs. Further, the k index, namely the number of documents ?doc without duplication for each researcher ?r, is computed by calling the statistical function ris:countDistinct in the EVALUATE clause.

```

@prefix rio: <http://omicspace.riken.jp/GRASQL/>
@prefix rip: <http://omicspace.riken.jp/GRASQL/predicate>
@prefix rix: <http://omicspace.riken.jp/GRASQL/procedure>
@prefix ris: <http://omicspace.riken.jp/GRASQL/statistics>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@let %keyword "type 2 diabetes"
@let %documentSet rio:MEDLINE
@let %geneSet rio:MouseGene
SELECT ?gene ?p
WHERE {
    ?gene                rip:hasDocument    ?docGene ;
                        rip:hasDocument    ?docIntersection ;
                        rdf:type           %geneSet .
    ?docKey              EXT:rix:hasWord    %keyword ;
                        rdf:type           %documentSet .
    ?docIntersection     EXT:rix:hasWord    %keyword ;
                        rdf:type           %documentSet .
    ?docGene             rdf:type           %documentSet .
    ?docAll              rdf:type           %documentSet .
}
EVALUATE ?p FOR ?gene {
    ?p = ris:statisticTest#FisherExactTest(?a,?b,?c,?d) ;
    ?a = count(DISTINCT ?docIntersection) ;
    ?b = count(DISTINCT ?docKey)-?a ;
    ?c = count(DISTINCT ?docGene)-?a ;
    ?d = count(DISTINCT ?docAll)-?a-?b-?c
}
ORDER BY ?p
    
```

Fig. 9. GRASQL query for a direct search using Fisher's exact test as a method of computing the statistical significance of the intersection ?docIntersection.

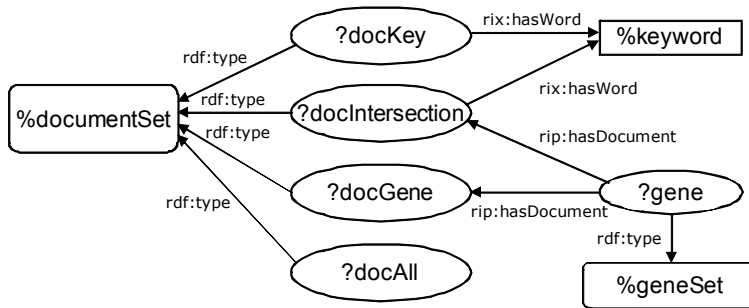


Fig. 10. RDF graph pattern satisfying the condition described in the WHERE clauses shown in Figs. 9 and 12.

3.2 Statistical tests in the PosMed search

Our method discovers entities significantly related to a user's keyword by using the documents associated with the entities. In this study, we use 'entity' (or 'document') to clearly denote that the RDF name is a biomedical entity (or a document). The simplest method for discovering entities is (1) full-text search over documents to find those containing the user's keyword, and then (2) obtaining entities associated with the documents found. This process is notated here as *user's keyword* $\rightarrow$ *document* $\rightarrow$ *entity*. To compute the significance of the association between each entity and a keyword, we have introduced a statistical test based on the number of shared documents. More concretely, for each entity, the search engine first generates a 2x2 contingency table consisting of the number of documents

- matching both the keyword and the entity,
- matching the keyword but not matching the entity,
- not matching the keyword but matching the entity and
- matching neither the keyword nor the entity.

Then, the engine applies a statistical test to the contingency table and computes a *P*-value, or the significance of the test. Finally, all resultant entities are ranked by their *P*-values. We call the discovery method described above a direct search, which is described by the query in Fig. 9. The statistical function `ris:statisticalTest#FisherExactTest` computes the

P-value by constructing a 2x2 contingency table $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with its four arguments *a*, *b*, *c* and *d*,

and applies Fisher's exact test to the table. The simple method of evaluating the query shown in Fig. 9 is a sequential evaluation of the WHERE, EVALUATE and ORDER BY clauses in this order. In this method, the WHERE clause is evaluated to obtain all RDF graphs satisfying the condition in the WHERE clause. Figure 10 shows the RDF graph pattern with all variables and constants appearing in the WHERE clause. In practice, since the number of RDF graphs matching the pattern in Fig. 10 may be huge, this simple method of evaluation requires the implementation of an optimisation mechanism to achieve a functional language processor. Figure 11 is a chart that includes a Venn diagram of MEDLINE abstracts; it shows the relationship between each subset of MEDLINE abstracts and other RDF entities. This figure shows the primitive data structure for query searching as a set of relationships between each subset of MEDLINE abstracts and other entities such as %keyword and ?gene, rather than the relationships between each MEDLINE abstract and the other entities. In our example, to compute ?p, only four subsets—?docAll, ?docKey,

?docIntersection and ?docGene –of MEDLINE abstracts are necessary; they can be obtained by a specialised document search method such as the full-text search technique. Since this approach does not require a huge RDF graph space for computing the statistical significance, it opens up a new possibility for realising a practical GRASQL language processing system. Furthermore, the results of statistical analysis can be stored as a named graph using the CONSTRUCT statement instead of the SELECT statement. Figure 12 shows a CONSTRUCT query that generates RDF graphs with blank nodes as shown in Fig. 11. The generated named graph can be efficiently used as input data in SPARQL as well as in GRASQL. Statistical tests can also be used in a search to indirectly generate the associations between entities and a keyword via entity–entity relationships associated with documents. A typical example of entity–entity relationships is the co-citation frequencies of the entities in documents. The significance of the association between two entities can be computed by a statistical test of the number of documents, similar to a direct search. That is, for each entity–entity relationship, a P -value is computed using a 2×2 contingency table that contains the number of documents

- matching both entities,
- matching the first entity but not matching the second,
- not matching the first entity but matching the second and
- matching neither entity.

The entity–entity relationship can be obtained as a set of RDF triples using the query shown in Fig. 13.

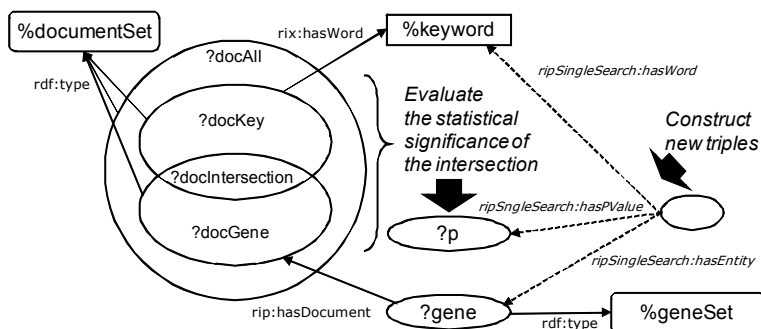


Fig. 11. Statistical diagram showing the relationships among the entities specified by the query in Fig. 12. New RDF triples are constructed by the query's CONSTRUCT statement.

We realise an inference search for the connection

user's keyword $\rightarrow$ *document* $\rightarrow$ *entity* $\rightarrow$ *document* $\rightarrow$ *entity* by applying entity–entity relationships to the entities resulting from a single association search. The P -value P_d of the associated entity is computed by

$$P_d = 1 - (1 - P_s)(1 - P_r) \quad (1)$$

where P_s is the P -value of the first direct search, and P_r is the P -value of the second association search of the entity–entity relationship.

Furthermore, several search connections,

user's keyword $\rightarrow$ *document* $\rightarrow$ *entity*₁ $\rightarrow$ *document* $\rightarrow$ *entity*₂, which reach the same entity *entity*₂ via a different entity *entity*₁, may be obtained. In this case, the P -value of the resultant entity *entity*₂ can be computed by

$$P_{entity_2} = \prod_i P_{i,entity_2} \quad (2)$$

where $P_{i,entity_2}$, ($1 \leq i \leq n$) are the P -values of n connections that finally reach $entity_2$. This model is based on the idea that a solution containing several connections may be more important than others. Another method is selecting the best connection by choosing the smallest P -value. In this case, the equation,

$$P_{entity_2} = \min_i (P_{i,entity_2}) \quad (3)$$

is applied for computing the P -value.

```
@prefix rio:      <http://omicspace.riken.jp/GRASQL/>
@prefix rip:      <http://omicspace.riken.jp/GRASQL/predicate>
@prefix rix:      <http://omicspace.riken.jp/GRASQL/procedure>
@prefix ris:      <http://omicspace.riken.jp/GRASQL/statistics>
@prefix rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@prefix ripSingleSearch:
                  <http://omicspace.riken.jp/GRASQL/singleSearch>

@let %keyword "type 2 diabetes"
@let %documentSet rio:MEDLINE
@let %geneSet rio:MouseGene
CONSTRUCT {
    [] ripSingleSearch:hasEntity      ?gene ;
      ripSingleSearch:hasWord %keyword ;
      ripSingleSearch:hasPValue    ?p .
}
WHERE {
    ?gene          rip:hasDocument      ?docGene ;
                  rip:hasDocument      ?docIntersection ;
                  rdf:type              %geneSet .
    ?docKey        EXT:rix:hasWord      %keyword ;
                  rdf:type              %documentSet .
    ?docIntersection EXT:rix:hasWord    %keyword ;
                  rdf:type              %documentSet .
    ?docGene       rdf:type              %documentSet .
    ?docAll        rdf:type              %documentSet .
}
EVALUATE ?p FOR ?gene {
    ?p = ris:statisticTest#FisherExactTest(?a,?b,?c,?d) ;
    ?a = count(DISTINCT ?docIntersection) ;
    ?b = count(DISTINCT ?docKey)-?a ;
    ?c = count(DISTINCT ?docGene)-?a ;
    ?d = count(DISTINCT ?docAll)-?a-?b-?c
}
```

Fig. 12. GRASQL query including a CONSTRUCT statement, which is used to save the results of the statistical analysis described in the WHERE and EVALUATE clauses into a set of RDF graphs. In the CONSTRUCT statement, as in SPARQL, a blank node [] is used to describe the relationships among ?gene, %keyword and ?p.

3.3 GRASQL representation of gene prioritisation in PosMed

To describe the semantics of gene prioritisation in PosMed more precisely, we will write the direct search and inference search patterns shown in Figs. 3(A) and 3(B), respectively, in GRASQL.

A GRASQL query for a direct search is written in Fig. 12. For convenience in enumerating examples, we first assume that the named graph

<http://omicspace.riken.jp/GRASQL/single/Mm/MEDLINE> from the direct search obtained by the query in Fig. 11 is generated. We also assume that the named graph <http://omicspace.riken.jp/GRASQL/relation/Mm/MEDLINE> of entity-entity relationships obtained by the query in Fig. 13 is generated.

Using these two named graphs, we write a query for an inference search of the connection *user's keyword* → *document* → *entity* → *document* → *entity*, as shown in Fig. 14.

```
@prefix rio:      <http://omicspace.riken.jp/GRASQL/>
@prefix rip:      <http://omicspace.riken.jp/GRASQL/predicate>
@prefix ris:      <http://omicspace.riken.jp/GRASQL/statistics>
@prefix rdf:      <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@prefix ripInference:
                  <http://omicspace.riken.jp/GRASQL/inference>

@let %documentSet rio:MEDLINE
@let %geneSet rio:MouseGene
CONSTRUCT {
  [] ripInference:hasEntity1 ?gene1 ;
    ripInference:hasEntity2 ?gene2 ;
    ripInference:hasPValue ?p .
}
WHERE {
  ?gene1          rip:hasDocument      ?docGene1 ;
                  rip:hasDocument      ?docIntersection ;
                  rdf:type              %geneSet .
  ?gene2          rip:hasDocument      ?docGene2 ;
                  rip:hasDocument      ?docIntersection ;
                  rdf:type              %geneSet .
  ?docGene1       rdf:type              %documentSet .
  ?docGene2       rdf:type              %documentSet .
  ?docIntersection rdf:type              %documentSet .
  ?docAll         rdf:type              %documentSet .
}
EVALUATE ?p FOR ?gene1 ?gene2 {
  ?p = ris:statisticTest#FisherExactTest(?a,?b,?c,?d) ;
  ?a = count(DISTINCT ?docIntersection) ;
  ?b = count(DISTINCT ?docKey)-?a ;
  ?c = count(DISTINCT ?docGene)-?a ;
  ?d = count(DISTINCT ?docAll)-?a-?b-?c
}
```

Fig. 13. GRASQL query that builds RDF triples of co-citation relationships of mouse genes from MEDLINE abstracts.

```

@prefix rio:          <http://omicspace.riken.jp/GRASQL/>
@prefix ris:          <http://omicspace.riken.jp/GRASQL/statistics>
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
@prefix ripSingleSearch:
                    <http://omicspace.riken.jp/GRASQL/singleSearch>
@prefix ripInference:
                    <http://omicspace.riken.jp/GRASQL/inference>

@let %keyword         "type 2 diabetes"
@let %geneSet         rio:MouseGene
SELECT ?gene2 ?gene1 ?p ?pTotal
FROM NAMED <http://omicspace.riken.jp/GRASQL/single/Mm/MEDLINE>
FROM NAMED <http://omicspace.riken.jp/GRASQL/relation/Mm/MEDLINE>
WHERE {
    ?x      ripInference:hasEntity2          ?gene2 ;
            ripInference:hasEntity1          ?gene1 ;
            ripInference:hasPValue           ?pInference .
    ?y      ripSingleSearch:hasEntity         ?gene1 ;
            ripSingleSearch:hasWord %keyword ;
            ripSingleSearch:hasPValue         ?pSingle .
}
EVALUATE ?p FOR ?gene1 ?gene2 {
    ?p = 1-(1-?pSingle) (1-?pInference)
}
EVALUATE ?pTotal FOR ?gene2 {
    ?pTotal = ris:multiPValue(?p)
}
ORDER BY ?pTotal ?p

```

Fig. 14. GRASQL query for inference search for connection %keyword $\rightarrow$?gene1 $\rightarrow$?gene2 using named graphs generated by CONSTRUCT statements in advance. In this query, the two EVALUATE clauses are evaluated sequentially in the order of their appearance. In the example, P -value ?p for each pair (?entity1, ?entity2) is computed, and then P -value ?pTotal Total for each entity ?entity2 is computed. Finally, by evaluating the ORDER BY clause, the solutions of 4-tuples (?entity1, ?entity2, ?p, ?pTotal) are sorted by ?pTotal and ?p.

The function ris:multiPValue in the second EVALUATE clause is an implementation of Equation 2. Furthermore, ris:minPValue is an implementation of Equation 3 that does not appear in this article.

4. Data preparation and implementation

4.1 Data sources

Currently, PosMed employs more than 20 million documents including MEDLINE (title, abstract and MeSH term), genome annotation, phenome information, PPI, co-expression, localisation, disease, drug and metabolite records (Table 1).

4.2 High-accuracy manual curation for generating semantic links from genes to documents

To develop a set of document databases for our original search engine for PosMed, we developed a method of mapping between genes and documents based on an NER (Leser & Hakenberg, 2005) technique that extracts named entities such as genes from a document.

A. PosMed

	No. of documents	Data sources	Data Contents	Reference
MEDLINE	18 295 132	MEDLINE	MEDLINE title, abstract and MeSH term	Coletti & Bleich, 2001
Mouse mutant	12 911	BRMM	Mouse phenotypes	Masuya et al., 2007
OMIM	21 136	OMIM	Genetic disorder descriptions	Amberger et al., 2009
HsPPI	35 731	HsPPI	Protein-protein interaction	Makino & Gojobori, 2007
REACTOME	10 761	REACTOME	Biological pathways	Matthews et al., 2009
Mouse gene record	58 768	MGI	Gene descriptions (annotations)	Blake et al., 2009
Rat gene record	36 634	RGD	Gene descriptions (annotations)	Dwinell et al., 2009
Human gene record	35 362	HGNC	Gene descriptions (annotations)	Wain et al., 2002
Metabolite record	18 045	KNapSAcK	Metabolite descriptions	Shinbo et al., 2006
Drug record	1 015	Original data	Drug descriptions	
Disease record	1 911	Original data	Disease descriptions	
RIKEN researcher record	8 603	Original data	Names of researchers appear as authors in MEDLINE	
Total	18 534 098			

B. PosMed-plus

	No. of documents	Data sources	Data Contents	Reference
MEDLINE	18 295 132	MEDLINE	MEDLINE title, abstract and MeSH term	Coletti & Bleich, 2001
At co-expression	44 082	ATTED-II	Microarray based co-expression prediction	Obayashi et al., 2009
At localisation	8 404	SUBA-2	Experimentally validated subcellular localisation	Heazlewood et al., 2007
At PPI	24 418	AtPID	Protein-protein interaction	Cui et al., 2008
	214	RAPID	RIKEN Arabidopsis Phenome Information DB	Kuromori et al., 2006
At phenotype	1 697	TAIR	Phenotype informations from TAIR	Swarbreck et al., 2008
	1 784	Literature	Manually collected original data	
Rice markers	1 712	RAP-DB	RFLP marker	Harushima et al., 1998
	15 623		SSR marker	McCouch et al., 2002
Homologus genes	1 553 922	Original data	Homologue genes between Arabidopsis and rice	Hanada et al., 2008
Arabidopsis gene record	33 003	TAIR, UniProt	Gene descriptions (annotations)	Swarbreck et al., 2008 ; UniProt Consortium, 2009
Rice gene record	29 389	RAP-DB	Gene descriptions (annotations)	Rice Annotation Project, 2008
Total	20 009 380			

Table 1. Data descriptions for (A) PosMed and (B) PosMed-plus.

Since false-positive relationships may arise from a primitive NER method that simply checks for the appearance of a name in a document, we instead employ a full-text search engine for NER, with logical queries defined as a list of names or words related to a gene concatenated with logical operators such as AND, OR and NOT. Specifically, as a base query we computationally collected all the synonym names for each gene from The Arabidopsis Information Resource (TAIR) and UniProt, connected these synonyms with the logical OR operation and added 'Arabidopsis' with the AND operation. Using these base queries, we performed a full-text search against a set of documents including MEDLINE title, abstract and MeSH terms (Coletti & Bleich, 2001). To reduce false-positive hits and true-negative hits, we carefully edited these queries manually through trial and error by performing a full-text search for each trial against the document set. For example, to detect all MEDLINE documents for the AT1G03880 (cruciferin B, CRB) gene while eliminating false-positive hits with the homonym 'CRB', which represents 'chloroplast RNA binding', we defined the following query: ('AT1G03880' OR 'CRU2' OR 'CRB' OR 'CRUCIFERIN 2' OR 'CRUCIFERIN B') AND ('Arabidopsis') NOT ('chloroplast RNA binding'). This curation method is effective for updating with the latest publications. Once we curate a query, the query can be reused to extract gene-document relationships by performing a full-text search against those new document sets.

4.3 Implementation

PosMed was developed as a web-oriented tool based on a client-server model in which users access the system with conventional web browsers. However, we recommend using Microsoft Internet Explorer 8 or later or Firefox 3 or later for Windows, and Safari 4 or later or Firefox 3 or later for Macintosh. The core software component GRASE must execute a search process by very rapidly interpreting a GRASQL query program. To develop GRASE, we employed Apache Lucene, a rapid full-text search engine with a rich query language, for testing the predicate `rix:hasWord`. Since a search process can be executed for each target entity in parallel, we use nine distributed computers to realise a high-throughput search. Therefore, we distributed the data for each entity; i.e. MEDLINE abstracts and mouse gene-mouse gene relationship data associated with each distributed mouse gene and researcher are distributed among the computers to achieve a parallel search.

5. Applications of PosMed

We describe examples illustrating the power of PosMed and PosMed-plus below.

5.1 General usage of PosMed

5.1.1 Search with user-specified keywords and chromosomal intervals

A typical application of PosMed is searching with user-specified keywords and chromosomal intervals suggested by linkage analysis. As an example, we retrieved diabetes- or insulin-related genes in the chromosomal interval from 90 Mbp to 140 Mbp on chromosome 1 in the mouse genome (Fig. 15(A)). In this example, PosMed retrieved candidate genes ranked by the statistical significance between the user's keyword and each gene. Although PosMed found > 470 000 documents, it returned results in 0,865 s. Users can download all the candidate genes together with the associated gene annotations by using the 'download rank list' button in the blue box on the left (Fig. 15(D)). PosMed also supports an expert mode that allows users to select possible search paths and confirm the number of resulting genes for each search path. Clicking on a gene name listed in the gene search result

(A) **Select genomic interval graphically**

search condition species **Pos Med**

Select Interval with OmicBrowse chromosome position from to clear version keyword gene name search clear recent 4 years show

(D) download annotation download rank list set as target add comment draw check all

- ☐ Insig2
- ☐ Il10
- ☐ Serpinb2
- ☐ Adora1
- ☐ Adipor1
- ☐ Ptprc
- ☐ Mapkapk2
- ☐ Capn10
- ☐ Ramp1
- ☐ Bcl2
- ☐ Plgr
- ☐ Lct
- ☐ Dbi
- ☐ Pdccl1
- ☐ Il19
- ☐ Tnfrsf2
- ☐ Cxcr4
- ☐ Myog
- ☐ Gpc1
- ☐ Sctr
- ☐ Ctse
- ☐ Ugt1a1
- ☐ Sox13
- ☐ Nr5a2
- ☐ Pam
- ☐ Ugt1a6a
- ☐ Bok
- ☐ Trpm8
- ☐ Ren2
- ☐ Avpr1b
- ☐ Chit1

(B) **All Hits** Total Hits: 120 (0.865 sec) Simple Mode

☒ mouse mutant ☒ HsPPI ☒ human gene record ☒ MEDLINE (sentence)
☒ mouse gene record ☒ OMIM ☒ REACTOME ☒ rat gene record

• Associate the keyword with
 • Further associate the entities with

(C) Page 1 next display mode: Simple

- Insig2, insulin induced gene 2**
 Insig2 ← 42 docs Position: [Mm:1:123200934-123224363](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 39 hits
 diabetes OR insulin
- Il10, interleukin 10**
 Il10 ← 24524 docs P value: 6.98E-3980 Position: [Mm:1:132916422-132921547](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 10470 docs P value: 6.98E-3980
 Tnf ← 107577 docs Position: [Mm:17:35336326-35338952](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 4394 hits P value: 1.39E-4602
 diabetes OR insulin
- Serpinb2, serine (or cysteine) peptidase inhibitor, clade B, member 2**
 Serpinb2 ← 854 docs P value: 1.69E-1179 Position: [Mm:1:109408000-109422169](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 735 docs P value: 1.69E-1179
 Serpine1 ← 6469 docs Position: [Mm:5:137537378-137548129](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 1327 hits P value: 1.64E-1390
 diabetes OR insulin
- Adora1, adenosine A1 receptor**
 Adora1 ← 2948 docs P value: 3.14E-1154 Position: [Mm:1:136095800-136132004](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 1413 docs P value: 3.14E-1154
 Adenosine ← 124232 docs Link to: [Original](#)
 ↑ ← 1343 hits P value: 3.04E-1407
 diabetes OR insulin
- Adipor1, adiponectin receptor 1** → Clicking here displays Fig. 16
 Adipor1 ← 278 docs P value: 1.81E-745 Position: [Mm:1:136312033-136329925](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 387 docs P value: 1.81E-745
 Adipoq ← 5574 docs Position: [Mm:16:23146692-23157813](#) Link to: [MGI](#) [RIKEN SciNeS](#)
 ↑ ← 5891 hits P value: 2.30E-6173
 diabetes OR insulin

Fig. 15. Example search result for mouse genes against the query keyword 'diabetes or insulin' and the genomic interval between 90 Mbp and 140 Mbp on chromosome 1 in the NCBI37 genome. Users can construct queries at the top of the output display (A). To select a genomic interval visually, PosMed cooperates with the Flash-based genomic browser OmicBrowse. The 'All Hits' tab (B) shows a list of selectable document sets to be included in the search. As a default parameter, PosMed sets 'Associate the keyword with entities co-cited within the same sentences'. If the total number of candidate genes is less than 20, PosMed will automatically change this to 'Associate the keyword with entities co-cited within the same document' to show more candidates (B). Search results are ranked in (C). Users can download at most 300 candidate genes and their annotations from (D).

(A)

Symbol: Adipor1
Name: adiponectin receptor 1
ID: MGI:1919924
Link to: [MGI](#) [RIKEN SciNeS](#)
Position: [Mm:1:136312033-136329925](#)

P value: 1.81E-745
238 mouse mutant 0
HsPPI: 0
MEDLINE: 238
mouse gene record: 0
REACTOME: 0

Symbol: Adipoq
Name: adiponectin, C1Q and collagen domain containing
Other aliases: adipoQ, adiponectin, GBP28, apM1, adipo, Acpr30, Acdc
ID: MGI:106675
Link to: [MGI](#) [RIKEN SciNeS](#)
Position: [Mm:16:23146692-23157813](#)

(B)

all (175/238) mouse mutant (0/0) HsPPI (0/0) MEDLINE (175/238) mouse gene record (0/0) REACTOME (0/0)

(C)

keyword: diabetes OR insulin year: all Find

documents

publication year

with keyword without keyword

(D)

show digest order by fact download

1-20 of 238 Page 1 next

(E)

Adipor1 related entities

any

check all

set as target download draw

Find

Adipoq

Adipor2

Ppara

non-insulin-dependent dia...

ADIPOR1

Adipor1

Deltagen and Lexicon Kno...

Mapk8

Sirt1

Map2k1

Nr3c1

Mapk14

Map2k2

Prkaca

Acs14

Hk2

Gad1

Cd44

Stk11

Crebbp

Irs1

Csnk2a2

Mc4r

Akt1

Pik3r1

Omp

Pomc1

Dgat1

Fpl1

Nfkbia

Frap1

Trglyd

1: Polymorphisms in adiponectin receptor genes ADIPOR1 and ADIPOR2 and insulin resistance.
[PMID:17716299](#)
Obesity reviews : an official journal of the International Association for the Study of Obesity 2007 Sep
Adiponectin plays an important role in **insulin** sensitivity via **adiponectin receptor 1** and 2 signalling.
? Adipor1 ,adiponectin receptor 1 ([references](#) , [MGI:1919924](#) , [RIKEN SciNeS](#) , [Mm:1:136312033-136329925](#))
? Adipoq ,adiponectin, C1Q and collagen domain containing ([references](#) , [MGI:106675](#) , [RIKEN SciNeS](#) , [Mm:16:23146692-23157813](#))

2: Gene-nutrient interactions in the metabolic syndrome: single nucleotide polymorphisms in ADIPOQ and ADIPOR1 interact with plasma saturated fatty acids to modulate insulin resistance.
[PMID:20032495](#)
The American journal of clinical nutrition 2010 Mar
CONCLUSIONS: A reduction in plasma SFAs could be expected to lower **insulin** resistance in MetS subjects who are minor allele carriers of rs266729 in **ADIPOQ** and rs10920533 in **ADIPOR1**.
? Adipor1 ,adiponectin receptor 1 ([references](#) , [MGI:1919924](#) , [RIKEN SciNeS](#) , [Mm:1:136312033-136329925](#))
? Adipoq ,adiponectin, C1Q and collagen domain containing ([references](#) , [MGI:106675](#) , [RIKEN SciNeS](#) , [Mm:16:23146692-23157813](#))

3: Adiponectin and AdipoR1 regulate PGC-1alpha and mitochondria by Ca(2+) and AMPK/SIRT1.
[PMID:20357764](#)
Nature 2010 Apr 29
Decreased levels of **adiponectin** and **AdipoR1** in obesity may have causal roles in mitochondrial dysfunction and **insulin** resistance seen in **diabetes**.
Adipor1 ,adiponectin receptor 1 ([references](#) , [MGI:1919924](#) , [RIKEN SciNeS](#) , [Mm:1:136312033-136329925](#))
Adipoq ,adiponectin, C1Q and collagen domain containing ([references](#) , [MGI:106675](#) , [RIKEN SciNeS](#) , [Mm:16:23146692-23157813](#))

Fig. 16. Detailed document screen in PosMed. This page shows document sets supporting both the *Adipor1* gene ranked fifth in Fig. 15(C) and the *Adipoq* gene. Gene descriptions are shown in (A). Users can select the type of documents from the mouse mutant, HsPPI, MEDLINE mouse gene record or REACTOME in (B). The bar chart represents the number of related documents per year. Red and blue indicate the number of documents with and without a user-specified keyword, respectively. All documents are shown at (D). The *Adipor1*-related genes are listed in (E).

page shown in Fig. 15(B) reveals the supporting evidence for each candidate gene. To confirm the expression pattern of candidate genes with a genome browser, we provide a link to our genome browser OmicBrowse (Matsushima et al., 2009; Toyoda et al., 2007) from the gene location (Fig. 15(C)). OmicBrowse covers genome versions for mouse, human, rat, Arabidopsis and rice, and each genome is mapped to omic-type databases and a total of 344 data sources.

5.1.2 Search with phenotypic keywords

PosMed also allows users to discover genes related to phenotypic keywords. For example, if users search on the keyword ‘rumpled leaves’ in Arabidopsis, PosMed-plus shows four known cases via the direct search and one new candidate gene via the inference search. For the four known cases, PosMed-plus shows the link to the RIKEN Arabidopsis Phenome Information Database (RAPID), and users can confirm the phenotypes by looking at pictures. PosMed-plus also shows the evidence documents in the inference path to the AT1G51500 candidate gene. In this case, AT1G51500 is retrieved via the AT1G17840 gene, which is one of the four known genes found in the direct search. They are highly connected with co-expression, PPI and co-citation data.

5.1.3 Reference search with gene IDs

It is difficult to retrieve all the appropriate references based on gene names because of the wide variation in synonyms. Moreover, sometimes the same abbreviated names are used for functionally different genes, causing false-positive hits. In PosMed, we carefully extracted these gene-reference relationships manually, as described above. Therefore, users can retrieve the curated results with the gene ID (e.g. MGI code and AGI code) even if the abstracts do not contain the gene ID itself.

5.1.4 Search for omics data

As shown in Fig. 16, PosMed integrates various data such as gene annotations, mouse mutant records and human PPIs. Users can select any document set (the default setting is to search everything) and retrieve the required data, all within the same interface. PosMed links not only to the original databases but also to OmicBrowse, which also assists users in accessing and downloading various omics data.

5.2 *In silico* positional cloning after QTL analysis in rice

To evaluate the efficiency of PosMed with a concrete example, we confirmed whether PosMed (PosMed-plus in this rice example) could successfully retrieve correct genes that have been identified by qualitative trait locus (QTL) analysis. Three examples are described below.

Ren et al. (2005) isolated the *SKC1* gene and through QTL analysis found that it encoded an Na⁺-selective transporter. In this example, we need to prioritise candidate genes without the functionally related keyword ‘transporter’. Instead of the functional keyword, we retrieved genes with the phenotypic keyword ‘salt tolerance’ and selected the genomic interval between the markers C955 and E50811 on chromosome 1. PosMed-plus returned the Os01g0307500 (cation transporter family protein) gene with a high ranking. This is because the keyword ‘salt tolerance’ was mapped to the sodium ion transmembrane transporter gene AT4G10310, and Os01g0307500 was suggested as a homologue of AT4G10310.

Using a no-pollen type of male-sterile mutant (*xs1*), Zuo et al. (2008) revealed that mutant microspores are abnormally condensed and agglomerated to form a deeply stained cluster at the late microspore stage. This halts the microspore vacuolation process, and therefore,

the mutant forms lack functional pollen. This mutation is controlled by a single recessive gene, *VR1* (vacuolation retardation 1), which is located between the molecular markers RM17411 and RM5030 on chromosome 4. We searched for candidate genes with the phenotypic keyword 'sterility' in the suggested chromosome region. PosMed-plus suggested the Os04g0605500 gene (similar to calcium-transporting ATPase) as the homologue of the Arabidopsis calcium-transporting ATPase, AT3G21180. Since Schiött et al. (2004) found that mutation of AT3G21180 results in partial male sterility, we conclude that PosMed-plus found an appropriate candidate.

Lastly, Zhang et al. (2008) found a male sterility mutant of anther dehiscence in advance, *add(t)*, between the markers R02004 and RM300 on chromosome 2. In this search, PosMed-plus returned RNA-binding region RNP-1, Os02g0319100 and disease-resistance protein family protein Os02g0301800, with strong homology with Arabidopsis genes. PosMed-plus retrieved the Os02g0319100 gene as a homologue of Arabidopsis mei2-like (AML) protein 5, AT1G29400. As supporting evidence, Kaur et al. (2006) showed that multiple mutants of all the AML genes displayed a sterility phenotype. The other candidate gene, Os02g0301800, was derived via an inference search. First, PosMed-plus retrieved the keyword 'sterility' in a document describing the AT2G26330 gene. Next, AT2G26330 was linked to AT5G43470 as supported by three co-citations. Finally, Os02g0301800 was returned as a homologue of AT5G43470. PosMed-plus originally suggested the Os02g0301800 gene because AT2G26330 is linked to the keyword 'sterility' in a document. However, this document states that AT2G26330 causes aberrant ovule development and female-specific sterility. Since Zhang et al. (2008) focused on male sterility, we conclude that Os02g0319100 is the appropriate candidate.

5.3 Other example results

In RIKEN's large-scale mouse ENU mutagenesis project, PosMed was used to prioritise genes and has contributed to the successful identification of more than 65 responsible genes (Masuya et al., 2007). PosMed is also used by researchers worldwide and has successfully narrowed the candidate genes responsible for a specific function after QTL analysis (Kato et al., 2008; Moritani et al., 2006).

5.4 Further usage

We here introduced PosMed as a web tool for assisting in the prioritisation of candidate genes for positional cloning. Using the search engine GRASE, we also implemented inference-type full-text search functions for metabolites, drugs, mutants, diseases, researchers, document sets and databases. For cross-searching, users can select 'any' for the search items at the top right on the PosMed web page. Since this system can search various omics data, we named it OmicScan. In addition to English, GRASE accepts queries in Japanese and French. More advanced usage of PosMed is explained in the PosMed tutorial available at http://omicspace.riken.jp/tutorial/HowToUseGPS_Eng.pdf.

6. Discussion and conclusion

To use not only well-formed knowledge in RDF but also non-well-formed document data on the Semantic Web, we have introduced statistical concepts into the existing RDF query language SPARQL using a literature mining technique for searching a vast number of documents written in a natural language. The core data structure in our method is that documents are linked with each entity accurately associated by NER with human refinement, namely manual curation. The advantages of this simple structure are as follows.

- Facility of keyword selection: An arbitrary keyword appears in at least one document. Thus, a user can choose a keyword that is not necessarily related to an entity the user wants to find.
- Open-ended extensibility of documents: A new document can be added to the system if it is associated with at least one existing entity. Documents about an entity written from various viewpoints enrich knowledge so that the entity can be linked to the user's keyword.
- Open-ended extensibility of entities: A new entity can be added if at least one document associated with it exists. Therefore, entities of different categories can be introduced, which allows association search among them.
- Open-ended extensibility of semantic knowledge: Existing biomedical data in RDF format can be introduced directly into a GRASQL query.

Thanks to these advantages, PosMed can support various types of heterogeneous omics knowledge.

PosMed has been widely used to prioritise candidate genes after QTL analysis in species including mouse and *Arabidopsis* and to successfully identify responsible genes. Our approach is novel compared to gene prioritisation systems such as BIOTLA (Hristovski et al., 2005), Manjal (Sehgal & Srinivasan, 2005) and LitLinker (Yetisgen-Yildiz & Pratt, 2006), since PosMed is based on *P*-values computed by Fisher's exact test via tables of numbers of documents and used as correlation scores between a user's keyword and the resulting genes for ranking.

Our future work will include data extension of PosMed with not only well-formed omics knowledge in RDF but also non-well-formed document data on the Semantic Web using the statistical concepts of GRASQL.

7. References

- Adie, E.; Adams, R.; Evans, K.; Porteous, D. & Pickard, B. (2006). SUSPECTS: enabling fast and effective prioritization of positional candidates. *Bioinformatics*, Vol. 22, 773-774
- Aerts, S.; Lambrechts, D.; Maity, S.; Van Loo, P.; Coessens, B.; De Smet, F.; Tranchevent, L.; De Moor, B.; Marynen, P.; Hassan, B.; Carmeliet, P. & Moreau, Y. (2006). Gene prioritization through genomic data fusion. *Nat. Biotechnol.*, Vol. 24, 537-544
- Amberger, J.; Bocchini, C.; Scott, A. & Hamosh, A. (2009). McKusick's Online Mendelian Inheritance in Man (OMIM). *Nucleic Acids Res.*, Vol. 37, D793-D796
- Ashburner, M.; Ball, C.A.; Blake, J.A.; Botstein, D.; Butler, H.; Cherry, J.M.; Davis, A.P.; Dolinski, K.; Dwight, S.S.; Eppig, J.T.; Harris, M.A.; Hill, D.P.; Issel-Tarver, L.; Kasarskis, A.; Lewis, S.; Matese, J.C.; Richardson, J.E.; Ringwald, M.; Rubin, G.M. & Sherlock, G. (2000). Gene ontology: tool for the unification of biology. *Nat. Gene.*, Vol. 25, 25-29
- Berners-Lee, T.; Hendler, J. & Lassila, O. (2001). The Semantic Web. *Sci. Am.*, Vol. 284, 34-43
- Blake, J.; Bult, C.; Eppig, J.; Kadin, J. & Richardson, J. (2009). The Mouse Genome Database genotypes::phenotypes. *Nucleic Acids Res.*, Vol. 37, D712-D719
- Coletti, M. & Bleich, H. (2001). Medical subject headings used to search the biomedical literature. *J. Am. Med. Inform. Assoc.*, Vol. 8, 317-323
- Cui, J.; Li, P.; Li, G.; Xu, F.; Zhao, C.; Li, Y.; Yang, Z.; Wang, G.; Yu, Q. & Shi, T. (2008). AtPID: *Arabidopsis thaliana* protein interactome database - an integrative platform for plant systems biology. *Nucleic Acids Res.*, Vol. 36, D999-D1008

- Dwinell, M.; Worthey, E.; Shimoyama, M.; Bakir-Gungor, B.; DePons, J.; Laudederkind, S.; Lowry, T.; Nigram, R.; Petri, V.; Smith, J.; Stoddard, A.; Twigger, S.; Jacob, H. & the RGD Team. (2009). The Rat Genome Database 2009: variation, ontologies and pathways. *Nucleic Acids Res.*, Vol. 37, D744-D749
- The Gene Ontology Consortium. (2006). The Gene Ontology (GO) project in 2006. *Nucleic Acids Res.*, Vol. 34, D322-D326
- GeneSniffer. Available From <http://www.genesniffer.org>
- Hanada, K.; Zou, C.; Lehti-Shiu, M.; Shinozaki, K. & Shiu, S. (2008). Importance of lineage-specific expansion of plant tandem duplicates in the adaptive response to environmental stimuli. *Plant Physiol.*, Vol. 148, 993-1003
- Harushima, Y.; Yano, M.; Shomura, A.; Sato, M.; Shimano, T.; Kuboki, Y.; Yamamoto, T.; Lin, S.Y.; Antonio, B.A.; Parco, A.; Kajiya, H.; Huang, N.; Yamamoto, K.; Nagamura, Y.; Kurata, N.; Khush, G.S. & Sasaki, T. (1998). A high-density rice genetic linkage map with 2275 markers using a single F2 population. *Genetics*, Vol. 148, 479-494
- Heazlewood, J.; Verboom, R.; Tonti-Filippini, J.; Small, I. & Millar, A. (2007). SUBA: the Arabidopsis subcellular database. *Nucleic Acids Res.*, Vol. 35, D213-D218
- Hirsch, J.E. (2005). An Index to Quantify an Individual's Scientific Research Output. *Proc. Natl. Acad. Sci. USA*, Vol. 102, 16569-16572
- Hristovski, D.; Peterlin, B.; Mitchell, J.A. & Humphrey, S.M. (2005). Using literature-based discovery to identify disease candidate genes. *Int. J. Med. Inform.*, Vol. 74, 289-298
- Kato, N.; Watanabe, Y.; Ohno, Y.; Inoue, T.; Kanno, Y.; Suzuki, H. & Okada, H. (2008). Mapping quantitative trait loci for proteinuria-induced renal collagen deposition. *Kidney Int.*, Vol. 73, 1017-1023
- Kaur, J.; Sebastian, J. & Siddiqi, I. (2006). The Arabidopsis-me12-like genes play a role in meiosis and vegetative growth in Arabidopsis. *Plant Cell*, Vol. 18, 545-559
- Kobayashi, N. & Toyoda, T. (2008). Statistical search on the Semantic Web. *Bioinformatics*, Vol. 24, 1002-1010
- Köhler, S.; Bauer, S.; Horn, D. & Robinson, P. (2008). Walking the interactome for prioritization of candidate disease genes. *Am. J. Hum. Genet.*, Vol. 82, 949-958
- Kuromori, T.; Wada, T.; Kamiya, A.; Yuguchi, M.; Yokouchi, T.; Imura, Y.; Takabe, H.; Sakurai, T.; Akiyama, K.; Hirayama, T.; Okada, K. & Shinozaki, K. (2006). A trial of phenome analysis using 4,000 Ds-insertional mutants in gene-coding regions of Arabidopsis. *Plant J.*, Vol. 47, 640-651
- Leser, U. & Hakenberg, J. (2005). What makes a gene name? Named entity recognition in the biomedical literature. *Brief Bioinform.*, Vol. 6, 357-369
- Makino, T. & Gojobori, T. (2007). Evolution of protein-protein interaction network. *Genome Dyn.*, Vol. 3, 13-29
- Makita, Y.; Kobayashi, N.; Mochizuki, Y.; Yoshida, Y.; Asano, S.; Heida, N.; Deshpande, M.; Bhatia, R.; Matsushima, A.; Ishii, M.; Kawaguchi, S.; Iida, K.; Hanada, K.; Kuromori, T.; Seki, M.; Shinozaki, K. & Toyoda, T. (2009). PosMed-plus: an intelligent search engine that inferentially integrates cross-species information resources for molecular breeding of plants. *Plant Cell Physiology*, Vol. 50, 1249-1259
- Manola, F. & Miller, E. (2004). RDF Primer. World Wide Web Consortium, Recommendation REC-rdf-primer-20040210. Available from <http://www.w3.org/TR/2004/REC-rdf-primer-20040210/>

- Masuya, H.; Yoshikawa, S.; Heida, N.; Toyoda, T.; Wakana, S. & Shiroishi, T. (2007). Phenosite: a web database integrating the mouse phenotyping platform and the experimental procedures in mice. *J. Bioinform. Comput. Biol.*, Vol. 5, 1173-1191
- Masuya, H.; Makita, Y.; Kobayashi, N.; Nishikata, K.; Yoshida, Y.; Mochizuki, Y.; Doi K.; Takatsuki, T.; Waki, K.; Tanaka, N.; Ishii, M.; Matsushima, A.; Takahashi, S.; Mizoguchi, R.; Kozaki K.; Furuichi, T.; Kawaji, H.; Wakana, S.; Nakamura, Y.; Yoshiki, A.; Murata, T.; Fukami-Kobayashi, K.; Mohan, S.; Ohara, O.; Hayashizaki, Y.; Obata, Y. & Toyoda, T. (2010). The RIKEN integrated database of mammals. *Nucleic Acids Res.*, (in press)
- Matsushima, A.; Kobayashi, N.; Mochizuki, Y.; Ishii, M.; Kawaguchi, S.; Endo, T.A.; Umetsu, R.; Makita, Y. & Toyoda, T. (2009). OmicBrowse: a Flash-based high-performance graphics interface for genomic resources. *Nucleic Acids Res.*, Vol. 37, Web Server Issue, W57-W62
- Matthews, L.; Gopinath, G.; Gillespie, M.; Caudy, M.; Croft, D.; de Bono, B.; Garapati, P.; Hemish, J.; Hermjakob, H.; Jassal, B.; Kanapin, A.; Lewis, S.; Mahajan, S.; May, B.; Schmidt, E.; Vastrik, I.; Wu, G.; Birney, E.; Stein, L. & D'Eustachio, P. (2009). Reactome knowledgebase of human biological pathways and processes. *Nucleic Acids Res.*, Vol. 37, D619-D622
- McCouch, S.; Teytelman, L.; Xu, Y.; Lobos, K.; Clare, K.; Waltam, M.; Fu, B.; Maghirang, R.; Li, Z.; Xing, Y.; Zhang, Q.; Kono, I.; Yano, M.; Fjellstrom, R.; DeClerck, G.; Schneider, D.; Cartinhour, S.; Ware, D. & Stein, L. (2002). Development and mapping of 2240 new SSR markers for rice (*Oryza sativa* L.). *DNA Res.*, Vol. 9, 199-207
- Moritani, M.; Togawa, K.; Yaguchi, H.; Fujita, Y.; Yamaguchi, Y.; Inoue, H.; Kamatani, N. & Itakura, M. (2006). Identification of diabetes susceptibility loci in db mice by combined quantitative trait loci analysis and haplotype mapping. *Genomics*, Vol. 88, 719-730
- Noruzi, A. (2005). Google Scholar: The New Generation of Citation Indexes. *Libli*, Vol. 55, 170-180
- Obayashi, T.; Hayashi, S.; Saeki, M.; Ohta, H. & Kinoshita, K. (2009). ATTED-II provides coexpressed gene networks for Arabidopsis. *Nucleic Acids Res.*, Vol. 37, D987-D991
- Prud'Hommeaux, E. & Seaborne, A. (2008). SPARQL Query Language for RDF. World Wide Web Consortium, Recommendation REC-rdf-sparqlquery-20080115. Available from <http://www.w3.org/TR/2008/REC-rdf-sparql-query-20080115/>
- Ren, Z.; Gao, J.; Li, L.; Cai, X.; Huang, W.; Chao, D.; Zhu, M.; Wang, Z.; Luan, S. & Lin, H. (2005). A rice quantitative trait locus for salt tolerance encodes a sodium transporter. *Nat. Genet.*, Vol. 37, 1141-1146
- Rice Annotation Project (2008). The Rice Annotation Project Database (RAPDB): 2008 update. *Nucleic Acids Res.*, Vol. 36, D1028-D1033
- Schiött, M.; Romanowsky, S.; Baekgaard, L.; Jakobsen, M.; Palmgren, M. & Harper, J. (2004). A plant plasma membrane Ca^{2+} pump is required for normal pollen tube growth and fertilization. *Proc. Natl. Acad. Sci. USA*, Vol. 101, 9502-9507
- Seelow, D.; Schwanz, J. & Schuelke, M. (2008). GeneDistiller - distilling candidate genes from linkage intervals. *PLoS ONE*, Vol. 3, e3874, 537-544
- Sehgal, A.K. & Srinivasan, P. (2005). Manjal - A Text Mining System for MEDLINE. *Proceedings of the 28th Annual International ACM SIGIR.*, Salvador, Brazil, 680

- Shinbo, Y.; Nakamura, Y.; Altaf-Ul-Amin, Md.; Asahi, H.; Kurokawa, K.; Arita, M.; Saito, K.; Ohta, D.; Shibata, D. & Kanaya, S. (2006). KNApSAcK: A Comprehensive Species-Metabolite Relationship Database, In: *Plant Metabolomics, Biotechnology in Agriculture and Forestry*, Vol. 57, Saito, K.; Dixon, R.A. & Willmitzer, L., 165-184, Springer Verlag, Berlin
- Swarbreck, D.; Wilks, C.; Lamesch, P.; Berardini, T.Z.; Garcia-Hernandez, M.; Foerster, H.; Li, D.; Meyer, T.; Muller, R.; Ploetz, L.; Radenbaugh, A.; Singh, S.; Swing, V.; Tissier, C.; Zhang, P. & Huala, E. (2008). The Arabidopsis Information Resource (TAIR): gene structure and function annotation. *Nucleic Acids Res.*, Vol. 36, D1009-D1014
- Thornblad, T.; Elliott, K.; Jowett, J. & Visscher, P. (2007). Prioritization of positional candidate genes using multiple web-based software tools. *Twin Res. Hum. Genet.*, Vol. 10, 861-870
- Toyoda, T.; Mochizuki, Y.; Player, K.; Heida, N.; Kobayashi, N. & Sakaki, Y. (2007). OmicBrowse: a browser of multidimensional omics annotations. *Bioinformatics*, Vol. 23, 524-526
- UniProt Consortium (2009). The Universal Protein Resource (UniProt) 2009. *Nucleic Acids Res.*, Vol. 37, D169-D174
- Van Driel, M.; Cuelenaere, K.; Kemmeren, P.; Leunissen, J.; Brunner, H. and Vriend, G. (2005). GeneSeeker: extraction and integration of human disease-related information from web-based genetic databases. *Nucleic Acids Res.*, Vol. 33, W758-W761
- Wain, H.; Lush, M.; Ducluzeau, F. & Povey, S. (2002). Genew: the human gene nomenclature database. *Nucleic Acids Res.*, Vol. 30, 169-171
- Yetisgen-Yildiz, M. & Pratt, W. (2006). Using statistical and knowledge-based approaches for literature-based discovery. *J. Biomed. Inform.*, Vol. 39, 600-611
- Yoshida, Y.; Makita, Y.; Heida, N.; Asano, S.; Matsushima, A.; Ishii, M.; Mochizuki, Y.; Masuya, H.; Wakana, S.; Kobayashi, N. & Toyoda, T. (2009). PosMed (Positional Medline): prioritizing genes with an artificial neural network comprising medical documents to accelerate positional cloning. *Nucleic Acids Res.*, Vol. 37, Web Server issue, W147-W152
- Zhang, Y.; Li, Y.; Zhang, J.; Shen, F.; Huang, Y. & Wu, Z. (2008). Characterization and mapping of a new male sterility mutant of anther advanced dehiscence (t) in rice. *J. Genet. Genomics*, Vol. 35, 177-182
- Zuo, L.; Li, S.; Chu, M.; Wang, S.; Deng, Q.; Ding, L.; Zhang, J.; Wen, Y.; Zheng A. & Li, P. (2008). Phenotypic characterization, genetic analysis, and molecular mapping of a new mutant gene for male sterility in rice. *Genome*, Vol. 51, 303-308

Artificial Neural Networks Technology to Model and Predict Plant Biology Process

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1. Introduction

The recent and significant technological advances applied to biology places the researchers in front of an unprecedented new influx of large data set from different levels as genomics, transcriptomics, proteomics, metabolomics and ionomics (Hirai et al., 2004; Belostotsky & Rose, 2005; Schauer & Fernie, 2006; Kliebenstein, 2010). Thousands of data sets including millions of measurements have been generated, and moreover, most are freely available for plant researchers worldwide from plant specific databases, as for example the whole sequencing of different plant genomes like rice, *Arabidopsis*, poplar, papaya, grapevine and others... (Jaillon et al., 2007; Ming et al., 2008; Brady & Provart, 2009). There is a wide concern of integrating molecular, cellular, histological, biochemical, genetic and physiological information in plant biology (Katagiri, 2003; Thum et al., 2003; Trewavas 2006; Boone et al., 2007; Álvarez-Buylla et al., 2007) and also in other related fields such as crop improvement (Hammer et al., 2002), ecology (Hilbert & Ostendorf, 2001; Jimenez et al., 2008) and biological engineering (Huang, 2009).

Biological processes are both time variant and nonlinear in nature, and their complexity can be understood as the composition of many different and interacting elements governed by non-deterministic rules and influenced by external factors (Coruzzi et al., 2009; Gago et al., 2009). Commonly, most of biological interactions cannot be elucidated by a simple stepwise algorithm or a precise formula, particularly when the data set are complex, noisy, vague, uncompleted or formed by different kind of data (Prasad & Dutta Gupta, 2008; Gago et al., 2010a). It is important to point out that many times the behaviour of a biological system over a time period is difficult to understand and interpret and additionally, genetic and environmental factors show a very high degree of intra- and inter-individual variability, yielding a wide spectrum of biological responses (Karim et al., 1997; Guégan et al., 1998).

The Scientific community agrees with the idea that plant biology requires more efforts in developing platforms to integrate multidimensional data and to derive models for describing biological interactions in plants (Kitano, 2002; Hammer et al., 2004; Struik et al., 2005; Tardieu, 2003; Yuan et al., 2008; Brady & Provart, 2009). In this sense, more efforts are recommended to shift our view from a reductionist way to a systems-level view. This concept can be illustrated by the Coruzzi & co-workers (2009) example of the painting "La Grande Jatte" by the

pointillist artist George Seurat. If one stands near the image each of its strokes can be distinguish but cannot appreciate the beauty of the whole composition. Only from far away is possible to admire the beautiful sunset over the Seine in Paris. Most researchers are really on top of their topic, but far away from the whole view of the global subject. However, the future in plant research requires a broader view of biological plant systems, from the new available molecular and cellular discoveries to the whole-plant improvements (Kirschner, 2005; Yuan et al., 2008) or the performance at the crop level (Kitano, 2002; Wang et al., 2002), and even to the agroecological level (Jiménez et al., 2008; Huang, 2009).

Performing a meta-analysis of the data set of a whole system is not an easy task. Hammer & collaborators (Hammer et al., 2004) expressed the present research requirement, comparing it to the paths of navigation the early sailors needed to determine to be able to arrive at their destiny with accuracy. In the 18th century this scientific problem was solved with the development of appropriate tools to predict the longitude and latitude in the middle of the oceans. Thus, these authors proposed an interesting conclusion: researchers need the equivalent to sailor's tools to navigate across the different levels of biological organization from gene to phenotype. As Tardieu (Tardieu, 2003) has pointed out that as in all "marriages of convenience" the first contact is always difficult between the ones having the tools, the modellers, with an essential mathematical and physical background, and the others, in this case the plant biologists, having the data and the knowledge in plant physiology (from molecular to whole plant and from individual cells to whole populations) who necessarily have to work together to obtain this exciting new challenge.

Since the spectacular development of computers, researchers have been attempting to create non-biological entities that can imitate human level of performance. Such attempts have manifested in the emergence of a cognitive approach termed as artificial intelligence (AI) (Legg & Hutter, 2007). In 1956, John McCarthy defined this term as "*the science and engineering of making intelligent machines*", and currently textbooks define this field of computer science as "*the study and design of intelligent agents*". AI achieved higher popularity in the 90s and early 21st century was introduced as a new tool in different scientific and technical fields (Russell & Norvig, 2003). Since then, successful studies have been carried out using different techniques, such as artificial neural networks, fuzzy logic and genetic algorithms, which can combine and complement in multiple ways and have been used in many industrial and commercial applications (Taylor, 1996) such as: character speech or image recognition (Hussain & Kabuka, 1998; Ma & Klorasani, 2004), chemical research (Cartwright, 1993; Zupan & Gasteiger, 1993), process modelling and control (Lennox et al., 2001), and in pharmaceuticals (Rowe & Roberts, 1998; Shao et al., 2006) or biomedicine (Hudson & Cohen, 2000). Finally, also since the late 90s artificial networks have been used in some biological areas, such as ecology or environmental sciences (Lek & Guégan, 1999; Hilbert & Ostendorf, 2001; Huang, 2009).

Since the basis for understanding the theoretical and practical approaches, of these technologies, to the development of models and their applications to specific problems, has been review elsewhere (Müller et al., 1995; Rowe & Roberts, 1998; Hudson & Cohen, 2000; Huang, 2009) and in other chapters of the present book. Therefore, the purpose of this chapter is to review the topics relevant to AI technology, mainly genetic algorithms and fuzzy logic, and more extensively, neural networks for the integration of multidimensional data into models (networks) and the application of these models in addressing questions (decision making) in plant biology.

As plant biology researchers are not commonly used to these new technologies; we will begin with a brief introduction for the readers on their fundamentals in order to facilitate the understanding of its applicability.

2. Analysis of biological data

Data from plant biology are inherently complex. The types of data must be taken into account since they influence the kind of analysis to be carried out. Normally, plant biology data can be classified as binary data, those with only two possible responses, generally yes/no, i.e. survival (alive–dead); discrete data, which have more than two responses (which can take one of a finite set of values), i.e. number of proliferated shoots or flowers in an inflorescence: 0, 1, 2, 3...; continuous data, which have any response (which can take any of an infinite number of values, i.e. weight rates of the proliferated shoots or the flowers...). Less frequent data in the plant biological process are image data (β -glucuronidase (GUS) and green fluorescence protein (GFP) histological analysis in transformation experiments); temporal data (a particular sequence of events: phenological development of fruits; duration of time for the fruit growth; or fruit fresh weight gain or loss per month), time series data (chilling hours in different seasons and their effect on bud formation) or fuzzy data (some processes or physiological states in plant science are described by linguistic tags. Using fuzzy data: for example, the embryo developmental stages such as the different sets can be explained as globular, torpedo, heart; or the callus colour, can be classified as: brown, brownish, yellowish during a plant in vitro culture callogenesis process; see fuzzy logic section for a complete description). All these types of data can be included in a neural networks systems database, but the precision and accuracy of the number of data must be taken into account.

Over years, experimental designs and statistics have been important research tools for the plant researchers. Conventional analytical tools including logistic regression (for binomial and multinomial); Poisson regression (for discrete data), analysis of variance (ANOVA) for data only normally or approximately normally distributed continuous data to extract conclusions from data and to understand biological process have generally been used (Mize et al., 1999; Gago et al., 2010a). These techniques have allowed many questions to be solved, however many shortcomings can also be pointed out. Relevant difficulties are found when researchers need to consider a large data set with different kinds of data at the same time, to model the whole process studied or when the non-idealities of plant science processes do not conform (Hammer et al., 2004; Prasad & Dutta Gupta, 2008; Gago et al., 2010a). Such problems need a different sort of intelligence and connectionism. The kind of approach that shows through nodes and connection diagrams of the interaction and integration of multiple components in organisms is possible at present using computational models (Rumelhart & McClelland, 1986; Yuan et al., 2008; Huang, 2009).

3. Modelling plant biology process

Since the purpose of modelling is to increase our understanding of a plant science process (for example: providing understanding of the regulatory networks controlling developmental, physiological or other processes in plants) plant biology models to approach the complexity of these processes are needed.

To solve this requirement, an abstraction is needed which is able to organize the factors (inputs) and the parameters measured (outputs) into a functional model. These principles are really useful to approach the biological complexity, by abstracting and focusing on the relevant factors to obtain a broad view of the whole system and become essential for the understanding principal questions and decision support. To explain the concept of how the plant biologists can approach the biological complexity, an interesting example of what is needed of the abstraction is as follows: in 1736 an old mathematical problem “The seven

bridges of Königsberg", a city in Prussia (actually Kaliningrado in Russia), was negatively resolved by Leonhard Euler, and meant the beginning of the mathematical graph theory. This city was crossed by the Pregel river and had two islands in the middle, connected by seven bridges to each other and to the mainland. Would it be possible to find a way through the city by crossing each bridge once? Euler reorganised the problem in abstract terms: giving no importance to the city or if the bridge was made of wood or stone... only to reducing all the information to nodes/vertex (land masses) and edges/link (bridges). The answer: "It would be not possible"; however, Euler's success was in reducing the problem to the significant inputs/factors and their relationships avoiding irrelevant data for the final purpose and to promote a suitable analysis.

Mathematical and computational models have been dramatically increasing in recent years in biological related sciences as in ecology (Stollenwerk et al., 2001; Anderson & Jensen, 2005), environmental and biodiversity conservation (Williams et al., 2004), epidemiology and pathogenesis (Brauer & Castillo-Chavez, 2001), genetic and biotechnology (Bar-Joseph et al., 2003), evolution (Nijhout et al., 2003) and animal (Schuster et al., 2005; Tracqui, 2006) or plant biology (Kovalenko & Riznichenko, 2007).

There is a wide variety of mathematical and computational models. Normally they are based in algorithms developed by theoretical and applied mathematicians, physics and/or bioinformatics, or for people with a firm background in theoretical biology, biological chemistry, mathematical biology in collaboration with informatics, computer science, physics, and/or engineering departments. In conclusion, mathematical modelling of biological systems generally requires a wide variety of methods and skills from multiple disciplines (de Vries et al., 2006).

For developing any mathematical model it is necessary to follow different steps: a) identification of the problem or process to be simulated, controlled and/or optimized; b) selection of data, variables (input and outputs), and very importantly, what is the model for; c) introduction of the accurate and precise data according from each variable, selection of equations (mainly algorithms) and the type of model: white box (all information is mostly available) or black box (no a priori information is available). If the black box model is chosen, some parameters can be used to fit the model to the system (in neural network the optimization of those parameters is called training); and d) model evaluation (normally cross-validation), to check the distances between the observed and predicted data (which should be as low as possible). The validity of the model is not only about if it fits well or not to empirical observations, but also about its ability to provide new insight which is partially occluded in the data, and which can not be known from direct observation or from statistical data analysis of the process. If the purpose is not achieved, the model is probably unnecessary, time consuming and useless. Another important issue to take into account is the quality of data. The model will be excellent if data are excellent. Models do not produce miracles. Bad quality data (non accurate, disperse or non precise) are not the best option to fit a model, especially if they are selected for a black box model, as neural networks are.

4. Artificial Neural Networks

Artificial Neural Networks (ANNs) are computational systems that simulate biological neural networks and they have been widely described in previous chapters of this book and in detail elsewhere (Russell & Norvig, 2003; Rowe & Roberts, 2005), but to better understand their application to plant biology it is pertinent to briefly review the way it is believed to function.

The main elements of biological neural networks are the neurons, electrochemical excitable cells that can receive signals or stimuli from other neurons via synapse connection (Fig. 1). The stimuli is received through the dendrites and transmitted to the cell body. If the stimulus is intense enough, the neuron generates another stimulus that is transmitted along the axon to the next neuron via synapses.

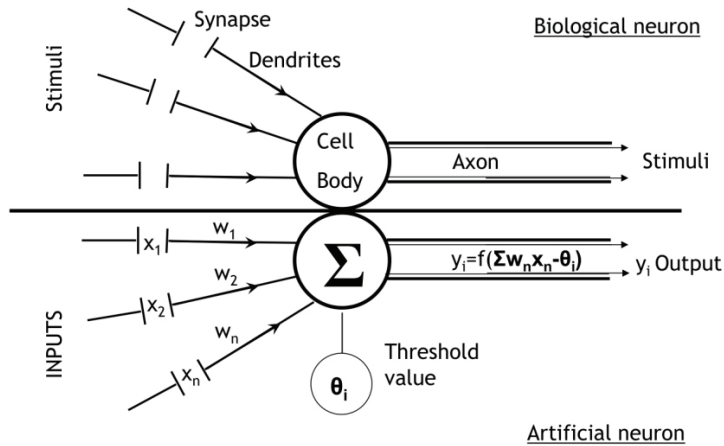


Fig. 1. Basic comparison between a biological neuron and an artificial neuron. X = input variable; W =weight of i_n input; θ = internal threshold value; f =transfer function.

The artificial neural network architecture is an interconnected assembly of individual processing elements called perceptron: "single nodes" or "artificial neurons" (Fig. 1). Each artificial neuron receives one or more inputs from neighbouring nodes, process the information and produces an output to be transmitted to the next node. The strengths of connections between two units are called "weights" which must be defined by the computational approach to solve or interpret a given problem (Takayama et al., 1999). While computing the output, the input information (X_i) is weighed either positively or negatively. The computational approach must also assign an internal threshold value (θ) to simulate the output action. At each node, the input values (X_n) are multiplied by their associate weight (W_n) to give a result, which is adjusted by its threshold value. The output is then determined using the non-linear weighted sum as the argument in a function " f " termed *transfer function* or *activation function* (Fig. 1; eq. 1).

$$y_i = f(\sum W_n X_n - \theta_i) \quad (1)$$

Among the functions that can be applied: linear, hyperbolic tangent or radial basis form, etc is the sigmoid function (eq. 2) and is the most commonly used. Sigmoid function, $f(y_i)$, is conducted to the following layer as an output value. Alpha is a parameter relating to the shape of the sigmoid function. Non-linearity of the sigmoid function is strengthened with an increase in α .

$$f(y_i) = 1 / [1 + \exp(-\alpha y_i)] \quad (2)$$

By connecting several artificial neurons (many hundreds of thousands) or simple nodes a complex artificial neural network can be obtained. Figure 2 illustrates the topology of one of the most popular and successful network architectures, a multilayer perceptron (MLP) networks, consisting of three simple layers; one input layer, one output layer and with just one hidden layer.

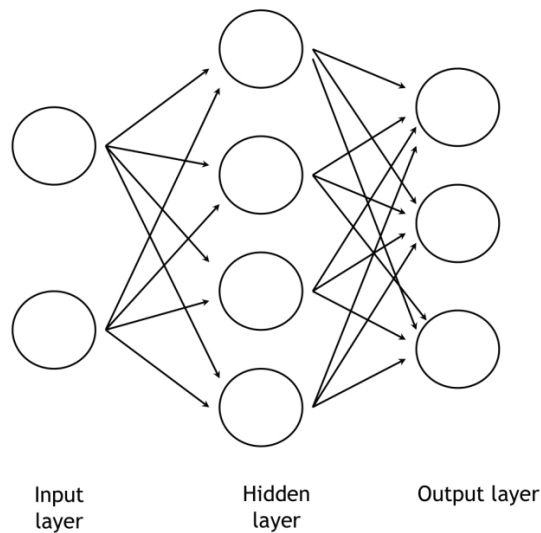


Fig. 2. A multilayer perceptron with one hidden layer.

ANNs is able to “learn” an approximate non-linear relationship between inputs and outputs using algorithms designed to alter the strength (weights) of the connections in the network to produce a desired signal flow. This “training” process is defined as a search process for the optimized set of weight values which can minimize the squared error between the data predicted by the model and the experimental data in the output layer (Takayama et al., 1999).

The ability of the network to memorize and process the information lies in the weights assigned to the inter-node connections, which determines the conductivity through the network. When the computed output is unacceptable, compared with experimental output, a back propagation process starts to modify several setting parameters (also called the learning rule) until the network attains a good generalization of the problem domain (Prasad & Dutta Gupta, 2008). The difference between actual and predicted outputs is usually quantified by means of an error function similar to those used in statistics.

In different research fields, it has been proposed that the performance of a well-designed MLP network is comparable to that achieved by classical statistical techniques (Rowe & Roberts, 1998) and therefore, suitable for a wide range of applications including: classification (Glezakos et al., 2010), pattern recognition (Frossyniotis et al., 2008), prediction on time series (Müller et al., 1995) interpolation (Gago et al., 2010a), and modelling complex systems with non-linear behaviour (Karim et al., 1997; Mehrota et al., 2008; Gago et al., 2010b, c)

It is important to point out that the strength of ANNs lays on its ability in detecting and quantifying complex non-linear relationships between inputs and outputs as well as its capability on generalizing distorted or partially occluded patterns (Taylor, 1996; Shao et al.,

2006). This powerful technology has also limitations, mainly related to the difficulties of interpreting the results in simple form or to elaborate formal reasoning or extract general rules (Colbourn, 2003). Complex “black box” models derived from ANNs technologies from a large data set with an important number of inputs could be difficult to analyze using general 2D plots or even 3D graphs. In order to avoid those limitations ANNs are usually combined with other AI techniques as genetic algorithms or fuzzy logic technology giving hybrid systems the results of which are easier to interpret and generalize (Plumb et al., 2005; Shao et al., 2006; Shao et al., 2007; Gago et al., 2010d).

4.1 Artificial Neural Networks as an alternative to traditional statistics

As pointed out by Mize and coworkers (1999), an extensive review of the literature shows that the use of correct statistical tools is not widespread in plant science. Many papers can be found where authors treat discrete or binomial variables as continuous variables by using the ANOVA. Moreover, when analyzed correctly, usually multiple comparison tests are performed to determine which of the factors studied have a significant effect on certain parameter or process.

The use of modeling and optimization techniques is even more restricted, generally being reduced to fit data to a specific functional form (linear or quadratic). This is because, in many cases, the plant researcher lacks the appropriate mathematical background for the analysis and his/her interaction with experts in statistics is not very fruitful. In this situation the use of a different technology as ANNs can be of great help.

Recent studies have demonstrated that AI technologies show the same or even better performance than traditional statistics for modelling complex non linear relationships hidden in the data and offer superior prediction powers (Landin et al., 2009 and references therein; Gago et al., 2010a).

From a formal point of view ANNs show several advantages over statistics: a) they can process different types of data together (continuous, binomial, discrete); b) they can be used to produce complex models without the previous knowledge of the functional form of dependence, so they can discover subtle relationships in the data; c) they do not require specific experimental design, being capable of using incomplete data, data acquired during a series of trial-and-error experiments or even historical data (Colbourn, 2003; Colbourn and Rowe, 2005).

Moreover, the use of ANNs does not require a specialized background and is a friendly technology, easy to use, that allows the modelling process with a limited number of experiments and costs, and makes inference of the combination of factors studied possible to obtain the best result (Gago et al., 2010a).

4.2 Applications of neural networks to plant biology

There are not many references in the literature on the applications of ANN to plant biology. This fact is more relevant, if we compare with other related areas as pharmaceutical science of important research in the last few years (Achant et al., 1995; Colbourn, 2003; Takayama et al., 1999; Shao et al., 2006; Landín et al., 2009), ecology (Guégan et al., 1998; Hilbert et al., 2001; Adriaenssens et al., 2004) or agriculture (Huang, 2009 and references therein).

Pioneer studies in plant science deal with the use of AI technology to improve and/or optimize biotechnology processes production. An early work (Fukuda et al., 1991) explored the use of artificial neural networks to recognize live or dead plant cells by image processing. Other authors have also investigated the capabilities of image analysis of

somatic embryos developmental stage by neural networks (Uozumi et al., 1993) and image analysis of shoot length of regenerated rice callus using a hybrid (artificial neural networks/fuzzy logic) technology (Honda et al., 1997). Additionally, works dealing with modelling the bioproduction of Ginjo sake were carried out using a fuzzy neural network in order to control the processes in bioreactors (Hanai et al., 1997).

During the last twenty years there has been an increasing interest on this technology in the agricultural and biological engineering fields (Huang, 2009). Firstly, it was applied to model food quality (Whittaker et al., 1991; Eerikäinen et al., 1993) or fruit colour (Thai & Shewfelt, 1991) and, more recently, to manage herbicide application (Yang, 2003), yield estimation (Kaul et al., 2005; Khazaei et al., 2008), and water stress (Ondimu & Murase, 2008).

Neural networks were used for modelling crop yields on the basis of environmental conditions and pest control treatments in order to improve production. The optimization of pesticide concentration and periods of treatments to be used has a great impact on the costs and toxic residual levels of agriculture products (Jiménez et al., 2008). Health and economy are the important issues in the agricultural production nowadays. Different authors also established the relationship between the factors and crop yield for corn, sugar beet, soybean and winter wheat in order to help on decision-making processes (Kehagias et al., 1998; Kaul et al., 2005; Green et al., 2007; Jiménez et al., 2008).

Other authors have described the usefulness of ANNs for modelling the distribution of vegetation in past, present and future climates coupled with GIS (geographic information system). They have provided worthy contributions to understanding and conservation of these areas, especially when more detailed biogeographical data were available (Hilbert & Ostendorf, 2001).

More recently, detection of plant viruses has been carried out through a Bioelectric Recognition Assay (BERA) method in combination with neural networks (Frossyniotis et al., 2008). The sensors monitor the electric signal of the cells interacting with viruses making their identification possible. ANNs were trained with the responses of the biosensors to obtain a classification model of the culture cells infected.

ANNs based modelling approaches have also been applied in cell culture practice (Prasad & Dutta Gupta, 2008). One of the most important topics in *in vitro* culture is related with the supply of carbon source, commonly sucrose, to determine the effect on plant growth and physiological parameters. Tani and co-workers (1992) developed a growth model for *in vitro* shoots of alfalfa describing the effects of CO₂ inside the culture vessel and the sucrose content. This model increases the understanding of *in vitro* processes in a non-deterministic way. Three years later, a comparison between the deterministic mathematical model Extended Kalman Filter approach and ANNs was performed (Albiol et al., 1995). Authors stated the usefulness of neural networks for modelling at less cost, time and a smaller dataset.

Pattern recognition and classification models are commonly applied in plant tissue culture studies (Prasad & Dutta Gupta, 2008). Usually, the selection of embryos inside the embryogenic cultures is laborious, cost intensive and time-consuming. An image analysis pattern recognition system was developed by Zhang and coworkers (1999) using ANNs to select embryos of Douglas fir: the contour of embryo images was segmented, digitalized and converted into numerical values after the discrete and fast Fourier transformation (values obtained were higher than 80% for normal embryos). In another study, embryos of sugarcane from callogenic culture were selected by image analysis using machine vision analysis (MVA) confirming the technique as a rapid, non-invasive method for qualitative evaluation and quantification of *in vitro* regenerated plantlets (Honda et al., 1999). Finally, regenerated

plantlets could be sorted using their photometrical behaviour of their leaves in Red, Blue and Green colour regimes combining image analysis and ANNs (Mahendra et al., 2004).

Transference to field the micropropagated plantlets is a typical bottleneck in micropropagation protocols. Optimizing rhizogenesis and subsequent acclimatization is highly recommended to improve efficient propagation yields (Gago, 2009). The quality and the functionality of the root system, the vigour of the plantlets and other physiological status will be responsible for the quality of the plantlets in *ex vitro* conditions (Gago et al., 2009). Recently (Gago et al., 2010b), ANNs were used to model *in vitro* rhizogenesis and subsequent acclimatization data simultaneously of grapevine *Vitis vinifera* L. cv. Albariño. Studied inputs/factors (cultivar, IBA concentration and exposure time to IBA) showed significant effects on root number, *ex vitro* leaves, number of nodes and height of the acclimatized plantlets, the exposure time to the synthetic auxin IBA being the more relevant. The model allowed optimal predictions for every studied cultivar. The knowledge derived through ANNs can be easily increased by training the model by adding to the database new inputs (salt concentration, type of medium, other plant hormone, etc.) and/or outputs (plantlets weight, chlorophyll and carotenes content, stomata analysis, etc.).

5. Genetic algorithms

Once an ANN model has been obtained it is easy to predict what will be the output for a specific set of inputs, or in other word to formulate “what if” questions obtaining accurate responses (Fig. 3). This consultation mode will provide insight into the process studied. However sometimes, the main research objective is to determine the combination of input parameters that will provide the optimum result, that is, an optimization process which means to formulate “how to get” questions on the best/highest...output. In these cases a different AI technique, the genetic algorithms, can be applied.

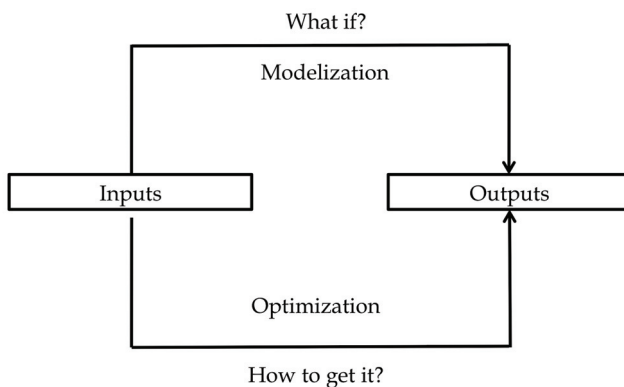


Fig. 3. The relationship between modeling and optimization from an ANN model (Modified from Rowe and Roberts, 2005).

“The genetic algorithm is an optimization technique based on evolutionary principles” (Cartwright, 1993). Genetic algorithms are based on the biological principles of genetic variation and natural selection, mimicking the basic ideas of evolution over generations. As Rowe & Roberts declare: “An optimization process evolves finding the best solution for a specific problem” (Rowe & Roberts, 1998). Genetic algorithm randomly generates a set of

candidate solutions to the problem. Solutions from one population are selected according to their fitness of evolving new populations to the problem. This is repeated until some condition is satisfied. After several generations an optimum must be achieved because the most suitable becomes the solution and therefore the more chances it has to reproduce. For a genetic algorithm to function it must possess several features. Firstly, a numerical description of how good a solution is to the problem. Secondly, a logical method of selecting individual solutions to become parents of the next generation must be fixed. And finally, a logical method of mixing the different elements to produce new solutions is necessary (Mitchell, 1998; Glezakos et al., 2010).

5.1. Applications of genetic algorithms to plant biology

Genetic algorithms have been used in plant science for different optimization processes. Noguchi & Terao (1997) have developed a mobile robot for harvesting fruit automatically and genetic algorithms were designed to find the optimal space solution for path planning. Hybrid systems combining neural networks and genetic algorithms have also been used for optimizing the quality of fruits stored under a controlled environment (Morimoto et al., 1997; Morimoto & Hashimoto, 2000) and for plant virus identification through a Bio-Electric Recognition Assay (BERA) (Glezakos et al., 2010).

More complex *in vitro* culture processes such as shoot proliferation, root formation (rhizogenesis) and plantlets acclimatization have been modeled by ANNs and successfully optimized by genetic algorithms in woody fruit plants, such as kiwifruit (Gago et al., 2010a) and grapevine (Gago et al., 2010b).

6. Neurofuzzy logic

Human knowledge is typically built on linguistic tags (characterized by uncertainty or imprecision) and not on quantitative mathematical data. Many times words have higher significance in the real world than a collection of numerical data (Fig. 4) the basis being to solve problems, make decisions or draw conclusions.

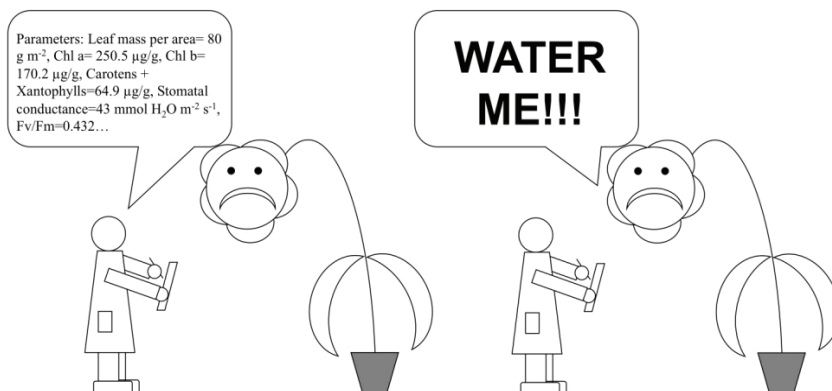


Fig. 4. Picture on the importance of precision and significance in the real world of plant research.

Neurofuzzy logic is a hybrid system technology developed to break in this topic. It combines the adaptive learning capabilities from ANNs with the generality of representation from fuzzy logic (Shao et al., 2006).

Fuzzy logic is an AI tool especially useful in problem solving. Fuzzy set theory was introduced by Zadeh (1965) as an extension of the classical set theory, which enables the processing of imprecise information using the membership concept (Adriaenssens et al., 2004). Prof. Zadeh illustrated the concept of fuzzy set and the degree of membership with the classical “tall man” example (Fig. 5). The conventional characteristic mapping of a classical logic set determines that a man is tall when his height is over 1.80 m. Zadeh extended the traditional definition of a logic premise from having just two extremes (either a man is over 1.80 m, so he is tall or lower 1.80 m, so he is not tall) to one in which there is a range in degree of truth from 0 to 1. For example new sets can be described qualitatively by terms as very low, low, average, tall or very tall. However, following the classical logic, a small difference of just 2 cm from 1.79 to 1.81 m, induces classification of the man in two completely different categories. This does not seem really “logic”. But, using the fuzzy set theory, an element of those sets can be assigned to a fuzzy set with its membership degree ranging from zero to one, so two men 1.79 m and 1.81 m tall belong to fuzzy set “tall man” with membership degrees of 0.70 and 0.90 respectively (Fig. 5).

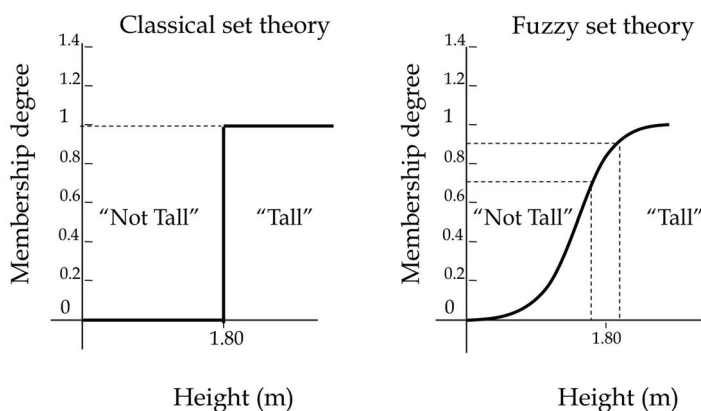


Fig. 5. Comparison between classical set theory and fuzzy set theory to illustrate the Zadeh's example of the “tall man” (Modified from Zadeh, 1965).

These kinds of fuzzy data or fuzzy variables can be numerically characterized, but a fuzzification process is necessary. For better understanding of this concept, we will choose a typical tissue culture proliferation experiment in order to evaluate the effect of light intensity on the success of the proliferation of kiwifruit shoots. Light conditions for an *in vitro* culture experiment can be expressed by the Photosynthetic Photon Flux Density (PPFD) parameter (Fig. 6). The x axis is the PPFD with ranges for the fuzzy sets low, medium or high. The y axis represents the membership function and ranges 0 to 1 (also could be expressed from 0 to 100 per cent). It can be seen that a PPFD light of $80 \text{ mmol}^{-2}\text{s}^{-1}$ can be regarded as both low and medium PPFD with membership functions of 0.7 and 0.4 respectively. In other words $80 \text{ mmol}^{-2}\text{s}^{-1}$ is low light in a greater degree than it is medium light.

Therefore, fuzzy logic sets labels qualitatively using linguistic terms and also assigns varying degrees of membership called membership functions. The membership function

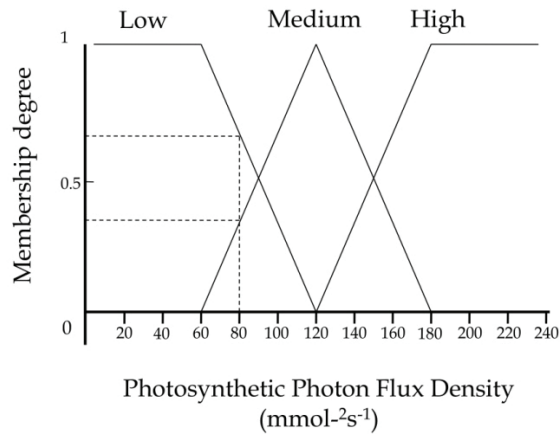


Fig. 6. Examples of fuzzy sets for light intensity (PPFD) for an *in vitro* plant tissue culture experiment.

then is subjective in nature and is a matter of definition rather than measurement. This process allows the interaction between linguistic terms (low, medium, high) and the membership functions making the terms meaningful to a computer. Additionally it makes expressing the behaviour of a system possible using natural language and enhancing the possibility of concise description of complex tasks or process.

In fuzzy-rule based systems, after modelling, knowledge is presented by IF-THEN rules. Fuzzy rules consist of two parts: an antecedent part stating conditions on the input variable(s), and a consequent part describing the corresponding values of the output variable(s). Given particularly values of the input variables, the degree of fulfilment of each rule is obtained by aggregating the membership degrees of these input values into the respective fuzzy sets. Going back to the example described previously if an evaluation of the effect of light conditions on the length of the plants in an *in vitro* culture experiment were carried out, the IF THEN rules could be similar to those presented in Table 1: IF PPFD is low THEN the plant length obtained is HIGH with a membership of 89% (more detailed information can be obtained in Gago et al., 2010d).

IF...THEN RULES		
IF PPFD is LOW	THEN Plant length is	HIGH (0.89)
IF PPFD is MID	THEN Plant length is	LOW (0.76)
IF PPFD is HIGH	THEN Plant length is	LOW (0.68)

Table 1. Examples of a fuzzy output using IF THEN rules describing the effect of the light intensity (PPFD) on the plant length in an *in vitro* culture experiment.

The fuzzy output is determined by the degrees of fulfilment and the consequent parts of the rules (Adriaenssens et al., 2004). The logical structure of rules facilitates the comprehension of a semi-qualitative manner, similar to that used by the human brain to analyze the real world (Babuska, 1998).

The major capabilities of fuzzy logic are the flexibility, the tolerance with uncertainty and vagueness and the possibility of modelling non linear functions, searching for consistent

patterns or systemic relationships between variables in a complex dataset, data mining and promoting deep understanding of the processes studied by generating comprehensible and reusable knowledge in an explicitly format (Setness et al., 1998; Shao et al., 2006; Yuan et al., 2008; Landin et al., 2009; Gago et al., 2010d).

Chen & Mynett (2003) have argued that definition of membership functions and induction of inference rules is the most difficult part in a fuzzy logic process. Fuzzy logic can be combined with neural networks to produce neuro-fuzzy techniques. Those hybrid systems combine the generality and flexibility of representation, a feature of fuzzy logic, with the powerful learning and adaptive capability of neural networks (Babuska, 1998; Adriaenssens et al., 2004).

6.1 Applications of neurofuzzy logic to plant biology

There are not many works in the literature on neurofuzzy logic in plant science. The neurofuzzy logic technology was used for controlling and modelling *Ginjo* sake brewing process: the interaction between sensory evaluation and the chemical composition of sake, beer and coffee was studied, showing this technology to have a high level of accuracy (Hanai et al., 1997). Also, there are some works related to the monitoring of herbicide sprayed in cornfields with a system that includes real-time image processing, weed identification, mapping of weed density, and sprayer control using a digital camera. Simulations using different fuzzy rules and membership functions indicated that the precision spraying has potential for reducing water pollution from herbicides needed for weed control in a corn field (Yang et al., 2003). Finally, hybrid systems as NUFZY involving a fuzzy approach and the training algorithm OLS (orthogonal least squares) has also been used to model accurately the lettuce growth and the greenhouse temperature (Tien & Van Straten, 1998).

Recently, Gago and coworkers (Gago et al., 2010d) compared the utility of the traditional statistical analysis and neurofuzzy logic technology for dataset highly complex and with great variability of direct rooting and subsequent acclimatization of grapevine. Neurofuzzy logic showed higher accuracy to identify the interaction effects between the factors: type of auxin (IBA, IAA and NAA), auxin concentration (1 to 50 mM) and sucrose concentration (0 to 9%) than conventional statistical analysis. Also, neurofuzzy showed a considerable potential for data mining and retrieve knowledge from the complex dataset. Understanding was increased thanks to IF-THEN rules generated from the model to facilitate researchers interpretation of the results, main effects and their consequences.

Considerable efforts have been made to understand artificial neural networks and neurofuzzy logic capacities in this sense, and many works are expected, in the near future, to provide a comprehensive insight into the expediency of processing networks in interpreting the database derived from plant biology research. In addition, one of the major advantages of these hybrid techniques is the capacity to model and estimate different complex processes.

7. Future perspectives

Biological systems are complex to understand. They have different scales of biological organization (genetic, biochemical, physiological...) and different factors influence them. Nowadays modern technology gives us the opportunity to generate a huge amount of biological data (Brady & Provart, 2009). This storm of information would be useless if at the same time the technology do not solve the problems associated of analysing, integrating and

extracting knowledge from those data. If the technology creates a problem, the technology should solve it.

AI technologies, the ones in use and the ones coming out in the future, will help the researchers with those difficult tasks: to integrate variable information, to explain what is going on, to model and finally to predict what will happen in a specific situation.

As described above neural networks can be used for a wide range of application domains in the general area of plant biology, and gives models as least as good as those obtained using statistical modelling. Neural networks combined with genetic algorithms can predict the combination of variables that would yield optimum solution when independent variables are fed into the network as it has been pointed out in this review. Neural network technologies have also spectacular advantages over other systems as the ability to capture non-linear relationships in the data (wherever their origin or type and even from incomplete data sets), without requiring prior knowledge from the user. In fact, the user does not need to have a deep mathematical or statistical background to employ effectively neural systems. We think that once having overcome the natural reluctance of scientists to these new technologies, they will impose as the usual way for dealing with biological results. For that purpose, we strongly recommend the use of software packages, which incorporates visualization and data manipulation capabilities, within an easy to use interface, so the users do not need to be experts in neural computer (ie those used in Gago et al., 2010a, d).

Finally, the knowledge derived through neural networks can be easily increased by adding new data (inputs and/or outputs) to the database, giving new insight to understand the regulatory process controlling developmental and physiological processes in plants, as a whole. The knowledge obtained in this way should be crucial for both basic and applied plant biology.

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9. References

- Adriaenssens, V.; De Baets, B.; Goethals, P.L.M. & De Pauw, N. (2004). Fuzzy rule-based methods for decision support in ecosystem management. *The Science of Total Environment*, 319, 1-12.
- Achanta, A.S.; Kowalski, J.G. & Rhodes, C.T. (1995). Artificial neural network: implications for pharmaceutical sciences. *Drug Development and Industrial Pharmacy*, 21, 119-155.
- Albiol, J.; Campmajo, C.; Casas, C. & Poch, M. (1995). Biomasa estimation in plant cell cultures: A neural network approach. *Biotechnology Progress*, 11, 8-92.
- Alvarez-Buylla, E.R.; Benitez, M.; Davila, E.B.; Chaos, A.; Espinosa-Soto, C. & Padilla Longoria P. (2007). Gene regulatory network models for plant development. *Current Opinion in Plant Biology*, 10, 83-91.
- Anderson, P.E. & Jensen, H.J. (2005). Networks properties, species abundance and evolution in a model of evolutionary ecology. *Journal of Theoretical Biology*, 232, 4, 551-558.

- Babuska, R. (1998). Fuzzy modelling for control, In: *International Series in Intelligent Technologies*, Babuska R. (Ed.), 1-8, Kluwer Academic Publishers, Massachusetts, USA.
- Bar-Joseph, Z.; Gerber, G.K.; Lee, T.I.; Rinaldi, N.J.; Yoo, J.Y.; Robert, F.; Gordon, D.B.; Fraenkel, E.; Jaakkola, T.S.; Young, R.A. & Gifford, D.K. (2003). Computational discovery of genes modules and regulatory networks. *Nature Biotechnology*, 21, 11, 1337-1342.
- Belostotsky, D.A. & Rose, A.B. (2005). Plant gene expression in the age of systems biology: integrating transcriptional and posttranscriptional events. *Trends in Plant Science*, 10, 347-353.
- Boone, C.; Bussey, H. & Andrews, B.J. (2007). Exploring genetic interactions and networks with yeast. *Nature Reviews Genetics*, 8, 6, 437-449.
- Brady, S.M. & Provart, N.J. (2009). Web-queryable large-scale data sets for hypothesis generation in plant biology. *Plant Cell* 21, 1034-1051.
- Brauer, F. & Castillo-Chavez, C. (2001). *Mathematical models in population biology and epidemiology* (Texts in Applied Mathematics 40), Springer Verlag, Berlin.
- Cartwright, H.M. (1993). *Applications of Artificial Intelligence in Chemistry*, Oxford University Press, Oxford.
- Chen, Q. & Mynett A.E. (2003). Integration of data mining techniques and heuristic knowledge in fuzzy logic modelling of eutrophication in Taihu Lake. *Ecological Modelling*, 162, 1-2, 55-67.
- Colbourn, E. (2003). Neural computing: enable intelligent formulations. *Pharmaceutical Technology Supplement*, 16-20.
- Colbourn, E. & Rowe, R.C. (2005). Neural computing and pharmaceutical formulation. In: *Encyclopaedia of pharmaceutical technology*, Swarbrick, J. & Boylan, J.C. (Eds), Marcel Dekker, New York.
- Coruzzi, G.M.; Burga, A.R.; Katari, M.S. & Gutiérrez, R.A. (2009). Systems biology: principles and applications in plant research, In: *Annual Plant Reviews*, Vol. 35, *Plant Systems Biology*, Gloria M. Coruzzi and Rodrigo A. Gutiérrez (Eds.), 3-40, Wiley-Blackwell, United Kingdom.
- De Vries, G.; Hillen, T.; Lewis, M.; Müller, J. & Schönfisch, B. (2006). *A course in mathematical biology: quantitative modelling with mathematical and computational methods*. SIAM, Society for Industrial and Applied Mathematics, Philadelphia, PA.
- Eerikäinen, T.; Linko, P.; Linko, S.; Siimes, T. & Zhu Y.H. (1993). Fussy logic and neural networks applications in food science and technology. *Trends in Food Science and Technology*, 4, 237-242.
- Frossyniotis, D.; Anthopoulos, Y.; Kintzios, S.; Moschopoulou, G. & Yialouris, C.P. (2008). Artificial neural network selection for the detection of plant viruses. *World Journal of Agricultural Sciences*, 4, 1, 114-120.
- Fukuda, T.; Shiotani, S.; Aral, F.; Takeuchi, N.; Sasaki, K. & Kimoshita, T. (1991). Cell recognition by image processing (1st report, recognition of dead or alive plant cells by neural network). *Nihon Kikaigakkai Ronbunshu*, 57, 77-84.
- Gago, J. (2009). *Bioteecnología de Vitis vinifera L.: Modelización mediante Inteligencia Artificial*. Doctoral Thesis, Universidade de Vigo, Vigo, Spain.

- Gago, J.; Martínez-Núñez, L.; Molist, P. & Gallego, P.P. (2009). Effect of three auxins on *in vitro* rooting in soil and acclimatization of *Vitis vinifera* L. Albariño. *Acta Horticulturae*, 812, 359-364.
- Gago, J.; Martínez-Núñez, L.; Landín, M. & Gallego, P.P. (2010a). Artificial neural networks as an alternative to the traditional statistical methodology in plant research. *Journal of Plant Physiology*, 167, 23-27.
- Gago, J.; Landín, M. & Gallego, P.P. (2010b). Artificial neural networks modeling the *in vitro* rhizogenesis and acclimatization of *Vitis vinifera* L. *Journal of Plant Physiology*, 167, 1226-1231.
- Gago, J.; Martínez-Núñez, L.; Landín, M. & Gallego, P.P. (2010c). Strengths of artificial neural networks in modelling complex plant processes. *Plant Signaling and Behavior* 5, 6, 1-3.
- Gago, J.; Landín, M. & Gallego, P.P. (2010d). A neurofuzzy logic approach for modeling plant processes: a practical case of *in vitro* direct rooting and acclimatization of *Vitis vinifera* L. *Plant Science*, 179, 241-249.
- Glezakos, T.J.; Moschopoulou, G.; Tsiligiridis, T.A.; Kintzios, S. & Yialouris, C.P. (2010). Plant virus identification based on neural networks with evolutionary preprocessing. *Computers and Electronics in Agriculture*, 70, 263-275.
- Green, T.R.; Salas J.D.; Martinez, A. & Erskine R.H. (2007). Relating crop yield to topographic attributes using Spatial Analysis Neural Networks and regression. *Geoderma*, 23-37.
- Guégan, J.F.; Lek, S. & Oberdorff T. (1998). Energy availability and habitat heterogeneity predict global riverine fish diversity. *Nature*, 391, 382-384.
- Hammer, G.L.; Kropff, M.J.; Sinclair, T.R. & Porter, J.R. (2002). Future contributions of crop modelling: from heuristics and supporting decision making to understanding genetic regulation and aiding crop improvement. *European Journal of Agronomy*, 18, 15-31.
- Hammer, G.L.; Sinclair, T.R.; Chapman, S.C. & van Oosterom E. (2004). On systems thinking, systems biology, and the *in silico* plant. *Plant Physiology*, 134, 909-911.
- Hanai, T.; Katayama, A.; Honda, H. & Kobayashi, T. (1997). Automatic fuzzy modelling for Ginjo sake brewing process using fuzzy neural network. *Journal of Chemical Engineering of Japan*, 30, 94-100.
- Hilbert, D.W. & Ostendorf B. (2001). The utility of neural networks for modelling the distribution of vegetation in past, present and future climates. *Ecological modelling*, 146, 311-327.
- Hirai, M.Y.; Yano, M.; Goodenowe, D.B.; Kanaya, S.; Kimura, T.; Awazuhara, M.; Arita, M.; Fujiwara, T. & Saito, K. (2004). Integration of transcriptomics and metabolomics for understanding of global responses to nutritional stresses in *Arabidopsis thaliana*. *Proceedings of The National Academy of Sciences USA*, 101, 10205-10210.
- Honda, H.; Takikawa, N.; Noguchi, H.; Hanai, T. & Kobayashi, T. (1997). Image analysis associated with fuzzy neural network and estimation of shoot length of regenerated rice callus. *Journal of Fermentation and Bioengineering*, 84, 342-347.
- Honda, H.; Ito, T.; Yamada, J.; Hanai, T. & Matsuoka, M. (1999). Selection of embryogenic sugarcane callus by image analysis. *Journal of Bioscience and Bioengineering*, 87, 5, 700-702.
- Huang, Y. (2009). Advances in artificial neural networks - Methodological development and application. *Algorithms*, 2, 973-1007.

- Hudson, D.L. & Cohen, M.E. (2000). *Neural networks and artificial intelligence for biomedical engineering*. IEEE Press, New York.
- Hussain, B. & Kabuka, M.R. (1998). A novel feature recognition neural network and its application to character recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 6, 98-106.
- Jaillon, O.; Aury, J.M.; Noel, B.; Policriti, A.; Clepet, C.; Casagrande, A.; Choisne, N.; Aubourg, S.; Vitulo, N.; Jubin, C.; Vezzi, A.; Legeai, F.; Huguency, P.; Dasilva, C.; Horner, D.; Mica, E.; Jublot, D.; Poulain, J.; Bruyere, C.; Billault, A.; Segurens, B.; Gouyvenoux, M.; Ugarte, E.; Cattonaro, F.; Anthouard, V.; Vico, V.; Del Fabbro, C.; Alaux, M.; Di Gaspero, G.; Dumas, V.; Felice, N.; Paillard, S.; Juman, I.; Moreoldo, M.; Scalabrin, S.; Canaguier, A.; Le Clainche, I.; Malacrida, G.; Durand, E.; Pesole, G.; Laucou, V.; Chatelet, P.; Merdinoglu, D.; Delledonne, M.; Pezzotti, M.; Lecharny, A.; Scarpelli, C.; Artiguenave, F.; Pe, M.E.; Valle, G.; Morgante, M.; Caboche, M.; Adam-Blodon, A.F.; Weissenbach, J.; Quetier, F. & Wincker, P. (2007). The grapevine genome sequence suggests ancestral hexaploidization in major angiosperm phyla. *Nature*, 449, 463-467.
- Jiménez, D.; Pérez-Urbe, A.; Satizábal, H.; Barreto, M.; Van Damme, P. & Marco, T. (2008). A Survey of Artificial Neural Network-Based Modeling in Agroecology. In: *Soft Computing applications in industry*, STUDFUZZ. B. Prasad (Ed), 247-269 Springer-Verlag, Berlin-Heidelberg Germany.
- Karim, M.N.; Yoshida, T.; Rivera, S.L.; Saucedo, V.; Eikens, B. & Oh, G. (1997). Global and local neural network models in biotechnology: application to different cultivation processes. *Journal of Fermentation and Bioengineering*, 83, 1, 1-11.
- Katagiri, F. (2003). Attacking complex problems with the power of systems biology. *Plant Physiology*, 132, 417-419.
- Kaul, M.; Hill, R.L. & Walthall, C. (2005). Artificial neural networks for corn and soybean yield prediction. *Agricultural Systems*, 85, 1-18.
- Khazaei, J.; Naghavi, M.R.; Jahansou, M.R. & Salimi-Khorshidi, G. (2008). Yield estimation and clustering of chickpea genotypes using soft computing techniques. *Agronomy Journal*, 100, 1077-1087.
- Kehagias, A.; Panagiotou, N.; Maslaris, V.; Petridis, V. & Petrou, L. (1998). Predictive modular neural networks methods for prediction of sugar beet crop yield. In: *IFAC Conference on Control Applications and Ergonomics in Agriculture*, Athens, Greece, pp 1-5.
- Kirschner, M.W. (2005). The meaning of systems biology. *Cell*, 121, 503-504.
- Kitano, H. (2002). Systems biology: a brief overview. *Science*, 295, 1662-1664.
- Kliebenstein, D.J. (2010). Systems biology uncovers the foundation of natural genetic diversity. *Plant Physiology*, 152, 480-486.
- Kovalenko, I.B. & Riznichenko, G.Y. (2007). Multiparticle direct simulation of photosynthetic electron transport processes. In: *Mathematical Modelling of Biological Systems, Volume I*, Deutsch A, Brush L, Byrne H, de Vries G & Herzel H (Eds), 3-9, Birkhäuser, Boston.
- Landín, M.; Rowe, R.C. & York, P. (2009). Advantages of neurofuzzy logic against conventional experimental design and statistical analysis in studying and developing direct compression formulations. *European Journal of Pharmaceutical Science*, 38, 325-331.

- Legg, S. & Hutter, M. (2007). Universal Intelligence: A Definition of Machine Intelligence. *Minds and Machines*, 17, 391-444.
- Lek, S. & Guégan, J.F. (1999). Artificial neural networks as a tool in *Ecological Modelling*, an introduction. *Ecological Modelling*, 120, 65-73.
- Lennox, B.; Montague, G.A.; Frith, A.M.; Gent, C. & Bevan, V. (2001). Industrial application of neural networks – an investigation. *Journal of Process Control*, 11, 497-507.
- Ma, L. and Khorasani, K. (2004). Facial expression recognition using constructive feedforward neural networks. *IEEE Transactions on Systems Man and Cybernetics part B-Cybernetics*, 34, 3, 1588-1595.
- Mahendra, V.; Prasad, V.S.S. & Dutta Gupta, S. (2004). Trichromatic sorting of in vitro regenerated plants of gladiolus using adaptive resonance theory. *Current Science*, 87, 348-353.
- Mehrota, S.; Prakash, O.; Mishra, B.N. & Dwevedi, B. (2008). Efficiency of neural networks for prediction of in vitro culture conditions and inoculums properties for optimum productivity. *Plant Cell Tissue and Organ Culture*, 95, 29-35.
- Ming, R.; Hou, S.; Feng, Y.; Yu, Q.; Dionne-Laporte, A.; Saw, J. H.; Wang, W.; Ly, B. V.; Lewis, K. L. T.; Salzberg, S. L.; Feng, L.; Jones, M. R.; Skelton, R. L.; Murray, J.E.; Chen, C.; Qian, W.; Shen, J.; Du, P.; Eustice, M.; Tong, E.; Tang, H.; Lyons, E.; Paull, R.E.; Michael, T.P.; Wall, K.; Rice, D.W.; Albert, H.; Wang, M.; Zhu, Y.J.; Schatz, M.; Nagarajan, N.; Acob, R.A.; Guan, P.; Blas, A.; Wai, C.M.; Ackerman, C.M.; Ren, Y.; Liu, C.; Wang, J.; Wang, J.; Na, J.; Shakirov, E.V.; Haas, B.; Thimmapuram, J.; Nelson, D.; Wang, X.; Bowers, J.E.; Gschwend, A.R.; Delcher, A.L.; Singh, R.; Suzuki, J.Y.; Tripathi, S.; Neupane, K.; Wei, H.; Irikura, B.; Paidi, M.; Jiang, N.; Zhang, W.; Presting, G.; Windsor, A.; Navajas-Perez, R.; Torres, M.; Feltus, F.A.; Porter, B.; Li, Y.; Burroughs, A.M.; Luo, M.; Liu, L.; Christopher, D.A.; Mount, S.M.; Moore, P.H.; Sugimura, T.; Jiang, J.; Schuler, M.A.; Friedman, V.; Mitchell-Olds, T.; Shippen, D.E.; dePamphilis, C.W.; Palmer, J.D.; Freeling, M.; Paterson, A.H.; Gonsalves, D.; Wang, L. & Alam, M. (2008). The draft genome of the transgenic tropical fruit tree papaya (*Carica papaya* Linnaeus). *Nature*, 452, 991-996.
- Mitchell, M. (1998). *An introduction to genetic algorithms*, MIT Press, Cambridge, USA.
- Mize, C.W.; Kenneth, J.K. & Compton, M.E. (1999). Statistical considerations for in vitro research: II – data to presentation. *In Vitro Cell & Developmental Biology – Plant*, 35, 122-126.
- Morimoto, T.; De Baerdemaeker, J. & Hashimoto, Y. (1997). An intelligent approach for optimal control of fruit-storage process using neural networks and genetic algorithms. *Computers and Electronics in Agriculture*, 18, 205-224.
- Morimoto, T. & Hashimoto, Y. (2000). An intelligent control for greenhouse automation, oriented by the concepts of SPA and SFA-an application to a post-harvest process. *Computers and electronics in agriculture*, 29, 3-20.
- Müller, B.; Reinhardt, J. & Strickland, M.T. (1995). *Neural Networks. An Introduction*. Springer, Berlin.
- Nemhauser, J.L.; Mallof, J.N. & Chory, J. (2003). Building integrated models of plant growth and development. *Plant Physiology*, 132, 2, 436-439.
- Nijhout, H.F.; Maini, P.K.; Madzvamuse, A.; Wathen, J.W. & Sekimura, T. (2003). Pigmentation pattern formation in butterflies: experiments and models. *Comptes Rendus Biologies*, 326, 8, 717-727.

- Noguchi, N. & Terao, H. (1997). Path planning of an agricultural mobile robot by neural network and genetic algorithm. *Computers and Electronics in Agriculture*, 18, 187-204.
- Ondimu, S.N. & Murase, H. (2008). Comparison of plant water stress detection ability of color and graylevel texture in sunagoke moss. *Transactions of the ASABE*, 51, 3, 1111-1120.
- Plumb, A.P.; Rowe, R.C.; York, P. & Brown, M. (2005). Optimisation of the predictive ability of artificial neural network (ANN) models: a comparison of three programs and four classes of training algorithm. *European Journal of Pharmaceutical Science*, 25, 395-405.
- Prasad, V.S.S. & Dutta Gupta, S. (2008). Applications and potentials of artificial neural networks in plant tissue culture. In: *Plant Tissue Culture Engineering*, Dutta Gupta, S. & Ibaraki Y. (Eds), 47-67, Springer-Verlag: Berlin, Germany.
- Rowe, R.C. & Roberts, R.J. (1998). Artificial intelligence in pharmaceutical product formulation: neural computing and emerging technologies. *Pharmaceutical Science & Technology Today*, 1, 5, 200-205.
- Rowe, R.C. & Roberts, R.J. (2005). *Intelligent Software for Product Formulation*, Taylor & Francis, London.
- Rumelhart, D.E. & McClelland, J.L. (1986). *Parallel Distributed Processing: Explorations in the Microstructures of Cognition*, MIT Press, Cambridge, USA.
- Russell, S.J. & Norvig, P. (2003). *Artificial Intelligence: A Modern Approach* (2nd ed.), Upper Saddle River, NJ: Prentice Hall, USA.
- Schauer, N. & Fernie, A.R. (2006). Plant metabolomics: towards biological function and mechanism. *Trends in Plant Science*, 11, 508-516.
- Setnes, M.; Babuska, R. & Verbruggen, H.B. (1998). Rule-based modelling: precision and transparency. *IEEE Transactions on Systems Man and Cybernet Part C-Applications and Reviews*, 28, 1, 165-169.
- Shao, Q.; Rowe, R.C. & York, P. (2006). Comparison of neurofuzzy logic and neural networks in modelling experimental data of an immediate release tablet formulation. *European Journal of Pharmaceutical Science*, 28, 394-404.
- Shao, Q.; Rowe, R.C. & York, P. (2007). Comparison of neurofuzzy logic and decision trees in discovering knowledge from experimental data of an immediate release tablet formulation. *European Journal of Pharmaceutical Science*, 31, 129-136.
- Schuster, S.; Knoke, B. & Marhl, M. (2005). Differential regulation of proteins by bursting calcium oscillations – a theoretical study. *BioSystems*, 81, 49-63.
- Stollenwerk, N.; Drepper, F. & Siegel, H. (2001). Testing nonlinear stochastic models on phytoplankton biomass time series. *Ecological Modelling*, 144, 261-277.
- Struik, P.C.; Yin, X. & de Visser, P. (2005). Complex quality traits: now time to model. *Trends in Plant Science*, 10, 513-516.
- Sweetlove, L.J.; Last, R.L. & Fernie, A.R. (2003). Predictive metabolic engineering: a goal for systems biology. *Plant Physiology*, 132, 2, 420-425.
- Takayama, K.; Fujikawa, M. & Nagai, T. (1999). Artificial neural network as a novel method to optimize pharmaceutical formulations. *Pharmaceutical Research*, 16, 1, 1-6.
- Tani, A.; Murase, H.; Kiyota, M. & Honami, N. (1992). Growth simulation of alfalfa cuttings *in vitro* by kalman filter neural network. *Acta Horticulturae*, 319, 671-676.
- Tardieu, F. (2003). Virtual plants: modelling as a tool for the genomics of tolerance water deficit. *Trends in Plant Science*, 8, 1, 9-14.

- Taylor, J.G. (1996). *Neural networks and their applications*. Taylor J. G. (Eds) John Wiley & Sons Ltd. England.
- Thai, C.N. & Shewfelt, R.L. (1991). Modelling sensory colour quality of tomato and peach: neural networks and statistical regression. *Transaction of the ASAE*, 34, 3, 950-955.
- Thum, K.E., Shasha, D.E.; Lejav, L.V. & Coruzzi, G.M. (2003). Light- and carbon-signaling pathways. Modeling circuits of interactions. *Plant Physiology*, 132, 440-452.
- Tien, B.T. & Van Staten, G. (1998). A neurofuzzy approach to identify lettuce growth and greenhouse climate. *Artificial Intelligence Review*, 12, 71-93.
- Tracqui, P. (2006). Mechanical instabilities as a central issue for insilico analysis of cell dynamics. *Proceedings of the IEEE*, 94, 4, 710-724.
- Trewavas, A. (2006). A brief history of systems biology. 'Every object that biology studies is a system of systems.' Francois Jacob (1974). *Plant Cell* 18, 2420-2430.
- Uozumi, N.; Yoshino, T.; Shiotani, S.; Suehara, K.; Arai, F.; Fukuda, T. & Kobayashi, T. (1993). Application of image analysis with neural network for plant somatic embryo culture. *Journal of Fermentation and Bioengineering*, 76, 505-509.
- Wang, E.; Robertson, M.J.; Hammer, G.L.; Carberry, P.S.; Holzworth, D.; Meinke, H.; Chapman, S.C.; Hargreaves, J.N.G.; Huth, N.I. & McLean, G. (2002). Development of a generic crop model template in the cropping system model APSIM. *European Journal of Agronomy*, 18, 121-140.
- Welch, S.M.; Roe, J.L. & Dong, Z. (2003). A genetic neural network model of flowering time control in *Arabidopsis thaliana*. *Agronomy Journal*, 95, 71-81.
- Whittaker, A.D.; Park, B.P.; McCauley, J.D. & Huang Y. (1991). Ultrasonic signal classification for beef quality grading through neural networks. In: *Proceedings of automated agriculture for the first 21st century*, Chicago, IL, USA, pp 116-125.
- Williams, J.C.; ReVelle, C.S. & Levin, S.A. (2004). Using mathematical optimization models to design nature reserves. *Frontiers in Ecology and the Environment*, 2, 98-105.
- Yang, C.C. (2003). Development of a herbicide application map using artificial neural networks and fuzzy logic. *Agricultural Systems*, 76, 561-574.
- Yuan, J.S.; Galbraith, D.W.; Dai, S.Y.; Griffin, P. & Stewart, N. Jr. (2008). Plant systems biology comes of age. *Trends in Plant Science*, 13, 4, 165-171.
- Zadeh, L. (1965). Fuzzy sets. *Information and Control*, 8, 338-353.
- Zhang, C.; Timmis, R. & Hu, W.S. (1999). A neural network based pattern recognition system for somatic embryos of Douglas fir. *Plant Cell Tissue and Organ Culture*, 56, 1, 25-35.
- Zupan, J. & Gasteiger, J. (1993). *Neural Networks for chemists - an introduction*. VCH Publishers, New York.

The Usefulness of Artificial Neural Networks in Predicting the Outcome of Hematopoietic Stem Cell Transplantation

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1. Introduction

1.1 Artificial neural networks in clinical medicine

In Medicine, several tools have been developed for the prediction of clinical outcomes following drug treatment and other medical interventions. The standard approach for a binary outcome is to use logistic regression (LR), however, this method requires formal training and a profound knowledge of statistics (Royston, 2000; Harrel et al., 1996). LR is used to predict a categorical (usually dichotomous) variable from a set of predictor variables; it has been especially popular with medical research in which the dependent variable is whether or not a patient has a disease.

Over the past years, artificial neural networks (ANNs) have increasingly been used as an alternative to LR analysis for prognostic and diagnostic classification in clinical medicine (Schwarzer et al., 2000). ANNs are composed of simple elements operating in parallel inspired by biological nervous systems. As in nature, the network function is determined largely by the connections between elements. After training with retrospective data ANNs are capable of making intelligent predictions given new, limited information. The growing interest in ANNs has mainly been triggered by their ability to mimic the learning processes of the human brain. However, the issue remains as to how these ANNs actually succeed in recognizing patterns within data that are too complex for the human brain. From here derives the so-called “black-box” aspect of ANNs. The network operates in a feed-forward mode from the input layer through the hidden layers (like in a black box) to the output layer. Exactly what interactions are modeled in the hidden layers is still a knot that remains untied. Each layer within the network is made up of computing nodes with remarkable data processing abilities. Each node is connected to other nodes of a previous layer through adaptable inter-neuron connection strengths known as synaptic weights. ANNs are trained for specific applications, such as pattern recognition or data classification, through a learning process and knowledge is usually retained as a set of connection weights. The backpropagation algorithm and its variants are learning algorithms that are widely used in neural networks. With backpropagation, the input data is repeatedly presented to the

network. Each time, the output is compared to the desired output and an error is computed. The error is then fed back through the network and used to adjust the weights in such a way that with each iteration it gradually declines until the neural model produces the desired output. ANNs have been successfully applied in the fields of mathematics, engineering, medicine, economics, meteorology, psychology, neurology, and many others. Indeed, in medicine, they offer a tantalizing alternative to multivariate analysis, although their role remains advisory since no convincing evidence of any real progress in clinical prognosis has yet been produced (Linder et al., 2006).

A systematic review on the use of artificial neural networks in decision support in cancer by Lisboa et al. showed that the number of clinical trials (CTs) and randomised controlled trials (RCTs) involving the use of ANNs in diagnosis and prognosis has increased from 1 to 38 over the past decade. However, out of 396 studies involving the use of ANNs in cancer, only 27 were either CTs or RCTs. Out of these trials, 21 showed an increase in benefit to healthcare provision and 6 did not. None of these studies however showed a decrease in benefit. Overall, the reviewed publications support the neural network approach but while on the one hand they identify trends in areas of clinical promise (particularly diagnosis, prognosis and therapeutic guidance for cancer), on the other they highlight the need for more extensive application of rigorous methodologies (Lisboa & Taktak, 2006).

Interesting, a review on the use of ANNs in the field of Gastroenterology over the last 10 years (their application in the field of gastroenterology has now entered the second decade) underlines that the increasing complexity of clinical data requires the use of mathematical models that are able to capture the key properties of entire ensembles, including their linkages and their hubs, abandoning the traditional statistical reductionistic approach, which tends to 'see' things individually, to simplify and to look at one single element at a time (Pace & Savarino, 2007). Some authors, for example, assessed the performance of ANNs in recognizing patients with chronic atrophic gastritis, a state of chronic inflammation that can eventually progress to gastric carcinoma, by using only clinical and biochemical variables (Annibale & Lahner, 2007).

In the field of urology, several papers have addressed the predictive efficacy of ANNs. In urological cancer, ANNs appear to be accurate and more explorative than traditional regression statistics artificial intelligence methods when used to analyze large data cohorts. Furthermore, they allow individualized prediction of disease behaviour. Each artificial intelligence method has characteristics that make it suitable for different tasks. The lack of transparency of ANNs hinders global scientific community acceptance, but this can be overcome by neuro-fuzzy modeling systems (Abbod et al., 2007).

New biomarkers within multivariate models have been analyzed with ANNs to improve early detection of prostate cancer (Stephan et al., 2007). Another field of application is the management of urolithiasis, a worldwide clinical challenge embracing a multitude of difficulties in diagnosis, treatment and prevention of recurrence. Recent reports have examined the role of ANNs in prediction of stone presence and composition, spontaneous passage, clearance and re-growth after treatment (Rajan & Tolley, 2005). The results suggest that ANNs may prove useful in clinician-led decision-making processes.

ANNs can identify important predictive variables and accurately predict treatment outcomes in clinical medicine but although the initial results appear promising, further prospective studies of larger patient cohorts will need to be performed in order to determine whether this mode of analysis can surpass standard statistical predictive methods, not only when solving problems concerning diagnosis and its classification into subtypes but also when predicting clinical outcomes of patients affected by diverse pathologies.

1.2 Artificial neural networks in hematology and bone marrow transplantation

Very few reports on this topic have been published in the field of hematology. The first computing devices based on artificial intelligence (AI) have been applied to routine laboratory data management whereas new innovative tools, based on neural networks trained with data from peripheral blood analysis, have been used for differential diagnosis in diseases such as anemias, thalassemias and leukemias. The introduction of the first microarray based and bio-informatic approach for molecular diagnosis of hematological malignancies can be considered a major step ahead. This approach is based on the monitoring of simultaneous expression of thousands of genes using DNA microarray, independently of previous biological knowledge, analyzed using AI devices (Zini, 2005).

In an attempt to create an application for discriminating different types of anemia, simply using data from peripheral blood, Zini & d'Onofrio (2003) collected data from peripheral blood of 1000 patients diagnosed mainly with hematopoietic disorders in 22 Italian Hematology Centers. The ANNs were trained with labeled samples and showed high capability of clustering signals according to the predefined normal as well as pathological profiles.

In 2002, Amendolia et al. investigated the use of ANNs for the classification of thalassemic pathologies, exclusively using the hematologic parameters resulting from hemochromocytometric analysis. Different combinations of ANNs made it possible to discriminate thalassemia carriers from normals with 94% classification accuracy, 92% sensitivity, and 95% specificity. Based on these results, an automated system for real-time support in diagnoses was proposed (Amendolia et al., 2002).

All these intriguing reports of ANNs in the field of hematology kindled our curiosity to discover whether ANNs were capable of predicting the outcome of hematopoietic stem cell transplantation (HSCT) after analyzing donor and recipient pre-transplantation clinical and immunogenetic variables.

1.3 The difficult setting of unrelated bone marrow transplantation in thalassemia.

Patients with chronic non-malignant genetic disorders, such as thalassemia, are faced with a dramatic decision: they can either undergo HSCT with a good possibility of cure but a high chance of death or continue the more conventional treatment with blood transfusions and iron chelation therapy. The important advances made in conventional treatment now allow transfusion-dependent thalassemia patients to live much longer (Caocci et al., 2006; Borgna Pignatti et al., 2004) but as a result these patients must cope with complications that occur over time. Treatment may be required for heart or liver diseases, infections, osteoporosis and other serious health problems.

On the other hand, although HSCT from an HLA-identical sibling can offer thalassemia patients a probability of cure that is close to 90% in children and adults in good clinical conditions (Lucarelli et al., 1990), this procedure is associated with a significant risk of mortality (Lucarelli et al., 1997), especially in patients with advanced age or poor clinical conditions. Moreover, the chance that any given sibling will be HLA matched with a potential recipient is one out of four which means that most patients will need to search for a compatible donor in the registries of voluntary donors worldwide. Transplantation from unrelated donors (UD) is burdened by an increased risk of acute and chronic graft-versus host disease (GVHD) with a consequent negative impact on overall survival (La Nasa et al., 2006). Therefore, every effort should be made to carefully evaluate the risk of GVHD before performing UD-HSCT (Hansen et al., 1998).

Graft-versus-host disease remains the major barrier to the successful outcome of HSCT in thalassemia. HLA disparity between the donor and the recipient is clearly the most powerful risk factor but also older age, gender mismatch, Pesaro risk class, cytomegalovirus (CMV) positivity as well as higher median infused hematopoietic stem cell doses have been shown to increase the risk for GVHD (Lucarelli et al., 1996). Evidence emerging from recent reports indicates a correlation between certain immunogenetic variables and the occurrence of GVHD: donor-recipient HLA-Cw ligand groups for killer immunoglobulin-like receptors (KIRs), KIR genotypes, the HLA-G 14-basepair (bp) polymorphism and HLA-DPB1 disparity (La Nasa et al., 2007; Littera et al., 2010; Fleischhauer et al., 2006). Although this information may contribute to our understanding of the pathogenesis of GVHD, it is difficult to apply in clinical practice. What we need is a simple prognostic tool capable of analyzing the most relevant predictive variables.



Fig. 1. To gaze into a crystal ball for a glimpse of the future has always been the dream of every doctor. Reliable assessment of the acute GVHD risk is crucial for making rational treatment decisions. During the process of donor selection and before discussing the choice of treatment with patients and their relatives, it is essential for physicians to integrate their knowledge with statistical or algorithmic tools capable of accurately predicting the likely incidence of GVHD. A more accurate prediction of acute and chronic GVHD would not only improve GVHD prophylaxis and conditioning regimens, but would also allow physicians to adapt their communication practices appropriately and to ensure that patients are supplied with effective

and comprehensive information on the pros and cons of transplantation, including the possibility of dying. This is particularly relevant for patients with chronic non malignant disorders, such as thalassemia. Maybe ANNs represent the crystal ball we are all looking for.

2. Patients and methods

2.1 Patients

We compared the prognostic performance of ANNs versus LR for predicting acute GVHD in a group of 78 beta-thalassemia major patients given UD-HSCT (Caocci et al, 2010). The following clinical and immunogenetic parameters were considered: recipient gender, recipient age, donor gender, donor age, the combination male recipient/female donor versus the other possible combinations, recipients and/or donors with positive CMV serology versus donor and recipient pairs with negative CMV serology, the Pesaro risk class at transplantation, HCV-RNA positivity, median infused CD34 cell dose, Treosulphan-containing conditioning regimen versus other regimens, HLA Class I mismatch, presence of HLA-A11, non permissive HLA-DPB1 disparity in GVHD direction, presence of the HLA-G 14-basepair deletion/deletion polymorphism in recipients, presence of the HLA-G 14-basepair deletion/deletion polymorphism in donors, heterozygosity for HLA-Cw ligand groups 1 and 2 in patients, recipient KIR ligand/donor activating KIR (recipient C1 absent/donor KIR2DS2 present versus the other 3 combinations; recipient C2 absent/donor KIR2DS1 present versus the other 3 combinations), recipient KIR ligand/donor inhibitory KIR (patient C1 absent/donor KIR2DL2 present versus the other 3 combinations; recipient C1 absent/donor KIR2DL3 present versus the other 3 combinations; patient C2 absent/donor KIR2DL1 present versus the other 3 combinations), donor homozygosity for KIR haplotype A (Uhrberg, 2002; Colonna, 1995; Bassi, 2007; Cook, 2004; Harrison, 1993).

2.2 Statistical analysis

Patient, disease, and transplantation-related variables were expressed as median and range or percentage, as appropriate. For the HSCT outcome, patients were censored at the time of rejection, death, or last follow-up. Probabilities of overall survival (OS) and survival with transfusion independence (thalassemia-free survival) were estimated by the Kaplan-Meier method.

The following 24 independent variables were analyzed for their potential impact on aGVHD: recipient gender, recipient age, donor gender, donor age, the combination male recipient/female donor versus the other possible combinations, recipients and/or donors with positive CMV serology versus donor and recipient pairs with negative CMV serology, the Pesaro risk class at HSCT, HCV-RNA positivity, median infused CD34 cell dose, Treosulphan conditioning regimen versus other regimens, HLA Class I mismatch, presence of HLA-A11, non permissive HLA-DPB1 disparity in GVHD direction, presence of the HLA-G 14-basepair deletion/deletion polymorphism in recipients, presence of the HLA-G 14-basepair deletion/deletion polymorphism in donors, heterozygosity for HLA-Cw ligand groups 1 and 2 in patients, recipient KIR ligand/donor activatory KIR (recipient C1 absent/donor KIR2DS2 present versus the other 3 combinations; recipient C2 absent/donor KIR2DS1 present versus the other 3 combinations), recipient KIR ligand/donor inhibitory KIR (patient C1 absent/donor KIR2DL2 present versus the other 3 combinations; recipient

C1 absent/donor KIR2DL3 present versus the other 3 combinations; patient C2 absent/donor KIR2DL1 present versus the other 3 combinations), donor homozygosity for KIR haplotype A.

2.3 Logistic regression

A binomial LR model with 24 independent variables (3 continuous and 21 categorical) was developed (Table 2). Acute GVHD was considered as a dichotomous dependent variable. Five consecutive random extractions were performed. For each extraction, patients were split into a learning data set (68 patients) and a test data set (10 patients). The independent variables were fitted into LR models via forward likelihood ratio test (chi-square difference) and stepwise selection. The chi-square test proposed by Hosmer and Lemeshow was used to analyze the goodness of fit: a finding of non-significance corresponds to the conclusion that the model adequately fits the data. Variables were retained only if their resulting p-value was ≤ 0.05 . The final equation, developed through parameter estimates with standard errors, odds ratios and asymptotic 95% confidence intervals for all significant variables calculated, was applied to each case of the data test. A cut-off value of 0.5 was established for assigning the probability of GVHD: "GVHD yes" (1) or "GVHD no" (0). Sensitivity and specificity were determined in the learning and test data sets of each random extraction, sensitivity being the ratio between true positive and true negative plus false negative and specificity the ratio between true negative and true negative plus false positive. Mean sensitivity and specificity of LR obtained in five consecutive extractions were compared to ANN, using the chi-square test with Yate's correction. Statistical analysis was performed using SPSS® software, version 12 (SPSS Inc., Chicago, IL, USA)

2.4 Artificial neural networks

ANNs are capable of learning from observed data or examples and under certain conditions are able to approximate nonlinear functions with arbitrary precision. The technique was originally inspired by perceptions of how the human brain learns and processes information and since then has successfully been applied in many different fields, including mathematics, engineering, medicine, economics, meteorology, psychology, neurology, and many others. Although the predictive power of ANNs is often superior to that of other more traditional methods, they are still regarded as black-boxes where it is difficult for the user to gain insight into the influence of the independent variables in the prediction process. While ANNs are capable of learning the relationship between the input parameters and the controlled and uncontrolled variables, they do not generate information on the causal relationship between the input and output patterns. Several studies are currently underway to overcome this problem.

The structure of ANN usually consists of three layers (Fig. 2). The input layer accepts data sets from an external source that constitute inputs to the next layer of neurons. The next layer is called the hidden layer because its neuron values are not visible outside the net. The use of one or more hidden layers increases the net's learning abilities. The final layer is the output layer. Each single neuron is connected to the neurons of the previous layer through adaptable synaptic weights. Knowledge is generally stored as distributed patterns of activation in weight matrices.

The key feature of neural networks is that they learn the input/output relationship through training. The training data set includes a number of cases, each containing values for a range

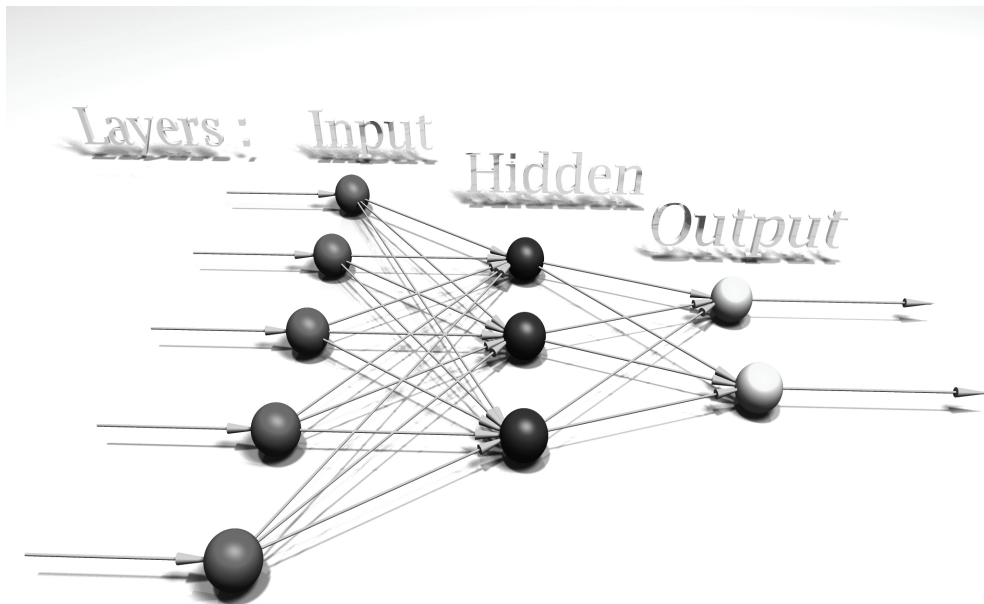


Fig. 2. The three layers structure of ANN

of well-matched input and output variables. The weighted connections between neurons in each layer are adjusted by a training algorithm to minimize error and provide accurate predictions on the training set. The outputs are the dependent variables that the network produces for the corresponding input. It is important that all learning data is supplied to the network as a data set. Once the input is propagated to the output neuron, this neuron compares its activation with the expected training output. If there is an error, the output neuron adjusts the connection weights to compensate the error by going backwards through the network. This step by step process is called backpropagation. The backpropagation (BP) algorithm and its variants are the most powerful learning algorithms in neural networks. By calculating the gradient vector of the error surface, the error gradually declines until all the expected outputs are correctly displayed.

The Neural Network Toolbox™ 6 of the software Matlab® 2008, version 7.6 (MathWorks, inc.) was used to develop a three layer feed forward neural network with the default tan-sigmoid transfer function in the hidden layer and linear transfer function in the output layer (Schwarzer et al., Demuth, 2008). The input layer of 24 neurons receives data that are processed in the hidden layer (30 neurons) and output layer (1 neuron). The output neuron predicts a number between 1 and 0 (goal), representing the event “GVHD yes” (1) or “GVHD no” (0), respectively. A cut-off value of 0.5 was established for assigning probability 1 or 0.

The architecture of ANN is schematized in Fig. 3. Input neurons receive data represented by the values of 24 independent variables processed in the hidden layer. The meaning of this process is to calculate interconnection weights between variables with the purpose of predicting an outcome and to calculate an error value by comparing this output value with the known outcome. The ANN attempts to minimize the error by adjusting the weights according to a learning algorithm (Linder et al., 2006). For the training procedure, we applied the ‘on-line back-propagation’ method on the same 5 sets of 68 patients previously analyzed by LR.

The 5 test phases utilized 10 patients randomly extracted from the entire cohort and not used in the training phase. A standard error of less than 10⁻² was required. Mean sensitivity and specificity of the 5 consecutive data sets were determined in the data test and compared to LR. Because sensitivity and specificity in the 5 learning tests always resulted to be 100%, they were considered not comparable to LR.

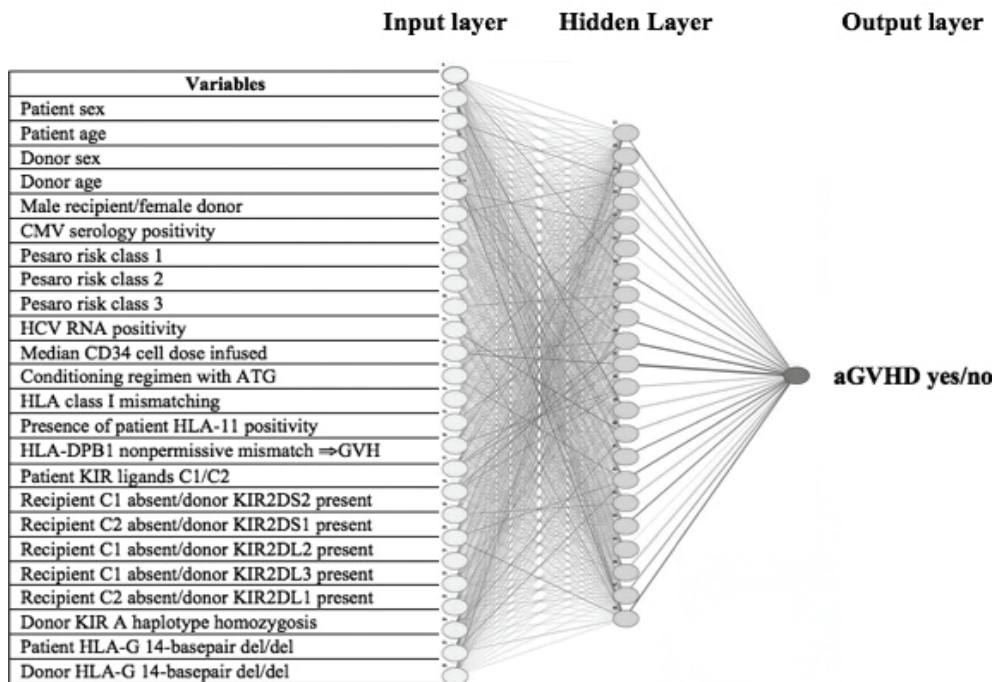


Fig. 3. Architecture of the three-layer artificial neural network. The input layer of 24 neurons (independent variables) receives data that are processed in the hidden layer and output layer (1 neuron). The output neuron predicts a number between 1 and 0 (goal), representing the event "GVHD yes" (1) or "GVHD no" (0), respectively.

3. Results

Three-year Kaplan-Meier estimates for the 78 patients studied were 89.7% for survival, 76.9% for thalassemia-free survival, 11.5% for the cumulative incidence of rejection and 10.3% for TRM. Nine patients rejected the allograft and 7 died of transplantation-related complications (Figure 4).

Twenty-six patients (33.3%) developed grade II-IV acute GVHD (Figure 5).

In multivariate analysis, only donor KIR AA haplotypes were independently significantly correlated to acute GVHD in our cohort of 78 patients ($p=0.037$). However, we found a

positive trend for donor age ($p=0.51$), patient heterozygosity (C1/C2) for the HLA-Cw KIR ligands ($p=0.56$) and donor homozygosity (deletion/deletion) for the HLA-G 14-bp polymorphism ($p=0.57$) (Table 2).

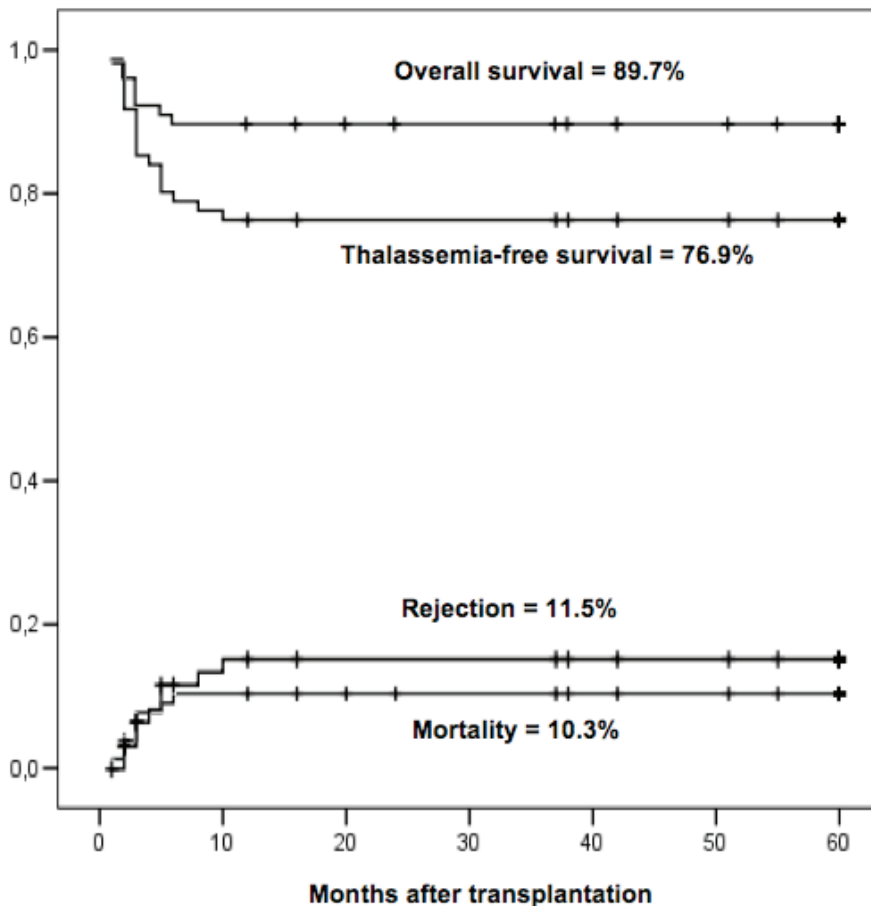


Fig. 4. Kaplan-Meier probabilities of overall survival, thalassemia-free survival, cumulative incidence of mortality and rejection in 78 thalassemia patients transplanted from an unrelated donor.

Table 3 shows the prognostic performance of LR and ANN in predicting acute GVHD in 5 consecutive randomly extracted training and test data sets. Sensitivity and specificity were determined in the learning and test data sets of each random extraction. Comparisons between LR and ANN on training data sets (5 consecutive extractions each composed of 68 patients) were not considered since ANN was able to recognize 100% of correct events by

means of its peculiar learning algorithm, whereas LR showed a mean value of 88.5% for specificity and 36.4% for sensitivity.

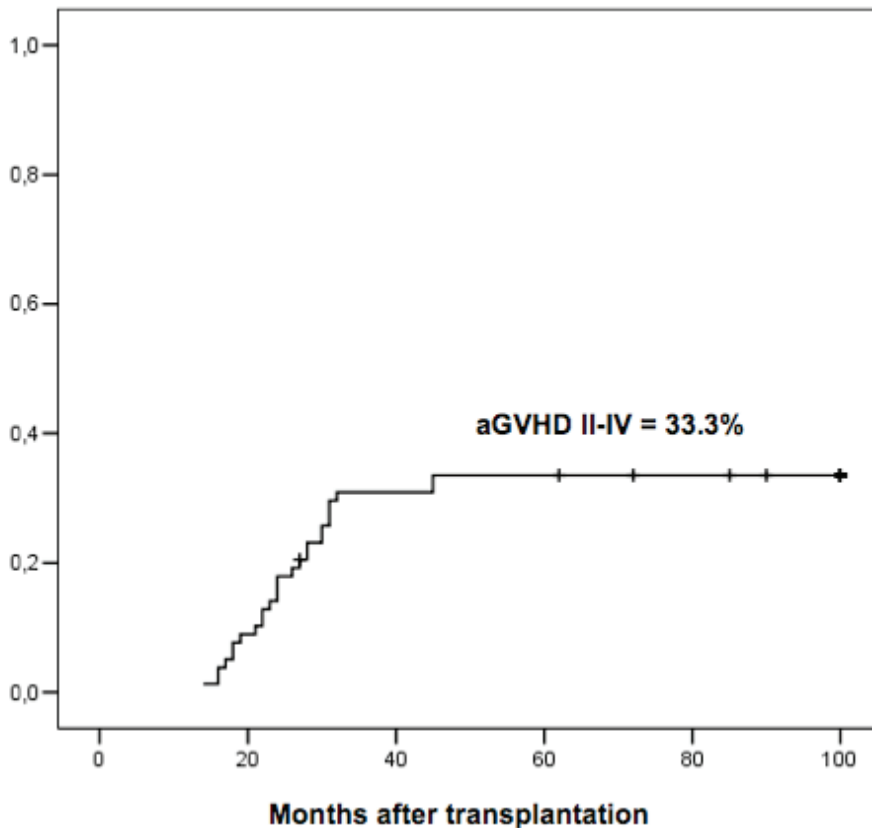


Fig. 5. Kaplan-Meier probabilities of cumulative incidence of acute GVHD in 78 thalassemia patients transplanted from an unrelated donor

In test data sets (5 extractions each composed of 10 patients), the mean specificity of LR was 80.5% compared to 90.1% of ANN (capability of predicting the absence of acute GVHD in patients who did not experience acute GVHD); this difference was not statistically significant. The mean sensitivity of LR was 21.7% compared to 83.3% of ANN (capability of predicting acute GVHD in patients who developed acute GVHD after HSCT). This difference was statistically significant ($p < 0.001$).

Variables	<i>p</i> value
Patient sex	.787
Patient age	.110
Donor sex	.496
Donor age	.051
Male recipient/female donor	.157
CMV serology, positivity	.834
Pesaro risk class 1	.907
Pesaro risk class 2	.721
Pesaro risk class 3	.799
HCV RNA positivity	.955
Median CD34 cell dose infused	.315
Conditioning regimen with ATG	.512
HLA class I mismatching	1.000
Presence of patient HLA-11 positivity	.067
HLA-DPB1 nonpermissive mismatch $\Rightarrow$ GVH	.510
Patient KIR ligands C1/C2	.056
Recipient C1 absent/donor KIR2DS2 present	.844
Recipient C2 absent/donor KIR2DS1 present	.127
Recipient C1 absent/donor KIR2DL2 present	.844
Recipient C1 absent/donor KIR2DL3 present	.799
Recipient C2 absent/donor KIR2DL1 present	.098
Donor homozygosity for KIR A haplotype	.037
Patient HLA-G 14-basepair del/del	.116
Donor HLA-G 14-basepair del/del	.057

Table 2. Multivariate analysis of 24 independent variables and onset of acute GVHD in 78 thalassemia patients transplanted from an unrelated donor.

4. Conclusion and future research

Artificial intelligence (AI) is a field in which computers or software perform complex tasks by means of programming techniques or programs called expert systems that allow computers to “make decisions” by interpreting data and selecting among alternatives. ANNs are considered to be a branch of AI but what distinguishes them from classical AI are their learning and adaptive capabilities. ANNs first emerged as a collection of small individual interconnected artificial neurons or nodes constructed to mimic the processing properties of the biological neurons of the human brain. The information flows through the network in one direction, from the input nodes, through the hidden layers, to the output nodes. ANNs operate like a “black box” model, requiring no detailed information about the system. In neural net architectures, the connection strengths between nodes are the storehouses of knowledge and the learning process is primarily a process of adjusting these connection strengths. The network learns by adjusting its weights until it identifies a set of weights that produce the correct output for every sample input. ANNs have been applied with success in many different sectors and recent years have registered a growing interest also in the field of medicine.

		Acute GVHD	Observed Cases	LR Expected cases (%)	ANN Expected cases (%)
Extraction_1	Training N=68	No	48	43 (89.6)	n.a.*
		Yes	20	8 (40)	
	Test N=10	No	6	5 (83.3)	5 (83.3)
		Yes	4	1 (25)	2 (50)
Extraction_2	Training N=68	No	48	44 (91.7)	n.a.*
		Yes	20	6 (30)	
	Test N=10	No	6	2 (66.7)	6 (100)
		Yes	4	2 (50)	4 (100)
Extraction_3	Training N=68	No	45	40 (88.9)	n.a.*
		Yes	23	6 (26.1)	
	Test N=10	No	9	3 (66.7)	9 (100)
		Yes	1	0 (0)	1 (100)
Extraction_4	Training N=68	No	47	41 (87.2)	n.a.*
		Yes	21	9 (42.9)	
	Test N=10	No	7	6 (85.7)	5 (71.4)
		Yes	3	0 (0)	3 (100)
Extraction_5	Training N=68	No	47	40 (85.1)	n.a.*
		Yes	21	9 (42.9)	
	Test N=10	No	7	7 (100)	7 (100)
		Yes	3	1 (33.3)	2 (66.7)
Specificity % (mean)	No aGVHD	Training		88.5	n.a.*
		Test		80.5	90.1 (NS)
Sensitivity % (mean)	YES aGVHD	Training		36.4	n.a.*
		Test		21.7	83.3 (p<0.001)

Table 3. Classification table and correct percentage of acute GVHD prediction for LR and ANN in five consecutive random extractions, splitting 78 thalassemia patients into a learning data set (68 patients) and a test data set (10 patients);

*n.a. = not applicable: comparisons between LR and ANN on training data sets were not considered since ANN was able to recognize 100% of correct events.

The process of clinical decision making that surrounds the choice of unrelated HSCT in a thalassemia patient is particularly difficult. The life span of patients with thalassemia is increasing, mainly attributable to a better control of iron overload.

HSCT is the only definitive curative approach to thalassemia. When the donor is an HLA-identical sibling, the probability of disease-free survival is between 80% and 90%. The worst results are obtained in the highest-risk category, particularly adult thalassemia patients, and in unrelated transplantation. Acute GVHD is the main cause of transplantation-related mortality in the setting of unrelated HSCT for thalassemia. Therefore, it becomes of fundamental importance to carefully evaluate the risk for the development of aGVHD before performing transplantation.

Several studies of unrelated HSCT in thalassemia patients have recently investigated a relatively large number of variables for their possible influence on the outcome and the

onset of acute GVHD. One of our recent reports shows that the HLA-G 14-bp polymorphism could be an important predictive factor for aGvHD; the HLA-CwAsn80 and HLA-CwLys80 molecules expressed by donor/recipient pairs as well as donor homozygosity for KIR haplotype A (AA) also seem to have a significant impact on transplantation outcome and acute GVHD incidence (Littera et al., 2010). In this study, multivariate analysis confirmed the impact of donor KIR haplotype AA on the incidence of acute GVHD ($p=0.037$). We also found a positive trend for donor age ($p=0.51$), patient heterozygosity (C1/C2), the HLA-Cw KIR ligands ($p=0.56$) and donor homozygosity (deletion/deletion) for the HLA-G 14-bp polymorphism ($p=0.57$).

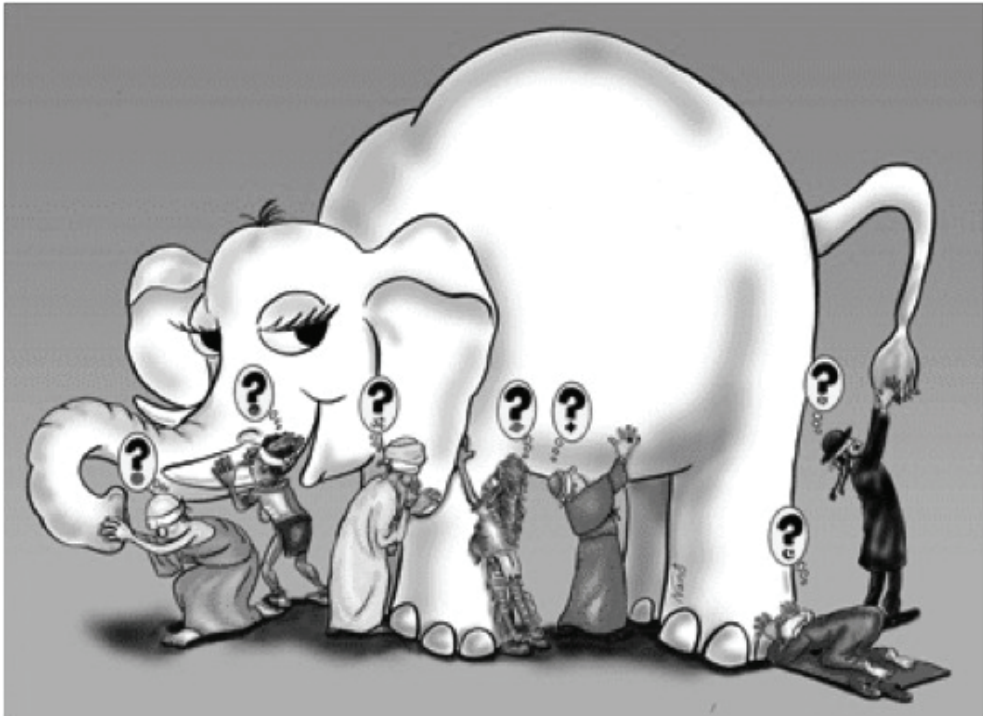


Fig. 6. Researchers are currently involved in the study of several variables that seem to have a role in GVHD onset. Unfortunately, a simple prognostic tool capable of analyzing the most relevant predictive variables and predicting GVHD is still missing. ANNs could represent this tool, capable of providing us with a more holistic vision of GVHD-phenomena.

In the field of medicine, several prognostic models have been developed for the prediction of outcome. The complicated clinical scenario of HSCT for chronic non malignant diseases would certainly benefit from a prognostic model based on a set of specific variables relevant to the development of GVHD. This report compared ANNs to LR in a cohort of 78 thalassemia patients transplanted from an unrelated donor. When comparing the prognostic performance of LR to ANN, the ability of predicting acute GVHD when the patient experienced acute GVHD (sensitivity) was 21.7% for LR versus 83.3% for ANN. This

difference was statistically significant ($p < 0.001$). Moreover, the mean specificity of LR was 80.5% compared to 90.1% of ANN (capability of predicting the absence of acute GVHD in patients who did not experience acute GVHD) but this difference was not significant.

The advantage of ANNs can theoretically be explained by their ability to recognize complex relationships that possibly exist between independent and dependent variables, a typical challenge when dealing with medical data. By contrast, ANNs are considered as “black boxes” because of their hidden layer, which remains an obstacle to their acceptance. Moreover, ANNs are unable to calculate the weight of a single variable on the outcome, while LR determines a relative risk for each variable, building a complex equation of outcome prediction. And finally, LR is a widely used statistical method while ANNs are still being developed in the medical field.

There are some limitations to this study: the results are based on a series of information obtained from a relatively small, albeit homogeneous group of patients, selected according to the diagnosis, clinical characteristics and transplantation procedure. Moreover, data were analyzed retrospectively and could have been biased by the small number of cases assigned to the test data set, despite the 5 consecutive random extractions performed to increase this number. Therefore, the results obtained here need to be verified in larger prospective studies of transplanted patients. It should be of major interest to extend the application of neural networks to patients transplanted for pathologies other than thalassemia and to include ulterior clinical parameters. Studies are currently underway to setup a neural network model capable of handling missing data in patient samplings.

In conclusion, ANN had a better prognostic performance than LR in predicting acute GVHD in our cohort of patients. This result is particularly important if we consider that GVHD is the major causative factor for TRM. Nevertheless, LR remains the “gold standard” of statistical predicting models in clinical settings.

A combination of the two approaches so that each method complements the other has the potential to significantly improve the clinical decision-making process and the overall outcome of HSCT.

5. References

- Abbod, MF.; Catto, JW.; Linkens, DA.; Hamdy, FC. (2007). Application of artificial intelligence to the management of urological cancer. *Journal of Urology*, 178, 4, (Oct 2007) 1150-1156, 00225347
- Amendolia, SR.; Brunetti, A.; Carta, P.; Cossu, G.; Ganadu, ML.; Golosio, B.; Mura, GM.; Pirastru, MG. (2002). A real-time classification system of thalassemic pathologies based on artificial neural networks. *Medical Decision Making*, 22, 1, (Jan 2002) 18-26, 0272-989X
- Annibale, B., Lahner, E. (2007). Assessing the severity of atrophic gastritis. *European Journal of Gastroenterology and Hepatology*, 2007; 19, 12, (Dec 2007) 1059-1063, 0954-691X
- Bassi, P.; Sacco, E.; De Marco, V.; Aragona, M.; Volpe, A. (2007). Prognostic accuracy of an artificial neural network in patients undergoing radical cystectomy for bladder cancer: a comparison with logistic regression analysis. *British Journal of Urology*, 99, 5, (May 2007), 1007-1012, 1464-4096
- Borgna-Pignatti, C.; Rugolotto, S.; De Stefano, P.; Zhao, H.; Cappellini, MD.; Del Vecchio, GC.; Romeo, MA.; Forni, GL.; Gamberini, MR.; Ghilardi, R.; Piga, A.; Cnaan A. (2004) Survival and complications in patients with thalassemia major treated with

- transfusion and deferoxamine. *Haematologica*, 89, 10, (Oct 2004) 1187-1193, 1592-8721
- Caocci, G.; Pisu, S.; Argioli, F.; Giardini, C.; Locatelli, F.; Vacca, A.; Orofino, MG.; Piras, E.; De Stefano, P.; Addari, MC.; Ledda, A.; La Nasa, G. (2006). Decision-making in adult thalassemia patients undergoing unrelated bone marrow transplantation: quality of life, communication and ethical issues. *Bone Marrow Transplantation*, 37, 2, (Jan 2006) 165-169, 0268-3369
- Caocci, G.; Baccoli, R.; Vacca, A.; Mastronuzzi, A.; Bertaina, A.; Piras, E.; Littera, R.; Locatelli, F.; Carcassi, C.; La Nasa, G. (2010). Comparison between an artificial neural network and logistic regression in predicting acute graft-vs-host disease after unrelated donor hematopoietic stem cell transplantation in thalassemia patients. *Experimental Hematology*, 38, 5, (May 2010) 426-433, 1873-2399
- Colonna, M.; Samaridis, J. Cloning of immunoglobulin-superfamily members associated with HLA-C and HLA-B recognition by human natural killer cells. (1995). *Science*, 268, 5209, (Apr 1995) 405-408, 0036-8075
- Cook, MA.; Milligan, DW.; Fegan, CD.; Darbyshire, PJ.; Mahendra, P.; Craddock, CF.; Moss, PA.; Briggs, DC. (2004). The impact of donor KIR and patient HLA-C genotypes on outcome following HLA-identical sibling hematopoietic stem cell transplantation for myeloid leukemia. *Blood*, 103, 4, (Feb 2004) 1521-1526, 0006-4971
- Demuth, H.; Beale, M.; Hagan, M. (2008). *Neural Network Toolbox™ 6. User's Guide*, The MathWorks, Inc., 0-534-95049-3, Natick, MA.
- Fleischhauer, K.; Locatelli, F.; Zecca, M.; Orofino, MG.; Giardini, C.; De Stefano, P.; Pession, A.; Iannone, AM.; Carcassi, C.; Zino, E.; La Nasa, G. (2006). Graft rejection after unrelated donor hematopoietic stem cell transplantation for thalassemia is associated with nonpermissive HLA-DPB1 disparity in host-versus-graft direction. *Blood*, 107, 7, (Apr 2006), 2984-2992, 0006-4971
- Hansen, JA.; Gooley, TA.; Martin, PJ.; Appelbaum, F.; Chauncey, TR.; Clift, RA.; Petersdorf, EW.; Radich, J.; Sanders, JE.; Storb, RF.; Sullivan, KM.; Anasetti, C. (1998). Bone marrow transplants from unrelated donors for patients with chronic myeloid leukaemia. *New England Journal of Medicine*, 228, 14, (Apr 1998) 962-968, 0028-4793
- Harrell, FE.; Lee, KL.; Mark DB. (1996). Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Statistics in Medicine*, 15, 4, (Feb 1996) 361-387, 0277-6715
- Harrison, GA.; Humphrey, KE.; Jakobsen, IB.; Cooper, DW. (1993). A 14 bp deletion polymorphism in the HLA-G gene. *Human Molecular Genetics*, 2, 12 (Dec 1993) 2200, 0964-6906
- La Nasa, G.; Caocci, G.; Argioli, F.; Giardini, C.; Locatelli, F.; Vacca, A.; Orofino, MG.; Piras, E.; Addari, MC.; Ledda, A.; Contu, L.; (2005) Unrelated donor stem cell transplantation in adult patients with thalassemia. *Bone Marrow Transplantation*, 36, 11, (Dec 2005) 971-975, 0268-3369
- La Nasa, G.; Littera, R.; Locatelli, F.; Giardini, C.; Ventrella, A.; Mulargia, M.; Vacca, A.; Orrù, N.; Orrù, S.; Piras, E.; Giustolisi, G.; Lisini, D.; Nesci, S.; Caocci, G.; Carcassi, C. (2007). Status of donor-recipient HLA class I ligands and not the KIR genotype is predictive for the outcome of unrelated hematopoietic stem cell transplantation in beta-thalassemia patients. *Biology of Blood and Marrow Transplantation*, 13, 11, (Nov 2007) 1358-1368, 1083-8791

- La Nasa, G.; Littera, R.; Locatelli, F.; Lai, S.; Alba, F.; Caocci, G.; Lisini, D.; Nesci, S.; Vacca, A.; Piras, E.; Bernardo, ME.; Di Cesare-Merlone, A.; Orrù, S.; Carcassi, C. (2007). The human leucocyte antigen-G 14-basepair polymorphism correlates with graft-versus-host disease in unrelated bone marrow transplantation for thalassaemia. *British Journal of Haematology*, 139, 2, (Oct 2007) 284-288, 0007-1048
- Linder, R.; König, IR.; Weimar, C.; Diener, HC.; Pöpl, SJ.; Ziegler, A. (2006). Two models for outcome prediction - a comparison of logistic regression and neural networks. *Methods of Information in Medicine*, 45, 5, (May 2006) 536-540, 0026-1270
- Lisboa, PJ.; Taktak, AF. The use of artificial neural networks in decision support in cancer: a systematic review. *Neural Networks*, 19, 4, (May 2006) 408-415, 0893-6080
- Littera, R.; Orrù, N.; Vacca, A.; Bertaina, A.; Caocci, G.; Mulargia, M.; Giardini, C.; Piras, E.; Mastronuzzi, A.; Vinti, L.; Orrù, S.; Locatelli, F.; Carcassi, C.; La Nasa, G. (2010). The role of killer immunoglobulin-like receptor haplotypes on the outcome of unrelated donor haematopoietic SCT for thalassaemia. *Bone Marrow Transplantation*, Feb 2010, (Epub ahead of print) 1476-5365
- Lucarelli, G.; Galimberti, M.; Polchi, P.; Angelucci, E.; Baronciani, D.; Giardini, C.; Politi, P.; Durazzi, SM.; Muretto, P.; Albertini, F. (1990) Bone marrow transplantation in patients with talassemia. *New England Journal of Medicine*, 322, 7, (Feb 1990) 417-421, 0028-4793
- Lucarelli, G.; Giardini, C.; Angelucci, E. (1997) Bone marrow transplantation in talassemia, In: *Blood Stem Cell Transplantation*, Winter JN, (Kluwer Academic), 305-315, 0-7923-4260-7, Boston, MA
- Pace, F.; Savarino, V. (2007). The use of artificial neural network in gastroenterology: the experience of the first 10 years. *European Journal of Gastroenterology and Hepatology*, 19, 12, (Dec 2007) 1043-1045, 0954-691X
- Rajan, P.; Tolley, DA. (2005). Artificial neural networks in urolithiasis. *Current Opinion in Urology*, 15, 2, (Mar 2005) 133-137, 0963-0643
- Royston, P. (2000). A strategy for modelling the effect of a continuous covariate in medicine and epidemiology. *Statistics in Medicine*, 19, 14, (Jul 2000) 1831-1847, 0277-6715
- Schwarzer, G.; Vach, W.; Schumacher, M. (2000). On the misuses of artificial neural networks for prognostic and diagnostic classification in oncology. *Statistics in Medicine*, 19, 4, (Feb 2000) 541-561, 0277-6715
- Stephan, C.; Cammann, H.; Meyer, H.; Lein, M.; Jung, K. (2007). PSA and new biomarkers within multivariate models to improve early detection of prostate cancer. *Cancer Letter*, 249, 1, (Apr 2007) 18-29, 0304-3835
- Uhrberg, M.; Parham, W.; Wernet, P. (2002). Definition of gene content for nine common group B haplotypes of the Caucasoid population: KIR haplotypes contain between seven and eleven KIR genes. *Immunogenetics*, 54, 4, (Jul 2002) 221-229, 0093-7711
- Zini G.; d'Onofrio, G. (2003). Neural network in hematopoietic malignancies. *Clinica Chimica Acta*, 333, 2 (Jul 2003) 195-201, 0009-8981
- Zini, G. (2005) Artificial intelligence in hematology. *Hematology*, 10, 5, (Oct 2005), 393-400, 1024-5332

Artificial Neural Networks and Retinal Ganglion Cell Responses

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1. Introduction

Our continuous perception of the world, our sensations about light, colour, music, speech, taste, smell is turned into and coded as binary data by the peripheral sensory systems, and sent by the corresponding nerves to the brain where this code is interpreted and coloured with emotions. The binary sensory data consists in sequences of identical voltage peaks, called action potentials or spikes. Seeing consists in decoding the patterns of these spike trains which are sent to the brain, via the optic nerve, by the visual transduction element: the retina. The external world object features, such as size, colour, intensity, are transformed by the retina in a myriad of parallel spikes sequences, which must describe with precision and robustness all the characteristics perceived. Getting insight into this population code is, nowadays, a basic question for visual science.

A considerable number of coding studies have focused on single ganglion cell responses. Traditionally, the spiking rate of single cells has been used as an information carrier due to the close correlation with the stimulus intensity in all sensory systems. There are, however some drawbacks when analysing single cell firings. Firstly, the response of a single cell cannot unequivocally describe the stimulus since the response from a single cell to the same stimulus has a considerable variability for different presentations. Moreover, the timing sequence differs not only in the time events but also in the spike rates, producing uncertainty in the decoding process. Secondly, the same sequence of neuronal events in an single cell may be obtained by providing different stimuli, introducing ambiguity in the neuronal response.

New recording techniques arisen from emerging technologies, allow simultaneous recordings from large populations of retinal ganglion cells. At this time, recordings in the order of a hundred simultaneous spike trains may be obtained. New tools for analysing this huge volume of data must be used and turn out to be critical for proper conclusions. FitzHugh used a statistical analyser which, applied to neural data was able to estimate stimulus features. Different approaches have been proposed on the construction of such a functional population-oriented analyzer, including information theory, linear filters, discriminant analysis and neural networks.

Analyzing the neural code, especially when this code is split by clustering algorithms in the search of certain levels of organization within it, implies to quantify the amount of

information each cell conveys. The goal of this study was to quantify the ganglion cells tendency to group themselves in sets of relatives according to their coding performance, using functional clustering (FC), information theory (IT) and artificial neural networks (ANN) as tools for providing an empirical value for the goodness of a coding capability. Therefore, a functional separation, or classification based on behaviour, was accomplished and the coding abilities of the subsets and the whole cluster of cells determined. Finally the strong relationship between stimulus reconstruction using artificial neural networks and mean cell information provided by information theory was proved.

In this chapter, we analyse the retinal population data looking at behaviour and exploring the contribution that single cells or population of cells make to the coding process. Two approaches have been used for this purpose: ANNs and IT. Each method required slightly different experimental paradigms that will be explained below.

Firstly, we have tested the ANN performances that small population of ganglion cells achieved at predicting stimuli of varying intensity and wavelength. We have compared at this point individual and population performances for each of the two experimental setups: intensity and wavelength variations. We have also looked at the ANN prediction performance of relevant parameters in the code such as firing rate, latency of first spike, latency of second spike and inter-spike interval. Two different network architectures were studied: Backpropagation and Kohonen Learning Vector Quantization (LVQ).

Secondly, entire populations of ganglion cells were reorganised in a varying number of subsets of cells (subpopulations). The method consisted of creating population subsets using the autocorrelograms of the cells and grouping them according to a minimal Euclidian distance. These subpopulations share functional properties (periodicity) and may be used for data reduction, extracting the relevant information from the code. Information theory and artificial neural networks have been used to quantify the coding goodness of every subpopulation, showing a strong correlation between both methods. All cells that belonged to a certain subpopulation showed very small variances in the information they conveyed while these values were significantly different across subpopulations, suggesting that the functional separation worked around the capacity of each cell to code different stimuli.

2. Methods

2.1 Experimental procedures

Extracellular recordings were obtained from ganglion cell populations in isolated superfused albino rabbit (*Oryctolagus cuniculus*) retina using a rectangular array of 100, 1.5 mm long electrodes, as reported previously (Fernandez et al., 2000; Normann et al., 2001a; Ortega et al., 2004; Bonomini et al., 2005). Briefly, after enucleation of the eye, the eyeball was hemisected with a razor blade, and the cornea and lens were separated from the posterior half. The retinas were then carefully removed from the remaining eyecup with the pigment epithelium, mounted on a glass slide ganglion cell side up and covered with a Milipore filter. This preparation was then mounted on a recording chamber and superfused with bicarbonate-buffered Ames medium at 35°C.

For visual stimulation we used a 17" NEC high-resolution RGB monitor. Pictures were focused with the help of lens onto the photoreceptor layer. The retinas were flashed periodically with full field white light whereas the electrode array was lowered into the retina until a significant number of electrodes detected light evoked single and multiunit responses. This allowed us to record with 60-70 electrodes on average during each

experiment. The electrode array was connected to a 100 channel amplifier (low and high corner frequencies of 250 and 7500 Hz) and a digital signal processor based data acquisition system. Neural spike events were detected by comparing the instantaneous electrode signal to level thresholds set for each data channel using standard procedures described elsewhere (Fernandez et al., 2000; Normann et al., 2001a; Shoham et al., 2003). When a supra-threshold event occurs, the signal window surrounding the event is time-stamped and stored for later, offline analysis. All the selected channels of data as well as the state of the visual stimulus were digitized with a commercial multiplexed A/D board data acquisition system (Bionic Technologies, Inc) and stored digitally.

For spike sorting we used a free program, NEV2lkit, which has been recently developed by our group (Bongard et al., 2004) and runs under Windows, MacOSX and Linux (source code and documentation is freely available at: <http://nev2lkit.sourceforge.net/>). NEV2kit loads multielectrode data files in various formats (ASCII based formats, LabView formats, Neural Event Files, etc) and is able to sort extracted spikes from large sets of data. The sorting is done using principal component analysis (PCA) and can be performed simultaneously on many records from the same experiment.

2.2 Visual stimulation

Initially light stimulation was applied using a halogen light lamp, selecting the wavelength by means of narrow bandpass interference filters. Intensity was fixed by using neutral density filters and a shutter provided the the stimuli flashes to the preparation. In order to place the array in the optimum recording situation, the retina was stimulated with full field flashes of constant intensity while the array was lowered towards the preparation. The multiarray was then fixed at the place of maximum retinal response.

For the ANN experimental paradigm, stimulation consisted of seven consecutive flashes, with 250 msec. length. In this case, the wavelength and the intensity was varied separately. This is, a first run of flashes with changing intensity was presented and afterwards, a second protocol of varying wavelengths showed flashes to the retina. The fifteen most responsive units were selected for analysis.

For the IT experimental paradigm, retinas were randomly stimulated with full field flashes with 16 different light intensities within the grey scale. In order to ensure that both the number of trials for each intensity and the probabilities of appearance of each intensity was equal, a lookup table with 16 light intensities equally distributed, ranging from black (RGB values: 0, 0, 0) to white (RGB values: 255, 255, 255), was constructed. Afterwards, the

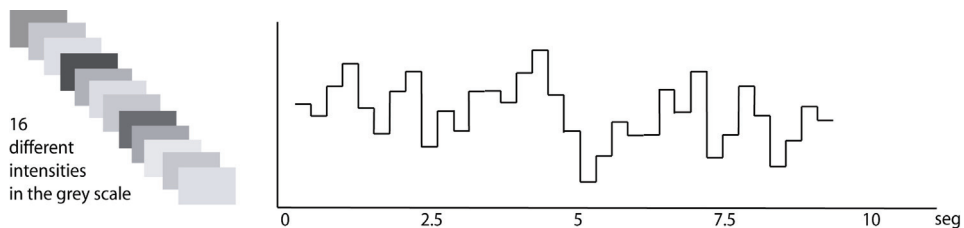


Fig. 1. Visual stimulation. Light intensity trace for 9 seconds of a sample trial containing 30 flashes, each lasting 0.3 ms. Ordinate axis represent the different intensity values (see methods).

elements of a list containing 20 repetitions for each of the intensities from the lookup table were relocated by changing their indexes according to a random entry chosen from a uniform distribution. The list was then loaded by a Python (<http://www.python.org>) script and embedded in VisionEgg (<http://www.visionegg.org>) for presentation of the flashes. Flashes were 300 ms long so that each trial lasted 96 seconds. Figure 1 shows one example of the light intensity trace.

2.3 Separation into subpopulations

In order to identify classes or groups of neurons that behave similarly we used DATAMEAns, a free open-source software for the classification and management of neural ensemble data (Bonomini et al., 2005) which is freely available from the following URL: <http://cortivis.umh.es>. We tested two spike train analysis methods, namely autocorrelations and post-stimulus time histograms (PSTHs). The outputs of this analysis fed then a non supervised clustering method, which used the nearest-neighbour or *k*-means approach for the creation of a varying number of autocorrelograms or PSTH groups. This is a simple clustering method that decomposes the data set into a set of disjoint clusters and then minimizes the dissimilarity in the samples within each cluster, while maximizes the dissimilarity of different clusters. As measure of dissimilarity we used the average squared distance of the data items from their nearest cluster centroids. This procedure defined a set of implicit decision boundaries for the separation of clusters or groups of units according to their periodicity. In this way, we ended up with several clusters or groups of ganglion cells that were a subset of the entire population.

In order to prove that functional clustering produced a reorganization of the cells around coding ability, two different control groups were defined as follows. The first control group was a set of subpopulations resulted from the clustering of the raw data (spike trains without any processing) instead of clustering the functional data (autocorrelograms or PSTHs). In the second control group, on the other hand, subpopulations were originated from a random arrangement. Each cell response was assigned a number and different subpopulations were constructed by picking up the spike train corresponding to a number originated from a random process, resembling a lottery process.

2.4 Neural Networks

For the analysis stage, 15 electrodes with highest activity (high level to noise ratio) were used, while the sixteenth channel was used for storing the original stimulus. For each electrode a 4-element vector was constructed using the number of spikes, the relative time of the first and second spike, and the interspike interval of these firings, building a 60-element vector with the concatenation of the 15 electrode 4-element vectors.

Two different neural networks were used. The first one was a three layer backpropagation, which receive the 60 elements or the four parameters of a single cell, with 20 nodes in the hidden layer, and the output layer consisting in the same number of neurons as the classes to be recognized. So each neuron just fires by applying a certain stimulus response, and the rest of the neurons of the output layer have no activation, acting as a winner-take-all network. The activation function used for all neurons, including the output layer was the hyperbolic tangent sigmoid transfer function given by:

$$f(x) = \frac{2}{1 + e^{-2x}} - 1 \quad (1)$$

using as initial momentum and adaptive learning rate the values established by default by Matlab Neural Network Toolbox, the initial weights randomly initialized and the network trained to minimize a sum square error goal of 1, to provide more generality in the recognition stage.

The other network used was the Kohonen Supervised Learning Vector Quantization (LVQ) with 16 neurons in the competitive map, and a learning rate of 0.05. This network is a competitive network, where the neurons with weights more similar to the input increase their strength, decreasing the rest of the nodes except a close neighborhood, establishing in this way topological relations in the map. The main advantage of using learning vector quantization is that it takes shorter to reach the convergence criteria.

Once the network has been trained, the recognition stage with extended data was accomplished, and the correlation coefficients between the stimulus and its estimation computed. It is important to mention that correlation gives a better estimation of the relation between the estimation and the known applied intensity or wavelength, than an absolute recognition rate, because the difference between two similar stimuli is very small, so the network can easily estimates any one of them. Other studies use their own concepts as mutual information in order to assess the overall quality of the reconstruction, but there no exist a common agreement about the measure that better estimate the goodness of the prediction.

2.5 Information theory

Information theory (Shannon, 1948) was used to assess the quality and reliability of the subpopulations obtained. This approach allows to answer questions about the relevant parameters that transmit information as well as addresses related issues such as the redundancy, the minimum number of neurons needed for coding certain group of stimuli, the efficiency of the code, the maximum information that a given code is able to transmit and the redundancy degree that exists in the population firing pattern (Borst and Theunissen, 1999; Amigo et al., 2003; Panzeri et al., 2003; Pola et al., 2003). We analysed the information about the stimulus that single cells conveyed, as well as the progression of the mutual information values after increasing the number of cells for each subpopulation. For it, the population responses of the retina under several repetitions of the stimuli were discretized into bins. Then, the firing rates from the cells of the population implemented a vector n of spikes counts, with an observed probability $P(n)$. Since the probability of the occurrence of different stimuli had a known probability $P(s)$, the joint probability distribution can be defined as the probability of a global response n and a stimulus s , $P(s,n)$. Thus, the information provided by the population of neurons about the stimulus is given by:

$$I(t) = \sum_{s \in S} \sum_n P(s,n) \log_2 \frac{P(s,n)}{P(s)P(n)} \quad (2)$$

From the above mentioned data, two informational indicators were constructed: the *mean cell information (MCI)*, calculated as the sum of the mutual information of each isolated cell divided by the total number of cells in a particular subpopulation, and the *subpopulation information (SI)*, defined as the overall mutual information for a given subpopulation (all the cells that belong to the subpopulation are taken into account). Note that these indicators are conceptually separated, since one points out to individual coding abilities (MCI) while the other refers to synergic coding abilities (SI).

3. Results

3.1 Population coding and ANNs

The fifteen electrodes with highest activities were selected for the analysis stage. Temporal coding or population coding are possible candidates for representing the stimuli. Another firing characteristic is that for a given cell, no unique response is obtained to the same stimulus. It can be observed that different presentations for the same stimuli evoke different responses, not only on the number of spikes but also on the relative timing of the firings, manifesting variability in their spiking behavior. This variability produces *uncertainty* for recognizing the right stimuli just using the spiking parameters, because there no exists an unequivocal function that associates the firing variables with the provided visual information in an aisle cell.

Ambiguity is another aspect noticed, a single cell can respond exactly the same to different stimuli, making more difficult, as for variability, their recognition. We point to a population coding as a candidate strategy for representing information in the visual system. The overall contribution of an aisle cell was compared with the recognition correlation scores for discriminating different intensities obtained using the fifteen cells. The firing parameters used were the number of spikes, the time of the first and the second spike during the ON stimulus, and the interspike interval. Figure 2 shows the correlation between the output of a trained backpropagation neural network and the correct stimuli, which consisted in 8 different intensities. The stimuli wavelength was fixed to 633, 546 and 450 nm. respectively in order to not influence the intensity transmitted. It can be seen that the scores show variability depending on the wavelength provided. The cells with higher recognition scores are cells 8, 10, 11 and 12 for all stimuli. While these cells had registered the maximum number of spikes, and there is a close relation between the number of spikes and the intensity, it seems reasonable that these cells obtain the higher scores. However using all cells the recognition index overpass 0.95 for all wavelengths.

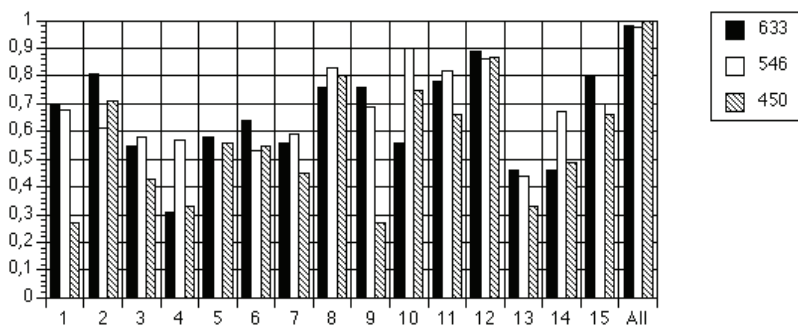


Fig. 2. Intensity recognition scores for aisle cells and the whole population using a BP network

Color recognition is more complex, and the recognition rates for aisle cell are much lower. Intensities were fixed to 1, 1.5 and 2 log units, and nine wavelengths had to be discriminated. Again the scores are variable according with the intensity. The population score, which lies in the range of 0.95, clearly surpass all the individual indexes for all kind of stimuli.

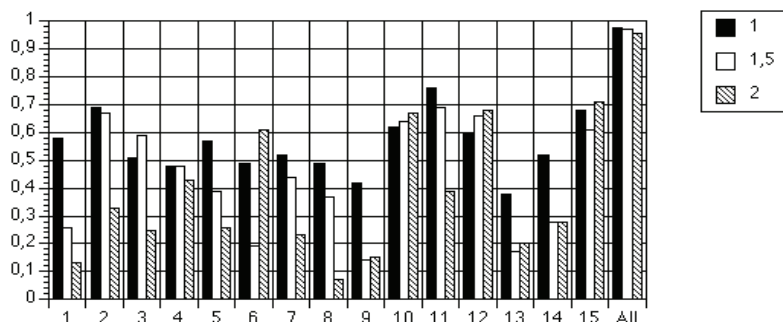


Fig. 3. Color recognition scores for aisle cells and the whole population using a BP network

For validating the prior results, the same data was presented to another kind of neural network, a supervised learning vector quantization (LVQ) with 20 nodes in the competitive layer. This network took lower time to converge than the prior one, and again, the correlation indexes between the correct stimulus and the estimation provided by the network were computed. The cells with higher recognition scores, cells 8, 10, 11 and 12, were the most successful in their estimations alone, and the population index was nearly the same that the one obtained using back-propagation networks for all kind of wavelength (633, 546 and 450 nm.), in the range of 0.95, confirming the recognizing capabilities of the population code.

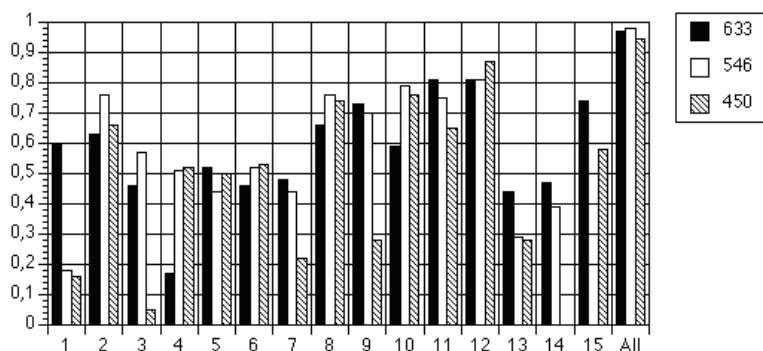


Fig. 4. Intensity recognition scores for aisle cells and the whole population using a LVQ network

The wavelength discrimination using competitive networks behaves similar to the prior feedforward network. Lower recognition scores are obtained, even for the neural population, (this may be due to the network difficulty to fix a decision border which divides the different clusters) however this value is clearly higher than the correlation rates obtained by the cells alone. Using different neural networks does not affect to the obtained results, in respect to the most discriminating cells, and the higher scores obtained by the population parameters.

Once the population code has been noted as the most discriminant element in the neural firing, it still remain to define which are the parameters embedded in the neural signal that are used by cells for transmitting the data. A backpropagation network with 20 nodes in the hidden layer was used, and the input to this layer consists in the 15 population cells aisle parameters in contrast to the whole set of parameters. Figure 6 shows the correlation indexes between the correct stimuli and the network estimations. It can be seen that using

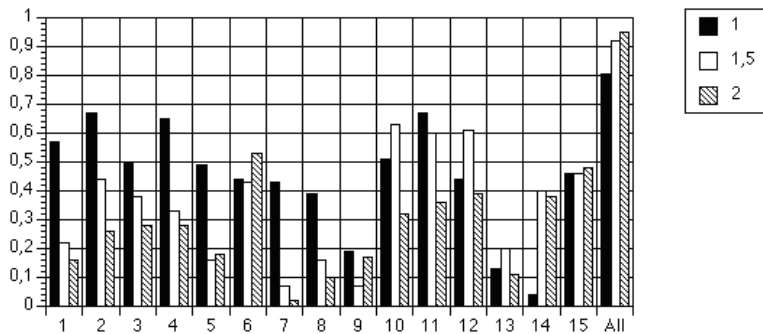


Fig. 5. Color recognition scores for aisle cells and the whole population using a LVQ network

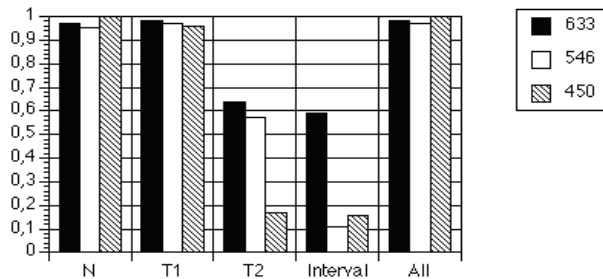


Fig. 6. Intensity recognition scores for the population using the parameters alone and the whole set in a BP network

only the number of spikes of the population is enough for recognizing successfully a given intensity (it has been described in literature the relation between the intensity level and the number of firing rate evoked), but also using only the exact timing of the first spike in all cells is enough for recognizing the stimuli for all wavelengths (lower intensities show longer delays). The second spike timing carries less information, and the interspike data is also a poor coding element.

The relative relevance of the aisle population parameters on recognizing colors was also computed. A backpropagation network was again used, showing similar results that the ones obtained for recognizing intensities. Just using the number of spikes of the whole population is enough for determining the provided color, and the time first spike occurs

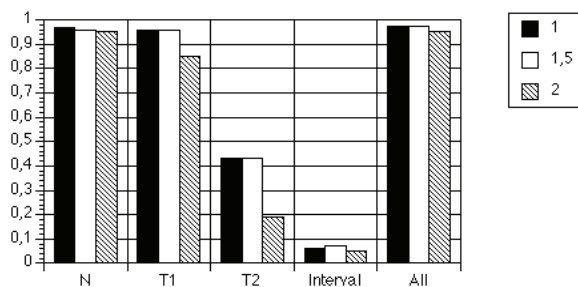


Fig. 7. Color recognition scores for the population using the parameters alone and the whole set in a BP network.

also discriminate between the different provided wavelengths for all kind of intensities used. The time of the second spike during the ON period or the interspike interval show lower recognizing scores than previous parameters.

3.2 Entire population and subpopulations obtained

The classification algorithm worked around the coding abilities of the cells. This is, the subsets emerged from the original population with very different MCI values. The clearest clustering strategy showed up when separating autocorrelograms. Although PSTHs performed well, the MCI separation across subpopulations suffered a little bit. Figure 8 illustrates this fact, and the fact that subpopulations originated randomly (first control group) did not evidence this feature. Neither did the second control group, the clustering of raw data, since no useful subpopulations were even generated. Notice that most of the points in the bottom-right panel are coloured green.

The generation of subpopulations using the autocorrelogram approach is illustrated in Figure 9 with an example where the maximum number of clusters was fixed to three. Top panel displays the raster plot of the whole population of ganglion cells while bottom panels show the raster plots of different subpopulations (named *s1*, *s2* and *s3*) obtained by applying a bin size of 10 ms and a maximum shift of 900 lags. Subpopulations are reordered so that *s1* is the subpopulation with fewer cells and subsequent subpopulations (*s2* and *s3*) contain an increasing number of cells. Clear differences among the different subpopulations can be perceived. These differences were related mainly to the firing patterns and to the number of cells in each subpopulation. For instance *s1*, contained very few cells that fired almost constantly during the presentation of the stimuli, *s2* contained a considerable number of cells with apparent temporal patterns and *s3* was integrated by a higher number of cells which showed a more randomised activity.

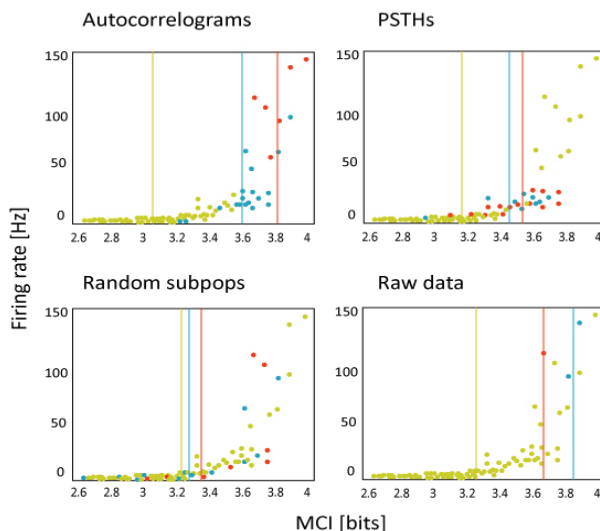


Fig. 8. Firing rate (Y axis) and MCI levels (X axis). Firing rates against MCI values for a collection of cells grouped into three subpopulations. Classes are indicated on different colours. Top left: autocorrelation approach. Top right: PSTH approach. Bottom left: random subpopulations. Bottom right: raw data.

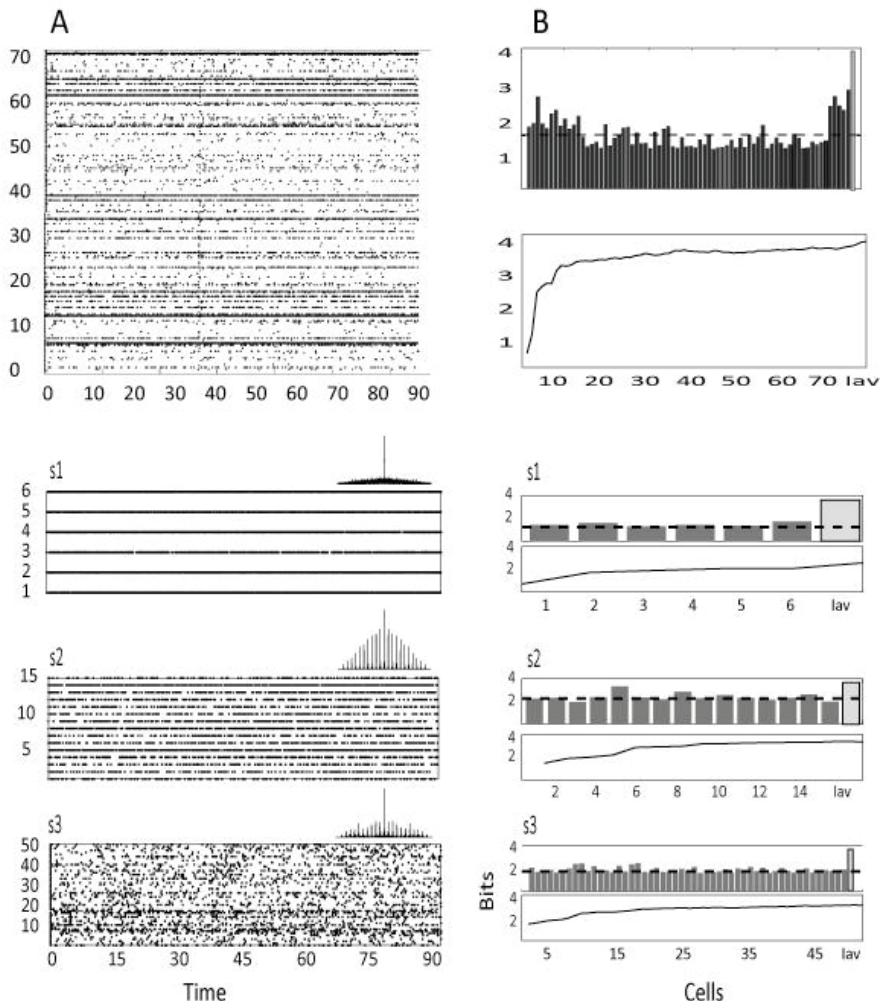


Fig. 9. *Top panel.* Example of simultaneously recorded extracellular responses from a population of rabbit ganglion cells to a trial of random full field flashes with 16 different intensities (see methods). (A) Original population raster plot. Each dot represents a single spike. (B) Mutual information values for each cell in the recording (bars) and for the whole population (last gray bar). The overall *mean cell information* is shown as a dashed line. (C) Accumulative mutual information for an increased number of cells. In this example the number of cluster was fixed to three. *Bottom panel, left.* Raster plots for each subpopulation (named s1, s2 and s3). *Bottom panel, right.* Information about the stimulus for the subpopulations. Mutual information values for each cell in the recording (bars) and for the whole subpopulation (last gray bar). The overall *mean cell information* in each case is shown as a dashed line. Below each bar graph it is displayed the accumulative mutual information for an increasing number of cells. Note the relationship between number of cells and *mean cell information*.

3.3 Quality of the subpopulations: information theory approach

Using the autocorrelogram approach the subpopulation information (SI), kept similar across classes (one-way ANOVA; $p=0.82$) while the informative value of the individual cells, summarised by the mean cell information (MCI) varied significantly (one-way ANOVA; $p<0.0005$). The PSTH approach was less effective at grouping subpopulations on an informational basis, even though the above trend was evident in the subpopulations separation.

Figure 9 (bottom panel, right) shows the information that each cell conveyed about the stimulus (black bars), their mean value (MCI, dashed line) and the SI value (grey bar) for different subpopulations arisen for k set to 3. Notice the difference in the MCI and SI values. As expected, s_1 , formed by the very few continuously firing cells gave the lowest *mean cell information* (1.48 ± 0.10 bits; $M\pm SE$), clearly above the overall value of the whole population (2.42 ± 0.08 bits; $M\pm SE$) whereas s_2 , kept the moderately informative cells (1.88 ± 0.07 bits; $M\pm SE$) and s_3 grouped the worst cells on a mean information basis (1.33 ± 0.02 bits; $M\pm SE$). On top of each raster plot there is a pattern of the autocorrelogram representative of the subpopulation.

Taking apart the noisy cells, a particular relation between number of cells and MCI was found. Although the number of cells did not affect the *subpopulation information*, this measure was inversely correlated with the *mean cell information*. Therefore, the subpopulations with fewer cells tended to have higher *mean cell information* values. Figure 10 illustrates this fact. Here, MCI values obtained with k ranging from 2 to 5 are collected and plotted versus the number of cells, n . It can be observed that the *mean cell information* decreases as the number of cells in a certain subpopulation increases, fitting this relationship to a linear equation, with square r going from major to minor through autocorrelograms, PSTHs and control group (raw data), respectively. It is important to notice that the latter behaviour does not hold for the *subpopulation information*, which keeps invariant for any cell number. Note the strength of this observation for the autocorrelogram approach (upper panel), and how this behaviour suffers with the PSTH approach (middle panel) until is completely faded when raw data, where no temporal feature was taken into account before the clustering procedure.

We also found out that there is a natural number of subpopulations which optimise the clustering strategy. Such critical value for the number of groups (k) can be appreciated on Figure 11, where the clustering is shown on an informational basis. MCI and SI values for subpopulations obtained by processing ten original populations by means of the autocorrelogram approach are collected and displayed for an increasing number of classes (k). In this figure, subpopulations belonging to a certain class are represented with the same marker through different values of k . On the x-axis are displayed the MCI values and on the y-axis the SI values. Note that although for $k=4$ the two central clusters start to overlap, this is, subpopulations start to share MCI levels, the goodness in the group separation when taking the autocorrelogram approach is still evident. However, for five sets of relatives, the clustering behaviour becomes totally blurred and redundant subpopulations start to generate. Left panels contain data obtained with the autocorrelogram approach while right ones display data from the PSTH approach. Unlike the autocorrelograms, the PSTH approach fails to produce a robust separation according to information features for k greater than two. Thus, for 3 classes on, clusters start to spread and overlap. This could imply that PSTHs have a natural k much lower than autocorrelograms, since they could tell less (extract less information) about the experimental paradigm proposed here. The k value is the

maximum number of predefined classes, but the algorithm finds the optimal number, lower or equal than this predefined value.

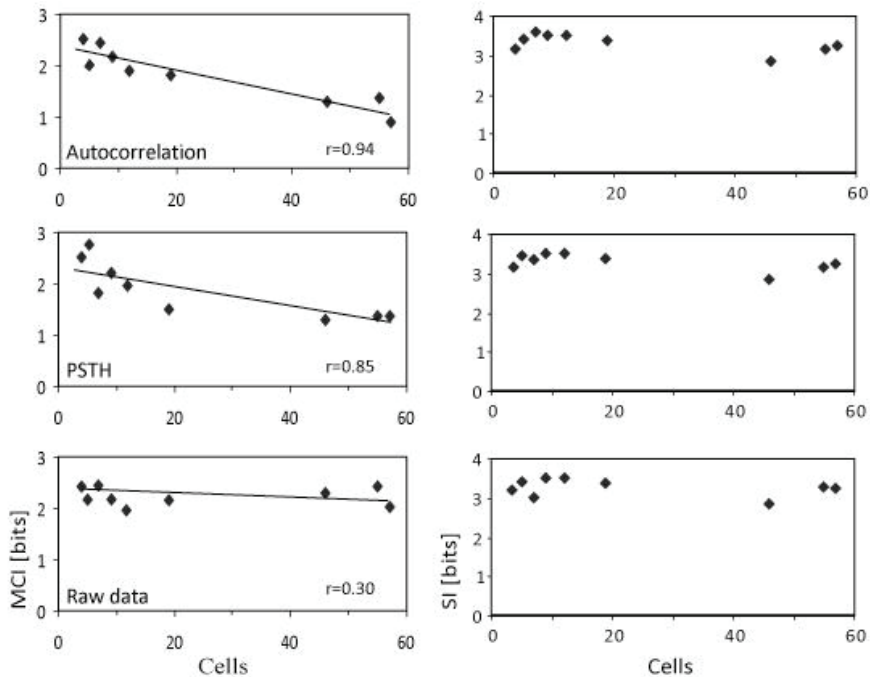


Fig. 10. Cell number versus coding quality. *Mean cell information (MCI)* against number of cells (n). Note that MCI decreases as the number of cells in a certain subpopulation increases. Upper panel: data obtained from the autocorrelogram approach. Middle panel: data obtained from the PSTH approach. Bottom panel: data obtained from raw data.

3.4 Correlations

We also looked for concerted activity within each subpopulation. In order to do so, a bin of 0.5 ms was used to be able to see peaks of concerted activity of 1 ms. Figure 12 describes how the method produces subpopulations with different levels of synchrony. Up to three main correlations widths were found: a narrow 5 ms peak, a 10 ms peak and a broad 30 ms peak. Subpopulations with highest MCI values showed either intermediate or broad correlation peaks and a smaller number of correlated cells. Continuously firing cells belonged to this latter group. Subpopulations containing the greatest number of correlated cells showed concerted activity around 5 ms but, surprisingly, the MCI values for these subpopulations were not the highest ones. This inverse relationship between synchronicity and individual informative value (MCI) is not surprising when looking at the overall information (SI), which kept constant across the different subpopulations, which in turn, presented SI values similar to the original population. This would suggest the coexistence of elements of both strategies, individual and population coding.

Raw data did not show any significant correlation. PSTH data, on the other hand, presented a weak separation of correlated cells with a narrow and a broad peak. Finally, autocorrelogram data, displayed a clear separation of the concerted activity giving place to a different peak for each subpopulation. Here, subpopulations were grouped according to their synchrony windows, from infinity (subpopulation with no correlation at all) to 30 ms.

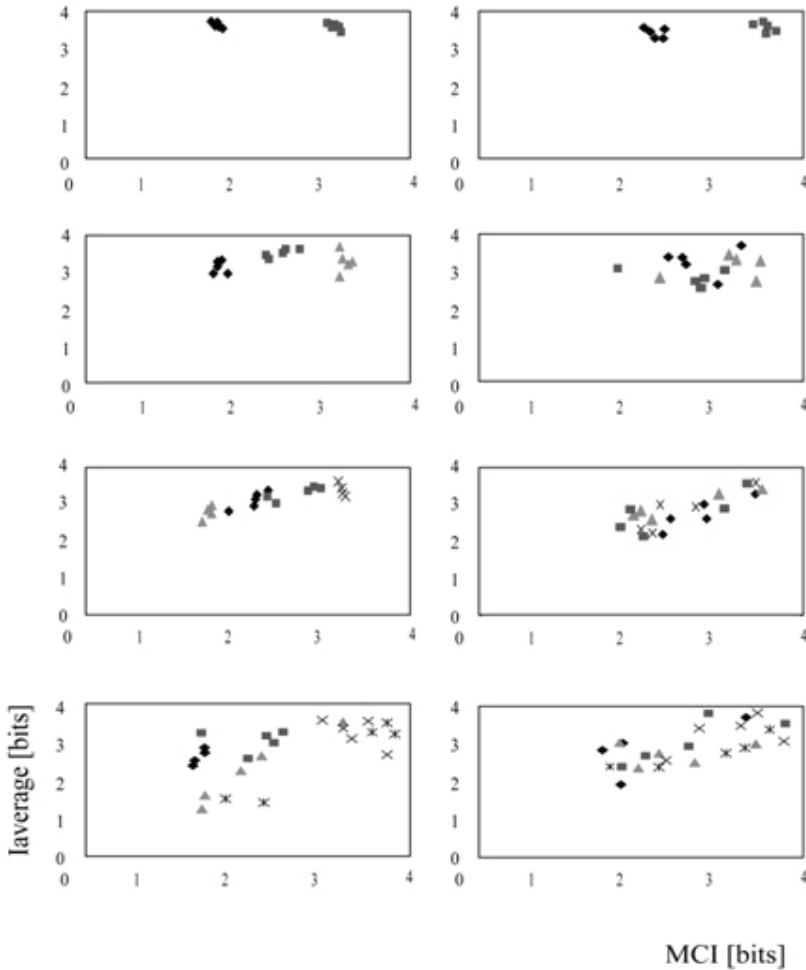


Fig. 11. Relationships between number of clusters and mean cell information values. Different trials are represented with different markers. Note that the individual components of each subpopulation have similar subpopulation information values and that there is a maximum number of clusters which optimise the classification strategy. Thus for $k=4$ although the two central clusters start to overlap, the group separation it is still apparent. For $k=5$, the clustering behaviour becomes totally blurred.

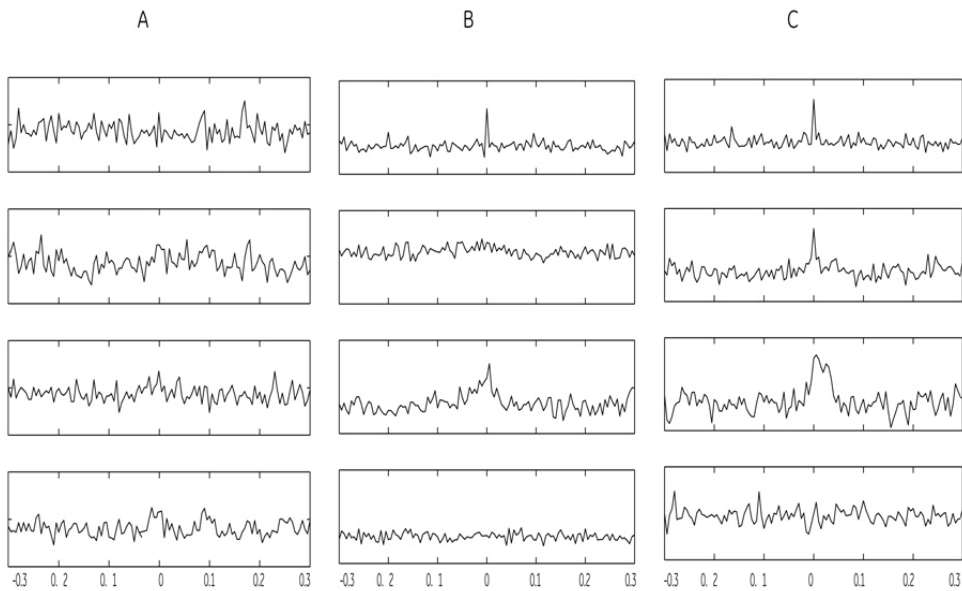


Fig. 12. Concerted activity of subpopulations. Each row represents the correlation signal of each subpopulation. Correlation were computed as the average of the correlation signals among all the cells belonging to that subpopulation in a pair wise process. Firing rate against MCI values. A), Data obtained from raw data. B), Data obtained from the PSTH approach. C), Data obtained from the autocorrelogram approach.

3.5 Artificial Neural Networks and information theory

In order to address quality differences across subpopulations, we used a feed-forward back-propagation neural network to study the contribution of the mean cell information indicator to group cells that are better encoders than others with respect to the stimulus applied.

Interestingly, the neural network performance obtained with the different sets of cells within each subpopulation was found to be related to the mean cell information. With the aim of quantifying this relationship, ANN performance percentages were transformed to arcsin values and the mean cell information values underwent a logarithmic transformation in order to study the linearity of its relationship. Afterwards, correlation coefficient and regression analysis was applied to the data. Fixing $k=5$ (see methods), a highly significant positive correlation was found for $s1$ ($r=0.9436$, $df=4$), $s2$ ($r=0.9785$, $df=4$) and $s3$ ($r=0.9261$, $df=4$). Figure 13 shows the lines fitted by means of regression analysis on each of the subpopulations generated from the population shown on Figure 9. Dots represent the samples from which regression analysis was calculated. Equations for the respective line fittings and square r are included in the figures. Here, it is clearly shown the strong relationship between stimulus reconstruction and mean cell information.

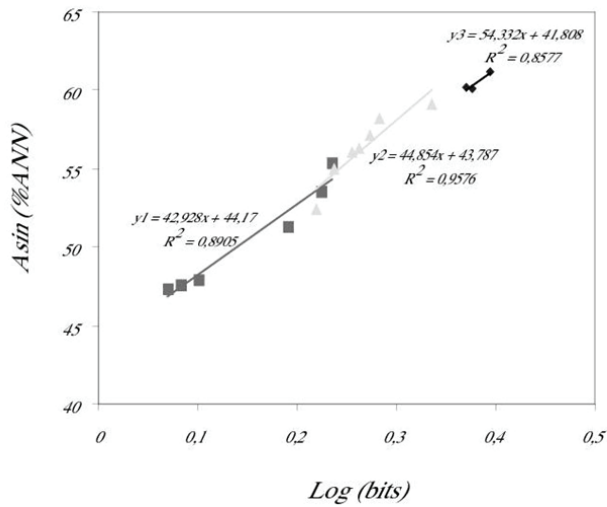


Fig. 13. ANN performance against mean cell information. Line fitting for s1 (squares), s2 (triangles) and s3 (rhombus). Notice that the best correlation coefficient belongs to the subpopulations with the best defined time patterns.

4. Discussion

The issue of classification has long been a central topic in the analysis of multielectrode data, either for spike sorting or for getting insight into interactions among ensembles of neurons (Fernandez et al., 2000; Nicolelis, 2003; Shoham et al., 2003; Carmenta et al., 2005; Fernandez et al., 2005; Nicolelis, 2005; Suner et al., 2005; Hochberg et al., 2006).

A major challenge in this context is to acquire meaningful data from different functional types of neurons, but there is not a standard way for addressing how many neuronal types are in a given multielectrode recording. In this work we have introduced a new approach to facilitate this task which is easy to implement and has proved to be useful for defining subsets of retinal ganglion cells which share similar temporal responses and coding capabilities.

The fact that reliable separation of clusters was achieved does not mean that finer separation utilizing additional variables is not possible, but a clear trend in the clustering strategy was present for all the generated subpopulations. This might be explained from a functional point of view. Thus, we speculate with a natural number of retinal ganglion cell classes, where every class contributes to coding different elements of the visual scenario such as intensity, colour, texture, orientation, shape, etc (Kang et al., 2004). If the number of classes is increased the coding process could lose effectiveness, starting to turn up redundant classes or subpopulations. In other words, the classes would enclose different kind of cells with similar coding behaviour, like the intensity coders, colour coders and so on. Taking into account that the stimulus applied was intensity variation of full field flashes, the best coder subsets would effectively code intensity in these particular cases. However, this should be addressed in future works by repeating the visual stimuli with other varying parameters, for instance using different colours, moving bars, and even natural scenes to confirm such a functional separation. Additionally more sophisticated unsupervised clustering algorithms, particularly on the crucial issue of assessing the proper number of

clusters, are required and should be developed. Finally, some degree of controversy can accompany any classification scheme but we should take into account that the controversy is not concerned with whether the groups exist, or whether they are important; rather it is related to the number of groups to be present and the functional semantics attributed to them.

5. Acknowledgements

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6. References

- Amigo JM, Szczepanski J, Wajnryb E, Sanchez-Vives MV. On the number of states of the neuronal sources. *Biosystems* 2003; 68:57-66.
- Bongard M, Micol D, Fernandez E (2004) Nev2lkit: a tool for handling neuronal event files. In: (<http://nev2lkit.sourceforge.net/>).
- Bonomini MP, Fernandez JM, Bolea JA, Fernandez E. DATA-MEAns: an open source tool for the classification and management of neural ensemble recordings. *J Neurosci Methods* 2005; 148:137-46.
- Borst A, Theunissen FE. Information theory and neural coding. *Nat Neurosci* 1999; 2:947-57.
- Cajal SR (1904) *Textura del sistema nervioso del hombre y de los vertebrados*. Madrid: Imprenta y librería de Nicolás Moya.
- Carmena JM, Lebedev MA, Henriquez CS, Nicolelis MA. Stable ensemble performance with single-neuron variability during reaching movements in primates. *J Neurosci* 2005; 25:10712-6.
- Costa L, Velte T. Automatic characterization and classification of ganglion cells from the salamander retina. *J Comp Neurol* 1999; 404:33-51.
- Diaz E. A functional genomics guide to the galaxy of neuronal cell types. *Nat Neurosci* 2006; 9:99-107.
- Egert U, Knott T, Schwarz C, Nawrot M, Brandt A, Rotter S, Diesmann M. MEATools: an open source toolbox for the analysis of multi-electrode data with MATLAB. *J Neurosci Methods* 2002; 117:33-42.
- Fernandez E, Fernandez J, Ammermüller J, Normann RA. Population coding in spike trains of simultaneously recorded retinal ganglion cells. *Brain Res* 2000; 887:222-9.
- Fernandez E, Pelayo F, Romero S, Bongard M, Marin C, Alfaro A, Merabet L. Development of a cortical visual neuroprosthesis for the blind: the relevance of neuroplasticity. *J Neural Eng* 2005; 2:R1-R12.
- Ferrández J, Bolea J, Ammermüller J, Normann R, Fernández E (1999) A Neural Network Approach for the Analysis of Multineural Recordings in Retinal Ganglion Cells: Towards Population Encoding. In: *Lecture Notes on Computer Sciences* (Verlag S, ed), pp 289-98.
- Golledge HD, Panzeri S, Zheng F, Pola G, Scannell JW, Giannikopoulos DV, Mason RJ, Tovee MJ, Young MP. Correlations, feature-binding and population coding in primary visual cortex. *Neuroreport* 2003; 14:1045-50.

- Hochberg LR, Serruya MD, Friehs GM, Mukand JA, Saleh M, Caplan AH, Branner A, Chen D, Penn RD, Donoghue JP. Neuronal ensemble control of prosthetic devices by a human with tetraplegia. *Nature* 2006; 442:164-71.
- Kang K, Shapley RM, Sompolinsky H. Information tuning of populations of neurons in primary visual cortex. *J Neurosci* 2004; 24:3726-35.
- Keat J, Reinagel P, Reid RC, Meister M. Predicting every spike: a model for the responses of visual neurons. *Neuron* 2001; 30:803-17.
- Kim SJ, Manyam SC, Warren DJ, Normann RA. Electrophysiological Mapping of Cat Primary Auditory Cortex with Multielectrode Arrays. *Ann Biomed Eng* 2006.
- Lehmkuhle MJ, Normann RA, Maynard EM. Trial-by-trial discrimination of three enantiomer pairs by neural ensembles in mammalian olfactory bulb. *J Neurophysiol* 2006; 95:1369-79.
- McClelland J, Rumelhart D (1986) *Explorations in Parallel Distributed Processing*. Cambridge.
- Meister M. Multineuronal codes in retinal signaling. *Proc Natl Acad Sci U S A* 1996; 93:609-14.
- Meister M, Pine J, Baylor DA. Multi-neuronal signals from the retina: acquisition and analysis. *J Neurosci Methods* 1994; 51:95-106.
- Migliore M, Shepherd G. Opinion: An integrated approach to classifying neuronal phenotypes. *Nat Rev Neurosci* 2005; 6:810-8.
- Nicolelis MA. Brain-machine interfaces to restore motor function and probe neural circuits. *Nat Rev Neurosci* 2003; 4:417-22.
- Nicolelis MA. Computing with thalamocortical ensembles during different behavioural states. *J Physiol* 2005; 566:37-47.
- Normann R, Warren D, Ammermuller J, Fernandez E, Guillory S. High-resolution spatio-temporal mapping of visual pathways using multi-electrode arrays. *Vision Res* 2001a; 41:1261-75.
- Normann RA, Warren DJ, Ammermuller J, Fernandez E, Guillory S. High-resolution spatio-temporal mapping of visual pathways using multi-electrode arrays. *Vision Res* 2001b; 41:1261-75.
- Ortega G, Bongard M, Louis E, Fernandez E. Conditioned spikes: a simple and fast method to represent rates and temporal patterns in multielectrode recordings. *J Neurosci Meth* 2004; 133:135-41.
- Panzeri S, Pola G, Petersen RS. Coding of sensory signals by neuronal populations: the role of correlated activity. *Neuroscientist* 2003; 9:175-80.
- Pola G, Thiele A, Hoffmann KP, Panzeri S. An exact method to quantify the information transmitted by different mechanisms of correlational coding. *Network* 2003; 14:35-60.
- Rieke F, Warland D, Van Steveninck R, Bialek W (1997) *Spikes: Exploring the Neural Code*. Cambridge.
- Shannon C. A Mathematical Theory of Communication. *Bell sys Tech* 1948; 27:379- 423.
- Shoham S, Fellows M, Normann R. Robust, automatic spike sorting using mixtures of multivariate t-distributions. *J Neurosci Meth* 2003; 127:111-22.
- Shoham S, Paninski LM, Fellows MR, Hatsopoulos NG, Donoghue JP, Normann RA. Statistical encoding model for a primary motor cortical brain-machine interface. *IEEE Trans Biomed Eng* 2005; 52:1312-22.

- Suner S, Fellows MR, Vargas-Irwin C, Nakata GK, Donoghue JP. Reliability of signals from a chronically implanted, silicon-based electrode array in non-human primate primary motor cortex. *IEEE Trans Neural Syst Rehabil Eng* 2005; 13:524-41.
- Warland D, Reinagel P, Meister M. Decoding Visual Information from a Population of Retinal Ganglion Cells. *J Neurophysiol* 1997a; 78:2336-50.
- Warland DK, Reinagel P, Meister M. Decoding visual information from a population of retinal ganglion cells. *J Neurophysiol* 1997b; 78:2336-50.
- Warren DJ, Fernandez E, Normann RA. High-resolution two-dimensional spatial mapping of cat striate cortex using a 100-microelectrode array. *Neuroscience* 2001; 105:19-31.
- Xia Y, Turken U, Whitfield-Gabrieli S, Gabrieli J (2005) Knowledge-based classification of neuronal fibers in entire brain. In: *Med Image Comput Assist Interv Int Conf Med Image Comput Comput Assist Interv*, pp 205-12.

Part 4

Medical Applications

Diagnosing Skin Diseases Using an Artificial Neural Network

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1. Introduction

The field of Artificial Neural Networks (ANNs) or Neurocomputing or Connectionists Theory (CT) was born out of a research conducted by McCulloch and Pitt in 1943 on a simple model of a neuron. Following this were further attempts to generate knowledge from the study of human nervous system and application of such knowledge to the design and implementation of systems, which mimic the way human beings process information. Most of the early works in ANNs attempted to simulate the human nervous system. An ANN is an information or signal processing system which is composed of a large number of simple processing elements called artificial neurons or nodes that are interconnected by direct links, called connections, and which cooperate to perform parallel distributed processing (PDP) operation in order to solve a given problem (Haykin, 1994). A major feature of ANNs is their ability to adapt to changes in the environment by changing their connection strength or structure. This feature enables neural network systems to learn automatically. ANNs in their present form are grossly simplified model of the human brain. This is because the operation involved in the behavior of the brain is much more complex than model proposed in the ANN concepts. It is more reasonable to compare ANN capabilities to simpler nervous systems found in primitive animals, like insects, which have the ability to adapt themselves to a complex environment. A simplified schematic diagram of a biological neuron is as shown in fig. 1.

The components of the biological neuron, which are of importance in neural network simulation study, are discussed as follows:

- i. **Soma or Cell Body:** This is the large round central body of the neuron in which all the logical functions are realized. The cell body contains the genetic and metabolic machinery necessary to keep the neuron alive. The neuron soma also contains the nucleus and the protein- synthesis machinery.
- ii. **Axon:** This is the nerve fibre attached to the soma and can serve as the final output channel of the neuron. An axon is usually highly branched. The initial segment of the axon is called an **axon hillock**. Here the signals are converted into sequence of nerve pulse (spikes), which are propagated without attenuation along the axon to the target cells (i.e. other neurons, receptors and muscles).
- iii. **Dendrites:** The dendrites (inputs) represent a highly branching tree of fibres. These long irregularly shaped nerve fibres are attached to the soma. There are about 10^3 to 10^4 dendrites per neuron. Dendrites connect a neuron to a set to other neurons. Dendrites either receive inputs from other neurons via specialized contacts called synapse or

connect other dendrites to the synaptic outputs. Dendrites are regarded as providing receptive surface for input signals to the neurons and as conducting signals with decrement to the body cell and the axon hillock.

- iv. **Synapses:** Synapses are specialized contacts on a neuron, which are the termination points for the axons from other neurons. Synapses play the role of interfaces connecting some axon of the neurons to the spines of the input dendrites. Synapses can also be found on the cell body. Synapses are capable of changing a dendrite's local potential in a positive or negative direction. Due to their function, the synapses can be of excitatory or inhibitory nature in accordance with their ability to increase or to damp the neuron excitation. Storing of information in a neuron is supposed to be concentrated in its synapses connections or more precisely in the pattern of these connections and strengths (weights) of the synaptic connections. The human synapses are mainly complex chemical nature while synapses in nervous system of primitive animals such as insects are predominantly based on electrical signal transmission.

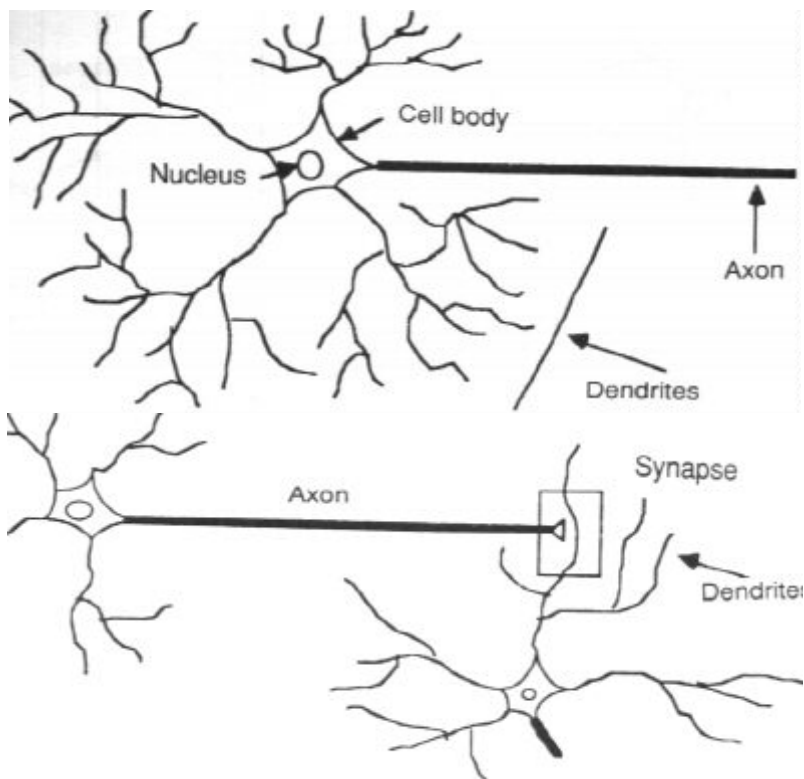


Fig. 1. Components of a neuron (Source: Haykin, 1999)

Neuron cell body (soma) receives inputs from other neuron through adjustable or adaptive synaptic connections to the dendrites. The output signal (consisting of nerve impulses) from a cell is transmitted along a branching axon to the synapses of other neurons. When a neuron is excited, it produces nerve impulses (i.e. a train of pulses), which are transmitted

along an axon to the synaptic connections of other neurons. The output pulse- rate (impulse density) depends on both the strength of the input signals and the weight/ strength of the input signals. The maximum firing rate is around 1000 pulses per second. The input signals at the excitatory synapse increase the pulse rate while the input signals at the inhibitory synapse reduce the pulse rate or even block the output signal. A neuron operates in mixed digital and analogue form. The neuron output signal is proportional, in some range, to a linear combination of the neurons input signal values. This information between neurons is transmitted in the form of nerve impulses, which can be considered as digital signals. However the encoded information has the form of the pulse density, which is an analogue signal. Each pulse (or nerve impulse) arriving at the synapse generates an analogue internal potential in some proportion to the synaptic strength, which has either a positive or negative value corresponding to an exciting or inhibitory synapses. These potentials are summed in a spatial-temporal way and when the total potential exceeds some value called the threshold, a train of pulse is generated and travels along its axon. The signals are changed when they travel along the connections: They are combined - usually by multiplication with the connection weights. A neuron gathers the input from all incoming connections to compute its activation value. In 1949 Hebb hypothesized about learning in networks of artificial neurons. The Hebb rule encodes the correlations of activations of connected units in the weights. A weight is increased, if the two neurons that are connected by it are active at the same time. The weight is decreased, if only one of the two connected neurons is active (Hebb, 1949). The structure of a neuron model is illustrated in figure 2.

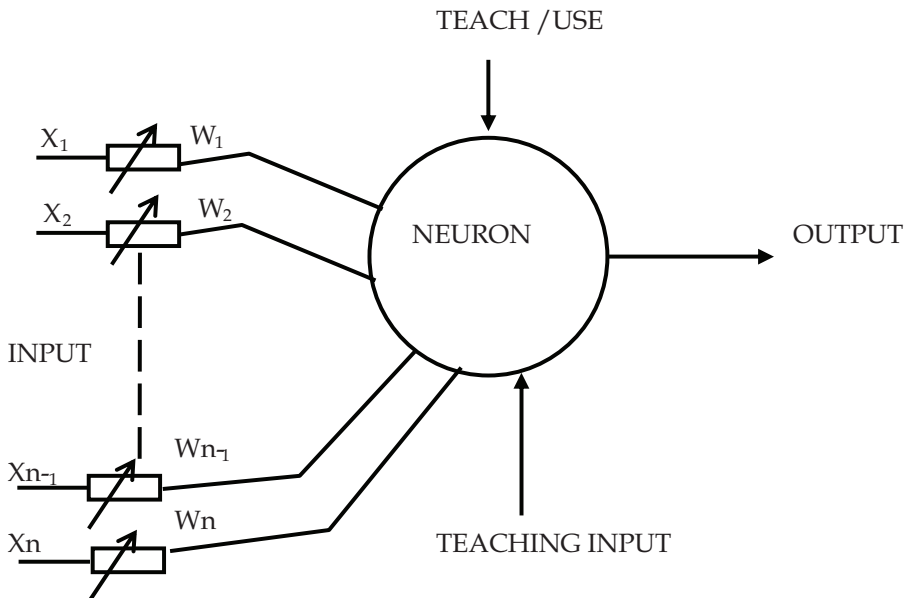


Fig. 2. The neuron model

The units of most types of neural networks are organized in layers, which are called input layer, hidden layers(s) and output layer, depending on their functionality. A neural network with n units in its input layer and m units in its output layer, implements a mapping $f: X^n \rightarrow Y^m$. The sets $X^n \subseteq \mathbb{R}^n$ and $Y^m \subseteq \mathbb{R}^m$ are the input and output domains. Hidden layers add to the complexity of a neural network, and are important, if arbitrary mappings have to be

represented. Neural networks differ from one another in the way the nodes are interconnected and in the rules they use to produce a given output signal for different sets of input signals. Some networks are also fully interconnected grids, with each processor directly connected to every other processor; others embody treelike or layered architectures. The capabilities of ANNs that are considered especially attractive include (DeCegama, 1989):

- i. Correct memory data retrieval, even if some individual processors (neurons) fail.
- ii. Retrieval of closest matching data if there is no exact match to the requested information.
- iii. The ability to retrieve original inputs from a degraded version, to connect one item with another, or to connect an item to a set of categories.
- iv. The capability to discover statistically salient features among the stored data.
- v. The ability to find solutions for problems that involve combinatorial explosion.
- vi. The ability to be trained (or taught) instead of programmed.

To achieve the above capabilities, the connection weights of ANNs are estimated by learning algorithms. Learning algorithms of neural networks use a learning problem, described by a set of training data and iteratively update the parameters of a network such that some error measure is decreased or some performance measure is increased. The training data can consist of input and output data (**supervised learning**), of input data and success or failure signals (**reinforcement learning**) or of input data alone (**unsupervised learning**). Among the many interesting properties of a neural network, the property that is of primary significance is the ability of the network to learn from its environment, and to improve its performance through learning; the improvement in performance takes place over time in accordance with some prescribed measure. A neural network learns about its environment through an iterative process of adjustments applied to its synaptic weights and thresholds. Ideally, the network becomes more knowledgeable about its environment after each iteration of the learning process.

In the context of neural networks, learning is defined by Haykin (1994) as follows: Learning is a process by which the free parameters of a neural network are adapted through a continuing process of stimulation by the environment in which the network is embedded. The type of learning is determined by the manner in which the parameter changes take place. This definition of the learning process implies the following sequence of events:

- i. An environment stimulates the neural network.
- ii. The neural network undergoes changes as a result of this stimulation.
- iii. The neural network responds in a new way to the environment, because of the changes that have occurred in its internal structure.

An essential ingredient of *supervised* or *active learning* evident in fig. 2 is the availability of an external teacher (Haykin, 1994).

In conceptual terms, we may think of the **teacher** as having knowledge of the environment that is represented by a set of *input-output* examples. The environment is, however, *unknown* to the neural network of interest. Suppose now that the teacher and the neural network are both exposed to a training vector (i.e. example) drawn from the environment. By virtue of built-in knowledge, the teacher is able to provide the neural network with a *desired* or *target response* for that training vector. Indeed, the desired response represents the optimum action to be performed by the neural network. The network parameters are adjusted under the combined influence of the training vector and the error signal; the *error signal* is defined as the difference between the actual response of the network and the desired response. This adjustment is carried out iteratively in a step-by-step fashion with the aim of eventually making the neural network *emulate* the teacher; the emulation is presumed to be optimum in some statistical sense. In other words, knowledge of the environment available to the

teacher is transferred to the neural network as fully as possible. When this condition is reached, we may then dispense with the teacher and let the neural network deal with the environment thereafter completely by itself (i.e. an unsupervised fashion). The form of supervised learning described is indeed the error-correction learning in a closed-loop feedback system, but the unknown environment is not in the loop. As a performance measure for the system, we may think in terms of the mean-squared error (i.e. the expected value of the sum of squared errors) defined as a function of the free parameters of the system. This function may be visualized as a multidimensional *error-performance* surface or simply *error surface*, with the free parameters as coordinates. The true error surface is averaged over all possible input-output examples (Haykin, 1999). Any given operation of the system under the teacher's supervision is represented as a point on the error surface. For the system to improve performance over time and therefore learn from the teacher the operating point has to move down successively toward a minimum point of the error surface; the minimum point may be a local minimum or a global minimum. A supervised learning system is able to do this by virtue of some useful information it has about the gradient of the error corresponding to the current behavior of the system. The gradient of an error surface at any point is a vector that points in the direction of steepest descent. In fact, in the case of supervised learning from examples, the system uses an *instantaneous estimate* of the gradient vector, with the example indices presumed to be those of time. The use of such an estimate results in a motion of the operating point on the error surface is typically in the form of a random walk. Nevertheless, given an algorithm designed to minimize the cost function of interest, and given an adequate set of input - output examples and enough time permitted to do the training, a supervised learning system is usually able to perform such tasks as pattern classification and function approximation satisfactorily. For this reason one is poised to expect more industrial applications of ANNs to human development. A recent example is the use of ANN in financial crime detection (Bakpo, 2008).

The use of expert system as a mean of conducting medical diagnosis and recommending successful treatments has been a highly active research field in the past few years. Development of medical expert system that uses artificial neural networks (ANN) as knowledge base appears to be a promising method for diagnosis and possible treatment routines. One of the major applications of medical informatics has been the implementation and use of expert systems to predict medical diagnoses based upon a set of symptoms (Eugene et al., 1997). Furthermore, such expert systems serve as an aid to medical professionals in recommending effective laboratory tests and treatments of diseases. An intelligent computer program assisting medical diagnosis could provide easy access to a wealth of information from past patient data. Such a resource may help hospitals reduce excessive costs from unnecessary laboratory test and ineffective patient treatment, while maintaining high quality of medical care. Klerfors (1998) argued that, so far these expert systems only served as aids to the physician and are not 100% reliable. Current expert systems do not provide enough value to the physician to justify their large-scale implementation. One major drawback of conventional medical expert systems is the use of static knowledge base developed from a limited number of cases and a limited population size, demographics, and geographic location. The knowledge base is inherently not dynamic and is not routinely updated to keep up with emerging trends such as the appearance or increased prevalence of unforeseen diagnoses. The result is that, after a given period of time this inflexibility limits the use of the knowledge base as it no longer reflects the current characteristics of the population at risk. Given these points, the development of a knowledge base using artificial neural network technology naturally lends itself towards the task of predicting medical diagnosis. In addition, the technology appears to be a promising method for recommending possible treatment

routines. It offers flexible and quick means of designing dynamic expert systems that consider different decision variables in their predictive routines. With its dynamic nature and on-line learning capability, a neural network knowledge base can also be updated with more recent patient data. Thus, once an initial knowledge base has been set up, it can never become obsolete with time. In this way, the system can effectively capture varying ailment trends in a given population while retaining its previous knowledge. One of the most important problems of medical diagnosis, in general, is the subjectivity of the specialist. Human being always makes mistakes and because of his limitation, errors do occur during diagnosis. It has been noted, in particular in pattern recognition activities, that the experience of the professional is closely related to the final diagnosis (Salim, 2004). This is due to the fact that the result does not depend on a systematized solution but on the interpretation of the patient's signals. Brause (2001) pointed out that almost all physicians are confronted during their formation by the task of learning to diagnose. Here, they have to solve the problem of deducing certain diseases or formulating a treatment based on more or less specified observations and past experience. For this task, certain basic difficulties have to be taken into account namely the basis for a valid diagnosis. A sufficient number of experienced cases is reached only in the middle of a physician's career and is therefore not yet present at the end of the academic formation. This is especially true for rare or new diseases where also experienced physicians are in the same situation as newcomers. Principally, humans can recognize patterns or objects very easily but fail when probabilities have to be assigned to observations. Brause (2001) gave, an example of a study in the year 1971 that showed these basic facts in the medical area. His study had shown that humans have many limitations in diagnosis. The results of this experiment were as follows:

- Best human diagnosis (most experienced physician): 79.7%
- Computer with expert data base: 82.2%
- Computer with 600 patient data: 91.1%

This result showed that humans cannot ad hoc analyze complex data without errors.

2. Skin diagnostic analysis and artificial neural network design

Skin diseases are diseases that may originate inside the body and manifest on the skin or start from the skin and manifest on the skin (Buxton, 1991). This session presents the analysis of these diseases. It also presents the structure of an artificial neural network for handling skin diseases diagnosis.

2.1 Analysis of skin disease diagnostic system

The human expert performs the diagnosis of skin diseases by collecting patient records and complaints. This list of patient complaints and observed skin conditions is then expanded into several Boolean symptoms. The symptoms are further subjected to knowledge matching with the knowledge already possessed by the human expert (knowledge base-experience). If there is a match, the doctor recommends the disease as a possible skin disease. In some cases, the human expert may subject the patient to further laboratory tests in order to ascertain the causative agent of the skin condition. The test could serve as a confirmatory test if the disease diagnosed is actually caused by microorganism such as bacteria, warts, virus, fungi, etc. When the human expert is inexperienced or has not come across such skin condition, he uses trial and error to diagnose. This is done by the combination of all the possible conditions, comparing them with known conditions and narrowing the judgment. During this process, learning is said to have taken place if the skin condition is properly diagnosed and treated.

Thus, the human expert depends largely on his experience and the patient complaint interpretation. Fig. 3 illustrates the stages of this analysis.

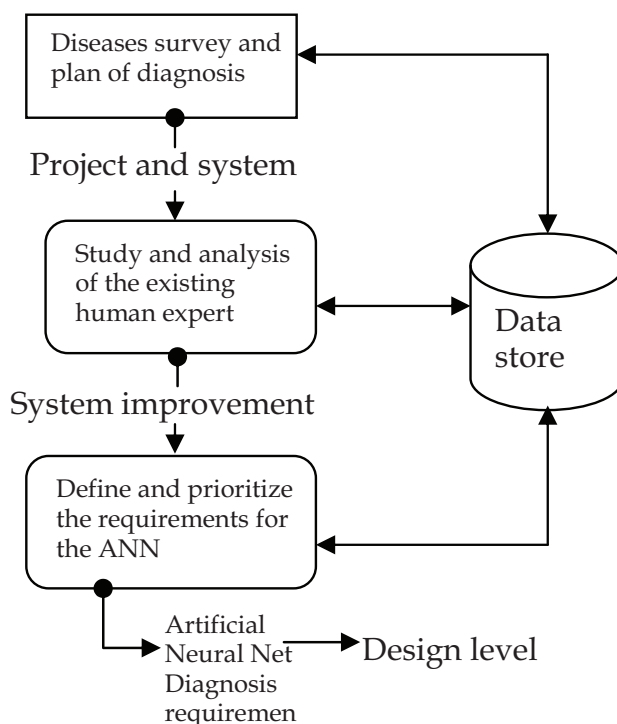


Fig. 3. Skin Disease Diagnostic System Analysis

In fig. 3, disease survey includes the general collection and listing of the skin diseases that is studied and used in the process of system development. This includes complaint, observation and lab test.

2.2 Disease survey and plan of diagnosis

Disease survey includes the general collection and listing of the skin diseases that will be studied and used in the process of development of the Artificial Neural Network for the system. Examples of several skin disease infections are shown in figure 4. Similarly, skin diseases and their symptoms are shown in Table 1.

The National Center for skin diseases Singapore was selected in this study for the symptomatic classification because the center uses state-of-the-art equipment in checking the diseases. It uses skin-scanning machines, which can better classify the skin colour variation due to the disease using sample human races. The particular skin lesion can also be better observed by the machine than by human sight which can be erratic. The center also offers online sample test in their Online Consultation using Camera on the particular skin portion to capture the symptom of a given case that can be passed to our Neural Network system when operational.



Fig. 4. Skin Disease Infections

Diseases	Symptoms/ Causes
Scabies	Tiny bumps, itching, scaly skin, mostly on fingers, caused by bacteria
Acne Vulgaris (pimples)	Spots on faces, small heads of pus, tiny bumps
Impetigo	Bacteria, irregular, spreading sores, shiny, yellow crusts.
Leishmaniasis	Small bump, crusty sore, on face or limbs, bacteria
Atopic Dermatitis	Itchy, dry, hypersensitive skin, hereditary, red rash, scaly, thick skin
Syringoma	Tiny skin coloured growth on eyelid, under- developed sweat glands, enlarged glands.
Benign Skin Tumors	Harmless growth, dark sketch on the skin, soft, rough and greasy surface
Leprosy	Sores, loss of feeling, eat up limbs
Mastitis	Bacterial, warm, painful lump in the breast of a woman
Viral warts	Virus, any body part, peeling o part affected.
Diaper Candidiasis	Fungi, diaper between two legs, maceration from urine, redness, swelling oozing and pus
Folliculities	Bacteria, small packets of pus, small red lumps
Carbuncle	Bacterial, low immune diabetes mellitus, tiny bumps at affected part.
Vitiligo	Colorless on hand, around the mouth or between the legs, net contagious, lighter patches
Eczema	Darker patches, blistering, peeling
Seborrhoeic Dermatitis (Dandruff)	Yellowish, greasy scales on the scalp, red scalp skin, patchy, white dusty scales.
Xanthelasma	Raised yellow patches, on eyelids, associated to high blood fat
Melasma	Brown spots or patches on both faces, upper lips, nose and chin, symmetrical, occur during pregnancy, darkening lesion
Psoriasis	Thick scales on skin, vascular birthmarks, discrete scaly plaques, non-itchy
Seborrhoeic keratosis	Black growth, stuck-on, very dark, people over 40 years, rough surface
Cancer of the skin	Small rings, small pit in the middle, found on temple, nose or neck, red edges, sometime clear in the center.
Urticaria (hives)	Itching, rash, with localized swellings of the skin
Intraepidermal Carcinoma	Solitary lesion growing slowly with sharp ureren outline, with itching, scaling
Lichen Plarus	White striae with ulcers sometimes on oral mucosa itching, blisters
Lichen Simplex	Itching, thickened hyperkeratic epidermis, due to rubbing.
Vasculitis Erythematosis	pulpable lesions, fixed, developing and recurring over days
Lupua Erythematosis	Similar to vasculitis but develops and recurring over weeks and months

Table 1. Skin Diseases and their Symptoms (National Skin Center Singapore) (NSC, 2005)

2.3 Skin disease data input design

The data used for the diagnostic system consist of the following components: Patient vital signs, Patient verbal complaints, Patient demographics and Presence of specific symptoms. The elements of each of the components serve as an input variable to the network. Thus, the task of the artificial neural network is to draw a correlation between the patient's presentation, using self-reported symptoms and vital sign. While patient demographics and vital signs are a key element in providing clues to an ailment, it has been observed that the patients' complaints provide greater insights for predicting medical diagnoses. However some skin conditions have to rely heavily on good scanning machines for an accurate prediction (NSC, 2005). Expanded complaint input variables are shown in Table 2.

Compliant type	Compliant type
Itching	Burous
Pus dripping	Inflamed
Hard inside	White striae
Painless	Sore
Painful	Warm
Rash	Redness
Irritation	Macerate
Swelling	Peeling
Bleeding	Blistering

Table 2. Expanded complaint input variables (*Source, NSC, 2005*)

Table 2, shows a list of input variables created by the expansion of the symptom string. The goal is to create an automated data processing system capable of taking as input a set of these records and using it to arrive at a correct diagnosis. The symptom input variables were created to draw a correlation between a specific complaint and the possible skin disease.

3. Artificial neural network structure for skin disease diagnostic

The design and architecture of ANN selected for the skin diseases is based on the feed - forward network. This means that the artificial neurons are organized in layers, and send their signals forward, (i.e., from input to output) and then the errors are propagated backwards (see, fig. 5).

The network receives input symptoms by neurons in the input layer, and the output of the network is given by the neurons on an output layer. There may be one or more

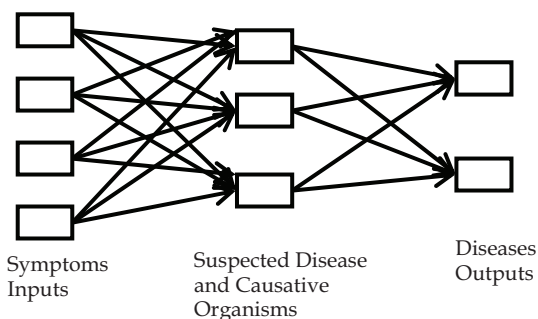


Fig. 5. ANN structure for skin disease

intermediate hidden layers. The backpropagation algorithm uses supervised learning, which means that we provide the algorithm with examples of the inputs and outputs we want the network to compute, and then the error (difference between actual and expected results) is calculated. The idea is to reduce this error, until the ANN learns the training data. The training begins with random weights, and the goal is to adjust them so that the error will be minimal. The activation function of the artificial neurons in ANNs implementing the backpropagation algorithm is given as follows (Haykin, 1999):

$$A_j(\bar{x}, \bar{w}) = \sum_{i=0}^n x_i \cdot w_{ji}. \quad (1)$$

The output function uses the sigmoidal function:

$$O_j(\bar{x}, \bar{w}) = 1/[1 + e^{-A_j(\bar{x}, \bar{w})}] \quad (2)$$

We defined the error function for the output of each neuron as:

$$E_j(\bar{x}, \bar{w}, d) = \Sigma (O_j(\bar{x}, \bar{w}) - d_j)^2 \quad (3)$$

The weights are adjusted using the method of gradient descent:

$$\Delta w_{ji} = -\eta (\partial E / \partial w_{ji}) \quad (4)$$

Where: x_i are the inputs, w_{ji} are the weights, $O_j(\bar{x}, \bar{w})$ are the actual outputs, d_j are the expected outputs and η - learning rate.

For the skin diagnostic system, we defined the following firing rules: take a collection of symptoms for a disease to a node, the presence of which causes it to fire (the 1-taught set of patterns) and the absence prevents it from firing (the 0-taught set). If there is a tie in symptoms, then the ANN remains in the undefined state (1/0). For instance, in scabies, which have 4 - input symptoms, the neuron is taught to fire (i.e., output 1), when the input (tiny bumps- X_1 , itching- X_2 , scaly- X_3 and on fingers- X_4) is 1111 or 0111 or 1101 and will not fire (i.e., output 0) when the input is 0000 or 1010 or 1110. The firing rule is depicted in Table 3.

It is important to note that the decision concerning each firing rule was derived directly from the Scabies condition. This condition specifies that: Scabies occurs only on the hand has to be Itchy. These conditions are weighed higher than other conditions. The values 1111 or 0111 or 1101, implies that all conditions must be met or at least there is either tiny bumps or scaly skin to meet the 50% threshold mark required for firing. This is a very important rule

X1:	0	0	0	0	0	0	0	0
X2:	0	0	0	0	1	1	1	1
X3:	0	0	1	1	0	0	1	1
X4:	0	1	0	1	0	1	0	1
OUTPUT	0	1/0	1/0	1/0	1/0	1/0	1/0	1
X1:	1	1	1	1	1	1	1	1
X2:	0	0	0	0	1	1	1	1
X3:	1	0	1	1	0	0	1	1
X4:	0	1	0	1	0	1	0	1
OUTPUT	1/0	1	0	1/0	1/0	1	0	1

Table 3. Truth table for the skin disease firing rule

formulation trick, which also requires some higher level knowledge of the disease for which rule is formulated. The rule provides a greater degree of flexibility for determining whether scabies is a skin disease or not. This is particularly true for a case where the patient has used cream to make the skin less scaly. Omission of the scaly symptom will not affect the decision of the neural network in determining whether to suspect scabies or not. This is a more realistic situation which definitely makes the neural network more desirable. The firing rule gives the neuron a sense of similarity and enables it to respond 'sensibly' to symptoms not seen during training.

4. System implementation

Once the system algorithm has been specified the coding of the system follows. The coding to a large extent is guided by the algorithm. Since the program is object oriented with classes, its implementation may not necessarily be procedural. The program coding is implemented in c++. Testing of the system was done module by module, code segment by code segment. Once a segment is in good order another segment is merged with it until the module is complete. In the testing, codes that are retested and proven correct were reused for system development. The stubs were tested on individual member functions; the main program, subroutine or subprogram. In this process the lower level modules were simulated to allow the test of the higher level modules. The diagnostic modules for instance were simulated to test the neural network selection of its actions when the selected operation is called in the main program module where the object is instantiated. When the response was proved correct the diagnostic modules itself was fully developed. This testing method made the program to work when they were merged into a single program. After the testing and merging, all the modules that have been coded and stub tested were retested as an integrated single unit.

4.1 Description of implementation data

The data used in the implementation and testing of the Artificial Neural Network were collected from the Olivet Clinic, Port Harcourt and from the National Skin Center for Dermatology. The center offered the access of their recorded data on the dermatological research carried out up to March 2005. The data contain information on various symptoms and their diseases and the various causative organisms. It was accessed on <http://www.nsc.gov.sg/>. The data collected were enormous but selected ones were used in testing the program. The selection was based on the skin diseases more common in Nigeria. The rectification of the commonality of the diseases was carried out at the Olivet Clinic, off Garison junction, Port Harcourt. Olivet is a Hospital dedicated to dermatology in Port Harcourt. Table 1 shows selected skin diseases collected from the net (<http://www.nsc.gov.sg/>). Table 4 shows the selected skin diseases from Olivet Clinic, Port Harcourt.

4.2 Training of the artificial neural network

The learning process proceeds by way of presenting the network with a training set composed of input patterns together with required response pattern. By comparing the actual output with the target output for given pattern, the error is computed using equation (3). The error can then be used to alter the connection strengths between layers in order to achieve a better network's response to the same input pattern in subsequent iterations. In the proposed ANN structure, a weight of 10 was initially assigned to all the symptoms,

Diseases	Symptoms
Scabies	Itching, tiny bumps on skin, scaly skin, finger area
Acne	Spot on the face, Small heads of pus
Impetigo	Spreading sores, shiny skin, yellow crust, infection irregular
Leishmaniasis	Small bumps, crusty sore on the skin, infection in the face or limbs
Atopicderm	Dry skin, hypersensitive skin, hereditary, red rash on the skin, thick skin on the affect spot
Syringoma	Underdeveloped sweat gland, enlarged skin gland, tiny skin coloured growth on eyelid
Benign	Harmless growth on the skin, dark sketch on the skin, rough and greasy skin
Leprosy	Mostly on the limbs, sores and loss of feeling
Mastitis	Warm skin portion, painful breast lumps
Diapercandi	Diapers with oozing, maceration from urine
Vitiligo	Colourless skin around hand , mouth or leg, lighter patches
Eczema	Blistering on the skin, darker patches(or lesion) on the skin, peeling of skin
Dandruff	White dusty scales in the scalp, red skin and patchy
Xanthelasma	Raised yellow patches on eyelids, raised yellow patches associated with blood fat
Lasma	Brown spot on face, brown spot on face occure during pregnancy
Cancer Small ring disease skin	Small ring disease skin, small pit in the middle of disease skin

Table 4. Selected Skin Diseases from Olivet Clinic, Port Harcourt.

while a threshold of 50, was chosen for the first layer. In the next stage a threshold of 70, was chosen and weights were assigned based on some probabilities. The network was trained and the result was not satisfied, although some diseases were diagnosed correctly. Following this, a threshold of 50 was reversed and weights assigned based on its peculiarity (but, all not 10 this time). Finally, a threshold value of 70 was chosen again, while higher weights are assigned based on its exclusiveness. Scabies was chosen as a test case because it has a large number of input symptoms which matches with the ANN structure implemented.

4.3 Results

Table 5, presents the results of testing the system. The result showed cases where suspected and diagnosed based symptoms were supplied to the system. From Table 5, a critical analysis of the results in test 1 to 4, clearly illustrates the effect of weight adjustments in the result of the diagnosis of the skin disease. In test 1, all the symptoms were used and the system confirmed with scabies. In test 2, tiny bumps symptom was dropped and the weight at 50 confirmed that bacteria is present in scabies. This was also true for test 3. However, in test 4 a symptom was dropped and the weight was below 50, indicating that an important symptom has been dropped. Thus the diagnosis indicated that bacteria are the causative organism of scabies.

Test #	Symptoms used	Causative organism	Diseases (Result)
1	Itching, tiny bumps, Scaly, fingers	Bacteria	Scabies (Success)
2	Itching, scaly, fingers	Bacteria	Scabies (Success)
3	Itching, tiny bumps, finger	Bacteria	Scabies (Success)
4	Itching, tiny bumps, scaly	Bacteria	Scabies (Success)
5	Face spot, pus, red	Non	Disease not covered (Fail)
6	Face spot, red	Non	Disease not covered (Success)
7	Tiny bumps, face spot, pus	Non	Acne Vulgar (Success)
8	Tiny bumps, face spot	Bacteria	Acne Vulgar(Fail)
9	Tiny bumps, face spot, pus	Fungi	Acne Vulgar(Success)
10	Harmless growth, dark sketch, rough and greasy skin	Bacteria	Benign (Success)
11	Soft, dark, smooth surface	Fungi	Disease not covered (Success)
12	Dry, thick skin, red rash hypersensitive, hereditary	Non	Atopic Dematitis(Fail)
13	Blister, dark patches, peeling	Fungi	Eczema (Success)
14	White dusty scales, red patchy scap	Fungi	Dandruff (Success)
15	Tiny colored growth on eyelid, underdeveloped sweat gland, enlarged skin gland.	Non	Syringoma(Success)
16	Thick skin on spot, sores and loss of feelings, disease on limbs	Non	Leprosy (Success)
17	Warm Skinny, painful breast lumps, harmless growth on skin	Non	Mastitis (Success)
18	Lighter patches, colorless skin around mouth or legs	Non	Vitiligo (Success)
19	Maceration from urine, diapers with oozing	Non	Diapercandi (Success)
20	Blistering, darker patches, skin peeling	Non	Ecema (Success)

Table 5. Results of ANN system

4.4 Performance analysis of the ANN system

To justify the performance of our diagnostic system, we conducted two analyses. The first is using a general performance scheme. Secondly, we carried out a number of tests at random using various symptom combinations.

A. Performance Benchmark

The proposed neural networks skin disease diagnostic system (NNSDDS) architecture relies on a piece of software for easy skin disease diagnosis. The principles underlying diagnostic software are grounded in classical statistical decision theory. There are two sources that generate inputs to the diagnostic software: disease (H_0) and no disease (H_1). The goal of the diagnostic software is to classify each diagnostic as disease or no disease. Two types of errors can occur in this classification:

- i. Classification of disease as normal (false negative); and
- ii. Classification of a normal as disease (false positive).

We define:

Probability of detection $P_D = \Pr(\text{classify into } H_1 | H_1 \text{ is true})$, or

Probability of false negative = $1 - P_D$.

Probability of false positive $P_F = \Pr(\text{classify into } H_1 | H_0 \text{ is true})$.

Let the numerical values for the no disease (N) and disease (C) follow exponential distributions with parameters λ_N and λ_C , $\lambda_N > \lambda_C$, respectively. Then we can write the probability of detection P_D and probability of false positive P_F as

$$P_D = \int_t^\infty \lambda_C e^{-(\lambda_C x)} dx = e^{-\lambda_C t} \quad (5)$$

$$P_F = \int_t^\infty \lambda_N e^{-(\lambda_N x)} dx = e^{-\lambda_N t} \quad (6)$$

Thus P_D can be expressed as a function of P_C as

$$P_D = P_C^r, \quad (7)$$

Where $r = \lambda_C / \lambda_N$ is between 0 and 1. Consequently, the quality profile of most diagnostic software is characterized by a curve that relates its P_D and P_C , known as the receiver operating curve (ROC) (Trees, 2001). ROC curve is a function that summarizes the possible performances of a diagnostic system. It visualizes the trade - off between false rates and success rates, thus facilitating the choice of a decision functions. Following the work done by Huseyin and Srinivasan, (2004), Fig. 6 shows sample ROC curves for various values of r .

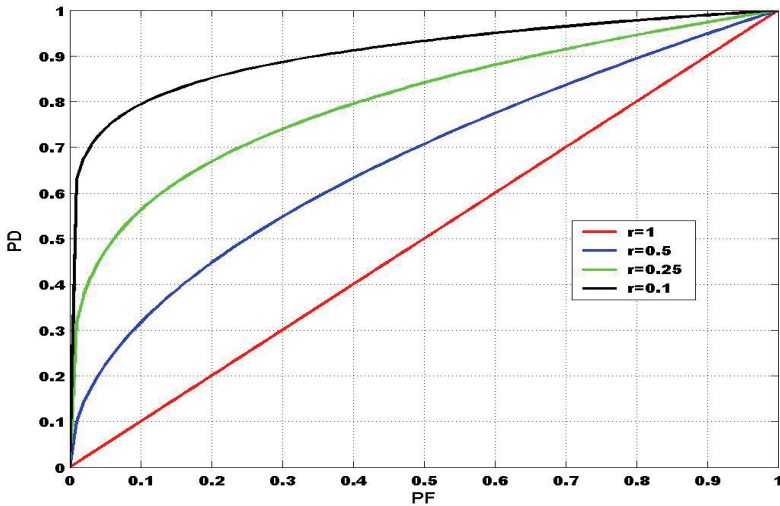


Fig. 6. ROC curves for different r

The smaller the value of r the steeper the ROC curve and hence the better the performance.

The performance analysis of the NNSDDS algorithms was carried out using MATLAB software package (MATLAB[®], 2009) and the results compared with the collected data for scabies, acne and vulgar are as shown in Fig. 7, Fig. 8 and Fig. 9, respectively.

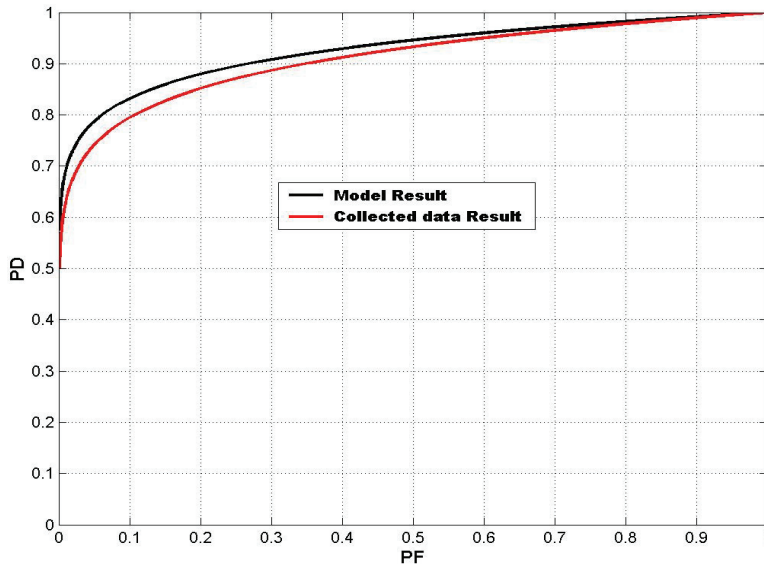


Fig. 7. ROC curves for Scabies diagnosis

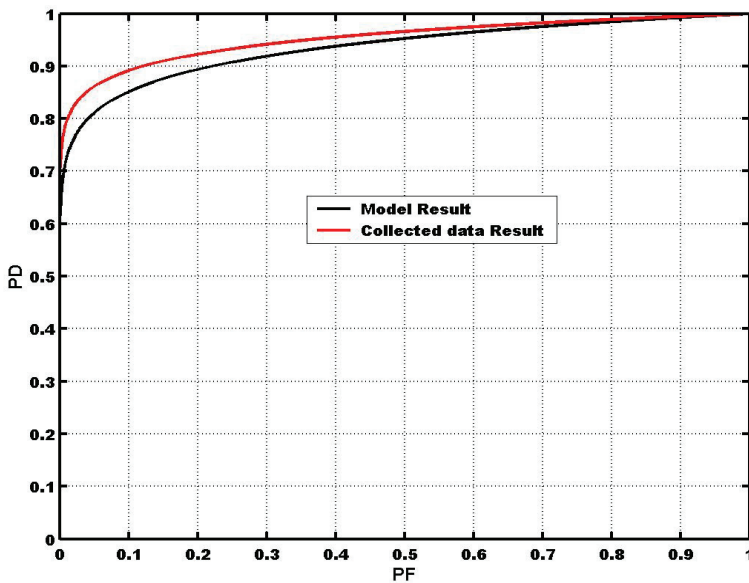


Fig. 8. ROC curves for Acne diagnosis

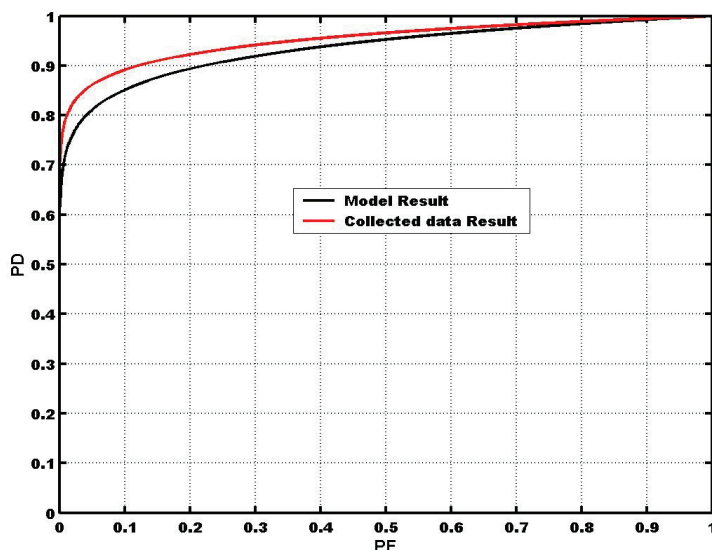


Fig. 9. ROC curves for Vulgar diagnosis

Explanation: In figures 7, 8 and 9, it can be seen that the model results compared satisfactorily well with the collected data results for all cases examined in the paper.

B. Overall Success Rating of the NNSDD System

In testing the NNSDDS, twenty different tests were carried out at random using various symptom combinations and the results matched with the expected result of the NNSDDS. Where there was a match, success was recorded. In situations where there was no match failure were recorded. The total number of success = 18. Total number of failure = 2. Total number of test was 20.

$$\text{Percentage Success} = \frac{18}{20} \times 100 = 90 \%$$

$$\text{Percentage Failure} = \frac{2}{20} \times 100 = 10 \%$$

5. Conclusion

The paper presented a framework for diagnosing skin diseases using artificial neural networks. The proposed system was able to achieve a high level of success using the artificial neural network technique. A success rate of 90% was achieved. This infers that ANN technique is an effective and efficient method for implementing diagnostic problems. The features of the ANN provided learning capability, which makes the system opened ended to new disease conditions or variation of a known skin disease due to the mutation of the causative organism. This provides great flexibility in the diagnostic system and makes the system to be opened ended. With this flexibility in the system, the level of coverage of skin conditions by the diagnostic system is limitless. This makes the application to be useful in many conditions even in unforeseen instances.

6. Recommendation

This work is recommended to human experts and dermatologist who specializes in diagnosing and treatment of skin and related diseases. The human experts will find it useful as an aid in the decision making process and confirmation of suspected cases. Also, a non-expert will still find the work useful in areas where prompt and swift actions are required for the diagnosis of a given skin diseases listed in the system. Medical practitioners who operate in areas where there are no specialist (dermatologist) can also rely on the system for assistance. The skin diseases covered include scabies, Acne, Vulgari, Impetigo, Lishmaniasis, Atopic Dermatitis, Syringoma, Benign, Skin Tumour, Leprosy, and Diaper Candidiasis. Others are Folliculities, Soborrhocic Dermatitis, Xantherlasma, Malasma, Urtticaria and Ichen simplex. All these diseases can be handled using the system developed in this research. The research work can also act as a pedestal for the advancement of research in neural network applications in medical diagnostic researches. The work is also recommended for developers of object-oriented system and decision-support systems.

7. References

- Bakpo, F. S. (2008). Application of Artificial Neural Networks in Detection of Financial Crimes in Nigeria, *PhD thesis*, Department of Computer Engineering, Enugu State, University of Science & Technology (ESUT), Agbani.
- Brause, R. W. (2001). Medical Analysis and Diagnosis by Neural networks. *Proceedings. Of IEEE International Conference On Neural Networks*, vol II, pp. 481 – 472, Computer Science Department, Frankfurt, Germany..
- Buxton, P. K. (1991). ABC of Dermatology. Jolly & Barber Ltd, Great Britain. Pp 1-30.
- DeCegama, A.L. (1989). The Technology of Parallel Processing: Parallel Processing Architectures and VLSI hardware. Prentice- Hall International, Inc., vol. 1, p478.
- Eugene, D.; Bubak, R.; & Derek, Z. (1997). Diagnosis Prediction using an Artificial Neural Networks Knowledge Base, Centre for Optimization and Semantic Control. Department of System Science and Mathematics. Washington University in St Louis. <http://www.cosc.Wustle.edu/mediinfo/html>
- Haykin, S. (1994). *Neural Networks: a comprehensive Foundation*. Macmillian College Publishing Company, New York. 696p.
- Haykin, S. (1999). *Neural Networks: a comprehensive foundation*. Second Edition. Prentice Hall. 842p.
- Hebb, D. O. (1949). *The Organization of Behavior: A Neuropsychological Theory*. Wiley, New York.
- Huseyin, C & Srinivasan, R. (2004). Configuration of detection software: a comparison of decision and game theory approaches. *Decision Analysis*, vol. 1, No. 3, pp131-148.
- Klerfors, D. (1998). Artificial Neural Networks. *Annals of Internal Medicine*, vol. 104, pp 410 – 471
- Salim, N. (2004). Medical Diagnosis Using Neural Network, Generation5article.
- MATLABR (2008). The MATLAB compiler user's guide. [In Mathworks Handbook; Math works.]
- National Skin Center (NSC). (2005). Singapore, available: www.nsc.gov.sg sponsored by UNESCO/US Foreign AID. <http://www.nsc.gov.sg/>
- Trees, H. V. (2001). *Detection, Estimation and Modulation Theory –Part I*. John Wiley, New York.

Artificial Neural Networks Used to Study the Evolution of the Multiple Sclerosis

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1. Introduction

The multiple sclerosis (ME) is an illness degenerative chronicle of the central nervous system. At the moment is considered that it doesn't have cure although exists effective medication and the investigation on their causes is an active field of investigation.

The objective of this article is to develop a model more precise, based on the use of the Artificial Neural network (ANN), to study the evolution of the illness in the patients with EM.

2. Important

This article presents a model to study the evolution of the illness ME using ANN. The implementation and validation of the pattern was carried out in the programs of Biomedical Engineering and of Systems of the ITM.

The article is organized in the following way: the first chapter presents an introduction to the ME, studying the possible forms of the evolution of the illness. In second chapter exposes the model of the evolution of the illness using ANN and the validation tests. The article concludes showing the main conclusions of this investigation.

3. Information

3.1 The multiple sclerosis

A special study on the ME is beyond the environment of this unit. In ((NATIONAL INSTITUTE OF NEUROLOGICAL DISORDERS AND STROKEN, 2010) offers an excellent treatment with specific references. In this section a brief revision is presented with the purpose of providing a basic knowledge on the matter.

The ME is an illness degenerative and chronicle of the central nervous system. At the moment is considered that it doesn't have cure although exists effective medication and the search on their exact causes are ignored and is a field of active investigation.

The ME can present a series of symptoms that appear in buds or that they progress slowly along the time. They are distinguished several subtypes of multiple sclerosis and many affected present forms different from the illness with the step of the time...

Because of their effects on the central nervous system, it can have as consequence a reduced mobility and disability in the most severe cases. Fifteen years after the appearance of the

first symptoms, if it is not treated, at least the patients' 50% conserves a high degree of mobility. Less than 10% of the sick persons they die because of the consequences of the multiple sclerosis or of their complications.

It is, after the epilepsy, the most frequent neurological illness among the young adults and the most frequent cause in paralysis in the western countries. It affects approximately at 1 of each 1000 people, in particular to the women. It is presented when the patients have between 20 and 40 years. (Noseworthy et to the., 2000; Rivera 2000).

Description. The ME is characterized by two phenomena:

Appearance of focuses spread in the brain and partially also in the spinal marrow caused by the attack of the system immune against the sheath of myelin of the nerves, like is illustrated in the figure 1.

The neurons, and especially their axons is damaged by diverse mechanisms.

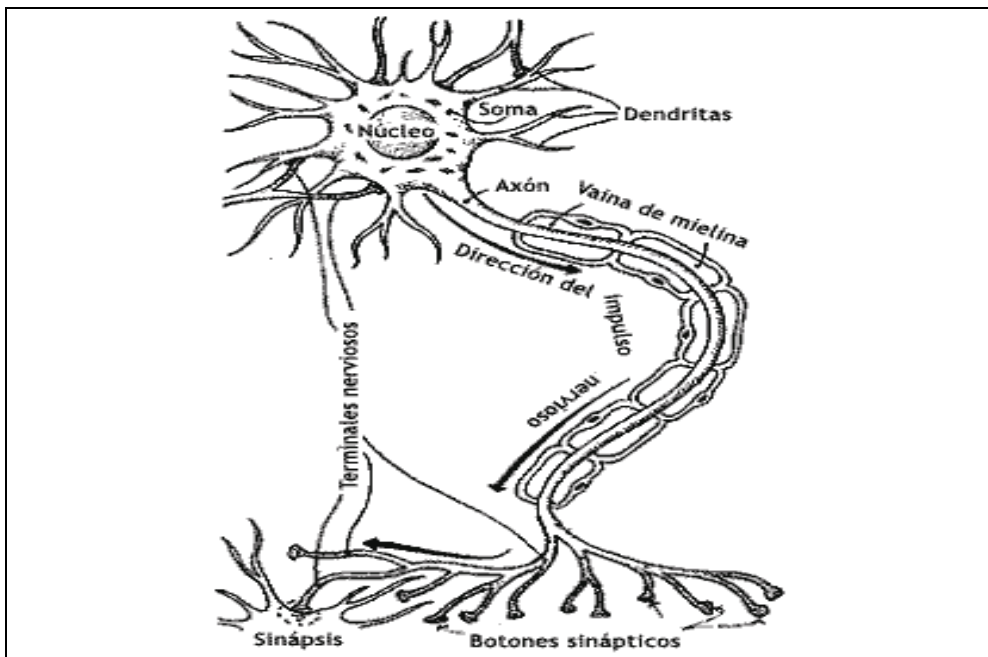


Fig. 1. Myelin

As a result, the neurons of the brain lose partial or totally their transmission capacity, causing the typical symptoms of drowsiness, tickling, spasms, paralysis, tires and alterations in the view.

In the variant Remittent-Recurrent has also been detected in the nervous tissue, or cut of the axons of the neurons, that makes permanent sequels.

Treatment. Cure doesn't exist for the ME. However we have been several medications that are effective in their treatment (Gutiérrez-Alvares et to the., 2001), braking the development of the illness and combatting the symptoms.

The remittent-recurrent variant only has treatments approved by the FDA (Administration of Foods, for its initials in English - Food and Drug Administration) and the EMEA

(European Agency of medications - European Medicines Agency). At the moment, they are three interferons (Blackish et al., 2003) (Avonex, -well-known Betaseron in Europe like Betaferon (The IFNB, 1993; Paty et al., 1993, Jacobs et al., 1996, (Yong et al., 1998) and Rebif (PRISMS, 1998; PRISMS, 2001), a group of called polipéptidos Copaxone, a called immunosuppressive Mitoxantrone and finally an antibody called monoclonal Natalizumab and marketed as Tysabri. The clinical rehearsals (Freedman, 1998; Durelli et al., 2002; Panitch et al., 2002; Clanet et al., 2002) on the different medications they demonstrate that they are not comparable because they correspond to different study designs.

The primary progressive ME is very difficult of trying. The corticosteroids to high dose every three months can have some effect. In principle a treatment preventive cash doesn't exist for the primary progressive ME. The treatment of the symptoms, and the rehabilitation by means of physiotherapy, occupational therapy, have an important paper. It is very important, equally, the evaluation on the part of a neuropsychologist to be able to approach any deficit cognitive that could be established.

Epidemiology. It is believed that MS occurs when a combination of environmental factors in genetically predisposed to buy it.

Environmental factors. In northern Europe, North America and Australasia continental one in 1,000 people suffers from MS. In Central Europe is the inflammatory disease most common central nervous system. In contrast, in the Arabian Peninsula, Asia, Central and South America continental frequency is much lower. Sub-Saharan Africa is extremely rare. With notable exceptions, there is a north-south gradient in the northern hemisphere and south-north in the southern hemisphere, with lower frequencies in the equatorial areas. The climate, diet, geomagnetism, toxins, sunlight, genetic factors and infectious diseases have been proposed as possible causes of these regional differences. It has been postulated that some environmental factor in childhood may play an important role in the development of MS in adult life.

Genetic factors. MS occurs mainly in Caucasians. It is 20 times less common among Canadian Inuit than among other Canadians living in the same region. It is also rare among American Indian tribes of North America, Australian Aborigines and the Maori of New Zealand. These examples indicate that genetics plays an important role in the development of the disease. (Noseworthy et al., 2000) MS is not hereditary. However, the disease is influenced by the genetic constitution of the individual and has been shown that there are genes that are associated with an increased risk of contracting the disease. These genes, which are being studied, are not sufficient to diagnose the disease.

Diagnosis. The diagnosis of MS is complex (Thompson et al., 2000; Poser et al., 1983, Fazekas et al., 1988; Offenbacher et al., 1993; Paty et al., 1988). It requires evidence of a dissemination of lesions over time and space in the CNS. That means that not only must have at least two separate injuries verifiable by clinical symptoms or MRI, in addition there must be evidence of new symptoms or injuries in a range of 30 days. (McDonald et al., 2001)

The conductivity studies of the optic nerve, sensory and motor also provide evidence of the existence of the disease, since the process of demyelination involves a reduction of conduction velocity of nerve signals. The study was done by comparing the reaction times with preset measurements.

Symptoms. The central nervous system lesions that cause MS does not always manifest as clinical symptoms directly detectable and clearly attributable to the disease, which sometimes tends to minimize the first signs. However, the origin of MS is present and begins to progress.

Although in some instances at the beginning of MS accumulates little disability and quality of life is not too concerned, the reality is that the substrate of the disease is developing. There is abundant clinical and scientific evidence indicate that what happens in the early stages of MS, largely depends on its further development. In other words, injuries today in the central nervous system are the cause of disability in the morning, so that if not prevented now, tomorrow will be too late for recovery. MS detection is key as soon as possible to act in time.

It has been shown that early treatment significantly reduced the number of shoots and their intensity. Affected individuals may exhibit a wide range of symptoms, but differ widely from each other, both in the type of symptoms and in their degree. In principle, can be classified according to the area affected the nervous system: derived from optic nerve damage, resulting from damage to the spinal cord (in particular those related to mobility are of this type) and derived from brain damage.

The following are the most common: asthenia (fatigue), muscle loss, muscle weakness, uncoordinated movements, dysphagia (trouble swallowing), dysarthria (speech problems), respiratory failure, dyspnea (breathing problems) spasticity (muscle stiffness), muscle spasms, cramps, muscle twitching (small but widespread muscle vibration), sexual dysfunction, vision problems (loss, double vision, nystagmus), cognitive problems (difficulty performing simultaneous tasks, to follow instructions, loss of short-term memory, depression), emotional lability (inappropriate laughing and crying without psychological effects), constipation secondary to immobility.

Types of MS: In most cases sclerosis begins with acute onset of symptoms in a space that varies from hours to days, usually called a flare, attack or episode. Later speaking of relapse. The first symptom is often the optic neuritis, an inflammation of the optic nerve that causes impaired vision and pain when moving the eye. However, not all patients with optic neuritis develop MS. Sensory disturbances such as numbness or tingling are also common early symptoms. In principle, MS can start with any of the symptoms associated with the disease.

Benign MS. In benign MS cases after one or two attacks, the recovery is complete. The disease worsens over time and usually less severe symptoms. These cases are identified only when it is a small permanent disability after 10 or 15 years after the first attack, which was identified at the time as relapsing MS. In this case, loss of vision or sensory symptoms (numbness or tingling) as initial symptoms are signs of a benign prognosis. Disturbances in gait and fatigue are signs of a negative prognosis.

Relapsing-remitting MS. Especially in the early stages of the disease, the symptoms diminish or disappear spontaneously over a period that can last from days to months. This type of course is called relapsing-remitting. New relapses may occur within weeks or several years and are unpredictable. These relapses may include the above symptoms and / or new ones. However, MRI studies show that nerve damage in these patients can continue even when symptoms have subsided. It has been known long ago that MS never sleeps, so the importance of preventive treatment is great. Many patients remain at this stage the rest of their life.

Secondary progressive MS. In many cases, the disease changes after several years and symptoms begin to slowly progress with or without relapses tax. It is not known yet but its etiology.

Primary progressive MS. 10% of all affected individuals have a chronic progress from the beginning without remission. It is called primary progressive and often appears with leg weakness and alterations in gait and urinary bladder. They appear to be degenerative and

inflammatory processes that play a dominant role in this type. In cases where the primary progressive form of relapse is usually imposed on talk of progressive relapsing. Other forms of multiple sclerosis, to be for many different diseases, which are grouped under the collective name of "border forms" of multiple sclerosis. (Noseworthy et al., 2000; Rivera, 2003)

An overview of different types of MS, is shown in Figure 2.

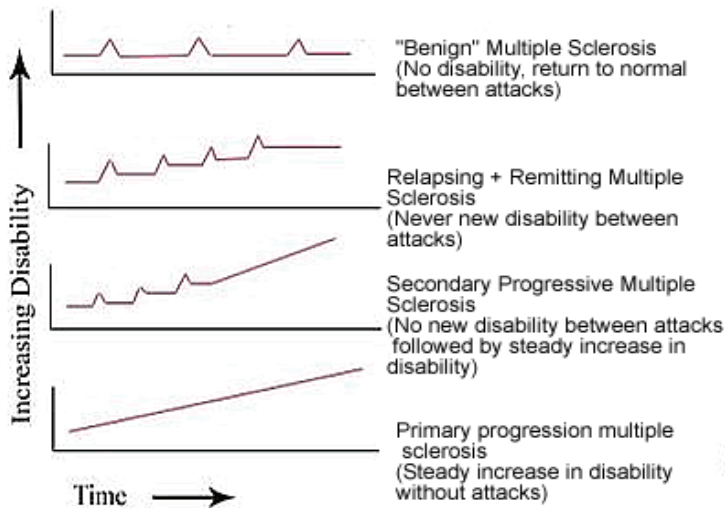


Fig. 2. Different types of evolution of MS

Demyelination. In health, there is a barrier between the central nervous system and the blood called blood-brain barrier, which consists of endothelial cells lining the blood vessel walls.

For unknown reasons, in patients with amyotrophic this barrier does not function well, and autoreactive T cells cross it.

Since that time, an inflammatory process appears. The inflammation is provided by other immune cells and soluble factors, such as cytokine and antibody. Because of this abnormal behavior of the immune system, MS is considered an autoimmune disease.

Widely accepted is that a particular subtype of lymphocytes, called CD4 cells, Th1-T, have a key role in the development of the disease. Under normal circumstances, these lymphocytes can distinguish between own and other cells. In a person with MS, however, the cells recognize healthy parts of the central system as foreign and attack them as they would a virus. In MS, the party attacked is myelin, a fatty substance that covers the axons of nerve cells and is important for proper nerve transmission. (Hafler et al, 1989)

The inflammation eventually leads to the opening of the blood-brain barrier, which can lead to problems such as edema. It also causes the activation of macrophages, metalloproteinases and other proteases and cytokines. Eventually lead to the destruction of myelin, a process called demyelination.

Remyelination. Original Oligodendrocytes form the myelin sheath are not able to recreate the cover once it is destroyed. However, the brain is able to recruit stem cells that migrate from other unknown regions of the brain, differentiate into mature oligodendrocytes and

recreate the myelin sheath. This new cover is often not as thick or effective as the original and repeated attacks remielinizaciones reaction will become less effective (Hauser et al., 1986, Trapp et al, 1998), until a plaque around the axons damaged (Tintoré et al, 2001; Tintoré et al, 2003), as illustrated in Figure 3.

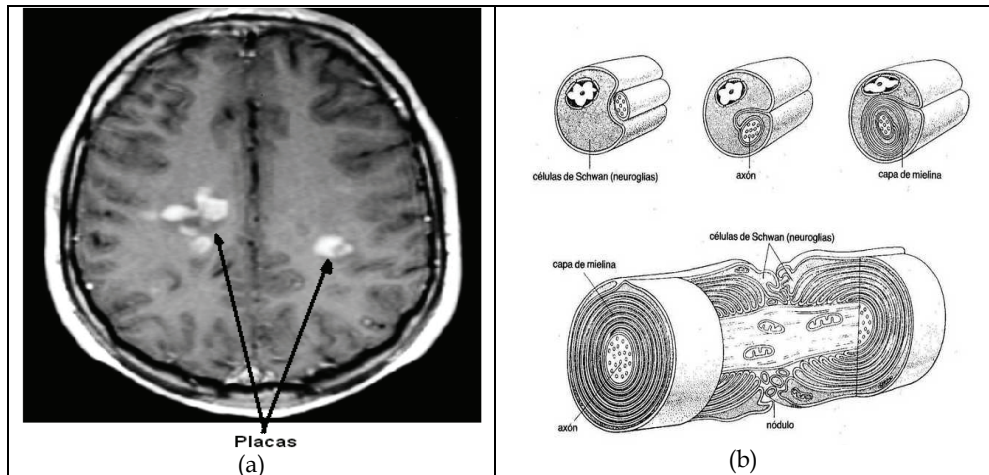


Fig. 3. Plaques (a) by the degeneration of myelin (b).

Remyelination is one of the reasons why, especially in the early stages of the disease, symptoms tend to diminish or disappear after days to months.

3.2 Study case

Respecting the confidentiality of a patient's medical information and confidentiality within ethical and legal framework in force, and himself as to the custody of the records of Magnetic Resonance, and not be subject to experimental research without their fully informed and free consent, is presented below for anonymous study of a patient diagnosed with MS since 2003, in control, receiving Betaferon Ampoules applied every 1.5 days. Her medical history (from the onset, January 2003 to March 2010), the number of relapses and MRI examinations performed, are illustrated in Figure 4.

The results of the MRI examinations are as illustrated below.

MR1 (June 2003). CERVICAL SPINE.

Symptoms: Patient 28, female, with three-month history of pain in the cervical region radiating to the upper limbs, back and neck.

Conclusion: One is transverse myelitis at the level of C2-C3 and C3-C4, involving mainly the left hemicord. Twelve months after the onset of the disease, the patient fully recovered with no sequelae.

MR2 (January 2006). BRAIN AND CERVICAL SPINE.

Symptoms: lower limb anesthesia.

Conclusion: EM Process signed demyelinating type of activity. Probable secondary left hippocampal sclerosis. Twelve months after the start of the second relapse, the patient is left with permanent recovery (lower extremities are slightly grassy).

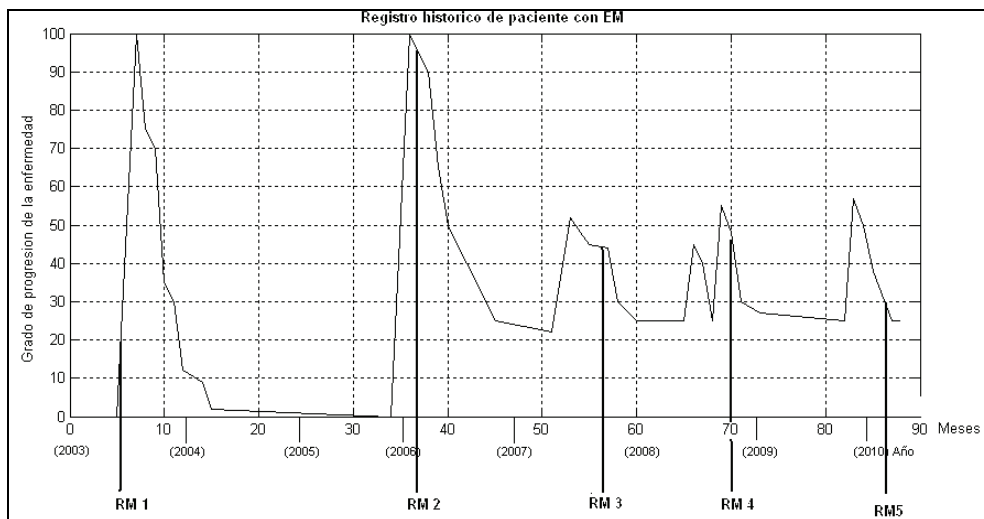


Fig. 4. Historical record of a patient with MS

MR3 (September 2007). BRAIN AND CERVICAL SPINE.

Symptoms: lower limb paresthesias.

Conclusion: Board demyelinating left side up to C3. Described small central disc protrusion at C3-C4 and C5-C6. MS stable with respect to previous control in January 2006.

MR4 (October 2008). CEREBRAL

Symptoms: lower limb paresthesias, twitching.

Conclusion: Injuries globular and plate engaging the white matter, periventricular, predominantly periatral, there are also other injury involving the left cerebral pendulum. These alterations are stable with respect to the cranial MRI September 2007, without showing enhancement postcontrast no increase in their sizes.

MR5 (March 2010). CEREBRAL

Symptoms: lower limb paresthesias, twitching.

Conclusion: Patients with MS DX unchanged compared to the 2008 study. There is emerging injury and alterations in the stable known time interval. Satisfactory.

Below in Table 1 summarizes the patient's history with MS.

Episode	Extent of disease progression.	Improvement achieved by the end of the episode.	Time between episodes
1 (Jun. 2003)	100%	100%	
2 (Ene. 2006)	100%	85%	30 meses
3 (Sep. 2007)	50%	85%	20 meses
4 (Oct. 2008)	50%	80%	13 meses
5 (Mar. 2010)	50%	80%	12 meses

Tabla 1. Registro histórico de recaídas.

3.3 Modeling the historical record of the disease ms with ANN.

Theory and modeling of the ANN is inspired by the structure and functioning of nervous systems, where the neuron is the fundamental element. There are neurons of different shapes, sizes and lengths, and attributes that determine their function and utility.

A detailed study of the ANN is beyond the scope of this unit. In [Row, 2000; Martin del Brio, 2002] offers an excellent study with specific references.

To solve the approximation problem posed in Figure 4, the use of a heuristic approach, based on intuition and experimentation to select the topology of the ANN. This will have an input neuron (time measured in months), one output neuron (Measurement of relapse) and two hidden layers, each with 15 neurons, as illustrated below.

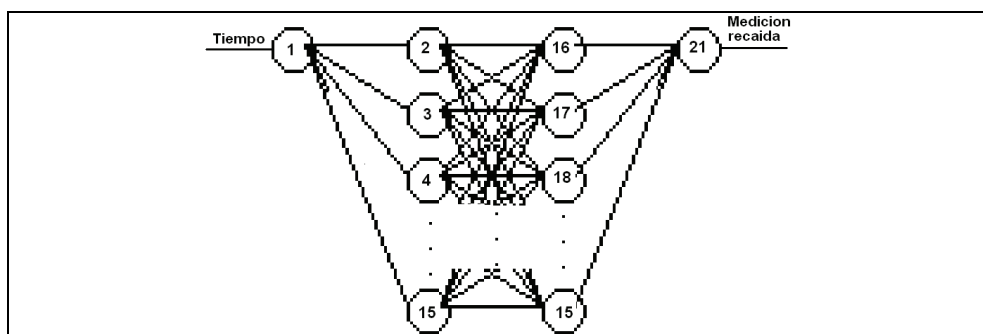


Fig. 5. Architecture of RNA used

The model validation tests were performed using the Neural Network Toolbox for MATLAB ®. The weights of the network is randomly initialized once. We use the training algorithm for second order Trainlm be considered the fastest, with a maximum 3000 iterations and a final error in the approximation of 0.01

Figure 6 shows the curve Approximation to the historical series of relapses That Had the patient with MS in the Period from January 2003 to March 2010. As Shown, the ANN Properly mapping the time series. Note That the final error in the Approximation Was 0.01, Which Is Considered a good measure.

4. Conclusions

MS is a chronic inflammatory demyelinating disease of the central nervous system that causes progressive disability of the individual. After an initial clinical syndrome develops the form of the disease relapsing-remitting (RRMS) that leads to a secondary phase of progressive disability. The goal of treatment is to reduce the frequency and severity of relapse and delay the onset of the secondary phase.

There are currently no established clinical evidence to allow a prognosis or to decide a therapeutic response to MS, although there is promising research that need to be confirmed, as the detection of anti-MOG antibodies (serum antibody against myelin glycoprotein of oligodendrocytes, Myelin oligodendrocyte glycoprotein) and anti-MBP (antibody against myelin basic protein, Myelin basic protein) as predictors of progression to disease introduced (Tintoré et al 2003). The uncertainty is one of the psychological aspects are more difficult to carry in MS.

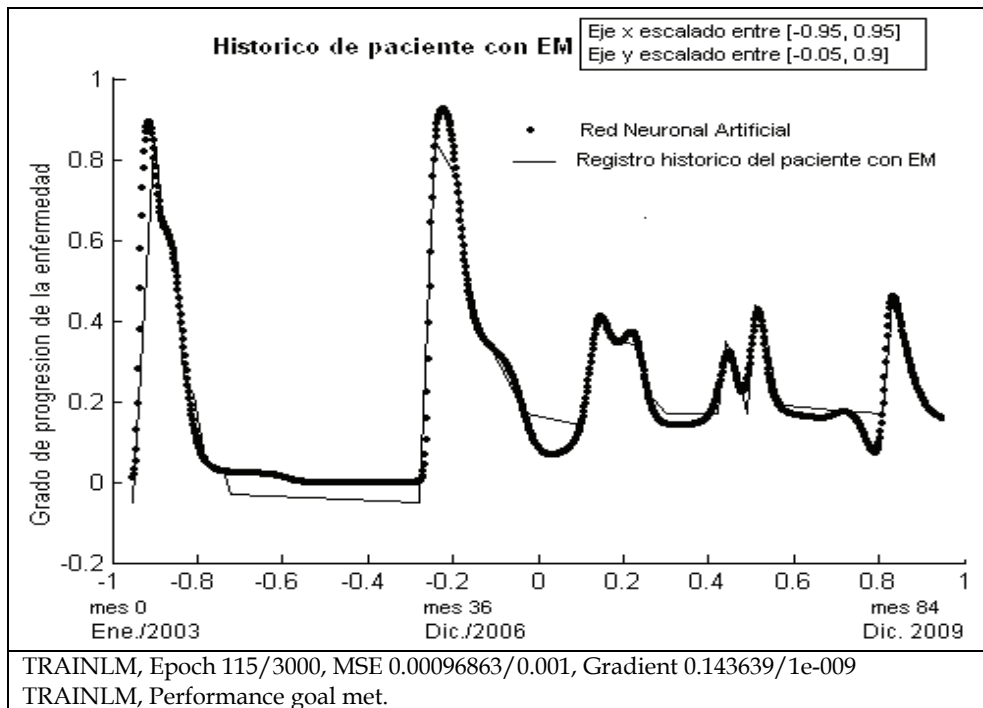


Fig. 6. Diagnosis of MS disease progression by using RNA.

The aim of this paper is to develop a more accurate model based on the use of Artificial Neural Networks, to study the evolution of the disease in patients with MS. The final error obtained in approximating the time series of relapses was 0.01, which is considered a good measure.

The method proposed to date to predict the efficiency of the treatment of relapsing-remitting MS is by clinical criteria and monitoring of active lesions by MRI examinations. As a way of contributing to the reliability of diagnosis and treatment efficiency is proposed in this paper to map the time series records of relapse using a ANN. Its use helps the doctor evaluate, more agile, the variation of the disease and its impact in the medium and long term.

Furthermore, long known that MS never sleeps, so the importance of preventive treatment is great. Many patients remain at this stage the rest of their life. In this case, using the model proposed in this paper helps the physician to improve their diagnoses in terms of disease trends. Likewise, the ANN used to evaluate the response has been the patient to the treatment of this disease, coupled with the clinical and MRI outcomes. With the ANN could be predicted more accurately in the first months of therapy if the patient responds or not to therapy.

Moreover, the drugs used in the treatment of MS are only "partially effective" (Anderson, 1997). Therefore, physicians need to evaluate their response by MRI, and clinical criteria that include measuring the frequency of relapses and measuring the degree of disability progression. In this case, the use of an ANN, helps the doctor to study the variation of the

disease and raise more swiftly changes when treatment is not effective before they produce a greater neurological damage.

The diagnosis of MS using ANN during treatment to detect changes in the state of neurological inflammation of the sick and to assess the effect of different drugs on injuries. The effectiveness of the diagnosis in ANN depend on the correct measurement of the frequency of relapses and the degree of progression of the disease.

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6. References

- Anderson PB, Waubant E, Goodkin D, (1997), How should we proceed with disease-modifying treatments for multiple sclerosis? *Lancet* 349:586-7.
- Clanet M, Radue EW, Kappos L, Hartung HP, Hohlfeld R. (2002) The European Interferon β -1a (Avonex) Dose-comparison Study Investigators, et al. A randomised, double-blind, dose-comparison study of weekly interferon α -1a in relapsing MS. *Neurology* 1507-1517.
- Durelli L, Verdun E, Barbero P, Bergui M, Versino E. (2002) The independent comparison of interferon (INCOMIN) Trial Study Group, et al. Every-other-day interferon beta-1b versus once-weekly interferon beta-1a for multiple sclerosis:results of a 2 year prospective randomized multicentre study (INCOMIN). *Lancet* 359:1453-1460.
- Fazekas F, Offenbacher H, Fuchs S, Schmidt R, Niederkorn K, Horner S, et. al. (1988) Criteria for an increased specificity of MRI interpretation in elderly subjects with suspected multiple sclerosis. *Neurology* 38:1822-5.
- Freedman MS (1998). Once-weekly interferon for MS (OWIMS) Study Group. Dose - dependent clinical and magnetic resonance imaging efficacy of interferon beta-1a (Rebif) in multiple sclerosis. *Ann Neurol* 44:992.
- Gutiérrez-Alvarez AM, González-Silva J, López-Forero P, Ojeda-Moncayo E, Sánchez-Múnera J, Toro-Gómez J, et.al. (2001) Esclerosis Múltiple. En: Zurek R, editor. Consensos en Neurología: guías de práctica clínica. Bogotá, Colombia: Exlibris Editores; 1a edición. p.1-13.
- Hafler DA, Weiner HL. (1989). MS a CNS systemic autoimmune disease. *Immunol Today* 10:104-107.
- Hauser SL, Bhan AK, Gilles F, Kemp M, Kerr C, Weiner HL. (1986) Immunohistochemical analysis of the cellular infiltrate in multiple sclerosis lesions. *Ann Neurol* 19:578-587.
- Hilera J. (2000), "Redes Neuronales Artificiales. Fundamentos, modelos y aplicaciones". Ed. Alfa-Omega Madrid. p.p 132-153.
- Jacobs LD, Cookfair DL, Rudick RA, Herndon RM, Richert JR, Salazar AM, et al. (1996) Intramuscular interferon β -1a for disease progression in relapsing multiple sclerosis. *Ann Neurol* 39:285-94.
- Martin del Brio, B (2002), "Redes Neuronales y Sistemas Difusos", Ed Alfa-Omega.

- McDonald WI, (2001) Compston A, Edan G, Goodkin D, Hartung HP, Lublin FD, *et. al.* Recommended diagnostic criteria for multiple sclerosis: guidelines from the International Panel on the diagnosis of multiple sclerosis. *Ann Neurol* 50:121-7.
- NATIONAL INSTITUTE OF NEUROLOGICAL DISORDERS AND STROKEN. "Esclerosis Multiple: Esperanza en la investigación". [En línea].
http://espanol.ninds.nih.gov/trastornos/esclerosis_multiple.htm#toc. [Citado el 9 de Marzo de 2010].
- Noseworthy JH, Lucchinetti C, Rodriguez M, Weinshenker BG. (2000) Multiple Sclerosis. *N Engl J Med* 343:938-52.
- Offenbacher H, Fazekas F, Schmidt R, Freidl W, Flooh E, Payer F, *et. al.* (1993) Assessment of MRI criteria for a diagnosis of MS. *Neurology* 43(5):905-9.
- Panitch H, Goodin DS, Francis G, Chang P, Coyle PK, O'Connor P, *et. al.* (2002) for the EVIDENCE (Evidence of interferon dose-response: European North American Comparative Efficacy) Study Group and the University of British Columbia MS/MRI research group. Randomised, comparative study of interferon beta-1a treatment regimen in MS: the EVIDENCE trial. *Neurology* 59:1496-1506
- Paty DW, Oger JJ, Kastrukoff LF, Hashimoto SA, Hooge JP, Eisen AA, *et. al.* (1988) MRI in the diagnosis of MS: a prospective study with comparison of clinical evaluation, evoked potentials, oligoclonal banding, and CT. *Neurology* 38(2):180-5.
- Paty DW, Li DKB, UBC MS/MRI Study Group, (1993) The IFNB Multiple Sclerosis Study Group. Interferon β -1b is effective in relapsing-remitting multiple sclerosis: II. MRI analysis results of a multicenter, randomised, double-blind, placebo-controlled trial. *Neurology* 43:662-667.
- Poser CM, Paty DW, Schleinberg L, McDonald WI, Davis FA, Ebers GC, *et. al.* (1983) New diagnostic criteria for multiple sclerosis: guidelines for research protocols *Ann Neurol* 13:227-231.
- Prieto JM, Lema M. (2003) Interferón α en la esclerosis múltiple. *Rev Neurol* 36:980-90.
- PRISMS Study Group. (1998) Randomized double-blind, placebo-controlled study of interferon β 1a in relapsing/remitting multiple sclerosis. *Lancet* 352:1498-1504.
- The PRISMS Study Group and the University of British Columbia MS/MRI Analysis Group. (2001) PRISMS-4: long term efficacy of interferon β -1a in relapsing MS. *Neurology* 56:1628-1636.
- Rivera VM (2003). Decisión del tratamiento en la esclerosis múltiple. *Rev Neurol* 36:80-85.
- Tintoré M, Rovira A, Brieva L, Grive E, Jardi R, Borrás C, *et. al.* (2001) Isolated demyelinating syndromes: comparison of CSF oligoclonal bands and different MR imaging criteria to predict conversion to CDMS. *Mult Scler* 7:359-63.
- Tintoré M, Rovira A, Rio J, Nos C, Grive E, Sastre-Garriga J, *et. al.* (2003) New diagnostic criteria for multiple sclerosis: application in first demyelinating episode. *Neurology* 60:27-30.
- Thompson AJ, Montalban X, Barkhof F, Brochet B, Filippi M, Miller DH, *et. al.* (2000) Diagnostic criteria for multiple sclerosis: guidelines for research protocols. *Ann Neurol* 47:831-835.
- Trapp BD, Peterson J, Ransohoff RM, Rudick R, Mörk S, Bö L. (1998) Axonal transection in the lesions of multiple sclerosis. *N Engl J Med* 338:278-85.

- The IFNB Multiple Sclerosis Study Group (1993). Interferon β -1b is effective in relapsing-remitting multiple sclerosis: I. Clinical results of a multicenter, randomized, double-blind, placebo-controlled trial. *Neurology* 43:655-61
- The IFNB Multiple Sclerosis Study Group and the University of British Columbia MS/MRI. (1995) Analysis Group Interferon β -1b in the treatment of multiple sclerosis: final outcome of the randomised controlled trial. *Neurology* 45:1277-85.
- Yong VW, Chabot S, Stuve O, Williams G. (1998) Interferon beta in the treatment of multiple sclerosis Mechanism of action. *Neurology* 51:682-689.

Estimation the Depth of Anesthesia by the Use of Artificial Neural Network

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1. Introduction

Anesthesia is used in surgery to minimize pain, shock, and discomfort for surgical patients. There are several types of anesthesia which can be used depending on the needs of the surgery: general, local, regional, and conscious sedation (Jones, 1994). When anesthesia works as expected, the patient feels no pain during a procedure, and often does not remember the proceedings either. Anesthesia increases patient comfort, which can in turn decrease recovery times. With the knowledge that they are not inflicting pain, it also makes it easier for a medical staff to work. When anesthesia comes to mind, most people think of general anesthesia. General anesthesia is a complete loss of consciousness in the patient accomplished through a combination of injected and inhaled drugs (Rampil, 1998). This type of anesthesia is often used for highly invasive surgeries or cases when total relaxation of the patient is required. General anesthesia carries the most surgical risk because of the state of complete unconsciousness. As a result, the anesthesiologist is essential in providing optimal working conditions for surgeons, and in ensuring patient safety and comfort. To achieve these endpoints, anesthesiologists employ a variety of drugs to alter cognitive processing, regulate cardiorespiratory functions, and block muscle movement. One major aspect of the practice is to use these drugs in quantities that warrant unconsciousness and the absence of response to surgical stress, while avoiding pharmacological toxicity (e.g., cardiac morbidity).

1.1 Depth Of Anesthesia (DOA)

One of the objectives of modern anesthesia is to ensure adequate depth of anesthesia (DOA) to prevent awareness without inadvertently overloading the patients with potent drugs (Welling, 1997). For this reason, the specialist must be able to monitor DOA. Various methods have been described to measure the DOA from time to time. The traditional monitoring of the DOA are based on automatic responses of patient body such as respiration pattern, blood pressure, body temperature, tearing, sweating and heart rate (Viertiö-Oja et al., 2004; Miller, 2005). However, considering the following major clinical problems at anesthesia discussion, i.e., sudden conscious during the surgery at the rate of 2-3% (in all under surgery patients) and the rate more than 4% for the patients with brain tumor and the patients under the heart surgery (Jameson & Sloan, 2006), and applying overdose anesthetic agents for the patients relying on the hemodynamic parameters as a criterion for the anesthesia conditions, encouraged the researchers to make an essential effort to introduce more reliable methods (Haddad et al., 2007). In this base, during the present two decades, some novel methods based on

electroencephalogram (EEG) signal processing instead of the traditional methods based on hemodynamic parameters have been used by medical engineers to estimate DOA. Since the central neural system (CNS) is the main aim of the anesthetic agents, the EEG signals have been significantly considered by the anesthesia experts (Sebel et al., 1997). There is not an exact and comprehensive theory, but it is generally accepted that making any change on the synaptic function of the neural cells results in change at the brain functions (Kreuer et al., 1997). Although, the electrical potential resulted from neural cell activities can be observed and registered by EEG signals, but the necessary information about DOA should be extracted from EEG signals using the special techniques. These techniques help experts to have accurate information about DOA and assist doctors during surgical practical in the operation room.

1.2 A short survey on EEG-based methods for measuring the DOA

Estimating the patient's anesthesia depth during the surgery has an important role to prescribe suitable doses of anesthetic agents; because there is the possibility of sudden awareness if the prescribed dose is not sufficient and the overdose agents can also increase the patient's DOA and result in irretrievable consequences. Therefore, it is necessary to have accurate information about DOA to balance the injected dose of anesthetic agents. It is shown that anesthetic agents have direct effects on synaptic activity of brain neurons (Welling, 1997). So, it is preferred to use an EEG analysis as a DOA estimator. Automatic and computer based signal processing methods are taken to evaluate DOA because of the difficulties in visual explanation of EEG. Therefore, EEG can be used as an objective quantitative measure of consciousness which could be taken as a performance variable for closed-loop control of anesthesia (Glass et al., 1997). However, EEG is considered as a complex of multiple time series that its analysis is a difficult task. One of the earliest methods proposed in (Traast & Kalkman, 1995) is based on the Fourier transform that determines the power of EEG in different frequency bands. Zikov et al. proposed a wavelet based anesthetic value for central nervous system monitoring (WAVCNS) that quantifies the depth of consciousness between awake and isoelectric state (Zikov et al., 2006). Ferenets et al. analyzed the performance of several new measures based on the regularity of the EEG signal. These measures consist of spectral entropy, approximate entropy, fractal dimension and Lempel-Ziv complexity. Their results show highly sensitive behavior of the mentioned measures on frequency content of signal and the dose of anesthetic agent (Ferenets et al., 2007). It has been shown in (Hagihira et al., 2001) that there is phase coupling between different frequency bands of EEG. This is known as quadratic phase coupling (QPC) and the amount of QPC changes in different DOA levels. Bispectral analysis (also referred to as the bispectrum) is a technique that incorporates the phase relationship between waves by determining the phase coupling between different frequencies.

1.2 Artificial Neural Network

In fact, there is a potential of artificial neural networks (ANNs) for various units of medicine, such as anesthesia and care unit. Neural networks (NNs) propose a proper framework for identifying on-line systems and the dynamical behaviours for them. Application of artificial neural networks (ANNs) and selecting proper parameters as neural network (NN) inputs in estimating DOA is reviewed in (Robert et al., 2002). In this work, various strategies of choosing the NN model were presented and discussed. In (Haddad et al., 2007), a neural adaptive output feedback control framework which is used to control the infusion of the anesthetic drug is developed. Knorr et al. proposed a NN classifier used to identify airway obstructions (Knorr

et al., 2006). Bailey et al. used neuro-adaptive feedback control algorithms for clinical pharmacology, and showed excellent regulation of unconsciousness allowing for a safe and effective administration of the anesthetic agent propofol (Bailey et al., 2006).

1.3 EEG-based monitors

According to the various methods proposed for analysis of acquired EEG signal (e.g., such as above techniques), different EEG monitors have been introduced. An essential effort has been done in the development of EEG-based monitors which analyze the raw EEG data to extract a single measure of the DOA. Fig. 1 shows the flow diagram procedure of these monitors. The best identified case of these monitors is the bispectral-based or BIS monitor, which calculates a single composite EEG measure with high correlation with the DOA (Bailey et al., 2006). Based on other different proposed methods, other EEG monitors are suggested such as Narcotrend™ monitor (Monitor Technik, Bad Bramsted, Germany) (Kreuer et al., 1997) that is based on pattern recognition of the raw EEG and classification of EEG into different stages, SEDLine™ EEG monitor that is able to calculate the PSI™ index using the shift in power between the frontal and occipital areas (Drover et al., 2002), Datex-Ohmeda™ s/5 Entropy Module (Viertiö-Oja et al., 2004) that uses the entropy of EEG waves to predict DOA.

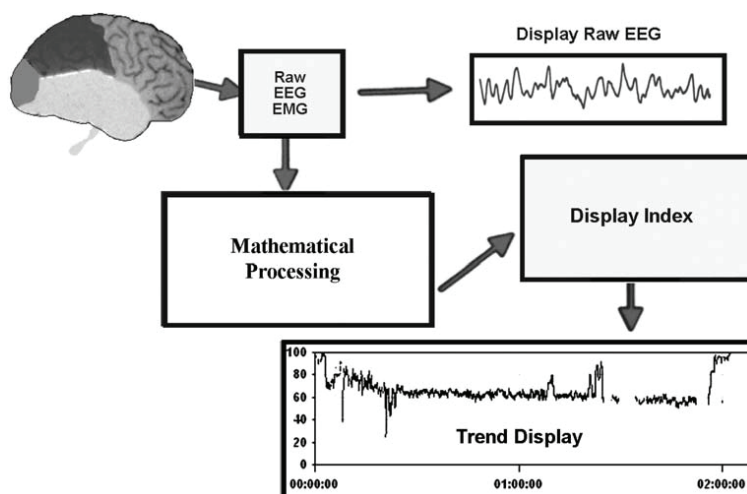


Fig. 1. Flow diagram Procedure of DOA monitors

1.3.1 BIS monitor

One of the most important equipments that is able to monitor the DOA based on analysis of EEG is BIS monitor (Fig. 2) confirmed by FDA (Bailey et al., 2006). The BIS™ index is a unitless number between 100 (awake state) and 0 (isoelectric state) and as the manufacturer claim, the BIS index ranging between 40 and 60 is considered as proper safe range for the operative objectives. Bispectral index is a complex parameter, including a combination of time domain, frequency domain and higher order spectral (HOS) sub-parameters respectively named Burst Suppression Ratio (BSR as defined in subsection 3.4), Relative Beta Ratio ($RBR = \log(P_{30-47\text{Hz}} / P_{11-20\text{Hz}})$, where $P_{30-47\text{Hz}}$ and $P_{11-20\text{Hz}}$ are the power spectrum

of frequency bands of 30Hz to 47 Hz and 11Hz to 20 Hz respectively) and Sych Fast Slow ($SFS = \log(B_{0.5-47Hz} / B_{40-47Hz})$ where $B_{0.5-47Hz}$ and $B_{40-47Hz}$ are Bi-Spectrum value of the windowed signal in the ranges 0.5 to 47 Hz and 40 to 47 Hz respectively). This monitor integrates these three sub-parameters to produce BIS index, however, a detailed algorithm of this monitor is not published yet (Jameson & Sloan, 2006). Between 1990 and 2006, approximately 450 peer-reviewed publications have been examined the clinical effectiveness and accuracy of the BIS™ monitor (Jameson & Sloan, 2006). Thus, besides the clinical usages, the BIS index can be as a proper criterion for comparing the achieved results with the new methods of the estimating anesthesia.

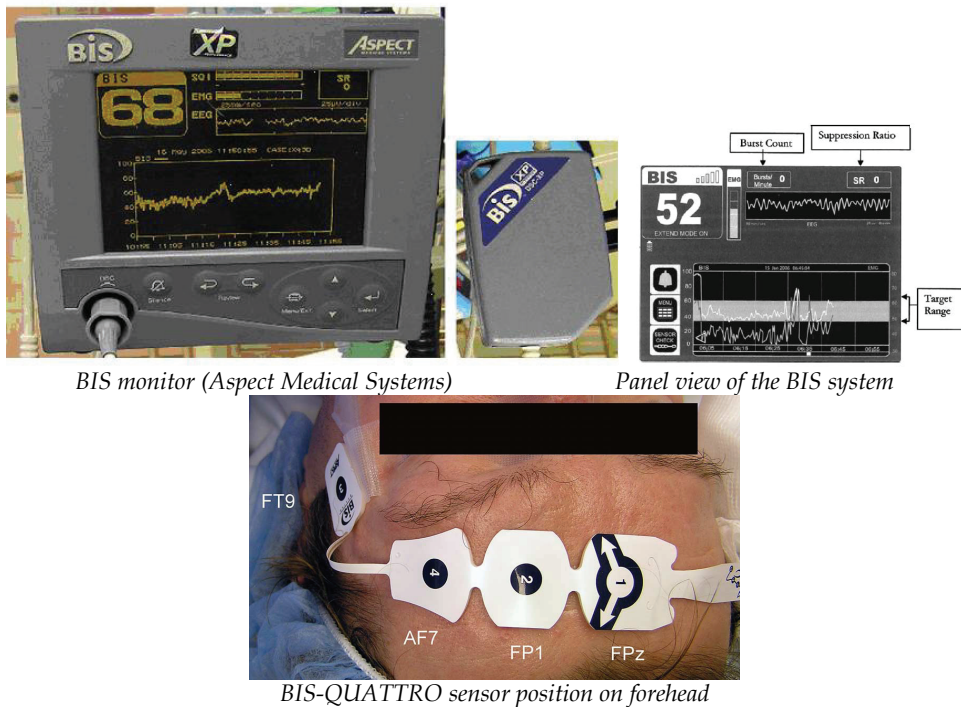


Fig. 2. BIS monitor

1.4 This chapter

In the data classification trials which the feature dimensionality of the data are big compared to the number of data samples, the conventional classifiers are not applicable, therefore the use of nonlinear classifiers such as NN maybe helpful in discrimination of different classes. Medical data such as EEG, Microarray extracts of gene expression or 1H NMR spectra of the normal and cancerous tissue are the data with high dimension. Classification of these type data to different categories is limited to the number of acquirable samples. The limitations on acquisition of these types of data depend on different causes such as cost, limited number of candidates, or limited technical resources. The solution is the reduction of the dimensionality of features to an optimum value. There would be some

dimensionality reduction methods for removing dependent features before applying the classifiers on the data. In the case of important independent features, the dimensionality reduction is limited to the categorization based on the most significant features such as PCA. In this study several features are extracted from EEG signal to obtain the awareness level of patients. These features are used as inputs of NNs to estimate the DOA by reducing the dimensionality of features.

More especially, in this study, we gathered EEG of the patients who were undergoing surgery, and then we used wavelet transform to extract an index indicate level of consciousness (WAI index). For this reason, wavelet coefficients calculated from the EEG signal are compared to three well-defined states, i.e., completely awake state, moderate state, and isoelectric state, and finally WAI quantifies the depth of consciousness between these three states. In addition, besides WAI obtained using wavelet transform, we extract 15 sub-parameters from the recorded EEG data. By applying these sub-parameters into neural and neuro-fuzzy networks, we obtain a new index to estimate the DOA. Finally, a comparison is done between the obtained results from the suggested methods and the obtained results from BIS monitor.

Work presentation is organized as follows: In Section 2, data acquisition procedure is presented. In Section 3, we explain about WAI index. The extracted features for estimation of DOA such as BSR, spectral edge frequency (SEF), signal quality index (SQI), and alpha, beta and theta ratio are used as inputs of NNs and an appropriate index for DOA are obtained in Section 4. The results are presented in Section 5, and finally this chapter is concluded in Section 6.

2. Data acquisition procedure

This study is performed on 33 under surgery coroner vessel patients. To limit the conditions of this research, the patients with equal kind of surgery and anesthesia diets are chosen. Having no records of diseases related to CNS is a criterion in this study (21 males, 12 females, with average age of 54 and the average weight of 67 kg). Anesthesia diets for all patients were the same. The anesthesia stage was obtained by the use of intravenous agents, Tupental sodium (5mg/kg), Pancronim Burmid (0/1 mg/kg), Fontanel (5mg/kg), and Lidokain (1/5 mg/kg). The next stage is following the anesthesia process by inhalation and using Isofloran agents (MAC). After finishing the surgery, the patients were taken to ICU. To relax them in ICU, morphine (2mg/kg) was injected if needed. BIS monitor was used for obtaining and registering EEG besides the related DOA evaluation by BIS software (these signals are transferred to computer). Two other series of data also related to awareness, including EEG signals of 10 persons (15 minutes for each one) and isoelectric states, including 40 minutes recorded EEG signals from one patient recorded from a brain death are recorded. Fig. 3 shows sample recorded EEG signals.

The EEG signal was collected by using a BISQUATTRO Sensor™ that bands holding four electrodes, applied to the forehead of the patients. The used EEG lead was Fpz-At1, and the reference lead was placed at FP1. A Sensor was connected to a BIS-VISTA Monitor and all binary data packets containing raw EEG data wave signals and BIS index were recorded via an RS232 interface on a laptop using a bispectrum analyzer developed by Hagihiro (Hagihiro et al., 2001). A sensor was attached to patient forehead at ICU and data were collected continuously until he/she awaked.

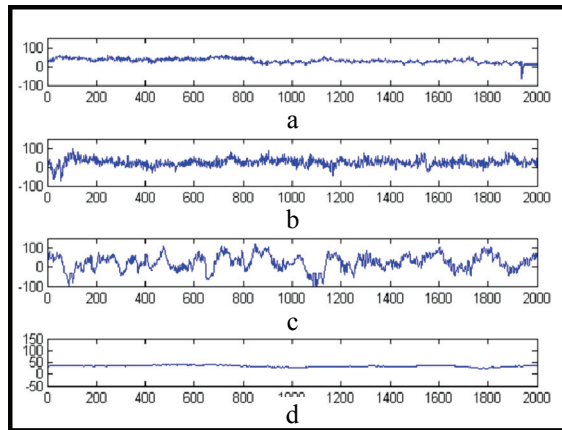


Fig. 3. Sample recorded EEG signals (15 sec. epochs). a) awake person, b) moderate anesthesia (recorded at ICU after surgery), c) deep anesthesia, d) isoelectric case

3. Anesthesia index based on wavelet (WAI)

The effect of anesthetic agents on EEG is studied by considering non aggressive state of this type of measurement. Besides increasing the blood pressure, most of the agents cause changes on EEG signal with low range and big frequency band-width with big range signal and quiet behaviour. Increasing the dose of anesthetic agents resulted to destruction of the recordable EEG activity at brain cortex and finally making isoelectric state (Smith et al., 1996). Therefore, it is supposed that EEG can be used for estimating the conscious level of the patient which shows the amount of this agents influence for saving time and complex interpretations of using EEG row signal to derive an index which shows the awareness state of the person. One of the processing instruments for obtaining information from EEG is as a result of giving an index to estimate the DOA through using wavelet transform. To show a method of computing the DOA, Zikov et al. supposed the patient as a system variant between the two states of a (completely awake) and b (totally isoelectric) and they identified an index by wavelet transform to show the state and features of this system (Zikov et al., 2006). To increase the precision, we suppose the patient as a system variant between three specified states of a (completely awake), b (moderate state or general anesthesia) and c (isoelectric state) as shown in Fig. 4. At complete awareness state, the patient has all his/her senses and he/she can give proper answers to all stimulations around him/her. Moderate state is a state when the patient has received anesthetic agents and is senseless to pain and other stimulations. In this state, he/she is in the best condition for going under surgery operation, meaning that he/she has received a suitable dose of anesthetic agent (neither too much nor too less). This state is determined by the anesthesia expert. Isoelectric state is related to brain death state or the least EEG signal activity state. According to Fig. 4, f is a function respect to x , where x is between a and c , and also the range of this function is between zero to one. In other words, x is a part of the signal with length n , that is mapping to m dimensional feature space extracted from x . h is scaled versions of f which indicates a number between zero and 100 representing the DOA.

$$x \in R^n \rightarrow f \in R^m \rightarrow h : z \in [a, c] \rightarrow i \in [0, 100] \quad (1)$$

To define f , it is important to identify a , b , and c points. To analyze EEG, we choose these points as we defined them in above, we can obtain 3 series of reference data which are related to our observations of this system at a , b , and c states.

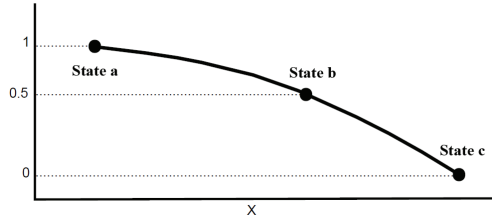


Fig. 4. f function for presenting the system features

$$\begin{cases} R_a = \{X_{a,k} & k = 1, 2, 3, \dots, M\} \\ R_b = \{X_{b,k} & k = 1, 2, 3, \dots, M\} \\ R_c = \{X_{c,k} & k = 1, 2, 3, \dots, M\} \end{cases} \quad (2)$$

$X_{a,k}$, $X_{b,k}$ and $X_{c,k}$ contain a finite number of samples (10 seconds of EEG to make it comparable with BIS index) and represent the k^{th} epoch of R_a , R_b and R_c data sets. In order to obtain reference data set, feature selection function f is applied to each data set. For one dimensional case, this feature function is defined as follows:

$$f: x \rightarrow f(x) = f \quad (3)$$

The ability of the index depends on f at $[a, c]$ to identify the states of the patients. As shown in Fig. 4, f function can be either linear, or non-linear. At any general state, the first order derivative of f should not have any sign change.

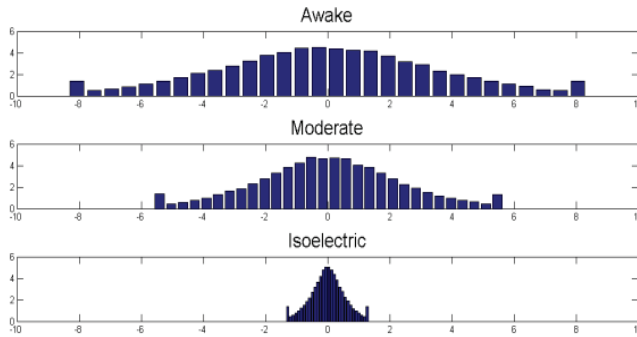


Fig. 5. PDF of awareness (awake), general anesthesia and isoelectric anesthesia

One choice for selecting f , is using probability density function (PDF) of d1 coefficients (details subband at first scale of Daubichies wavelet (Zikov et al., 2006)). In Fig. 5, the PDF of d1 coefficients are presented where histogram is used for estimating PDF. It is shown that

this histogram has more concentration around zero in isoelectric state while it is more scattered at awareness state. So, the variance of this histogram can be used as an appropriate feature:

$$f_{t,k} = \text{var}(\text{hist}(d1)), \quad t = a, b, c \quad k = 1, 2, \dots, M \quad (4)$$

By using three series of reference signals, the awareness reference feature vector, general anesthesia, and isoelectric states are measured through the following equation:

$$\overline{f_t} = \frac{1}{M} \sum_{k=1}^M f_{t,k} \quad t = a, b, c \quad (5)$$

where $f_{a,k}$, $k = 1, 2, \dots, M$ are obtained by applying 10-second awake reference data sets to f function, and $f_{b,k}$ and $f_{c,k}$ are characterized respectively by the same manner for b and c states. $\overline{f_a}$, $\overline{f_b}$ and $\overline{f_c}$ are average of this feature in the corresponding states and are the reference features. Now, this feature on every 10-second period for unidentified EEG signal (e.g., state d) can be measured by comparing this feature with $\overline{f_a}$, $\overline{f_b}$ and $\overline{f_c}$, and obtaining the similarity between the patient's condition with these amounts. As a result, we define the following quantities as below:

$$j_t = \|f_d - \overline{f_t}\| \quad , t = a, b, c \quad (6)$$

The norm vector V , $\| \cdot \|_1$ is defined as:

$$\| \cdot \|_1 = \sum_{n=1}^N |v(n)| \quad (7)$$

j_a , j_b and j_c indexes measure the distance of the system from a , b , and c states. High norms can be applied for this analysis but in this case j_a , j_b and j_c may increase and this increase might cause noisy state of the output index.

Since we have three points of reference, it's possible to use either a second order function or a partial-linear function to show properties of function f . First, using the concept of interpolation a second order function was designed in MATLAB. The data were fitted to the reference, but in all cases first order derivative of this curve changed, which shows the intermediate stats are not diagnosed clearly with this function. Instead, we can define this function linearly between a and b , also linearly between b and c (partially-linear function). To maximize signal to noise ratio, we can combine j_a , j_b and j_c , to define two descriptors $j_1 = j_b - j_a$ and $j_2 = j_c - j_b$. For this reason, we define functions $g_1(x)$, $g_2(x)$ as follows:

$$g_1 : x \rightarrow g_1(x) = j_1 = \|f(x) - \overline{f_b}\| - \|f(x) - \overline{f_a}\| \quad (8)$$

$$g_2 : x \rightarrow g_2(x) = j_2 = \|f(x) - \overline{f_c}\| - \|f(x) - \overline{f_b}\| \quad (9)$$

In order to scale the final index namely wavelet anesthetized index (WAI) such that $h = \text{WAI} = 1$ for state a , $\text{WAI} = 0.5$ for state b , and $\text{WAI} = 0$ for state c , the function g_1 is applied to the entire reference data sets a & b , and the function g_2 is applied to the entire reference data sets b & c . This yields four sets J_a , J_{b1} , J_{b2} and J_c :

$$\begin{cases} h=1 \rightarrow J_a = \{j_{a,k}, & k=1,2,3,\dots,N\} \\ h=0.5 \rightarrow J_{b1} = \{j_{b1,k}, & k=1,2,3,\dots,T\} \end{cases} \quad (10)$$

$$\begin{cases} h=0.5 \rightarrow J_{b2} = \{j_{b,k}, & k=1,2,3,\dots,N\} \\ h=0 \rightarrow J_c = \{j_{c,k}, & k=1,2,3,\dots,T\} \end{cases} \quad (11)$$

Now, assume that d is a state between a and c , WAI is defined as follows (the scaled version of g_1 and g_2):

$$\begin{aligned} WAI : x \rightarrow WAI(x) = i \\ = \begin{cases} g_1(x) \cdot \frac{1}{\bar{J}_a - \bar{J}_{b1}} - \frac{\bar{J}_{b1}}{\bar{J}_a - \bar{J}_{b1}} & \text{if } b < d < a \\ g_2(x) \cdot \frac{1}{\bar{J}_{b2} - \bar{J}_c} - \frac{\bar{J}_c}{\bar{J}_{b2} - \bar{J}_c} & \text{if } c < d < b \end{cases} \end{aligned} \quad (12)$$

where

$$\bar{J}_t = \frac{1}{M} \sum_{k=1}^M J_{t,k}, t = a, b1, b2, c \quad (13)$$

3.1 WAI vs. BIS

This algorithm is applied for the data obtained from a sample patient and the obtained amounts of WAI along with BIS index amount are figured in Fig. 6.

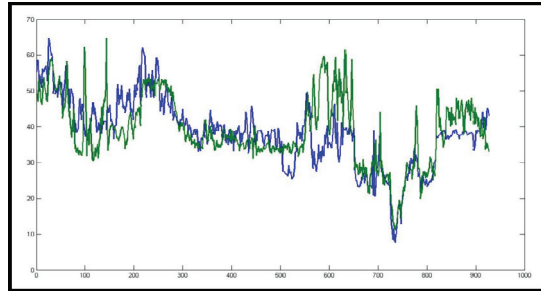


Fig. 6. WAI index vs. BIS index

Number of reference states	Correlation With BIS	Time Complexity (second)
2	45%	5
3	77%	7
4	73%	15

Table 1. Correlation and time complexity compared using two, three and four reference state to obtain WAI

To extract WAI, we also tested two, three and four reference states and compared the results. First, by using awake and isoelectric states, only 45% correlation with BIS were achieved (Table 1), total time needed to calculate the reference vectors and extracting the desired index from samples was almost five seconds. Using three reference states (as explained in section 3), correlation increased to 77 percent and total operation time increased to seven seconds. The main reason of this improvement is using 3 states instead of two states. So, we examined using 4 references, but the correlation decreased to 73 percent and the time increased to 15 seconds. Considering the time complexity and amount of correlations we can conclude that choosing 3 references is the best choice.

3.2 Further discussions in time-frequency domain

The mentioned results are obtained using Daubechies wavelet. We also used other wavelet transforms such as complex wavelet transform (Selesnick et al., 2005), however considerable improvement didn't obtain in results. In addition, we employed other joint time-frequency tools such as spectrogram, scalogram, Wigner-Ville transform, and Choi-Villiams transform (Mallat, 2009). The results for isoelectric, awake and moderate states are illustrated in Fig. 7. Our simulations show that wavelet transform is superior for estimation of the DOA.

4. Index of the DOA using ANN

ANN is basically a wide processor with parallel structure which is able to save experimental data and use them in the next processor. There is a potential of ANN for various units of medicine, such as anesthesia and care unit. We divide the collected signals to 10-second periods, and extract 15 features for each epoch. The calculated parameters are classified as follows.

4.1 Spectral edge frequency of 95 and 90

The spectral edge frequency 95 (SEF95) is the frequency that 95% of the power in the spectrum exists within. It's clear that in the case of anesthesia SEF95 becomes small. A similar definition can be also used for SEF90.

4.2 Alpha, beta and theta ratio

Alpha, beta and theta ratios indicate logarithmic relative power of two distinct frequency bands. Alpha-ratio decreases as anesthesia becomes deeper as follows:

$$\alpha_{ratio} = \log \frac{E(30 - 42.5\text{Hz})}{E(6 - 12\text{Hz})} \quad (14)$$

Beta-ratio is defined as follows:

$$\beta_{ratio} = \log \frac{E(30 - 42.5\text{Hz})}{E(11 - 21\text{Hz})} \quad (15)$$

The difference between alpha and beta ratios is considered as theta ratio:

$$\theta_{ratio} = \beta_{ratio} - \alpha_{ratio} = \log \frac{E(6 - 12\text{Hz})}{E(11 - 21\text{Hz})} \quad (16)$$

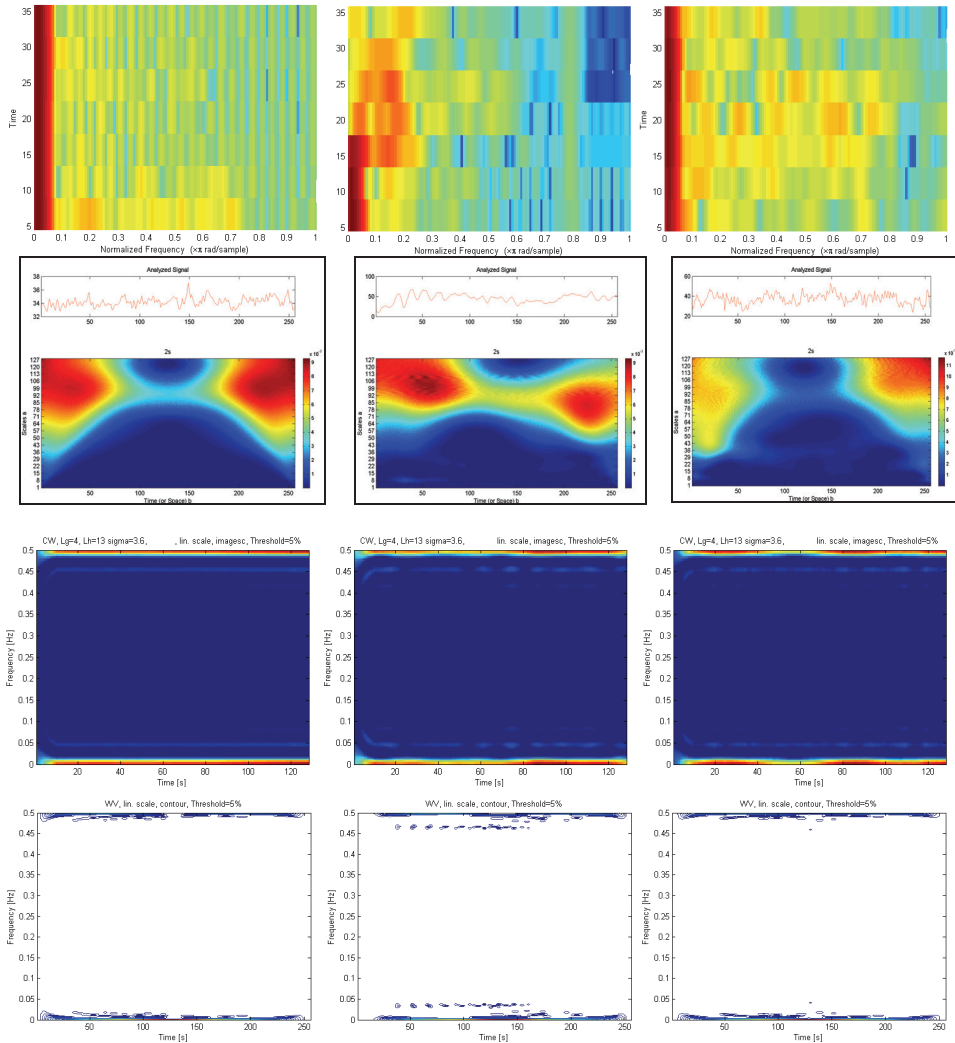


Fig. 7. First, second, third and fourth rows respectively show the spectrogram (2 sec. window), scalogram (2 sec. window), Choi-Villiams transform (1024 samples), and Wigner-Ville transform (256 samples). First, second and third column respectively illustrate isoelectric, awake states, and general anesthesia.

4.3 Burst Suppression Ratio (BSR)

We define burst suppression (BS) as amplitude less than 5mV lasts more than 0.5 sec in the processed wave. Fig. 8 illustrates EEG signal with burst and without burst suppression. Note that the small amplitude waves that have fast frequency also appear in the light anesthetic state causes a false BS detection. The BSR is the summation of the times with burst suppression divided to whole windowed time range which includes burst events (Fig. 9).

$$BSR = \frac{T_{s1} + T_{s2} + T_{s3}}{T_{total}} \quad (17)$$

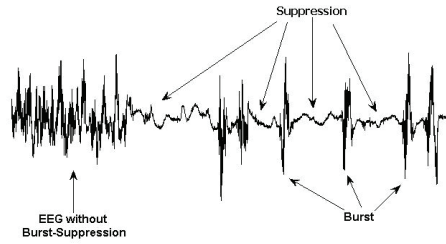


Fig. 8. EEG signal with burst and without burst suppression

It shows that BSR alone is not an appropriate feature for DOA in all cases.

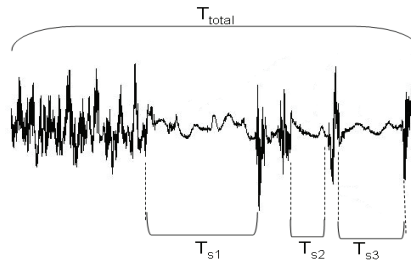


Fig. 9. The EEG windows with burst suppression

4.4 Signal Quality Index (SQI)

The ratio of the number of epochs that are without artifact to the number of total epochs is known as SQI.

4.5 Wavelet coefficients

For the γ -band, an optimal result for classifying different anesthetics state can be obtained by using Daubechies wavelet coefficients ($w[Db_6]$). We extract the amplitude of PDF of wavelet coefficients, mean and variance of these coefficients in each epoch as features for estimation of DOA (since the mean of wavelet coefficients tends to zero we can exclude this feature).

4.6 Applying Back Propagation (BP) algorithm

In addition to the mentioned features in this section, other features like mean-amplitude, mean and variance of each epoch, are also extracted and all of the obtained fifteen features (see caption of Fig. 11 for a list of these features) set in a 15×7440 matrix to train the ANN using BP algorithm with sigmoid function (Fig. 10). To train the ANN, the 20 hours collected data taken from 25 patients were used (data from 8 patients were held to use for testing the ANN). In this way, we divided the 20 hours collected data into 10-second epochs and extracted 15 features mentioned in previous part for every 10-second epochs. The all 15 features were applied to an ANN. At first, we designed a structure with 15 input neurons,

20 neurons at hidden layer and one neuron in output layer (which we prepared besides a bias for hidden and output layers). We applied all 7440 data in the proposed structure, and set BIS index as the desired output in the output of this network. After training step, we used the data taken from patient number 7 (who hadn't taken part in network training process) for testing our network and comparing the output of network with the output of

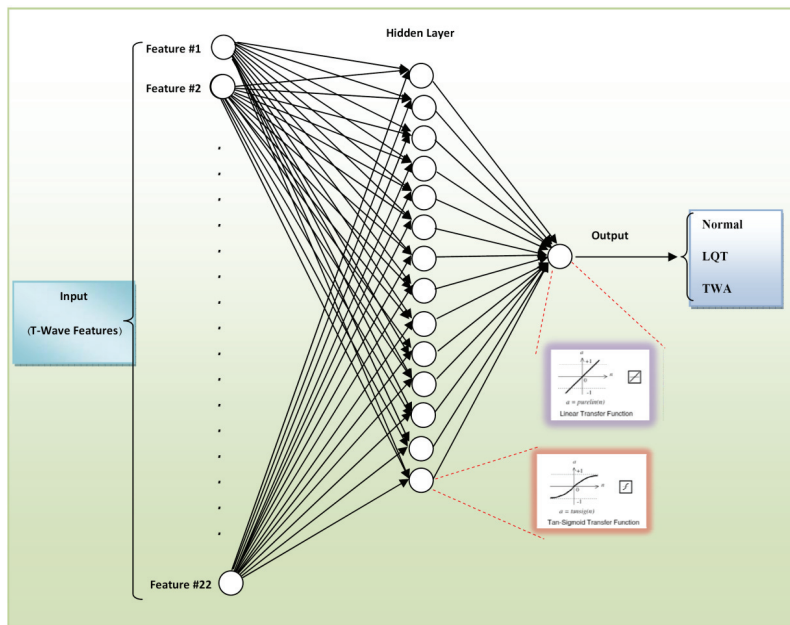


Fig. 10. Multi-layer perceptron (MLP) NN architecture.

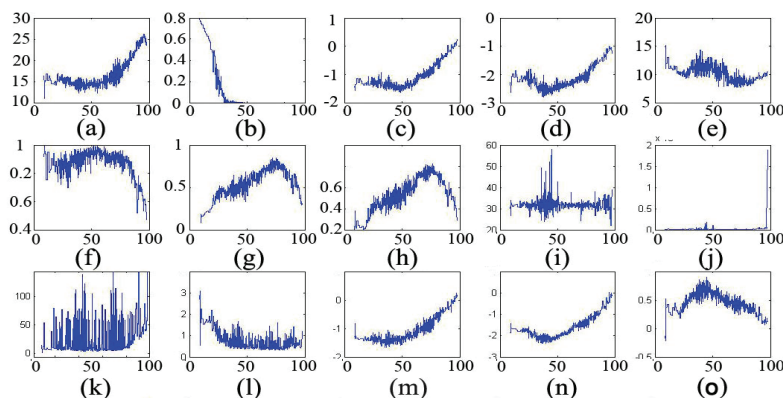


Fig. 11. BIS Index versus : a) SEF95, b) BSR, c) SEF90, d) SEF50, e) Mean Amplitude, f) SQI of time domain, g) SQI of frequency domain, h) SQI of bispetral domain, i) Mean of each epoch, j) Variance of each epoch, k) Amplitude of pdf of wavelet coefficient, l) Variance of pdf of wavelet coefficient, m) Alpha ratio, n) Beta ratio, p) Theta ratio

BIS monitor. We can also see how these 15 features change during the changes of BIS index in Fig. 11. As represented in this figure, most of these features are changing corresponding to the changes of BIS index.

In second proposed structure for ANN, based on results of Fig. 11, we omitted 5 features which didn't have regular changes with the changes in BIS and had poor correlation with BIS index (features e, g, i, j, k in Fig. 11). As a result, we designed a new network (named second structure) with 10 input neurons, 20 neurons in first hidden layer, 10 neurons in the second hidden layer and one neuron in output. Using BP algorithm, we trained our proposed structure. After 30000 iteration, our network was trained properly and the data from patient number 23 and 7 were applied to the network, and achieved a correlation more than 89% (95%, $\text{corr}=[88.9\%, 93.44\%]$) with BIS index. The result is represented in Fig. 12.

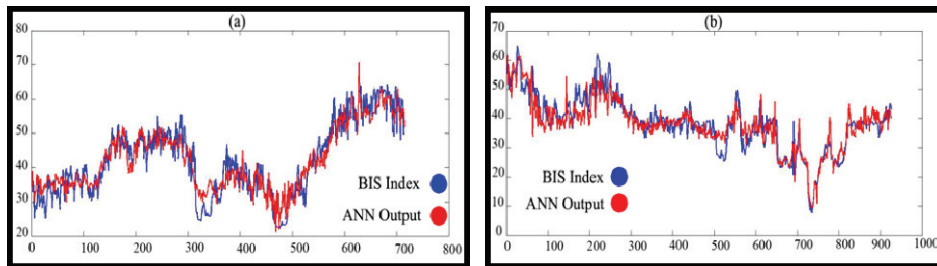


Fig. 12. a) BIS versus ANN output with correlation 93% for patient no 7, b) BIS versus Ann output with correlation 90% for patient no 23

4.7 Additional works

We use complex wavelet transform (Selesnick et al., 2005) to estimate the DOA by using second, third, fourth central moments (μ_2, μ_3, μ_4), skewness (S), kurtosis (K), and the difference between maximum and minimum values of PDF (Δ) (Papoulis & Pillai, 2001). These features extracted from the real part of first scale (high frequency subband d_1), and our simulations show that similar results can be obtained from imaginary part. We use a BP-based NN with 6 input neurons, 6 neurons (with Gaussian function) at hidden layer and one neuron (with linear function) in output layer. For training we classified the data into 10 classes according to Table 2. For each class we have 15 minutes recorded data that after 10-second windowing we would have 90 training data. We used 35 test data for each class and finally the correlation between the output of this NN and BIS index was calculated 65.7%. We can see how these 6 features change during the changes of BIS index in Fig. 13. From this figure we can conclude that third and fourth central moments don't differ as good as other features.

Regarding to ANN-based method, other networks such as ANFIS (Jang, 1993) was also tested. For this reason, at first complex wavelet transform with four stages was applied on data. Then, maximum, minimum, mean, standard deviation and the $VP=\text{var}(\text{hist}(w))$ were extracted where w shows the wavelet coefficients for real and imaginary parts. We use this features for all subbands d_1, d_2, d_3, d_4 , and low-pass subband a_4 . So, we would have 50 features for each class. Table 3 illustrates the values of these features for various BIS ranges.

The range of data is [BIS-5, BIS+5]	Extracted features					
	Δ	K	S	μ_4	μ_3	μ_2
BIS=95	80	3.94	0.057	31.45	0.27	2.82
BIS=85	60	2.81	-0.092	14.7	-0.32	2.29
BIS=75	76	3.92	0.095	4.68	0.1092	1.092
BIS=65	62	2.01	0.16	0.85	0.063	0.53
BIS=55	122	23.05	-1.8	1.44	-0.23	0.25
BIS=45	244	49.17	-2.95	5.09	-0.53	0.31
BIS=35	365	83.98	-1.98	55.9	-1.46	0.81
BIS=25	359	95.7	51.0	44.5	0.29	0.68
BIS=15	400	305.3	-14.2	27.3	-2.33	0.29
BIS=5	230	125.8	7.49	0.34	0.089	0.052

Table 2. Extracted features from 10 classes of anesthesia for the real part of first scale (high frequency subband d_1) of complex wavelet coefficients

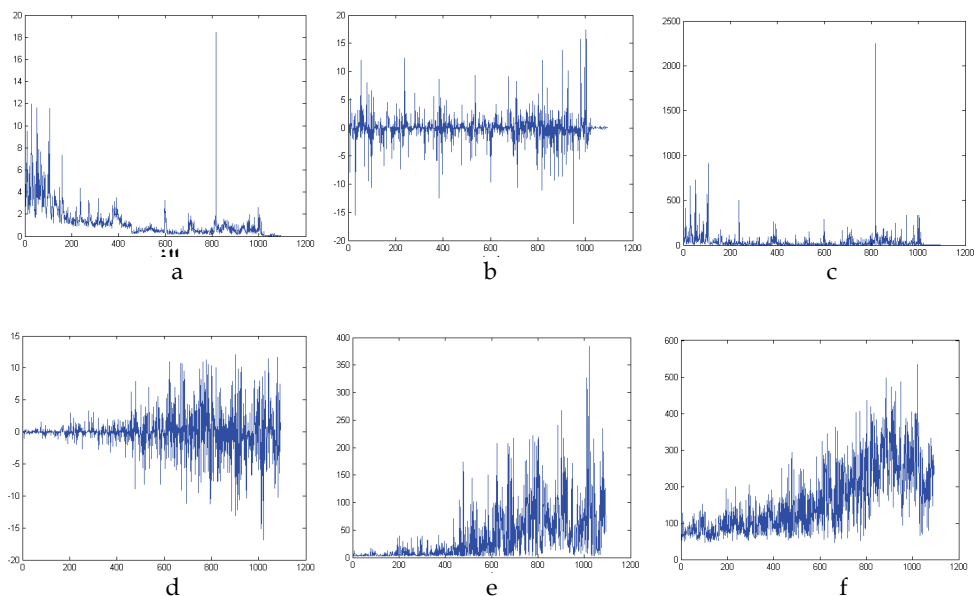


Fig. 13. BIS index versus : a) second central moment (μ_2), b) third central moment (μ_3), c) fourth central moment (μ_4), d) skewness (S), e) kurtosis (K), and f) the difference between maximum and minimum values of PDF (Δ)

Although it seems that all features have a reasonable changes for various BIS ranges, but using all of them results in an ANFIS network with high computational complexity. Computing the correlation of these features with BIS index we extract 16 features with highest correlation as shown in Table 4. Applying 4 features with highest positive correlation to ANFIS network achieves 88.7% correlation with BIS index. For this method (that is a combination of complex wavelet transform and neuro-fuzzy network) the specificity, sensitivity, and the total classification accuracy were calculated. The results have been concluded in Table 5. It's clear from this table that for $20 < \text{BIS} < 30$ we have highest sensitivity and for $40 < \text{BIS} < 60$ we have the least. For all levels of anesthesia the specificity is higher than 90% and except $50 < \text{BIS} < 60$, in all cases the total classification accuracy is higher than 90%.

The range of data is [BIS-5, BIS+5]	features	subband				
		a_4	d_4	d_3	d_2	d_1
BIS=95	Max.	123.7	13.72	10.59	8.68	5.85
	Min.	33.78	-9.72	-12.22	-9.44	-6.68
	mean	93.76	0.442	0.328	0.097	-0.036
	std.	18	4.72	3.77	2.61	1.9
	VP	1.15	1.05	3.69	11.07	41.95
BIS=65	Max.	150.63	34.18	19.14	11.49	7.8
	Min.	51.09	-32.51	-21.34	-8.1	-7.33
	mean	85.66	-0.39	-0.15	-0.12	0.007
	std.	18.74	12.006	7.35	2.59	1.07
	VP	1.33	1.09	3.21	16.8	52.18
BIS=45	Max.	153.2	21.74	18.9	6.99	71.01
	Min.	10.97	-25.52	-14.26	-8.1	-11.47
	mean	95.87	-0.29	0.14	-0.012	0.005
	std.	8.07	8.24	6.35	1.57	0.7
	VP	0.96	1.27	3.23	23.93	486
BIS=5	Max.	118.5	4.3	3.3	1.58	2.13
	Min.	80.98	-5.59	-2.8	-1.8	-0.4
	mean	95.8	0.13	0.005	0.007	0.009
	std.	6.07	1.4	0.8	0.33	0.17
	VP	1.37	1.75	5.91	27.19	156.15

Table 3. Extracted features from 4 classes of anesthesia for complex wavelet coefficients

Features	Correlation with BIS (-)	Features	Correlation with BIS (+)
VP of real part of d_1	-67.74	-std of imaginary part of d_1	73.22
VP of imaginary part of d_1	-65.09	std of real part of d_1	68.3
VP of real part of d_2	-64.4	Max. of imaginary part of d_1	62.22
VP of real part of d_3	-62.3	std of imaginary part of d_2	62.09
VP of imaginary part of d_2	-57.4	std of real part of d_2	48.2
VP of real part of d_4	-56.8	Max. of real part of d_2	40.08
VP of imaginary part of d_3	-55.9	std of real part of d_3	39
VP of imaginary part of d_4	-55.13	Max. of imaginary part of d_2	36.8

Table 4. Features with highest correlation with BIS

5. Results

Our aim is to maximize the correlation between BIS index and output of our proposed methods. For this reason several methods based on wavelet transform and ANN have been proposed and we can conclude from the correlation between BIS and the obtained index from methods that second structure of BP-based NN (section 4.6) is the best in most cases. For this network, the correlation between BIS index and the extracted sub-parameters was investigated with the model-independent prediction probability (PK) (Smith et al., 1996). These results represented that among the extracted feature, BSR, Beta Ratio and SEF90 features respectively had the highest weight for calculating the outputs of ANN. In this way, BSR had individually proper results for 44% cases for the depth below 40, and for the cases above 40, Beta ratio and SEF95 had individually proper results for 35% of cases.

The obvious outcome of the algorithms is that, Alpha and Beta ratio decreases and BSR increases with the increasing DOA (decreasing BIS index). As Fig. 12 shows, there is a high correlation 90% between BIS index and output values of proposed ANN in the moderate anesthetized range.

The results indicate that the SEF95, Alpha and Beta ratio decreases and BSR increases as BIS decreases. Other parameters are weak and somehow unrelated to BIS index. This confirms the results of (Haddad et al., 2007), which indicate that "bispectral analysis adds nothing but complexity". Due to equations (14, 15), Alpha and Beta ratio quantifies the activity of β waves, and this confirms the fact that anesthetic drugs affect these waves. The burst-suppression pattern usually occurs in deep anesthesia (Fig. 12-b). Our results demonstrate a high correlation between BSR and BIS index. Our findings show that in deep anesthesia (below 40), BSR could predict BIS index lonely.

The level of anesthesia	sensitivity	specificity	total classification accuracy
90<BIS<100	99.22	75.4	97.89
80<BIS<90	97.98	60	94.5
70<BIS<80	96.5	58.2	91.0
60<BIS<70	92.08	68.2	91.2
50<BIS<60	93.6	49.47	89.7
40<BIS<50	93.85	49.54	90.6
30<BIS<40	96.71	58.93	94.63
20<BIS<30	98.84	93	96.8
10<BIS<20	97.85	70.8	97.07
0<BIS<10	99.51	64.6	97.4

Table 5. The specificity, sensitivity, and total classification accuracy of complex-wavelet-neuro-fuzzy system for estimation of DOA

6. Conclusion

In this work, we used wavelet transform (WAI index) for extracting DOA index, and also employed several sub-parameters for assessing DOA based on ANN with the back propagation (BP) learning algorithm and ANFIS algorithm. The output of second structure of proposed BP-based NN has a high correlation with the output of the BIS index in compared with the WAI. However, using wavelet transform to extract WAI has less complexity (computational time) than ANN method. The combination between complex wavelet transform and neuro-fuzzy network also achieves a new method with high sensitivity and total classification accuracy between various level of anesthesia.

We can also test this algorithm by adding other features extracted from other clinical devices or combined features (as inputs of ANN). In addition, since the extracted EEG may be corrupted with noise and interferences, we can use some pre-processing steps in wavelet domain to improve the results of wavelet-based method without considerable increscent in computational complexity.

7. References

- Bailey, J.; Haddad, W.; Im , J.; Hayakawa, T. & Nagel, P. (2006). Adaptive and Neural Network Adaptive Control of DOA During Surgery, *Proc. American Control Conference*, pp. 3409-3414, Minneapolis, Minnesota, USA, June 14-16, 2006
- Drover, D. R.; Lemmens, H. J.; Pierce, E. T.; Plourde, G.; Loyd, G.; Ornstein, E.; Prichep, L. S.; Chabot, R. J. & Gugino, L. (2002). Patient State Index: titration of delivery & recovery from propofol, alfentanil, nitrous oxide anesthesia, *Anesthesiology*, pp. 82-89

- Ferenets, R.; Vanluchene, A.; Lipping, T., Heyse, B. & Struys, M. (2007). Behavior of Entropy/Complexity Measures of the Electroencephalogram during Propofol-induced Sedation, *Anesthesiology*, vol. 106, pp. 696-706
- Glass, P. S.; Bloom, M.; Kears, L.; Rosow, C.; Sebel, P. & Manberg, P. (1997). Bispectral analysis measures sedation and memory effects of propofol, midazolam, isoflurane, and alfentanil in normal volunteers, *Anesthesiology*, vol. 86, no. 4, pp. 836-847
- Haddad, W. M.; Bailey, J. M.; Hayakawa, T.; & Hovakimyan, N. (2007). Neural Network Adaptive Output Feedback Control for Intensive Care Unit Sedation and Intraoperative Anesthesia, *IEEE Trans. Neural Networks*, vol. 18, no. 4, pp. 1049-1066
- Hagihira, S.; Takashina, M.; Mori, T.; & Yoshiya, I. (2001). Practical issues in bispectral analysis of electroencephalographic signals, *Anesth. Analg.*, vol. 93, no. 4, pp. 966-70
- Papoulis, A. & Pillai, S. U. (2001). *Probability, Random Variables and Stochastic Processes*. McGraw-Hill Science/Engineering/Math.
- Jameson, L.C. & Sloan, T.B. (2006). Using EEG to monitor anesthesia drug effects during surgery, *J. Clin. Monit. Comput.*, pp. 445-472
- Jang, J.-S. R. (1993). ANFIS: Adaptive-Network-based Fuzzy Inference Systems, *IEEE Trans. on Systems, Man, and Cybernetics*, vol. 23, pp. 665-685
- Jones J. G. (1994). Perception and memory during general anesthesia, *British Journal of Anesthesia*, pp. 31-37
- Knorr B. R.; McGrath S. P., and Blike G. T., (2006). Using a Generalized Neural Network to Identify Airway Obstructions in Anesthetized Patients Post-Operatively based on Photoplethysmography, *Proc. IEEE EMBS Annual International Conf.*, pp. 6765-6768, New York City, USA, Aug 30-Sept 3, 2006
- Kreuer, S.; Biedler, A.; Larsen, R.; Schoth, S.; Altmann, S. & Wilhelm, W. (1997). The Narcotrend - a new EEG monitor designed to measure the depth of anaesthesia. A comparison with bispectral index monitoring during propofol-remifentanyl-anaesthesia, *Anaesthetist*, vol. 50, no. 12, pp. 921-925
- Mallat S. G., (2009). A wavelet tour of signal processing: The sparse way, Acad. Press, San Diego
- Miller R. D. (2005), *Miller's Anesthesia*, Sixth edition, Elsevier Churchill Livingstone
- Rampil, I. J. (1998). A primer for EEG signal processing in anesthesia, *Anesthesiology*, vol. 89, no. 4, pp. 980-1002
- Robert, C.; Karasinski, P.; Arreto, C. D.; and Gaudy, J. F. (2002). Monitoring Anesthesia using neural networks: A survey, *J. Clin. Monit. Comput.*, vol. 17, pp. 259-267
- Sebel, P. S. ; Lang, E.; Rampil, I. J. ; White, P.; M. Jopling, R. C. ; Smith, N. T. ; Glass, P. S. & Manberg, P. (1997). A multicenter study of bispectral electroencephalogram analysis for monitoring anesthetic effect, *Anesth. Analg.*, vol. 84, no. 4, pp. 891-899
- Selesnick I. W.; Baraniuk R. G.; and Kingsbury N. G., (2005). The Dual-Tree Complex Wavelet Transform, *IEEE Signal Processing Magazine*, vol. 22, no. 6, pp. 123-151
- Smith W. D., Dutton R. C., Smith N. T., (1996). Measuring the performance of anesthetic depth indicators, *Anesthesiology*, vol. 84, pp. 38-51
- Traast, H. S. & Kalkman, C. J. (1995). Electroencephalographic characteristics of emergence from propofol/sufentanil total intravenous anesthesia, *Anesth. Analg.*, vol. 81, pp. 336-371

- Viertiö-Oja, H.; Maja, V.; Särkelä, M.; Talja, P.; Tenkanen, N.; Tolvanen-Laakso, H.; Paloheimo, M.; Vakkuri, A.; Yli-Hankala, A. & Meriläinen, P. (2004). Description of the Entropy™ algorithm as applied in the Datex-Ohmeda S/5™ Entropy Module, *Acta Anaesthesiologica Scandinavica*, vol. 48, no. 2, pp. 154-161
- Welling P. G. (1997). *Pharmacokinetics: Processes, Mathematics, and Applications*, 2nd ed. Washington DC: American Chemical Society.
- Zikov, T.; Bibian, S.; Dumont, G.A.; Huzmezan, M. & Ries, C.R. (2006). Quantifying Cortical activity During General Anesthesia Using Wavelet Analysis, *IEEE Transaction on Biomedical Engineering*, vol. 53, no. 4, pp. 617- 632

Artificial Neural Networks (ANN) Applied for Gait Classification and Physiotherapy Monitoring in Post Stroke Patients

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1. Introduction

1.1 Overview of the problem

Humans have an innate predisposition for ambulation (walking). The motor neuron stimulation involved in ambulation is generated by a natural neural network located in the spinal cord, known as the central pattern generator for locomotion. This network is strongly influenced both by super-spinal structures situated mainly in the hypothalamus and brainstem, and by signals coming from various types of peripheral receptors (Carter & Page 2009).

To facilitate research and analysis, free gait in humans is traditionally divided into phases and cycles. Each full gait cycle comprises two individual steps; a single step consists of a stance phase and a swing phase. The gait cycle includes a stage of single limb stance (when the body rests on a single lower extremity) and a double limb stance (on both lower extremities).

Kinematic gait analysis assumes a simplified, 15-segment model of the human body (feet, shins, thighs, forearms, upper arms, hands, head, torso, and pelvis) (Błaszczyk 2004). There are two kinds of basic parameters adopted for gait modelling and routine testing of ambulation in healthy and disabled individuals: spatial values of motion (including step length, velocity of the body mass centre, progressions of changes in joint angles, body mass oscillations) and dynamic values of gait mechanics (most often including ground reaction forces in 3 planes and the distribution of foot forces on the ground). These physical values are measured in parallel with bioelectric muscle activity (EMG), registered by surface electrodes as a subject walks (Perry & Burnfield, 2010).

Correct ambulation requires the precise integration of practically all the systems of the human body. When one of the elements, especially a motor organ, is damaged as a consequence of injury, degeneration, or deformation, this immediately finds reflection in divergences of the above parameters from normative values, which is in practice described as pathological gait (Perry & Burnfield, 2010). The field of clinical biomechanics therefore

seeks to identify normative values and limits for a broad range of parameters describing the mechanics of free gait, to help in detecting divergences in individual patients and in monitoring the process of gait rehabilitation following serious injury, surgical intervention, or neurological disorders – such as stroke.

In the remainder of Section 1 we will first outline the specifics of pathological gait classification in post-stroke patients and consider the link between brain lesion parameters and gait classification, then offer an overview of the application of artificial neural networks (ANNs) to physiotherapy in general and to gait classification in specific. In Section 2 we will present various findings from our own research dealing with the application of ANNs to post stroke gait classification (Kaczmarczyk et al., 2009; Kaczmarczyk et al., in preparation). In Section 3 we will more broadly discuss these findings in the light of other approaches to rehabilitation and in relation to other work dealing with ANNs.

1.2 Classification of pathological gait in post stroke patients

The traditional definition of stroke, devised by WHO in the 1970s, is a "neurological deficit of cerebrovascular cause that persists beyond 24 hours or is interrupted by death within 24 hours". Clinical symptoms of stroke depend on the type of stroke (ischemic caused by blockage in an artery that supplies blood to the brain, resulting in a deficiency in blood flow, and hemorrhagic caused by the bleeding of ruptured blood vessels in the brain), the size of the lesion, the location of the arterial blockage or hemorrhage, previous stroke damage, collateral circulation, and variability in areas supplied by individual arteries (Brust 2004). One consequence of stroke may be hemiparesis (weakness of one side of the body), which can have a profound effect upon the capacity for ambulation (Kinsella & Moran, 2008).

Recovery of functional ability after stroke is variable, with between 30% and 60% of subjects remaining dependent on others for some activities of their daily living (Duncan et al., 1994). Key functional tasks, such as regaining the ability to walk, have been identified by patients with hemiparesis as being of great significance in stroke rehabilitation (Bohannon et al., 1988, 1991).

The gait of post stroke patients is generally characterized by what is known as the Wernicke-Mann posture – the upper limb is adducted in the shoulder joint, flexed and rotated in the elbow joint, flexed in the brachioradial joint and the finger joints, whereas the lower limb is extended in the knee joint. However, a wide variety of gait deviation is observed in post stroke patients (Voigt & Sinkjaer, 2000; Burdett et al., 1988; Rodda et al., 2004). Many post stroke patients suffer from a foot drop problem during walking (an inability to move the ankle and toes upward). They might adopt different hemiplegic gait patterns with large variations, such as circumduction gait, high stepping pattern, etc. (Hermann 1987, Hong-yin et al., 2009). This variety of gait deviation poses a clinical problem, making it difficult to deliver targeted treatment. Clinical practice has therefore sought to develop methods for the appropriate (and early) classification of post-stroke gait dysfunction – the general idea being that once a given patient has been correctly classified as having a certain type of gait dysfunction, therapists may provide adopt a strategy of treatment best suited for their rehabilitation.

The observation-based classification of gait in neurological patients described by Hermann (1987) remains the primary method of gait diagnostics used for clinical purposes, supplying information that facilitates the qualitative evaluation of a given patient's dysfunction. Seeking to ensure better uniformity of evaluation methods, various forms of qualitative or

point-based ways of recording observations of pathological gait have been devised – such as various scales, questionnaires, gait evaluation indexes, etc. However, such widely used observation-based methods of classifying pathological gait quality, not being underpinned by objective measurement of mechanical gait parameters, are unfortunately fraught with a large degree of subjectivity and their effectiveness varies greatly depending on how experienced the observer is and on how systematically the criteria are applied. This shortcoming has given rise to various attempts at providing formal descriptions and classifications of various types of pathological gait.

In terms of post stroke dysfunction, numerous authors (Mulroy et al., 2003; Olney & Richards, 1996; Knutsson & Richards, 1979; Perry et al., 1995; Kramers de Quervain et al., 1996; Kinsella & Moran, 2008) have attempted to identify homogeneous subgroups of post stroke walking patterns. Knutsson & Richards (1979) used EMG signals to distinguish three types of pathological gait. Kramers de Quervain et al. (1995) used the Mahalanobis distance statistical technique on five temporal distance parameters to distinguish four gait patterns. A similar study was carried out by Mulroy et al. (2003), analyzing gait based on temporal distance and sagittal plane joint kinematics, using a non-hierarchical cluster analysis to categorise four subgroups of walking patterns. Kinsella & Moran (2008) used hierarchical cluster analysis to identify three gait patterns in hemiplegics with equinus deformity of the foot based on temporal distance parameters and joint kinematic and kinetic measures in the sagittal and coronal planes.

Overall, irrespective of the parameters measured or technique used, this line of investigation seems to show some convergence of results: suggesting that post stroke patients can be usefully classified into more or less three-four types of dysfunctional gait based on quantitative data.

Wong et al. (2004) proposed another, simple gait classification technique based on evaluating foot position at ground contact. This study looked at 65 post stroke patients and distinguished between three gait types by analyzing the motion of the point of application of the resultant reaction force on the foot. Wong et al. (2004) found a correlation between the results of their classification, the neurological condition of the patients, and the temporal-spatial data obtained from kinematic analysis. However, they did not attempt to find a link between foot position at ground contact and the progression of lower limb angle values over the gait cycle. The work of Wong et al. (2004), one of the few papers in the literature attempting to classify post stroke patients based on the analysis of foot motion during ambulation, served as the inspiration for the methodology used in our research.

None of these studies discussed above attempted (as we have) to categorize walking patterns based on the full progression of joint angle changes as a function of the gait cycle in post stroke subjects – an avenue of research that is made possible by the use of Artificial Neural Networks (ANNs), as described in detail in the next section below.

1.3 Association between brain lesion and gait classification

Relatively little is known about the specific association between the parameters of the brain lesion causing a stroke and the gait type found in post stroke patients. Elucidating this relationship could enhance our understanding of the neural circuitry involved in locomotion and could have important clinical implications, underscoring the need to prioritize gait retraining for patients in the early stages after stroke.

In view of this need for early rehabilitation, many studies (Jørgensen et al., 1995; Viosca et al., 2005; Dominkus et al., 1990) have discussed the association between motor recovery and

the most predictive factors that can be identified. However, none of these authors attempted to identify any link between CT scan parameters and the characteristic gait patterns seen in post-stroke patients – as we have explored in the work described below.

The various types of pathological gait have a neurophysiological basis, due to post stroke changes in the brain. Clinical evidence suggests that the site of damage of the sensorimotor cortex influences the pattern of motor deficits (Glymour et al., 2007). This raises the possibility of exploring CT scan parameters as a way to predict gait patterns. Studies mostly evaluated the association of only one parameter of brain lesion with motor and functional outcomes after stroke. While several studies have suggested that brain lesion parameters correlate with final outcomes (Bear & Smith 2001; Dominkus et al., 1990; Alexander et al., 2009; Laufer et al., 2003; Pérennou et al., 1999; Turney et al., 1984; Kwolek & Splawinski 1996), other studies have found no such association (Chen et al., 2003; Viosca et al., 2005; Binkofski et al., 2001; Nakayama et al., 1994; Chae et al., 2000). The reason for this controversy might be that outcomes actually correlate with some combination of brain lesion factors together, rather than individually. Chen et al. (2000) showed that recovery and functional outcomes correlate with “brain lesion profiles” that combine two factors: size and location. However, all the cited authors have concentrated on finding an association between brain lesion parameters and motor recovery and functional outcome in hemiplegic stroke patients, rather than objectively identified gait patterns – this led us to further investigate the possibility of the latter link using Artificial Neural Networks (ANNs), as described in detail below.

1.4 Introduction to Artificial Neural Networks

Artificial neural networks (ANN), also known as connectionist systems or parallel distributed processing models, are computer-based, self-adaptive models that were first developed in the 1960s, but gained broad popularity only in the 1980s after the development of the backpropagation algorithm by Rumelhart et al. (1986). It is hard to ascertain today what sort of motivation initiated the development of neural network theory, but we can assume that a fascination with the human brain was a fundamental factor. Neural network research can be traced back to the work of McCulloch and Pitts (1943), who put forward the first formal arithmetic-logical model of the neuron.

ANNs are simulations of the nervous system: a computational model consisting of an interconnected group of artificial neurons, often situated in distinct layers, which can be used for processing information. An ANN system is adaptive, responding to information that flows through the network during a learning phase. In a testing phase, ANNs generate on output signals as a response to previously unknown inputs. ANNs offer an extraordinarily flexible tool for inductive, nonlinear modelling of complex input-output relationships and finding complex patterns in data. The effectiveness of generalization can be expressed as the ratio of correctly recognised input patterns to all of the presented patterns in the test phase. The advantage of ANNs is that they can process large numbers of data simultaneously and because of their internal structure the pieces of data do not have to be isolated from each other, preserving the inherent relationships amongst the data set. The attractive features of simultaneous data handling and the concept of contextuality make ANNs potentially useful tools in the automated recognition of various gait patterns.

Three types of neural network architecture can be distinguished: feedforward networks, recurrent networks, and cellular neural networks. In the feedforward type of network

mainly discussed below, the information moves in only one direction – forward – from the input nodes, through any hidden layers of nodes to the output nodes, and there are no cycles or loops in the network. In a recurrent network, in turn, some connections between units form a directed cycle, enabling the network to exhibit dynamic temporal behaviour.

1.5 Application of Artificial Neural Networks in physiotherapy

The use of neural networks in medicine and rehabilitation has grown enormously in the past decade (Carter 2007). They have been applied as statistical tool to solve problems including the following: i) prediction of diagnosis, e.g. for several types of cancer (Rogers et al., 1994), ii) prognoses, e.g. for heart disease (Katz et al., 1993), iii) the interpretation of diagnostic tests, e.g. for pancreatic enzymes (Kazmierczak et al., 1993), and iv) decision support, e.g. (Doornewaard et al., 1999).

Neural networks are especially useful if the main goal of building a model is to predict outcomes for new cases. Grigsby et al. (1994) attempted to predict functional outcome, length of stay, and cost for patients with hip fractures who were undergoing inpatient rehabilitation. This study was one of the first to apply ANN methodology to the analysis of patients undergoing rehabilitation. The input data included age and selected Function Independence Measure scores at admission. An accurate prediction was defined as a value within $\pm 15\%$ of the actual outcome: the functional outcome model predicted the mean rating of the 13 FIM motor items score at discharge with an accuracy of 86%; the length of stay model prediction was 87% accurate, the cost model 91% accurate. Oczkowski & Barreca (1997) attempted to predict functional outcomes and discharge placement for moderately impaired stroke patients. The input data used to predict outcome included age, days since stroke, motor recovery of leg strength and postural control, the presence of sensory loss, neglect, a care-giver, and the admission FIM score. The network demonstrated an accuracy of 88% for the prediction of discharge FIM score, and 75% for the prediction of discharge placement. Both studies used separate training and test sets to evaluate model performance and back propagation methods for error reduction. Ohno-Machado et al. (1999) created neural network models to predict early mortality and ambulation for patients with spinal cord injuries. Their model included 15 variables for example, day from injury to admission, age gender level of presented neurologic function, and American Spinal Injury Association (ASIA) impairment score (Rowland et al., 1998), achieving an 97% rate of accuracy.

Neural networks have also been applied to the study of functional electrical stimulation (FES). FES is one of the most used technologies for restoring the functions of patients affected by neurological pathologies. By electrically activating the muscular system, FES is increasingly recognized as a method of therapy and treatment for subjects impaired by stroke, multiple sclerosis and cerebral palsy (Popovic et al., 2002; Galen & Granat 2002). Electrical stimulation generates control signals that effectively activate paralyzed upper limb muscle for standing up, walking and maintaining body balance. A radial-basis neural network was used by (Popovic, Radulovic et al., 2003) for controlling FES in eight muscles in six subjects with paraplegia as a result of spinal cord injury. Their input data consisted of a function of hip, knee, ankle angles, flexors and extensors activation and ground reaction force (GRF) during the gait cycle in able-bodied individuals. Another study by (Muniz, Liu et al., 2010) evaluated three different models, including a probabilistic neural network (PNN) for discriminating between normal and Parkinson disease subjects in terms of ground reaction force during walking. Their experimental protocol included medications

and deep brain stimulation of the subthalamic nucleus. They found that neural networks, as well as the other models, showed high performance indexes for classifying ground reaction forces of normal and Parkinson subjects.

ANNs can be also used for gleaning a better understanding of the mechanisms of motor control. Bernabucci, Conforto et al. (2007) simulated a simplified version of the biomechanical arm model, constructed with two mono-articular pairs of muscles for each joint (elbow and shoulder) and a bi-articular third pair of muscles connecting the two joints. The proposed system was only able to make ballistic plantar movements. As a crucial part of the system, ANNs were engaged for synchronizing the muscle activation during arm movement. Such theory-based investigations are important because they commonly open up wide fields of application. For instance the system developed by Bernabucci, Conforto et al. (2007) can be adapted to FES and could enhance paretic patients' capacity to control their arm movements with reduced effort during therapy.

Encouraging results achieved in overcoming simple functional limb substitution (Liberson et al., 1961) and successful therapy both in lower (Bogataj et al., 1995) and in upper limb movements (Wang et al., 2002) have recently led to the development of FES-assisted rehabilitation programs for hemiplegic patients (Gritsenko & Prochazka 2004; Goffredo et al., 2008). Restoration of upper limb movements in post stroke patients is one of the keystones of rehabilitative practices. Rehabilitation of arm movements is usually more difficult than for the lower extremities (Goffredo et al., 2008). Goffredo et al. introduced a non-invasive FES-assisted rehabilitation system for the upper limb, called smartFES (sFES). The system includes a markless motion estimation algorithm and a biologically inspired neural inverse dynamics model, fed by the kinematic information that drives a biomechanical arm model, which could be used to drive an sFES. The algorithm is based on the design of a specific ANN, which works on a two-step basis: ANN1 was used to estimate the dynamics of shape and ANN2 to generate the muscular activation that will make the artificial muscles produce the forces necessary to drive the arm model. ANN1 is a two-hidden-layers network. The network inputs consisted of horizontal and vertical components of position, velocity and acceleration of all the contour points n ($n=1,...,N$) in the current frame ($i-1$), which means the number of the input neurons is $N*6$. The ($N*2$) outputs are given by the horizontal and vertical components of the position of the contour points in the subsequent frame i . ANN2 is also a two-hidden-layer network, fed by four inputs, representing the coordinates of the starting and the ending points of the movement to be generated. The output layer yields: the time of contraction of the muscle pairs of both the shoulder and the elbow joint, together with the duration of the overall neural activations. In the stimulation of movements of a synthetic arm with a length of ± 20 cm, the model has shown an unbiased angular error, and a mean position error of about 1.5 cm, thus confirming the ability of the system to reliably drive the model to the desired targets.

A similar study, this time concerning the lower limb, was undertaken by Chen et al. (2004). Hemiplegic patients very often suffer from the drop-foot problem. Chen et al. (2004) established a neural network and fuzzy feedback control FES system for adjusting the optimum electrical stimulating current to control the motion of an ankle joint. Three hemiplegics and a healthy individual participated in their study. Their three-level neural network structure used back-propagation. The network was used to estimate the current volume input into the body for electrical stimulation of the tibialis anterior of the hemiplegic patients. The fuzzy controller solved the nonlinear problem by compensating for the motion trace errors between neural networks control and actual system. Locomotion of the

hemiplegics was effectively improved by applying the neural network to electrical stimulation therapy.

On the whole, therefore, we can conclude that ANNs have proven to be effective tools in medicine and rehabilitation.

1.6 Gait classification using ANNs

Artificial neural networks have a lot to offer to gait analysis in particular, facilitating the study of complicated gait variable relationships that have traditionally been difficult (Chau, 2001). A standard three-layered feedforward network works as a universal function approximator (Lapedes & Farber, 1988). This property allows one to model any relationship among gait variables, provided adequate data are available and the requisite network complexity is computationally feasible.

Another notable aspect of neural networks is that they can analyze great amounts of gait data, as evidenced in the studies of Holzreiter & Kohle (1997) and Lafuente et al. (1997). Other benefits to gait analysis include their inherent non-linear mapping ability, demonstrated aptitude at capturing temporal dependence (Savelberg & Herzog, 1997), and short processing time (Miller, 2009). Another advantage is the ability to capture patterns in the data within their internal parameters known as weights and biases which may influence gait patterns (Hinton, 1992).

A significant number of papers (Holzreit & Kohle, 1993; Barton & Lees, 1997, 1995; Gioftsos & Grieve, 1995; Lafuente et al., 1998; as well as our own research outlined in the next section) have shown ANNs to be useful in distinguishing gait patterns. Recent efforts generally fall into three categories of application: i) classification of human gait, ii) biomechanical modelling, iii) prediction of gait variables and parameters (Chau, 2001). It is the first and third of these applications – classification and prediction – that we will explore in detail below, reporting the results of our own research.

One of the first attempts at classifying gait in patients using ANNs was made by Holzreiter and Kohle (1993), using the standard network structure (i.e. with one hidden layer) and showing a 95% rate of successfully distinguishing gait patterns of healthy individuals from those of physically disabled individuals. Gioftsos & Grieve (1995) investigated the application of ANNs (again with one hidden layer) to the recognition of temporal gait parameters associated with altered gait patterns. Their network had a mean accuracy of 73% in correctly recognizing the unknown patterns. Similar studies were undertaken by Barton & Lees (1995, 1997) – their first study expanded the ANN classification to three categories and achieved a successful classification rate of 77% to 100% based on foot position, whereas their second study concluded that angle changes in the hip and knee joints offer a good basis for automatic identification of gait types (with an average correct classification rate of 83.3%). In both tests, Barton & Lees (1995, 1997) used a complex neural network with two hidden layers concealed between the input and output cells. Lafuente et al. (1997) reverted to the standard network structure (i.e. with one hidden layer); here data concerning gait rhythm, speed and five kinetic values were fed into the neural network, based on which four gait types were correctly distinguished at a rate of 80%. The review of Miller et al. (1992) described other neural networks applied in different areas of investigation, with similar results. In turn, Aminian et al. (1995) used accelerometer to recognize the speed and incline of walking. The neural networks were first "trained" by known patterns of treadmill walking. Then the inclines, the speeds, and the distance covered during overground

walking (outdoor circuit) were estimated. The results show a good agreement between actual and predicted variables. All the above studies showed rates of correct classification within the 77%-100% range, reaffirming the great potential of neural networks in distinguishing gait patterns.

The various approaches and findings of these studies are summarized in Table 2.

Author	Measured parameters	Inputs	Network type	Outputs	Conclusions
Holzreiter & Kohle (1993)	Ground reaction forces	128 FFT coefficients	Feed forward (one hidden layer)	1) Able-bodied gait 2) Pathological gait	Up to 95% accuracy
Barton & Lees (1995)	Maximum pressure prints	1316 maximum pressure values	Feed forward (two hidden layers)	1) Healthy feet 2) Pes cavus 3) Hallux vagus	77-100% accuracy
Gioftsis & Grieve (1995)	Duration of the double, right and left support phases	three temporal measurements	Recurrent network	Three walking conditions; 7 walking speeds	73% accuracy
Barton & Lees (1997)	Hip and knee angles	30 FFT coefficients	Feed forward (two hidden layers)	1) Normal walking 2) 20 mm thick sole 3) 3.5 kg mass	83.3% accuracy
Lafuente (1997)	Cadence Velocity 5 kinetic parameters	Kinetic and kinematic parameters	Feed forward (one hidden layer)	1) Healthy 2) Ankle arthrosis 3) Knee arthrosis 4) Hip arthrosis	80% accuracy
Aminian et al. (1995)	10 parameterizations of accelerations of the trunk and right heel	Forward, vertical and lateral accelerations	Two feed forward networks	Speed and incline	

Table 1. Summary of studies applying neural networks for gait classification

2. Example application of ANNs in post stroke gait rehabilitation

Overall, therefore, ANNs have been shown to be a promising avenue of research in post stroke gait rehabilitation. As an illustration of this, in the remainder of this chapter we will describe in detail the findings of our own research in this direction, which may be viewed as a single study reported in two separate papers, Kaczmarczyk et al. (2009, in preparation).

As we noted above, the ability to correctly classify gait patterns in post stroke patients is now recognized as a crucial aspect for improving the rehabilitation process and providing targeted therapy tailored to the patient's individual needs. ANNs offer an advantage over

existing methods for characterizing post stroke gait types – which are mainly based on verbal descriptions, hence their use in clinical practice involves considerable risk and is strongly dependent upon the experience of the observer. At the same time, the use of traditional statistical methods, based on analysis of min/max values, may involve a certain error: two individuals may exhibit similar ranges of motion, yet have significantly different progressions of joint angle values over the course of their gait cycles.

The aim of our work, reported in two separate papers but described in conjunction in the remainder of this section, was to compare methods for classifying post stroke patients into gait pattern types, taking as a point of departure Wong et al.'s (2004) three types of foot position. The methods we considered were: 1) min/max joint angle values and 2) the full progression of joint angle changes analyzed with ANNs, concluding that the ANN method yielded superior results (as reported in Kaczmarczyk et al., 2009). We also used these ANN gait classification predictions as a reference against which to test the association between the simultaneous impact of four brain lesion parameters and different pathological gait patterns (as reported in Kaczmarczyk et al., in preparation)

2.1 Materials and methods

2.1.1 Participants

Seventy-four hemiplegic patients (31 females and 43 men) participated in this study; their characteristics are shown in Table 2. The inclusion criteria were: 40-70 years old, first incidence of hemorrhagic or ischemic stroke within the past six months, capable of walking independently (without any orthopedic walking aid), with no other disorders of an orthopedic, rheumatologic, etc., nature that could affect gait kinematics, with no cognitive disorders and with consent from their physician and physiotherapist for their participation in this study.

Group	Height [cm]	Body mass [kg]	Age [years]	Time since stroke [weeks]	Type of stroke
Women (n = 31)	162 ± 5 (150 ÷ 172)	69,6 ± 11,6 (51 ÷ 105)	55,6 ± 9,4 (40 ÷ 70)	7 ± 2,9 (4 ÷ 12)	I = 25 H = 6
Men (n = 43)	173,8 ± 5,2 (164 ÷ 186)	78,7 ± 9,9 (59 ÷ 100)	58,9 ± 9,3 (40 ÷ 70)	7,2 ± 3,0 (3 ÷ 12)	I = 30 H = 14

I – ischemic, H – hemorrhagic (from Kaczmarczyk et al., 2009)

Table 2. Average (±SD) values and ranges (min÷max) for selected characteristics of study participants

Both the men and women subjects had experienced their stroke incident at a similar timeframe (7 months prior to the study), with the scatter coefficient of this parameter in both groups being 41%. One difference between the male and female groups was that hemorrhagic stroke accounted for 32% of the cases in the male group (n = 44) involved but 19% of the cases in the female group (n = 31) – consistent with WHO data indicating that hemorrhagic stroke accounts for about 20% of all stroke incidents.

Approval for the experiment was sought and obtained from the Institute's Research Ethics Commission and all the participants provided written informed consent.

2.1.2 Data collection and processing

Firstly, gait analysis was performed once for each subject using the Ariel Performance Analysis System (APAS). Participants walked unassisted at a self-selected speed along a 10 m walkway being recorded by two analogue cameras set perpendicularly to one another, 7 m from the subject (one standard setup for 3D analysis). 18 markers were placed on each patient, at selected characteristic points (following a standard protocol for full body motion analysis using the APAS system): the base of the first metatarsal bone, the calcaneal tuberosity, the lateral malleolus, the articular space of the knee joint, the greater trochanter of the femur, the radiocarpal joint, the elbow joint, the greater tubercle of the humerus, the jugular notch of the sternum and the root of the nose to define joint centres and the axes of rotation.

Gait was evaluated quantitatively, based on local extremes (min/max) angle values and absolute angle values in the leg joints as a function of time. Because foot position was considered a dependent variable, the values collected were for the knee joint and for the hip joint in the frontal and sagittal planes.

Secondly, all the patients had CT brain scans performed within 7 days after stroke. The images were evaluated in line with accepted diagnostic criteria, following American Stroke Association guidelines. The scans were performed without using a contrast medium, with slice thickness of 8 to 10 mm. A stroke lesion is treated as being a hypodense ischemic area situated in the cortex, subcortically, or in the deep structures of white or grey matter, according to vascularization from one of the cerebral arteries. Lesion number was evaluated for each patient as follows: 1) one, 2) two, 3) multiple. Lesion size was classified according to Brott et al. (1989) as: 1) lacunar – up to 1 cm, 2) small lesion – less than $\frac{1}{2}$ lobe but more than 1 cm, 3) medium-size lesion – between $\frac{1}{2}$ and 1 lobe, 4) large lesion encompassing more than 1 lobe. Lesion location was defined according to the criteria of Damasio (1984) by vascularization area: 1) middle cerebral artery, 2) anterior cerebral artery, 3) posterior cerebral artery, 4) basilar artery, and by structures occupied: 1) deep structures, 2) frontal lobe 3) temporal lobe, 5) occipital lobe, 6) cerebellum, 7) brainstem.

2.1.3 Statistics

The data from the study were put through detailed statistical analysis using the STATISTICA software, adopting a significance level of $\alpha = 0.05$. The methods used for gait classification were discriminant function (DF) analysis and ANNs.

DF analysis was used to classify patients and to identify parameters that make a significant contribution to distinguishing between gait types in post stroke patients. To illustrate the progression of the analysis, the forward stepwise method was used.

ANNs were used to assign each case, as represented by the corresponding set of input data, to one of the selected gait pattern types. The input variables were discrete (continuous) values on the progression of knee and hip angle changes. The input variables were joint angles previously normalized for the gait cycle (expressed in percent), resampling to 50 points. The input signals from each of the subjects, a total of 74 cases, were coded on a scale from 0 to 1. The output cell was a dependent variable of a nominal value (GROUP), represented using the "one-of-N" technique. In the "one-of-N" coding, one neuron corresponds to only one of 3 possible values of the GROUP variable, containing information about 3 types of gait in post stroke patients. The classification was implemented with the STATISTICA™ v7.0 Neural Networks software, using the Multi Layer Perceptron (MLP)

network type, to establish a network of 51 input cells, one hidden layer of 27 cells and one three-level output cell (MLP 51:51-27-3:1). For the three different network-creation subsets, i.e. the training, validation and test subsets, different quality measures were selected. Cases (subjects) were assigned to the individual subsets automatically and randomly.

To identify the weight, in the range of 0 to 1, for each of the predicting values, one of the multivariate exploratory techniques was applied: analysing the robustness of the classification tree for the three gait types as dependent variables and for the four independent variables, lesion size and lesion location (as opposed to lesion number or lesion type). The classification tree was built performing discriminant-based split, bottom up running, generalizing Chi-square goodness of fit measures. Prior and misclassification cost were specified as equal, estimated from the data.

2.2 Results

We performed two stages of gait assessment using quantitative methods. The first such stage was based on local min/max angle values in the knee joint and the hip joint in the sagittal and frontal planes during the stance phase. This stage of discriminant analysis showed that none of these variables were significant ($p > 0.05$) in predicting classification into groups (Tab. 2).

Summary of discriminant function analysis (local min/max)						
Variable	Wilks Lambda	Partial Wilks	F to remove (2.63)	p-level	Toler.	1-Toler. (R-Sqr.)
$\alpha_{\min}$	0.839	0.955	1.554	0.219	0.204	0.796
$\alpha_{\max}$	0.807	0.993	0.232	0.793	0.194	0.806
$\beta_{\min}$	0.813	0.986	0.478	0.622	0.224	0.776
$\beta_{\max}$	0.811	0.988	0.404	0.669	0.227	0.773
$\delta_{\min}$	0.831	0.964	1.243	0.295	0.196	0.804
$\delta_{\max}$	0.859	0.932	2.394	0.099	0.202	0.798

Table 3. Significance of min/max parameters in building a model of gait types in post stroke patients ($n = 74$) (from Kaczmarczyk et al., 2009)

The Lambda Wilks values in the second column close to 1.0 indicate that the variables do not make a significant contribution to discriminating between groups. This is because the proposed variables do not individually carry sufficient information to build a good model of three gait type groups based on the proposed kinematic gait parameters (extreme range values of lower limb joints).

In the next stage in our research, we tested the use of discriminant analysis for gait classification. The results are shown in Table 4.

This unsatisfactory result of classification for discriminant analysis (below 50% both for each gait classification group and overall) can be explained in terms of the fact that the peak values characteristic for a specific gait phase may be distorted by random variations in extreme values or by the data processing itself, especially the filtering and smoothing of raw values in kinematic analysis. That is why our next step was to check whether it was possible to analyze the entire progression of angle values as a function of the cycle of a single step.

GROUP	Classification matrix (min/max)			
	Percent correct	forefoot p=.405	flatfoot p=.3562	heel p=.2329
forefoot (n=30)	66.7	20	8	2
flatfoot (n=26)	38.5	13	10	3
heel (n=18)	38.9	3	8	7
Total	48.3	30	26	18

Table 4. Percentage breakdown of correct classification of post stroke patients into gait types (n = 74) (from Kaczmarczyk et al., 2009)

The next stage of our research, therefore, involved classifying the gait of patients based on the progression of angle values in the knee and hip joints in the frontal and sagittal planes using artificial neural networks. For ANN training, the input cells were the successive values of the joint angle progression $\alpha=f(t)$ during gait, sampled with step $k = 2$. Networks were built manually (i.e. not using the automatic option), and multi layer networks were ultimately selected for training. The network that most faithfully recreated the real values involved angle changes in the knee joint during walking. Figure 1 illustrates the configuration of the network we used for classifying post stroke patient gait on the basis of the progression of knee joint values during walking – a feedforward network using a single layer of hidden nodes. A network of identical configuration was built for post stroke gait classification based on the progression of hip joint values in both the frontal and sagittal planes.

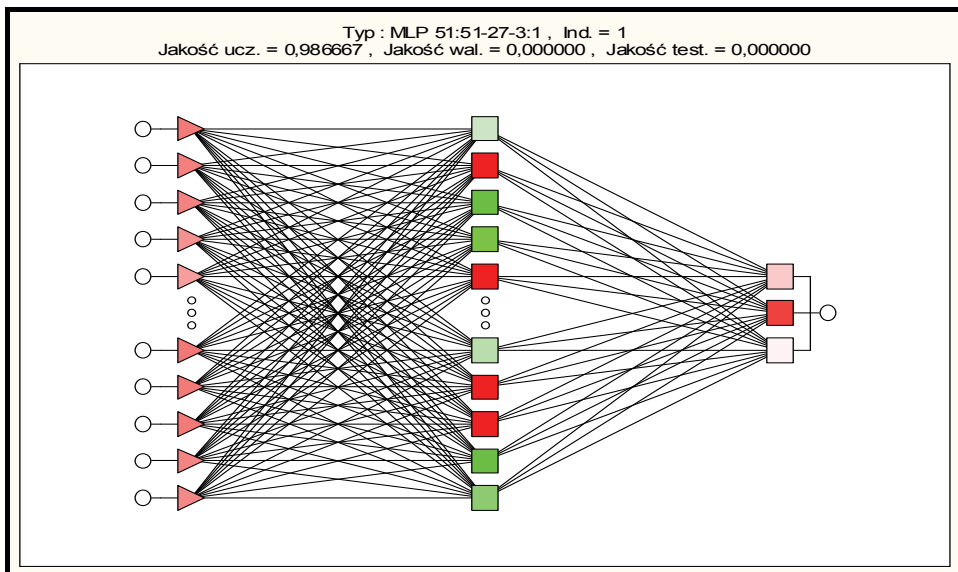


Fig. 1. Configuration of ANN for post stroke gait classification based on the progression of knee joint values during walking (from Kaczmarczyk et al., 2009)

The main measure of the quality of the configured network during the learning process were errors in the training, validation, and test sets. The sets we used were of equal size and each constituted 1/3 of all patients; the selection of patients for each of the three sets was random. Table 5 shows data concerning the quality of the best sets accepted for further analysis.

Joint	Training error	Validation error	Testing error
Knee	0.07	0.00	0.00
Hip sagittal	0.01	0.03	0.19
Hip frontal	0.39	0.00	0.00

Table 5. Error report for training of MLP51:51-27-3:1 neural network configured for 3 joint angles (from Kaczmarczyk et al., 2009)

The critical error level is assumed to be 0.1, and the smaller the error value the better adapted the network is. The data in Table 5 shows that a fully satisfactory network was built solely on the basis of knee joint angle changes. For the other two hip joint angles, the program was unable to create a network so well adjusted to the measurement data set. As a result, classification of individual patients based on information from hip joint angle progressions is not as accurate as for the knee joint.

The outcome of the ANN classification for individual joints – which we believe to be impressive given the unsatisfactory results obtained using the quantitative methods described so far – is presented in Table 5. Classification based on changes in knee joint angle values as a function of time placed all subjects correctly for all three gait types. Analysis of hip joint angle values in the sagittal plane placed all the subjects into the appropriate groups for two gait types (with a rate of nearly 97% for the third). For hip joint angle values in the frontal plane, successful classification rates were around 95% for two gait types, and 85% for the third gait type.

Joint	forefoot (n= 30) correct [%]	flatfoot (n = 26) correct [%]	heel (n = 18) correct [%]
<i>Knee</i>	100	100	100
<i>Hip, sagittal</i>	96.7	100	100
<i>Hip, frontal</i>	96.7	84.6	94.4

Table 5. Classification obtained using a neural network, with knee and hip joint angle values during gait as input parameters (from Kaczmarczyk et al., 2009)

The above findings were reported in Kaczmarczyk et al. (2009), drawing the major conclusion that ANN analysis is superior to qualitative variable analysis for classifying post stroke patients' gait patterns into three gait types, as well as superior to the analysis of min/max joint angle values. Next, as reported in Kaczmarczyk et al. (in preparation), we used this ANN-derived classification method (which placed post stroke patients into one of three groups – F1 (*forefoot*), F2 (*flat foot*), H (*heel*)) as a reference gait classification against which we looked at the influence of four individual brain lesion parameters on the nature of gait in the same set of early stage post stroke patients. The results of comparing CT scan parameters against this ANN-derived gait classification are presented in Table 6.

Observed	Predicted by CT scan parameters			Correct classified (%)
	<i>F1(forefoot)</i>	<i>F2 (flat foot)</i>	<i>H (heel)</i>	
<i>F1(forefoot)</i> (n=30)	22	4	4	73.3
<i>F2 (flat foot)</i> (n=26)	5	18	3	69.2
<i>H (heel)</i> (n=18)	2	3	13	72.2

Table 6. The relationship between CT scan and gait pattern classification (from Kaczmarczyk et al., in preparation)

The best correspondence between CT scan classification and the reference gait pattern from the previous study (listed as "observed" in Table 6) was seen in the *F1(forefoot)* group (of the 30 individuals to be correctly placed into this group, CT scan parameters classed 22 individuals correctly, 8 incorrectly). In general, all three groups showed similar levels of correct classification based on CT scan parameters, with the overall average result of 71.3% correct classification for all patients – a result that may be described as relatively good.

Our next analysis looked into the classification trees techniques. The analysis followed a pairwise principle, comparing each pair to identify how each of the individual CT parameters contributes to classification into the different gait types. Table 7 lists the results of the comparisons – the values stated are weights that range from 0 to 1. The comparisons indicate that the influence of the individual parameters varies and depends on the configuration (accompanying parameter). The parameter whose contribution is analyzed is listed in the first column of Table 7, and the accompanying parameters are listed in order in the first row of the table 7.

	Number	Size	Location	Type
Number		0.60	0.56	0.79
Size			0.33	0.8
Location				0.89

Table 7. The individual CT parameters' contribution to classification into the three different gait types (from Kaczmarczyk et al., in preparation)

The data in Table 6 indicate that considering only two stroke parameters derived from CT scans may be used to identify the importance (influence) of each CT scan parameter for the type of patient gait. Comparisons of all parameters against each other indicate a clearly lower importance of type of stroke, whereas for "number-type", "size-type", and "location-type" comparisons the first parameter is always of dominant importance in predicting the type of gait in the future course of rehabilitation.

The results of this analysis seem to indicate that to obtain a full picture of the influence of each of the parameters on the result of gait classification, analysis should be carried out considering all the CT scan parameters simultaneously. Therefore, next the individual significance of the four lesion geometry parameters was considered. Of these four, lesion size and lesion location (as opposed to lesion number or lesion type) appeared to influence gait pattern in post stroke patients most strongly (Fig. 2).

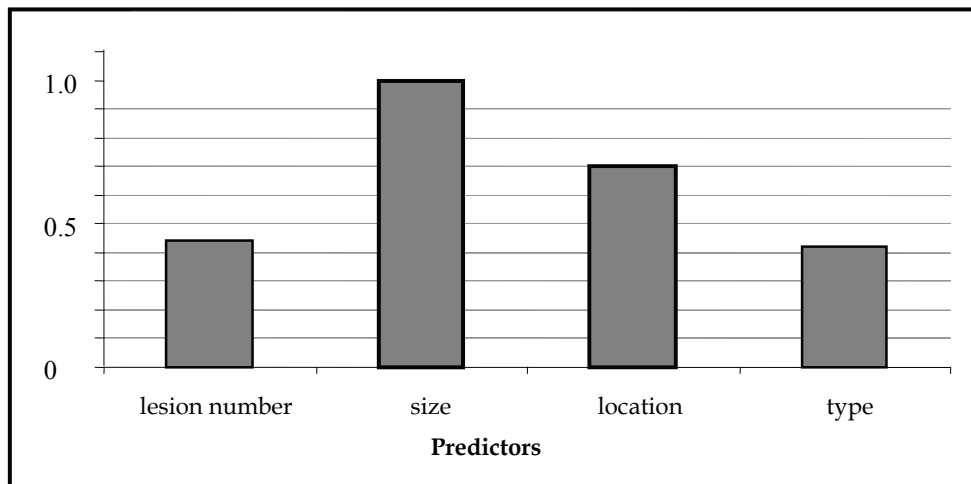


Fig. 2. Weight of the influence of individual CT parameters on the type of post stroke gait disorders in patients studied (from Kaczmarczyk et al., in preparation)

These main findings obtained by recursive partitioning methods suggest that CT scan parameters, specific lesion size and lesion location may serve as a useful early gait classification strategy, nearly as accurate as the ANN method previously presented by Kaczmarczyk et al. (2009), which provided a useful reference classification.

In the next section, we will discuss these various findings of ours in the broader context of other studies.

3. Discussion

The process of post stroke rehabilitation is protracted and costly. There are many papers arguing that post stroke patients engaged in a rehabilitation program achieve higher degrees of independence than patients without rehabilitation, and that the former do not acquire pathological gait patterns (Prescott et al., 1982). Even so, the effectiveness of such programs continues to raise many doubts. For the rehabilitation process to be effective, it has to be initiated as early as possible and properly targeted. One of the criteria for including patients in our research, reported above, was the relatively short time that had elapsed since their incidence of stroke.

Functional reorganization after stroke is a commonly-hypothesized phenomenon. It is thought to play an essential role in the functional recovery that occurs during the first 3 to 6 months after stroke through the recruitment of alternative neural paths (Fujii & Nakada 2003). This opinion is confirmed by the results of research on various populations of post stroke patients using such tests and scales for evaluating neurological condition as the Fugl-Meyer test and the Bartel scale. Using them it has been shown that the largest improvement in functional condition is achieved during the first 6 months after the occurrence of the stroke incident (Kwolek i Sławiński 1996). After that the process of improvement slows down, although in certain cases progress has been noted even up to 5 years later (Bach-y-Rita 1981). Richards & Olney (1996) confirmed previous research that the time of rehabilitation onset is important, showing at the same time that rehabilitation during the

first 6 weeks is the most effective. All these findings serve to confirm the premise of early rehabilitation.

Accurate gait classification in post stroke patients could aid the effectiveness of therapy. Performing qualitative gait evaluation in post stroke patients (especially early stage post stroke patients) based on kinematic parameters, ground reaction force and muscle activity is not an easy procedure in clinical practice. Wong et al. (2004) therefore proposed a simple classification of post stroke patients based on the analysis of foot movement during gait. Their classification corresponds to the criterion we adopted and verified with cluster analysis, based on evaluating foot position at the onset of the single stance phase.

Wong et al. (2004) concluded that patients with hemiplegia have a tendency not to place their heel on the ground at the moment of first foot contact and to experience disturbances in the propulsion mechanism. Depending on the degree of neurological deficiency, they are observed to shift the trajectory of the centre of pressure towards the front of the foot, which is consistent with our results. Different results were obtained by von Schroeder et al. (1995), who observed only one case involving first contact of the toes with the ground, the remaining subjects seen to position their whole foot. This was confirmed by Karsznia et al. (2004), who in turn observed flatfoot position in their subjects both during first contact and during the propulsion stage. We can surmise that in both these experiments this finding resulted from the small degree of group differentiation in terms of neurological deficiency, although the authors did not present detailed data on this in their papers. However, Karsznia (2004) attempted to identify a link between foot position on the ground and angle progression in leg joints during the gait cycle.

Burdett et al. (1988) found certain leg joint angle values at certain gait phases to be most important qualitative traits distinguishing gait in post stroke patients from that of able-bodied subjects. Other authors concluded that the greatest differences between pathological and normal gait involve the maximal and minimal angles in the knee and ankle joints during the toe-off stage and at first foot contact with the ground (Knutsson & Richards 1979, Intiso et al., 1996). With respect to post stroke patients, in particular, impressive classification results (98%) were obtained by Mulroy et al. (2003), utilizing the maximal and minimal values of only three kinematic parameters. Kim and Eng (2004) attempted to classify gait in post stroke patients using the extreme values of angles in selected leg joints, successfully distinguishing two types of gait.

Our study, however, did not find min/max angle values of the leg joints to serve as a useful indicator for classifying post stroke patients into gait types, showing a correct classification rate of below 50 percent. The unsatisfactory result in our case might be explained by the fact that peak values characteristic for a specific phase of pathological gait are subject to random fluctuation. The very procedure of filtering and normalizing the registered positions in the kinematic analysis could also be a source of additional error. In view of this unsatisfactory result, the next stage of our study analyzed the full progression of joint angle changes as a function of the gait cycle, using artificial neural networks as the method of analysis.

One of the first attempts at classifying gait in patients using ANNs was made by Holzreiter and Kohle (1993). Holzreiter and Kohle measured two successive ground reaction forces during normal walking from 94 subjects with various lower limb conditions, including calcaneus fracture and limb deficiencies. The authors computed fast-Fourier transforms (FFT) of vertical components of the two ground reaction forces. The FFT coefficients served as inputs to a standard network with one hidden layer achieved 95% rate of successfully distinguishing gait patterns of healthy individuals from those of physically disabled

individuals. This early work demonstrated simple two-category gait classification with a fairly large number of input parameters. Gioutsos & Grieve (1995) measured the duration of the double support and right and left single support phases at seven speeds under three walking conditions (normal walking, walking with a 3.5 kg mass strapped to the right ankle, walking with the right knee fixed in an extended position) in 20 subjects. The network they applied showed a mean accuracy of 73% in correctly recognizing gait patterns.

Similar studies were undertaken by Barton & Lees (1995), who expanded the ANN classification to three categories: healthy feet, pes cavus (high arch) and hallux valgus (bunion). They achieved a successful classification rate of 77% to 100% based on foot position. Below-foot pressure patterns were recorded from 18 subjects during normal walking. The patterns were rotated to a common orientation, scaled to a common size and normalized to the interval [0,1]. The network inputs consisted of a great number of 1316 measured pressure values. Unlike the three-layer network used by Holzreiter & Kohle (1993) and Gioutsos & Grieve (1995), Barton & Lees (1995) developed a more complex neural network with two hidden layers. In 1997, Barton & Lees concluded that hip-knee joint angle diagrams offer a good basis for automatic identification of gait types. The diagrams show the changes in the knee-joint angle as a function of the hip-joint angle. These curves combine the temporal changes of two joint angles, which allow interpretation of the relationships between the two angles, although the time dimension is lost on the graphical representation. The hip-knee joint angle diagrams represent the movement of nearly the entire body, and could be representative of the subject's gait pattern and serve as a basis for distinguishing different gait patterns (Barton & Lees, 1997). These authors distinguished three gait patterns, which were normal walking, a simulation of lower limb length and a simulation of lower limb weight asymmetry. The angles were normalized in time, fast -Fourier transformed and normalized to the interval [0, 1]. As in their previous work, Barton & Lees again used a two-hidden-layer neural network. The average accuracy of classification rate among the three walking conditions was 83.3%.

Lafuente et al. (1998) reverted to the standard network structure (i.e. with one hidden layer), utilized likewise by Holzreiter and Kohle (1993) when attempting a classification into four gait categories. 148 subjects with ankle, knee or hip arthrosis and 88 control subjects without lower limb pathology participated in the study. Data concerning gait rhythm, speed and five kinetic values were fed into the neural network, based on which four gait types were correctly distinguished at a rate of 80%.

All the above authors obtained rates of correct classification within the 77%-100% range, reaffirming the great potential of neural networks in distinguishing gait patterns. Among the studies cited, only Barton & Lees (1997) utilized kinematic parameters similar to those we used in our study. The rates of correct classification they obtained (83.3%) were poorer than the average result of our study (92.5%), most likely due to the small size of the group analyzed ($n = 8$).

Our study, as reported in Kaczmarczyk et al. (2009), found ANN analysis to be superior to qualitative variable analysis for classifying post stroke patients' gait patterns into three gait types, as well as superior to the analysis of min/max joint angle values. Moreover, it substantially decreases data processing time for clinical gait labs (Miller 2009). The detection of gait events is essentially a classification problem; an application for which artificial neural networks are well suited. Miller (2009) used a single-hidden-layer, feedforward network for the purpose of classifying foot-contact and foot-off events using the sagittal plane coordinates of heel and toe markers. The timing of events detected using this method was

compared to the timing of events detected by measuring the ground reaction force using a force plate for a total of 40 pathologic subjects divided into two groups: barefoot and shod/braced. On average, the neural network detected foot-contact events 7.1 ms and 0.8 ms earlier than the force plate for the barefoot and shod/braced groups respectively. The average difference for foot-off events was 8.8 ms and 3.3 ms. Given that motion capture data were collected at 120 Hz, this implies that the force plate method and neural network method generally agreed within 1–2 frames of data.

There are currently no robust methods available for the identification of gait events in pathologic gait, although several methods of identifying gait events based on kinematic data have been developed and successfully validated in normal walking (Hreljac & Marshall 2000, Zeni et al., 2008, O'Connor et al., 2007, Ghousayni et al., 2004). For example, the maximum vertical and horizontal components of the acceleration of foot markers can be used to identify foot contact and foot-off events (Hreljac & Marshall 2000). However, this method has been invalidated for toe walkers (Hsue et al., 2009). In addition, others have used accelerometers to identify gait events (Lau & Tong 2008). Even though these methods have shown reasonable accuracy when used on normal subjects, none have been validated for use in pathologic gait. Consequently, ANNs have been shown to be an accurate, autonomous method for detecting gait events in pathologic gait. More generally, our study has helped confirm the appropriateness of using neural networks in gait research.

Aside from gait classification, early prediction of functional outcome and motor recovery remains a crucial factor in client-centred practice, discharge planning, and utilization of rehabilitation resources. According to Oczkowski & Barreca (1999) traditional predictive models, utilizing standard population statistics, appeared unable to predict the degree of disability or place of discharge for individual stroke survivors. These statistical methods of prediction incorporated the most recurrent or powerful variables so that some specific patient information was inevitably lost. ANNs modelling offers another methodologic approach to predicting outcome. It has been successfully used to determine outcome, length of stay, disease reoccurrence, and costs in other medical conditions by identifying patterns based on a group of input variables and their resultant outcomes. The patterns are not preconceived but learned from experience. After learning occurs, the neural network can then classify new cases having the same or similar defining characteristics.

Furthermore, neural network modelling differs from standard regression analysis by maintaining and processing all available information in the clinical data base. This method does not discard variables that may be critical in determining outcome for an individual stroke survivor. Consequently, neural network modelling better handles the heterogeneity found in the stroke population. Neural network modelling imitates the brain's biological features of learning, association, and memory without the addition of human judgment errors.

Numerous studies seem to confirm the superiority of ANNs over traditional statistical methods like regression analysis and rough sets (Edwards, et al.; 1999, Rowland et al., 1998). Edwards et al. (1999) showed that the ANN was superior to the logistic regression model and correctly classified all patients (100%) as alive or dead compared with 85% correct classification for the logistic regression model. ANN analysis seems to use information for prediction of mortality more effectively in this sample of patients with ICH. Rowland et al. (1998), in turn, developed and compared 3 models (logistic regression, neural networks, and rough sets) in the prediction of ambulation at hospital discharge following spinal cord injury. All models had sensitivity, specificity, and accuracy greater than 80% at ideal

thresholds; however, the ANNs performed better than the other traditional methods. Similar results were obtained by Li et al. (2000), who compared three different mathematical models for building a traumatic brain injury (TBI) medical decision support system (MDSS). The results showed that, assuming equal importance of sensitivity and specificity, the logistic regression model had a (sensitivity, specificity) of (73%, 68%), compared to (80%, 80%) from the RBF model and (88%, 80%) from the MLP model. This study demonstrated the feasibility of applying neural networks as the mechanism for TBI decision support systems based on clinical databases. The results also suggest that ANNs may be a better solution for complex, non-linear medical decision support systems than conventional statistical techniques such as logistic regression.

One important aspect of our research, as profiled above, is the relationship between CT scan results and the pathological gait pattern in post stroke patients. A better awareness of these interdependencies, gleaned through the use ANNs, will enable the identification of CT scan criteria to be used in strategies for early rehabilitation of post stroke patients.

The post stroke patients were classified into three groups according to pathological gait patterns rather than brain lesion parameters and showed similar levels of correct classification based on CT scan parameters, with the overall average result of 71.3% correct classification for all patients – a result that may be described as relatively good.

Our study, as reported in Kaczmarczyk et al. (in preparation), showed that the weight of the individual parameters depends on the configuration (accompanying parameter), and thus does not have a fixed effect on the classification. In specific, we found that of the four lesion geometry factors considered, lesion size and location are the most dominant factors. Evidence from previous studies lends support to the importance of these factors. Alexander et al. (2009) identified an association between a focal subcortical structure and gait asymmetry. This finding is similar to that of Miyai et al. (2000), who proved that injury to the putamen was associated with poor functional outcome in chronic stroke patients. In contrast, Dominkus et al. (1990) showed that better results were observed in those with subcortical than with cortical lesions. Perennou et al. (1999), in turn, found that lesion size has the greatest influence on motor recovery. However, Binkofski et al. (2001) showed no significant correlation between initial lesion size and recovery of upper-limb motor function. There was also no significant correlation between motor recovery and functional outcome with stroke pathology (infarction or hemorrhage) in the study of Chen et al. (2000). This result was compatible with the report of Nakayama et al. (1994), although Allen (1984) reported that patients with intracerebral hemorrhage have a worse outcome in the acute stage. Some studies reported there was no link between walking recovery and the hemiplegic side (Viosca et al., 2005, Laufer et al., 2003, Chen et al., 2000) while other studies found such association (Kwolek & Sławiński 1996, Mcciocchi et al., 1998). The reason for the disparity of findings may lie in the application of only one parameter describing stroke in the analysis.

Our work studied the effect of simultaneous impact of four brain lesion factors on gait patterns in post stroke patients. This concept is supported by the findings of several previous studies (Chen et al., 2000, Fries et al., 1993). According to Chen et al. (2000) motor recovery and functional outcome after stroke correlate with “brain lesion profiles” that combined the delimiting size and location of lesions, rather than the absolute or relative lesion size only. When the delimiting sizes were set at 75cm³ for cortical, 4cm³ for CR, 0.75cm³ for IC, 22cm³ for putaminal, and 12cm³ for thalamic lesions, BLPs could determine motor and functional outcomes. Fries et al. (1993) did not identify the delimiting size,

although they demonstrated that small capsular lesions can selectively disrupt the output of distinct motor areas, while large capsular lesions cause more severe deficits. These authors analyzed two factors simultaneously: size and location. However, because numerous studies have confirmed the influence of side of stroke and type of stroke, in our analysis we also took those factors into account.

Although many studies have assessed the influence of brain lesion parameters on motor and functional outcomes, to the best of our knowledge, none so far has investigated the association between lesion parameters and pathological gait patterns in hemiplegic stroke patients. Only Giroud and Dumas (1995) have attempted, as we have, to identify a link between stroke location and gait disturbances. These authors described a characteristic type of gait in patients with lesions near the corpus callosum. The patients used a wide base, with feet rooted to the ground. Gait consisted of shuffling with short steps, without upper limb movement and slight extension of the trunk. These authors used evaluative scales in quantifying neurological deficit and improvement in locomotive function that involves a subjective element. Clinical practice thus has a need for objective gait classification in post stroke patients, as a basis for predicting the degree of rehabilitation progress that is achievable within a specific timeframe, in each of the gait type groups distinguished.

Our observations indicate that finding the relationship between an objective patient classification based on ANN technique into a specific subgroup of pathological gait and CT scan parameters may serve as a relatively good predictor of future functional condition and degree of improvement. Given that CT data are beginning to have more influence on rehabilitation practices (Boyd et al., 2007), our results may have considerable clinical implications. For instance, during the early stages after stroke when the most recovery may be possible (Jørgensen et al., 1995) long before kinematic gait tests may be performed, the results presented herein may lay the groundwork for implementing an early diagnostic and therapeutic procedure by providing an early prediction of pathological gait classification. This procedure would moreover provide predictions about prospects for gait improvement, but would be subject to further verification by other gait analysis methods as patient motor function returns.

The results of our study may provide the basis for implementing diagnostic and therapeutic procedures that take account of predictions concerning future gait pattern, which would then be gradually verified as locomotive function returns. Future research in this direction should concentrate on the more precise specification of the distinguished gait patterns and on the development of effective programs of procedure in individual cases.

The ANN methodology, here used successfully for classifying gait types in post stroke patients, seems to have the potential to distinguish pathological gait types stemming from other disorders. The scheme followed by our research, as described in detail in the previous section, could be productively applied in such cases for developing an objective diagnostic method based on quantitative data, associating symptoms with therapeutic procedures and predicted rehabilitative effects.

4. Conclusions

The work by various authors reviewed in this chapter, including our own, has shown that ANNs are very promising as a potential tool for use in gait classification in various types of disabilities, offering numerous advantages over other methods. ANN classification may allow for more effective treatment with appropriately targeted, early intervention.

Moreover, ANNs are also proving useful as a state-of-the-art tool in the monitoring and planning of rehabilitation strategies. In general, the various studies described in this chapter have helped confirm the appropriate use of neural networks in gait classification, and so ANNs can be expected to continue to be a productive field in physiotherapy research.

5. References

- Aminian, K.; Robert, P. & Jequier, E. (1995). Incline, speed, and distance assessment during unconstrained walking. *Med Sci Sports Exerc*, 27, 226-234, ISSN 0195-9131
- Allen, CMC. (1984). Predicting the outcome of acute stroke: a prognostic score. *J Neurol Neurosurg Psychiatry*, 47, 475-80, ISSN 0022-3050
- Alexander, LD.; Black, SE.; Patterson, KK.; Gao, F.; Danells, CJ. & McIlroy, WE. (2009). Association between gait asymmetry and brain lesion location in stroke patients. *Stroke*, 40, 537-544, ISSN 0039-2499
- Bach-y-Rita, P. (1981). Brain plasticity as a basis of the development of rehabilitation procedures for hemiplegia. *Scand J Rehabil Med*, 13, 73-83, ISSN 0036-5505
- Barton, JG. & Lees, A. (1995). Development of a connectionist expert system to identify foot problems based on under-foot pressure patterns. *Clinical Biomechanics*, 10, 7, 385-391, ISSN 0268-0033
- Barton, JG. & Lees, A. (1997). An application of neural networks for distinguishing gait patterns on the basis of hip-knee joint angle diagrams. *Gait and Posture*, 5, 28-33, ISSN 0966-6362
- Bear, G. & Smith, M. (2001). The recovery of walking ability and subclassification of stroke. *Physiotherapy Research International*, 6, 135-144, ISSN online 1471-2865
- Bernabucci, I., S. Conforto, et al. (2007). A biologically inspired neural network controller for ballistic arm movements. *Journal of NeuroEngineering Rehabilitation* 3, 33, 1-17, ISSN 1743-0003
- Blaszczuk, J. (2004). *Biomechanika kliniczna*. Wydawnictwo Lekarskie PZWL, Warsaw, ISBN 9788320042023
- Bogataj, U.; gros, M.; Kljajic, M.; Acimovic, R. & Malezic, M. (1995). The rehabilitation of gait in patients with hemiplegia - a comparison between conventional therapy and multichannel functional electrical -stimulation therapy. *Physical Therapy*, 75, 490-502, ISSN 1083-3196
- Bohannon, RW. Andrews, AW. & Smith, MB. (1988) Rehabilitation goals of patients with hemiplegia. *Int J Rehabil Res*, 11, 181-183, ISSN 0342-5282
- Bohannon, RW. Horton, MG. & Wikholm, JB. (1991). Importance of four variables of walking to patients with stroke. *Int J Rehab Res*, 14, 246-250, ISSN 0342-5282
- Boyd, LA.; Vidoni, ED. & Daly, JJ. (2007). Answering the call: The influence of neuroimaging and electrophysiological evidence on rehabilitation. *Phys Ther*, 87, 684-703, ISSN 1083-3196
- Brott, T.; Marler, JR.; Olinger, CP.; Adams, HP Jr.; Tomsick, T.; Barsan, WG.; Biller, J.; Eberle, R.; Hertzberg, V. & Walker, M. (1989). Measurements of acute cerebral infarction: lesion size by computed tomography. *Stroke*, 40, 871-875, ISSN 0039-2499
- Brust, JCM. (2004). *Neurologia Merita*, Lewisa P. Roland. Wydawnictwo Medyczne Urban & Partner, Warsaw, ISBN 9788360290347

- Burdett, RG.; Borello-France, D.; Blatchly, C. & Poptter, C. (1988). Gait comparison of subjects with hemiplegia walking unbraced, with ankle-foot orthosis, and with airstirrup brace. *Phys Ther.*, 68, 1197-1203, ISSN 1083-3196
- Carter, M. (2007). *Minds and computers: An introduction to the philosophy of artificial intelligence*, Edinburgh University Press, Edinburgh, ISBN 9780748620999
- Carter, R. & Page, M (2009). *The human brain book*, Barnes & Noble, ISBN 9780756654412
- Chau, T. (2001). A review of analytical techniques for gait data. Part 2: neural network and wavelet methods. *Gait and Posture*, 13, 102-120, ISSN 0966-6362
- Chen, CL.; Chen, HC.; Tang, SF.; Wu, CY.; Cheng, PT. & Hong, WH. (2003). Gait performance with compensatory adaptation in stroke patients with different degrees of motor recovery. *Am J Phys Med Rehabil*, 82, 925-935, ISSN 0894-9115
- Chen, CL.; Tang, SF.; Chen, HC.; Chung, CY & Wong, MK. (2000). Brain lesion size and location: effects on motor recovery and functional outcome in stroke patients. *Arch Phys Med Rehabil*, 81, 447-452, ISSN 0003-9993
- Chen, Y-L.; Chen, S-H.; Chen, W-L.; Hsiao, CH-CH.; Kuo, T-S. & Lai J-S. (2004). Neural network and fuzzy control in FES-assisted locomotion for the hemiplegic. *J Med Eng Tech*, 28, 1, 32-38, ISSN 0309-1902
- Damasio, HA. (1983). Computed tomography guide to the identification of cerebral vascular territories. *Arch Neurol*, 40, 138-142, ISSN 0003-9942
- Dominkus, M.; Grisold, W. & Jelinek, V. (1990). Transcranial electrical motor evoked potentials as a prognostic indicator for motor recovery in stroke patients. *J Neurol Neurosurg Psychiatry*, 53, 745-748, ISSN 0022-3050
- Doornewaard, H.; van der Schouw, YT.; van der Graaf, Y.; Bos, AB.; Habbema, JD. & van der Tweel, JG. (1999). The diagnostic value of computer-assisted primary cervical smear screening: a longitudinal cohort study. *Mod Pathol*, 12, 1, 995-1000, ISSN 0893-3952
- Duncan, PW.; Goldstein, LB.; Horner, RD.; Landsman, PB.; Samsa GP. & Matchar, DB. (1994). Similar motor recovery of upper and lower extremities after stroke. *Stroke*, 25, 1181-1188, ISSN 0039-2499
- Edwards, DF.; Hollingsworth, H.; Zazulia, AZ. & Diringer, MN. (1999). Artificial neural networks improve the prediction of mortality in intracerebral hemorrhage. *Neurology*, 53, 351, ISSN 0028-3878
- Fries, W.; Danek, A.; Scheidtmann, K. & Hamburger, C. (1993). Motor recovery following capsular stroke: role of descending pathways from multiple motor areas. *Brain*, 16, 369-82, ISSN 0006-8950
- Fujii, Y. & nakada, T. (2003). Cortical reorganization in patients with subcortical hemiparesis: neural mechanisms of functional recovery and prognostic implication. *J Neurosurg*, 98, 1, 64-73, ISSN 0022-3085
- Galen, SS. & Granat, MH. (2002). Study of the effect of functional electrical stimulation (FES) on walking in children undergoing botulinum toxin A therapy, *Proceedings of the first FESnet Conference*, 31-32, University of Strathclyde, September 2002, Glasgow
- Ghoussayni, S.; Stevens, C.; Durham, S. & Ewins, D. (2004). Assessment and validation of a simple automated method for the detection of gait events and intervals. *Gait and Posture*, 20, 266-72, ISSN 0966-6362
- Gioftsos, G. & Grieve, DW. (1995). The use of neural networks to recognize patterns of human movement: gait patterns. *Clin Biomech*, 10, 4, 179-183, ISSN 0268-0033

- Giroud, M. & Dumas, R. (1995). Clinical and topographical range of callosal infarction: a clinical and radiological correlation study. *J Neurol Neurosurg Psychiatry*, 59, 238-42, ISSN 0022-3050
- Glymour, MM.; Berkman, LF.; Ertel, KA.; Fay, ME.; Glass, TA. & Furie, KL. (2007). Lesion characteristics, nih stroke scales, and functional recovery after stroke. *Am J Phys Med Rehabil.*, 86, 725-733, ISSN 0894-9115
- Grigsby, J.; Kooken, R. & Hershberger, J. (1994). Simulated neural networks to predict outcomes, costs, and length of stay among orthopedic rehabilitation patients. *Arch Phys Med Rehabil*, 75, 1077-1081, ISSN 0003-9993
- Gritsenko, V. & Prochazka, A. (2004). A functional electric stimulation-assisted exercise therapy system for hemiplegic hand function, *Arch Phys Med Rehabil*, 85, 6, 881-885, ISSN 0003-9993
- Goffredo, M.; Bernabucci, I.; Schmid, M. & Conforto, S. (2008). A neural tracking and motor control approach to improve rehabilitation of upper limb movements. *J Neuroeng Rehabil*, 5, 5-16, ISSN 1743-0003
- Herman, EJ. (1987) *Diagnostyka chorób układu nerwowego*. Wydawnictwo Lekarskie PZWL, Warszawa, ISBN 8320006538
- Hong-yin, L.; Kai-yu, T. & Hailong, Z. (2009) Support vector machine for classification of walking condition of persons after stroke with dropped foot. *Human Movement Science*, 28, 504-514, ISSN 0167-9457
- Holzreiter, SH. & Kohle, ME. (1993). Assessment of gait patterns using neural networks. *J Biomech*, 26, 645-651, ISSN 0021-9290
- Hreljac, A. & Marshall RN. (2000). Algorithms to determine event timing during normal walking using kinematic data. *J Biomech*, 33, 783-786, ISSN 0021-9290
- Hsue, B-J.; Miller, F.; Su, F-C.; Henley, J. & Church, C. (2007). Gait timing event determination using kinematic data for the toe walking children with cerebreal palsy: XXI Congress, International Society of Biomechanics. *J Biomech*; 40 (Suppl. 2), S529
- Intiso, D.; Santilli, V.; Grasso, MG.; Rossi, R. & Caruso, I. (1994). Rehabilitation of walking of electromyographic biofeedback in foot-drop after stroke. *Stroke*, 25, 1189-1192, ISSN 0039-2499
- Jørgensen, HS.; Nakayama, H.; Raaschou, HO. & Olsen, TS. (1995). Recovery of walking function in stroke patients: The Copenhagen stroke study. *Arch Phys Med Rehabil*, 76, 27-32, 0003-9993
- Kaczmarczyk, K.; Wit, A.; Krawczyk, M. & Zaborski, J. (2009). Gait classification in post-stroke patients using artificial neural networks. *Gait & Posture*, 30, 207-210, ISSN 0966-6362
- Kaczmarczyk, K.; Wit, A.; Krawczyk, M.; Zaborski, J. & Gajewski, J. (in preparation). Association between gait patterns and brain lesion factors in stroke patients.
- Karsznia, A.; Oberg, T. & Dworak, L. (2004). Gait in subjects with hemiplegia. *Acta of Bioengineering and Biomechanics*. 6, 2, 65-76, ISSN 1509-409X
- Kazmierczak, SC.; Catrou, PG. & Van Lente, F. (1993). Diagnostic accuracy of pancreatic enzymes evaluated by use of multivariate data analysis. *Clin Chem*, 39, 1960-1965, ISSN 1434-6621

- Katz, AS.; Katz, S.; Wickham, E. & Quijano, RC. (1993). Prediction of valve-related complications for artificial heart valves using adaptive neural networks: a preliminary study. *J Heart Valve Disease*, 2, 504-508, ISSN 0966-8519
- Kim, CM. & Eng, JJ. (2004). Magnitude and pattern of 3D kinematic and kinetic gait profiles in person with stroke: relationship to walking speed. *Gait and Posture* 20: 140-146, ISSN 0966-6362
- Kinsella, S. & Moran, K. (2008) Gait pattern categorization of stroke participants with equinus deformity of the foot. *Gait and Posture*, 27, 144-151, ISSN 0966-6362
- Knutsson, E. & Richards, C. (1979). Different types of disturbed motor control in gait of hemiparetic patients. *Brain*, 102, 405-430, ISSN 0006-8950
- Kramers de Quervain, IA.; Simon, SR.; Leurgan, S.; Pease, WS. & McAllister, D. (1996). Gait pattern in the early recovery period after stroke. *J Bone Jt Surg*, 78, 1506-1514, ISSN 0021-9355
- Kwolek, A. & Sławiński, J. (1996). Usprawnianie chodu chorych z niedowładem połowicznym spastycznym po udarze mózgu. *Advances in Rehabilitation*, 2, 382-421, ISSN 0860-6161
- Lafuente, R.; Belda, JM.; Sanchez-Lacuesta, J.; Soler, C. & Prat, J. (1998). Design and test of neural networks and statistical classifiers in computer-aided movement analysis: a case study on gait analysis. *Clin Biomech*, 13, 3, 216-20, ISSN 0268-0033
- Lapedes, A. & Farber, R. (1988). How neural networks work. *Neural Information Processing Systems*, 442-456, American Institute of Physics, New York
- Lau, H. & Tong K. (2008). The reliability of using accelerometer and gyroscope for gait event identification on persons with dropped foot. *Gait and Posture*, 27, 248-57, ISSN 0966-6362
- Laufer, Y.; Sivan, D.; Schwarzmann, R. & Sprecher, E. (2003). Standing balance and functional recovery of patients with right and left hemiparesis in the early stages of rehabilitation. *Neurorehabil Neural Repair*, 17, 207-213, ISSN 1545-9683
- Liberson, WT.; Holmquest, HJ.; Scot, D. & Dow, M. (1961). Functional electrotherapy: stimulation of the peroneal nerve synchronized with the swing phase of the gait of hemiplegic patients. *Arch Phys Med Rehabil*, 42, 101-105, ISSN 0003-9993
- Mcciocchi, SN.; Diamond, PT.; Alves, WM. & Mertz, T. (1998). Ischemic stroke: relation of age, lesion location, and initial neurological deficit to functional outcome. *Arch Phys Med Rehabil*, 79, 1255-1257, ISSN 0003-9993
- Miller A. (2009). Gait event detection using a multilayer neural network. *Gait and Posture*, 29, 542-545, ISSN 0966-6362
- Miller, AS.; Blott, BH. & Hames, TK. (1992). Review of neural network applications in medical imaging and signal processing. *Med Biol Eng Comput*, 30, 449-464, ISSN 0140-0118
- Miyai, I.; Suzuki, T.; Kang, J. & Volpe, BT. (2000). Improved functional outcome in patients with hemorrhagic stroke in putamen and thalamus compared with those with stroke restricted to the putamen or thalamus. *Stroke*, 31, 1365-1369, ISSN 0039-2499
- Mulroy, S.; Gronley, J.; Wiess, W.; Newsam, C. & Perry, J. (2003) Use of cluster analysis for gait pattern classification of patients in the early and late recovery phases following stroke. *Gait and Posture*, 18, 114-125, 0966-6362
- Muniz, A. M. S.; H. Liu, et al. (2010). Comparison among probabilistic neural network, support vector machine and logistic regression for evaluating the effect of

- subthalamic stimulation in Parkinson disease on ground reaction force during gait. *J Biomech*, 43, 4, 720-726, ISSN 0021-9290
- Nakayama, H.; Jørgensen, HS.; Raaschou, HO. & Olsen, TS. (1994). Compensation in recovery of upper extremity function after stroke: Copenhagen Stroke Study. *Arch Phys Med Rehabil*, 75, 852-857, ISSN 0003-9993
- O'Connor, CM.; Thorpe, SK.; O'Malley, MJ. & Vaughan, CL. (2007). Automatic detection of gait events using kinematic data. *Gait and Posture*, 25, 469-74, 0966-6362, ISSN 0003-9993
- Ohno-Machado, L. & Rowland T. (1999). Neural network applications in physical medicine and rehabilitation. *Am J Phys Med Rehabil*, 78, 4, 392-398, ISSN 0894-9115
- Olney, SJ. & Richards, C. (1996). Hemiparetic gait following stroke. Part I: Characteristics. *Gait and Posture*, 4, 136-148, ISSN 0966-6362
- Pérennou, D.; Bénéim, C.; Rouget, E.; Rousseaux, M.; Bland, JM. & Pélissier, J. (1999). Postural balance following stroke: towards a disadvantage of the right brain-damaged hemisphere. *Rev Neurol*, 155, 281-290, ISSN 1759-4758
- Perry, J. & Burnfield, J. (2010). *Gait analysis: normal and pathological function*, SLAC Incorporate, Thorofare, ISBN 9781556421921
- Perry, J.; Garrett, M.; Gronley, JK. & Mulroy, SJ. (1995). Classification of walking handicap in the stroke population. *Stroke*, 26, 6, 982-989, ISSN 0039-2499
- Popovic, MB.; Popvic, DB.; Sinkjaer, T.; Stefanovic, A. & Schwirtlich, L. (2002). Restitution of reaching and grasping promoted by functional electrical therapy. *Artif Organs*, 26, 271-275, ISSN 0391-3988
- Popovic, D., M. Radulovic, et al. (2003). Automatic vs hand-controlled walking of paraplegics. *Medical Engineering & Physics*, 25, 1, 63-73, ISSN 1350-4533
- Prescott, RJ.; Garraway, WM. & Akhtar, AJ. (1982). Predicting functional outcome following acute stroke using a standard clinical examination. *Stroke*, 13, 641-647, ISSN 0039-2499
- Richards, CL. & Olney, SJ. (1996). Hemiparetic gait following stroke. Part II: Recovery and physical therapy. *Gait and Posture*, 4, 149-162, ISSN 0966-6362
- Rodda, JM.; Graham, HK.; Carson, L.; galea, MP. & Wolfe, R. (2004). Sagittal gait patterns in spastic diplegia. *J BoneJoint Surg*, 86, 251-258, ISSN 0021-9355
- Rogers, SK.; Ruck, DW. & Kabrisky, M. (1994). Artificial neural networks for early detection and diagnosis of cancer. *Cancer Lett*, 77, 79-83, ISSN 0304-3835
- Rowland, TR.; Ohno-Machado, L. & Ohrn, A. (1998). Comparison of Multiple Prediction models of ambulation following spinal cord injury. *Proceedings of American Medical Informatics Association (AMIA) Annual Symposium*, Orlando, Nov. 7-11
- Savelberg, HHCM. & de Lange, ALH. (1999). Assessment of the horizontal, fore-aft component of the ground reaction force from insole pressure patterns by using artificial neural networks. *Clin Biomech*, 14, 8, 585-592, ISSN 0268-0038
- Turney, T.; Garraway, W. & Whisnant, J. (1984). The natural history of hemispheric and brainstem infarction in Rochester, Minnesota. *Stroke*, 15, 790-794 ISSN 0039-2499
- Viosca, E.; Lafuente, R.; Martínez, JL.; Almagro, PL.; Gracia, A. & González, C. (2005). Walking Recovery After an Acute Stroke: Assessment With a New Functional Classification and the Barthel Index. *Arch Phys Med Rehabil*, 86, 1239-1244, ISSN 0003-9993

- Voigt, M. & Sinkjaer, T. (2000). Kinematic and kinetic analysis of the walking patterns in hemiplegic patients with drop foot using a peroneal stimulator. *Clin Biomech*, 15, 340-351, 0268-0033
- von Schroeder, HP.; Coutts, RD.; Lyden, PD. & Billings, E Jr. (1995). Gait parameters following stroke: a practical assessment. *J Rehabil Res Dev*, 32, 25-31, ISSN 0748-7711
- Wang, RY.; Yang, YR.; Tsai, MW.; Wang, WTJ. & Chan, RC. (2002). Effects of functional electric stimulation on upper limb motor function and shoulder range of motion in hemiplegic patients. *Am J Phys Med Rehabil*, 81, 4, 283-290, ISSN 0894-9115
- Wong, AM.; Pei, Y-C.; Hong, WH.; Chung, Ch-Y.; Lau, Y-Ch. & Chen, CP. (2004). Foot contact pattern analysis in hemiplegic stroke patients: an implication for neurologic status determination. *Arch Phys Med Rehabil*, 85, 1625-1630, ISSN 0003-9993
- Zeni, Jr JA.; Richards, JG. & Higginson, JS. (2008). Two simple methods for determining gait events during treadmill and overground walking using kinematic data. *Gait and Posture*, 27, 710-714, 0966-6362

Part 5

Clinical and Other Applications

Forecasting the Clinical Outcome: Artificial Neural Networks or Multivariate Statistical Models?

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1. Introduction

The field of prognostics has grown rapidly in the last decade and clinicians have been provided with numerous tools to assist with evidence-based medical decision-making. Most of these included nomograms, classification and regression tree analyses, and risk group stratification models [Grossberg JA, et al., 2006] and Artificial Neural Networks (ANN) [Djavan B, et al., 2002]. Nomograms are graphic representation of statistical model, which incorporate multiple continuous variables to predict a patient's risk of developing a specific endpoint (recurrence, survival, complications) [Kattan MW, 2005]. Each variable is assigned a scale of points according to its prognostic significance. The total score for all the variables is converted to an estimated probability of reaching the endpoint [Akl A, et al., 2008]. Statistical approaches require guesses as to how outputs functionally depend on inputs. Artificial neural networks have been used for evaluation of clinical data to provide results similar to conventional modeling methods [Freeman RV, et al., 2002]. They do not require the articulation of such a mathematical model. ANNs are complex computational systems that can provide a nonlinear approach for data analysis.

The ANNs forms a mapping from input to output nodes (simulated neurons) by extracting features from input patterns, assigning them weights, summing weights with activation functions, and propagating decisions to output nodes if activation thresholds are exceeded. Typical networks are organized into three layers of computational units (nodes) in which input/output layers are linked by hidden layers of nodes. Subject factors determine the number of input units, and the classification complexity determines the number of output units. The number of hidden units is determined by trial and error (training). Common routines start with one hidden unit and assign small arbitrary weights to all nodal connections. The network is fed sample data with known outcomes, and an error term is calculated by means of differences between known and predicted outputs. Learning consists of adjusting weights by backward pass of errors through the connections to network nodes in response to input data. Hidden units are added to achieve minimum error criteria, while constraining the number to promote generalization of input patterns and prevent over-fitting (memorization). Interconnection density determines the network's ability to correctly discriminate the outcomes. In statistical parlance, ANN models are a form of nonlinear

discriminant analysis with input units, weights, and activation functions resembling covariates, coefficients, and generalized additive models, respectively. To generate an accurate prediction model, several conditions should be fulfilled: use of a robust dataset that represents a large patient population, and the incorporation of prognostically significant variables into the model [Kattan MW, 2005]. In addition, the generated model should be validated using an independent testing group [Graefen BM, et al., 2002]. Herein, we have developed two predictive models using ANN. The first is to predict graft outcome after renal transplantation. The second is to predict patient outcome after radical cystectomy for invasive bladder carcinoma. The results of these models were compared with those of statistically-based multivariate models.

2. Artificial intelligence versus Nomogram in renal transplantation:

The importance of having a possibility to predict the graft outcome after renal transplantation does not need emphasis. This would allow the choice of the best possible kidney donor and the optimum immunosuppressive therapy for a given patient. Several methods have been used to construct a prognostic model. A multivariate analysis was used to predict the outcome of renal transplantation from deceased donor in an attempt to optimize the allocation of the recovery of organs [Poli F, et al. 2000]. In another study, multivariate analysis was used to predict creatinine levels in recipients of kidneys from living donors [Zapletal C, et al. 2004]. The probability of deceased donor-graft survival was studied using tree regression model [Goldfarb-Rumyantzev AS, et al. 2003]. ANNs were used to predict the possibility of delayed graft function after deceased donor renal transplantation [Brier ME, et al. 2003]. For the development of our ANNs, we have opted to use a feed-forward with back-propagation model since it is known for its stability and tendency not to over fit [Ripley RM, et al. 2004]. The algorithm is often described as a decision making process functioning like the human brain [Cross SS, et al. 1995]. In statistical parlance, ANN models are a form of nonlinear discriminant analysis with input units, weights, and activation functions resembling covariates, coefficients, and generalized additive models, respectively. Thus, it is not surprising that ANN applications are undermined by similar limitations and misuses afflicting conventional discriminant analysis. Schwarzer identified four frequent mistakes when applying ANNs [Schwarzer et al., 2000]:

- Overfitting models by training large, multilayer networks with small data sets.
- Neglecting traditional statistical methods due to inadequate bench marks or lack of significance testing.
- Applying naïve approaches to survival data, sometimes ignoring censorship.
- Claiming overly optimistic generalization properties.

In our study only significant univariate variables were incorporated as input units, but in multivariate cases, insignificant univariate variables sometimes become relevant confounders or effect modifiers. Since ANNs are touted as having the ability to select those items most important in performing classifications, this prior variable selection seems unnecessary [Figure 1].

Our study was designed to predict 5-year survival using cases with complete data (81 with missing data were dropped) divided into training (n=1500) and test (n=319) sets for building and validating models, respectively. We followed steps to guard against these problems. To avoid over-fitting, the ANN was restricted to one hidden layer, and the number of hidden

nodes was controlled by reasonable stopping criteria. Also, the ratio of number of observations in the training set (1500) to number of parameters in the model (361) was greater than 2, a recommended guideline [Schwarzer et al., 2000] [Figure 2, 3].

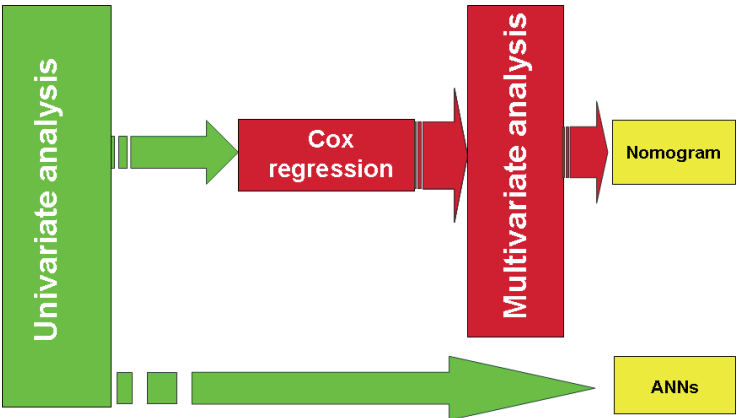


Fig. 1. Artificial neural networks model construction for predicting graft survival.

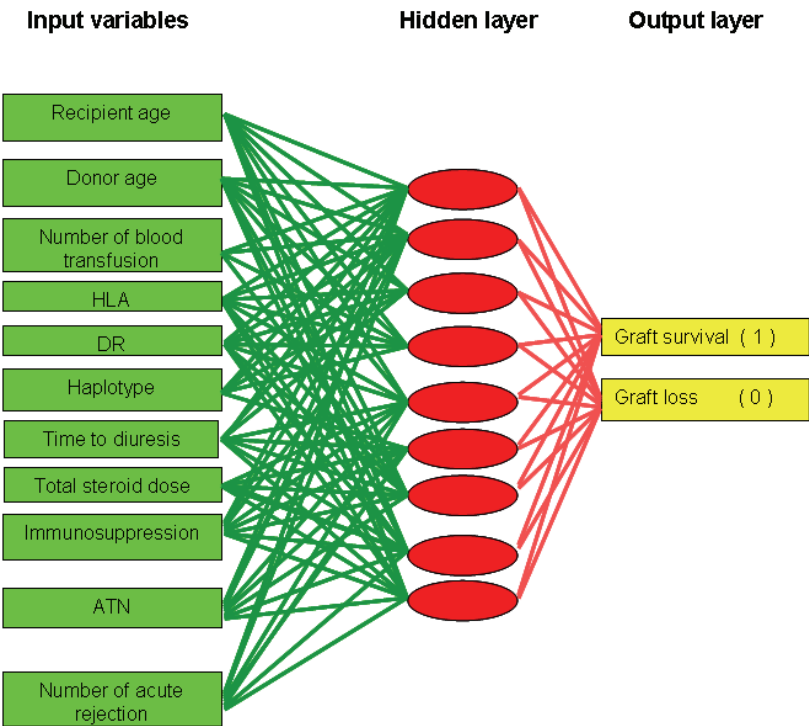


Fig. 2. Artificial neural networks architecture for predicting renal graft outcome.

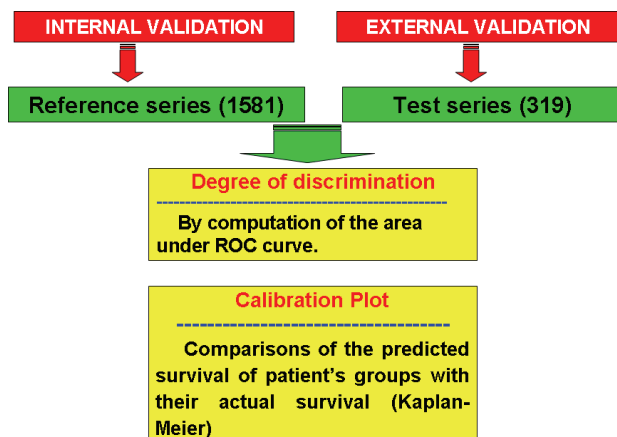


Fig. 3. Validation strategy of transplantation Ann model.

Preprocessing or normalizing data entering the ANN was done. This is an important step that usually requires a floating-point value between 0 and 1 to be assigned for each input node, with special consideration for missing values. The ANN model was tested against a nomogram built on a standard statistical approach (Cox regression) [Figure 4].

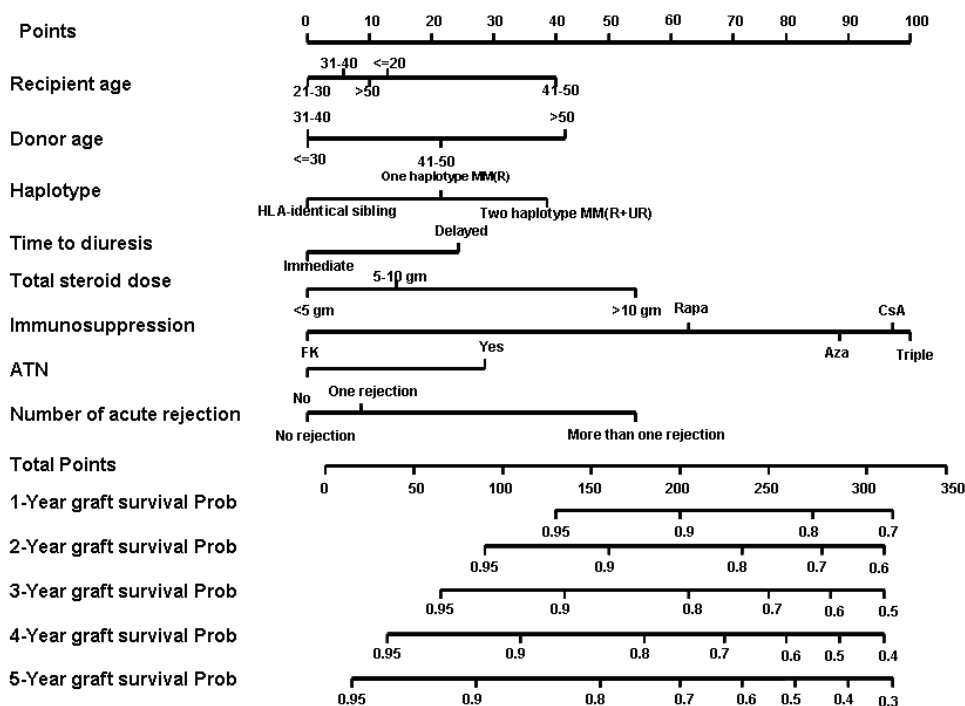


Fig. 4. Cox based nomogram for predicting renal graft outcome.

Although this nomogram was simple to interpret by clinicians, but the linear nature of Cox regression may limit its handling power of complex tasks. The nomogram was not discarded altogether but credited for easy interpretation. The imputation strategy seemed to be sufficient since calibration points between average predicted (ANN) and observed (Kaplan-Meier) 5-year survival rates for various patient groups produced virtually a 45° line. The ANNs sensitivity was 88.43 (86.4 –90.3) %, specificity was 73.26 (70 –76.3) %, and its predictions was 16% significantly more accurate than the Cox regression-based nomogram ($P<0.001$). The Cox regression-based nomogram sensitivity was 61.84% (50 –72.8) with 74.9% (69–80.2) specificity and area under ROC curve was 72% (67–77) [Table 1].

	Nomogram	ANNs
Area under ROC curve	0.72	0.88
Sensitivity at 95% specificity	0.40	0.77
Sensitivity at 90 % specificity	0.48	0.58
Specificity at 95% sensitivity	0.86	0.90
Specificity at 90% sensitivity	0.30	0.48
Hosmer and lemeshow Test X ² (P-value)	11.6 (0.2)	14.5 (0.1)
Prediction error %	10	5

Table 1. Comparison of the predictive accuracy of renal graft survival between the Cox regression-based Nomogram and ANN.

The predicted graft survival probabilities of both models responded well to the observed graft survival. The negative predictive value for the ANNs and Cox regression-based nomogram was 82% and 86.3%, respectively. Eighty-two percent of the ANNs predicted graft survival probabilities were responding to observed graft survival, whereas only 43.5% of the Cox regression-based nomogram predicted graft survival probabilities were responding to their observed graft survival. The predictive accuracy of the ANNs prognostic model was superior to that of the nomogram in predicting 5-year graft survival. In a future project, we shall implement the ANNs model that was developed in this study in a form of software to make it available to estimate survival and prognosticate individual transplant recipients outcomes.

3. ANNs versus Nomograms and risk groups in cancer bladder:

Radical cystectomy has evolved as the primary therapeutic modality for localized or regionally advanced invasive carcinoma of the bladder. Treatment outcomes after radical cystectomy have been based on TNM staging or pathological grouping [Ghoneim M, et al. 1997]. Although this prognostic grouping yielded useful estimates of recurrence risk and survival outcomes, significant variation within each group has been observed due to the heterogeneity of the tumor biology and patient characteristics. To circumvent these limitations several mathematically based prognostic models were developed which appear to outperform clinical judgment in estimating outcome probabilities. Accurate estimates of the likelihood of success, patient outcomes and long-term morbidity are essential for patient counseling, informed decision making and to assist in clinical trial design. Clinical judgment might be biased due to subjective and objective confounders that exist at all stages of the prediction process. The

clinical TNM staging system is routinely used in decision making and patient counseling. The ease of using this tool is offset by its limited accuracy. Additional factors that can provide a predictive value, e.g. age, gender, time to surgery and other pathological features such as lymphovascular invasion, are missed. The influence of tumor stage, grade and lymph node involvement has been investigated and extensively reported [Pollack A, et al. 1995]. To improve the accuracy of pathological staging molecular markers such as P53, P21, pRb, angiogenesis and growth factor changes have been identified as having potential prognostic value [Ghoneim M, et al., 2008; Chatterjee SJ, et al., 2004]. It could be legitimately argued that the presence of a significant number of patients with bilharziasis may result in conclusions different from those of patients with cancer with typical nonbilharzial transitional cell cancer. However, on multivariate analysis the presence of bilharziasis was not an independent prognostic factor. Statistical methods based on logistic regression were traditionally used. Unfortunately complex interactions within medical data do not allow the easy use of these methods by clinicians. To overcome the shortcomings of conventional statistical methods, new models were developed to provide a means which incorporates multiple clinical, pathological properties into 1 system and achieves the best possible risk assessment. In our study the risk group stratification model identified 4 risk groups which were prospectively validated as in a previous work [Kiemeny LA, et al. 1993]. One of the potential advantages of such a system is the identification of a high risk group of patients for whom adjuvant therapy is indicated and justified [Ghoneim M, et al. 2000]. Another advantage of this model is its versatility. In the future additional prognostic variables that may influence survival significantly and independently could be included in the model once their regression estimate is determined. Construction of nomograms for survival prediction had been performed by several groups. The principle advantage of this approach is that it provides a survival probability for individual cases. Accordingly it could be maintained that they are more relevant to the patient than the group/level probabilities.

In 2006 the IBCNC published the first postoperative nomogram predicting the risk of recurrence at 5 years after radical cystectomy [Stockle M, et al. 1995]. In the same year the BCRC reported their nomogram [Bochner B, et al. 2006]. The nomogram developed by the IBCNC had a predictive index of 75% while the BCRC nomogram offered a 4% accuracy advantage. It is clear that adding more variables would improve the accuracy of any prognostic model. Shariat et al demonstrated that an abnormal profile of 2 or more abnormal markers among a panel of 5 cell cycle regulator biomarkers increased the overall predictive accuracy of the nomogram from 72.6% to 83.4% [Shariat S, et al., 2006]. Both groups demonstrated that their nomograms were more accurate and discriminating than the TNM system, solving some of the heterogeneity of outcome prediction within each stage. Nevertheless these nomograms have inherent limitations. The extent of pelvic lymph node dissection and number of lymph nodes removed were not included, although these are reported as important prognostic indices [Herr HW, et al. 2004]. In our study the operative procedure was standardized and the pathological material was revised by a single pathologist. The positive predictive index was 82.9%, which is superior to that of previously mentioned studies. Artificial neural networks are complex computational systems that can perform a large number of mental tasks. They have many advantages compared to models based on logistic regression in that they do not require statistical training, they can deal with complex nonlinear relationships and detect possible interactions among predictor variables [Tu JV, 1996]. In a recent investigation Bassi et al compared the prognostic accuracy of ANNs and logistic regression analysis for patients undergoing radical cystectomy [Bassi P,

et al. 2009]. They concluded that the ANNs accurately predicted survival and had a prognostic performance comparable to that of the logistic regression. In our study the 3 models of risk group stratification, nomogram and ANNs were compared to assess accuracy and performance characteristics.

4. Study layout

Significant variables by multivariate analysis were used for building the risk group stratification model as well as for construction of the nomogram. On the other hand all the studied factors were entered directly into the ANNs [Figure 5, 6].

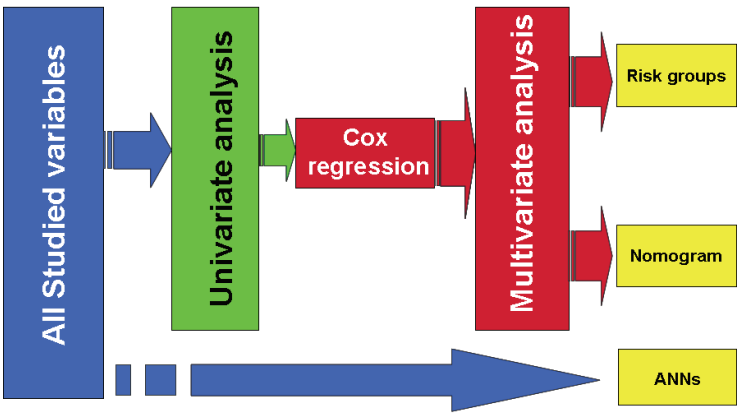


Fig. 5. Artificial neural networks model construction for predicting patient survival after radical cystectomy.

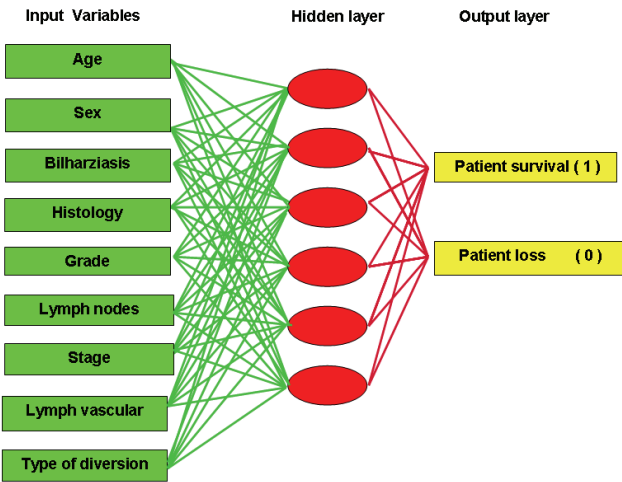


Fig. 6. Artificial neural networks architecture for predicting patient survival after radical cystectomy.

5. Risk group stratification

5.1 Model: construction and validation

The regression coefficient (B) of the significant factors on multivariate analysis was used to construct the model for risk group stratification. The algebraic sum of the regression coefficients for each case represented a proportional hazard score (Y). The possible range of Y values of patients in the reference series was divided equally into 4 groups, whereby those with the lowest Y score represented the lowest risk category and those with the highest Y score the highest risk category. Survival curves were then constructed for each of the 4 groups using the Kaplan-Meier method. Patients in the test series were classified into 4 groups according to the previously mentioned proportional hazard score (Y). For external validation survival curves were constructed for each risk category of the test series and compared with the corresponding curves of the reference series. Statistical differences were determined using the log rank test with $p < 0.05$ considered significant. In addition, the degree of discrimination was determined by computing the ROC curve [Figure 7].

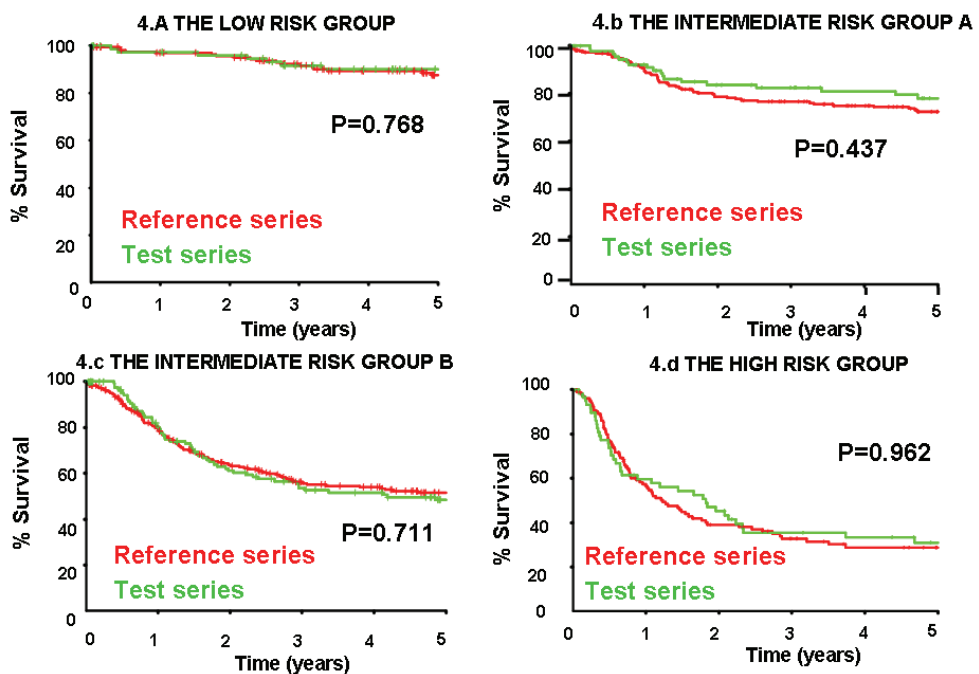


Fig. 7. External validation of risk grouping for predicting patient survival after radical cystectomy.

5.2 Nomogram: construction and validation

For construction of the nomogram the R software package version 2.7.0 was used (R foundation for statistical computing, Bell Laboratories, Lucent technologies, with Frank Harrel's Design and Hmisc libraries included). Each variable was assigned a scale of points according to prognostic significance which ranged from 0 to 100. The point values

determined for each individual case were added to give a total sum. The total sum thus calculated was correlated to the 5-year patient survival probability of the same case. Validation of the nomogram involved internal as well as external validation. Internal validation was performed by discrimination as well as calibration. The degree of discrimination was quantified by computation of the area under the ROC curve. The calibration was assessed by grouping patients with respect to nomogram predicted survival probabilities. The group means were then compared with the observed Kaplan-Meier estimates. For the discrimination and calibration steps a total of 200 bootstrap re-sampling was used to obtain less biased estimates. The test series was used for external validation. The degree of discrimination was again achieved by determination of the area under the ROC curve [Figure 8].

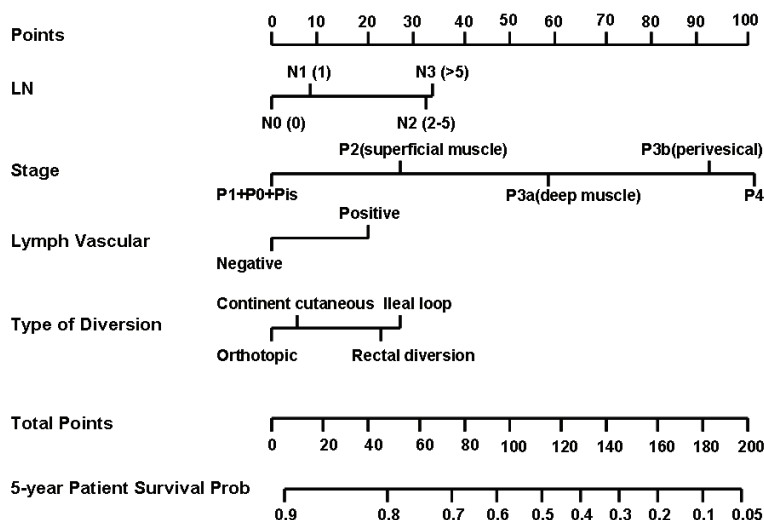


Fig. 8. Cox based nomogram for predicting patient survival after radical cystectomy.

5.3 ANNs: Construction and validation

All studied variables were used as input variables for ANN construction. The model used is commercially available software (SPSS version 16). The basis of this network is multilayer perceptrons that have feed-forward back-propagation topology. The network consists of 3 layers.

1. The input layer is of 31 neurons into which variables were entered following their normalization through a process process of standard rescaling (subtract the mean and divide by the standard deviation).
2. A hidden layer is composed of 6 neurons in which computation and differential weighing of the different variables are performed.
3. The output layer is 2 neurons into which the end results of therapy are entered (patient survival or loss).

Training was performed by a conjugate gradient descent optimization algorithm. The target was to achieve a sum of squared error of 0.01. The model starts with 1 neuron in the hidden layer, and when that neuron fails to improve the square error at the end of the training cycle

another neuron is automatically added to the hidden layer to improve the model performance. This process is terminated once a sum of square error of 0.01 has been achieved or 1,000 training cycles are completed. Six neurons in the hidden layer and 200 epochs (training cycles) were necessary to achieve this goal. After training the system was tested by internal and external validation using the same methodology as that used for the nomogram.

6. Comparison of risk group

Stratification, nomogram and ANNs

The ability to discriminate between patient survival or loss using the 3 models was compared by estimation of the area under the ROC curves (MedCalc® software version 9.3.7.0) Using these curves the sensitivity and specificity of the 3 models were also obtained at certain cutoff points. Comparisons between the performance and accuracy of the three models are provided in Figure 8 and Table 4. The ANNs outperformed the risk group stratification and nomogram models in predicting 5-year patient survival probabilities. ANNs were more sensitive and specific with a larger area under ROC curve than the other two models. The area under ROC curve for the ANNs was 0.86 while it was 0.72 for the risk groups and 0.74 for nomogram. Further more the ANNs demonstrated a superior positive as well as negative predictive values [Table 2].

	Risk groups (A-D)	Nomogram	ANNs
Area under ROC curve (95% CI)*	0.72 (0.67-0.77)	0.74 (0.69-0.78)	0.86 (0.82-0.89)
Sensitivity (95% CI)*	61.06 (54.4 - 67.5)	60.44 (53.7 - 66.9)	78.7 (72.7-83.9)
Specificity(95% CI)*	75.38 (67.1 - 82.5)	78.46 (70.4 - 85.2)	81.25 (73.4-87.6)
Positive LR#	2.48	2.81	4.20
Negative LR#	0.52	0.50	0.26
Positive predictive value	81.2	82.9	87.9
Negative predictive value	52.7	53.4	68.9
Hosmer and lemeshow Test X2 (P-value)	0.16 (0.921)	10.7 (0.219)	13 (0.112)

Table 2. Comparison of the predictive accuracy of patients survival following radical cystectomy between Risk groups, nomogram and ANNs.

Predictive models for invasive bladder carcinoma can accurately stratify patients according to risk of recurrence, progression and treatment tolerability. This enables the clinicians to make the best decision in patient counseling, selecting the optimal adjuvant therapy and follow up schedule. Among the available models our comparative study provides evidence that ANNs have several advantages compared to risk groups and nomogram models, the most important being the higher predictive accuracy, better performance and clinical applicability. However, risk groups and nomograms rely on methodologically sound and valid alternatives that cannot be ignored. Despite their advantages predictive tools cannot replace clinical judgment. Their input has to be weighed against several other considerations

such as comorbidity, cost, and social and psychic factors. Finally improvements in the current models can be attained by incorporating modern imaging tools and novel biomarkers. Moreover validation in large patient cohorts and prospective data acquisition need no emphasis.

7. Conclusions

The Artificial neural networks outperformed the risk groups and nomogram models in predicting the 5-year patient survival. Furthermore, ANN have a more satisfactory prognostic performance in terms of specificity and sensitivity compared to the other models.

8. References

- Akl A, Mostafa A, Ghoneim MA. (2008). Nomogram that predicts graft survival probability following living-donor kidney transplant. *Exp Clin Transplant*; 6: 30.
- Bassi P, Sacco E, DeMarco V, Aragona M, Volpe A. (2007). Prognostic accuracy of an artificial neural network in patients undergoing radical cystectomy for bladder cancer: a comparison with logistic regression analysis. *BJU*; 99 : 1007 – 12.
- Bochner B, Kattan M, Vora K. (2006). Postoperative nomogram predicting risk of recurrence after radical cystectomy for bladder cancer. *J Clin Oncol*; 24 (24): 3967 – 72.
- Brier ME, Ray PC, Klein JB. (2003). Prediction of delayed renal allograft function using an artificial neural network. *Nephrol Dial Transplant*; 18: 2655.
- Chatterjee SJ, Datar R, Youssefzadeh D, George B, Goebell PJ. (2004). Combined effects of p53, p21, and pRb expression in the progression of bladder transitional cell carcinoma. *J Clin Oncol*; 22 (6) : 1007 – 13.
- Cross SS, Harrison RF, Kennedy RL. (1995). Introduction to neural networks. *Lancet*; 346: 1075.
- Djavan B, Remzi M, Zlotta A, et al. (2002). Novel artificial neural network for early detection of prostate cancer. *J Clin Oncol*; 20: 921.
- Freeman RV, Eagle KA, Bates ER, et al. (2002). Comparison of artificial neural networks with logistic regression in prediction of in-hospital death after percutaneous transluminal coronary angioplasty. *Am Heart J*; 140: 511.
- Ghoneim M, Abdel-Latif M, El-Mekresh M, Abol-Enein H, Mosbah A. (2008). Radical cystectomy for carcinoma of the bladder: 2720 consecutive cases, 5 years later. *J Urol*; 180 : 121 – 7.
- Ghoneim M, El-Mekresh M, Mokhtar A, Gomha M, El-Baz M, El-Attar I. (2000). A predictive model of survival after radical cystectomy for carcinoma of the bladder. *BJU*; 85 : 811 – 6.
- Ghoneim M, El-Mekresh M, El-Baz M, El-Attar I, Ashamallah A. (1997). Radical cystectomy for carcinoma of the bladder: critical evaluation of the results in 1026 cases. *J Urol*; 158 : 393 – 9.
- Goldfarb-Rumyantzev AS, Scandling JD, Pappas L, et al. (2003). Prediction of 3-year cadaveric graft survival based on pre-transplant variables in a large national dataset. *Clin Transplant*; 17: 485.
- Graefen BM, Karakiewicz PI, Cagiannos I, et al. (2002). Validation study of the accuracy of a postoperative nomogram for recurrence after radical prostatectomy for localized prostate cancer. *J Clin Oncol*; 20: 951.

- Grossberg JA, Reinert SE, Monaco AP, et al. (2006). Utility of a mathematical nomogram to predict delayed graft function: A single-center experience. *Transplantation*; 81: 155.
- Herr HW, Faulkner JR, Grossman HB, Natale RB, deVere White R, (2004). Surgical factors influence bladder cancer outcomes. A cooperative group report. *J Clin Oncol*; 22 : 2781 – 9.
- Kattan MW.(2005). When and how to use informatics tools in caring for urologic patients. *Nat Clin Pract Urol*; 2: 183.
- Kiemeny LA, Witjes JA, Verbeek AL, Heijbroek RP, Debruyne FM.(1993). The clinical epidemiology of superficial bladder cancer. Dutch South-East cooperative urological group. *B J Cancer* ; 67 (4): 806 – 12.
- Poli F, Scalapogna M, Cardillo M, et al. (2000). An algorithm for cadaver kidney allocation based on a multivariate analysis of factors impacting on cadaver kidney graft survival and function. *Transpl Int*; 13(suppl 1): S259.
- Pollack A, Zagars G, Cole C, Dinney C, Swanson D, (1995). The relationship of local control to distant metastases in muscle invasive bladder cancer. *J Urol*; 154 : 2059 – 64.
- Ripley RM, Harris AL, Tarassenko L. (2004). Non-linear survival analysis using neural networks. *Stat Med*; 23: 825.
- Schwarzer G, Vach W, Schumacher M. (2000). On the misuses of artificial neural networks for prognostic and diagnostic classification in oncology. *Stat Med*; 19: 541.
- Shariat S, Karakiewicz P, Palapattu GS, Amiel GE, Lotan Y, et al. (2006). Nomograms provide improved accuracy for predicting survival after radical cystectomy. *Clin Cancer Res*; 12 : 6663 – 76.
- Stockle M, Meyenburg W, Wallek S, Voges GE, Rossmann M, et al. (1995). Gertenbach U, Thurof JW, Huber C, Hohenfellner R. Adjuvant poly-chemotherapy of non-organ-confined bladder cancer after radical cystectomy. *J Urol*; 153 : 47-52.
- Tu JV . (1996). Advantages and disadvantages of using artificial neural networks versus logistic regression for predicting medical outcomes. *J Clin Epidemiol* ; 49: 1225-31.
- Zapletal C, Lorenz MW, Woeste G, et al. (2004). Predicting creatinine clearance by a simple formula following live- donor kidney transplantation. *Transpl Int*; 17: 490.

Telecare Adoption Model Based on Artificial Neural Networks

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1. Introduction

Telecare will become a trend in the twenty-first century (Haux, 2006; Miller, 2007). With respect to telecare, evaluation studies are few in number. Furthermore, few studies were carried out using nonlinear structural models (such as the nonlinear neural network model) to explore the users' adoption. Therefore, the purpose of this study is to utilize the healthcare information adoption model (HIAM) that it is created first time by Huang (2010), and use it to establish telecare adoption by artificial neural networks (ANNs).

According to the healthcare information adoption model (HIAM) Huang, 2010), the research structure underlying this study is shown in Fig. 1.

H1. An individual's attitude toward using (ATT) and behavioral intention to use (BI) with respect to telecare are found to be positively associated.

H2a. Perceived ease of use (PEOU) has a direct effect on the ATT of telecare.

H2b. PEOU has a direct effect on PUB (perceived usefulness and benefits).

H3. The stronger the perceived usefulness and benefits (PUB) of telecare, the stronger is the ATT of telecare.

H4. The stronger the perceived disease threatens (PDT) (which includes perceived susceptibility and perceived severity), the stronger is the ATT of telecare.

H5. The stronger is the perceived barriers of taking action (PBTA) of telecare, the weaker is the ATT of telecare.

H6. The stronger the cues to action (CUES), the stronger is the ATT of telecare.

H6a. The stronger the external cues to action (ECUE), the stronger is the ATT of telecare.

H6b. The stronger the internal cues to action (ICUE), the stronger is the ATT of telecare.

2. Research methodology

2.1 Data collection

This study used face-to-face interview to collect senior citizens aged over 60 years, who lived in a village in Taiwan. A total of 295 valid copies of a questionnaire were obtained with males accounting for 57% of the respondents. Most of them were age groups of 60–69 (amounting to 58%). In terms of the educational level, 49% of the subjects had completed primary schools, and the average monthly income ranged between NT\$20,000 and NT\$50,000 (amounting to 71%) (US\$1 ≈ NT\$31.61).

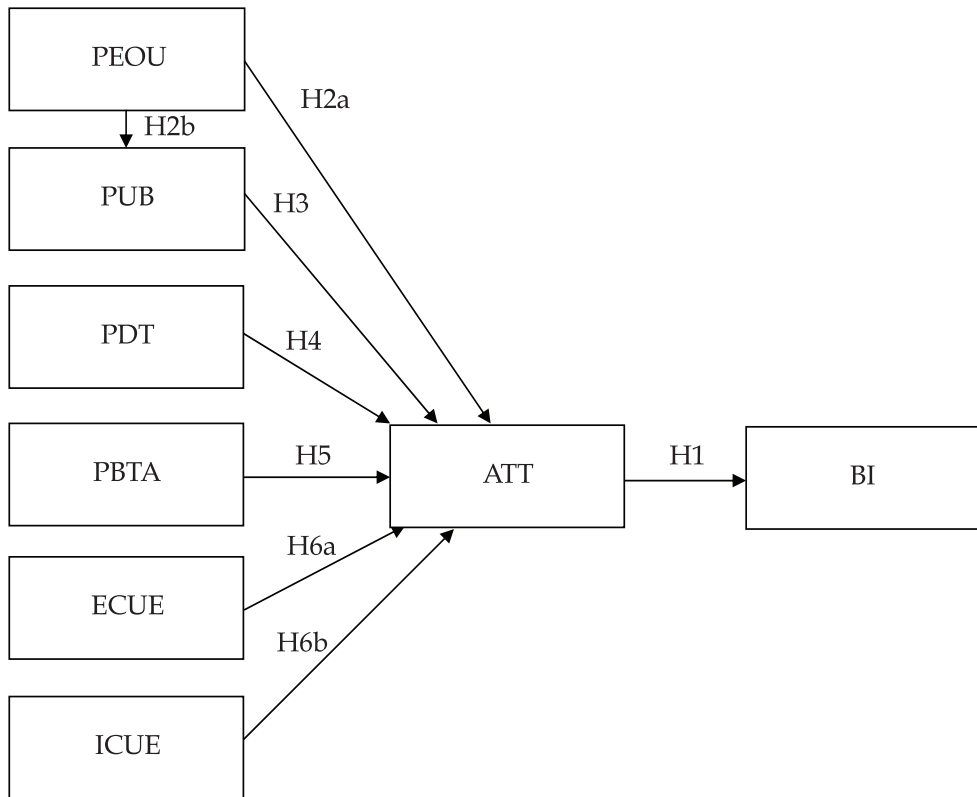


Fig. 1. Research structure. Note: perceived ease of use (PEOU), perceived usefulness and benefits (PUB), perceived disease threat (PDT), perceived barriers of taking action (PBTA), external cues to action (ECUE), internal cues to action (ICUE), attitude toward using (ATT), behavioral intention to use (BI).

2.2 Measures of the constructs

This study is based on the definition as well as on the constructs related to TAM (Davis, 1989; Davis, Bagozzi, & Warshaw, 1989) and HBM (Rosenstock, 1966; Rosenstock, 1974). The healthcare information adoption model (HIAM) includes eight constructs: perceived ease of use (PEOU), perceived usefulness and benefits (PUB), perceived disease threatens (PDT), perceived barriers of taking action (PBTA), external cues to action (ECUE), internal cues to action (ICUE), attitude toward using (ATT), and behavioral intention to use (BI). The operationalization and sources of measurement items in this study are shown in Table 1. All evaluation items employ a five-point Likert-type scale for measurement, where 1, 2, 3, 4, and 5 indicate "strongly disagree," "disagree," "fair," "agree," and "strongly agree," respectively.

2.3 Reliability and validity analysis

The internal consistency was assessed by using Cronbach's alpha. The alpha coefficients of all the seven constructs were higher than the benchmark of 0.6, as suggested by Bagozzi and Yi (1988). This demonstrated the reliability of each construct. Besides, the questionnaires of this study have considerable content validity (refer to Table 1).

Categories	Measure
Perceived ease of use (Davis, 1989; Davis, Bagozzi, & Warshaw, 1989)	
PEOU1	I find that using telecare is simple.
PEOU2	I find that telecare is easy to learn.
PEOU3	I find that telecare is easily understandable and clear.
PEOU4	Overall, I find that using telecare is convenient.
Perceived usefulness and benefits (Davis, 1989; Davis, Bagozzi, & Warshaw, 1989; Rosenstock, 1966; Rosenstock, 1974)	
PUB1	I find that using telecare is helpful in monitoring health.
PUB2	I find that using telecare makes me safer in my daily life.
PUB3	Telecare can enhance my level of convenience in accessing medical care service.
PUB4	Telecare can enhance the quality of my life.
PUB5	Overall, I find telecare highly useful.
Perceived disease threat (Rosenstock, 1966; Rosenstock, 1974)	
PDT1	I find that I can fall ill easier than others.
PDT2	I find that I can suffer from high blood pressure, diabetes, heart disease and other chronic diseases in the future.
PDT3	I find that my health is deteriorating.
PDT4	I find that I can suffer from high blood pressure, diabetes, heart disease, and other chronic diseases in the future and could be forced to change my previous way of life.
Perceived barriers of taking action (Rosenstock, 1966; Rosenstock, 1974)	
PBTA1	For me, the cost of telecare is a very heavy burden to bear.
PBTA2	I am concerned that telecare is not adequately secure

Categories	Measure
	and that it might lead to the leak or abuse of my personal information.
PBTA3	I am concerned that telecare would violate my privacy.
PBTA4	I am concerned that the accuracy and reliability of the instruments of telecare are not high enough.
PBTA5	I am concerned that I might forget to use the telecare.
PBTA6	I am concerned that I might be too busy to use the telecare equipment.
External cues to action (Rosenstock, 1966; Rosenstock, 1974)	
ECUE1	Relatives encourage and support me to use telecare.
ECUE2	Friends encourage and support me to use telecare.
ECUE3	Medical care personnel encourage and support me to use telecare.
ECUE4	Media endorses the use of telecare.
ECUE5	I have always obtained health-related information from TV, newspapers, and other media.
ECUE6	I have always obtained health-related information from the Internet.
Internal cues to action (Rosenstock, 1966; Rosenstock, 1974)	
ICUE	How many times did you fall sick in the last three months?
Attitude toward using (Taylor & Todd, 1995)	
ATT1	I like using telecare.
ATT2	Overall, I consider telecare to be just right.
ATT3	In my old age, using telecare would be ideal.
Behavioral intention to use (Taylor & Todd, 1995)	
BI1	Overall, I am highly willing to use telecare.
BI2	If necessary, I would use telecare often.
BI3	In my old age, I am willing to use telecare.
BI4	In my old age, I would use telecare often.

Table 1. Measuring items used in this study

2.4 Data analysis methods

2.4.1 Research framework of the Artificial Neural Network (ANN)

Of the various Artificial Neural Network (ANN) models, the Back-Propagation Network (BPN) is the simplest model and the easiest to understand. The feed-forward neural network (FFNN) has a general architecture, as shown in Fig. 2 (Kumar, 2007). The development procedure of ANN can be divided into two processes: learning process, and recalling process. A large number of examples are required in both processes as the input data, in order for the ANN operation to complete at each process. During the learning process, the errors of trained examples with targets are used to adjust the weights of ANN by various efficient learning algorithms. The weights are updated after processing all examples. The final weights are adjusted to meet the minimization of errors we desired. Then, the trained weights are stored in the network after learning is completed. In the second process, testing examples are used to verify the validity of the network. Furthermore, neural network is utilized in case study to infer the results.

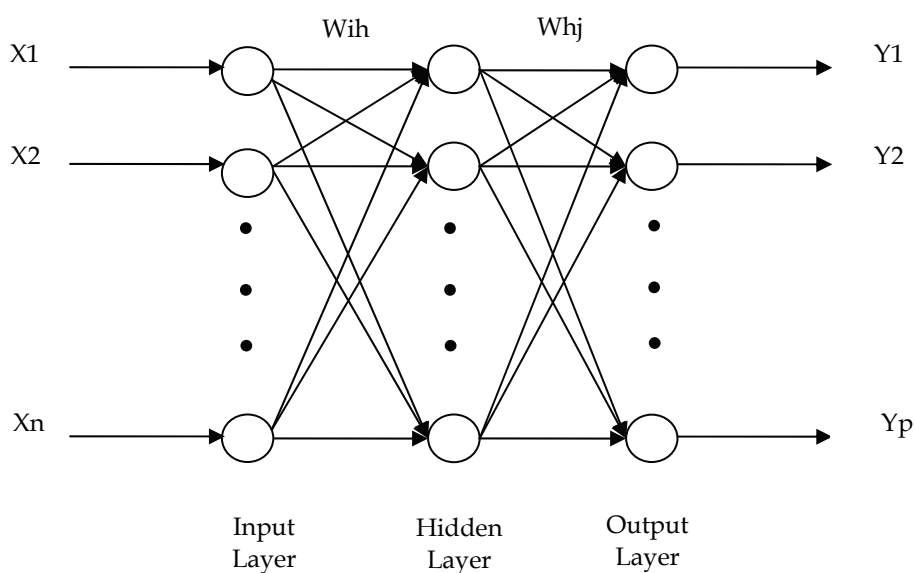


Fig. 2. Feed-forward neural network with n inputs and p outputs

This study adopted three ANN frameworks that use a single hidden layer. They are shown in Figs. 3~5, and Back-Propagation Network (BPN) was used to adjust the network connection weights. The parameters of three ANNs are summarized in Tables 2~4. In Fig. 3, PEOU, PUB, PDT, PBTA, ECUE, and ICUE are input layers, ATT is the output layer. In Fig. 4, PEOU is the input layer, and PUB is the output layer. In Fig. 5, ATT is the input layer, and BI is the output layer.

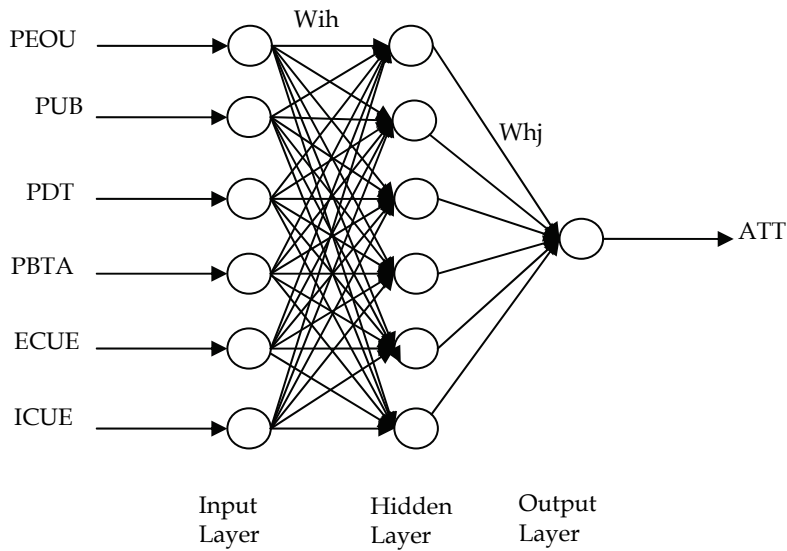


Fig. 3. ANN research framework diagram (A). Note: perceived ease of use (PEOU), perceived usefulness and benefits (PUB), perceived disease threat (PDT), perceived barriers of taking action (PBTA), external cues to action (ECUE), internal cues to action (ICUE), attitude toward using (ATT).

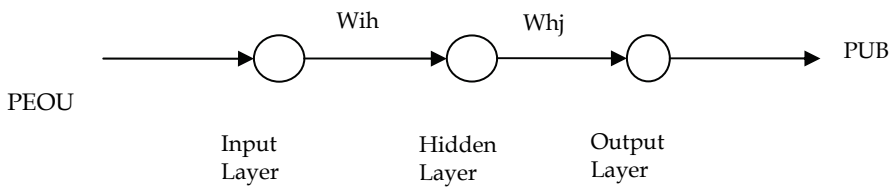


Fig. 4. ANN research framework diagram (B). Note: perceived ease of use (PEOU), perceived usefulness and benefits (PUB).

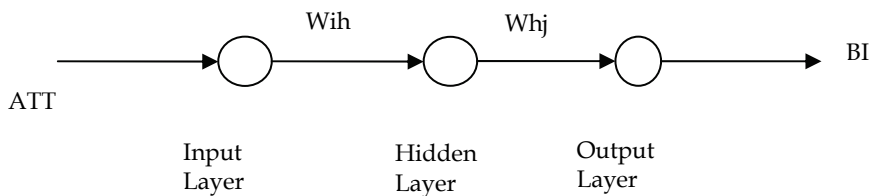


Fig. 5. ANN research framework diagram (C). Note: attitude toward using (ATT), behavioral intention to use (BI).

	Input	Hidden	Output
Number of neurons	6	6	1
Signal function	linear	sigmoidal	linear
Neuron index range	$i=0, \dots, 5$	$h=0, \dots, 5$	$j=0$
Weights (including bias)	$\rightarrow W_{ih} \rightarrow$		$\rightarrow W_{hj} \rightarrow$

Table 2. The parameter for ANN research framework diagram (A)

	Input	Hidden	Output
Number of neurons	1	1	1
Signal function	linear	sigmoidal	linear
Neuron index range	$i=0$	$h=0$	$j=0$
Weights (including bias)	$\rightarrow W_{ih} \rightarrow$		$\rightarrow W_{hj} \rightarrow$

Table 3. The parameter for ANN research framework diagram (B)

	Input	Hidden	Output
Number of neurons	1	1	1
Signal function	linear	sigmoidal	linear
Neuron index range	$i=0$	$h=0$	$j=0$
Weights (including bias)	$\rightarrow W_{ih} \rightarrow$		$\rightarrow W_{hj} \rightarrow$

Table 4. The parameter for ANN research framework diagram (C)

2.4.2 Data analysis steps

In this study, we constructed, performed training on and tested the ANN model with Matlab Neural Network Toolbox.

- Step 1. Normalize all data into $[0, 1]$, including PEOU, PUB, PDT, PBTA, ECUE, ICUE, ATT, and BI.
- Step 2. The trained algorithm of the Levenberg-Marquardt backpropagation (Hagan and Menhaj, 1994) is used to adjust the weight with MSE (Mean Square Error).
- Step 3. Randomly choose the initial weights.
- Step 4. Proceed with the ANN model training, 2/3 of the total samples (196 samples) are chosen as training examples. Each training session for all 196 data is deemed as an epoch. Parameters are updated during this period to obtain the optimal ANN models.
- Step 5. The final optimal ANN model is acquired after completing the training. Tests (a) and (b) are employed to examine the stability and reliability of the ANN; tests (c) and (d) are employed to estimate the effect of individual input variables on output variables.
 - a. Input the 196 samples which have been used in training to obtain the 196 estimated output values. The errors between the estimated and raw data are shown in Fig. 6. The statistical result of errors, including mean, standard deviation (S.D.), and Root of Mean Square (RMS) are shown in Table 5.
 - b. Input the remaining 1/3 of the samples (99 samples), which have not been used in training to obtain the 99 estimated output values. The errors between the estimated and

raw data are shown in Fig. 6. The statistical result of errors, including mean, standard deviation (S.D.), and Root of Mean Square (RMS) are shown in Table 5.

- c. Take the dependence of ATT on PEOU as an example, the correlation with ATT could be obtained by setting PEOU to 1 and other input variables to zero. Thus, the test vector of input is [1 0 0 0 0]. Similarly, the correlation between all input variables and output variables could be acquired (Table 5).
- d. Besides test (c), there is another approach to identify the effect of one input variable on output variables, and it is by fixing other input variables to zero. Thus, continuous discrete input is used with one input variable increasing from 0 to 1 at 0.01 steps. Similarly, the correlation between all input variables and output variables could be acquired. The results are presented in Figs. 7~9. Therefore, the dependence of all output variables on input variables can be identified.

3. Results

Table 5 shows the dependence of ATT on PEOU, PUB, PDT, PBTA, ECUE, and ICUE. As presented in the ANN research framework diagram (A) of Fig. 3, the error mean (S.D) and Root of Mean Square (RMS) for the training examples are 0.1304 (0.0793) and 0.1525, respectively; the error mean (S.D) and Root of Mean Square (RMS) for the testing examples are 0.1282 (0.0722) and 0.1468, respectively. As to the dependence of PEOU on PUB, as presented in the ANN research framework diagram (B) of Fig. 4, the error mean (S.D) and Root of Mean Square (RMS) for the training examples are 0.1035 (0.0962) and 0.1411, respectively; the error mean (S.D) and Root of Mean Square (RMS) for the testing examples are 0.0475 (0.0438) and 0.0643, respectively. As to the dependence of ATT on BI, as presented in the ANN research framework diagram (C) of Fig. 5, the error mean (S.D) and Root of Mean Square (RMS) for the training examples are 0.1091 (0.0847) and 0.1380, respectively; the error mean (S.D) and Root of Mean Square (RMS) for the testing examples are 0.1211 (0.0948) and 0.1532, respectively. As shown, the errors are acceptable. Furthermore, it could also be inferred from Fig. 6 that the errors for the training and testing examples in both ANN research framework diagram (A), (B) and (C) are acceptable. And, it is means that the model of ANN constructed in this study is stable and reliable.

Path	Correlation	Train error		Test error	
		Mean (S.D.)	RMS	Mean (S.D.)	RMS
PEOU → ATT	0.1571				
PUB → ATT	0.2606				
PDT → ATT	0.4319				
PBTA → ATT	-0.0119	0.1304 (0.0793)	0.1525	0.1282 (0.0722)	0.1468
ECUE → ATT	Minimum				
ICUE → ATT	Minimum				
PEOU → PUB	0.9867	0.1035 (0.0962)	0.1411	0.0475 (0.0438)	0.0643
ATT → BI	0.9628	0.1091 (0.0847)	0.1380	0.1211 (0.0948)	0.1532

Table 5. Analysis results of the ANN model.

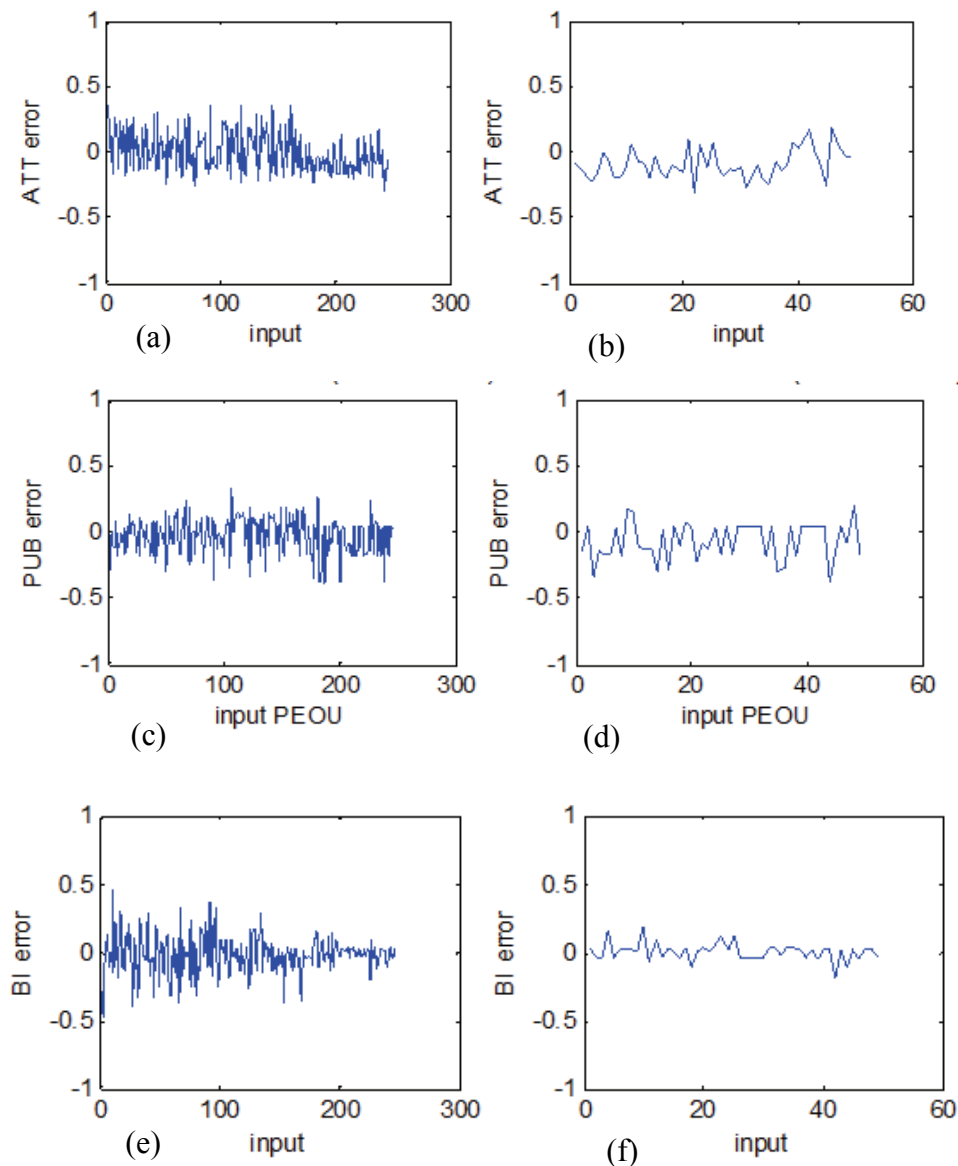


Fig. 6. Errors of the training and testing examples in both ANN research framework diagram (A), (B) and (C). Note: (a) Errors of the training examples in ANN research framework diagram (A), (b) Errors of the testing examples in ANN research framework diagram (A), (c) Errors of the training examples in ANN research framework diagram (B), (d) Errors of the testing examples in ANN research framework diagram (B), (e) Errors of the training examples in ANN research framework diagram (C), (f) Errors of the testing examples in ANN research framework diagram (C).

As shown in Fig. 7, ATT varies positively with BI and thus H1 is supported. As shown in Fig. 8, PEOU varies positively with PUB and thus H2b is supported. As shown in Fig. 9, PEOU, PUB, PDT and ECUE have a significantly positive effect on ATT, and thus H2a, H3, H4, and H6a are supported. PBTA has a significantly negative effect on ATT which indicates that H5 is also supported. However, ICUE has no significant effect on ATT which leads to the rejection of H6b.

The correlation coefficients in Table 5 showed that, among the effects of PEOU, PUB, PDT, PBTA, ECUE and ICUE on ATT, that of PDT is the most significant (with a correlation coefficient 0.4319) and positive; followed by PUB and PEOU (with a correlation coefficient of 0.2606 and 0.1571, respectively) which are also positive. The following is the effect of PBTA (with a correlation coefficient of -0.0119) on ATT, which is negative. However, ATT does not vary significantly with ECUE and ICUE.

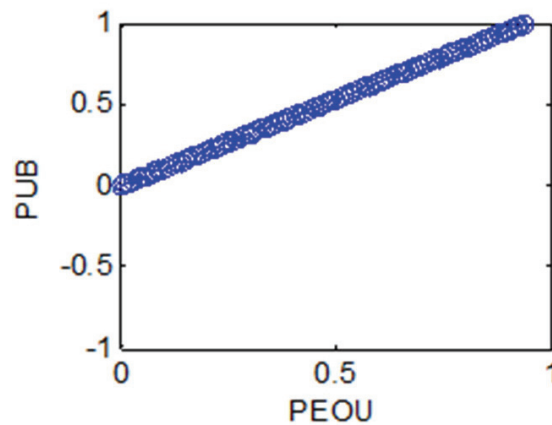


Fig. 7. Effects of ATT on BI. Note: attitude toward using (ATT), behavioral intention to use (BI).

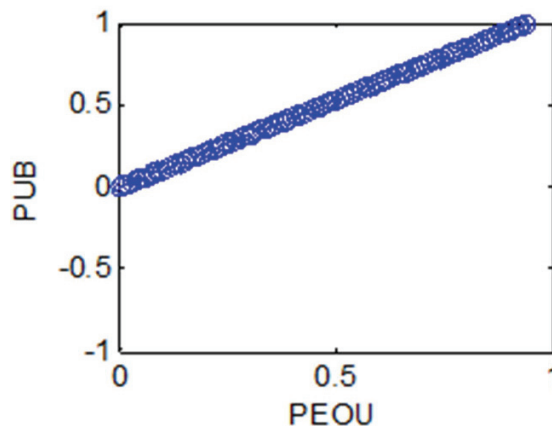


Fig. 8. Effects of PEOU on PUB. Note: perceived ease of use (PEOU), perceived usefulness and benefits (PUB).

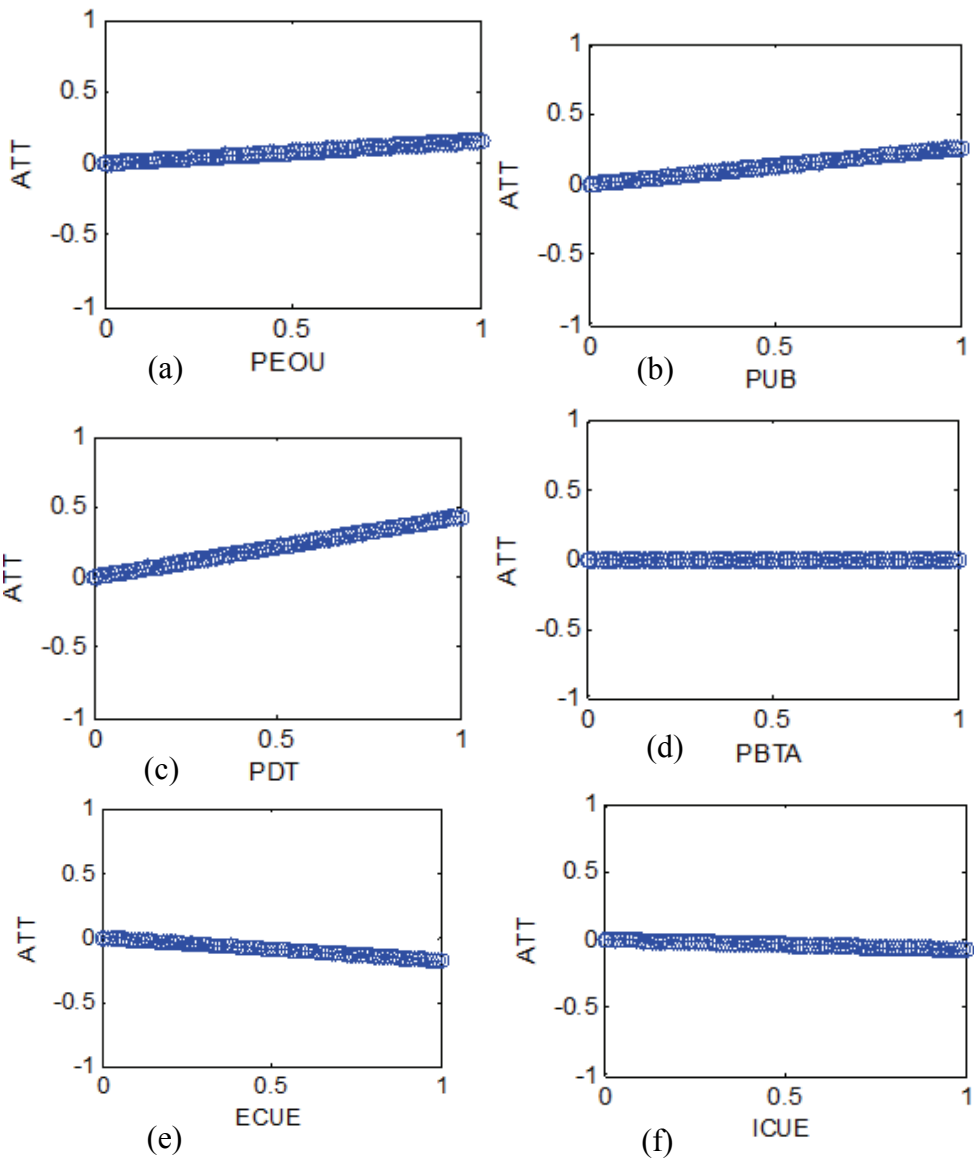


Fig. 9. Effects of PEOU, PUB, PDT, PBTA, ECUE and ICUE on ATT. (a) Effects of PEOU on ATT, (b) Effects of PUB on ATT, (c) Effects of PDT on ATT, (d) Effects of PBTA on ATT, (e) Effects of ECUE on ATT, (f) Effects of ICUE on ATT. Note: perceived ease of use (PEOU), perceived usefulness and benefits (PUB), perceived disease threat (PDT), perceived barriers of taking action (PBTA), external cues to action (ECUE), internal cues to action (ICUE), attitude toward using (ATT).

4. Conclusions

The results indicated that it is feasible to construct a model with ANN, and identify telecare adoption model by using ANN based on the healthcare information adoption model (HIAM) that is created first time by Huang (2010). The finding may offers significant reference for subsequent studies. Besides, the most effective way to enhance the adoption of telecare is to improve the perceived disease threat (PDT) for potential users.

5. References

- Bagozzi, R.P., & Yi, Y. (1988). On the Evaluation of Structural Equation Models. *Journal of Academic Market Science*, 16 (1), 74 – 94.
- Davis, F.D. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Quarterly*, 13 (3), 319 – 339.
- Davis, F.D., Bagozzi, R.P., & Warshaw, P.R. (1989). User Acceptance of Computer Technology: A Comparison of Two Theoretical Models. *Management Science*, 35 (8), 982 – 1003.
- Hagan, M.T. and Menhaj, M. (1994) Training feedforward networks with the Marquardt algorithm, *IEEE Transactions on Neural Networks*, 5 (6), 989–993.
- Haux, R. (2006). Health Information Systems—Past, Present, Future. *International Journal of Medical Informatics*, 75, 268 – 281.
- Huang J.C. (2010), Remote health monitoring adoption model based on artificial neural networks. *Expert Systems with Applications*, 37 (1), 307–314.
- Kumar, S. (2007). *Neural Networks: A Classroom Approach*, 4rd ed., Tata McGraw-Hill, New Delhi.
- Miller, E.A. (2007). Solving the Disjuncture between Research and Practice: Telehealth Trends in the 21st Century. *Health Policy*, 82 (2), 133 – 141.
- Rosenstock, I.M. (1966). Why People Use Health Services. *Milbank Memorial Fund Quarterly*, 44, 94 – 124.
- Rosenstock, I.M. (1974). Historical Origins of the Health Belief Model. *Health Education Monographs*, 2, 1 – 8.
- Taylor, S., & Todd, P.A. (1995). Understanding Information Technology Usage: A Test of Competing Models. *Information Systems Research*, 6 (2), 144 – 176.

Effectiveness of Artificial Neural Networks in Forecasting Failure Risk for Pre-Medical Students

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1. Introduction

The registration department in universities usually regulates a set of rules in order to accept new students. These rules are set to select applicants who have the abilities and skills to pursue and succeed in their academic career in a particular field of studies. Acceptance to the College of Medicine and Medical Sciences at the Arabian Gulf University is no exception. The admission body of the college firstly makes sure that qualifications and other particulars of the individual students fulfil the requirements of the college admission according to its rules and conditions. Secondly, they arrange acceptance tests that result in accepting, or rejecting applicants. However, it has been noticed throughout the years that the decisions based solely on the results of these acceptance tests, including an additional personal interview, are not sufficient. A number of students still fail in the first year of education (premedical stage). This creates many problems both to the University and to the students; whether administrative, financial and/or psychological. The Artificial Neural Networks techniques can play a role in this process. However, it should use the same set of entrance rules but with different weightings. This research aims at modelling an effective and intelligent tool that can help the decision makers to have better judgment on possible ability of each individual student to succeed in the premedical stage and hence in his/her subsequent studies in the medical school.

2. Artificial Neural Networks approach

Artificial Neural Networks are sophisticated modeling techniques; capable of modeling extremely complex functions. Cascade Correlation Network (CCN) architecture of Artificial Neural Networks is used to deal with this research problem. It is a supervised learning algorithm developed by Fahlman and Lebiere [1]. It is an attempt to overcome certain drawbacks and limitations of popular Back Propagation Learning Algorithm developed by Rumelhart et al. [2]. It is trained using the Quick Propagation Algorithm (QPA) which is the enhanced version of Back Propagation (BP). It is a convenient approach to solve problems since it is an ontogenic neural network that generates its own topology during training. It overcomes the limitation of BP which is slow at learning from examples. A BP network may

require many thousands of *epochs* to learn the desired behaviour from examples. An *epoch* is defined as one pass through the entire set of training examples. Previous studies involve some researches close to this research. Predicting MBA student performance by evaluating the ability of three different models, namely, logistic regression, probability analysis, and neural networks; is reported by Fahlam and Lebiere [3]. The result was that the neural network model performs better than statistical models. Back-Propagation Neural Network was used in selecting surgical residents via the National Residency Matching Program applied to medical students interested in surgery in their fourth year of study. It is used to facilitate the surgical residents' selection process using 36 variables. The study showed that neural networks are capable of producing significantly better results than traditional statistical approaches [4]. Many other studies (Entwistle, 1988; Ardila, 2001; Busato et al., 1999, Furnham et al., 1999) were undertaken in order to try to explain the academic performance or to predict the success or the failure of students; they highlighted a series of explanatory factors associated to the student [5]-[8].

3. Implementation of ANN in forecasting the failure risk of applicants to Medical College

Premedical students' data including their grades at the end of first academic years were collected from both registration department and Medical College Acceptance Committee. Data for the academic years: 2003/2004, 2004/2005, and 2005/2006 were available at the beginning of this research work. Students' academic results of years 2003/2004 and 2004/2005 are used as training/ validation sets. While those of years 2004/2005 and 2005/2006 are used for forecasting (testing).

3.1 Neural inputs and output

The main variables affecting the process of selecting medical students were identified. The main input variables used and involved in the acceptance process are shown in Table 1. The notations column is used for abbreviation.

	Variable	Description	Notation
1	High School Grade	Average grade in high school certificate.	Sec. GPA
2	Science Grade	Average grade of science subjects at high school.	Sec. Sc.
3	English Grade	Student grade at English test carried out at AGU's English department.	ENG
4	MCAT	Medical College Acceptance Test in science provided by AGU medical staff.	MCAT
5	Interview	Measure stability and test the personal quality of applicant	Interview

Table 1. Input variables for ANN forecasting model

The following input variables were added in order to test if they have any significant effect on the forecasted results. The added variables are shown in Table 2, and the Output variable is shown in Table 3.

	Variable	Description	Notation
1	Gender	Student's sex	Gender
2	Age	Student's age	Age
3	Marital Status	Student's marital status	Marital
4	Graduate Year/ Batch	Current or previous year graduate (0 or 1)	Batch

Table 2. Added input variables for forecasting model

	Variable	Description	Notation
1	Accumulative GPA	Forecasted student's Grade (Out of 4.00 points)	Ac. GPA

Table 3. Output variable for ANN forecasting model

As there are many variables involved in the acceptance process, Cascade Correlation Network technique was chosen to build the forecasting model. Two types of networks have been built for each data set: One network with only five inputs as in Table 1. The other network with nine inputs is formed by including those in Table 2.

3.2 Forecasting model

Neural network software tool used in this research is "Forecaster XL' (v. 2.3)". It uses the Cascade Correlation supervised *learning algorithm* - trained using Quick-Propagation. It divides the rows of inputs x_i and their related output y_i into training and validation subsets. 83% of rows are used for training, while 17% of rows are used for validating the model. Network training is designed to adjust network weights for maximizing predictive ability and minimizing forecasting error. Validation subset is that part of data used to tune the network topology or network parameters other than weights. For example, it is used to define the number of hidden units to detect the moment when predictive ability of neural network started to deteriorate. Test subset is a part of data set used to test how well the trained neural network forecasts using new data. Test subset is used after the network is trained, and hence ready to forecast. This subset is not used during training and thus consists of a new data to the model. The regular Sigmoidal activation function is used. It is expressed in relation to the input variable "x" as follows: $1 / (1 + \exp(-x))^{0.5}$, (Where x is the input).

3.3 Testing the performance of the trained model

After developing the model through training and validating, goodness of model fitting is examined statistically. Mean Square Error (MSE) is calculated after each training. The predictive model is then identified as a good one if the MSE is sufficiently small i.e. close to 0. The model with minimum MSE is identified as the best predictive model. A brief analysis of this study is presented in the following.

4. Results and discussions

The Artificial Neural Networks model is built based on students' historical data. The input variables are identified together with their related output variable (Input / Output variables

pair). A typical example of the prepared data Inputs for a particular student are: Gender (F), Age (18), marital (S), Batch (0), Sec. GPA (93.90), Sec. Sc. (91.50), MCAT (60.00), ENG (95.00), and Interview (88.10). A typical related output is Ac. GPA (3.104). The proposed architecture for the Cascade Correlation Network forecasting model is illustrated in Figure 1.

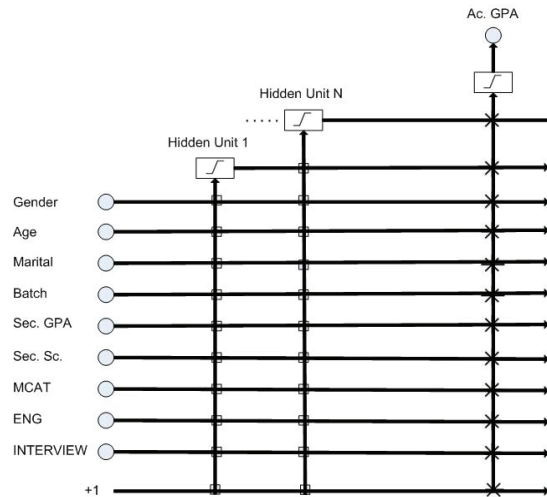


Fig. 1. The architecture of Cascade correlation network to forecast academic performance

Data reprocessing is done by analyzing the input and output variables. Data was cleaned by removing any column of data that was identified as unsuitable for neural network (e.g. that contains text or too many missing values or repeated values). Numeric columns are marked with *numeric* mark. Categorical columns are marked as *categorical*. Table 4 shows input and output variables used in this research work as recognized by the used software.

INPUT/OUTPUT Columns	Location	TYPE
Gender	INPUT	Category
Age	INPUT	Number
Martial	INPUT	Category
Batch	INPUT	Number
Sec. GPA	INPUT	Number
Sec. Sc.	INPUT	Number
MCAT	INPUT	Number
ENG	INPUT	Number
Interview	INPUT	Number
Ac. GPA	OUTPUT	Number

Table 4. Types of Input/ Output columns

Table 5 shows an analysis of the data for training process. The data has been classified into training set and validation set. Some columns (input variables) were excluded because it proved that they have no effect on the training process. Specifically, namely; marital and Batch input variables. 'Marital' was excluded since all the students were 'single' and hence it

has no contribution to the network output. 'Batch' variable was also excluded since all students enrolled are fresh graduates, and hence the variable is not an informative one.

Analysis parameters	2003/2004 (9) inputs	2003/2004 (5) inputs	2004/2005 (9) inputs	2004/2005 (5) inputs
No. of students Tested	76	76	77	77
Training Set (83%) of total rows	63	63	64	64
Validation Set (17%) of total rows	13	13	13	13
Variables Excluded/ Uninformative or too noisy	Marital Batch	-	Batch	-

Table 5. Analyzing data for training

Different network models have been built using each of the four academic year data as training data. The target is to study the importance of the input to the network being built. The number of training iterations made by each network in order to achieve the minimum possible error of the output result is shown in Table 6. The minimum error reached within few seconds in the software. This is when the percentage of error becomes very low and generalization loss stays within acceptable limit. A performance report then gives the average errors as well as tolerance and number of *Good* and *Bad* forecasts for training and test sets. The quality of the model could be determined by analyzing R-Squared and the Correlation values.

Academic year (no. of variables)	2003/2004 (9)	2003/2004 (5)	2004/2005 (9)	2004/2005 (5)
Number of training iterations	5070	2260	5029	5009
Training stop reason	Generalization loss & Maximum no. of iterations achieved.	Error reduction became too low.	Generalization loss & Maximum no. of iterations achieved.	Generalization loss & Maximum no. of iterations achieved.

Table 6. Neural networks Information

The progressive improvement of network Mean Square Error (MSE) during training is shown in Figures 2 and 3. Figure 2 shows the training MSE using data of the academic year 2003/2004 with 9 input variables, the lowest MSE reached at the early stages of training. Figure 3 shows the training MSE using data of academic year 2003/2004 with 5 input variables. The network restores the structure that resulted in the lowest error (this is at iteration number 1956).

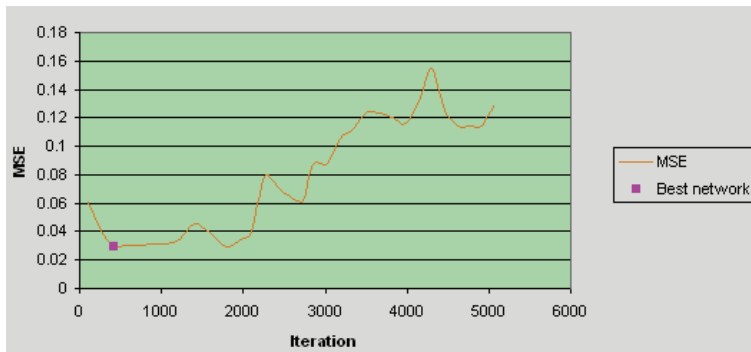


Fig. 2. Training errors (local minima errors) for academic year 2003/2004 with 9 inputs

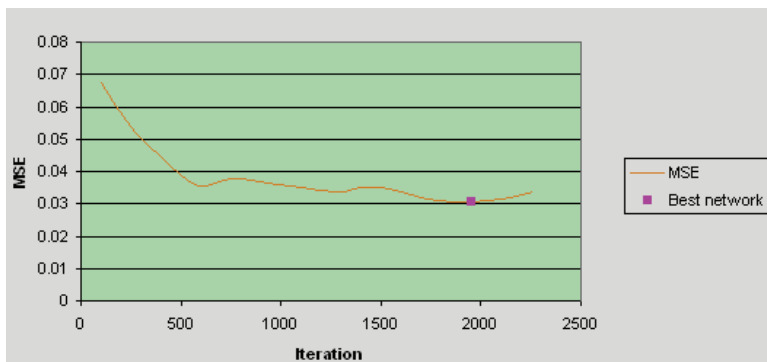


Fig. 3. Training errors (local minima errors) for academic year 2003/2004 with 5 inputs

The resulted network structure is presented in Table 7 for each academic year. Since input data are numerical and categorical; numerical inputs would be recognized as one input in the training set, while categorical inputs would be recognized as two inputs. The table shows the number of hidden units in each network structure. The categorical inputs such as (male/ female) is represented by $\{-0.4, 0.4\}$, while the forecasted output (target) is the GPA. Hidden units vary when using a different academic year as training set in order to achieve the lowest error.

(no. of variables)	2003/2004 (9)	2003/2004 (5)	2004/2005 (9)	2004/2005 (5)
Number of inputs	8	5	10	5
Number of hidden units	2	11	5	12
Number of outputs	1	1	1	1

Table 7. Neural networks Information

In academic year 2003/2004 with 9 inputs, the input variables Marital and Batch were excluded as discussed before. Therefore 7 inputs were left. And since Gender is categorical, it had been recognized as 2 inputs and may take 'Male' or 'Female' values. Because of that, the final number of input variables would be 8. In academic year 2004/2005 with 9 inputs, the input variable Batch was excluded, hence, the final number of input variables was 10. In Table 7, the number of hidden units varies among the different networks to achieve the lowest error for each network. Inputs importance (Sensitivity Analysis) is shown in Table 8. Table 9 shows the obtained results in each trained network.

INPUT (no. of variables)	2003/2004 (9)	2003/2004 (5)	2004/2005 (9)	2004/2005 (5)	2005/2006 (9)	2005/2006 (5)
Gender	2.195	Excluded	5.724	Excluded	0.374	Excluded
Age	4.293	Excluded	1.435	Excluded	6.406	Excluded
Martial	0	Excluded	0.175	Excluded	0	Excluded
Batch	0	Excluded	0	Excluded	2.956	Excluded
Sec. GPA	63.070	67.429	8.531	17.121	51.137	42.584
Sec. Sc.	1.207	4.168	52.540	65.124	16.407	10.052
MCAT	16.171	13.492	12.629	3.972	10.658	13.910
ENG	11.913	13.744	13.087	9.859	5.875	21.747
Interview	1.151	1.167	5.879	3.924	6.188	11.706

Table 8. Input Importance percentages for trained networks

The previously trained networks are used to forecast the academic results (Ac. GPA) for a different academic year. The trained network using data of academic year 2003/2004 was used to forecast Ac. GPA for academic year 2004/2005, and then to forecast students' Ac. GPA for academic 2005/2006. The trained network using data of academic year 2004/2005 was then used to forecast the GPA for academic year 2005/2006. A third network was trained using the combined data of both academic years 2003/2004 and 2004/2005, and then used to forecast the Actual GPA for academic year 2005/2006. As seen from Table 8, the Secondary GPA (Sec. GPA) is the most important input in both mentioned years. When comparing the forecasted and actual grades of year 2005/2006, the lowest MSE resulted when trained network using grades of 2003/2004 (as in Table 9 below).

	Forecasted sets (test sets)		
Training sets	2003/2004	2004/2005	2005/2006
2003/2004	-	0.526	0.430
2004/2005	-	-	0.453
2003/2004 and 2004/2005	-	-	0.495

Table 9. Mean Square Error (MSE) for forecasting Ac. GPA for the following academic year

5. Conclusion

The target of this research work is to study the effectiveness of Artificial Neural Networks in forecasting failure risk for pre-medical students at the Arabian Gulf University. The model structure for forecasting uses the Cascade Correlation Networks technique, and trained using the Quick Propagation algorithm. Different models with different input variables were built. The data of the academic years 2003/2004, 2004/2005, and 2005/2006 were available. The best accuracy was achieved with the model that had five input variables and trained using the academic data of the year 2003/2004 data to forecast GPA for academic year 2005/2006. MSE is 0.430, and was enhanced to 0.354 when increasing number of training iterations. Statistical analysis is carried out to ensure the validity of the forecasted GPA. Accepted MSE in forecasting the results of the students in the academic year 2005/2006 were reached using trained network with data of academic year 2003/2004. In this case, there is no significant mean differences for those academic years forecasted results with $p < 0.000$. *This proved that ANN is a valid tool for forecasting the results of the students.* Good result obtained because both data of academic years 2003/2004 and 2005/2006 had the same input importance, therefore there were similarity in the characteristics of their data. It has also been seen that Secondary GPA has the highest influence on the GPA for the premedical students in the academic years 2003/2004 and 2005/2006. A future research work has been suggested to take into consideration added input parameters such as psychological and physiological issues.

6. References

- Fahlman, S. E. and Lebiere, C. "The cascade-correlation learning architecture" *Advances in Neural information Processing Systems 2*, D. S. Touretzky, Ed. Morgan Kaufmann Publishers, San Francisco, CA, 524-532, 1990.
- Rumelhart, D. E., Hinton, G. E., and Williams, R. J. "Learning Internal Representations by Error Propagation", *Parallel Distributed Processing: Explorations in the Microstructure of Cognition*, MIT Press, 1986.
- Naik B. and Ragothaman S., "Using neural network to predict MBA student success", *College Student Journal*, 38(1):143-149, 2004.
- Aggarwal A. K., Travers S. D., and Scott-Conner C. E. H. "Selection of surgical residents: A neural network approach." *Cybernetics and Systems* 31:417-430, 2000.
- Entwistle N. , "Motivational factors in students' approaches to learning", In R.R. Schmeck (Ed.), *Learning strategies and learning styles*. 21-51. New York: Plenum, 1988.
- Ardila A., "Predictors of university academic performance in Colombia", *International Journal of Educational Research*, vol. 35, pp. 411-417, 2001.